MODULE 1:

INSTRUMENTATION

DRAWINGS & SYMBOLS

INSTRUMENTATION DRAWINGS & SYMBOLS

Objectives

At completion of this module, the trainee will have understanding of:

1. Instrumentation symbols and abbreviations,
2. Structure of instrument codes (Tag Numbers),
3. Process Block Diagram
4. Process Flow Diagram (PFD)
5. Piping and Instrumentation Drawing (P&ID)
6. Electrical Loop Drawing
7. DCS (I/O) Input & Output Loop Drawing
8. Pneumatic Loop drawing
9. Cause and Effect Diagram
10. Functional Logic Diagram
11. Instrument Installation Hook-Up Diagram (Pneumatic or Process)

Introduction

This manual has been written to provide the reader with an understanding of the various codes and symbols used to illustrate instrumentation in facilities designed for the production of oil, gas and associated hydrocarbon products.

Instrument codes and symbols are graphically represented in technical diagrams such as Process Flow Schemes (PFD) and in Pipeline and Instrumentation Drawings (P&ID).

Such drawings are of particular importance to operation and maintenance technicians who are required to understand the process control systems associated with an installation.

However, difficulties are often experienced primarily due to the existence of several systems of instrument codes and symbols which have been developed over the years by owners and contractors who carry out the engineering design, construction and operation of processing installations.

Purpose of Codes and Symbols

The primary purpose of using codes and symbols is to enable the various instrument functions required in a process to be clearly and concisely represented on Process Flow Diagrams (PFD) and on Pipeline and Instrumentation Drawings (P&ID).

The measuring instrument and control device function codes and symbols indicate which process parameter is being measured, the relative locations of the measurement and control devices and the permissible limits applicable to certain variable process conditions.

In cases where supervisory computer systems are installed in a system, special symbols are used to indicate the computer and the instruments, which are connected to it.

For instance, letter codes and symbols permit the following instrument; functions to be graphically represented.

Process Monitoring Instrument Codes

- Flow rate (F)
- Level (L)
- Pressure (P)
- Quality (Q)
- Speed (S)
- Temperature (T)

These codes are integrated with various symbols to distinguish between indicators, recorders and in certain cases, their geographical locations.

At the end of this section, there are several sheets contain wide range of the applicable instrument symbols and abbreviations.

Emergency or Safety Instrument Codes

A list is given below, for the abnormal conditions, which must be measured by function qualification instruments. State display or alarm signals from such instruments are for the purpose of alerting the process operator, thus enabling corrective action to be taken. In cases of emergency or to safeguard vital equipment, the instruments automatically initiate trip or shutdown actions.

- High level (H) initiates an alarm.

- Extreme high level (HH) trips the inlet valve shut.

- Low level (L) initiates an alarm.

- Extreme low level (LL) trips the outlet valve shut.

Low flow (L) initiates an alarm and may also open a minimum flow spill-back or recycle valve to prevent the pump from overheating. Extreme low flow (LL) trips the pump motor to prevent damage.

High-pressure (H) increases the overhead condenser coolant flow. Extreme high pressure (HH) initiates an alarm and opens a vent valve to flare.

High temperature (H) initiates an alarm. Extreme high temperature (HH) trips the fuel inlet valves to protect the furnace coil from overheating.

Structure of the Instrument Codes

In general, every conventional measuring or controlling instrument Installed in a process unit is identified by three separate codes as follows.

- A location number code indicates the specific process unit in which the instrument is installed.
- A function letter code indicates the property or process variable being measured or controlled.
- A serial number code identifies the specific instrument and therefore prevents confusion when there are several Instruments In a single process unit, each having the same function letter code.

The combination of the three codes is known as the Instrument tag number, which has the basic format -

xx a - yyy

TAG NUMBERS

"**xx**" is a two-digit number used to identify the process unit.

'a' is a letter code containing two or more capital letters and is used to identify the instrument function.

'yyy' is a three-digit number used to identify the particular instrument.

When the instrument code or tag number is written on a drawing or document, a dash is inserted between the '**a**' and the '**yyy**' sections of the format. For example, a pressure indicating controller installed in a process unit coded **10** and identified by serial number **101**, is described in written form as **10 - PIC - 101**.

In the case of the same tag numbers, the process pressure correcting element, usually a control valve, often has the same tag number as the control instrument.

However, when the controller operates two valves in a split range mode, the valves are tagged and numbered consecutively, for example,

10 - PIC - 101 ----------→ **10 – PCV – 101-1**

10 – PCV – 101-2

NOTE:

Refer to the Following reference documents in the next pages:

1- List of General Abbreviations
2- List of Instrument Identification Code
3- Instrument Symbols (ISA S501)
4- Legend of Symbols

GOW	Gas/Oil/Water
gpm	gallons per minute
GT	Gas Turbine
H_2S	Hydrogen Sulphide
HHP	High-High Pressure
HP	High Pressure
hr	hour
HSE	Health, Safety and Environment
HVAC	Heating/Ventilation/Air-conditioning
in	inch
IR	Infra Red
KO	Knock-out
lb	pound (weight)
LC	Locked Closed
LCR	Local Control Room
LCV	Level Control Valve
LED	Light Emitting Diode
LEL	Lower Explosive Limit
LER	Local Equipment Room
LI	Level Indicator
LIC	Level Indicator Controller
LNG	Liquefied Natural Gas
LO	Locked Open
LP	Low Pressure
LPG	Liquefied Petroleum Gas
LT	Level Transmitter

M	Thousand
m^2	square metres
m^3	cubic metres
MAC	Manual Alarm Callpoint
MAWP	Maximum Allowable Working Pressure
MGL	Main Gas Line
MM	Million
mmscfd	million standard cubic feet per day
MOL	Main Oil Line
MOOCS	Modernization of Offshore Control Systems
MOV	Motor Operated Valve
MSF	Main Structure Frame
MTU	Master Telemetry Unit
N_2	Nitrogen
NGL	Natural Gas Liquids
NO	Normally Open
NRV	Non-return Valve
NTP	Normal Temperature and Pressure
OCCP	Original Central Collection Platform
O/I	Output/Input
OIS	Operator Interface Station
P&ID	Piping and Instrument Diagram
PAH	Pressure Alarm High
PAHH	Pressure Alarm High-High
PAL	Pressure Alarm Low
PALL	Pressure Alarm Low-Low
PC	Personal Computer

PCS	Process Control System
PCV	Pressure Control Valve
PDIC	Pressure Differential Indicator Controller
PDRC	Pressure Differential Recorder Controller
PFD	Process Flow Diagram
PGP	Power Generation Platform
PIC	Pressure Indicator Controller
PLC	Programmable Logic Controller
ppm	parts per million
psia	pounds per square inch (absolute)
psig	pounds per square inch (gauge)
PSLL	Pressure Switch Low-Low
PSU	Power Supply Unit
PSV	Pressure Safety Valve
PTB	Pounds per Thousand Barrels
RFIS	Remote Flare Ignition System
RO	Restriction Orifice
RTU	Remote Telemetry Unit
RV	Relief Valve
SCADA	Supervisory Control and Data Acquisition
SDV	Shutdown Valve
sec	second(s)
SG	Specific Gravity
SOV	Solenoid Operated Valve
SSV	Surface Safety Valve
TAH	Temperature Alarm High
TAHH	Temperature Alarm High-High

TAL	Temperature Alarm Low
TALL	Temperature Alarm Low-Low
Temp	Temperature
TIC	Temperature Indicator Controller
TSLL	Temperature Switch Low-Low
UPS	Uninterruptible Power Supply
US	Umm Shaif
US gal	United States Gallons
USSC	Umm Shaif Super Complex
UTC	Unique Tracking Commodity (used in PASSPORT)
UV	Ultra Violet
VDU	Visual Display Unit
wg	water gauge (inches pressure)
WI	Water Injection
WIP	Water Injection Platform
WO	Work Order
WV	Wing Valve
XV	Emergency Shutdown Valve
ZCC	ZADCO Central Complex
°C	degrees Centigrade
°F	degrees Fahrenheit

FUNCTION IDENTIFICATION CODE

Term	Code
AMMETER	II
ANALYSER ALARM HIGH	AAH
ANALYSER ALARM HIGH-HIGH	AAHH
ANALYSER ALARM LOW	AAL
ANALYSER ALARM LOW-LOW	AALL
ANALYSER ALARM MULTIFUNCTION	AAU
ANALYSER MULTIFUNCTION	AU
ANALYSER ELEMENT	AE
ANALYSER INTEGRATE ELEMENT (AUTO-SAMPLER)	AQE
ANALYSER INTEGRATE ALARM HIGH	AQAH
ANALYSER INTEGRATE ALARM LOW	AQAL
ANALYSER INDICATOR CONTROLLER	AIC
ANALYSER INDICATING TRANSMITTER	AIT
ANALYSER RECORDER	AR
ANALYSER RECORDER CONTROLLER	ARC
ANALYSER RECORDING SWITCH	ARS
ANALYSER SWITCH	AS
ANALYSER SWITCH LOW	ASL
ANALYSER SWITCH HIGH	ASH
ANALYSER TRANSMITTER	AT
BURNER ALARM LOW	BAL
BURNER CONTROLLER	BC
BURNER ELEMENT	BE
BURNER GLASS	BG
BURNER INDICATOR	BI
BURNER RECORDER	BR
BURNER SCAN CONTROLLER	BJC
BURNER SCAN INDICATOR	BJI
BURNER SCAN RECORDER	BJR
BURNER SWITCH	BS
BURNER TRANSMITTER	BT
COMBINATION ELEMENT	UE
COMBINATION SWITCH	US
COMBINATION SCAN CONTROLLER	UJC
COMBINATION SCAN INDICATOR	UJI
COMBINATION SCAN RECORDER	UJR
CONDUCTIVITY ALARM	CA

Term	Code
CONDUCTIVITY ALARM HIGH	CAH
CONDUCTIVITY ALARM LOW	CAL
CONDUCTIVITY ELEMENT	CE
CONDUCTIVITY INDICATOR	CI
CONDUCTIVITY INDICATING CONTROLLER	CIC
CONDUCTIVITY RECORDER	CR
CONDUCTIVITY RECORDING CONROLLER	CRC
CONDUCTIVITY SCAN CONTROLLER	CJC
CONDUCTIVITY SCAN INDICATOR	CJI
CONDUCTIVITY SCAN RECORDER	CJR
CONDUCTIVITY SWITCH	CS
CONDUCTIVITY TRANSMITTER	CT
CURRENT ALARM	IA
CURRENT ALARM HIGH	IAH
CURRENT ALARM LOW	IAL
CURRENT CONTROLLER	IC
CURRENT ELEMENT	IE
CURRENT INDICATOR	II
CURRENT INDICATOR CONTROLLER	IIC
CURRENT INDICATING INTEGRATOR	IQI
CURRENT RECORDER	IR
CURRENT RECORDING CONTROLLER	IRC
CURRENT SCAN CONTROLLER	IJC
CURRENT SCAN INDICATOR	IJI
CURRENT SCAN RECORDER	IJR
CURRENT TRANSMITTER	IT
CURRENT TRANSFORMER	IE
DENSITY ELEMENT	DE
DENSITY SWITCH	DS
FLOW ALARM HIGH	FAH
FLOW ALARM LOW	FAL
FLOW DIFFERENTIAL ALARM HIGH	FDAH
FLOW DIFFERENTIAL ALARM HIGH-HIGH	FDAHH
FLOW DIFFERENTIAL ALARM LOW	FDAL
FLOW DIFFERENTIAL SWITCH LOW	FDSL
FLOW ELEMENT	FE
FLOW INDICATOR	FI
FLOW INDICATOR CONTROLLER	FIC
FLOW INTEGRATE CONTROLLER	FQC
FLOW INTEGRATE INDICATOR	FQI
FLOW INTEGRATE RECORDER	FQR
FLOW INDICATING TRANSMITTER	FIT

FUNCTION IDENTIFICATION CODE (Contd.)

Term	Code
FLOW MIXER	FX
FLOW QUANTITY INDICATING CONTROLLER	FQIC
FLOW RELAY	FY
FLOW RECORDER	FR
FLOW SCAN CONTROLLER	FJC
FLOW SCAN INDICATOR	FJI
FLOW SCAN RECORDER	FJR
FLOW SWITCH	FS
FLOW SWITCH LOW	FSL
FLOW SWITCH HIGH	FSH
FLOW TRANSMITTER	FT
FLOW TEMPERATURE ALARM	FTA
FLOW VALVE	FV
HAND CONTROL STATION	HK
HAND LIGHT SWITCH	HLS
HAND RELAY	HY
HAND SAFETY SWITCH	HSS
HAND SCAN SWITCH	HJS
HAND SWITCH	HS
LEVEL ALARM HIGH	LAH
LEVEL ALARM HIGH-HIGH	LAHH
LEVEL ALARM LOW	LAL
LEVEL ALARM LOW-LOW	LALL
LEVEL CONTROL	LC
LEVEL CONTROL SWITCH	LCS
LEVEL GAUGE	LI
LEVEL INDICATOR	LI
LEVEL INDICATING TRANSMITTER	LIT
LEVEL SCAN CONTROLLER	LJC
LEVEL SCAN INDICATOR	LJI
LEVEL SCAN RECORDER	LJR
LEVEL SWITCH HIGH	LSH
LEVEL SWITCH HIGH-HIGH	LSHH
LEVEL SWITCH LOW	LSL
LEVEL SWITCH LOW-LOW	LSLL
MOISTURE ALARM	MA
MOISTURE ALARM HIGH	MAH
MOISTURE ALARM LOW	MAL
MOISTURE ELEMENT	ME
MOISTURE SCAN CONTROLLER	MJC
MOISTURE SCAN INDICATOR	MJI
MOISTURE SCAN RECORDER	MJR
MOISTURE SWITCH	MS
MOISTURE TRANSMITTER	MT
MULTIFUNCTION SCAN ALARM	UJA
MULTIFUNCTION SCAN INDICATOR	UJI
MULTIFUNCTION SCAN RECORDER	UJR
MULTIVARIABLE MULTIFUNCTION	UU
MULTIVARIABLE SCAN TRANSMITTER	UJT
POSITION ALARM	ZA
POSITION CONTROLLER	ZC
POSITION ELEMENT	ZE
POSITION INDICATOR	ZI
POSITION RECORDER	ZR
POSITION INDICATING CONTROLLER	ZIC
POSITION RECORDING CONTROLLER	ZRC
POSITION SCAN CONTROLLER	ZJC
POSITION SCAN INDICATOR	ZJI
POSITION SCAN RECORDER	ZJR
POSITION SWITCH	ZS
POSITION TRANSMITTER	ZT
POWER ALARM	JA
POWER ALARM HIGH	JAH
POWER ALARM LOW	JAL
POWER ELEMENT	JE
POWER CONTROLLER	JC
POWER INDICATOR	JI
POWER RECORDER	JR
POWER INDICATING CONTROLLER	JIC
POWER RECORDING CONTROLLER	JRC
POWER SCAN CONTROLLER	JJC
POWER SCAN INDICATOR	JJI
POWER SCAN RECORDER	JJR
POWER SWITCH	JS
POWER TRANSMITTER	JT
PRESSURE ALARM HIGH	PAH
PRESSURE ALARM LOW	PAL
PRESSURE DIFFERENTIAL ALARM HIGH	PDAH
PRESSURE DIFFERENTIAL SWITCH HIGH	PDSH
PRESSURE DIFFERENTIAL INDICATOR	PDI
PRESSURE DIFFERENTIAL INDICATING CONTROLLER	PDIC
PRESSURE INDICATOR	PI
PRESSURE INDICATING TRANSMITTER	PIT
PRESSURE POINT	PP
PRESSURE RECORDER CONTROLLER	PRC
PRESSURE SCAN INDICATOR	PJI
PRESSURE SCAN RECORDER	PJR
PRESSURE SWITCH	PS
PRESSURE SWITCH HIGH	PSH
PRESSURE SWITCH LOW	PSL
PRESSURE VALVE	PV
QUANTITY ALARM	QA
QUANTITY ALARM HIGH	QAH
QUANTITY ALARM LOW	QAL
QUANTITY CONTROLLER	QC

FUNCTION IDENTIFICATION CODE (Contd.)

Term	Code
QUANTITY INDICATOR	QI
QUANTITY RECORDER	QR
QUANTITY INDICATING CONTROLLER	QIC
QUANTITY INDICATING INTEGRATOR	QQI
QUANTITY RECORDING CONTROLLER	QRC
QUANTITY SCAN CONTROLLER	QJC
QUANITTY SCAN INDICATOR	QJI
QUANTITY SCAN RECORDER	QJR
QUANITY SWITCH	QS
QUANTITY TRANSMITTER	QT
SPEED ALARM SWITCH LOW	SASL
SPEED SWITCH	SS
SPEED TRANSMITTER	ST
TEMPERATURE ALARM HIGH	TAH
TEMPERATURE ELEMENT	TE
TEMPERATURE INDICATOR	TI
TEMPERATURE SCAN INDICATOR	TJI
TEMPERATURE SCAN RECORDER	TJR
TEMPERATURE SWITCH HIGH	TSH
TEMPERATURE TRANSMITTER	TT
THERMOWELL	TW
TIME ALARM	KA
TIME ALARM HIGH	KAH
TIME ALARM LOW	KAL
TIME CONTROLLER	KC
TIME INDICATOR	KI
TIME RECORDER	KR
TIME INDICATING CONTROLLER	KIC
TIME INDICATING INTEGRATOR	KQI
TIME RECORDING CONTROLLER	KRC
TIME RELAY	KY
TIME SCAN CONTROLLER	KJC
TIME SCAN INDICATOR	KJI
TIME SCAN RECORDER	KJR
TIME SWITCH	KS
TIME TRANSMITTER	KT
VISCOSITY ALARM	VA
VISCOSITY ALARM HIGH	VAH
VISCOSITY ALARM LOW	VAL
VISCOSITY ELEMENT	VE
VISCOSITY CONTROLLER	VC
VISCOSITY INDICATOR	VI
VISCOSITY RECORDER	VR
VISCOSITY INDICATING CONTROLLER	VIC
VISCOSITY RECORDING CONTROLLER	VRC
VISCOSITY SCAN CONTROLLER	VJC
VISCOSITY SCAN INDICATOR	VJI
VISCOSITY SCAN RECORDER	VJR
VISCOSITY SWITCH	VS
VISCOSITY TRANSMITTER	VT

INSTRUMENTATION SYMBOLS

INSTRUMENT LINE SYMBOLS

ALL LINES TO BE FINE IN RELATION TO PROCESS PIPING LINES.

(1) INSTRUMENT SUPPLY * OR CONNECTION TO PROCESS

(2) UNDEFINED SIGNAL

(3) PNEUMATIC SIGNAL **

(4) ELECTRIC SIGNAL ---------- OR

(5) HYDRAULIC SIGNAL

(6) CAPILLARY TUBE

(7) ELECTROMAGNETIC OR SONIC SIGNAL *** (GUIDED)

(8) ELECTROMAGNETIC OR SONIC SIGNAL *** (NOT GUIDED)

(9) INTERNAL SYSTEM LINK (SOFTWARE OR DATA LINK)

(10) MECHANICAL LINK

OPTIONAL BINARY (ON-OFF) SYMBOLS

(11) PNEUMATIC BINARY SIGNAL

(12) ELECTRIC BINARY SIGNAL ---\-----\--- OR

NOTE: 'Or' means user's choice. Consistency is recommended.

* The following abbreviations are suggested to denote the types of power supply. These designations may also be applied to purge fluid supplies.

AS - Air Supply
IA - Instrument Air } Options
PA - Plant Air
ES - Electric Supply
GS - Gas Supply
HS - Hydraulic Supply
NS - Nitrogen Supply
SS - Steam Supply
WS - Water Supply

The supply level may be added to the instrument supply line, e.g., AS-100, a 100-psig air supply; ES-24DC, a 24-volt direct current power supply.

** The pneumatic signal symbol applies to a signal using any gas as the signal medium. If a gas other than air is used, the gas may be identified by a note on the signal symbol or otherwise.

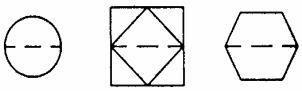
*** Electromagnetic phenomena include heat, radio waves, nuclear radiation, and light.

GENERAL INSTRUMENT OR FUNCTION SYMBOLS

	PRIMARY LOCATION *** NORMALLY ACCESSIBLE TO OPERATOR	FIELD MOUNTED	AUXILIARY LOCATION *** NORMALLY ACCESSIBLE TO OPERATOR
DISCRETE INSTRUMENTS	1	2	3
SHARED DISPLAY, SHARED CONTROL	4	5	6
COMPUTER FUNCTION	7	8	9
PROGRAMMABLE LOGIC CONTROL	10	11	12

* Symbol size may vary according to the user's needs and the type of document. A suggested square and circle size for large diagrams is shown above. Consistency is recommended.

** Abbreviations of the user's choice such as IPI (Instrument Panel #1), IC2 (Instrument Console #2), CC3 (Computer Console #3), etc., may be used when it is necessary to specify instrument or function location.

*** Normally inaccessible or behind-the-panel devices or functions may be depicted by using the same symbols but with dashed horizontal bars, i.e. 

GENERAL INSTRUMENT OR FUNCTION SYMBOLS (Contd.)

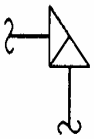
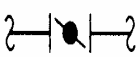
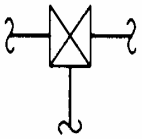
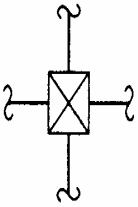
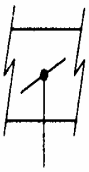
13	14 6TE 2584-23 INSTRUMENT WITH LONG TAG NUMBER	15 INSTRUMENTS SHARING COMMON HOUSING*
16 PILOT LIGHT	17 C 12 PANEL MOUNTED PATCHBOARD POINT 12	18 P ** PURGE OR FLUSHING DEVICE
19 R ** RESET FOR LATCH-TYPE ACTUATOR	20 DIAPHRAGM SEAL	21 I ** *** UNDEFINED INTERLOCK LOGIC

* It is not mandatory to show a common housing.

** These diamonds are approximately half the size of the larger ones.

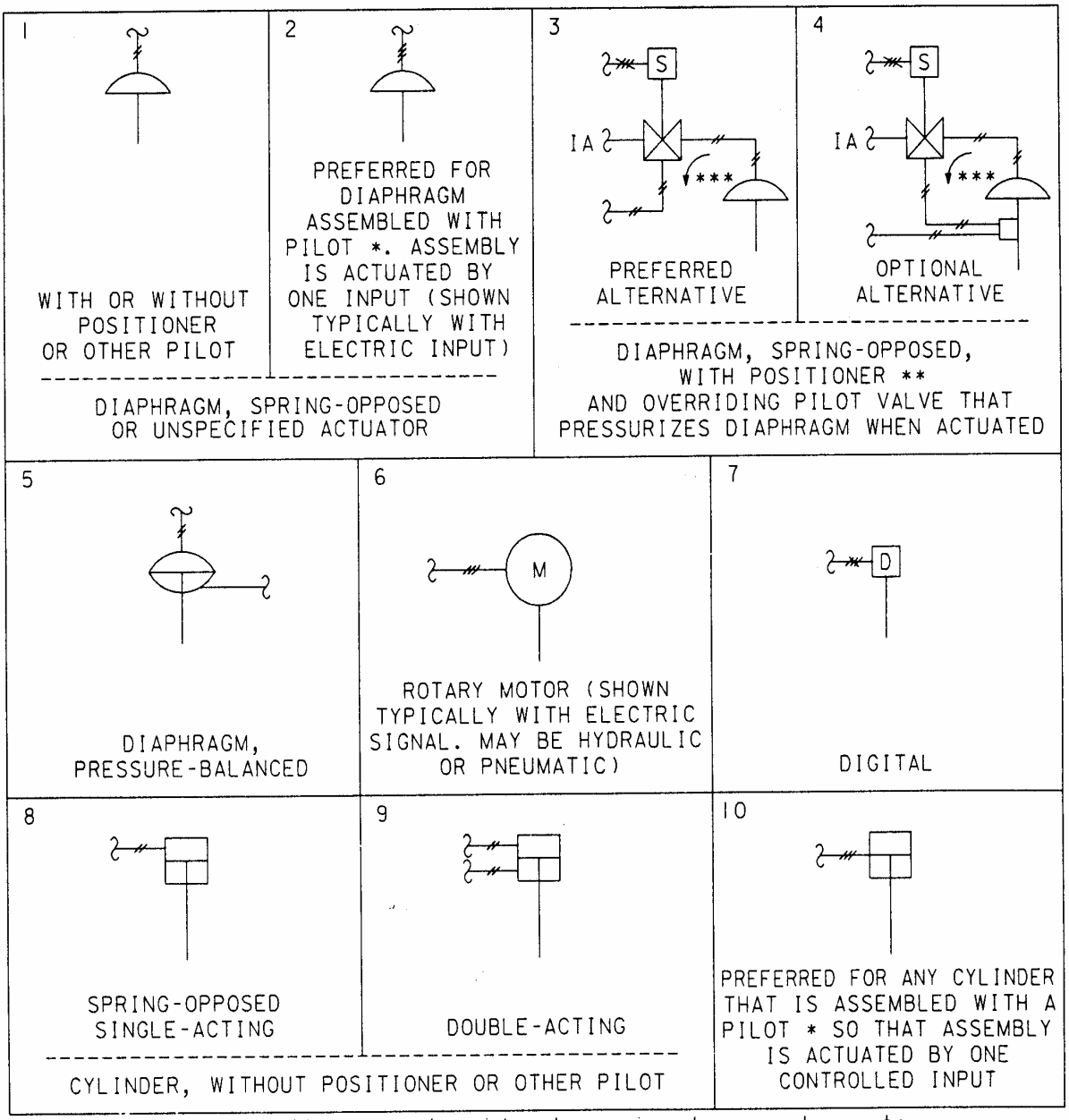
*** For specific logic symbols, see ANSI/ISA Standard S5.2.

CONTROL VALVE BODY SYMBOLS, DAMPER SYMBOLS

1 GENERAL SYMBOL	2  ANGLE	3  BUTTERFLY	4 ROTARY VALVE
5  THREE-WAY	6  FOUR-WAY	7 GLOBE	8
9 DIAPHRAGM	10	11	12 
	DAMPER OR LOUVER		

Further information may be added adjacent to the body symbol either by note or code number.

ACTUATOR SYMBOLS

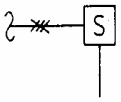
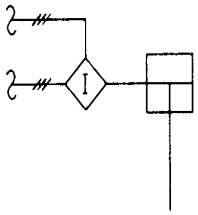
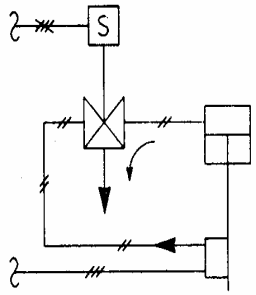
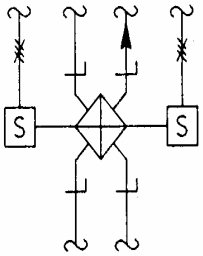
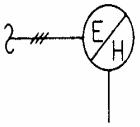
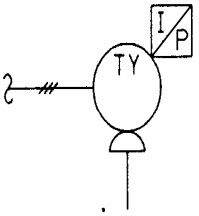
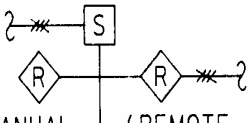


* Pilot may be positioner, solenoid valve, signal converter, etc.

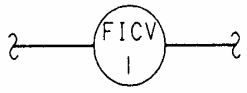
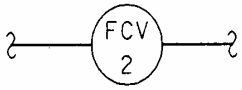
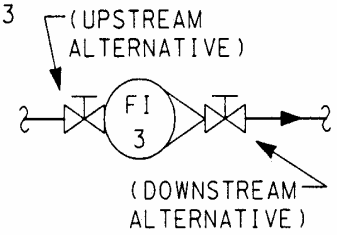
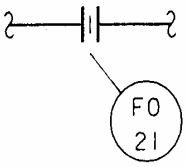
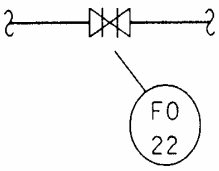
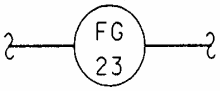
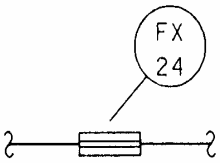
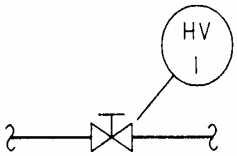
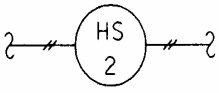
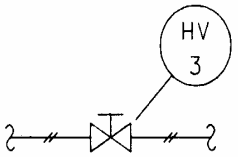
** The positioner need not be shown unless an intermediate device is on its output. The positioner tagging, ZC, need not be used even if the positioner is shown. The positioner symbol, a box drawn on the actuator shaft, is the same for all types of actuators. When the symbol is used, the type of instrument signal, i.e., pneumatic, electric, etc., is drawn as appropriate. If the positioner symbol is used and there is no intermediate device on its output, then the positioner output signal need not be shown.

*** The arrow represents the path from a common to a fail open port. It does not correspond necessarily to the direction of fluid flow.

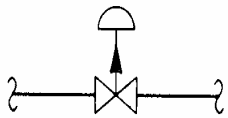
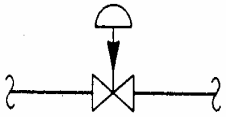
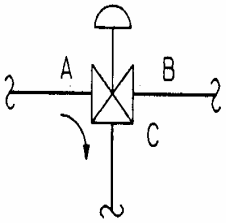
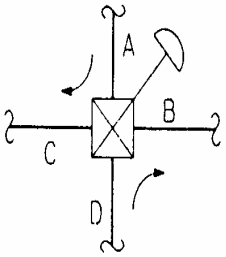
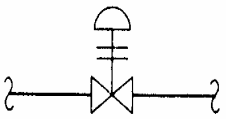
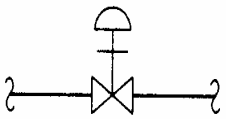
ACTUATOR SYMBOLS (Contd.)

11  SOLENOID	12  PREFERRED ALTERNATIVE. A BUBBLE WITH INSTRUMENT TAGGING, E.G., TY-1, MAY BE USED INSTEAD OF THE INTERLOCK SYMBOL CYLINDER WITH POSITIONER AND OVERRIDING PILOT VALVE	13  SINGLE-ACTING CYLINDER (IMPLIED I/P)
14  DUAL SOLENOIDS SWITCHING 4-WAY HYDRAULIC VALVE	15  ELECTROHYDRAULIC	16  VALVE ACTUATOR WITH ATTACHED ELECTRO-PNEUMATIC CONVERTER
17  (MANUAL RESET) (REMOTE RESET) LATCH-TYPE ACTUATOR WITH RESET (SHOWN TYPICALLY FOR SOLENOID ACTUATOR AND TYPICALLY WITH ELECTRIC SIGNAL FOR REMOTE RESET, WITH MANUAL RESET ALTERNATIVE)	18 FOR PRESSURE RELIEF OR SAFETY VALVES ONLY: DENOTES A SPRING, WEIGHT, OR INTEGRAL PILOT	19 HAND ACTUATOR OR HANDWHEEL

SYMBOLS FOR SELF-ACTUATED REGULATORS, VALVES, AND OTHER DEVICES

FLOW	1  AUTOMATIC REGULATOR WITH INTEGRAL FLOW INDICATION	2  AUTOMATIC REGULATOR WITHOUT INDICATION	3  INDICATING VARIABLE AREA METER WITH INTEGRAL MANUAL THROTTLE VALVE
	4  RESTRICTION ORIFICE (ORIFICE PLATE, CAPILLARY TUBE OR MULTI-STAGE TYPE, ETC.) IN PROCESS LINE	5  RESTRICTION ORIFICE DRILLED IN VALVE (INSTRUMENT TAG NUMBER MAY BE OMITTED IF VALVE IS OTHERWISE IDENTIFIED)	6  FLOW SIGHT GLASS, PLAIN OR WITH PADDLE WHEEL, FLAPPER, ETC.
	7  FLOW STRAIGHTENING VANE (USE OF TAG NUMBER IS OPTIONAL. THE LOOP NUMBER MAY BE THE SAME AS THAT OF THE ASSOCIATED PRIMARY ELEMENT)	8	9
HAND	1  HAND CONTROL VALVE IN PROCESS LINE	2  HAND-ACTUATED ON-OFF SWITCHING VALVE IN PNEUMATIC SIGNAL LINE	3  HAND CONTROL VALVE IN SIGNAL LINE

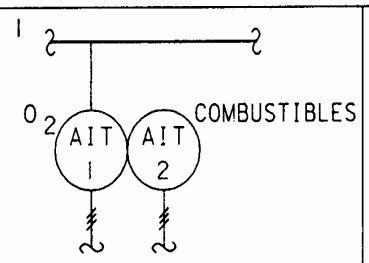
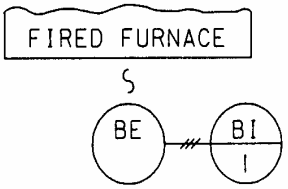
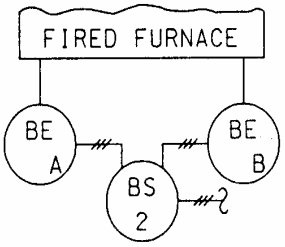
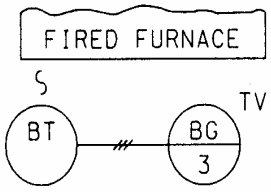
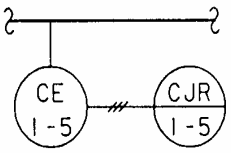
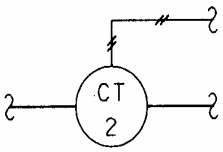
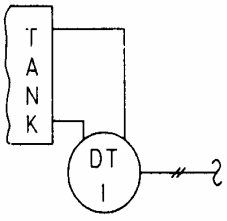
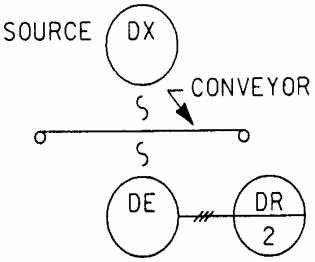
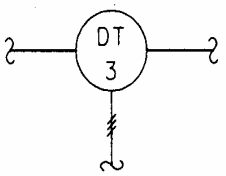
SYMBOLS FOR ACTUATOR ACTION IN EVENT OF ACTUATOR POWER FAILURE. (SHOWN TYPICALLY FOR DIAPHRAGM-ACTUATED CONTROL VALVE)

1  TWO-WAY VALVE, FAIL OPEN	2  TWO-WAY VALVE, FAIL CLOSED	3  THREE-WAY VALVE, FAIL OPEN TO PATH A-C
4  FOUR-WAY VALVE, FAIL OPEN TO PATHS A-C AND D-B	5  ANY VALVE, FAIL LOCKED (POSITION DOES NOT CHANGE)	6  ANY VALVE, FAIL INDETERMINATE

The failure modes indicated are those commonly defined by the term, "shelf-position". As an alternative to the arrows and bars, the following abbreviations may be employed:

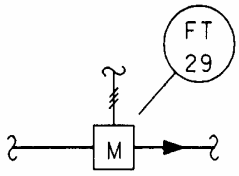
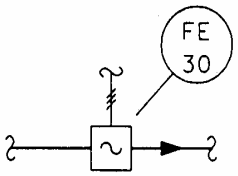
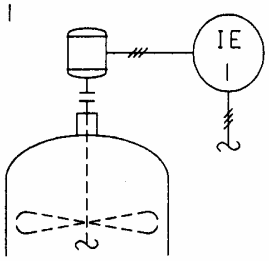
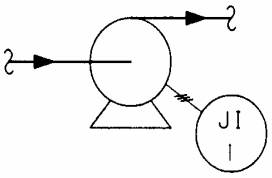
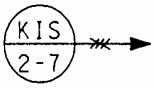
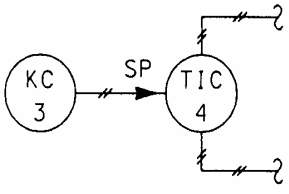
FO - Fail Open
FC - Fail Closed
FL - Fail Locked (last position)
FI - Fail Indeterminate

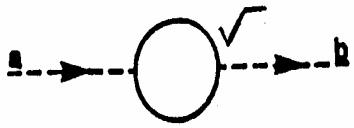
PRIMARY ELEMENT SYMBOLS

	1	2	3
A ANALYSIS	 DUAL ANALYSIS INDICATING TRANSMITTER FOR OXYGEN AND COMBUSTIBLES CONCENTRATIONS		
B BURNER/COMBUSTION	 ONE BURNER ULTRA-VIOLET FLAME DETECTOR CONNECTED TO ANALOG-TYPE FLAME INTENSITY INDICATOR	 TWO BURNER FLAME ROD SENSORS CONNECTED TO COMMON SWITCH	 TELEVISION CAMERA AND RECEIVER TO VIEW BURNER FLAME
C USER'S CHOICE	 CONDUCTIVITY CELL CONNECTED TO POINT 5 OF MULTIPOINT SCANNING CONDUCTIVITY RECORDER	 INLINE CONSISTENCY TRANSMITTER	
D USER'S CHOICE	 DENSITY TRANSMITTER, DIFFERENTIAL-PRESSURE TYPE, EXTERNALLY CONNECTED	 RADIOACTIVE-TYPE DENSITY ELEMENTS CONNECTED TO RECORDER ON PANEL	 SPECIFIC GRAVITY TRANSMITTER, FLOW-THROUGH TYPE

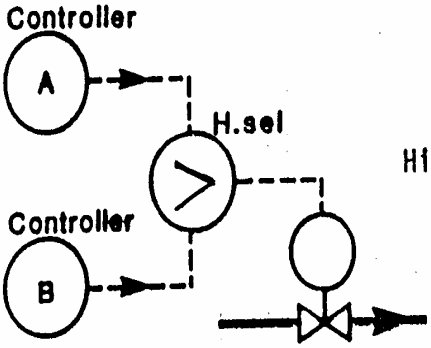
USE OF LETTER C AND D DEFINED ON USER'S LEGEND SHEET

PRIMARY ELEMENT SYMBOLS (Contd.)

F FLOW RATE (Contd.)	22  MAGNETIC FLOWMETER WITH INTEGRAL TRANSMITTER	23  SONIC FLOWMETER "DOPPLER" OR "TRANSIT TIME" MAY BE ADDED	24
I CURRENT	1  CURRENT TRANSFORMER MEASURING CURRENT OF ELECTRIC MOTOR	2	3
J POWER	1  INDICATING WATTMETER CONNECTED TO PUMP MOTOR	2	3
K TIME OR TIME-SCHEDULE	1 CLOCK	2  MULTIPOINT ON-OFF TIME SEQUENCING PROGRAMMER POINT 7	3  TIME-SCHEDULE CONTROLLER, ANALOG TYPE, OR SELF-CONTAINED FUNCTION GENERATOR



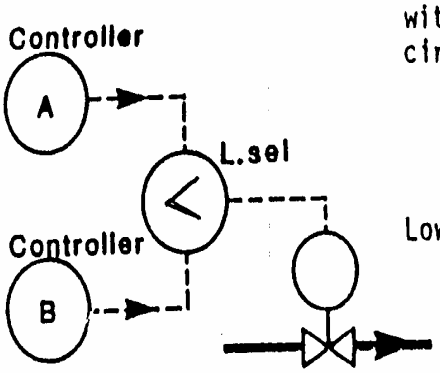
Square root relay



High selector relay

chooses highest signal

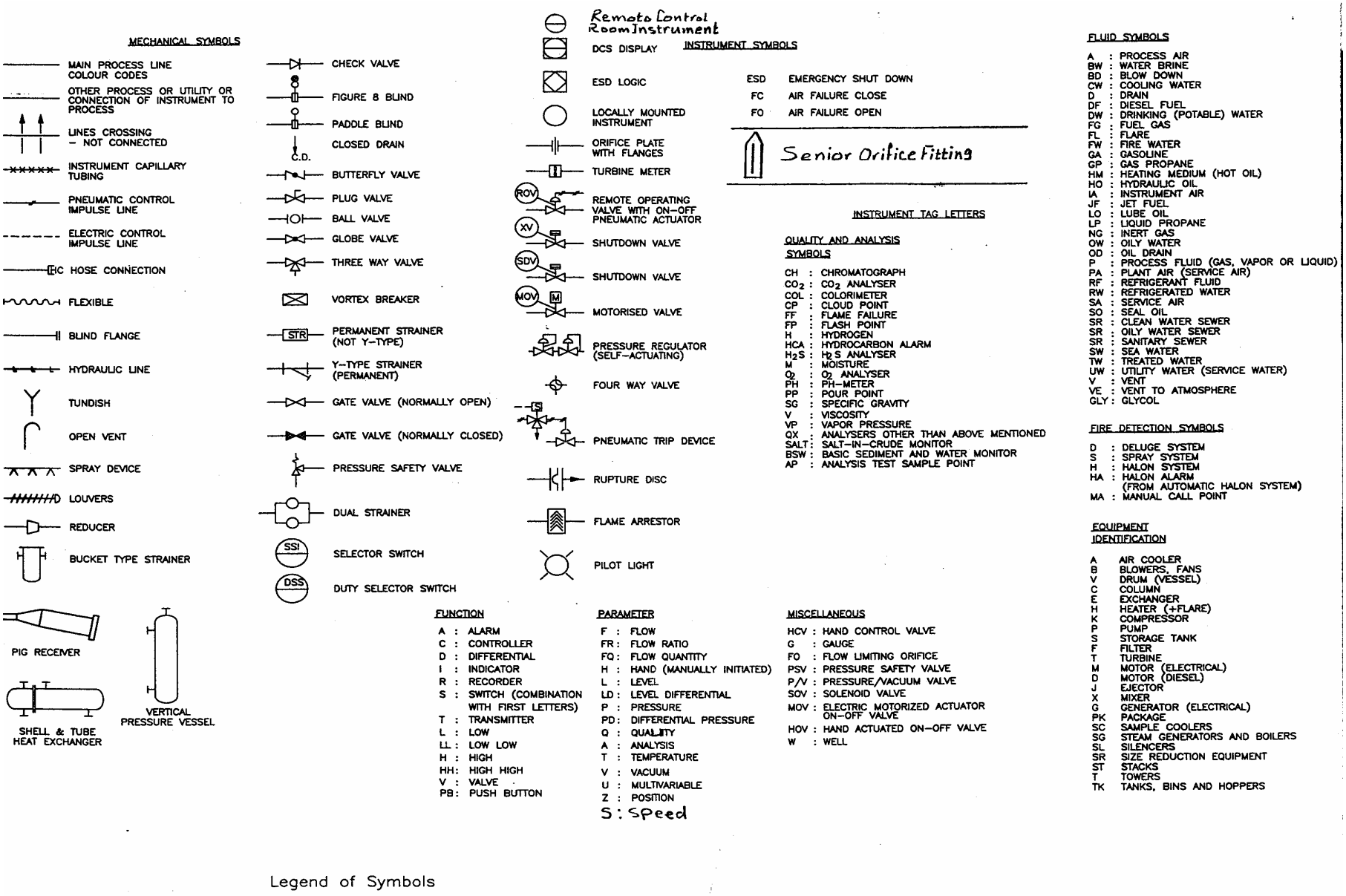
The high selector relay chooses the highest signal from two control instruments and passes this on to a third controller or to a control valve actuator. The Shell symbol for this relay is a 12 mm circle with the abbreviation 'H. sel' written outside the circle and close to its upper right hand (RH) side.



Low selector relay

chooses lowest signal

The low selector relay chooses the lowest signal from two control instruments and passes this on to a third controller or control valve actuator. The Shell symbol for this relay is a 12 mm circle with the abbreviation 'L. sel' written outside the circle and close to its upper RH side.



Legend of Symbols

MODULE 2:

PRESSURE MEASUREMENTS

PRESSURE MEASUREMENT

Objectives

- At Completion of this module, the trainee will have understanding of:
- Pressure definition, types and units.
- Pressure sensing elements.
- Principles of pressure sensing elements; bourbon tubes, bellows, diaphragms, vibrating wires, strain gauges and capacitance sensors.
- Protection devices for pressure measuring elements.
- Pressure measurement devices.
- Function of pressure measurement devices.
- Select a pressure device for a service.
- Identify the types of pressure gauge errors.
- Identify the parts and function of pneumatic and electronic pressure transmitters.
- Describe the difference between electronic and smart transmitters.
- How to Convert 4-20 mA signal to 1-5 vdc signal and why.
- Calculate an output signal of a pressure transmitter at certain input.
- Definition of range and span of a transmitter.
- Field-wiring connection methods of the electronic pressure transmitter.
- Pressure switches types and function.
- Pressure regulators construction parts and function.

Related Safety Regulations for Module I-1: PRESSURE MEASUREMENT
Trainee have to be familiarized with the following **SGC HSE** regulations, while studying this module:
Regulation No. 6: Permit to Work system.
Regulation No. 7: Isolation
7.18 (1-10) control systems procedures and isolations.
Regulation No. 22: Hot and Odd Bolting.
Regulation No. 23: General Engineering Safety.

Pressure Measurement

Oil and gas production operations require that system operating pressures be regulated to specific pressures in order for the system(s) to work properly.

In addition, safety considerations dictate that system operating pressures be monitored and controlled to ensure that the pressure limitations of equipment and piping are not exceeded. In order to meet these objectives, the industry relies on a variety of devices to generate an output signal which may be used to adjust or change the observed pressure, The devices used by the oil and gas industry for sensing operating pressures and generating the needed output signals are described in this manual. The purpose of this document is to provide the reader with an understanding of how the different types of device functions and how they should by applied, in order to satisfy the requirements of system monitoring and control.
Pressure is defined as the force exerted per unit area of surface.

$P = F/A$

P = pressure

F = force

A = surface area exposed to the force

In processing plants the hydrocarbon gases and liquids handled in pipes and vessels exert pressure on the surface area.

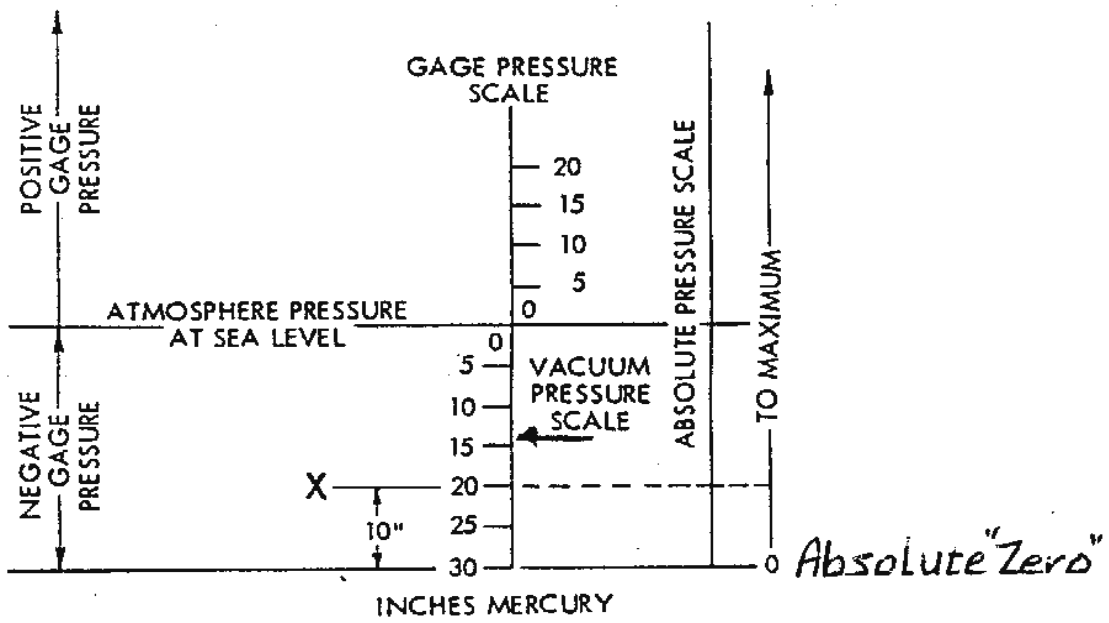
Types of Pressure

In order to understand various types of pressure the following will be considered: Pressure Scale reference points, there are two reference points, the zero point of pressure which is assumed to a perfect vacuum, another point is atmospheric pressure which varies with altitude above sea level and with weather conditions.

Absolute pressure scale starts from a zero reference point representing the full vacuum and extends through atmospheric pressure to the highest limit of measurable pressure.

Gauge pressure scale starts zero reference point representing the local atmospheric pressure and extends to a chosen limit applicable to the specific process system.
Vacuum scale starts from the absolute zero reference point and extends to a maximum represented by atmospheric pressure.

The above can be expressed as following:
Zero of absolute pressure = perfect vacuum
Absolute Pressure = Pressure above Absolute zero
Gauge Pressure = Absolute Pressure – Atmospheric pressure
Vacuum gauge Pressure = Atmospheric Pressure – Fluid Pressure

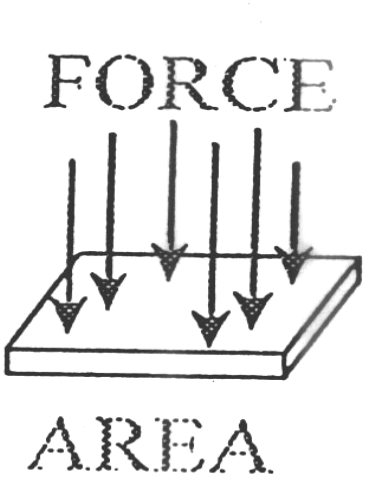


* The pressure at point X can be expressed as 10 inches of mercury absolute, or as 20 inches of mercury vacuum, or as —20 inches of mercury gage.

Pressure Units:

SGC uses a variety of pressure units but the two main systems are the Imperial (British and American) units and the S.I. (System International).

As pressure can be expressed as FORCE divided AREA then the units of pressure can by expressed as the units of force divided by the units of area.



a) Imperial units

In the Imperial system the unit of force is pound force (lbf) and the unit of area is the inch square (in^2). It follows that the unit of pressure in the Imperial system is the pound force divided by the inch square (lbf/Sq. in) (pounds per square inch). This is often abbreviated to PSI.

b) S.I. Unit

In the S.I. system the unit of force is the Newton (N) and the unit of area is the meter square (m2). Therefore the unit of force in the S.I. system is the Newton per square meter (N/m2). This is a very small unit of pressure and the S.I. unit that is more commonly used on the plant is bar. One bar is equal to 100000 N/m^2.

c) Liquid Column

Pressure can also be expressed in terms of liquid column height. The Imperial units are inches water column (in Wc) and the S.I. units are millimeters water column (mm Wc). Imperial units are inches Wc (or Hg) S.I. unitus are mm Wc (or Hg).

Pressure Conversions

The table below gives a few examples of different pressure:

IMPERIAL		S.I.	
Lbf/in^2	lnch Wc	bar	mm Wc
1	27.73	0.06895	703.1
0.03613	1	2.487x10^{-3}	25.4
14.504	402.1	1	10.22x10^3
1.422x10^{-3}	0.03937	97.98x10^{-6}	1

Examples:

1. Change 20 psi to bar

1 psi = 0.06895 bar

20 psi = 20x0.06895 bar

20 psi = 1.379 bar

2. Change 1.6 bar to psi

1 bar = 14.504 psi

1.6 bar= 1.6 x 14.504 psi

1.6 bar= 23.2064 psi

3. Change 100 in Wc to mm Wc

1 inch = 25.4 mm

100 inch = 100 x 25.4 mm

100 in Wc = 2540 mm Wc

4. Change 520 mm Wc to in Wc

1 mm Wc = 0.03937 in Wc

520 mm Wc = 520 x 0.03937 in Wc

520 mm Wc = 20.4724 in Wc.

Primary measuring Elements for the Process Pressure

Bourdon Tubes

Bourdon tubes are the most common type of pressure sensors. A bourdon tube is a metal tube with a flattened circular cross section bent into a C-shape, Spiral, or Helix. When pressure is applied through the open end, the increased pressure causes the flattened cross section to become more circulars and the shape to straighten. This moves the closed end. The device is illustrated in figure 1.

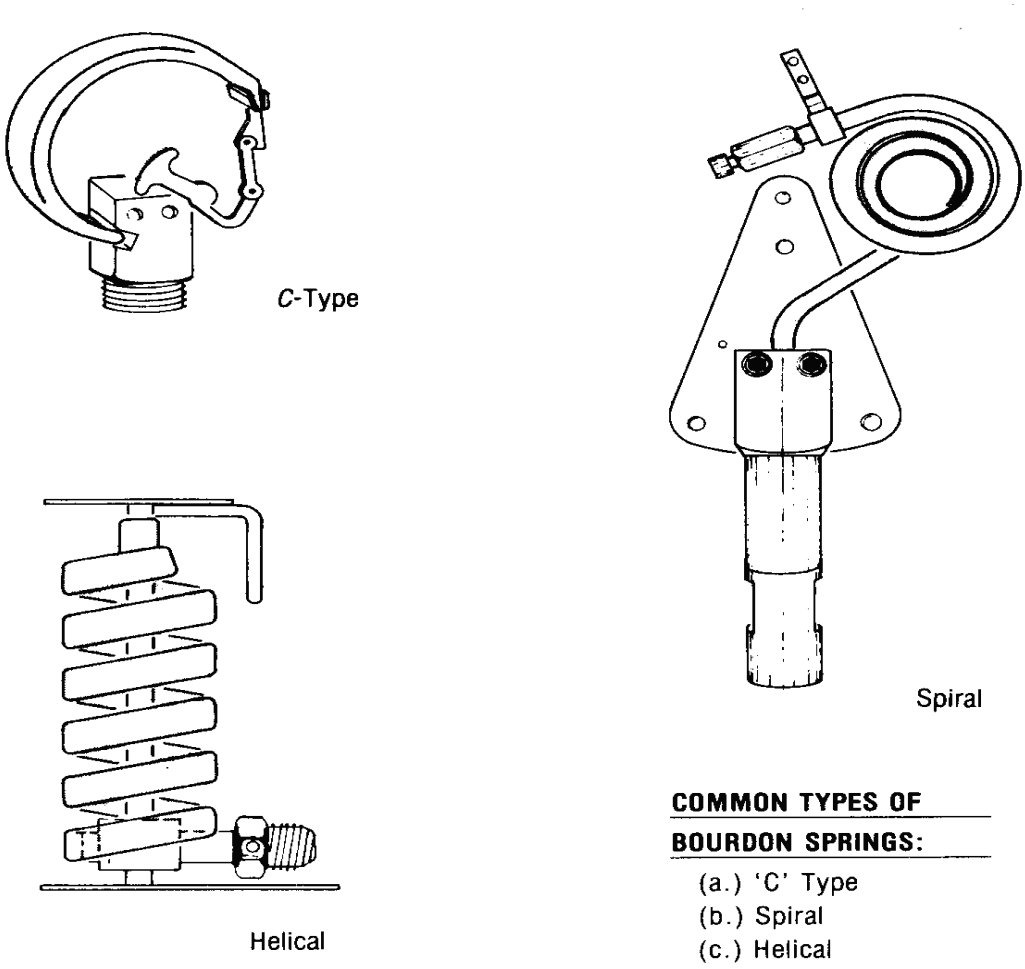


Figure 1, Bourdon Tube Configurations

The closed end of the bourdon tube is attached to a mechanical linkage. The linkage is connected to a pointer or other output device, see figure 2. There are three common types of bourdon tubes, the C-shape, the spiral, and the helix.

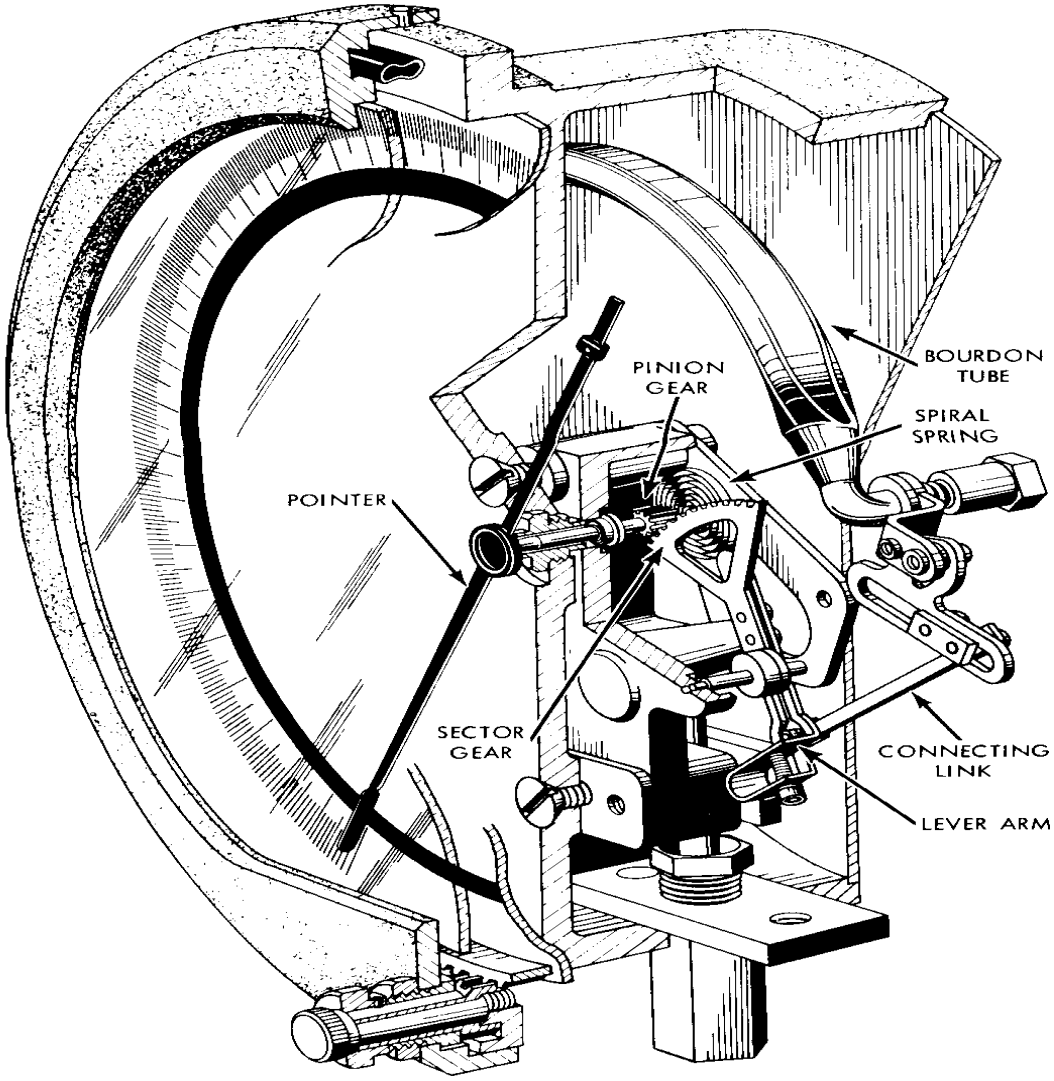


Figure 2, Bourdon Pressure Element Linkage

C- Type Bourdon

C-type bourdon tubes are used for ranges as low as 0 - 15 psig (0 - 100 kPa) and as high as 0 - 1500 psig (0 - 10,000 kPa). They are simple, accurate, and have good repeatability, but they are bulky and highly subject to damage from over-ranging. Most C-type bourdon tubes will tolerate only minimal overpressure.

Helical Bourdon

Helical bourdon tubes are used for ranges as low as 0 - 200 psig (0 - 1300 kPa) up to 0 - 6000 psig (0 - 40,000 kPa). Heavy-duty helical bourdons can sometimes tolerate as high as ten times the maximum range pressure.

Spiral Bourdon

Spiral bourdon tubes are used for both very low ranges and very high ranges. Very sensitive units are manufactured to measure as low as 0 -10 psig (0 - 65 kPa). Heavy-duty units can measure up to 0 -100,000 psig (0 -700,000 kPa).

Bellows Sensors

A bellows sensor is an axially flexible, cylindrical enclosure with folded sides. When pressure is applied through an opening, the closed end extends axially as shown in figure 3.

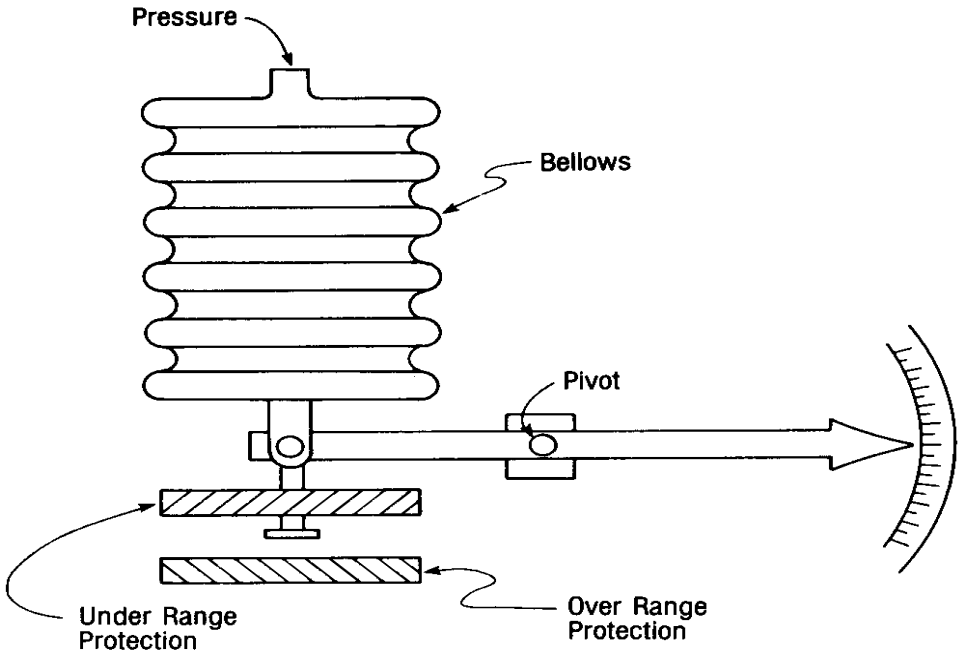


Figure 3, Bellows Gauge with under/over range protection

The movement rotates a pointer by a mechanical linkage. Movement of the bellows is opposed by the spring action of the bellows material, the pressure surrounding the bellows, and usually, the force of an external spring or another bellows.

Figure 4 shows an absolute pressure gauge. Bellows A is evacuated and the process pressure is connected to bellows B. The gauge will read zero when bellows B is at perfect vacuum and increase as the pressure is increased in bellows B and the low pressure to bellows A.

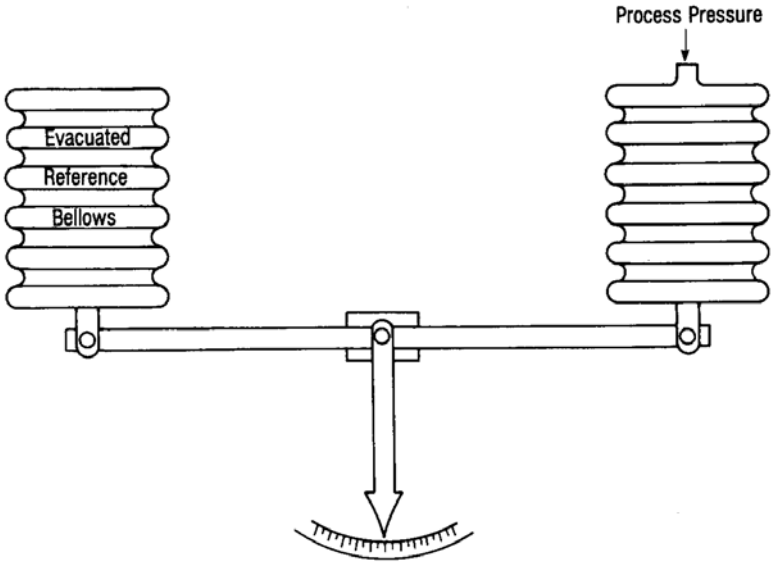


Figure 4, Beam-balanced Bellows Sensor.

Figure 5 shows a variation and adds a calibrated spring. The pressure outside the bellows compresses the bellows against the combined action of the bellows, the force of the calibrated spring, and the pressure within the bellows. Other variations are shown in figures 6 and 7.

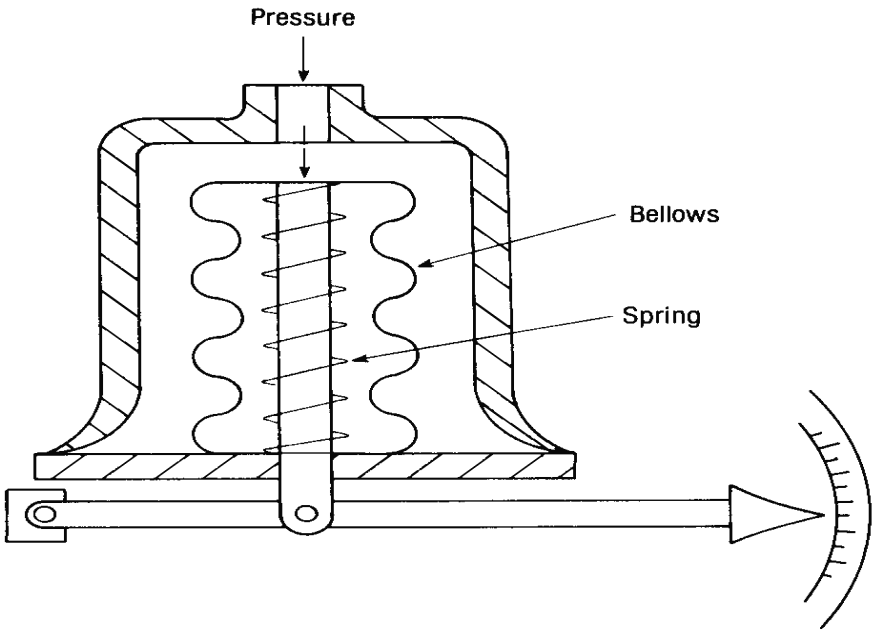
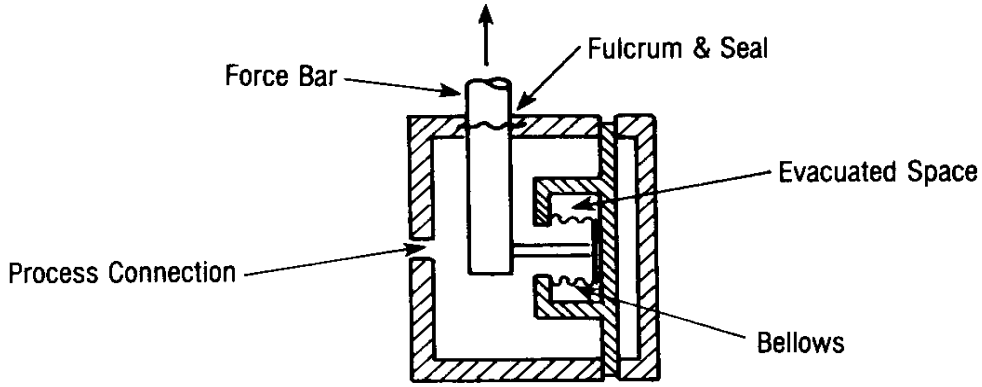


Figure 5, Bellows Sensor with a Calibrated Spring

A bellows sensor can accurately measure much lower pressures than a bourdon tube. Absolute pressure ranges as low as 0 -100 mm Hg and gauge pressure ranges as low as 0 -5 inches H_2O (0 -125 mm H_20) are available. Bellows elements can measure absolute pressure, gauge pressure, vacuum, or differential pressure.



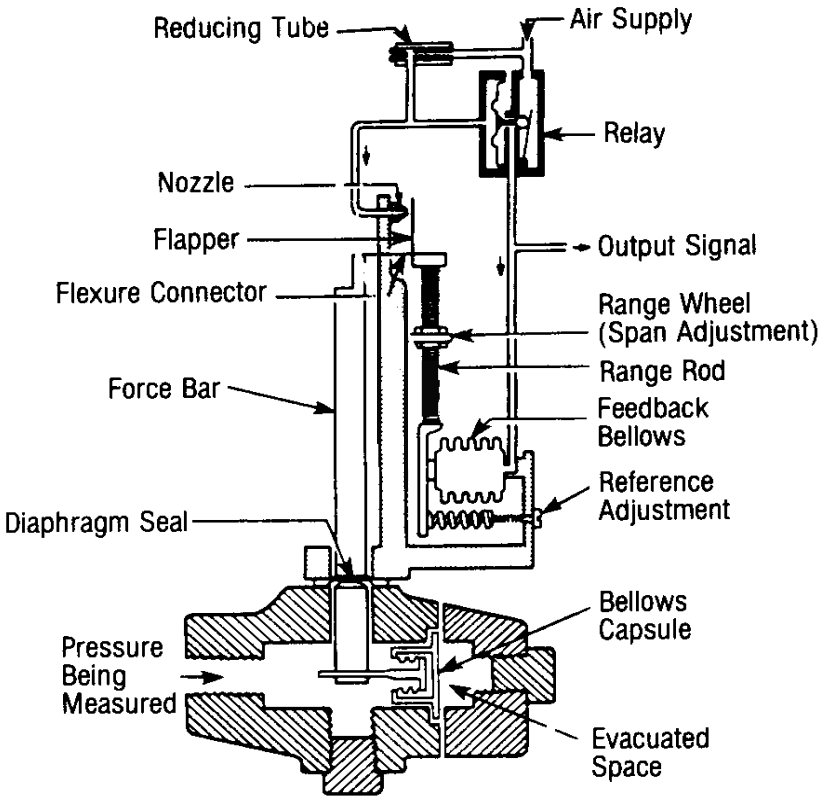


Figure 6, Force-Balanced, Absolute-Pressure Sensor

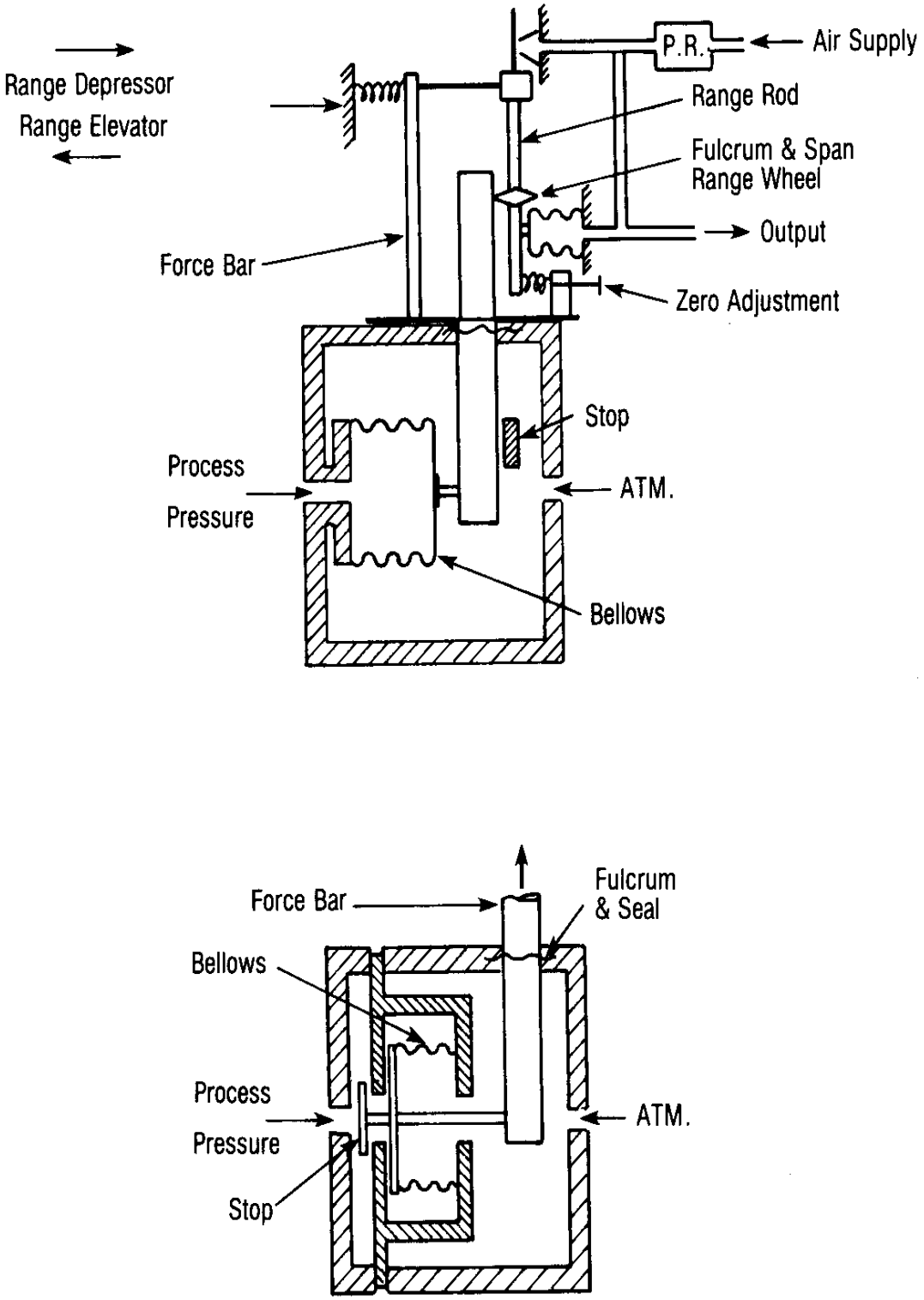
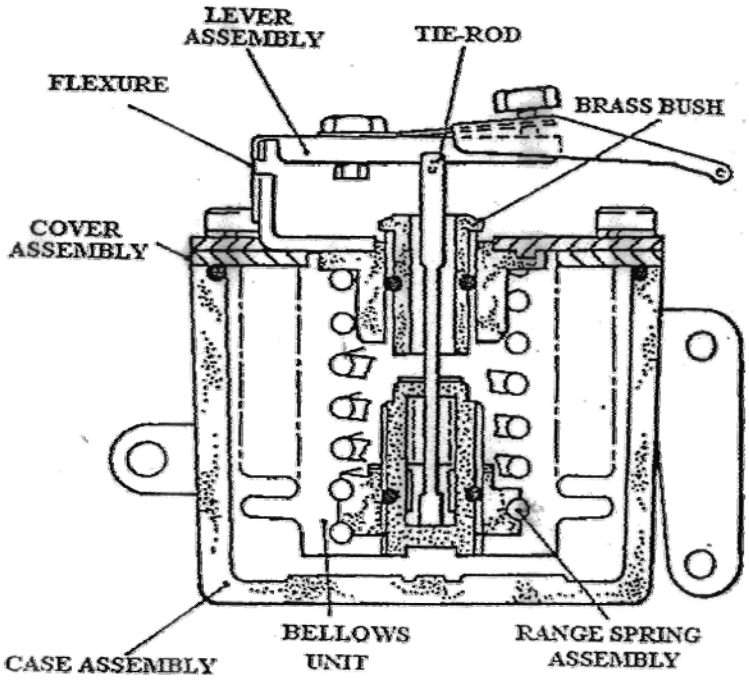
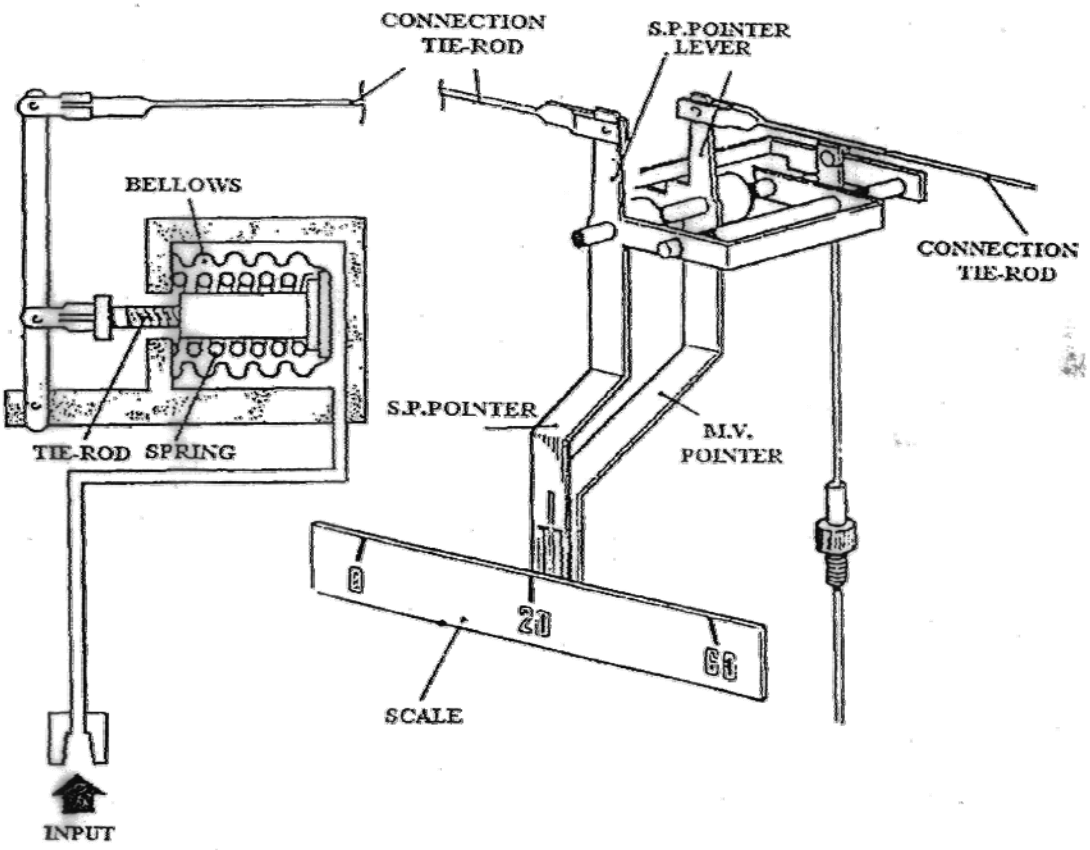


Figure 7, Two Types of Force-Balance, Gauge Pressure Sensors



Internal workings of the bellows



Diaphragm Sensors

A diaphragm is a thin, flexible, flat or corrugated disk, held in place so that it is axially flexible. When pressure is applied to one side of the diaphragm it will deflect. Deflection is proportional to the pressure. The force opposing the pressure is the sum of the spring constant of the diaphragm, the pressure on the opposing side of the diaphragm, and the spring constant of any opposing spring. The sensitivity of a diaphragm increases as the diameter increases. The axial movement of the diaphragm can rotate a pointer or actuate a controller or transmitter by attaching the free end to a mechanical linkage.

There are two types of diaphragm elements, elastic and limp. The elastic type uses the stiffness of the diaphragm to oppose the pressure applied. It is usually metallic and comes in two different configurations; single and capsular.
The single diaphragm is, as its name implies, a single diaphragm either flat or with concentric corrugations.

The capsular diaphragm consists of two diaphragms welded together at their perimeters as shown in figure 8. Capsules can be either convex or nested as illustrated. Capsules can be mounted in multiples to give more deflection for a given pressure as shown by figure 9. Evacuated capsules are used for absolute pressure reference and single diaphragms for very sensitive measurements.

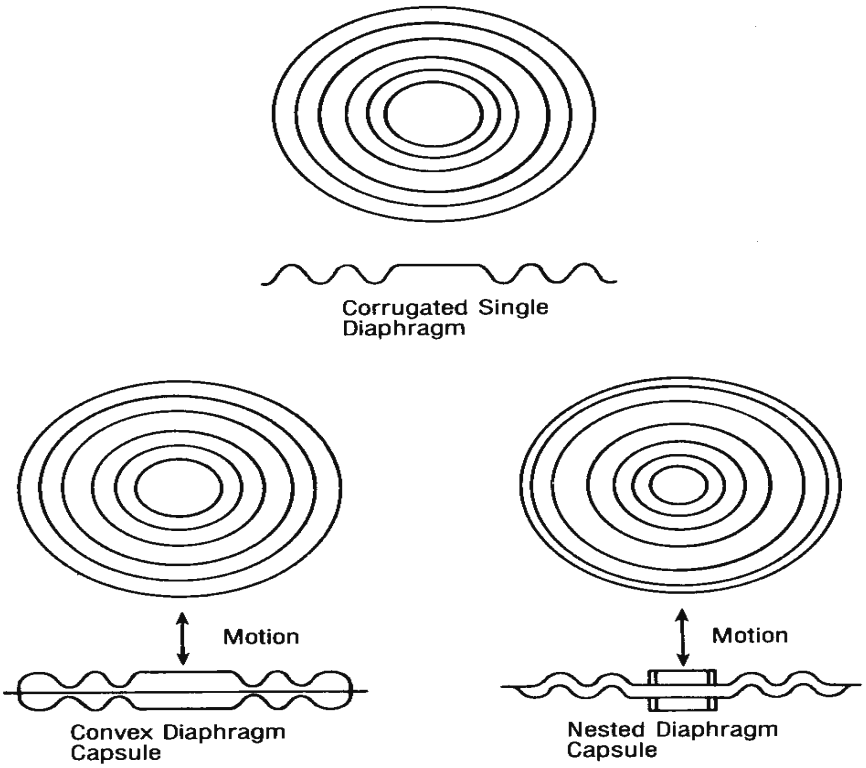


Figure 8, Typical Diaphragm Elements

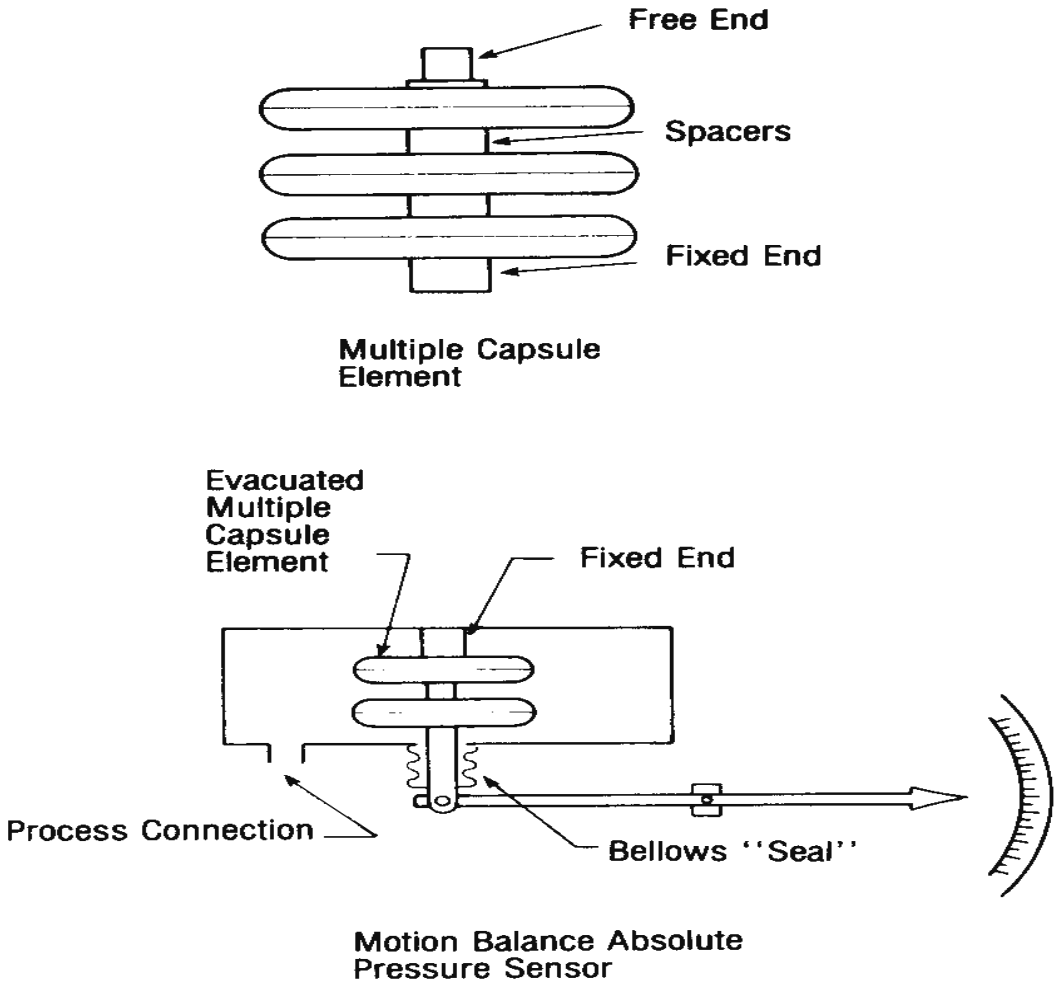
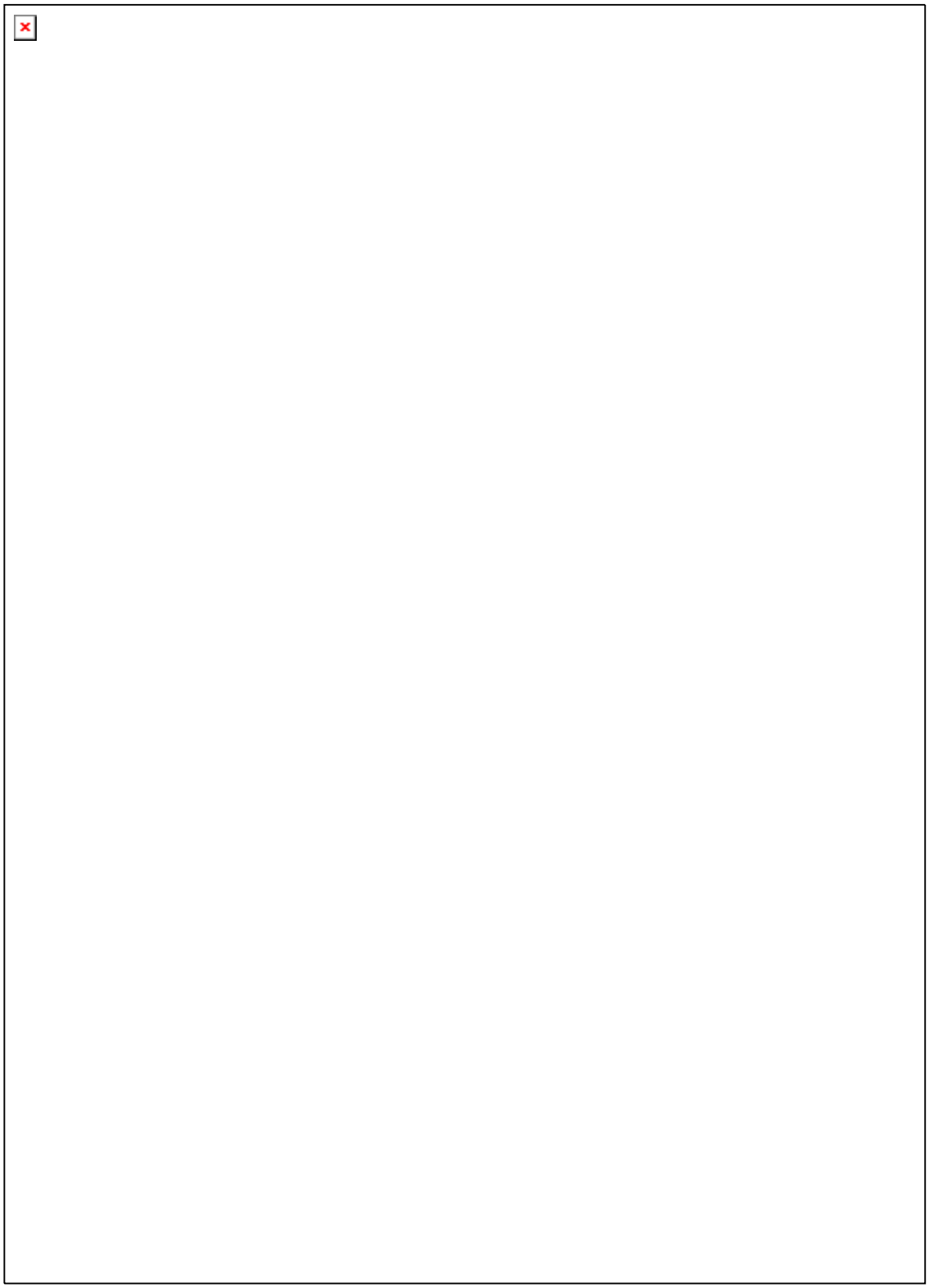


Figure 9, Examples of Capsule-Type Pressure Sensors



Resonant-Wire Sensors

Resonant-wire sensors are used in electronic pressure transmitters. The resonant frequency of a vibrating wire is a function of the length, the square root of the tension, and the mass of the wire. When the length and mass are constant, it can be said that wire's tension is proportional to pressure then the resonant frequency will be a function of pressure.

In resonant-wire pressure transmitters, a wire or ribbon under tension is located in the field of a permanent magnet. The tension on the wire is proportional to the pressure. An electrical signal with a frequency proportional to the square root of the tension will be generated. This signal is converted to a 4-20 ma transmitter output.
This principle is illustrated in figures 10 and 11. Figure 10 is a diagram of the sensor assembly for a medium-range, gauge pressure transmitter. This sensor uses a taut wire surrounded by fluid. One end of the wire is connected to the closed end of a metal tube, which is fixed to the sensor body. The other end of the wire is connected to a bellows.

Initial tension is applied to the wire by the spring connected between the bellows and the zero-adjustment screw. The fill fluid transfers the force of process pressure on the diaphragm assembly to the bellows. This force on the bellows changes the tension on the wire and thus its resonant frequency.

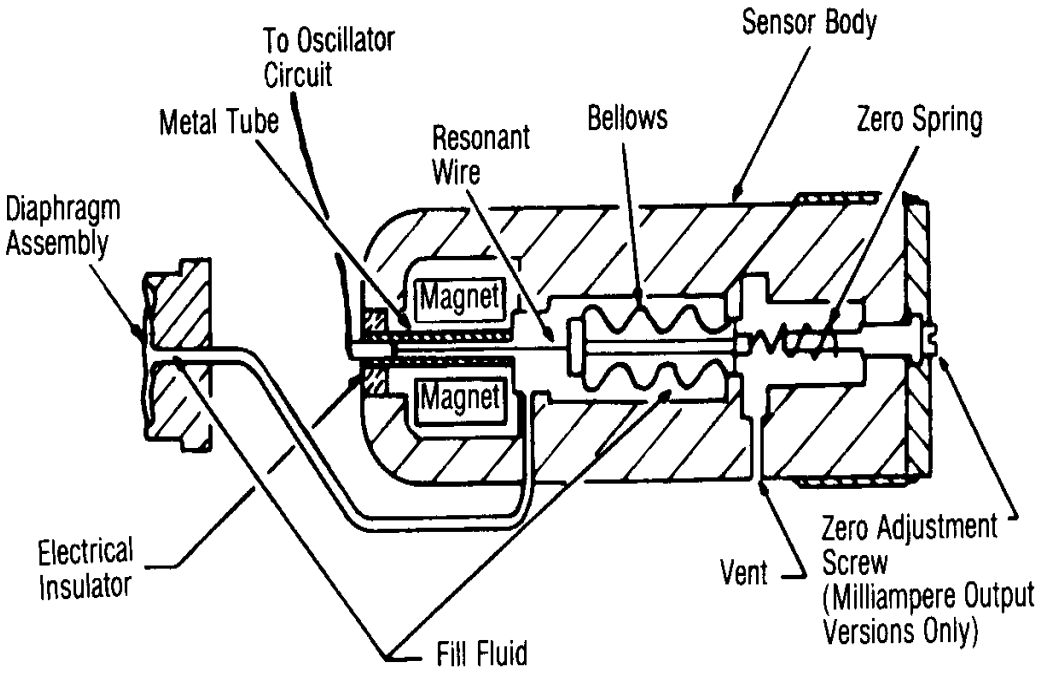


Figure 10, Resonant-Wire, Medium-Pressure

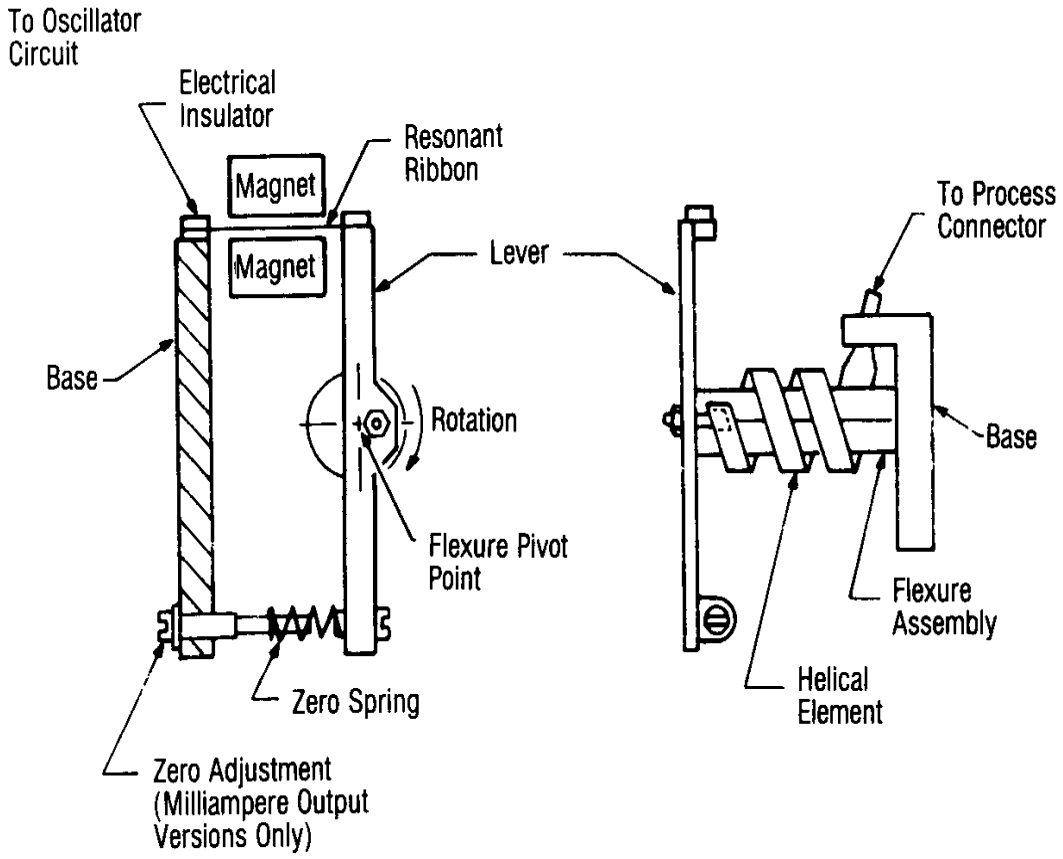


Figure 11, Resonant-Wire, High-Range Pressure Sensor

Strain-Gauge Sensors

Strain-gauge pressure sensors are used in most brands of electronic pressure transmitters. When metallic conductors or semiconductors are subjected to mechanical strain, a change in resistance will occur. This resistance is then electrically converted into a 4-20 mA signal proportional to the pressure.

There are many different designs of strain-gauge pressure sensors. The most common designs use a metallic diaphragm to isolate the process fluid and exert a force on a force bar as shown in figure 12. This force bar transfers the diaphragm movement to the strain gauge. Most of the strain elements in current use are semiconductor type.

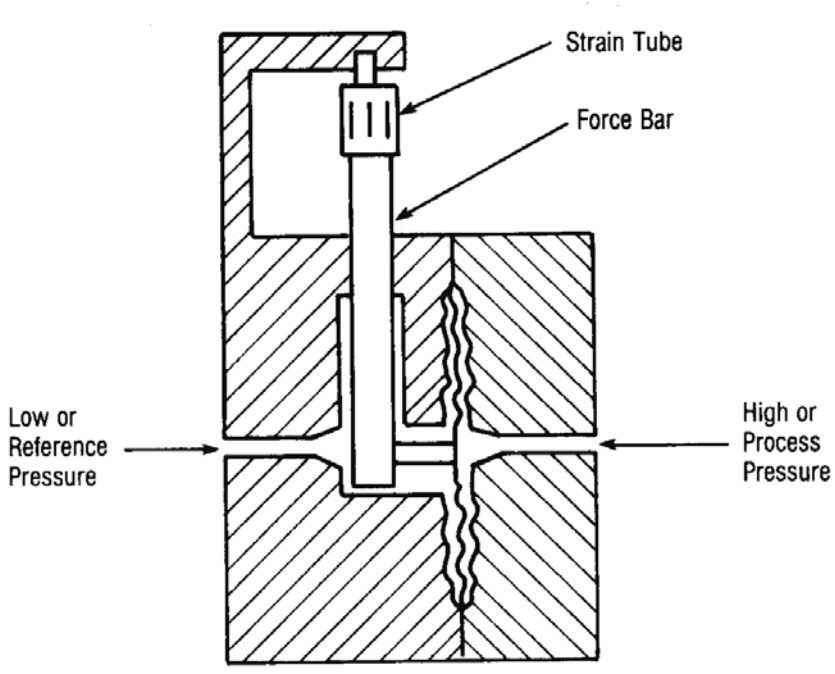


Figure 12, Force Balance D/P Cell with Strain Gauge Elements

Temperature-compensated Whetstone bridges circuits as shown in figure 13 measures the resistance change. The bridge imbalance is converted electronically to a 4-20 mA signal.

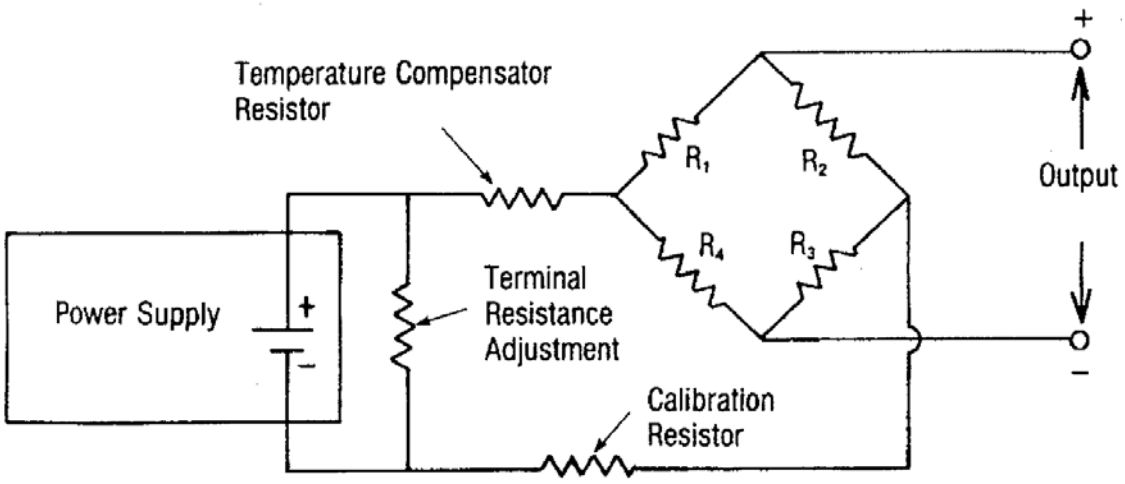


Figure 13, Whetstone Circuit for Strain Gauges

Gauge pressure is measured with the backside of the diaphragm left open to the atmosphere. Absolute pressure is measured by evacuating and sealing the backside of the diaphragm. Strain gauge pressure sensors can be used for ranges from 0-30 inches H_2O (0 -750 mm H_20) to 0 -10,000 psig (0 -66,000 kPa). These devices are stable with high speeds of response and are relatively small. Strain gauge accuracy falls between 0.2 and 0.5 percent of span. Special designs can handle process temperature to 600°F (316°C).

Capacitance Pressure Sensors

Capacitance pressure sensors are also used in electronic pressure transmitters. These devices operate on the principle that the change in capacitance resulting from the movement of an elastic element is proportional to the pressure applied to the elastic element. The elastic element usually is a stainless steel diaphragm. Other materials are available if stainless steel is not suitable for the process fluid. As shown in figure 14, the capacitor plates. A high-frequency oscillator is controlled by the sensing

element. Changes in pressure deflect the diagram and the resultant change in capacitance changes the oscillator frequency. The variation in oscillator frequency is converted to a 4-20 mA signal proportional to the pressure.

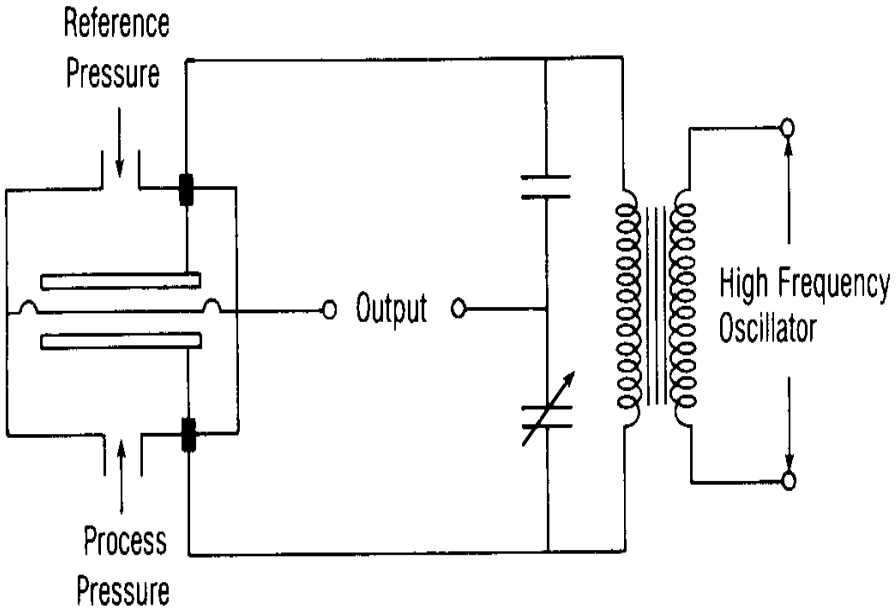


Figure 14, Capacitance Pressure Sensor

Spring-Loaded Piston Sensors

Spring-loaded piston sensors are used for both pneumatic and electric pressure switches. These devices are usually called pressure switches when companies who fit either electric or pneumatic output modules to their sensors. Companies who manufacture devices, which are only pneumatic refer to their products as pressure sensors or pressure pilots. Heavy-duty pressure sensors such as the one shown in figure 15 are often called stick pilots.

Stick pilots are manufactured so that they can serve as either a high-pressure sensor or low-pressure sensor as required. The terms high-pressure pilot and low-pressure pilot refer to the way the sensor is connected rather than being two different devices.

Figure 15 shows a stick pilot with no process pressure applied. When installed as a high-pressure pilot, instrument air is connected to the high-inlet port and the shutdown system is connected to the outlet port. The low-inlet port is left open. Notice that the high-inlet port and the outlet port are connected.

A stick pilot installed as a low-pressure pilot will have the instrument air connected to the low-inlet port and the shutdown system connected to the outlet port.

The high-inlet port will be left open. The shutdown system is vented when the process pressure is below the set point and pressured when it is above the set point.

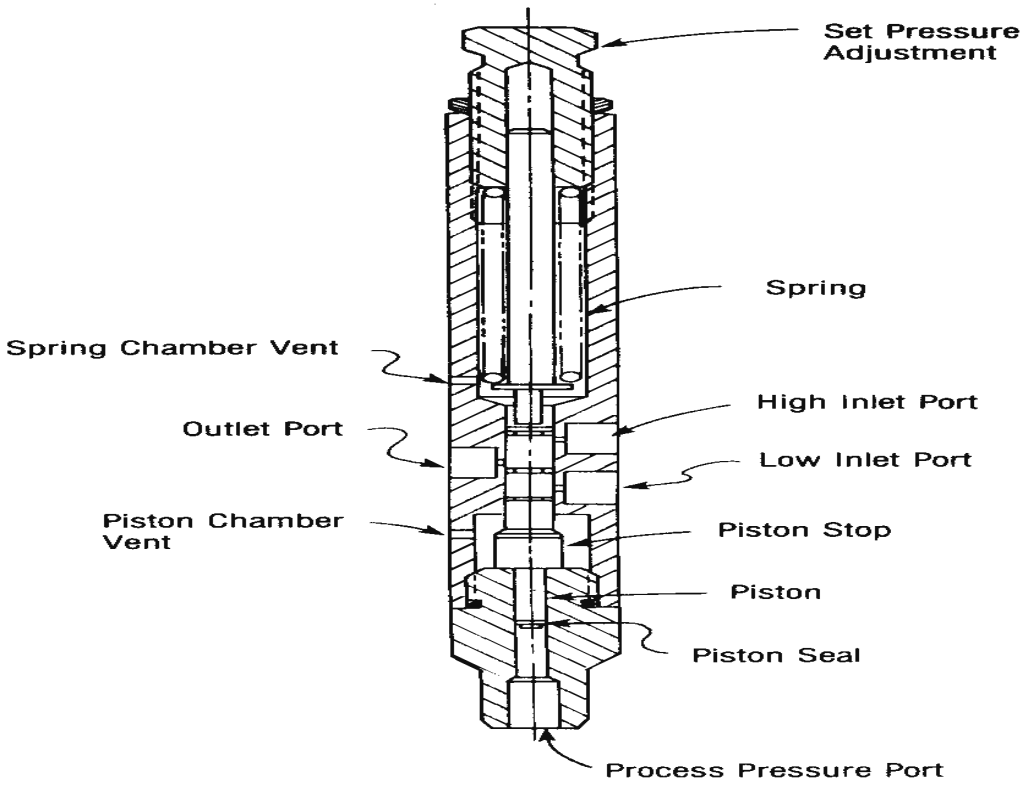


Figure 15, Typical Stick Pilot

Pressure Sensor's Protection

Certain applications will be so severe the pressure sensor will not remain functional for any reasonable amount of time. For these cases the devices described in the following sections can be used to protect the pressure sensor.

Diaphragm Seals

Diaphragm seals are used to isolate the pressure sensor from the process fluid. This is done when the fluid is toxic, corrosive, dirty (has entrained solids or mud that may plug the instruments), solidifies at ambient temperature, or is extremely cold and may freeze the instrument. The diaphragm seal is a thin, flexible disk, which separates the pressure sensor from the process media.

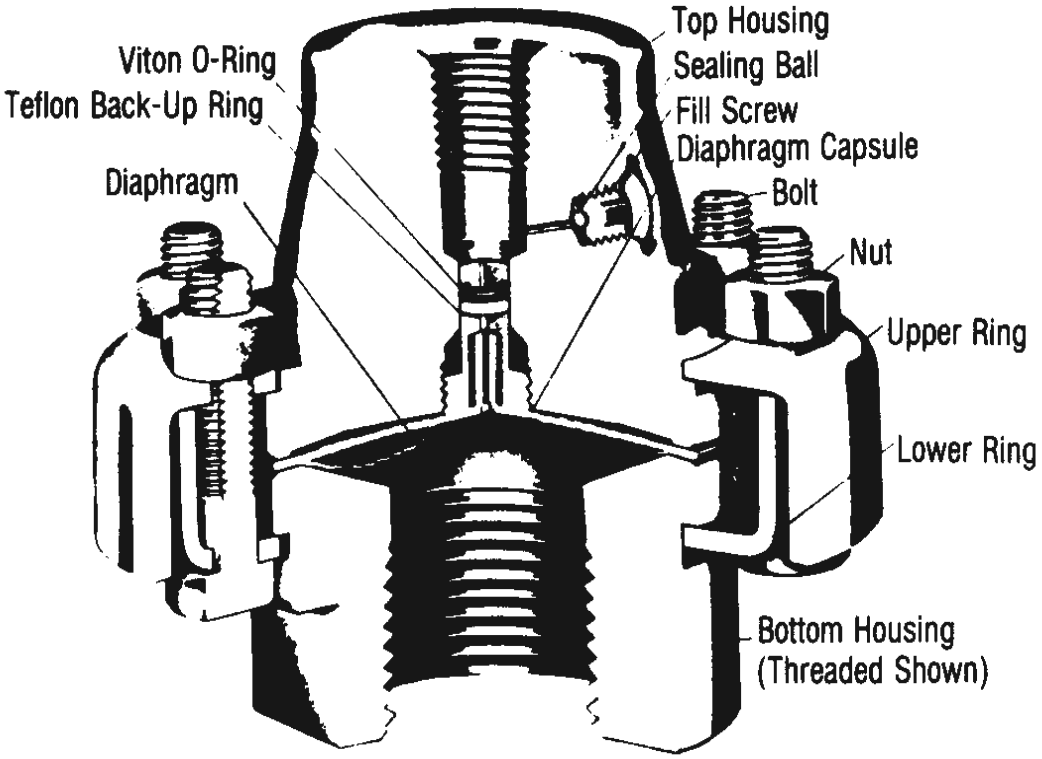
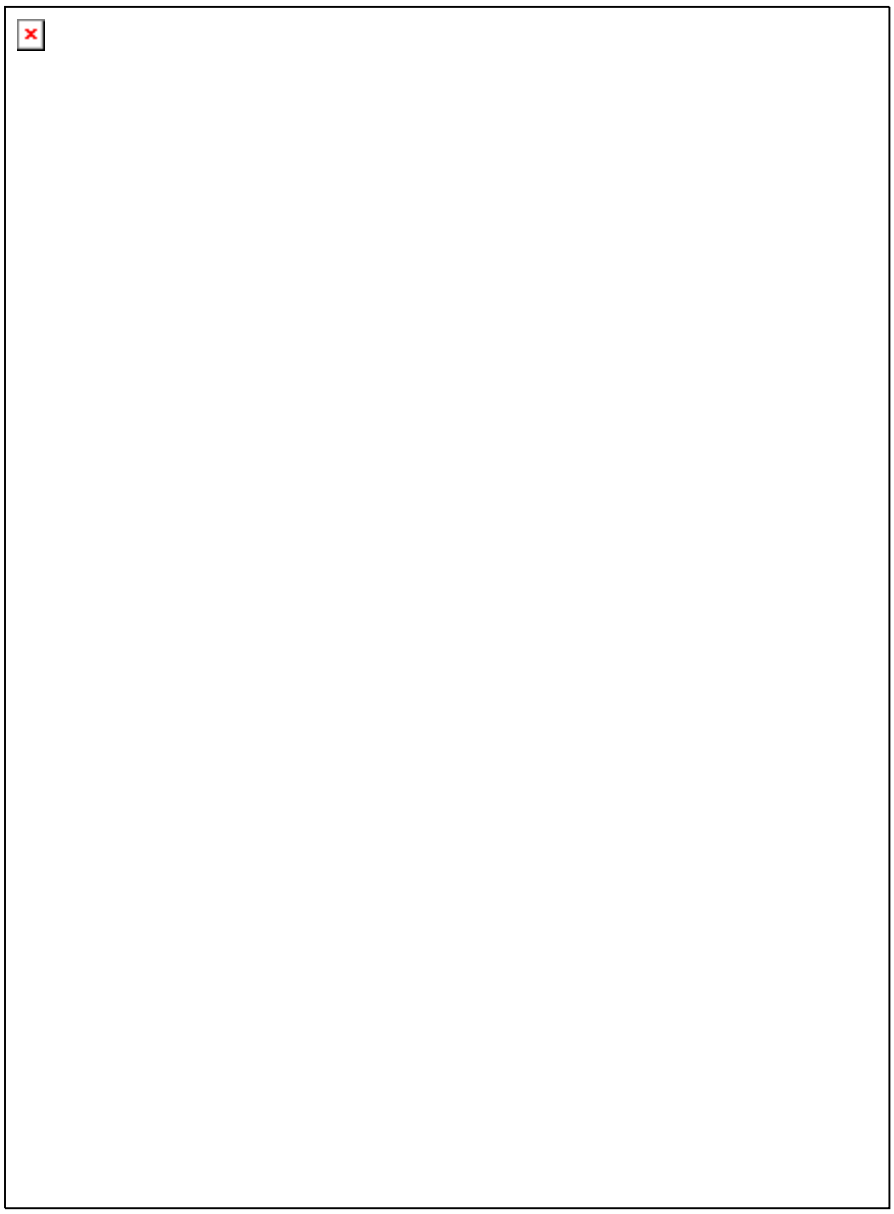


Figure 16, Diaphragm Seal

The connecting space on the sensor side of the diaphragm is completely filled with a non-compressible liquid. When process pressure is applied, the diaphragm is displaced sufficiently to transmit an equal pressure to the pressure sensor.

The three main components of a diaphragm seal are the top housing, bottom housing, and diaphragm as shown in figure 16.



Siphons

Siphons are generally used to isolate a hot-process media from the pressure sensor. The siphon is a metal, tubular device shaped to form a plumber's loop, (a low pocket in the tube). It can either be filled with a high-boiling-point liquid or process condensate which acts as a barrier to the heat contained in the hot gases or steam as shown in figure 17. In addition, these devices will act as a pulsation dampener.

The path the hot vapor takes to the pressure sensor is relatively long and narrow with a lot of surface area for cooling siphons.

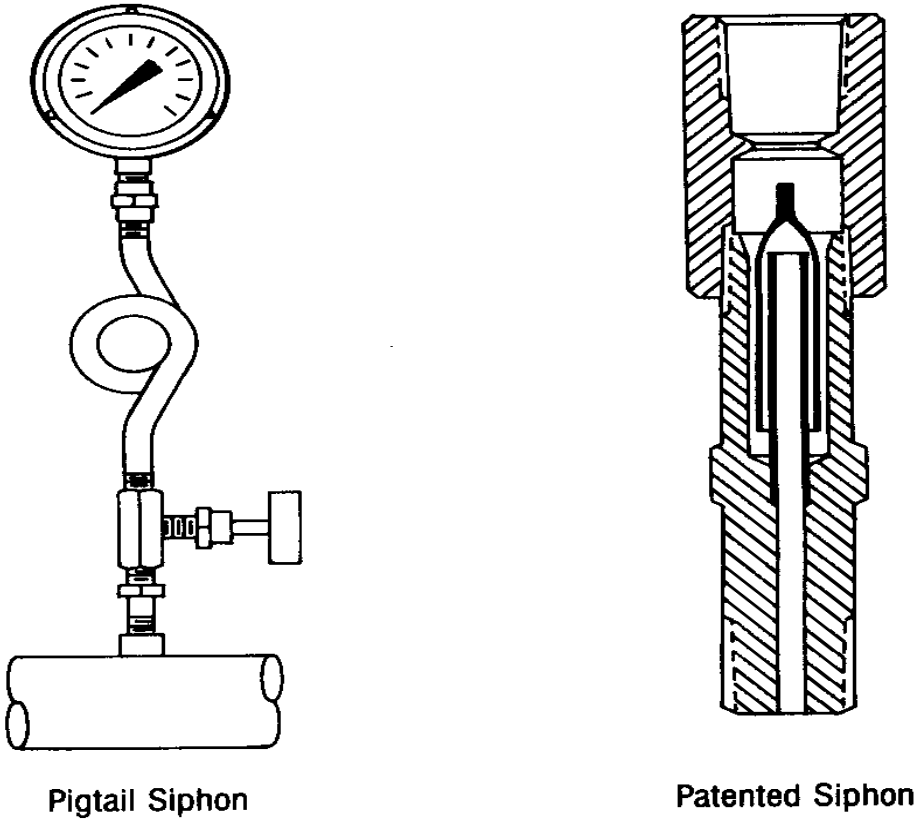


Figure 17, Two Types of Siphon Pressure Sensors

Throttling Devices

Throttling devices are commonly used to dampen high-frequency pressure fluctuations by putting a restriction in the inlet to the pressure sensor.

Throttling screws are the simplest means of providing a restriction, Throttling screws are a special screw that comes in several orifice sizes and are inserted into a tapped hole in the base (socket) of the pressure sensor to provide a flow restriction as shown in figure 18.

Pressure snubbers are very common for attenuating pressure fluctuations and filtering the media. Snubbers are compact fittings with a porous element, which both restricts the velocity and filters the fluid as shown in figure 19.

The pulsation dampener is another commonly used device. This device is also sometimes called a pressure snubber, but does not have a filtering element. There are several designs of pulsation dampeners.

The most common design consists of a bar-stock fitting, (sometimes two fittings screwed together), as shown in figure 20. As the pressure pulse comes through the dampener, the piston is forced up and restricts the flow from the large chamber by closing the outlet of the chamber.

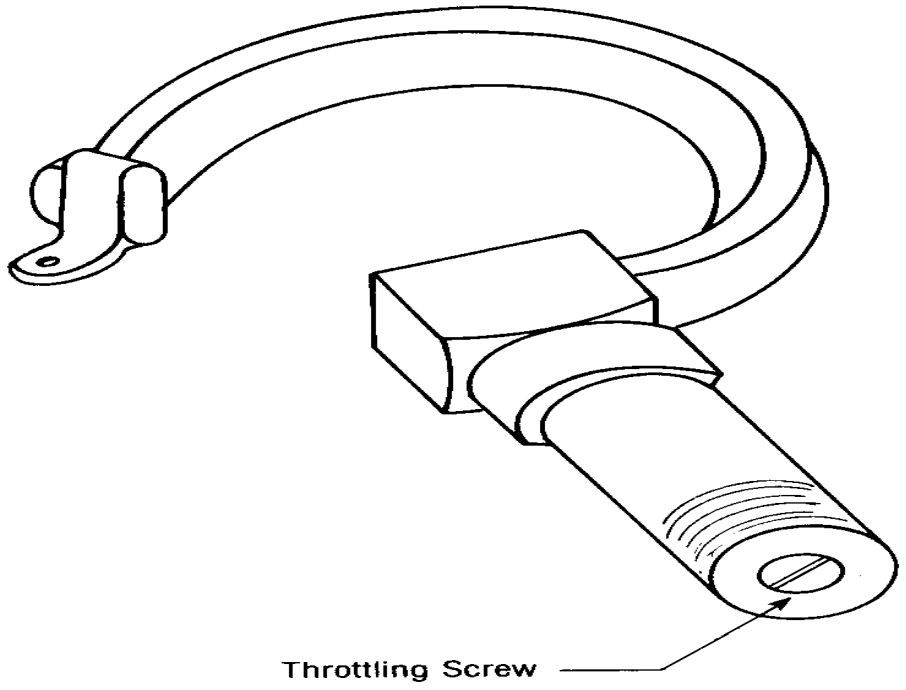


Figure 18, Gauge Borden Assembly with Throttling Screw

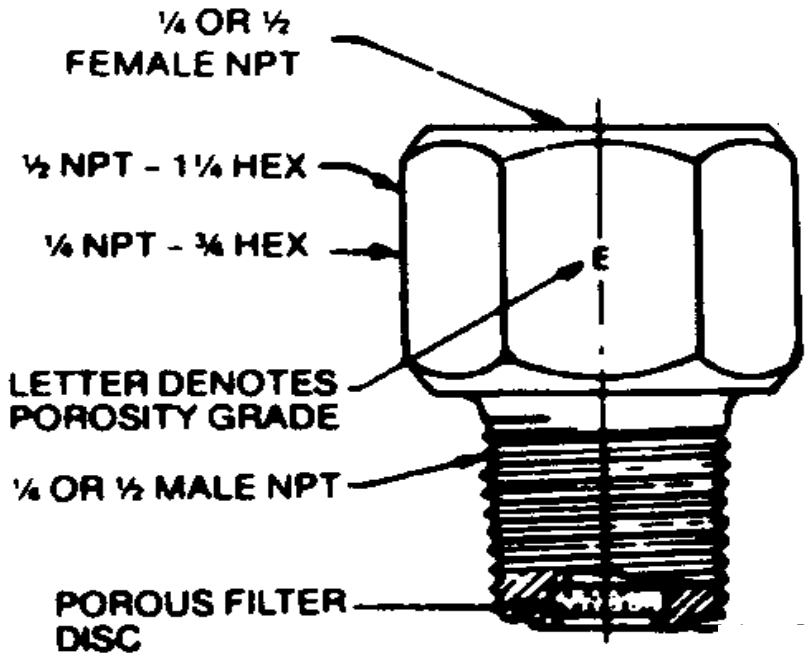


Figure 19, A Typical Pressure Snubber

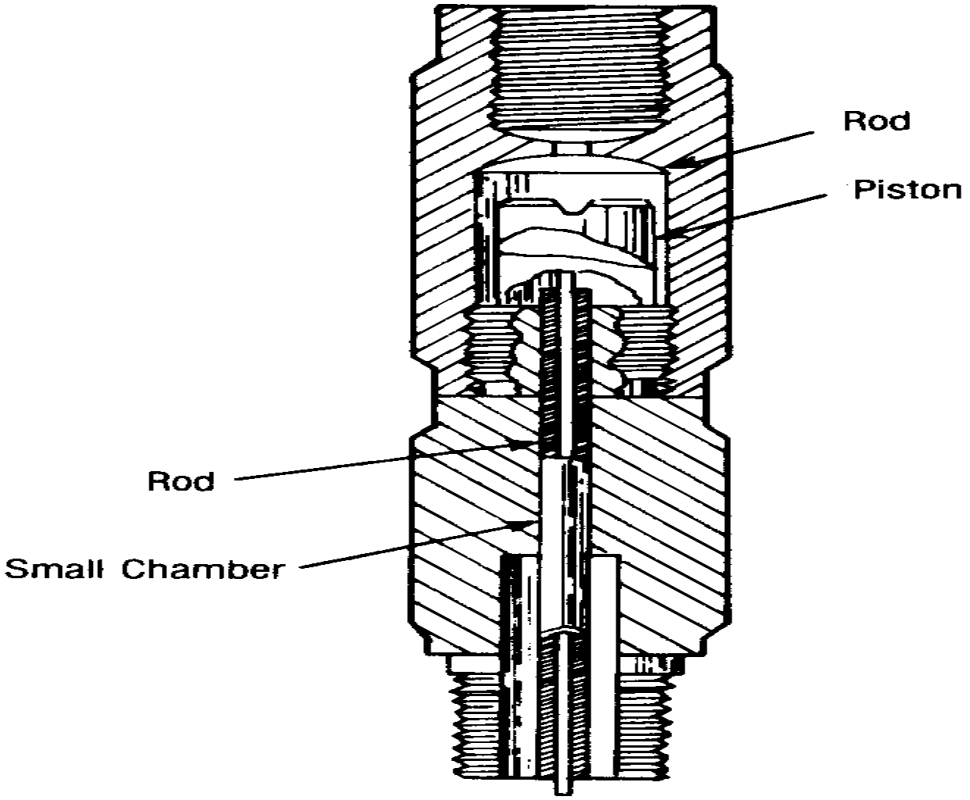


Figure 20, Typical Pulsation Dampener

Pressure-Limiting Valves

Pressure-limiting valves protect the pressure sensor from overpressure by blocking the process fluid at a preset limit. There are several designs of pressure-limiting valves. One common design has the fluid coming in the inlet, passing around a piston, and out to the pressure sensor as shown in figure 21. The piston has process pressure on the bottom and atmospheric pressure on the top. A spring opposes the process pressure. As the pressure increases, it exerts greater force on the piston and moves the O -ring up to seal the area around the piston and isolate the pressure sensor. The set point is adjusted by compressing or releasing the spring and thereby changing the force required to move the piston.

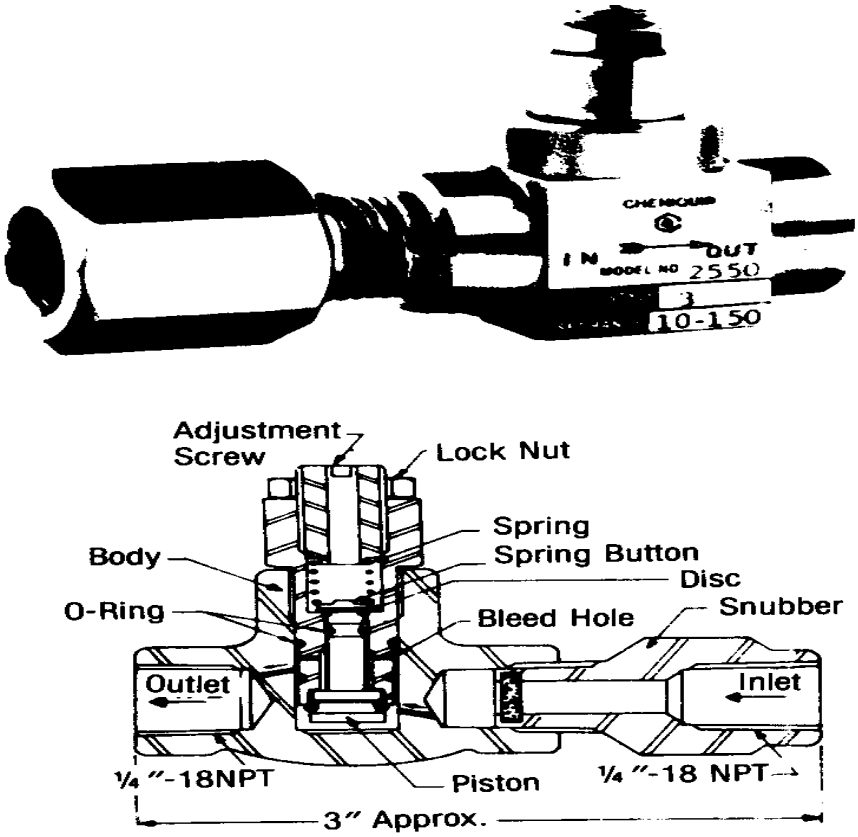


Figure 21, Pressure-Limiting Valve

Pressure Measurement Devices

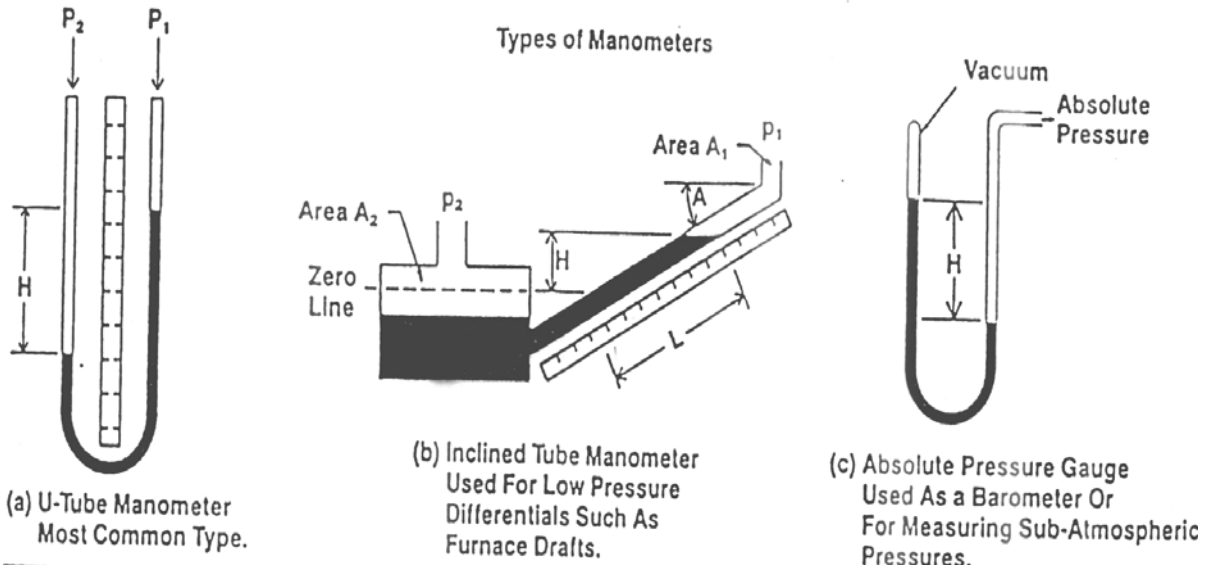
Manometers

Manometers work on the principle of balancing an unknown pressure against a known pressure produced by a column of liquid in a vertical or inclined tube.

The typical pressure range covered by manometers is from absolute zero pressure to approximately 1.5 bar depending upon the length of the tube and the liquid used within the manometer. Some indicating liquids for use in manometers are shown in the following table.

Liquid	Relative Density
Transformer Oil	0.864
Water	1.000
Dibutyl Phthalate	1.048
Carbon Tetrachloride	1.606
Mercury	13.560

It is important to use the correct relative density of liquid in the manometer, as the wrong fluid will result in incorrect readings. If transformer oil were used instead of water in a manometer the resulting pressure reading would be too high, due to the oil being less dense than water.

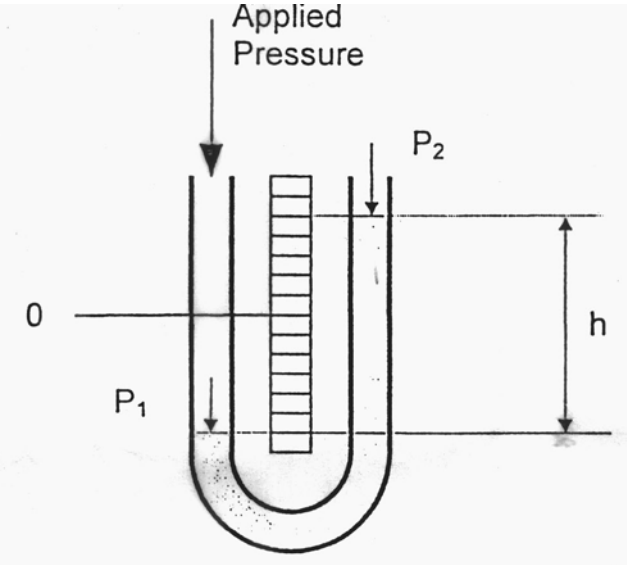


The U tube or Double Limb

The U tube manometer is widely used as a simple means of measuring low pressures. Provided a reading is taken between the levels in each limb, the shape and size of the glass tube play no part in the accuracy. In use the tube has to be mounted vertically.

In practice it is common to have an adjustable scale graduated from a centre zero line and read off from both sides of the scale.

It is essential that the U tube is of uniform bore, otherwise the readings from the left and right scales will disagree. The applied pressure is equal to the sum of the tow scale readings.



The U tube manometer has the following advantages:

Simplicity and no mechanical moving part

Accuracy and repeatability

However it also has a number of disadvantages:

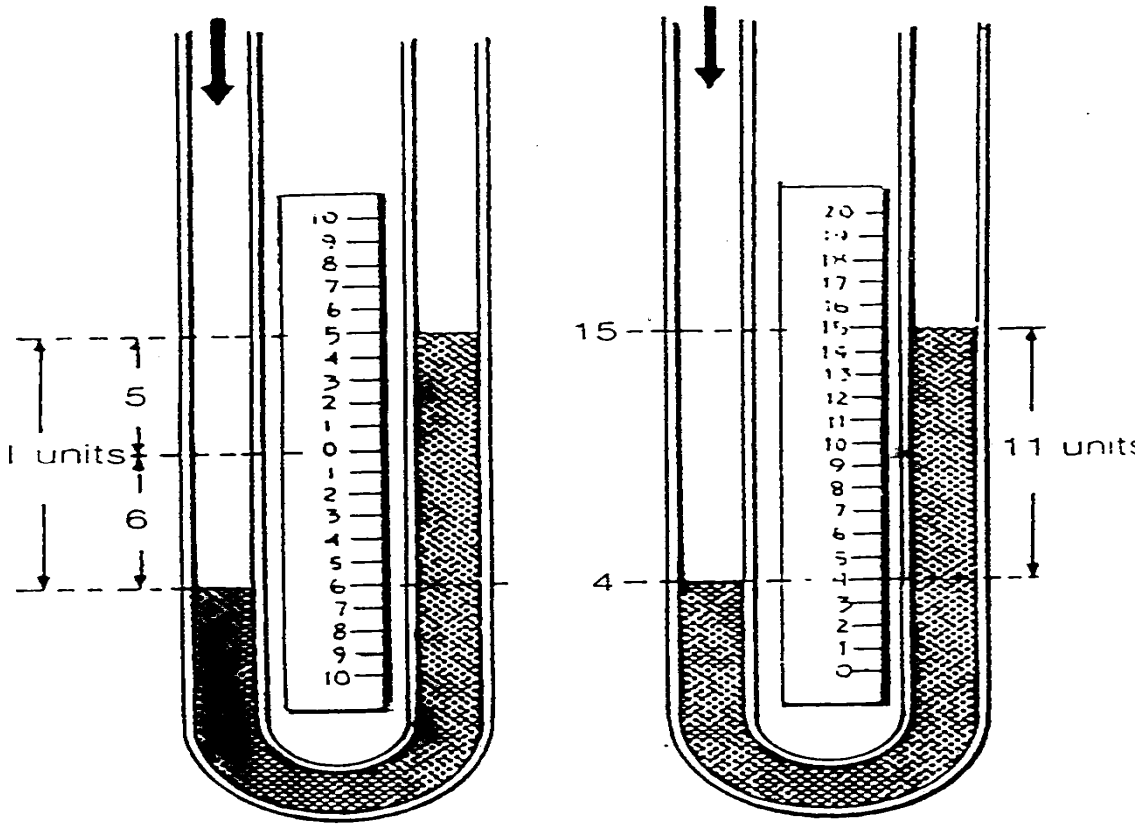
It must be carefully positioned

The range is limited otherwise the tube becomes too long and cumbersome.

Note:

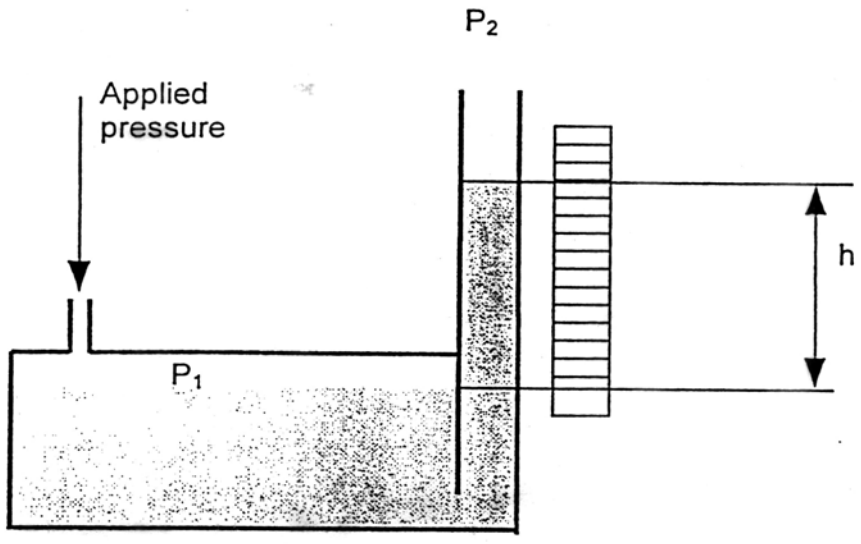
If mercury is used as the liquid in a manometer then care must be used when reading the manometer. The meniscus of mercury is convex, by comparison with other liquids that have a concave meniscus. The reading has to be taken the top of the meniscus and should always be read at its centre.

Consideration should be made in to the dangers of using mercury in such a fragile glass container and the proper precautions made available in the event of a spillage.



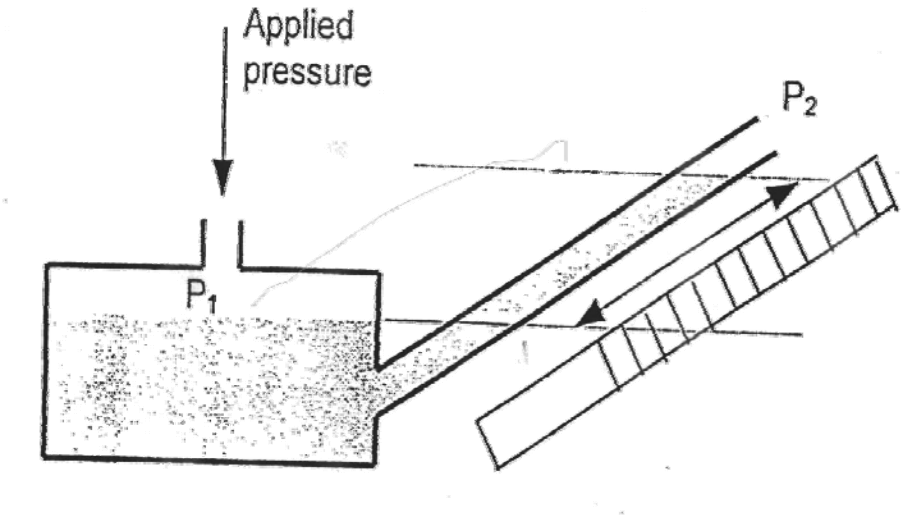
Single limb or well-manometer:

This is essentially a u tube manometer with one limb very much larger in diameter then the other and is widely used because of the convenience of having to read only a single leg.



Inclined Manometer

By using an inclined manometer a greater sensitivity can be achieved. This instrument is used for measuring very low pressures such as the draft in a furnace, a chimney or a ventilation duct.



The inclined manometer enables small pressure differentials to measure more conveniently and more accurately than using the U tube or well type.

The inclined manometer is a modification of the well manometer. Instead of being vertical the single leg of the inclined manometer is sloped at small angle above the horizontal. This produces a larger movement and results in a more easily read length of liquid.

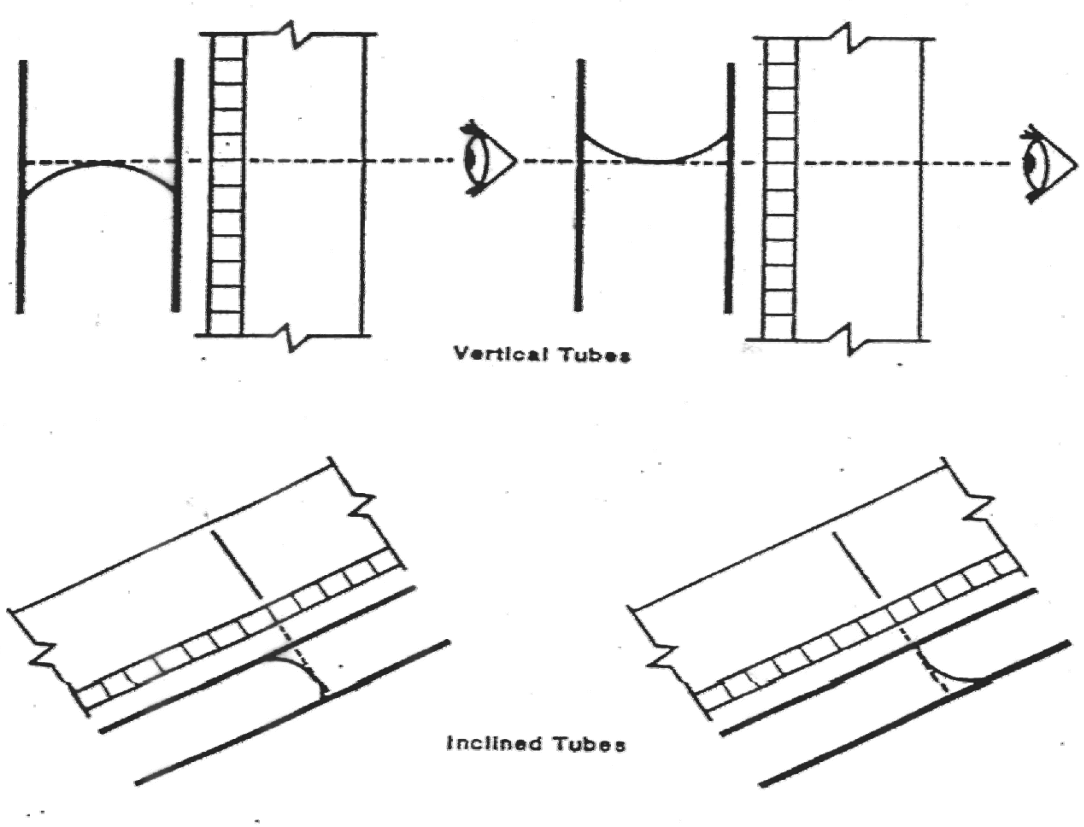
Because the reading of the manometer is very sensitive to any change in angle the instrument is usually mounted on leveling screws and fitted with a spirit level, so that it can be accurately set up before use.

Note:

With all types of manometer care must be taken to avoid a parallax error by ensuring that your eye is in line with the meniscus.

Other sources of error when using a manometer include:

- The effect variation in local gravity
- The effect of temperature



Manometric Errors

Meniscus

When tube contains a liquid, the surface of the liquid is not flat, but curved this surface is called the meniscus with mercury the meniscus is convex and with other liquids it is concave.

When reading a manometer it should from the centre of the meniscus. This is caused by not viewing the liquid at right angles with the scale.

Parallax

Parallax error can be minimized by viewing the manometer at right angles and by putting the scale as close to the manometer as possible.
Pressure Gauges

A pressure gauge is a device, which senses pressure and provides a visual representation of that pressure. Most pressure gauges have bourdon tube sensors. Vacuum gauges and low-range gauges often use bellows sensors. Differential-pressure gauges can use piston or bellows sensors. The preferred manufacturer and the required range usually dictate the sensor type.

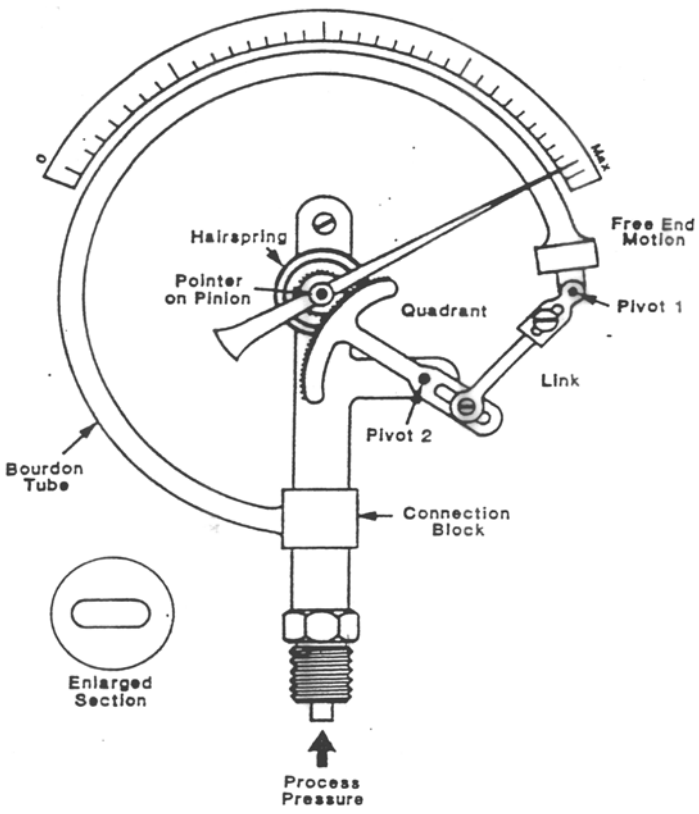
Selection of Pressure Gauge

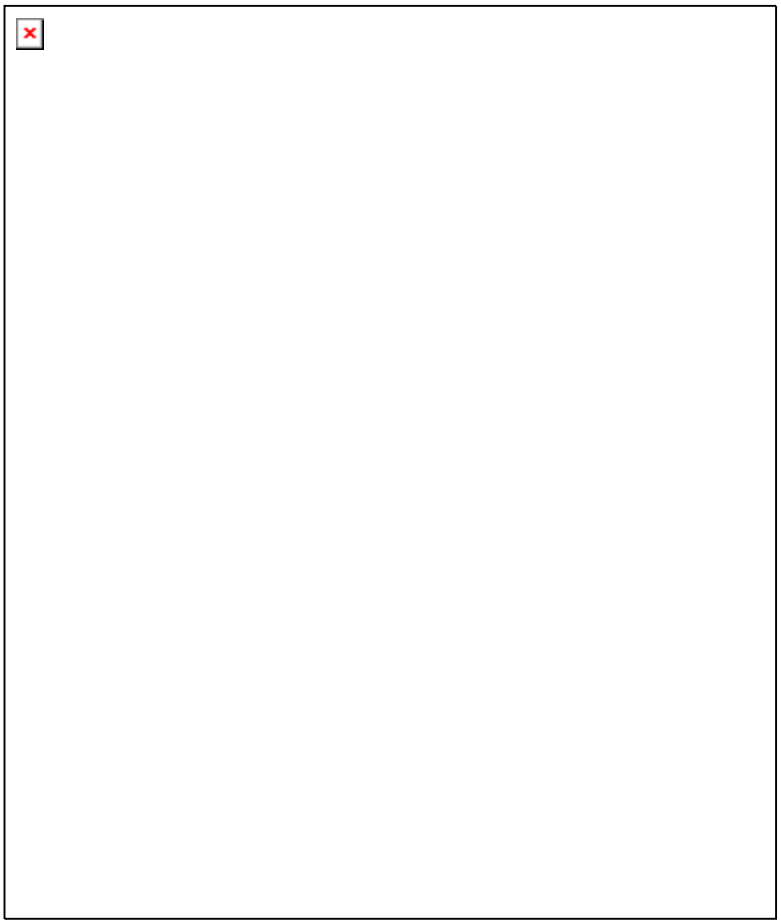
Pressure gauges should be selected so that the expected operating pressure is in the centre third of the gauge range. It is also important that the highest pressure that will ever be applied to the gauge be below the maximum reading. Usually, the gauge shall be selected so that the gauge maximum is above the set pressure of the system relief valve and the normal pressure is in the readable range.

Pressure gauges are sometimes liquid filled. This is to protect the gauge dial and movement from the atmosphere. The liquid fill also provides some pulsation or vibration dampening. Glycerine is the most common fill liquid.

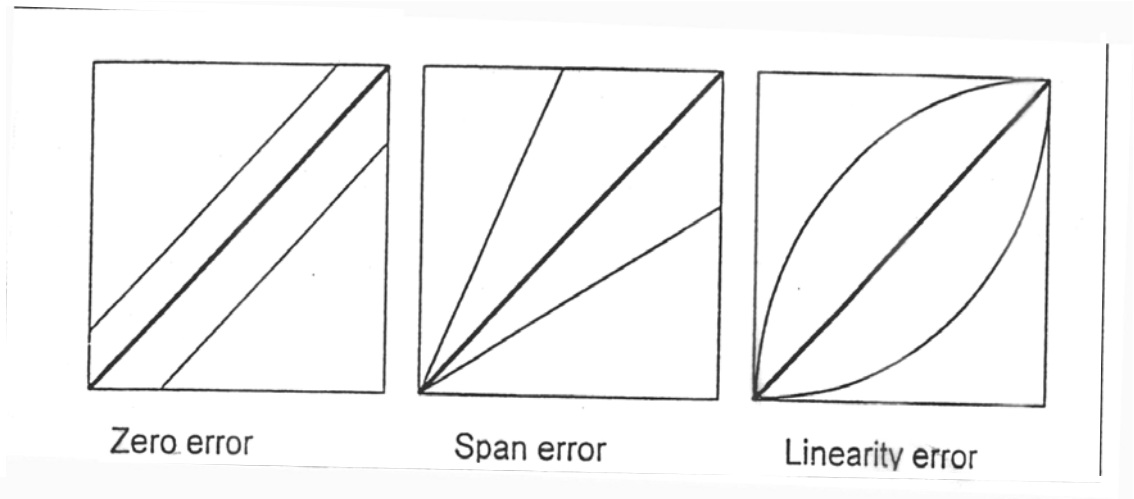
Pressure gauges lose accuracy when exposed to hot fluids. When the process temperature is above approximately 180^{o} F (82^{o} C) a siphon should be installed. If the process fluid will not condense, at ambient temperature, the siphon can be filled with a suitable fluid such as ethylene glycol or glycerine.

Differential-pressure gauges are useful when a pressure difference that is small compared to the static pressure needs to be measured. Differential-pressure gauges differ from static-pressure gauges in that they have two pressure connections. Differential gauges must be installed with an equalizing valve so that they will not be over-ranged while disconnecting.





Gauge Errors



Pressure gauges may suffer form several types of errors.

A gauge with a zero error will always read high or low by a constant amount.

A gauge with a span error suffers from an internal magnification error therefore the gauge reading will by out by different amounts at each point.

A gauge with a linearity error may read correctly at 0 and 100% but will not follow a linear path between these points. This is one reason why it is not sufficient to just check a gauge at its tow and points but to carry out a three or five point check Checks should be made on both rising and falling pressures.

Precautions

Tubes for gauge to be used on Acetylene must be made of steel Gauges.

Associated fittings for use on oxygen must be kept entirely free of oil.

Gauges used on Hydrogen plants need to by gold plated.

Pressure Transmitters

Pressure transmitters are used when the controller, recorder, or indicator needs to be located in a control room or panel where it is undesirable to pipe the process fluid. They are also used when several devices are to be operated from a single measurement or when elevated zero is required. The output is usually 4-20, mA for electronic transmitters or 3-15 psig (20-100 kPa) for pneumatic transmitters. Other signals can be used if required by the receiver, but these are the most common and should be used if possible. Pneumatic pressure transmitter shown in figure 22

Suppressed Zero

Suppressed zero occurs when the base value of the measured variable is above the atmospheric pressure. Most transmitters have this as an option. Elevated zero is used when the pressure range of interest is to be narrowed for accurate monitoring and control (better resolution).

Elevated Zero

Elevated zero where the base value of the measured variable is below atmospheric pressure, is sometimes available. Usually, the zero is as near to perfect vacuum as possible and the unit is called a absolute pressure transmitter.

The use of pneumatic transmitters is decreasing; however, a number of manufacturers still make them for the replacement market and some new installations are still being made. Pneumatic transmission may be advantageous when existing equipment is pneumatic with which operating personnel are already familiar.

Electronic transmitters with 4-20 mA outputs are the most common. Typical electronic pressure transmitter is shown in figure 23.

Pressure transmitters are available in a variety of ranges. The ranges available vary from one manufacturer to another. Read carefully the manufacturer's literature before selection.

The range and the span are two different parameters. The span is the actual pressure range to be measured after the transmitter has been adjusted. The range is the pressure range within which the span can be adjusted.

Most transmitters have two adjustments, zero and span.

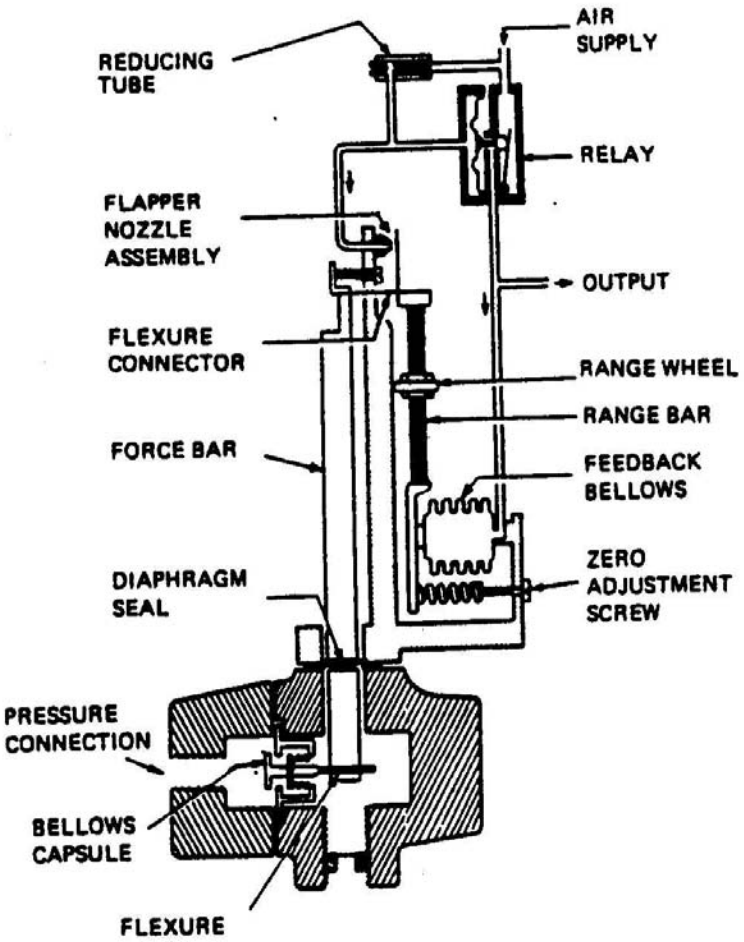


Figure 22, Pneumatic Pressure Transmitter

Differential pressure transmitters often referred to DP Cells are used to provide a pneumatic or electronic output for use in a remote indication panel or as an input signal to a control loop.

Typical Ranges:

Pneumatic transmitter Output: 0.2 to 1.0 Bar OR 3 to 15 psi

Electronic transmitter Output: 4 to 20 mA.

Pneumatic DP Cell

A diaphragm that is deflected by the applied differential pressure separates the HP and LP chambers of a DP cell.

A force bar at the top moves a flapper closer or further away from the nozzle depending on the pressure difference between the high and low signals. This movement results in a change in the output pressure from the transmitter that is proportional to the applied pressure difference.

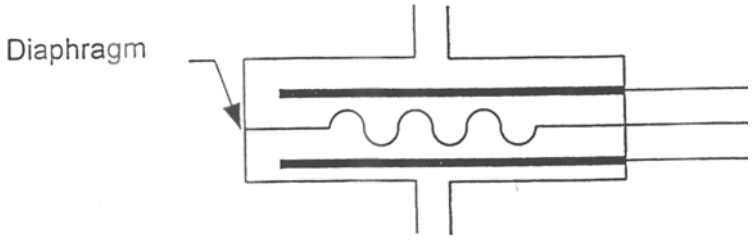
If the LP side is open to atmosphere, the cell will measure gauge pressure.
If the LP chamber is evacuated and sealed, the cell will measure absolute pressure.

Electronic DP cells

Electronic DP cells provide a higher level of accuracy then their pneumatic counterparts. Two sensor systems have gained popularity.

The capacitance type

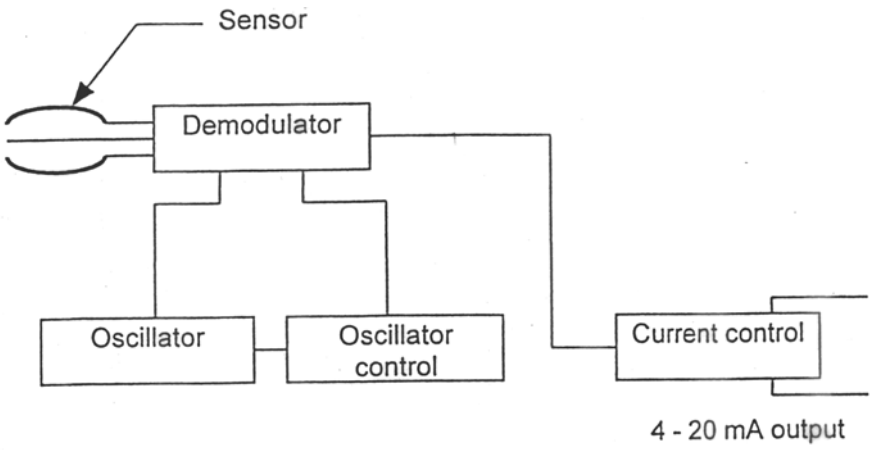
The resonant (vibrating) wire type



Capacitance Type

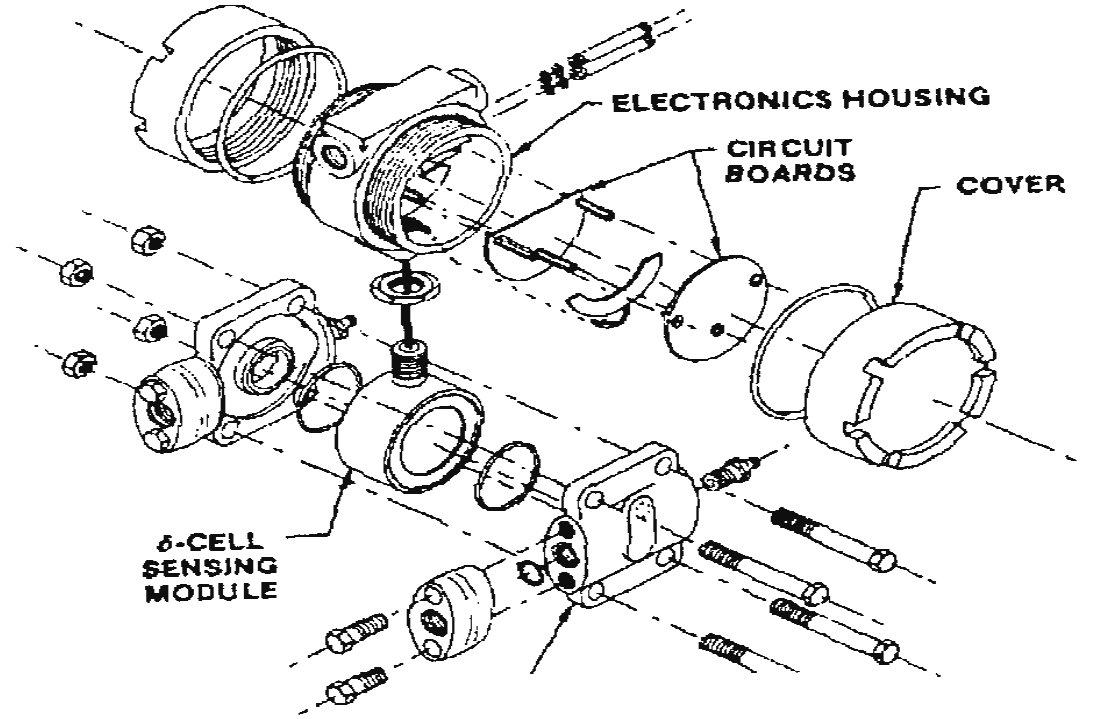
In this type of sensor a movable diaphragm is fixed between two capacitance plates. As the differential pressure is applied the diaphragm will move changing the capacitance between the plates. This change in capacitance can be used to change the frequency of on oscillator system where by the change in frequency is directly related to the pressure applied.

This gives excellent response, resolution, linearity, repeatability, and stability properties to the instrument.



Resonant (Vibrating) Wire Type

This system uses a pre- tensioned wire suspended in a magnetic field. The wire is forced to oscillate at its natural frequency. When a differential pressure is applied the tension in the wire changes changing the natural frequency of the wire. This can be easily detected and used to control an output current directly proportional to the applied pressure.



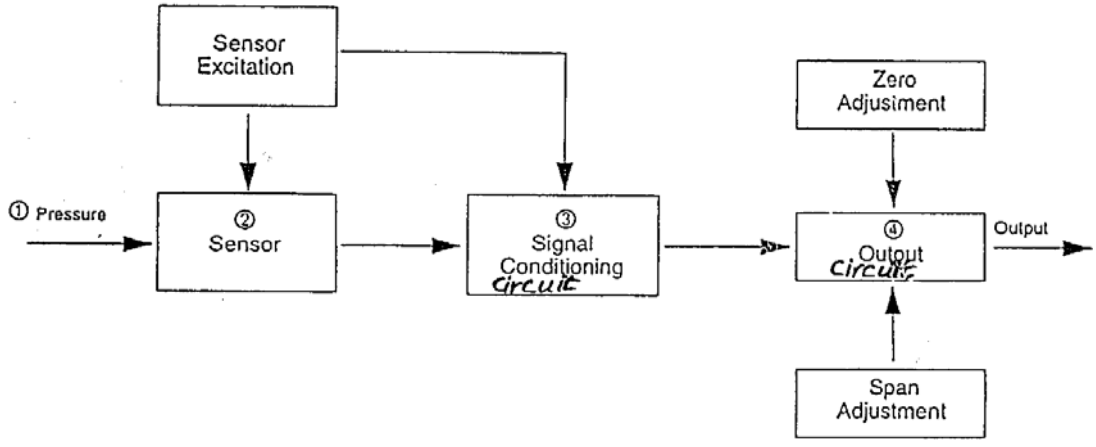


Figure 23, Electronic Pressure Transmitter

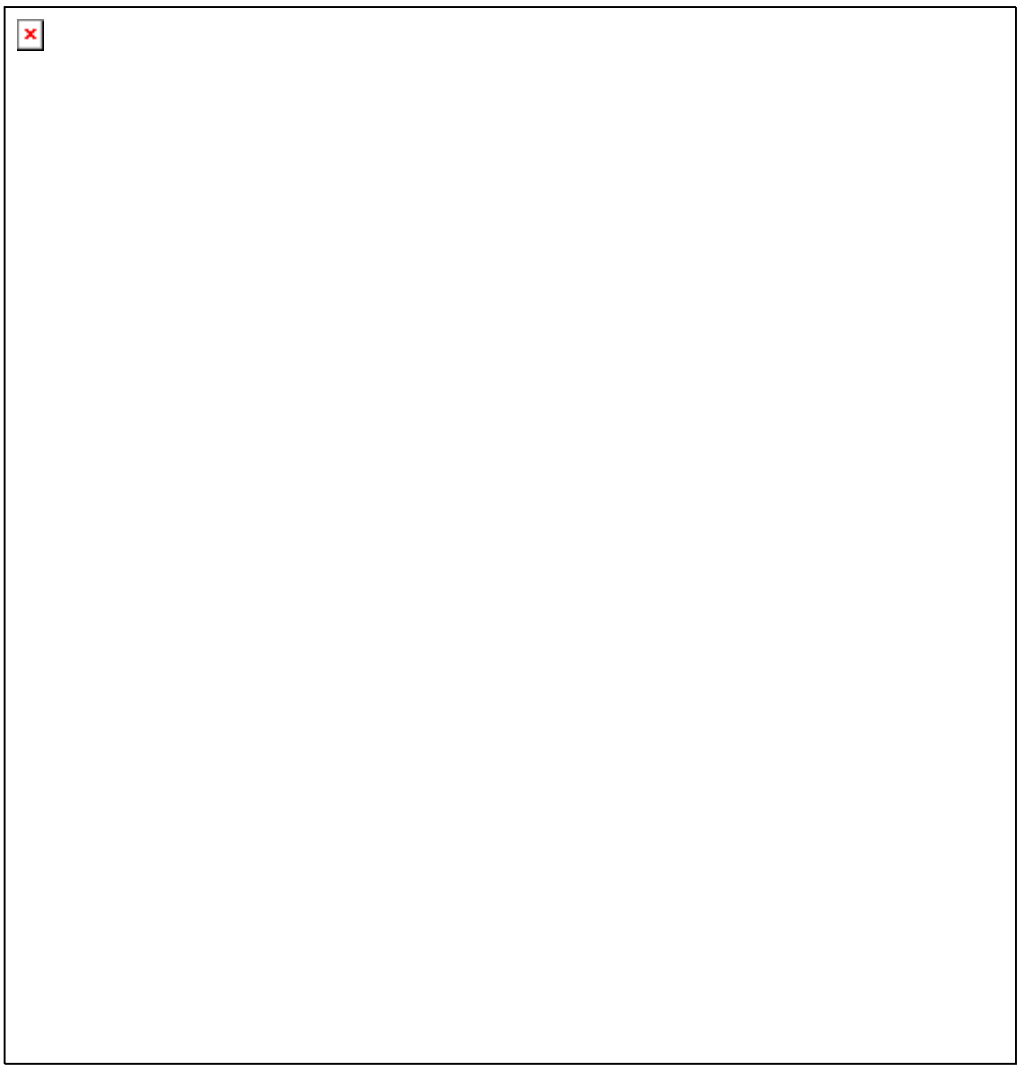
As shown in figure 23, the data flow can be summarized in four major steps:

Pressure is Applied to the Sensor.

A change in pressure is measured by a change in the sensor output.

The sensor signal is conditioned for various parameters.

The conditioned signal is converted to an appropriate analogue output (i.e. 4 – 20 mA)



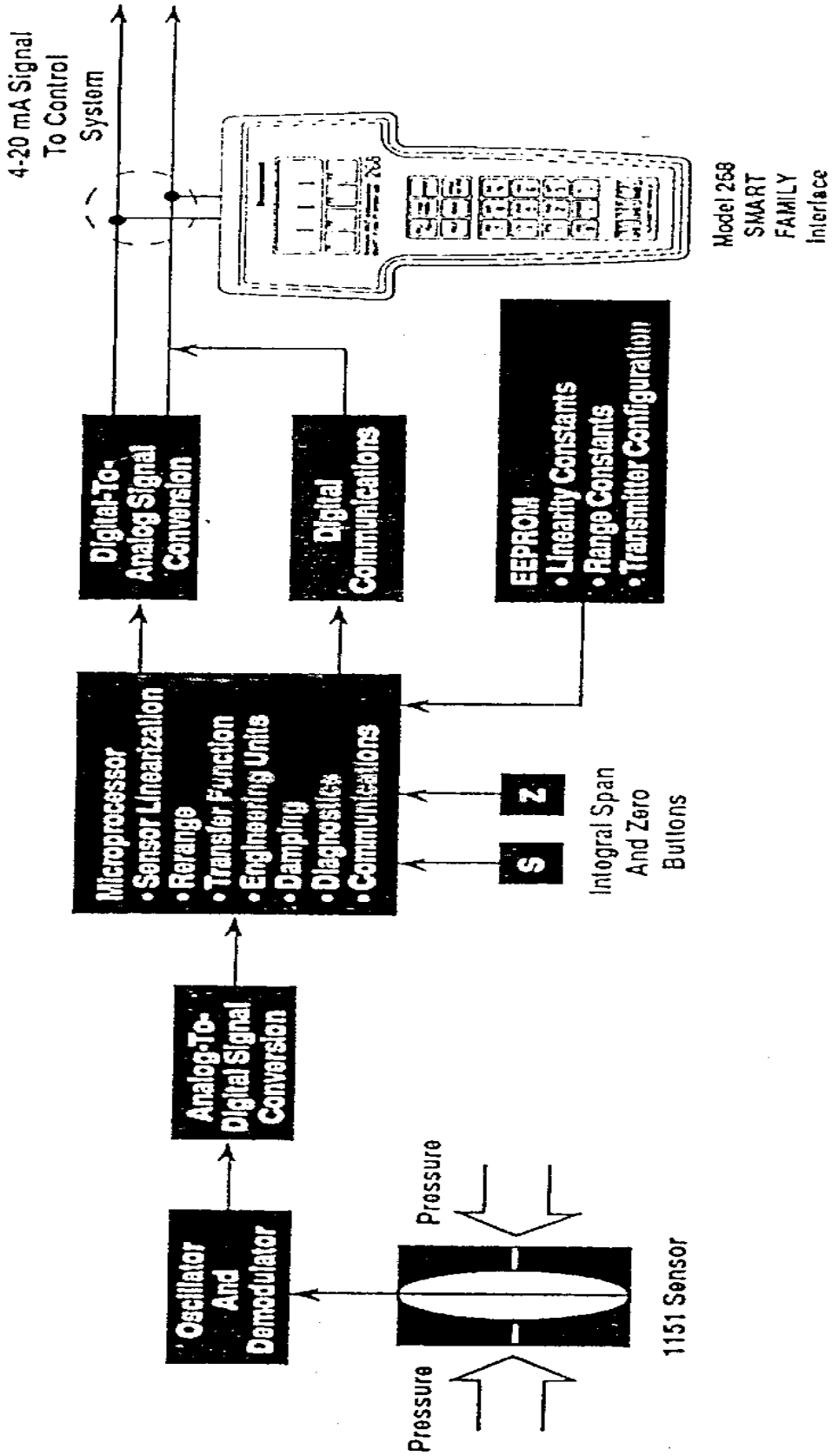


Figure 24, Smart Transmitter Functional Block Diagram

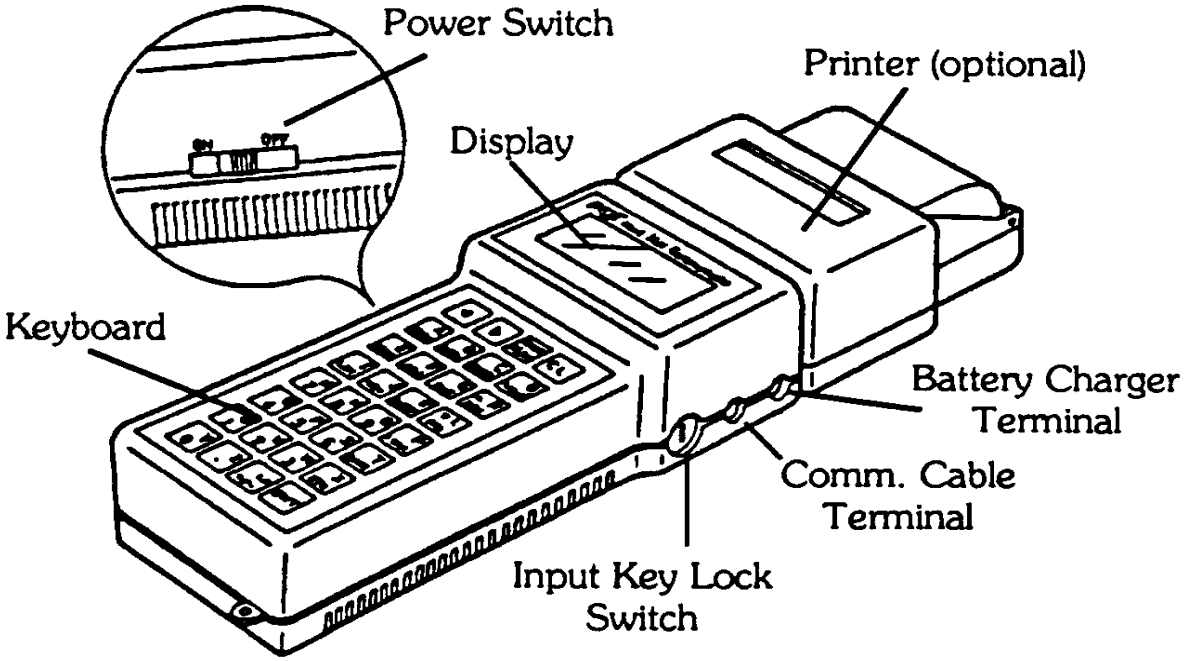


Figure 25, HHC Components

Pressure Controllers

A pressure controller is a device, which senses the process pressure in the process and develops an output, which controls a device to regulate that pressure. The control device, or end element, is usually a pneumatic-control valve. The controller output is usually either a 3-15 or 6-30 psig (20-100 or 40-200 kPa) pneumatic signal.

Pressure controllers can be categorized either as indicating or blind. The indicating controller has a mechanism so that the operator can read the process pressure directly on the controller. The blind controller has no direct-reading mechanism and the operator must rely on an adjacent pressure gauge or other device to know the process pressure. The indicating controller set point is usually marked on the indicator, thus it is easy to adjust to the desired point. Adjustment of the blind controller is more of a trial and error process. Indicating controllers are somewhat more expensive than blind controllers, but the cost difference is moderate if a pressure gauge can be eliminated.

Pressure controllers must provide an output to control the end element. This can be an electric or pneumatic signal, but is most often pneumatic for field-mounted controllers. The pneumatic signal is usually 3-15 psig (20-100 kPa), but it can be 6-30 psig (40-200 kPa) if required to reduce the control valve actuator size.

The control action needed for pressure control is proportional plus integral, or P and I (Integral is also referred to as reset by some manufacturers).

The proportional action varies the output in proportion to the difference between the measured pressure and the set pressure. The integral or reset action gradually increases the amount of the correction until the measured pressure is returned to the set point. A more extensive discussion of control modes and controller tuning can be found in the manual Controllers and Control Theory.

A common option for pressure controllers is an auto/manual switch. This is a valve, which allows the output of a manual regulator to be directed to the end element (valve actuator) instead of the controller's automatic output. The transfer can be either bump-less where the outputs are automatically matched to each other when the auto/manual switch is transferred or manual balance where the operator must match the manual regulator output to the automatic output transfer to manual or the set point to the process variable before transfer to automatic.

Pressure controllers are surface, panel, pipe-stand, or yoke mounted. Surface-mounted controllers are fastened to a wall or other vertical surface. Panel-mounted, also called flush-mounted, controllers are mounted in a cut-out in a control panel.
Pipe-stand mounting occurs where a vertical or horizontal pipe support is constructed and the controller is provided with a bracket and U-bolts to attach it to a two-inch pipe-stand. It is not a good idea to support controllers on process piping. Yoke mounted controllers are fastened to the valve yoke with special brackets. Yoke mounting is convenient when the valve is accessible.

Electric Pressure Switches

An electric pressure switch senses pressure and opens or closes an electrical switch element at a set pressure to signal another electrical device. Electric pressure switches are available in a wide variety of styles.

Most pressure switches trip at a pressure above atmospheric, and are called gauge pressure or simply pressure switches. Switches can also be manufactured to trip at a pressure referenced to a complete vacuum and is called absolute pressure switches. Those set to trip below atmospheric pressure are called vacuum switches and those, which can be set either above or below atmospheric pressure, are called compound switches. Some switches are manufactured so that the trip point is factory set, while others are field adjustable.

Pressure switches are set to trip at a certain point with rising or falling pressure. When the pressure is returned to within the acceptable range, the switch does not reset at exactly the same point that it tripped. The difference in the trip point and the set point is called dead band or reset or switch differential. The electrical switch is usually single-pole, double-throw or double-pole, double-throw. Figure 27 shows these types, as well as others less frequently used. The number of poles determines the number of separate circuits that can be controlled by the switch, single pole for one circuit and double-pole for two circuits. The double-throw term means that a common terminal is connected to either of two other terminals normally open or normally closed.

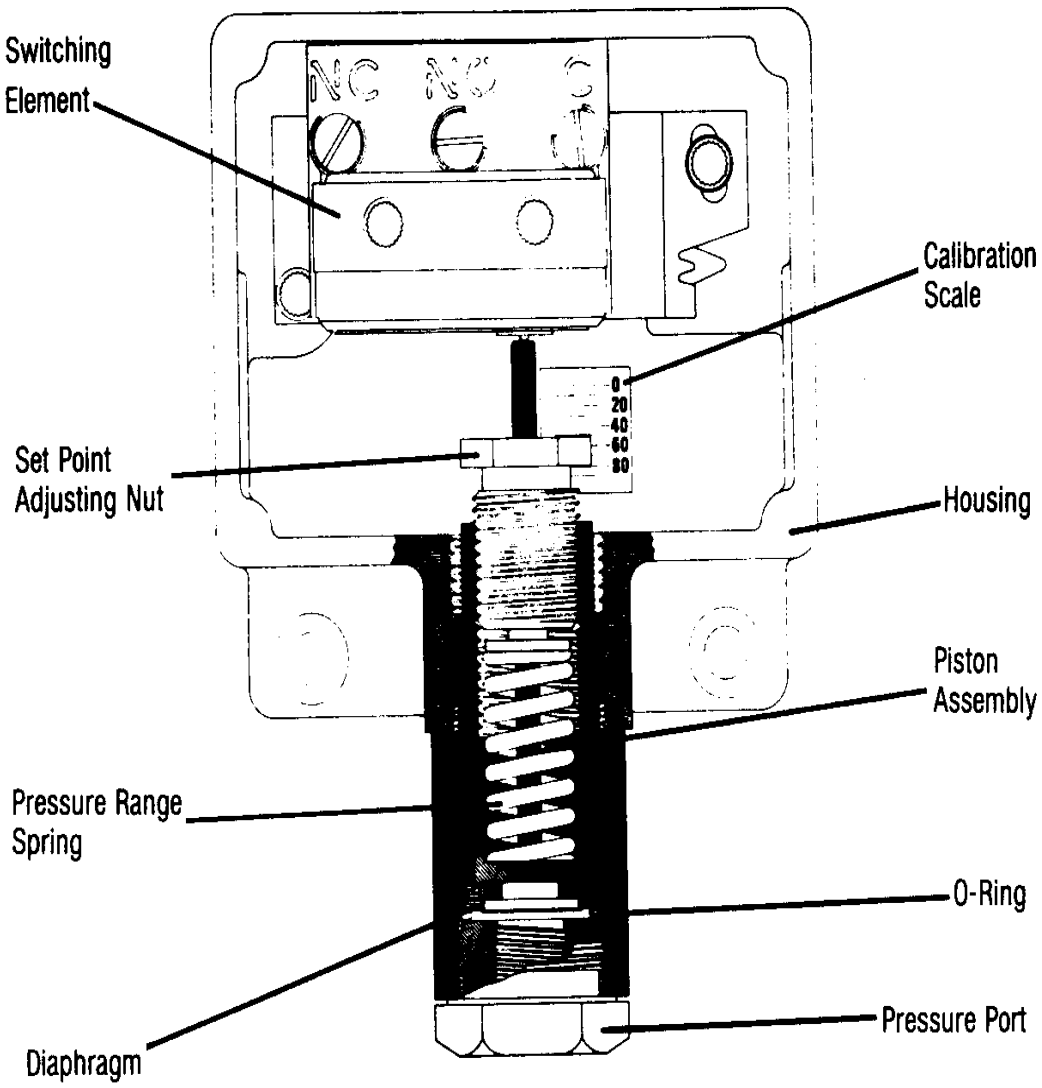


Figure 26, Spring-Loaded Piston Pressure Switch

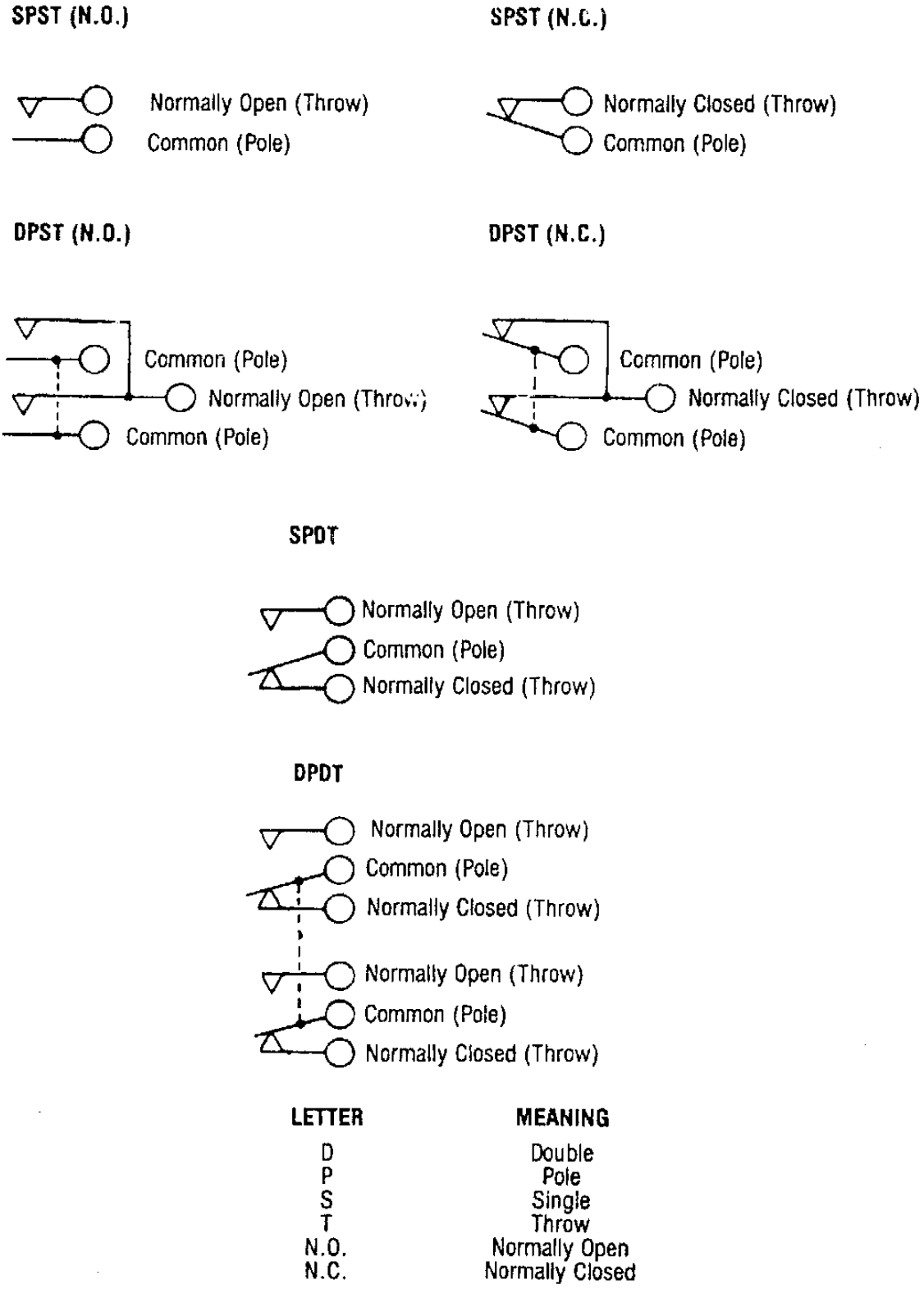


Figure 27, Diagram Showing the Types of Electrical Switches

Pneumatic Pressure Switches/Pressure Pilots

A pneumatic pressure switch senses pressure and opens or closes a small valve at a set pressure to supply or vent a pneumatic signal to another pneumatic device. Pneumatic pressure switches are commonly known as pressure pilots. They are frequently used when pneumatic shutdown and control systems are selected. Often, pressure pilots are used in Division 1 areas, such as on wellheads even when the primary process control is electronic. Devices, which are similar to electric pressure switches, are called pneumatic pressure switches. Pneumatic pressure switches are equipped with a two-way or three-way valve instead of an electrical switch. The two-way valve is either open or closed.

A three-way valve connects a common port with one of two other ports, depending on whether the switch is tripped or not. Devices, which have been designed to be pneumatic, are usually called pressure pilots. The most common types of pilots are the piston-actuated, known as stick pilots, and the bourdon tube actuated pilots. Stick pilots are more often used on wellheads and bourdon tube pilots are more often used on process equipment. Dead band or reset is equally important for pneumatic pressure switches /pressure pilots as for electric pressure switches. Pneumatic devices tend to have an even larger dead band than electric devices because more movement is required for actuation. Most pressure pilots are equipped with three-way pneumatic valves so that they can be used either as a high-pressure pilot or a low-pressure pilot depending on how they are connected.

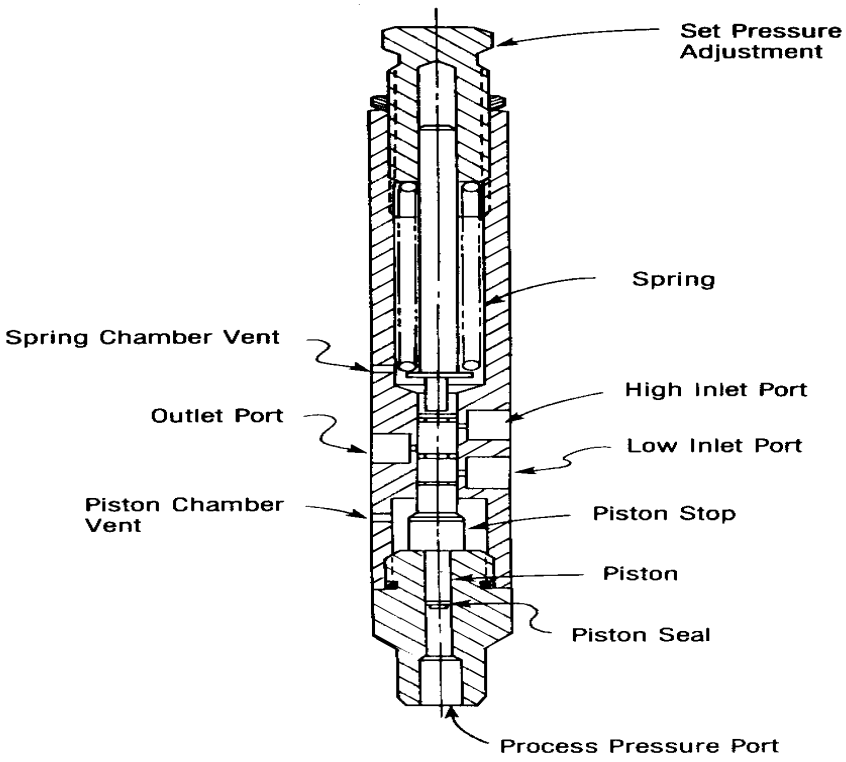


Figure 27, A spring action of Spring-loaded Piston Pressure pilot

Pneumatic Switching Valves

To understand the purpose of using these types of valves and its construction details; INVALCO model CDM is an example which is a diaphragm operated pilot valve for pneumatic or hydraulic control. The unit is equipped with one, two or three snap-acting 3-way MICRO VALVES to provide on-off output to one or more controlled circuits.

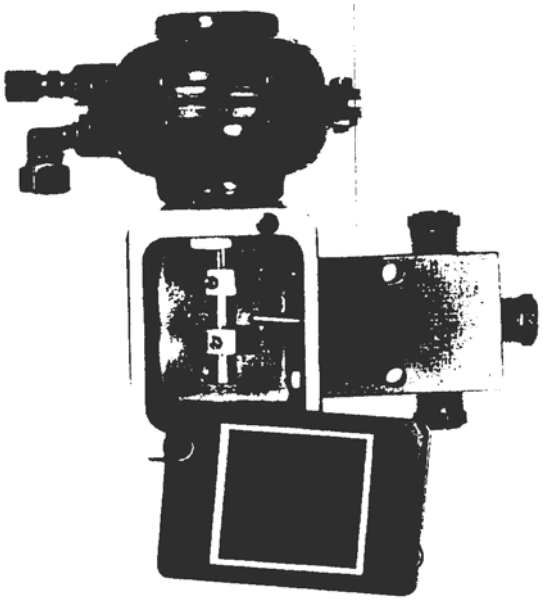


Figure 28, INVALCO Switching Valve

Operation

Process pressure is applied to the upper diaphragm chamber, causing the stem to lower against the spring. When the upper drive collar has been dropped sufficiently, the MICRO VALVE will snap, thereby reversing the control circuit. As process pressure decreases, the lower drive collar will raise, contacting the toggle arm which causes the MICRO VALVE to snap to its' "normal" position.

Adjustment

Operating adjustments are very simple on the CDM pilot. The range spring is fixed and requires no adjustment. The process pressure required to trip and release the MICRO VALVE will depend upon the spacing of the drive collars. The adjustable drive collars provide a convenient means of adjusting the span of output valve action within the operating range. Tripping pressure can be adjusted from approximately 6 to 20 psi; with release pressure varying from approximately 2 to 8 psi.

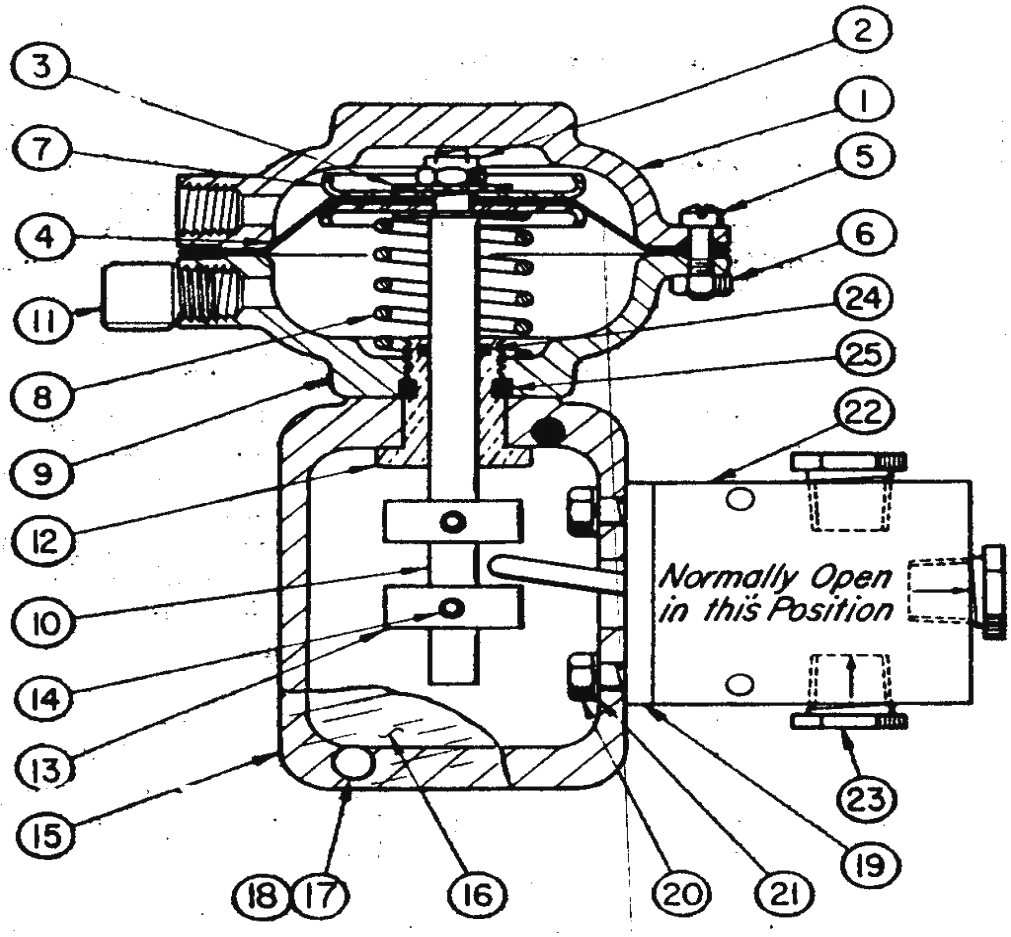
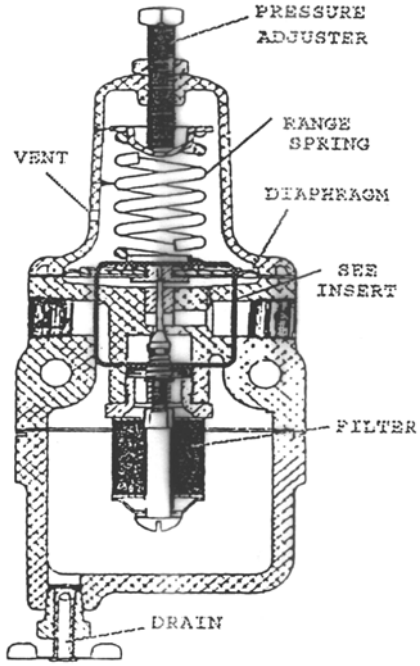


Figure 29, INVALCO Switching Valve Partial View.

Where manual reset is required, it is necessary merely to remove one of the adjustable drive collars. The CDM pilot lends itself to lock-up or alarm service, since either application or loss of process pressure can cause the MICRO VALVE to trip, with manual reset required by removing either the lower or upper drive collar. The clear plastic cover permits visual indication of the MICRO VALVE position as well as the approximate value of process pressure.

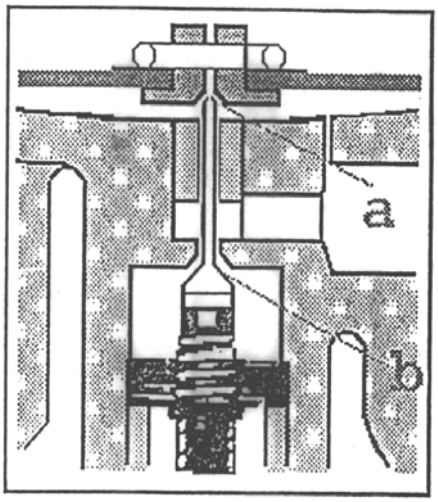
Pressure Regulator

The following paragraph is a description of the operation of a pressure regulator: When the diaphragm is balanced both 'a' and 'b' are closed. With pressure applied to the spring (fully unwound) 'b' is closed and 'a' is open. Any pressure in the output leaks through the hole in the diaphragm and bleeds to atmosphere, through the vent. When all the air from the output has been vented to atmosphere the diaphragm is balanced and both 'a' and 'b' are closed.



Fisher Regulator

The spring is compressed to increase the pressure. The diaphragm and valve are pushed down. This opens 'b' ('a' is still closed) and allows the inlet air to pass, through the filter, to the output. As the pressure in the output builds up it will force the diaphragm up. When the diaphragm is balanced both valves is closed. The diaphragm is balanced when the pressure applied by the spring (applied on top of the diaphragm) is the some as the output pressure (applied to the bottom of the diaphragm). If the adjustment is decreased the diaphragm moves up b closes a opens and some output bleeds to atmosphere. This continues until the diaphragm is balanced again.



The valve on a Fisher Regulator

SGC HSE Regulation

Module I-1: PRESSURE MEASUREMENT

Refer to HSE Regulation No. 6 "Work to Permit System"

Dealing with pressure is sometimes covered under Cold work permit

Actual or possible breaking of containment of systems under pressure or which contain substances which are flammable, toxic or corrosive.

Pressure testing of plant and equipment.

No permit required when: Adjustments to separator pressures and to separator levels.

Refer to HSE Regulation No. 22 "Hot and Odd Bolting"

C. For odd bolting, pressure gauges must be suitable for reading reduced line pressure to allow monitoring of the system during the work.

Refer to HSE Regulation No. 23 "General Engineering Safety"

Pressure

Pressure is the main process fluid condition in a process which can create hazard with respect to work on control engineering hardware.

Operational and maintenance work on control engineering hardware associated with high pressure fluids or hazardous fluids must be accorded particular respect. The job method must be clearly written and the procedure rigorously applied.

When pressure gauges are to be removed from running machinery, the gauge and associated pipe-work must be correctly vented down. Pressure gauge pipe work should be plugged off immediately when the gauge is removed.

Leaking hydraulic oil or fluid under pressure can easily penetrate a person's skin and cause serious injury. Should a person be struck by escaping hydraulic oil/fluid at high pressure, they should inform their supervisor and then immediately seek medical attention.

Refer to HSE Regulation No. 7 "Isolations"

CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

4. Isolation of Hardware

Isolation of control engineering hardware may be necessary to enable maintenance work to be done or permit removal of the hardware to effect repairs (either locally or remotely). Isolation of hardware can take several forms, for example isolation from:

Process Plant

Utilities (electric, pneumatic, hydraulic, cooling media etc).
Larger system of which the hardware is a subsystem or component.

5. Isolation from Process

Isolation of instruments which, are connected to or form a part of the process is usually achieved by valving. It is important that, where isolation of an instrument is required for maintenance purposes, correct venting/draining and valve closure procedures are adhered to.

Where instruments have local isolating valves in addition to the primary process isolating valves, the local valves may be used for some routine in-situ testing at the discretion of the Senior Control Engineer. If an instrument is to be removed from site, the process isolating valves must be used and any impulse pipe work must be drained or vented completely.

Where the process fluids are of a hazardous nature (eg toxic, flammable etc), particular care should be taken to ensure correct venting and draining, and also to clean or flush the instrument carefully, prior to effecting work or removal of the hardware from site for maintenance or repair. Gas testing may be required. On large items, e.g. control valves, a certificate of cleanness is necessary prior to delivery to workshops.

On removal of a directly mounted instrument, from a process line containing hazardous fluids, e.g. pressure gauges etc, isolation by the primary isolation valve only is NOT acceptable. The valve outlet shall be blanked off, capped or plugged with a blank flange, solid screwed plug or cap, whichever is appropriate.

Refer to HSE Regulation No. 7 "Isolations"

CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

6. Isolation from Electrical/Pneumatic Supplies

If practical, equipment must be made safe before any work is done on it. The operation of making the equipment safe must be done by a Competent Control Engineering Person. Care shall be taken when working on live equipment to ensure avoidance of contact with live electrical components (refer to Regulation No 19 Working with Electricity).

Pneumatically operated equipment must be isolated before it is disconnected or removed for repair by closing the valve at the supply manifold for the individual instrument and venting through the drain/vent of the pressure regulators.

7. Isolation from Utilities

Control engineering equipment may be connected to utilities (other than electrical associated with the hardware e.g. stream cooling water, hydraulic fluid, chemicals, carrier gases (analysis) and air supplies. It is important that attention is given to rendering the utilities safe when the control engineering hardware is being serviced or removed.

Utilities should be isolated at the point of distribution to the control engineering equipment being removed (e.g. isolating valve at distribution head) and not solely at the hardware itself.

Where utility fluids are 'piped' to an instrument, the pipe work should be drained down or vented if the instrument is removed.
It is important that removal of a utility from a specific piece of hardware does not influence any other hardware to which the utility may also be connected (eg cooling water may have been series connected to more than one item of hardware).

Review Questions

Q1. Define pressure.

Q2. Define the following:

Gauge pressure

Absolute pressure

Perfect vacuum

Q3. What are the two main purposes of measuring pressure in a processing facility?

Q4. Sketch and label the main parts of a Bourdon tube gauge.

Q5. Define the following pressure errors:

Zero error

Range error

Angularity error

Q6. What are the approximate operating pressure range of a helical type Bourdon tube?

Q7. Describe the operation of a C- type Bourdon pressure gauge.

Q8. Describe the operation of a differential pressure gauge.

Q9. What is the purpose of protective diaphragms in a Bourdon element?

Q10. Define the principles of operation of a strain gauge.

Q11. Describe the operation of a dead-weight tester.

Q12. Describe electronic pressure transmitter components and connections.

Q13. What is the output signal range of pneumatic, electronic and smart pressure transmitters?

Q14. Describe the principle of operation of the electronic pressure transmitter.

Q15. Demonstrate how to perform bench calibration of an electronic pressure transmitter?

Q16. What are the routine maintenance required for a pneumatic pressure transmitter

Q17. What are the main parts of a pressure controller?

Q18. Describe the correct procedure to be followed to a switch a controller from manual to auto mode?

Q19. What are the PID values setting of controller to achieve better control function?

Model Answers

A1. Pressure is the force per unit area, that is:

$$\text{Pressure } \frac{\text{Force}}{\text{Area}}$$

A2. Gauge pressure: This is the pressure measured above the atmospheric pressure, that is, gauge pressure is the difference between the pressure being measured and the atmospheric pressure.

Absolute pressure: Absolute pressure uses zero pressure as its datum and is the total pressure above zero.

Vacuum: Vacuum is a state where the pressure being measured is below atmospheric pressure and above absolute pressure.

A3. The main purpose for measuring pressure are: Safety and Process control.

A4. See accompanying diagram.

A5. Zero error: Constant error over the entire scale. Requires realignment of the pointer on its shaft.

Range error: Constant percentage error over the entire scale. Corrected by adjustment to the shoulder screw.

Angularity error: Either widens or narrows from the scale center mark. Corrected by adjusting the connecting link.

A6. 0 to 700 bar.

A7. One end is closed and the other joined to a connection block by soldering, brazing or welding. When the tube is subjected to internal pressure the stresses imposed cause the cross section to become slightly more circular in

shape. The tube tends to straighten and the free moves in proportion to the applied pressure.

The small free end movement is magnified by a rack in the form of a quadrant and a pinion. A hair spring, under tension, is fitted to bias the teeth of the rack and pinion and eliminates hysteresis due to lost motion in this region.

A8 The common form of Bourdon differential pressure gauge consists of two separate tubes one of which has the high pressure connected to it; the lower pressure is connected to the other tube. The tips of pointer in opposite directions. Therefore, the difference in pressure will be indicated.

A9. The purpose of protective diaphragms in a Bourdon element is to protect the element from direct contact with highly corrosive, viscous or very dirty process fluids.

A10. If an electric conductor is stretched so that its length increases and its diameter decreases, a corresponding increase in the electrical resistance results.

FLOW MEASUREMENTS

FLOW MEASUREMENTS

Objectives

At completion of this module, the developee will have an understanding of:

1. Major factors affecting the flow of fluids through the pipes
2. Classification of flow meters
3. Orifice plates construction
4. The purpose of using orifice plates
5. Different types of orifice fitting
6. Fluid profile when passing through an orifice bore
7. Relationship between fluid flow through an orifice and its differential pressure
8. Venturi tube construction and principle of operation
9. Difference between orifice meter and Venturi tube meter
10. Pitot tubes (Annubars) construction and function.
11. Principle of operation of Rotameter
12. Turbine flow meter parts, function and maintenance
13. Magnetic flow meters construction and principle of operation
14. Principle of operation of Vortex flow meters
15. Positive displacement meters construction and function
16. PDM advantages and disadvantages
17. Ultrasonic flow meters principle of operation
18. Mass flow metering methods and the instruments used
19. Flow switches types and pre-setting procedure

Related Safety Regulations for Module I-2: FLOW MEASUREMENT

Developees have to be familiarised with the following **SGC HSE** regulations, while studying this module:

Regulation No. 6: Work to permit system.

Regulation No. 7: Isolation

7.18 (1-10) control systems procedures and isolations.

Regulation No. 22: Hot and Odd Bolting.

Regulation No. 23: General Engineering Safety.

Regulation No. 27: General Services,

Safe use of hand tools and powered tools/equipment.

Flow Measurement

Fluid flow measurements in oil and gas production operations are used as the basis for revenue payment, determining well allocations, and controlling the process for certain systems. There are many types of instruments for measuring liquid and/or gas flow. The accuracy of flow measurement will vary from instrument to instrument and the desired accuracy will vary from application to application.

Measuring flow is one of the most important aspects of process control. It is one of the most frequently measured process variables. Flow tends to be the most difficult variable to measure. No single flow meter can cover all flow measurement applications.

The physical properties of fluids are important factor in flow metering accuracy. The major factors affecting the flow of fluids through pipes are:

The Velocity of the Fluid

The velocity of a flowing fluid is its speed in the direction of flow. Fluid velocity depends on the head pressure that is forcing the fluid through the pipe. Greater the head pressures, faster the fluid flow rate.

Pipe Size

Pipe size also affects the flow rate. Larger the pipe the greater the potential flow rate.

Friction due to contact with the pipe

Pipe friction reduces the flow rate through the pipe. Because of the friction due to the fluid in contact with the pipe, flow rate of the fluid is slower near walls of the pipe than at then the centre.

The Viscosity of the Fluid

The viscosity of a fluid refers to its physical resistance to flow. Higher the viscosity the fluid, the slower fluid flow.

The Specific Gravity of the Fluid

Specific gravity of liquid is the density of the liquid/density of water. The specific gravity of gas is the density of the gas / the density of air. At any given operating condition, higher the fluid's specific gravity, lower the fluid's flow rate.

Fluid Condition

The condition of the fluid (clean or dirty) also limitations in flow measurement. Some measuring devices become blocked/plugged or eroded if dirty fluids are used.

Velocity Profiles

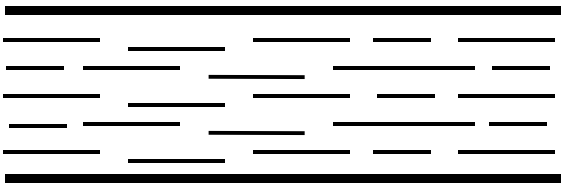
Velocity profiles have major effect on the accuracy and performance of most flow meters. The shape of the velocity profile inside a pipe depends on:

- The momentum or internal forces of the fluid, that moves the fluid through the pipe

- The viscous forces of the fluid, that tend to slow the fluid as passes near the pipe walls.

There are three types of flow profile.

- **Laminar or Streamlined**
- **Transition**
- **Turbulent**
- **Laminar or Streamlined**

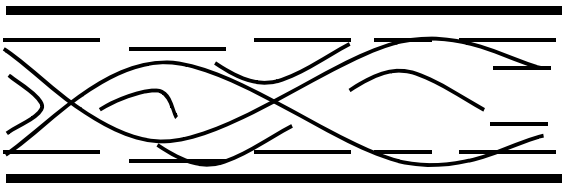


Laminar or streamlined flow is described as liquid flowing through a pipeline, divisible into layers moving parallel to each other.

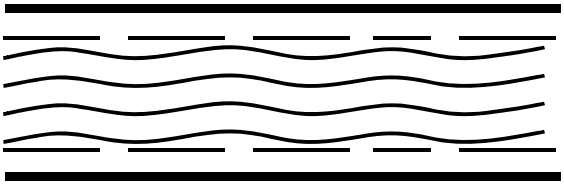
Laminar Flow pattern

Turbulent

Turbulent flow is the most common type of flow pattern found in pipes. Turbulent flow is the flow pattern which has a transverse velocity (swirls, eddy current).



Turbulent Flow pattern



Transitional

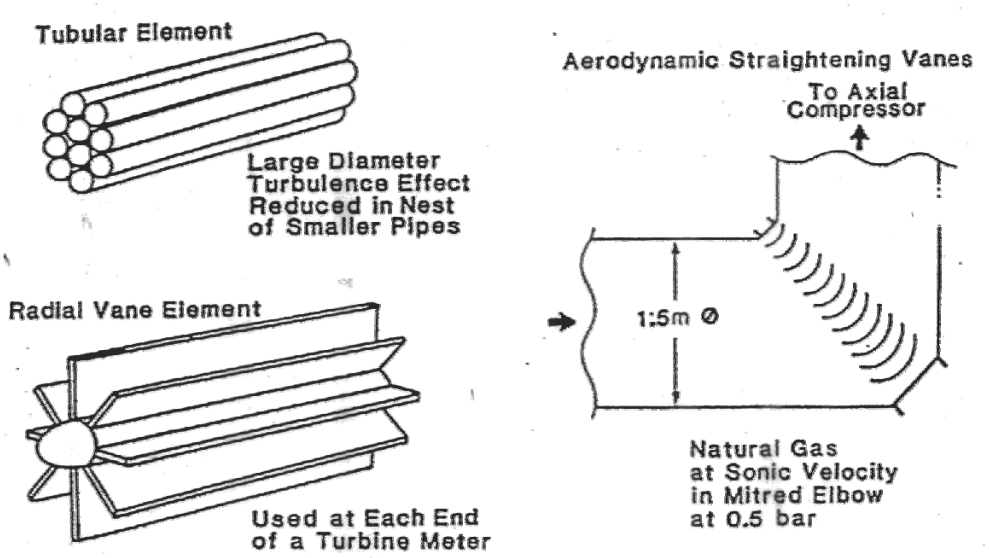
Transitional flow profile exists which is between the laminar and turbulent flow profiles. Its behaviour is difficult to predict and it may oscillate between the laminar and turbulent flow profiles.

Transition Flow pattern

Flow-Straightening Devices

These devices are used to improve the flow-pattern from turbulent to transition or even to laminar in order to improve the accuracy of the flow measurement.

There are three common elements; tubular element, radial Vane element and aerodynamic straightening vanes.



There are two kinds of flow measurement:

Rate of Flow

The rate of flow of a fluid is defined as the amount of fluid that passes a given point in a set time.

Total Flow

The total flow of a fluid can be defined as the total amount of fluid that passes a given point over an extended period of time.

Note

Most flow meters measure volumetric flow, but some types measure mass flow. Volume is related to mass by the density of the liquid.

Classification of Flow Meters

Flow meters operate according to many different principles of measurement although this can by broadly classified into four areas:

1. Flow meters with wetted non-moving parts
2. Flow meters with wetted moving parts
3. Obstruction less Flow meters
4. Flow meters with sensors mounted externally

Flow meters can further classified into four types:

- **Volumetric flow meters** that measure volume directly

  Positive displacement meters

- **Velocity**

  Magnetic, turbine and ultrasonic

- **Inferential flow meters**

  Differential pressure, target, and variable area flow meters

- **Mass flow meters** that measure mass directly

  Coriolis

Flow meters with wetted non-moving parts

These devices with no moving parts that gives them an advantage. However excessive wear plugged impulse lines and excessively dirty fluids may cause problems.

Types of Differential Pressure Flow Meters

Differential pressure type flow meters provide the best results where the flow conditions are turbulent. Some of the most common types of differential pressure flow meters are:

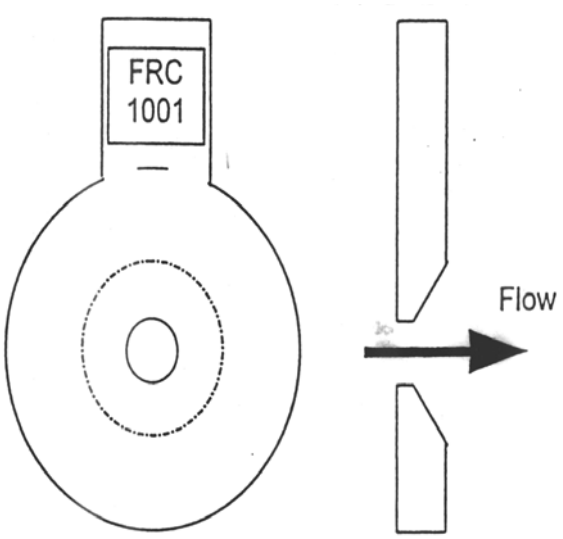
- Orifice plate
- Venture tube
- Elbow
- Pitot tube

Components of a typical orifice metering loop are:

- Orifice plate and holder
- Orifice taps
- Differential pressure transmitter
- Flow indicator / recorder / controller

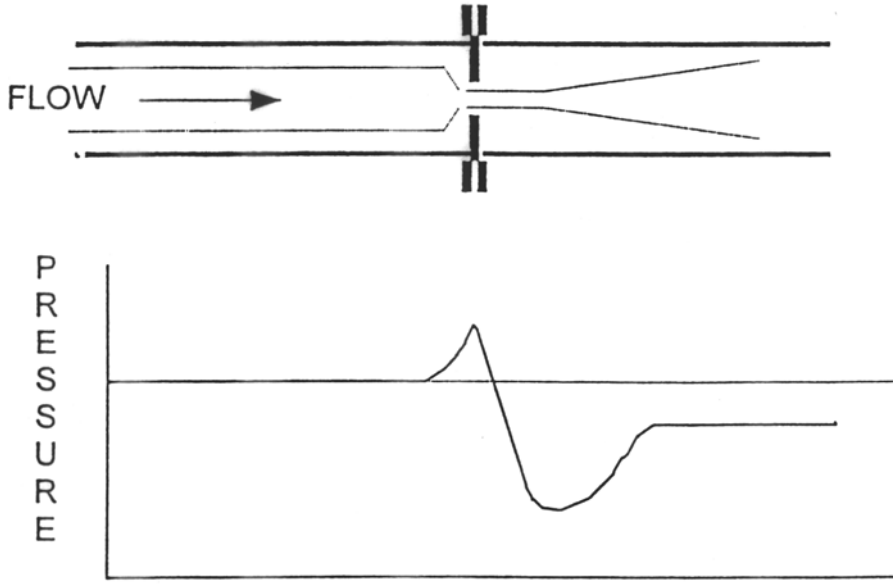
Orifice Plates

Orifice plates in various forms are the most widely used primary elements and consist of a flat piece of metal with a sized hole bored in to it.

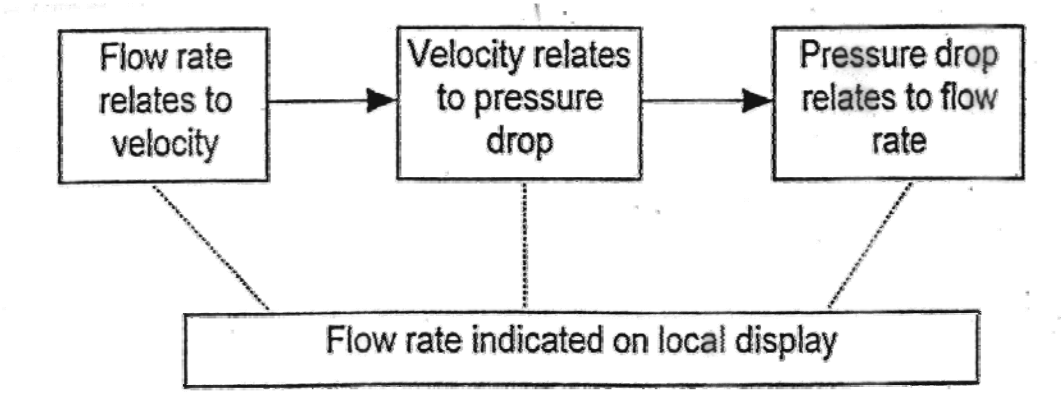


When fluid through the orifice its velocity increases, resulting a drop in pressure and an increase in turbulence. After the fluid has passed through the orifice its velocity decreases again, causing an increase in pressure although only some of the pressure loss is recovered. The amount of pressure recovery can be up to 50% of the total pressure drop across the orifice plate.

The flow of liquid through the orifice plate creates a differential pressure across it, in such a way that the faster the flow the larger the pressure drop.



Pressure Profile Through the Orifice Plate



Orifice Plates and Holders

Orifice plates are usually installed between special flanges in a horizontal pipe run. The flanges are thicker than normal to accommodate two small bore tapings for connection to a DP cell. The plate is positioned concentrically within the flange bolt circle with the tap protruding near the top of the flanges. The tab has a hole in to indicate that it is an orifice plate and not a pipe blank. Orifice plates have information engraved on them to indicate the correct upstream / downstream orientation.

Orifice taps

There are 4 common arrangements of pressure taps:

- **Flange taps**

  Flange taps are the most popular because their distance from the orifice plate is precisely controlled.

- **Vena Contracta**

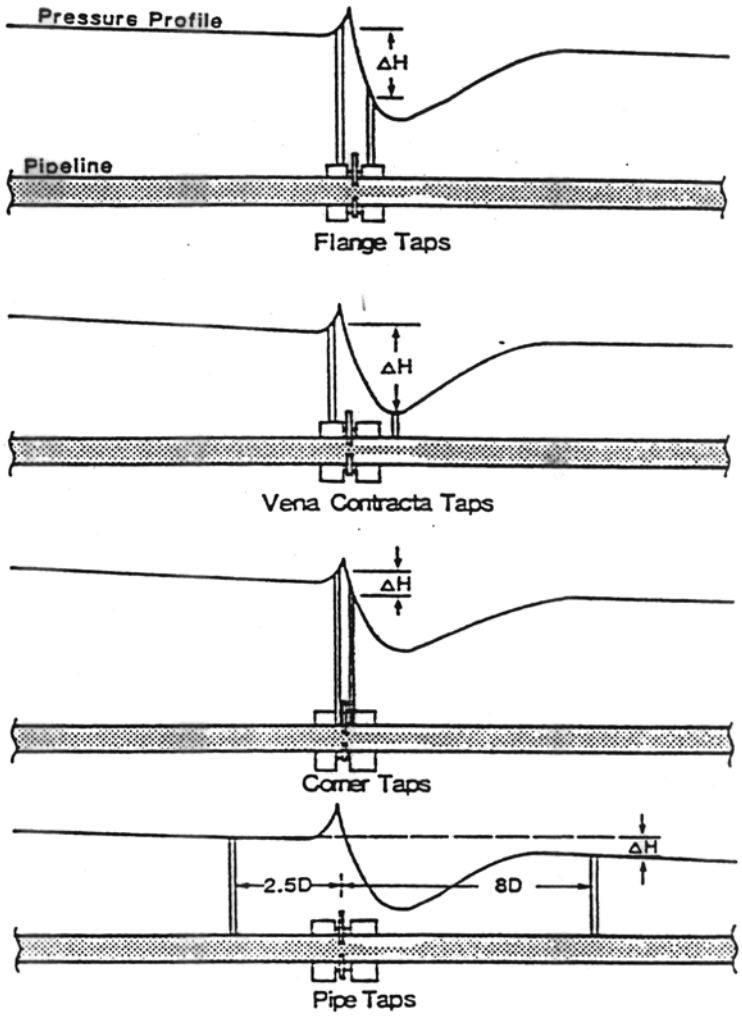
  Vena contracta taps are located to obtain the maximum differential pressure across the orifice.

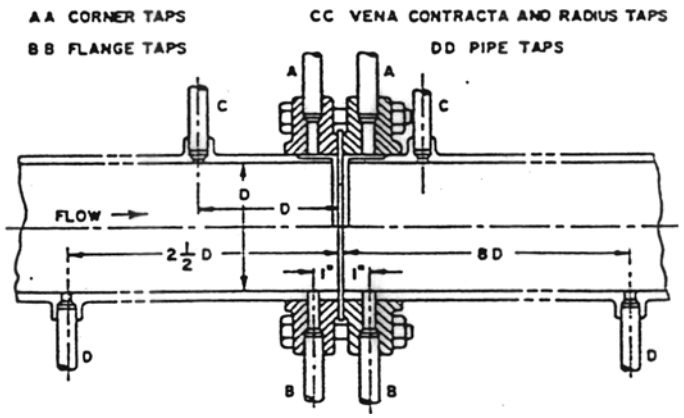
- **Corner taps**

  Corner taps are located at each side of the orifice plate and are good for pressure measurements in pipes less then 50 mm diameter.

- **Pipe taps**

  Pipe taps measure the permanent loss of pressure across an orifice.


Pressure Profile
ΔH
Pipeline
Flange Taps
ΔH
Vena Contracta Taps
ΔH
Corner Taps
ΔH
2.5D
8D
Pipe Taps


AA CORNER TAPS
BB FLANGE TAPS
CC VENA CONTRACTA AND RADIUS TAPS
DD PIPE TAPS
A
A
C
C
D
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FLOW
2½ D
8D
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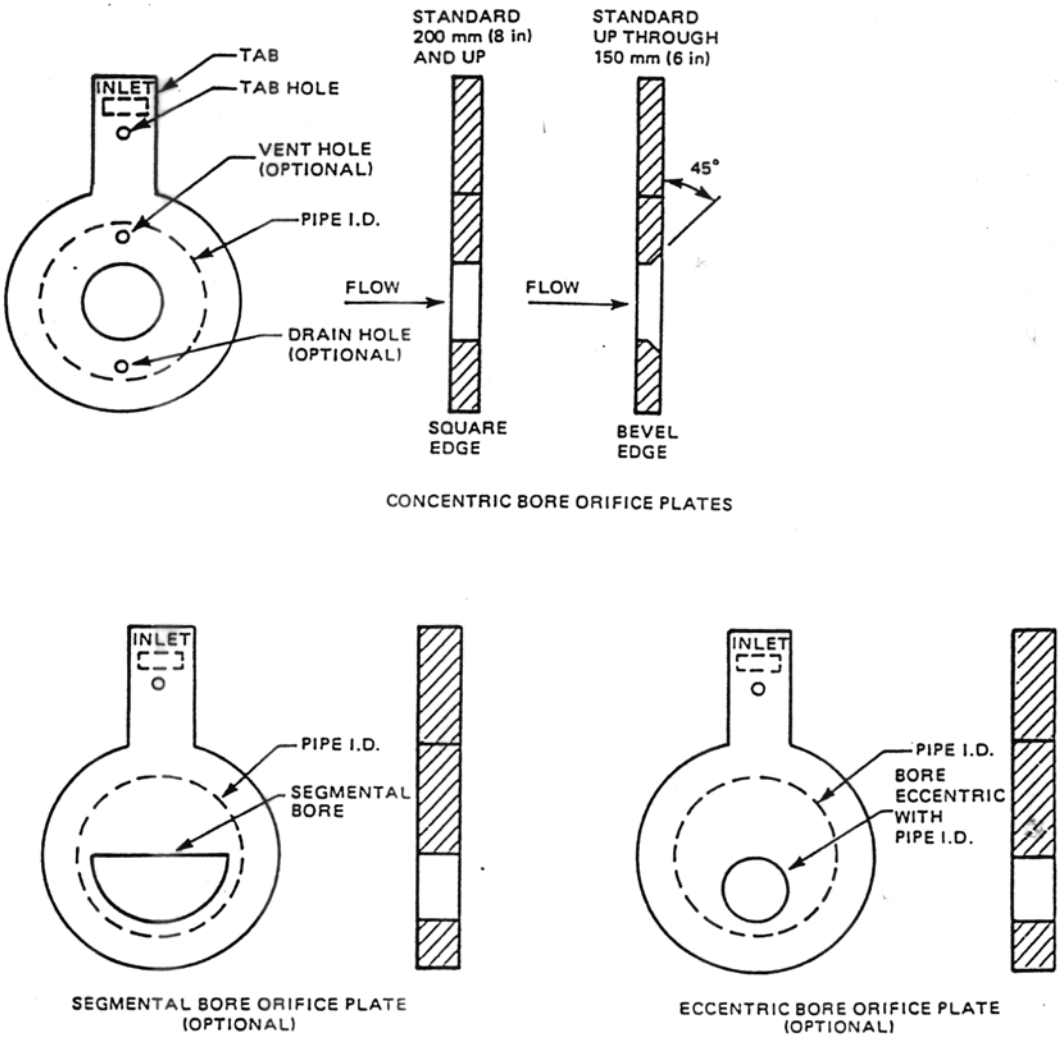
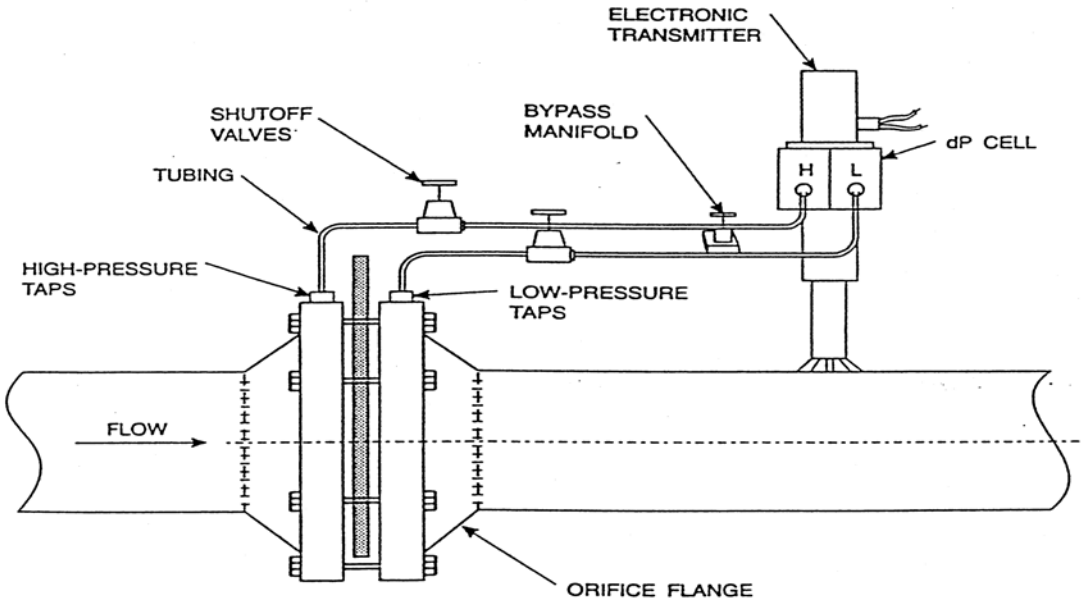
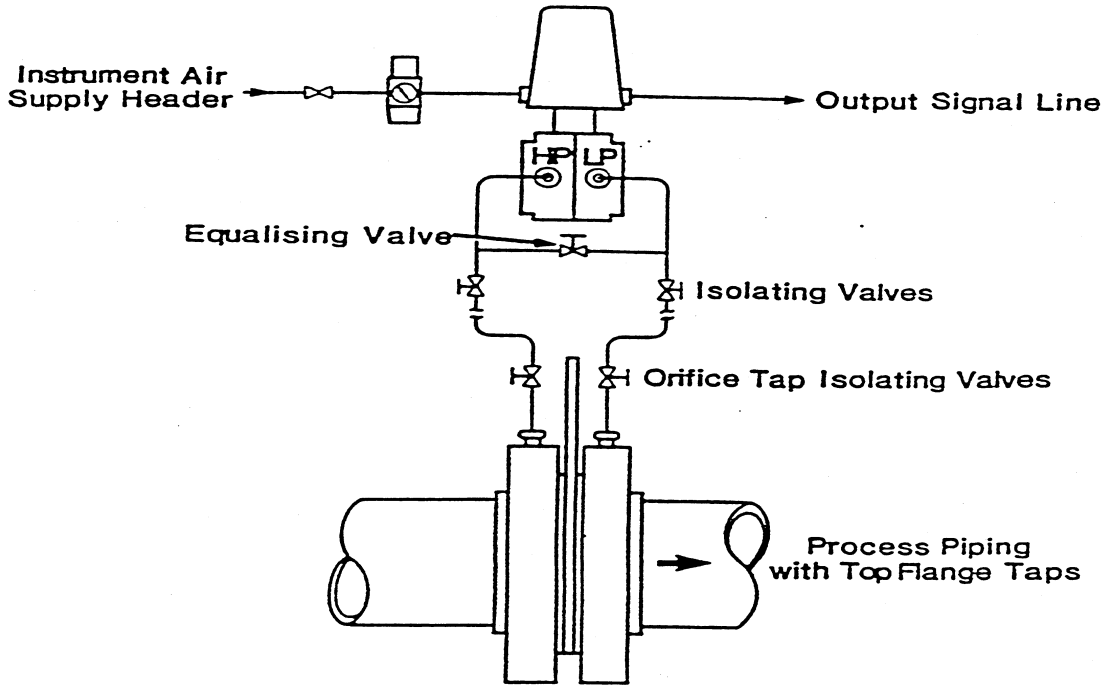
4- Types of Orifice Taps



Types of the Orifice Plates

Note that Orifice plates typically have a drain hole located at the bottom for steam and gas applications and a vent hole at the top for liquid applications.



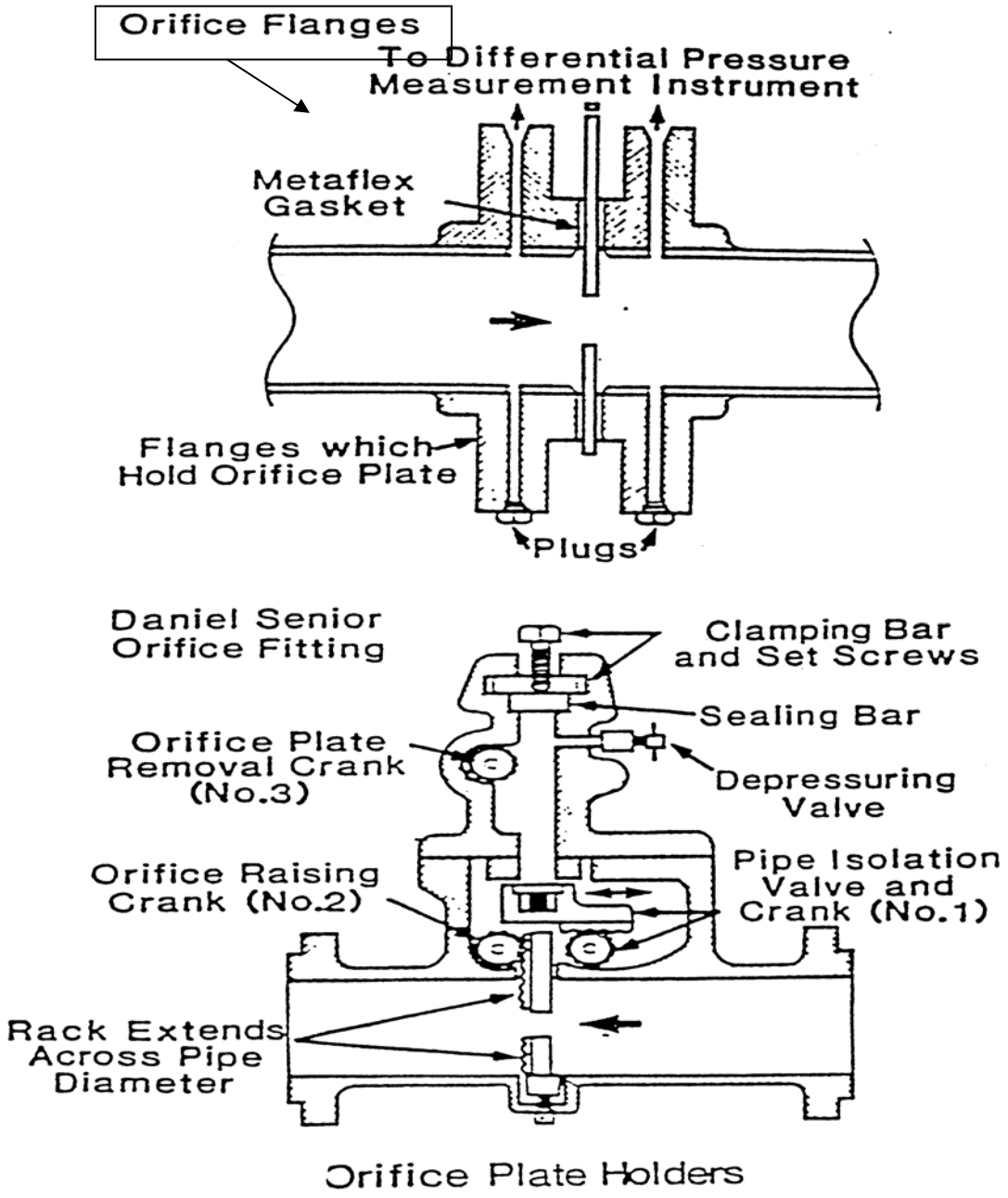


Differential Pressure Transmitter Line Connections

Differential Pressure Transmitters Line Connections

-Upper: Sample of Electronic Transmitter connection

-Lower: Sample of Pneumatic Transmitter connection



Orifice Plate Holders

Differential Pressure Measuring instruments

This type of measurement uses a differential pressure Instruments such as:

1. D.P. transmitter that are used to send a signal to remote controller, indicator, or recorder or to a DCS includes indicating controller and trend recording function.
2. Local D.P. Indicator that gives a direct indication in the field (near to the line).
3. D.P. Recorder that gives a direct recording on a chart in the field.

Relationship between Differential pressure and flow

When a DP (differential pressure) cell is used to transmit a flow measurement the output of the transmitter is not linear. If the square root of differential pressure is plotted against flow, a straight line is obtained showing that the rate of flow is in direct proportion to the square root of differential pressure (see the curves below). This based on the basic mathematical equation that is used in the orifice flow calculations:

For liquid flow: $Q = C\sqrt{hw}$

This equation is used to calculate the liquid flow (incompressible fluids) **where:**

Q: Flow Rate,

C: constant, to obtained from some factors (about 4 factors),

hw: the differential pressure measured by the DP element.

There is another equation for calculating the gas flow (compressible fluids) which includes more correction factors:

For Gas flow: $Q = C\sqrt{hwPf}$

where:

Q: Flow Rate,

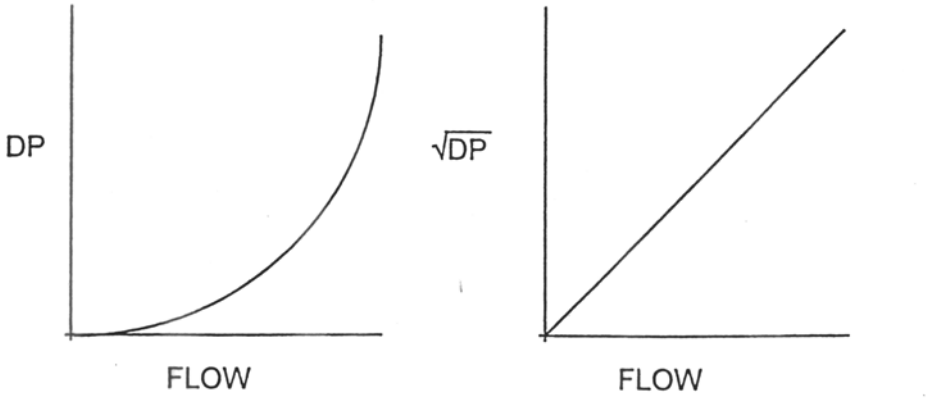
C: constant, to obtained from some factors (about 11 factors),

hw: the differential pressure measured by the DP element,

Pf : Flowing Pressure

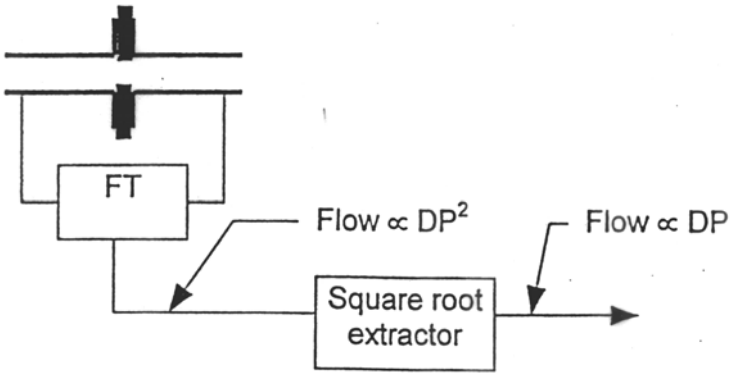
To solve this problem some form of signal conditioning is needed to condition the signal for use with a linear scaled indicator. Therefore, in many flow measurement installations a Square Root Extractor is fitted to the output of a differential pressure transmitter.

Most of the modern electronic transmitters have the option of integral square root function. Modern control systems using the DCS, contains the square root function within their computation modules A simple alternative to this is to use a square root scale on the local indicator (in the conventional systems).



Transmitter output curve Square root extractor output curve

When the differential pressure is obtained experimentally and plotted against flow, the resulting graph is a square function.

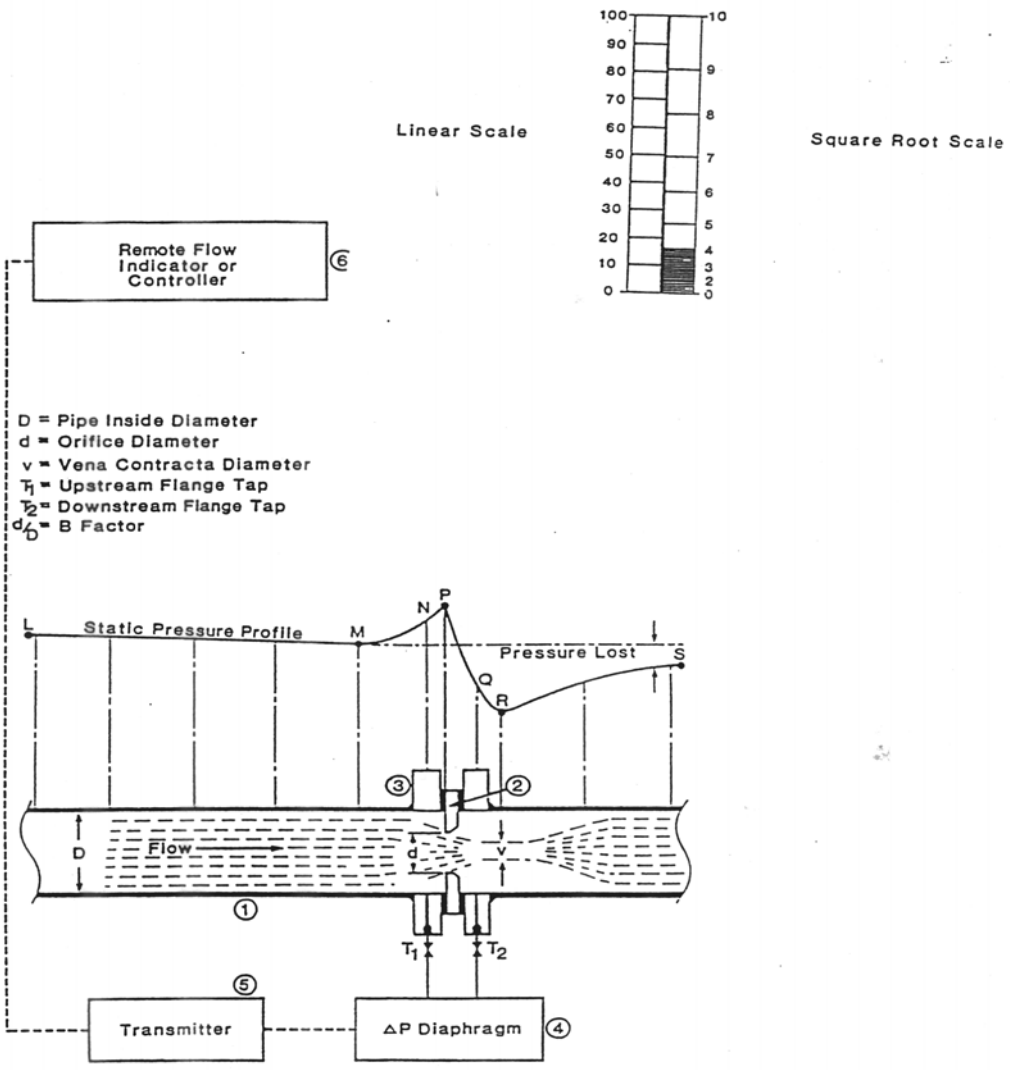


Calculation of the transmitter output with reference to the flow span:

Transmitter output = ($\sqrt{}$Input flow value / Input flow span) X output span + Bias

Where: The **output span** = 16 mA for Electronic Transmitters,
= 12 psi for Pneumatic Transmitters.

Bias = 4 mA for Electronic Transmitters,
= 3 psi for Pneumatic Transmitters.



Orifice Flow Metering Advantages

- They are easy to install.
- One differential pressure transmitter applies for any pipe size.
- Many DP sensing materials are available to meet process requirements. Type 316 stainless steel is the most common material used in orifice plates unless material of higher quality is required by the process conditions.
- Orifice plates have no moving parts and have been researched extensively; therefore, application data well documented (compared to other primary differential pressure elements).

Orifice Flow Metering Disadvantages

- The process fluid is in the impulse lines to the differential transmitter may freeze or block (plug).
- Their accuracy is affected by changes in density, viscosity, and temperature.
- They require frequent calibration

Segmental and Eccentric orifice plates

The eccentric orifice plate is typically used for dirty liquids, gases, liquids containing vapour (bore above pipeline flow axis) or vapour containing liquid (bore below pipeline flow axis).

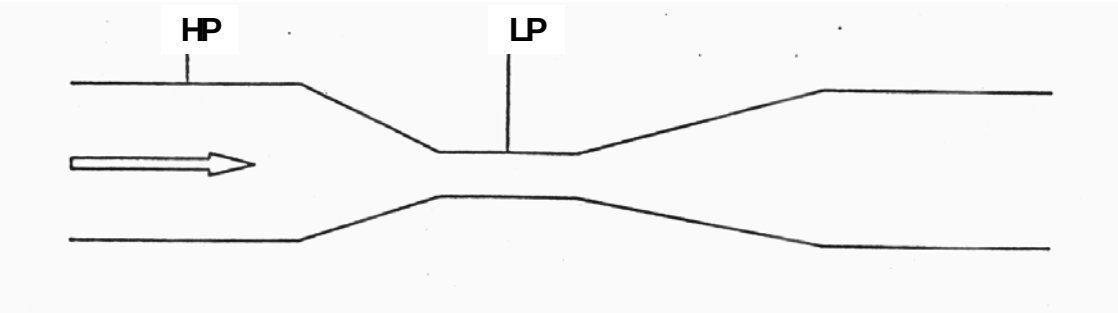
The segmental orifice plate is the same as the square edged orifice plate except that the hole is bored tangentially to a concentric circle with a diameter equal to 98% that of the pipe inside diameter. They are used for dirty fluids, in preference to eccentric bore plates, because it allows more drainage around the circumference of the pipe. During installation, care must be taken that no portion of the gasket or flange covers the hole.

Advantages

- This type of orifice plate is less subject to wears than the square edged orifice plate, however it is good for low flows only.
- For slurry applications where differential pressure devices are required, segmental orifice plates provide satisfactory measurements.

Venturi Tube

Venturi tube consists of a section of pipe with a conical entrance, a short straight throat, and a conical outlet. The velocity increases and the pressure drops at the throat. The differential pressure is measured between the inlet (upstream of the conical entrance) and the throat.



Venture Tube

Advantages

- It can handle low-pressure applications
- It can measure 25 to 50% more flow than a comparable orifice plate
- It is less susceptible to wear and corrosion compared to orifice plates
- It is suitable for measurement in very large water pipes and very large air/Gas ducts.
- Provides better performance than the orifice plate when there are solids in Suspension.

Disadvantages

- It is the most expensive among the differential pressure meters
- It is big and heavy for large sizes
- Its has considerable length

Pitot Tube

Pitot tube consists of two parts that senses two pressures:

- The impact pressure (dynamic)
- The static pressure

The impact pressure is sensed with either one-impact tube bent towards the flow. Sometimes four or more pressure taps (averaging type) are used.

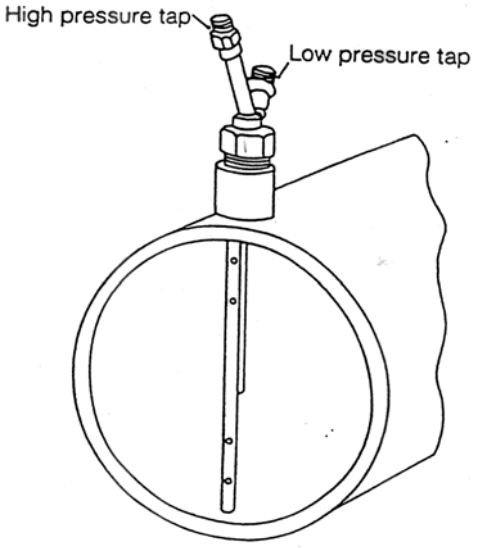
The non-averaging type is extremely sensitive to abnormal velocity distribution profiles (because it does not sample the full stream) hence the advantage of the averaging.

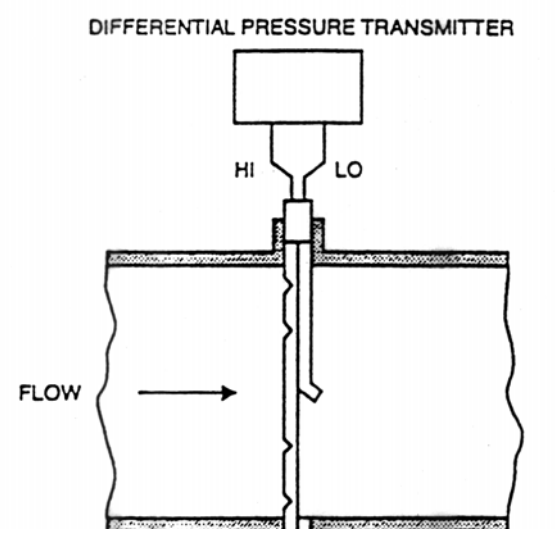
Advantages

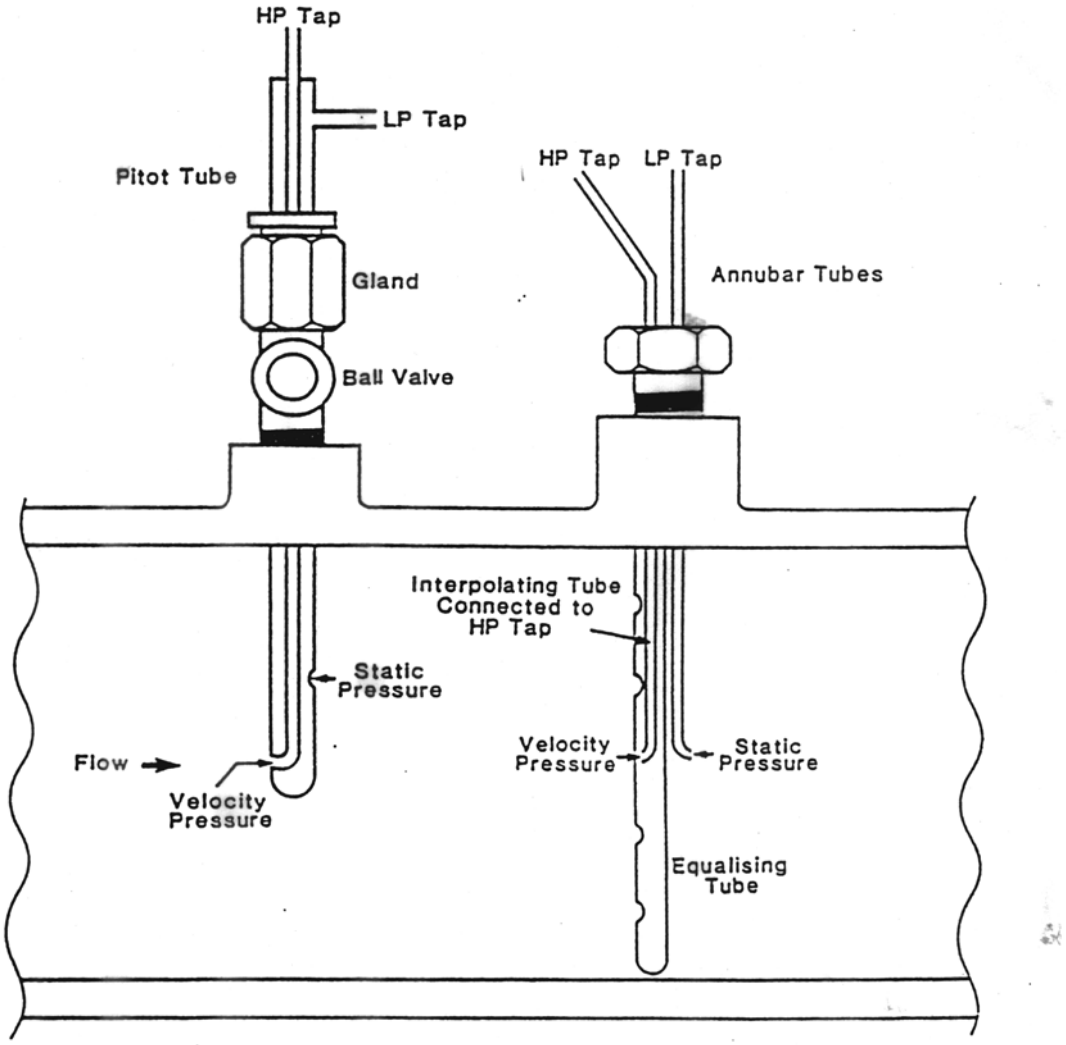
- Pitot tubes are easy and quick to install, especially in existing facilities.
- They can be inserted and removed from the process without shutting down.
- They are simple in design and construction
- They produce energy savings when compared to equivalent orifice (low-permanent pressure loss)
- They are suitable for measurement in large water pipes and large air/gas ducts

Disadvantages

- Their low differential pressure for a given flow rate
- They tend to block/plug in the process lines, unless provision is made for purging or flushing





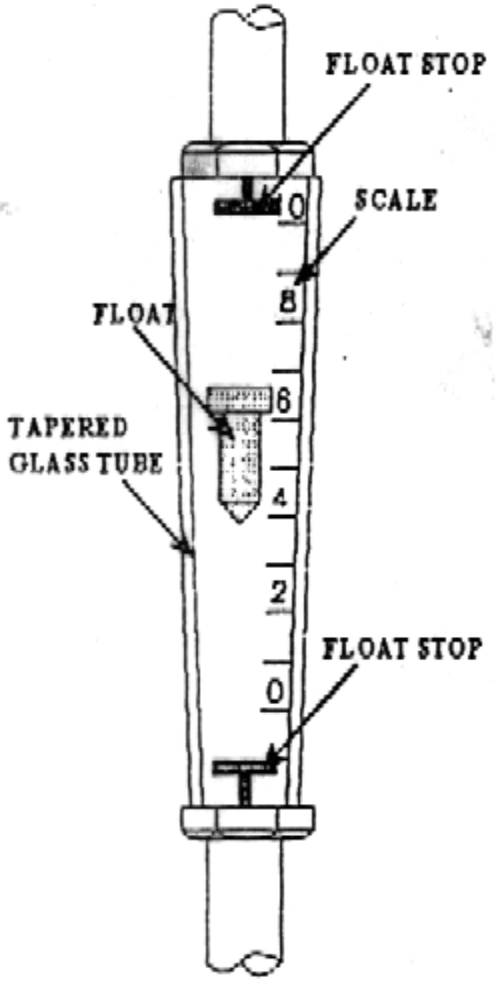


Typical Installation of single and multiple taps Pitot Tubes

Variable Area Flow meter (Rotameter)

The variable area flow meter or Rotameter is the simple, low cost, direct reading indicator for measuring flow of liquids or gases It is used on clean, low viscosity fluids, such as light hydrocarbons.

In a Rotameter, a moving body called the float represents a restriction in the line. Since the float moves freely within the tube, the pressure drop across the float remains constant as the flow rate changes.



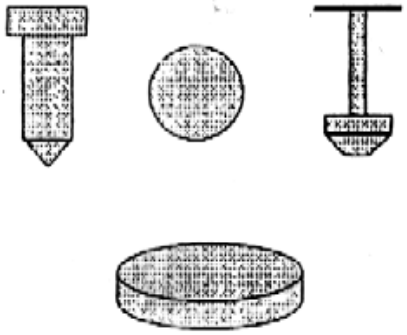
The tube is designed so that the area of the annulus is proportional to the height of the float in the tube. The scale can be very nearly linear over the range.

Conventional Rotameters permit flow measurement as low 0.1 cm^3/ min of water or an equivalent gas flow.

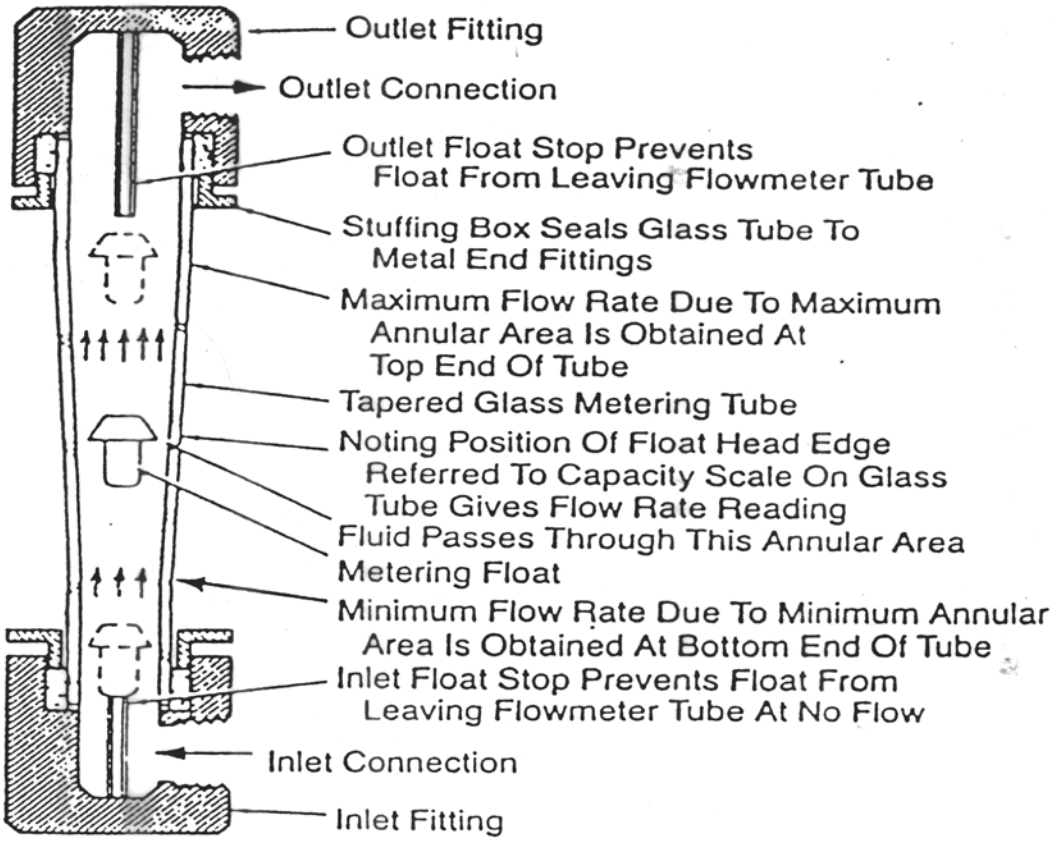
For measuring very small flows, down to 0.05 cm^3 / min, variable area meters are available with glass tube having non-circular cross-sections. Metal tube Rotameters are used for measuring low flows of liquids or gases of high temperatures and pressure.

These instruments can be used to determine liquid flows as low as 10cm^3 / min. at 500° F and pressure above 300 psi.

Typical Rotameter



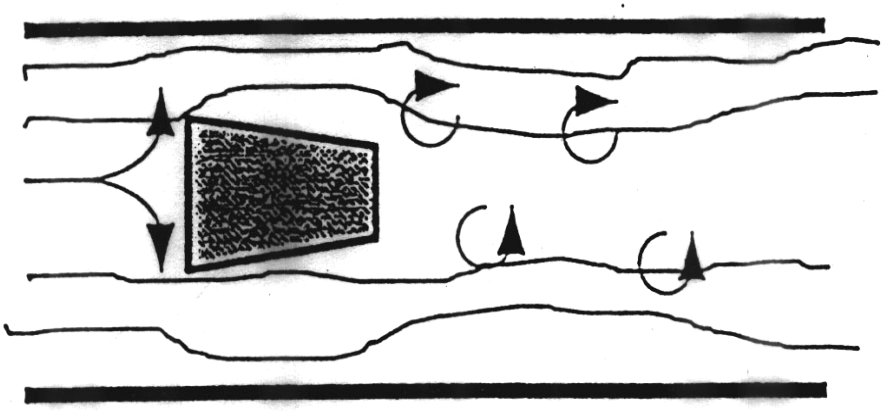
Typical Rotameter Floats



Bluff Body or Vortex Shedding Flow Meters

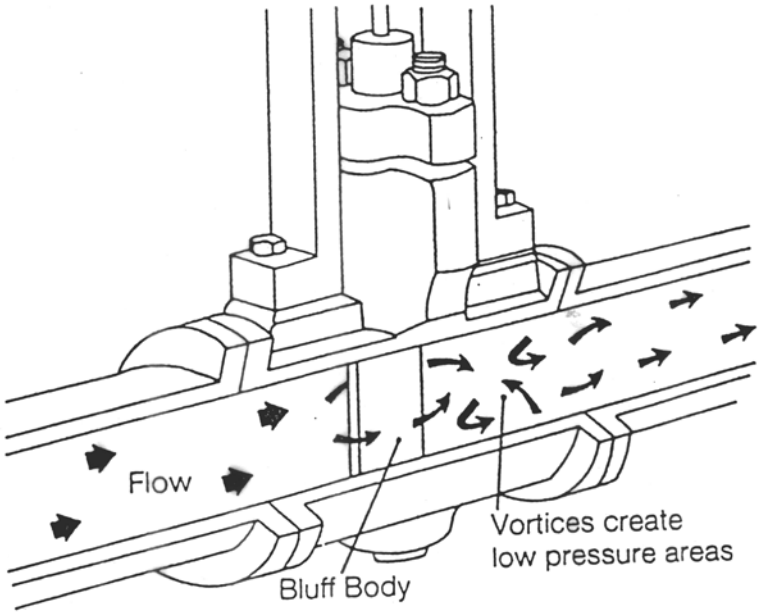
This type flow meter is suitable for measuring liquid flows at high velocity.

Its output is linear and maintains accuracy when fluid velocity, temperature or pressures varies. The



vortex-producing meter consists of a smooth bore pipe across which an obstruction called a bluff body fitted to cause turbulence in the flow stream. Vortices are produced from alternate edges of the bluff body at a frequency proportional to the volumetric flow rate without the use of any moving parts.

This makes this type of flow meter inherently more reliable than a flow meter with moving parts.

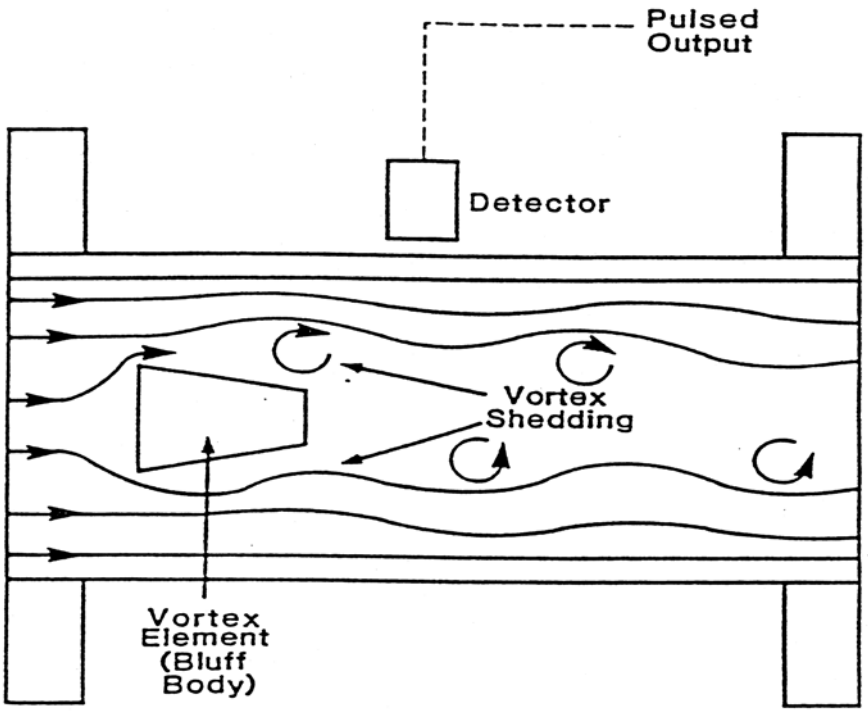


Advantages

- Since the vortex meter has no moving parts it can be installed vertically, horizontally, or in any position, although, when used in a liquid line the pipe must be kept full to avoid gas bubbles.
- It does not suffer from zero drift and requires minimal maintenance.
- It is suitable for many types of fluids, has excellent price for performance ratio.
- Its frequency output is linearly proportional to the to volumetric flow.

Disadvantages

- Unfortunately the meter's bluff body obstructs the centre of the pipe, and if it wears it may cause a calibration shift. The meter should not be used where the fluid viscosity may vary significantly.



Flow Meters with Wetted Moving Parts

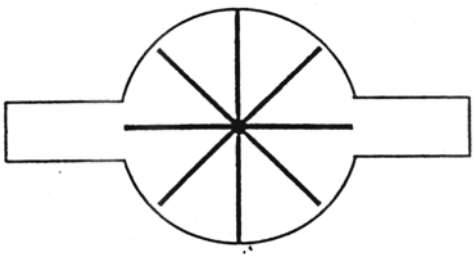
Performance of these types of flow meters depends on the precision machining of its moving parts. These moving parts are subject to mechanical wear and therefore are best suited to clean fluids only.

Turbine Flow Meter

In a turbine flow meter a rotor with a diameter almost equal to the pipe internal diameter is supported by two bearings to allow free rotation. A magnetic pickup, mounted on the pipe detects the passing of the rotor blades generating a frequency output. Each pulse represents the passage of a calibrated amount of fluid. The angular velocity (i.e. the speed of rotation) is proportional to the volumetric rate of flow. There is a minimum flow below which accuracy cannot be guaranteed due to liquid slippage. When the flow ceases, the liquid itself provides sufficient damping to stop the rotor rotating.

Advantages

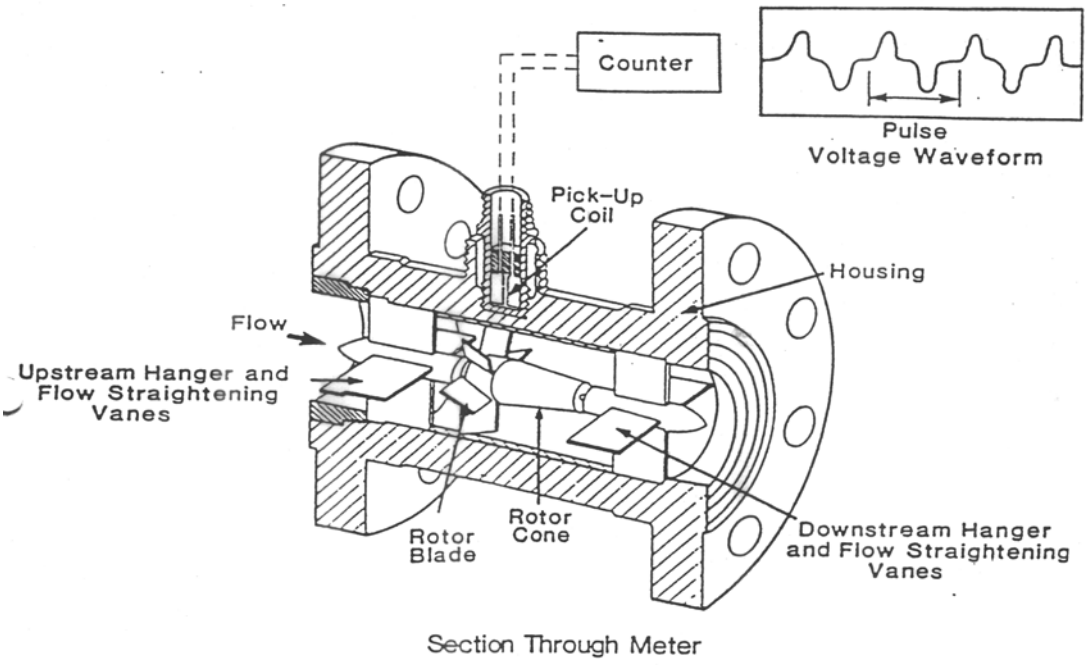
- The turbine meter is easy to install and maintain. They:
- Are bi-directional
- Have fast response
- Are compact and light weights



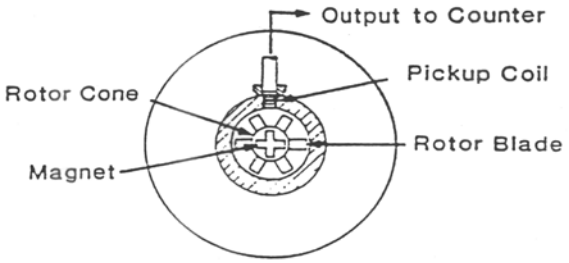
The device is not sensitive changes in fluid density (but at very low) specific gravity's, range ability may be affected), and it can have a pulse output signal to directly operate digital meters.

Disadvantages

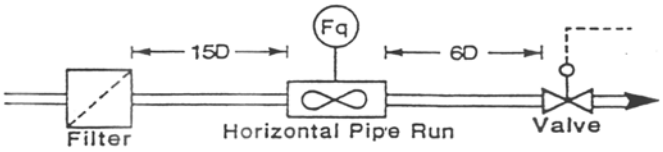
- They generally are not available for steam measurement (since condensate does not lubricate well.
- They are sensitive to dirt and cannot be used for highly viscous fluids or for fluids with varying.
- Flashing or slugs of vapour or gas in the liquid produce blade wear and excessive bearing friction that can result in poor performance and possible turbine damage.
- They are sensitive to the velocity profile to the presence of swirls at the inlet; therefore, they require a uniform velocity profile (i.e. straight upstream run and pipe straightness may have to be used).
- Air and gas entrained in the liquid affect turbine meters (in amounts exceeding 2% by volume: therefore, the pipe must be full).
- Strainers may be required upstream to minimise particle contamination of the bearings (unless special bearings are used), finely divided solid particles generally pass through the meter without causing damage.
- Turbine meters have moving parts that are sensitive to wear and can be damaged by over speeding. To prevent sudden hydraulic impact, the flow should increase gradually into the line.
- When installed, bypass piping may be required for maintenance. The transmission cable must be well protected to avoid the effect of electrical noise. On flanged meters, gaskets must not protrude into the flow stream.



Section Through Meter



Detail of Rotor Multimagnet



Typical Meter Installation

The volumetric flow rate is:

$$Q = t * f / k$$

Where:

Q = Volumetric Flow rate

t = Time Constant = 60 for flow rate per minute

f = Frequency, hertz

k = Pulse per unit volume, also referred to as the meter's **K** factor, pulses/gal

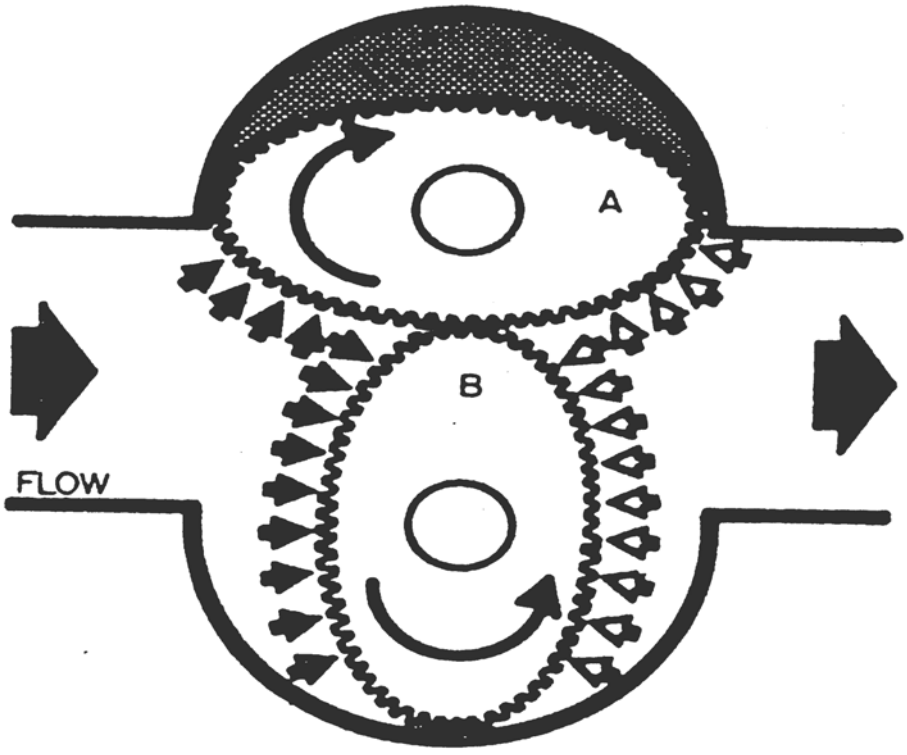
Positive Displacement Flow Meters

Principle of measurement

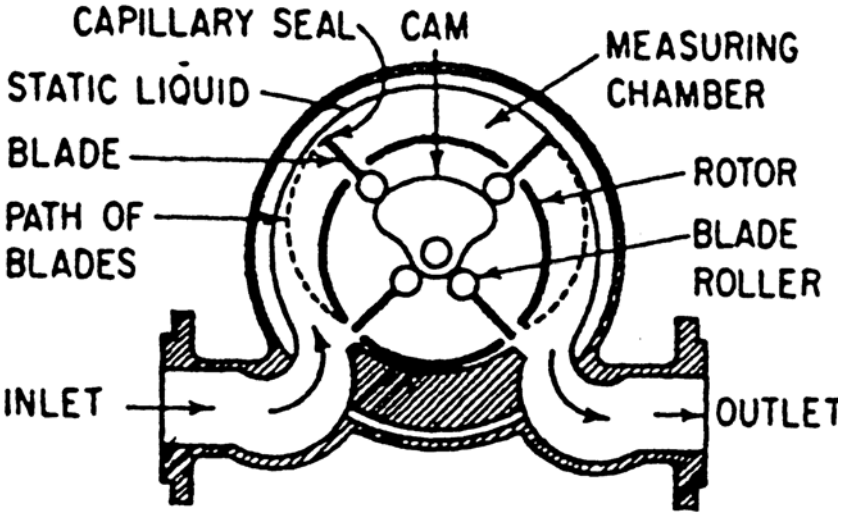
The positive displacement meter separates the incoming fluid into a series of known discrete volumes then totalises the number of volumes in a known length of time.

The common types of positive-displacement flow meters include:

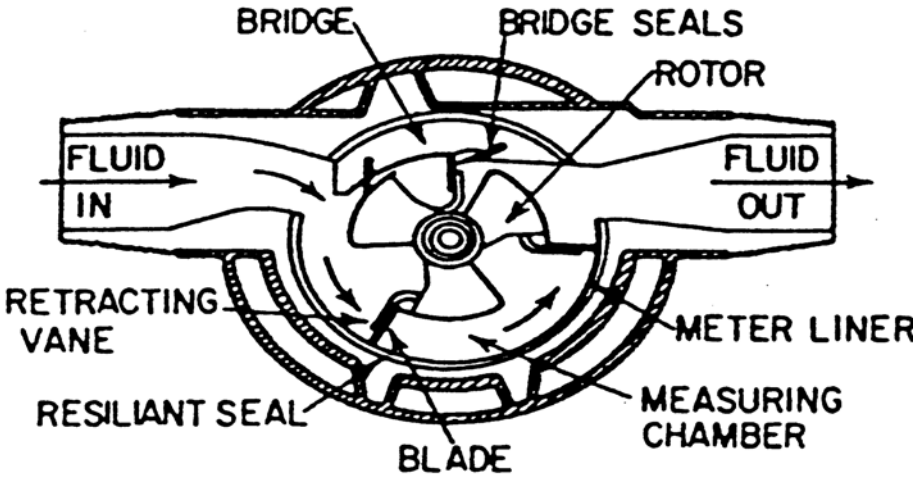
- **Rotary piston**
- **Rotary vane**
- **Reciprocating piston**
- **Nutating disk**
- **Oval gear**



This figure is a sectional schematic of Oval gear flowmeter, showing how a crescent-shaped gap captures the precise volume of liquid and carries it from inlet to outlet.



This figure shows the sliding-vane rotary meter, vanes are moved radially as cam followers to form the measuring chamber.



This figure shows another version, the retracting-vane type

Positive displacement meters are selected mainly according to the type of fluid and the rate of flow to be measured and are normally used for clean liquids where turbines cannot be used. When installed, the following should be avoided to prevent damage to the meter:

- Over speeding
- Back flow
- Steam or high-pressure cleaning

Advantages

- Positive Displacement Meter (PDM) has many advantages.
- Simple versions require no electrical power.
- They are unaffected by upstream pipe conditions
- Direct local readout in volumetric units is available.
- The highly engineered versions are very accurate.
- The low cost mass produced versions are commonly used as domestic water meters.

Disadvantages

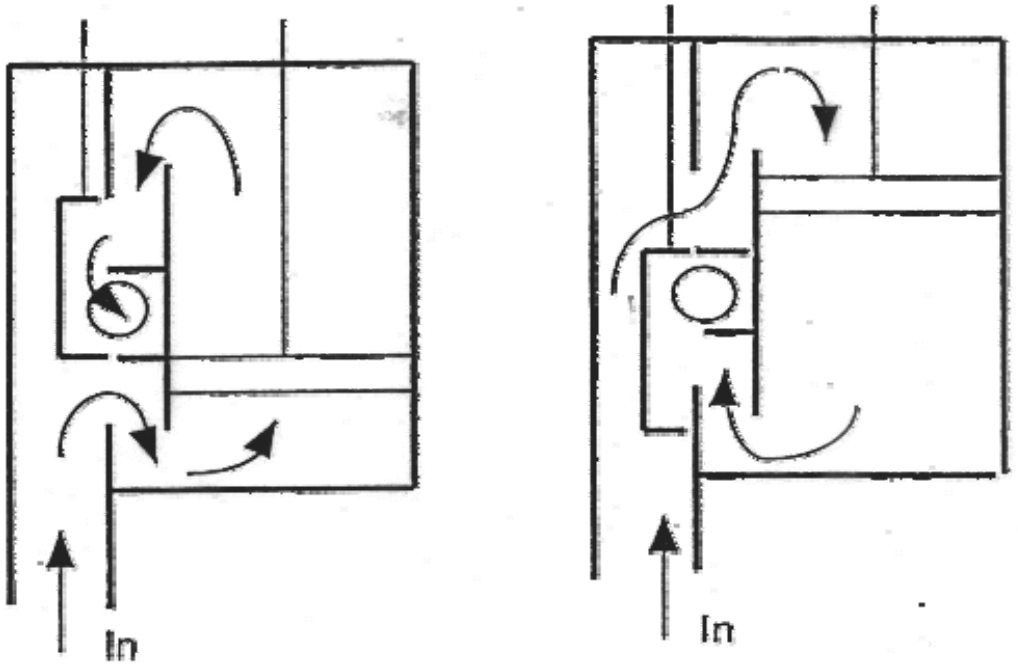
- They have many moving parts
- Clearances are small (and dirt in the fluid is destructive to the meter).
- Depending on the application, seals may have to be replaced regularly since they are subject to mechanical wear, corrosion, and abrasion.
- Periodic calibration and maintenance are required
- They are they are sensitive to dirt (and may require upstream filters).
- PD meters are large in size (and thus heavy and expensive).
- They cannot be used for reverse flow or for steam (since condensate does not lubricate well).
- Viscosity variations have a detrimental effect on performance.
- These meters have a high maintenance cost

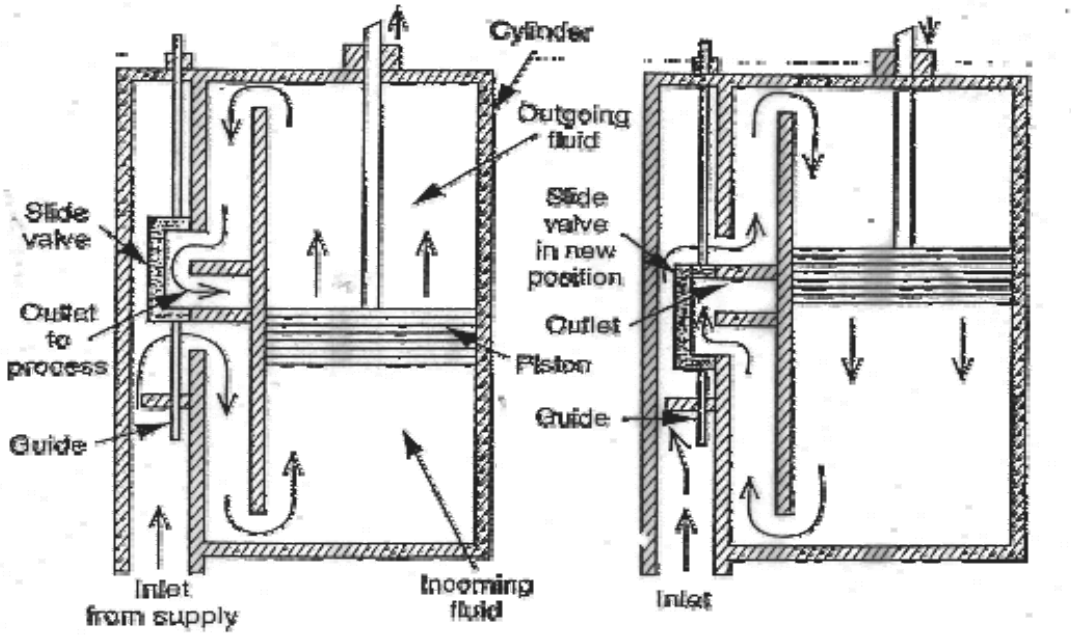
Mechanical failure the meter can block the flow in the line.

Reciprocating Piston

Fluid enters the meter, fills a compartment of fixed size and then continues on its way to the pipe work system. The number of compartments filled are counted and registered by means of a gear train and pointers operating over dials or by a cyclo-meter dial.

The system is very accurate, provided that the compartment size does not change and that there is on leakage. Fouling of the mechanism may slow down the meter operation limiting the throughput but the accuracy remains unaffected.





Obstruction Less Flow Meters

These meters allow the fluid to pass through undisturbed and thus maintain their performance while handling dirty and abrasive fluids.

Magnetic Flow Meter

The magnetic flow meter is a volumetric device used for electrically conductive liquids and slurries.

The magnetic flow meter design is based on Faraday's law of magnetic induction, which states that:

"The voltage induced across a conductor as it moves at right angles through a magnetic field proportional to the velocity of that conductor." That is, if a wire is moving perpendicular to its length through a magnetic field, it will generate an electrical potential between its two ends.

Based on this principle, the magnetic flow meter generates a magnetic field perpendicular to the flow stream and measures the voltage produced the fluid passing through the meter. A set of electrodes detects the voltage.
The voltage produced is proportional to the average velocity of the volumetric flow rate of the conductive fluid.

The tube is constructed of non-magnetic material (to allow magnetic field penetration) and is lined with a suitable material to prevent short-circuiting of the generated voltage between the electrodes. The tube is used to support the coils and transmitter assembly.

Generally the electrodes are of stainless steel but other materials are also available. These electrodes have to be chosen with care to avoid corrosion.
Dirty liquids may foul the electrodes, and cleaning methods such as ultrasonic may be required.
Theoretically, it can measure flow down to zero, but in reality its operating velocity should less than 3 ft / s (1 m/s). A velocity of 6 to 9 ft/s (2 to 3 m/s) is preferred to minimise coating.
It should be noted that at velocities greater than 15 ft/s (5 m/s) accelerated liner wear could result.
This meter has no moving parts; and is unaffected by changes in

- Fluid
- Viscosity
- Pressure

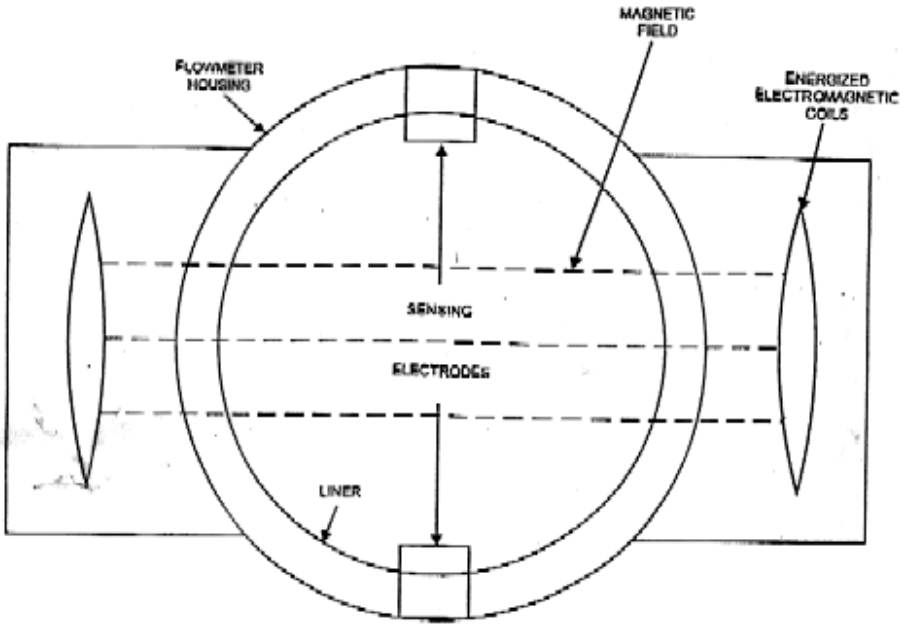
Advantages

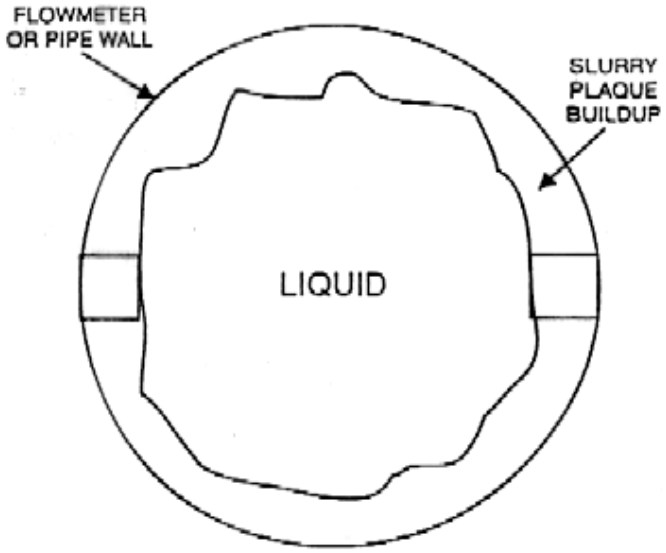
- Are bi-directional
- Have no flow obstruction
- Are easy to re-span
- Are available with DC or AC power
- It can measure pulsating and corrosive flow.

- It can measure multiphase; however, all components should be moving at the same speed; the meter can measure the speed of the most conductive component.
- It can install vertically or horizontally (the line must be full, however) and can be used with fluids with conductivity greater than 200 umhos/cm.
- Changes in conductivity value do not, affect the instrument performance.

Disadvantages

- It's above average cost
- It's large size
- Its need for a minimum electrical conductivity of 5 to 20 umhos / cm
- Its accuracy is affected by slurries containing magnetic solids (some meters can be provided with compensated output in this case).
- Electrical coating may cause calibration shifts
- The line must be full and have no air bubbles (air and gas bubbles entrained in the liquid will be metered as liquid, causing a measurement error).
- Vacuum beakers may require in some applications to prevent the collapse of the liner under certain process conditions
- In some applications, appropriate mechanical protection for the electrodes must be provided.



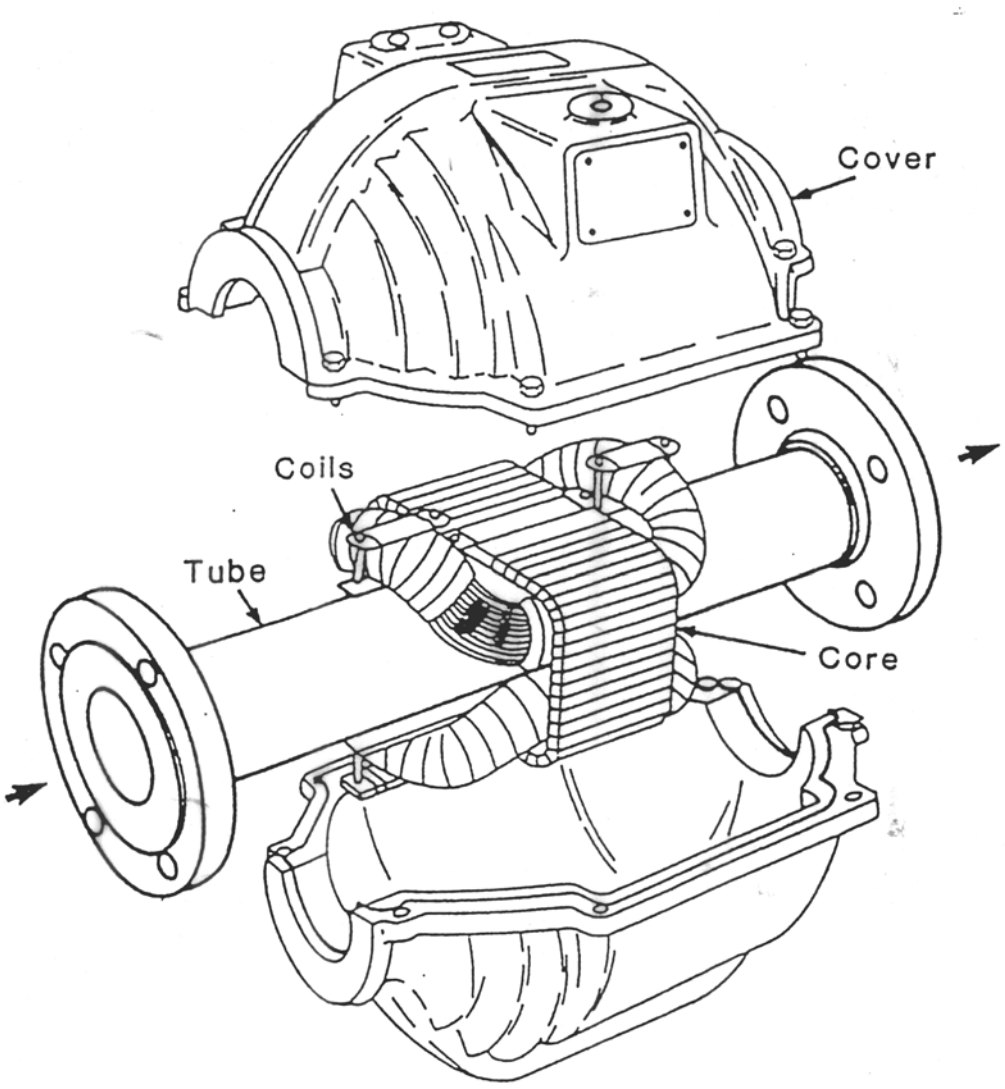


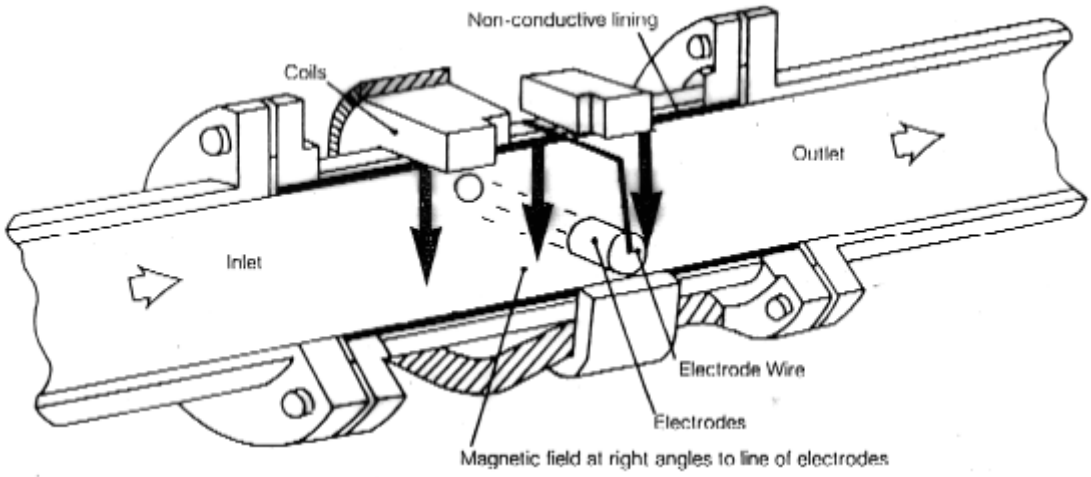
DC types are unaffected by variations fluid conductivity and thus are generally preferred. However, AC types are used for.

- Pulsating flow applications
- Flow with large amounts of entrained air
- Applications with spurious signals that may be generated from small

Electro-Chemical Reactions

- Slurries with non-uniform particle size (they may clamp together)
- Slurries with solids not g well mixed into the liquid.
- Quick response.





Mass Flow Measurement

Traditionally fluid flow measurement has been made in terms of the volume of the moving fluid even though the meter user may be more interested in the weight (mass) of the fluid. Volumetric flow meters also are subject to ambient and process changes, such as density, which changes with temperature and pressure. Viscosity changes also may affect volumetric flow sensors.

Thus for a number of years there has been much interest in finding ways to measure mass directly rather than to use calculating means to convert volume to mass. As of the early 1990s, there are three ways to determine mass flow:

1. The application of microprocessor technology to conventional volumetric meters.
2. Use of Coriolis flow meters, which measure mass flow directly.
3. The use of thermal mass flow meters that infer mass flow by way of measuring heat dissipation between two points in the pipeline.

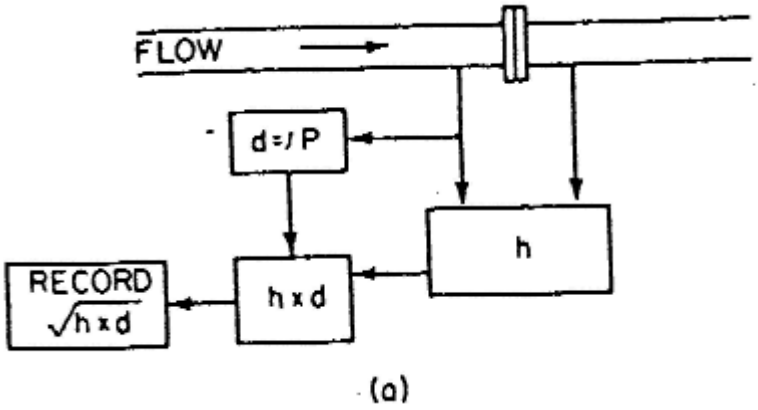
Microprocessor-Based Volumetric Flow Meters

As shown in the figure below, with microprocessors it is relatively simple to compensate a volumetric flow meter for temperature and pressure. With reliable composition (density) information, this factor also can be entered into a microprocessor to obtain mass flow readout. However, when density changes may occur with some frequency, and particularly where the flowing fluid is of high monetary value (for example, in custody transfer), precise density compensation (to achieve mass) can be expensive.

For example, a gas mass flow meter system may consist of a vortex gas velocity meter combined with a gas densito-meter. The densito-meter can be located upstream of the flow device and produce a pressure difference that is linearly proportional to the density of the flowing gas at line conditions. This unit will automatically correct for variations in pressure, temperature, specific gravity, and super-compressibility.

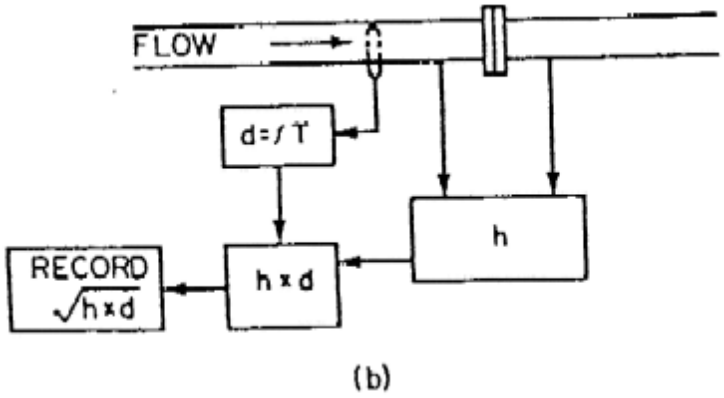
The gas sample from the pipeline passes across a constant-speed centrifugal blower and returns to the pipeline. The pressure rise across the blower varies directly with the gas density. A differential-pressure signal from the densito-meter is combined with a flow-rate signal from the gas meter. The cost of such instrumentation can be several times more than an uncompensated meter.

The relatively high cost of this instrumentation, combined with an increasing need for reliable mass-flow data, established the opportunity for direct mass-flow instruments of the Coriolis and thermal types.



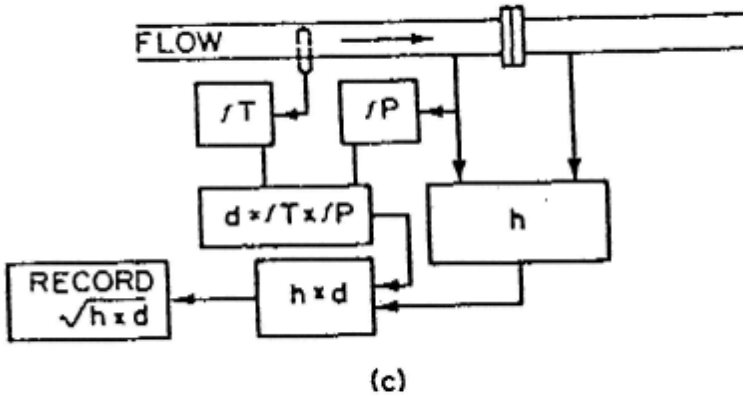
Pressure compensated Flow loop

Pressure-compensated meter wherein the differential pressure is measured by an appropriate sensor and the signal is fed into a combining module, along with a signal representing the pressure correction. The output from the combining module is used for display and to regulate the meter integrator.



Temperature compensated Flow loop

Temperature-compensated meter wherein the differential pressure is measured by an appropriate sensor and the signal is fed into a combining module, along with a signal representing the temperature correction. The output from the combining module is used for display and to regulate the meter integrator.

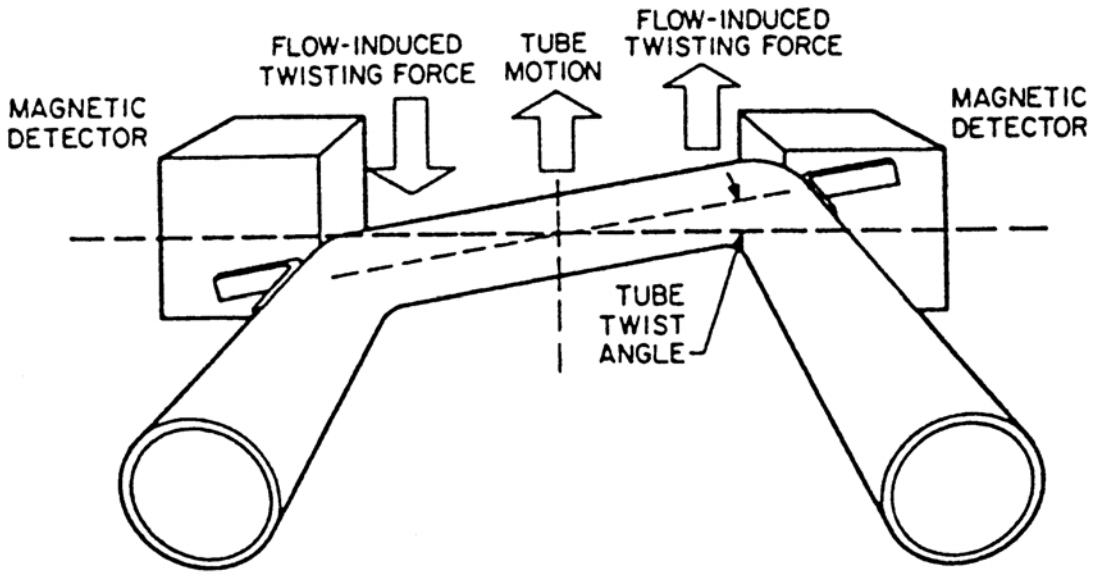


Pressure and Temp compensated Flow loop

Flow measurement where the flow is compensated for any change in the operating temperature and pressure.

Coriolis Flow Meters

The complete Coriolis unit consists of (1) a Coriolis force sensor and (2) an electronic transmitter. The sensor comprises a tube (or tubes) assembly, which is installed in the process pipeline. As shown in the below figure, an U-shaped sensor tube is vibrated at its natural frequency. The angular velocity of the vibrating tube, in combination with the mass velocity of the flowing fluid, causes the tube to twist. The amount of twist is measured with magnetic position detectors, producing a signal, which is linearly proportional to the mass flow rate of every parcel and particle passing through the sensor tube.



Typical Coriolis Meter

The output is essentially unaffected by variations in fluid properties, such as viscosity, pressure, temperature, pulsation, entrained gases, and suspended solids.
The detectors are not in contact with the flowing fluid, except the fluid at the inside wall of the tube. The tube is usually made of stainless steel. In some other application it is made of corrosion and erosion resistant material. Two magnetic position detectors, one on each side of the U-shaped tube, generate signals that are routed to the associated electronics for processing into an output.

There are two common tube types:

- **Straight**
- **Curved**

Straight Tube

The straight tube is used mainly for multiphase and for fluids that can coat or clog (since the straight type can be easily cleaned). In addition, the straight tube requires less room, that can be drained, has a low-pressure loss. Straight tube reduces the probability of air and gas entrapment, which would affect meter performance. However, the straight tube must be perfectly aligned with the pipe.

Curved Tube

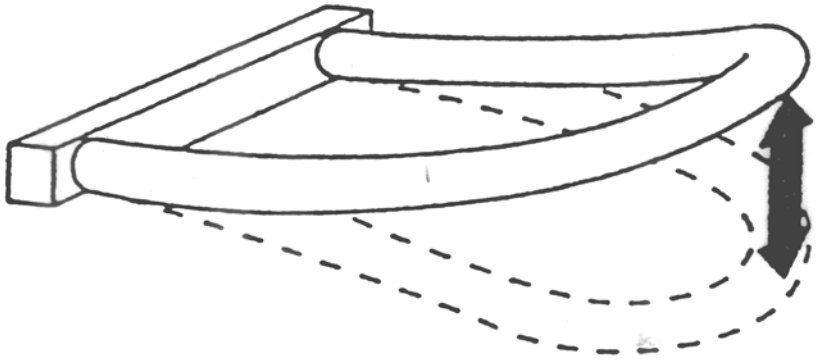
Compared to the straight tube, the curved tube has a wider operating range measures low flow more accurately, is available in larger sizes, tends to be lower in cost (due to low cost of materials), and has a higher operating temperature range. However it is more sensitive to plant vibrations than the straight type.

Advantages

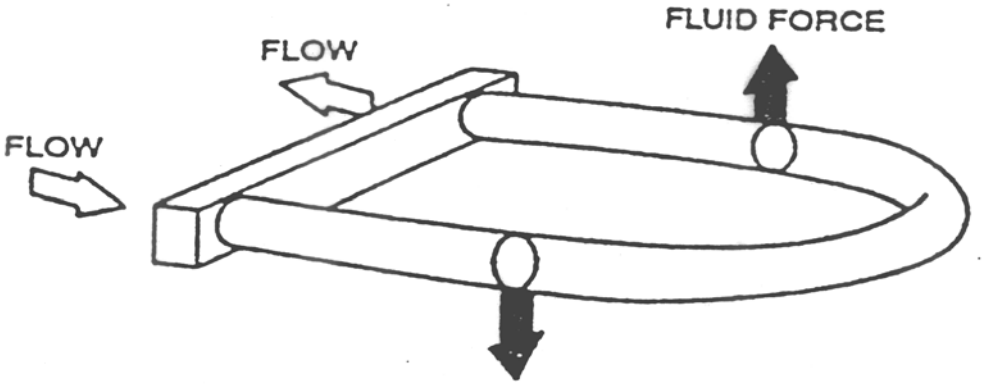
- It measures mass flow directly.
- One device that measures flow and density. Some Coriolis meter also measures temperature.
- It can handle difficult applications.
- It is applicable most fluids that has no Reynolds number limitation.
- It is not affected by minor changes in specific gravity or by viscosity.
- This type of device requires low maintenance.
- It is not sensitive to velocity profiles
- It can be used bi-directional
- It can handle abrasive fluids

Disadvantage

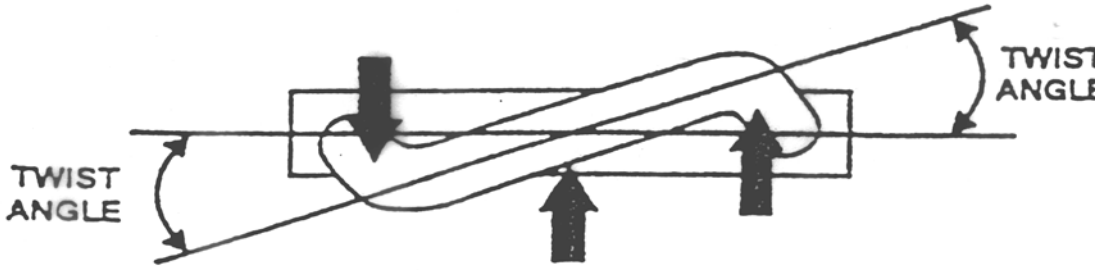
- Its purchase cost is high
- Inaccurate measurement when air and gas pockets in the liquid and by slug flow.
- The pipe must be full and must remain full to avoid trapping air gases inside the tube.
- A high-pressure loss is generated due to the small tube diameters
- It needs re-calibration if the density of the liquid being measured is very different from the one for which calibration was performed.
- Coating of the tube affects the density measurement (since it will affect the measured frequency), but not the flow measurement (since the degree of tube twist is independent of tube coating).



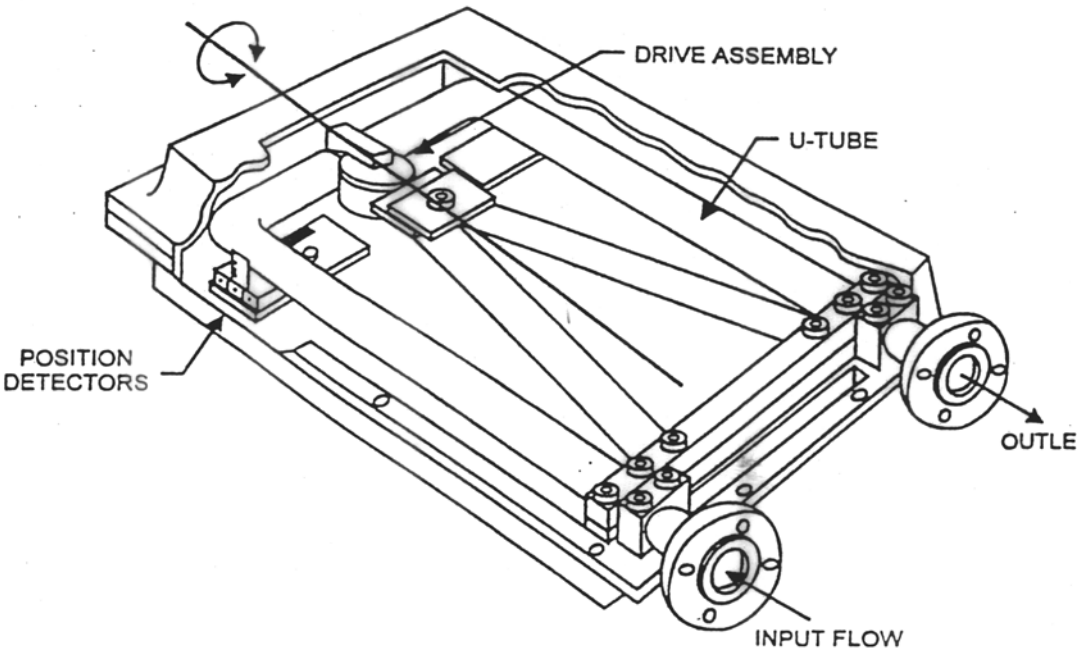
VIBRATING FLOW TUBE



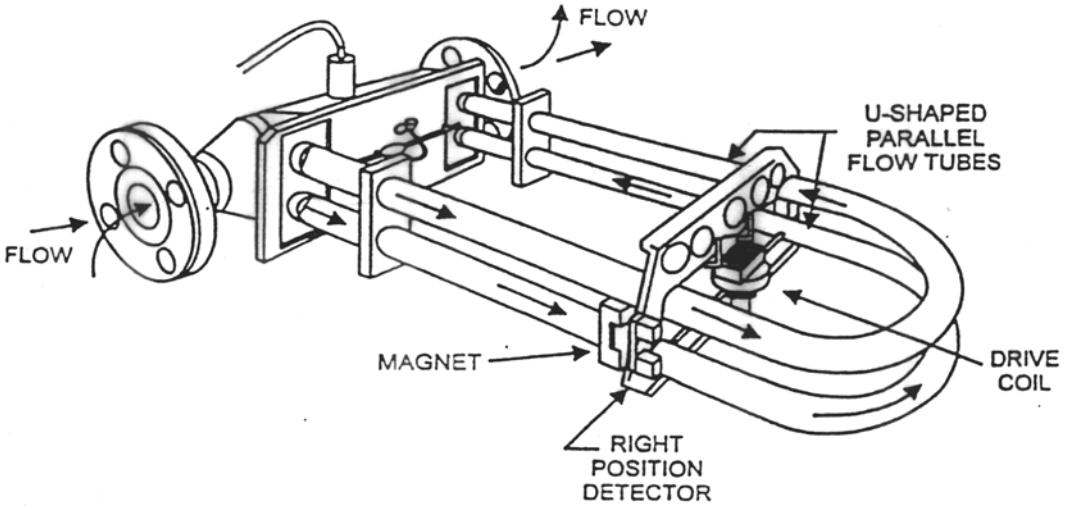
FLUID FORCES REACTING TO VIBRATION OF FLOW TUBE



END VIEW OF FLOW TUBE SHOWING TWIST



(A) SINGLE U-TUBE DESIGN



(B) DUAL U-TUBE DESIGN

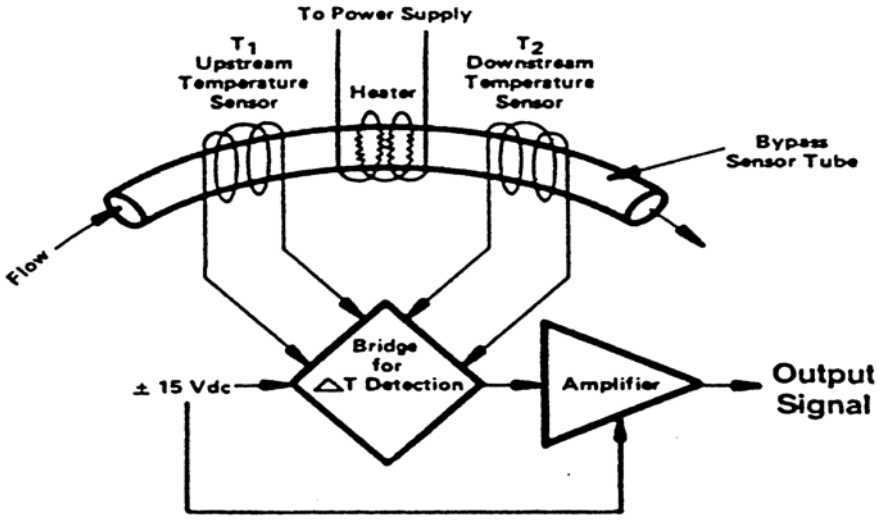
Thermal Mass Flow Meters

Like the Coriolis flow meter, after many years of design work and limited applications, the thermal mass flow meter did not become widely accepted until the late 1970s and early 1980s. In Thermal Mass Flow Meter's thermodynamic operating principle is applied.

As shown in the below figure, a precision power supply directs heat to the midpoint of a sensor tube that carries a constant percentage of the flow. On the same tube at equidistant two temperature elements (RTD) are installed upstream and downstream of the heat input. With no flow, the heat reaching each temperature element (RTD) is equal. With increasing flow the flow stream carries heat away from the upstream element T1 and an increasing amount toward the downstream element T2. An increasing temperature difference develops between the two elements.

This temperature difference detected by the temperature elements is proportional to the amount of gas flowing, or the mass flow rate.

A bridge circuit interprets the temperature difference and an amplifier provides the 0- to 5-volt dc and 4- to 20-mA output signal.



Typical Thermal Mass Flow meter construction

Thermal Mass Flow is measured by the following formula:

$$W = \frac{Q}{C_p * (T_2 - T_1)}$$

Where:

W = Massflow rate of fluid (lbm/hour)

Q = Heat transferred (BTU/hour)

$\mathbf{C_p}$ = Specific Heat of fluid (BTU lbm F°)

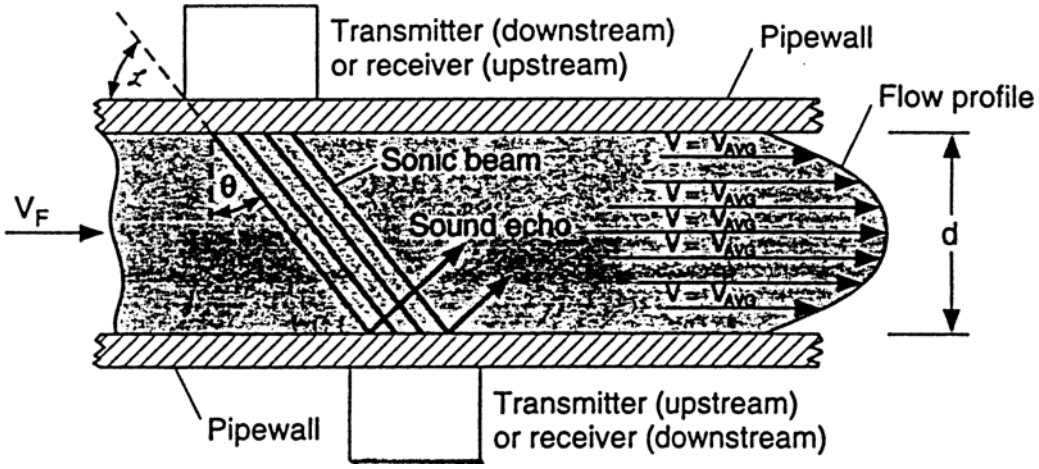
Flow Meters with Sensors Mounted Externally

These offer no obstruction to the fluid and have no wetted parts. They cannot be used in all applications due to their inherent limitations.

Ultrasonic Flow meters

Transit Time, Time-of-Travel, Time-of-flight

In an ultrasonic (transit time) flow meter two transducers are mounted diametrically opposite, one upstream of the other. Each transducer sends an ultrasonic beam at approximately 1 MHz generated by a piezoelectric crystal. The difference in transit time between the two beams is used to determine the average liquid velocity. The beam that travels in the direction of the flow travels faster then the opposite one.



Ultrasonic Flow Meter

This figure shows the principle of transit-time ultrasonic flow meter, clamp-on type. Transducers alternately transmit and receive bursts of ultrasonic energy.
Each transducer acts as a transmitter and receiver. Two transducers are used to cancel the effect of temperature and density changes on the fluid sound transmission properties. The speed of sound is not a factor since the meter looks at differential values.

The crystals producing the ultrasonic beam can be in contact with the fluid or mounted outside the piping (clamp-on transducers).

Advantages

- It does not cause any flow obstruction
- It can be installed bi-directional
- It is unaffected by changes in the process temperature
- It is suitable to handle corrosive fluids and pulsating flows.
- It can be installed by clamping on the pipe and is generally suited for measurements in very large water pipes.

Disadvantages

- This type of meters are highly dependent on the Reynolds number (the velocity profile)
- It requires nonporous pipe material (cast iron, cement and fibreglass should be avoided)
- It requires periodic re-calibration
- It is generally used where other metering methods are not practical or applicable.

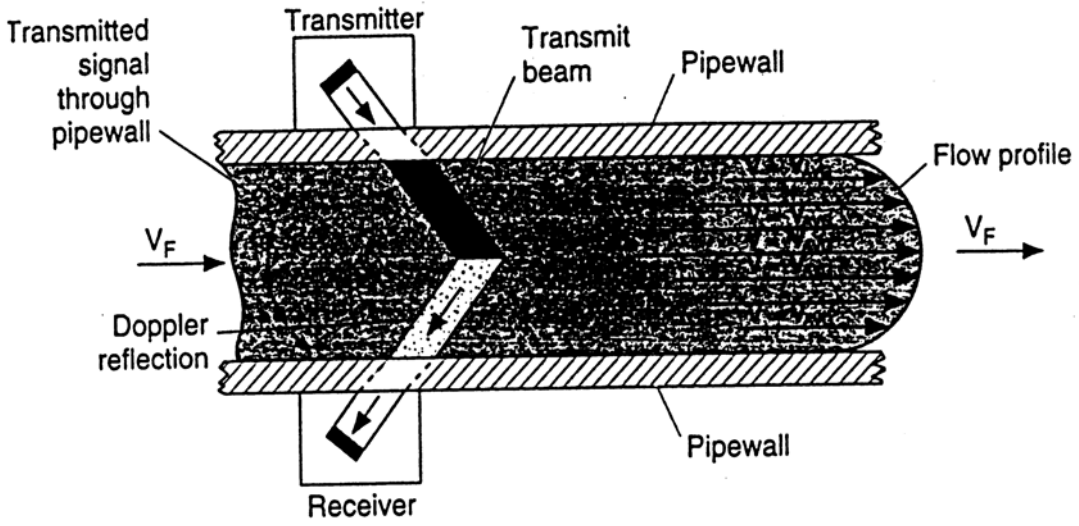
Doppler-Effect Flow Meter

The configuration shown utilizes separated dual transducers mounted on opposite sides of the pipe.

It is mandatory in a Doppler-Effect Flow Meter the flowing stream contains sonically reflective materials, such as solid particles or entrained air bubbles. Without these reflectors, the Doppler system will not operate. In contrast, the transit-time ultrasonic flow meter does not depend on the presence of reflectors.

Doppler-effect flow meters use a transmitter that projects a continuous ultrasonic beam at about 0.5 MHz through the pipe wall into the flowing stream. Particles in the stream reflect the ultrasonic radiation, which is detected by the receiver. The frequency reaching the receiver is shifted in proportion to the stream velocity. The frequency difference is a measure of the flow rate. When the measured fluid contains a large concentration of particles or air bubbles, it is said to be sonically opaque. More opaque the liquid, greater the number of reflections that originate near the pipe wall, a situation exemplified by heavy slurries. It may be noted from the flow profile, that the fluid velocity is greatest near the centre of the pipe and lowest near the pipe wall.

The Doppler Flow meter works satisfactorily for only some applications and is generally used when other metering methods are not practical or applicable. It should not be treated as a "universal" portable meter.



Doppler-effect Ultrasonic Flow Meter

This figure shows the principle of Doppler-effect ultrasonic flow meter with separated opposite-side dual transducers.

Advantage

- The common clamps-on versions are easily installed without process shutdown.
- It can be installed bi-directional
- Flow measurement is not affected due to change in the viscosity of the process.
- Generally suitable for measurements in large water pipes
- The meter produces no flow obstruction
- Its cost is independent of line size.

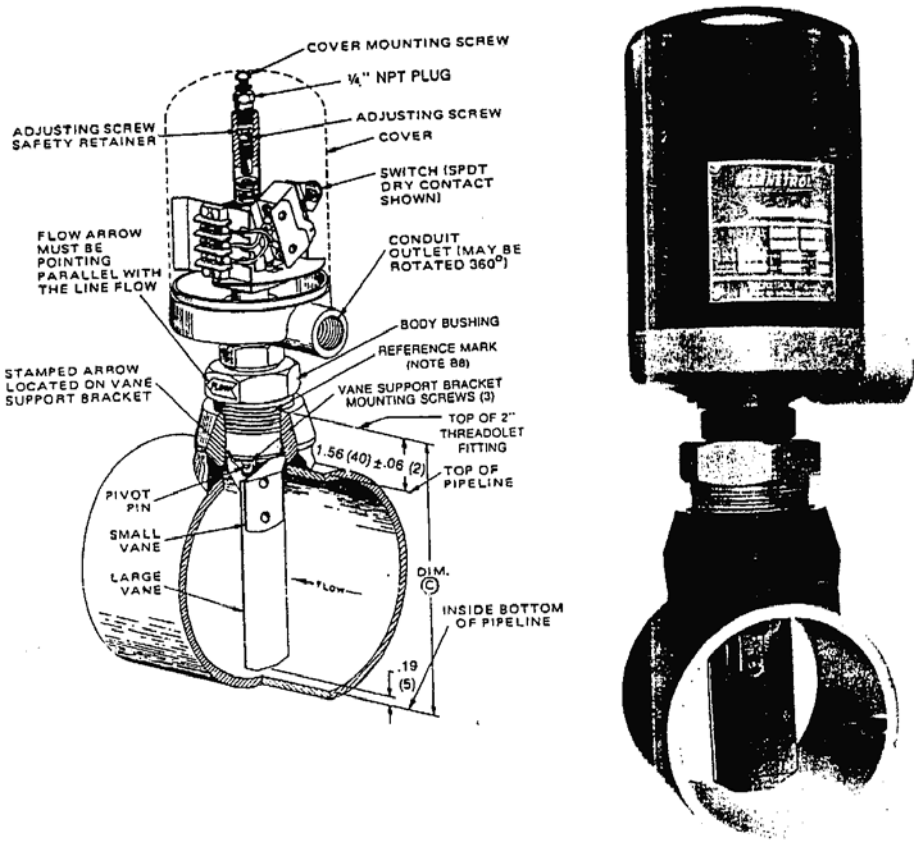
Disadvantage

- The sensor may detect some sound energy travelling in the causing interference reading errors.
- Its accuracy depends on the difference in velocity between the particles, the fluid, the particle size, concentration, and distribution.
- The instrument requires periodic re-calibration.

Flow Switches

Flow switches are devices, which indicate either the presence or the absence of flow. Any of the primary flow elements can have associated switches that may be a part of the controller circuitry or operated from the pneumatic or electronic output signal. Dedicated flow switches are available which operate by a paddle or vane inserted into the flow. When flow is present the paddle or vane is moved and a switch mechanically tripped.

Thermal flow switches use a heater and a heat sensor. When flow is present the heat sensor is cooled by the flow and the switch activated a thermal flow switch.



Flow Switch

Flow Glasses

Flow glasses are windows in the pipe, which allow the fluid to be directly observed. Usually a fitting is provided with a glass on either side of the pipe so that one can see the flow of the liquid. A paddle wheel, float or other device is often used so that movement in the fluid is more readily observed.

SGC HSE Regulation

Refer to HSE Regulation No. 7 "Isolation"

7.18 CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

1. Preparation for Work

When work is to be done on control engineering hardware, it is essential that the work site is prepared correctly. The work permit system (Regulation No 06) provides the mechanism for ensuring that essential preparatory activity is documented and witnessed. It is important that the Control Engineering Person doing the work also has responsibility for effecting the task safely. This is particularly important in the control engineering discipline where the instrument may still be connected to the process, or may require to be serviced with power-on for fault finding.

2. Interaction.

Care must be taken to ensure that the work to be carried out on a specific item of instrumentation will not cause a hazard due to interaction with other protection systems or operational process controls.

3. Preparatory Work.

Prior to a work permit being issued all appropriate preparatory work at the site must be completed. The following items are some examples.

a. Removal of potential hazards from the area. Particular vigilance is needed for enclosed areas (refer to Regulation No 09 Confined Space Entry).
b. Gas testing the area for flammable, toxic or suffocating gases.
c. Construction of scaffolding to permit safe access.
d. Provision of additional fire fighting apparatus.
e. Provision of necessary protective clothing.
f. Isolation of the control engineering hardware from the process.
g. In the case of equipment removal, isolation of the hardware from utilities (e.g. electricity, air supplies etc).

On completion of all necessary preparatory work (defined by the Senior Control Engineering Person and the Area Authority), a hot or cold work permit/entry permit will be issued signed by the appropriate responsible authorities.

Refer to HSE Regulation No. 7 "Isolation"

7.18 CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

4. **Isolation of Hardware.**

Isolation of control engineering hardware may be necessary to enable maintenance work to be done or permit removal of the hardware to effect repairs (either locally or remotely). Isolation of hardware can take several forms, for example isolation from:

a. Process plant.
b. Utilities (electric, pneumatic, hydraulic, cooling media etc).
c. Larger system of which the hardware is a subsystem or component.

5. **Isolation from Process.**

a. Isolation of instruments, which are connected to or form a part of the process is usually achieved by valving. It is important that, where isolation of an instrument is required for maintenance purposes, correct venting/draining and valve closure procedures are adhered to.
b. Where instruments have local isolating valves in addition to the primary process isolating valves, the local valves may be used for some routine in-situ testing at the discretion of the Senior Control Engineer. If an instrument is to be removed from site, the process isolating valves must be used and any impulse pipe work must be drained or vented completely.
c. Where the process fluids are of a hazardous nature (e.g. toxic, flammable etc), particular care should be taken to ensure correct venting and draining, and also to clean or flush the instrument carefully, prior to effecting work or removal of the hardware from site for maintenance or repair. Gas testing may be required. On large items, e.g. control valves, a certificate of cleanness is necessary prior to delivery to workshops.
d. On removal of a directly mounted instrument, from a process line containing hazardous fluids, e.g. pressure gauges etc, isolation by the primary isolation valve only is NOT acceptable. The valve outlet shall be blanked off, capped or plugged with a blank flange, solid screwed plug or cap, whichever is appropriate.

Refer to HSE Regulation No. 7 "Isolation"

7.18 CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

6. Isolation from Electrical/Pneumatic Supplies.

a. If practical, equipment must be made safe before any work is done on it. A Competent Control Engineering Person must do the operation of making the equipment safe. Care shall be taken when working on live equipment to ensure avoidance of contact with live electrical components (refer to Regulation No 19 Working with Electricity).

b. Pneumatically operated equipment must be isolated before it is disconnected or removed for repair by closing the valve at the supply manifold for the individual instrument and venting through the drain/vent of the pressure regulators.

7. Isolation from Utilities.

a. Control engineering equipment may be connected to utilities (other than electrical associated with the hardware e.g. streams, cooling water, hydraulic fluid, chemicals, carrier gases (analysis) and air supplies. It is important that attention is given to rendering the utilities safe when the control engineering hardware is being serviced or removed.

b. Utilities should be isolated at the point of distribution to the control engineering equipment being removed (e.g. isolating valve at distribution head) and not solely at the hardware itself.

c. Where utility fluids are 'piped' to an instrument, the pipe work should be drained down or vented if the instrument is removed.

d. It is important that removal of a utility from a specific piece of hardware does not influence any other hardware to which the utility may also be connected (e.g. cooling water may have been series connected to more than one item of hardware).

Review Questions

Q1) Give three reasons for measuring fluid flow rates on an oil platform.

Q2) Describe how you would calculate the quantity of a product made in 24 hours from a control room strip chart flow rate recorder.

Q3) Name three process operating variables, which influence the accuracy of an orifice type flow meter.

Q4) Name three types of flow patterns.

Q5) Describe what is meant by the terms:

Rate of flow meter, Total flow meter.

Q6) What types of instrument are normally found in mass flow metering systems?

Q7) Name three types of flow rate metre?

Q8) Sketch the pressure profile in pipeline upstream and downstream of an orifice plate.

- Show the pressure on the profile at the following points
- Upstream flange tap
- Downstream flange tap

Q9) Describe the operating principle of an orifice meter

Q10) What is the purpose of the booster relay in a pneumatic differential pressure transmitter?

Q11) State the output range of a differential pressure transmitter in terms of the following units:

Bar, psi, mA, volts

Q12) Draw a 0 – 100% linear flow scale side – by side with a square root scale and state the problems of the latter scale?

Q13) How does the operating principle of the orifice plate differ from that of the Rotameter in terms of the basic flow equation components?

Q14) Describe the operating principle of a turbine meter with an Electro- magnetic pick – up?

Q15) Sketch an oval gearwheel meter showing the direction in which the gearwheels rotate?

Q16) Sketch a vortex flow meter?

Q17) Name one disadvantage of an electromagnetic flow meter?

LEVEL MEASUREMENTS

LEVEL MEASUREMENTS

Objectives

At completion of this module, the developee will have understanding of:

1. Level measurement definition.
2. Principles that level devices operate under.
3. Level measurement units.
4. Direct methods used to measure the level of a liquid.
5. Dipsticks, weighted gauge tape and floats function.
6. Types of sight gauges; tubular, reflex, armoured and magnetic.
7. Safety feature of external sight glasses.
8. Pressure (hydrostatic method) as a level measurement.
9. Differential pressure method as level measurement.
10. DP system for open tank applications.
11. DP system for closed tank; dry leg and wet leg applications.
12. Zero suppression and zero elevation requirements.
13. DP level transmitter span, range calculation and calibration procedure.
14. Bubble tube (purge) systems working principle.
15. Bubbles tube zero and span adjustment.
16. Purge system applications.
17. Displacement devices principle of operation.
18. Displacer apparent weight calculations.
19. Displacement devices advantage.
20. Interface level measurement using displacer type.
21. Calibration procedure of interface level measurement using displacer type.
22. Capacitance probes as a level sensor working principle.
23. Capacitance level sensor applications as continuous level measurement, point-level and interface level.

24. Conductivity level sensors principle and applications as high and low alarm level.
25. Ultrasonic level sensors principles and applications.
26. Factors affecting the performance of ultrasonic level meters.
27. Electronic level Transmitter (Fisher model 2390).
28. Pneumatic level Transmitter (Fisher Leveltrol).
29. Automatic Tank Gauge parts, operation and application.
30. Level Switches parts, operation and application.

Related Safety Regulations for Module I-3: LEVEL MEASUREMENT
Juniors have to be familiarised with the following **SGC HSE** regulations, while studying this module:
Regulation No. 6: Work to permit system.
Regulation No. 7: Isolation
7.18 (1-10) control systems procedures and isolations.
Regulation No. 22: Hot and Odd Bolting.
Regulation No. 23: General Engineering Safety.
Regulation No. 27: General Services:
Safe use of hand tools and powered tools/equipment.

Level Measurement

In the Oil and Gas industry level is an important process parameter that needs proper measurement and control. Level measurement is defined as the measurement of the position of an interface between two media such as gas and liquid or between two liquids. Level is a key parameter used for accounting needs and for control. Level measurement may be expressed in units of length or percentage level. In some cases the level measurement is converted to a volume to give a more meaningful indication.

Level measurement is a single dimension from a reference point. Figure 1 shows tank level measurement, either by Inage method or Outage method.

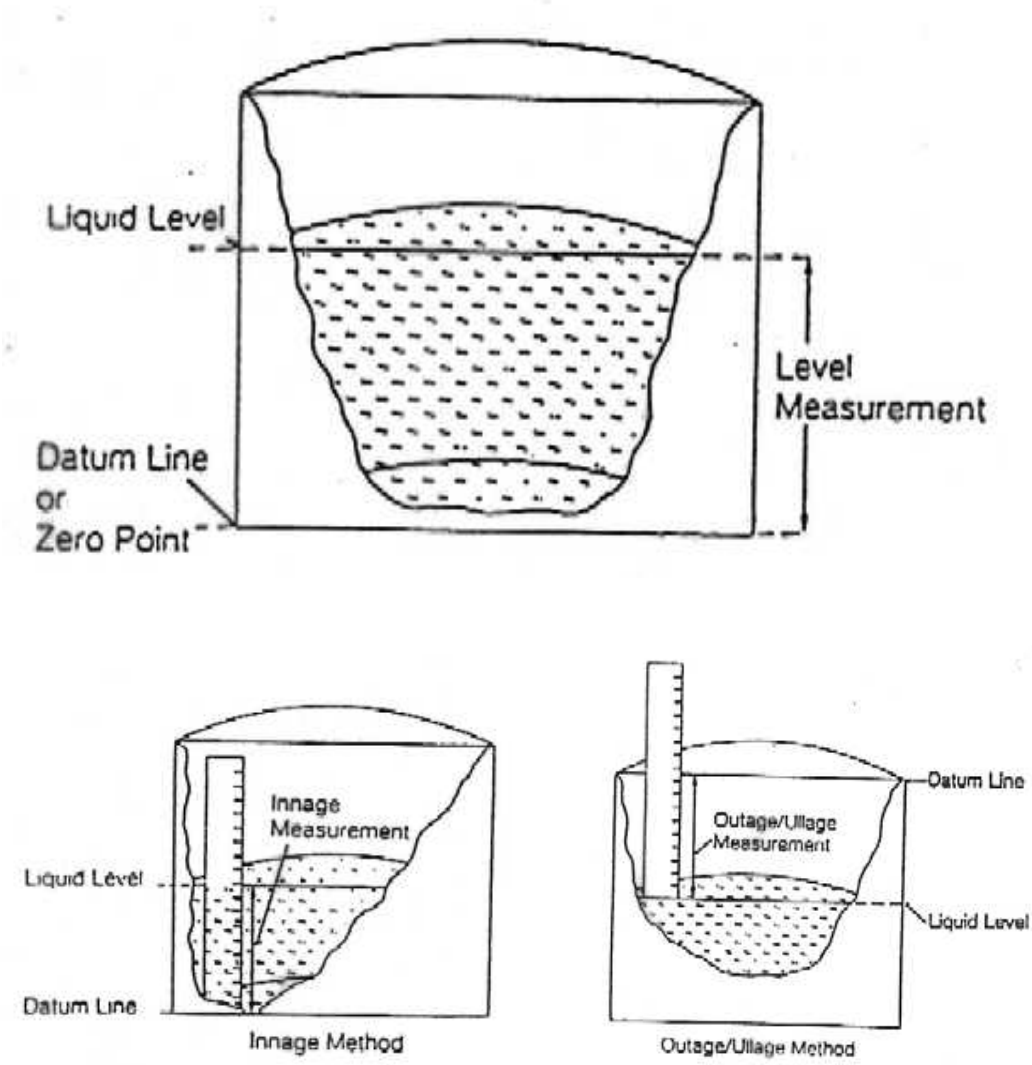


Figure 1

Level Measurement Principle

Level devices operate under three main different principles:

a. The position (height) of the liquid surface
b. The pressure head
c. The weight of the material

There are two methods used to measure the level of a liquid:

1. **Direct Methods**
2. **Indirect or inferential Methods**

Direct Methods (Visual Methods)

The direct method measures the height above a zero point by any of the following methods. Direct methods for level measurement are mainly used where level changes are small and slow such as; Sump tanks and Bulk storage tanks.

Direct methods are simple to use, reliable, low cost items and generally well suitable to hazardous areas. There are four types of direct level measurement devices:

1. Dip-sticks & Dip-Rods
2. Weighted gauge tape
3. Sight Glasses, and
4. Floats.

Dip-Sticks & Dip-Rods

The dipstick needs little explanation. The liquid wets the lower end of the rod that has been dipped into it. The rod is stopped either at the top of the vessel by a protruding flange on the rod, or at the bottom of the vessel when the tip of the rod touches it.

When the rod is withdrawn, the wet/dry interface can be clearly seen and the level determined from a scale on the rod. Figure 2 shows some styles of dipsticks and dip-rods.

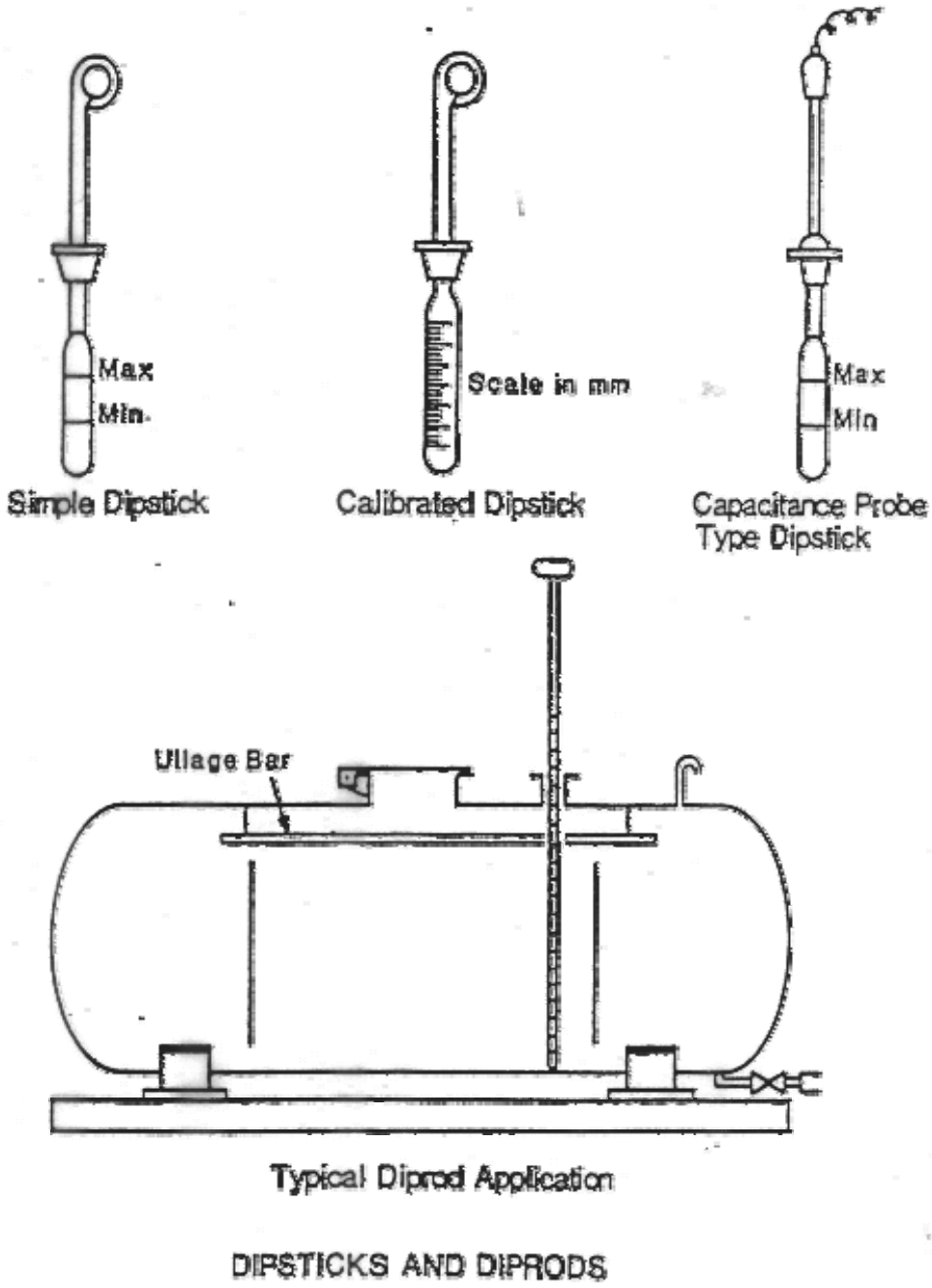

Max
Min
Simple Dipstick
Scale in mm
Calibrated Dipstick
Max
Min
Capacitance Probe
Type Dipstick
Ullage Bar
Typical Diprod Application
DIPSTICKS AND DIPRODS

Figure 2

Weighted Gauge Tape

Another variation is the weighted gauge tape illustrated in figure 3. This is used in a similar fashion to the dipstick, but on deep vessels and tanks where a solid rod would be inappropriate.

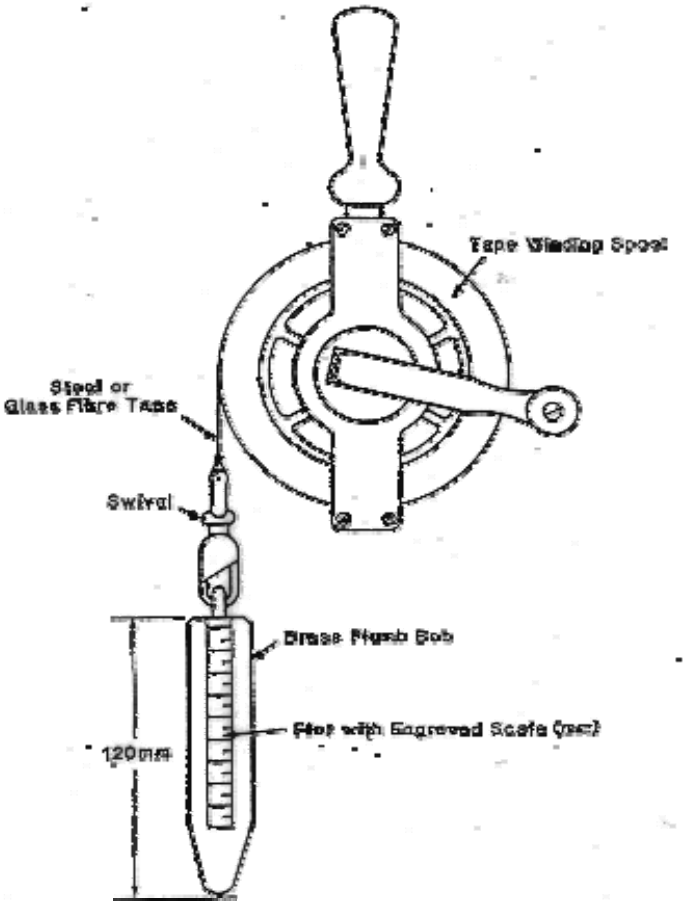


Figure 3. Weighted Gauge Tape

Sight Glasses

There are various types of sight glass, the two most common types being used are:

1. The flat glass tubular (or reflex)
2. Magnetic.

The Flat Glass

The flat glass type, as shown in figure 4, is used for non pressurised vessels, It consists of a glass window or windows that forms part of the vessel. A typical application is in hot oil tanks, where

excessive foam contaminated oil may be easily detected. In such applications the glass would be heat resistant.

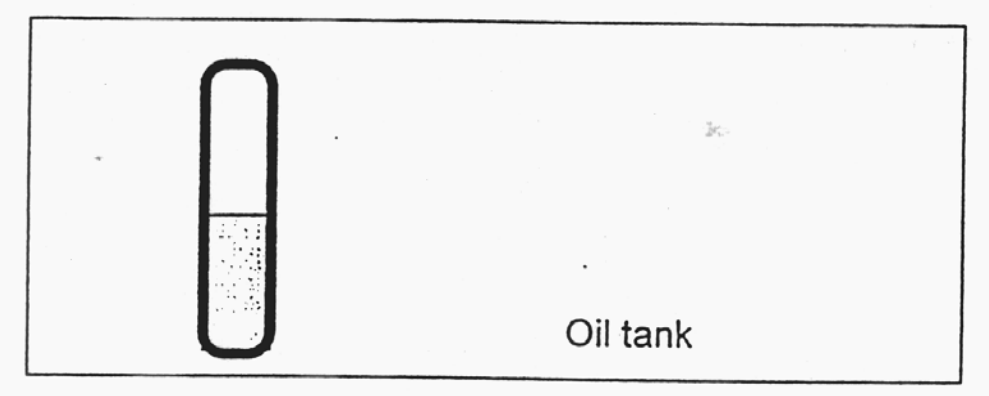
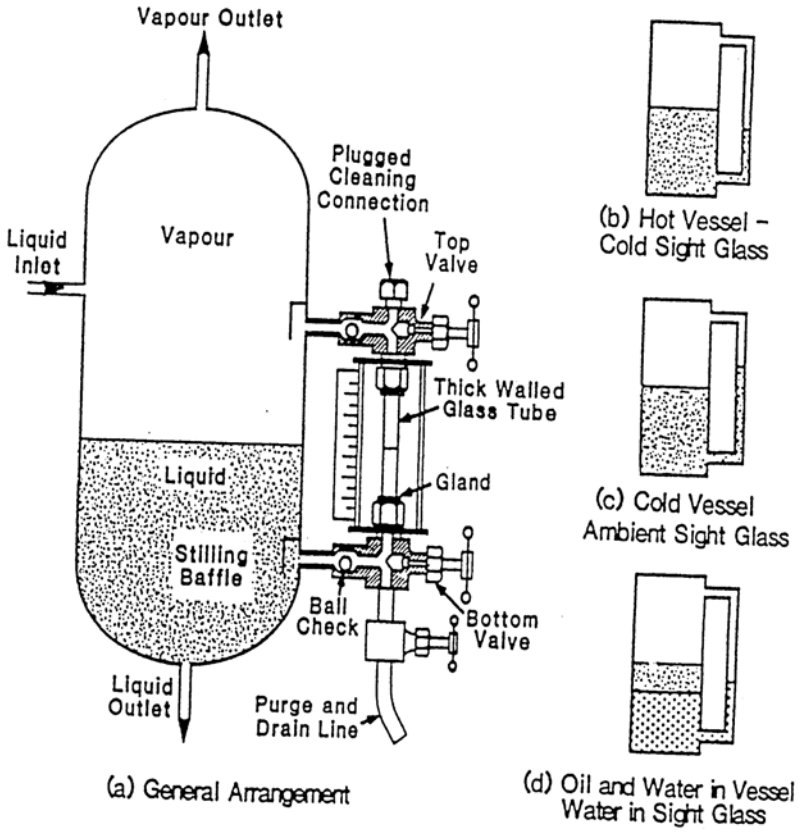


Figure 4. Flat Glass Type



Tubular or Reflex

Tubular or reflex sight glasses, as illustrated in figures 5 and 6, consist of a single glass with cut prisms. Light is refracted from the vapour portion of the column and is shown generally as white colour. Light is absorbed by the liquid portion in the column and is shown generally as a dark colour. They are used mainly for non-corrosive, non-toxic inert liquids at moderate temperatures and pressures. The tube may be made of glass or transparent plastic and must be rated for the operating pressure of the vessel.

Figure 5. Tubular Sight Glasses

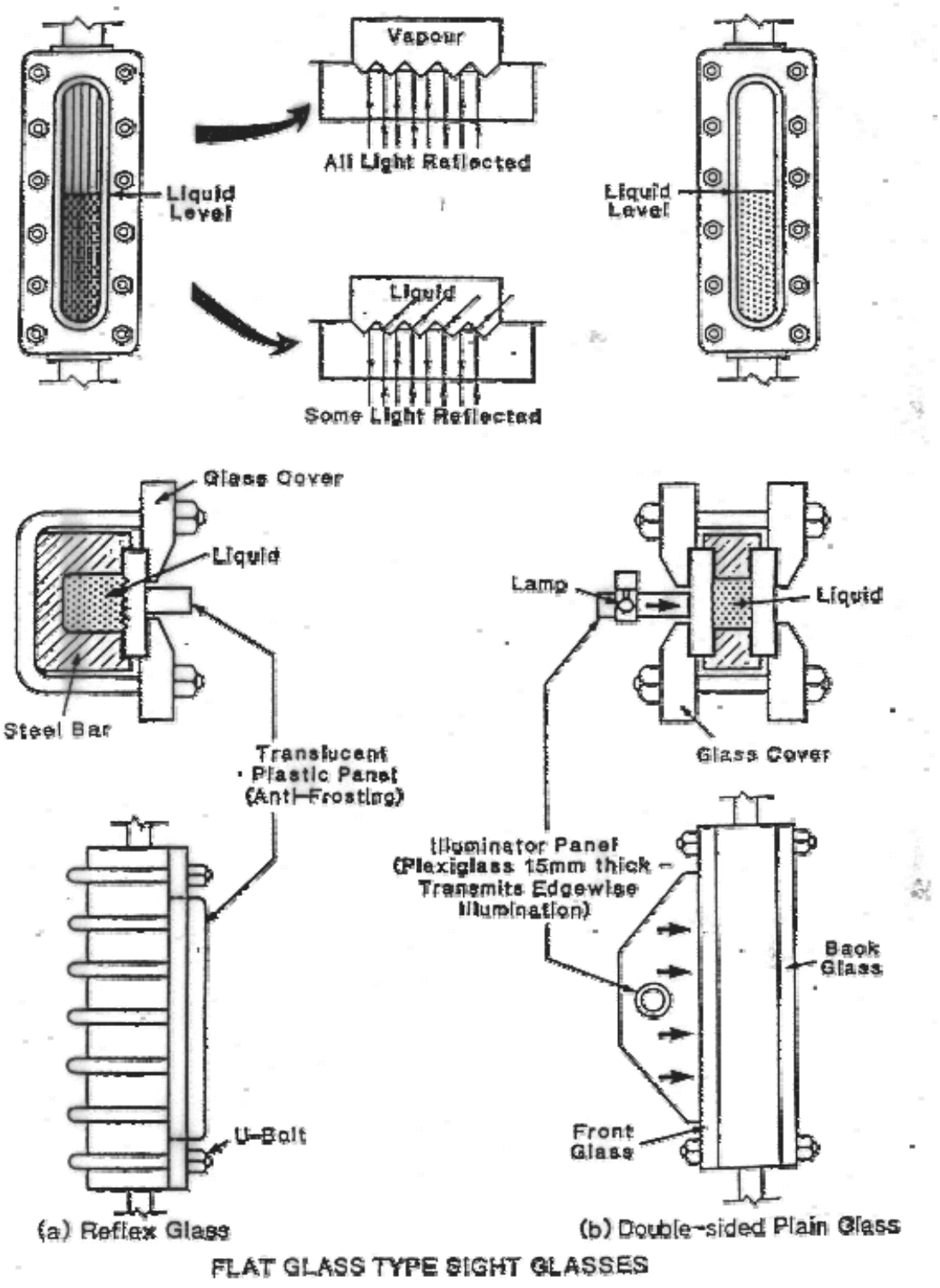
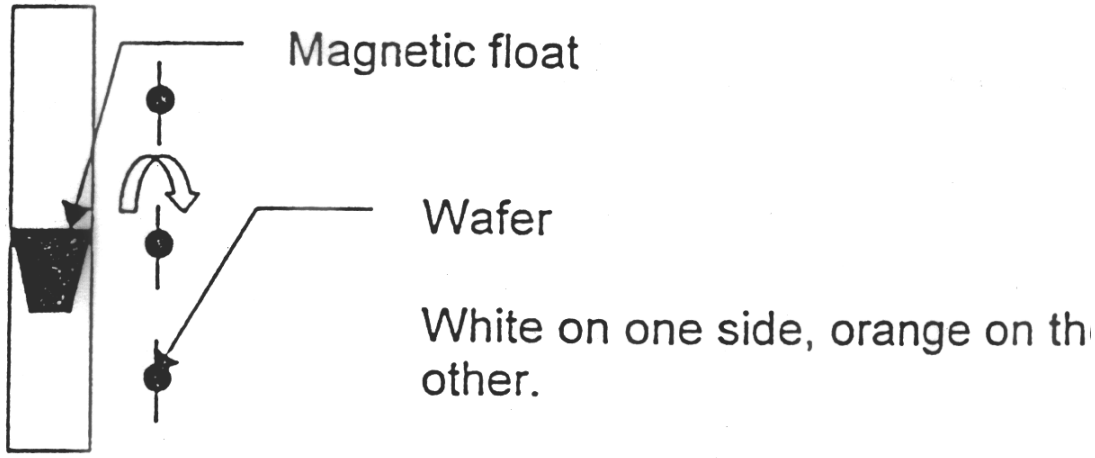


Figure 6. Reflex Sight Glasses

Magnetic Type Sight Gauges

Magnetic type Sight gauges, as illustrated in figure 7, have a float inside a non-magnetic chamber. The float contains a magnet, which rotates wafers over as the surface level increases or decreases. The rotating wafers present the opposite face, which has a different colour. It is more suitable for severe operating conditions where liquids are under high pressure or contaminated.



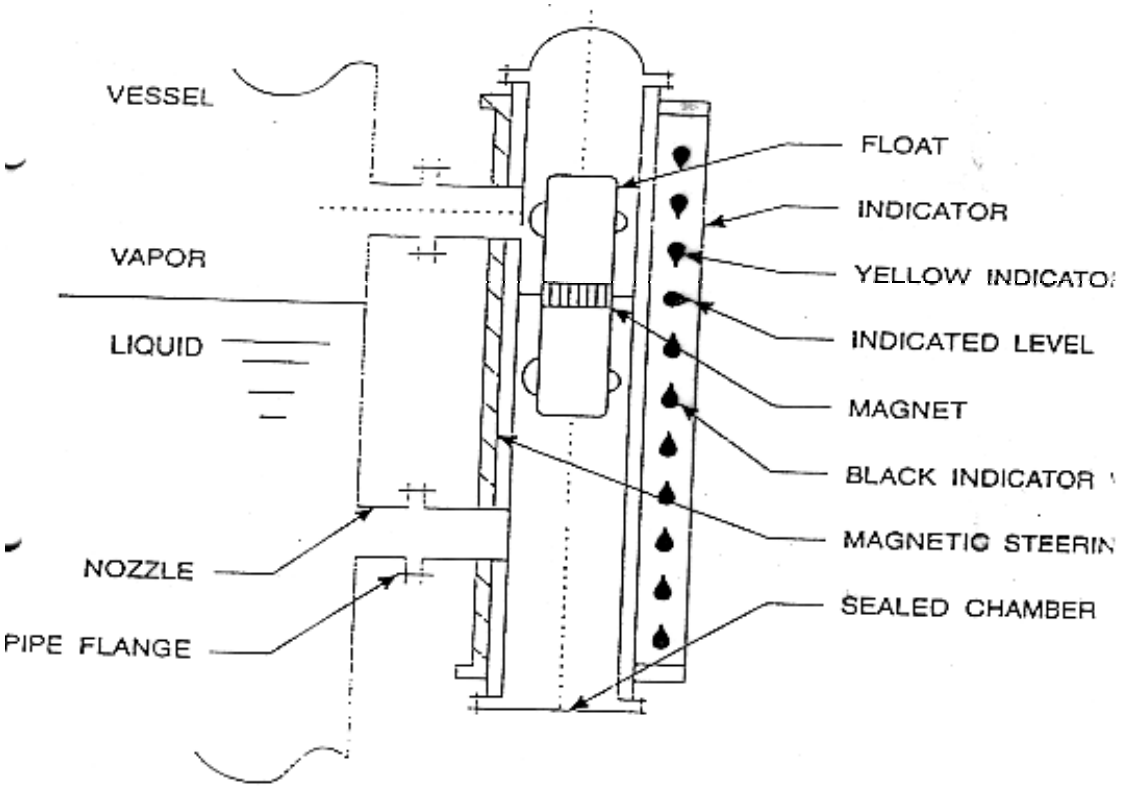


Figure 7. Magnetic type Sight gauges

Sight glasses are usually installed with shutoff valves and a drain valve for the purpose of maintenance, repair and replacement. An important safety feature of these external sight glasses is the inclusion of ball check valves within the isolation valves. The purpose of these check valves is to prevent the escape dangerous fluids if the glass breaks. Therefore it is important that the isolation valves are left fully open when the sight glass is in use, otherwise the operation of the check valves may be inhibited.

Operational considerations for Sight Glasses

The gauge must be accessible and located within visual range. They are not suitable for dark liquids. Dirty liquids will prevent the viewing of the liquid level. Glass has the obvious disadvantage of being fragile and easily damaged or broken. Therefore, this type should not be used for measuring hazardous liquids. On safe applications, tubular gauge glasses can be used.

Reflex gauges are permissible for low and medium pressure applications. For high-pressure applications or where the fluid is toxic, armored gauges with magnetic dials should be used for safety reasons, gauge glass lengths between process connections should not exceed 4 ft.

With this type of device, good lighting is required and sometimes an illuminator may be required in dark areas. In installations where the gauge is at a lower temperature than the process, condensation may occur on the walls, making reading difficult.

Floats

Floats give a direct readout of liquid level when they are connected to an indicating instrument through a mechanical linkage. A simple example of this is the weighted tape tank gauge, illustrated in figure 8. The position of the weighted anchor against a gauge board gives an indication of the liquid level in the tank. The scale of the gauge board is in reverse order, i.e. the zero level indication is at the top and the maximum level indication is at the bottom of the gauge board.

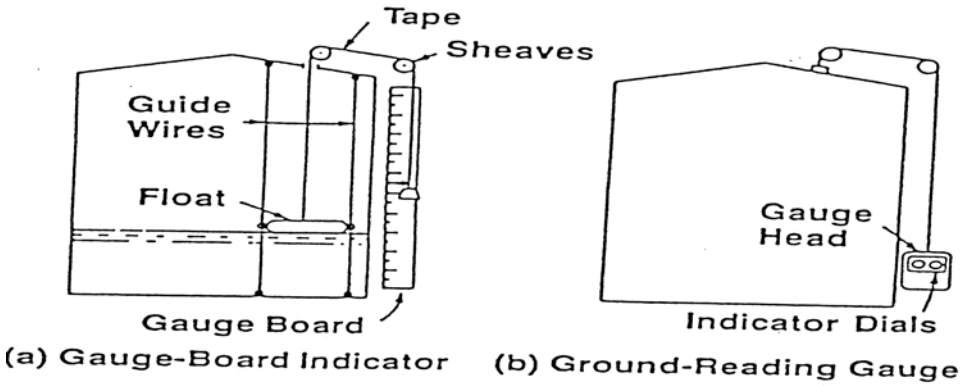


Figure 8. Weighted Tank Gauge using Float Type

Floats can be used in level systems. They give an indication of the actual level, as illustrated in figure 9.

It is used also to drive level switches and level transmitters in different designs.

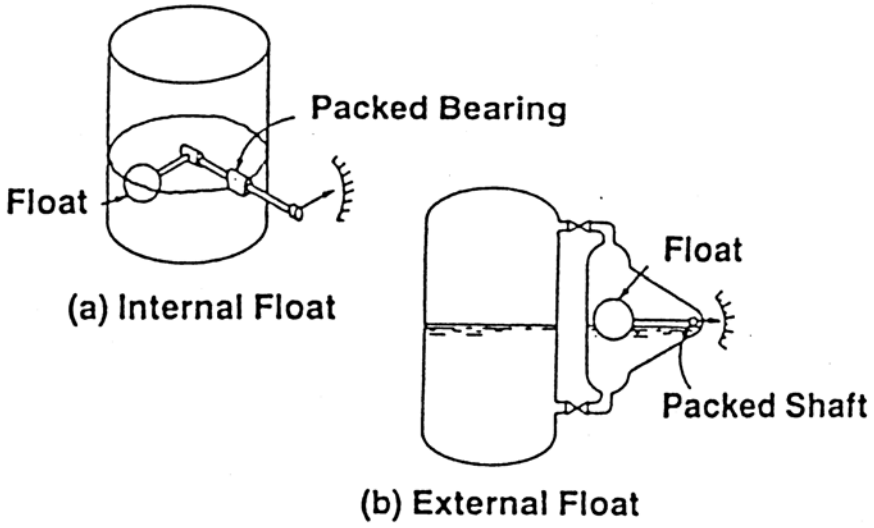


Figure 9. Floats for Level Indication

Indirect (Inferential) Methods

The indirect or inferential method of measurement uses the changing position of the liquid surface to determine level with reference to a datum line. It can be used for low & high levels where the use of the direct method instruments is impractical.

Hydrostatic Pressure Methods

Level measurement involving the principles of hydrostatics has been available for many years. These gages have taken numerous forms, including:

a. The diaphragm-box system
b. Hydrostatic differential-pressure meters
c. The air-bubble tube or purge system

Hydrostatic head may be defined as the weight of liquid existing above a reference or datum line. It can be expressed in various units, such as pounds per square inch (psi), grams per square centimetre, and feet or meters of liquid measured. As shown in Fig. 10, the head is a real force, due to liquid weight and it is exerted equally in all directions. It is independent of the volume of liquid involved or the shape of the

containing vessel. A depth/height of liquid has a particular static pressure or head may be expressed by the relationship:

Where **P = p g h**

P is the static head pressure,

ρ is the density of the liquid,

G is the specific gravity of the liquid,

H is the height of the liquid

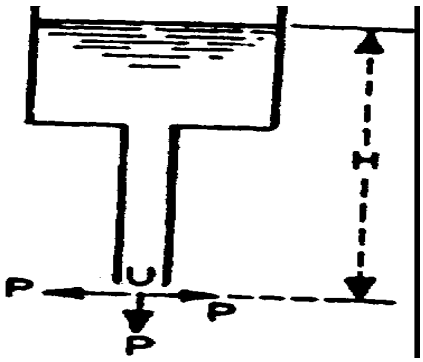


Figure 10. Basic elements of hydrostatic head

From this relationship it is seen that a measurement of pressure P at the datum or reference point in a vessel provides a measure of the height of the liquid above that point, provided the density or specific gravity of the liquid is known. Also, this relationship indicates that changes in the specific gravity of the liquid will affect liquid-level measurements by this method, unless corrections are made for such changes. A compensation for environmental changes, which affect measurement accuracy, may be automated in some systems through the use of microprocessors and sensors that would continuously or intermittently detect the changes in such factors as liquid temperature or density.

When a pressure greater than atmospheric is imposed on the surface of the liquid in a closed vessel, this pressure adds to the pressure due to the hydrostatic head and must be compensated for by a pressure measuring device which records the liquid level in terms of pressure.

$P_{\text{liquid head}} = (P_{\text{total at vessel bottom}} - P_{\text{overhead}})$

The Diaphragm-box System

Pressure sensor / transmitter can make use of this principle for liquid level measurement. As illustrated in figure 11, the transducer can be connected to the bottom the vessel so that it's input is related the hydrostatic pressure within the tank.

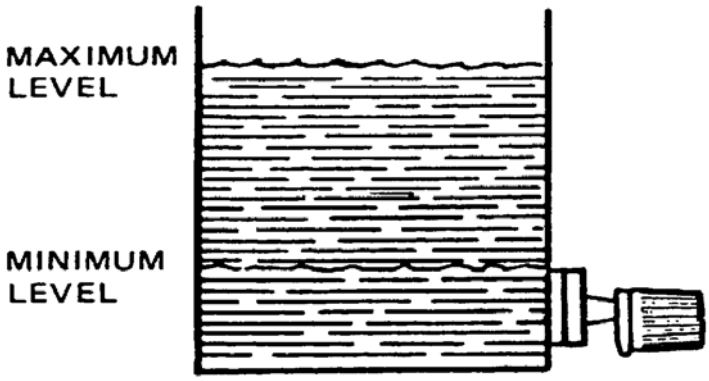


Figure 11. Diaphragm-box Transmitter

As shown in figure 12, A level can be measured using a pressure gauge**.** These systems are employed on open vessels. They operate by giving an indication of the pressure produced by the static head of the liquid that is related to the actual level in the tank. In this case the gauge is calibrated in units relating to the liquid level in the tank. in %.

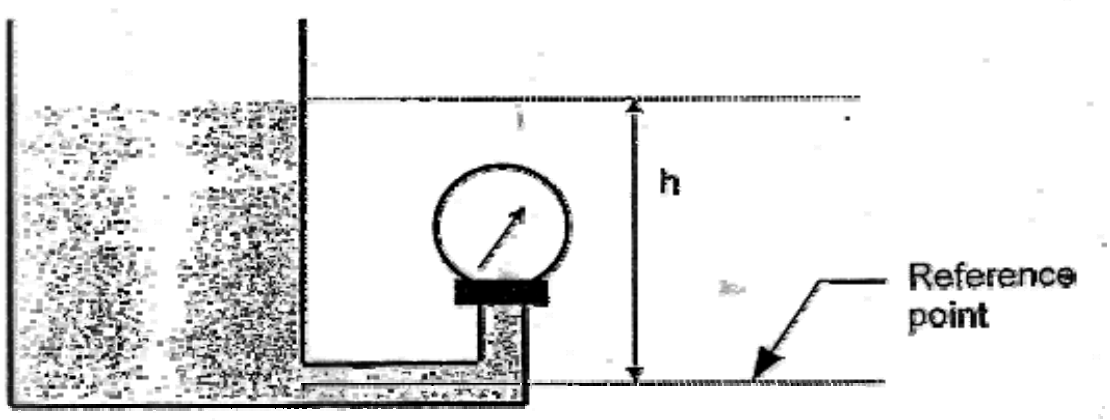


Figure 12. Usage a Pressure Gauge for hydrostatic-head indication

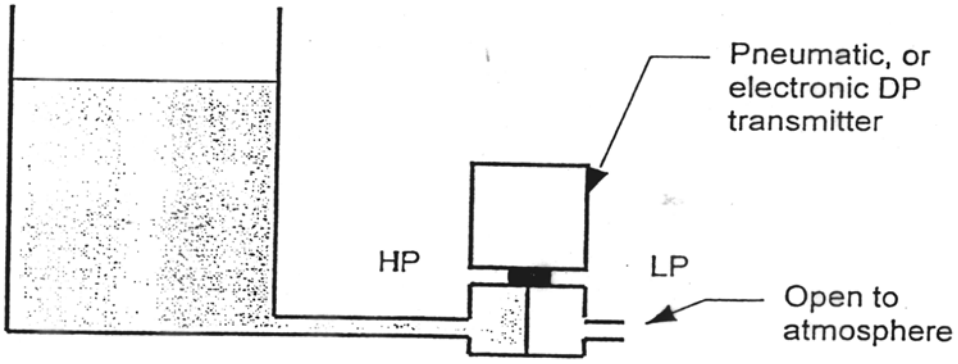
Hydrostatic Differential-Pressure Meters

Open Vessels

Differential pressure measurement is easy to install, has a wide range, and provides a fast response time. With the use of external diaphragm seals and flange connections, these instruments can be used to measure fluids such as slurries and hot or corrosive liquids.

This system is based on the same principle as the hydrostatic pressure gauge method, but uses a DP transmitter to provide a signal to a remote indicator or controller.

As shown in figure 13, the hydrostatic pressure exerts its force against the diaphragm on the HP side (high pressure). Any differential pressure detected between the HP and LP side is converted to a signal that is directly proportional to the level in the tank.



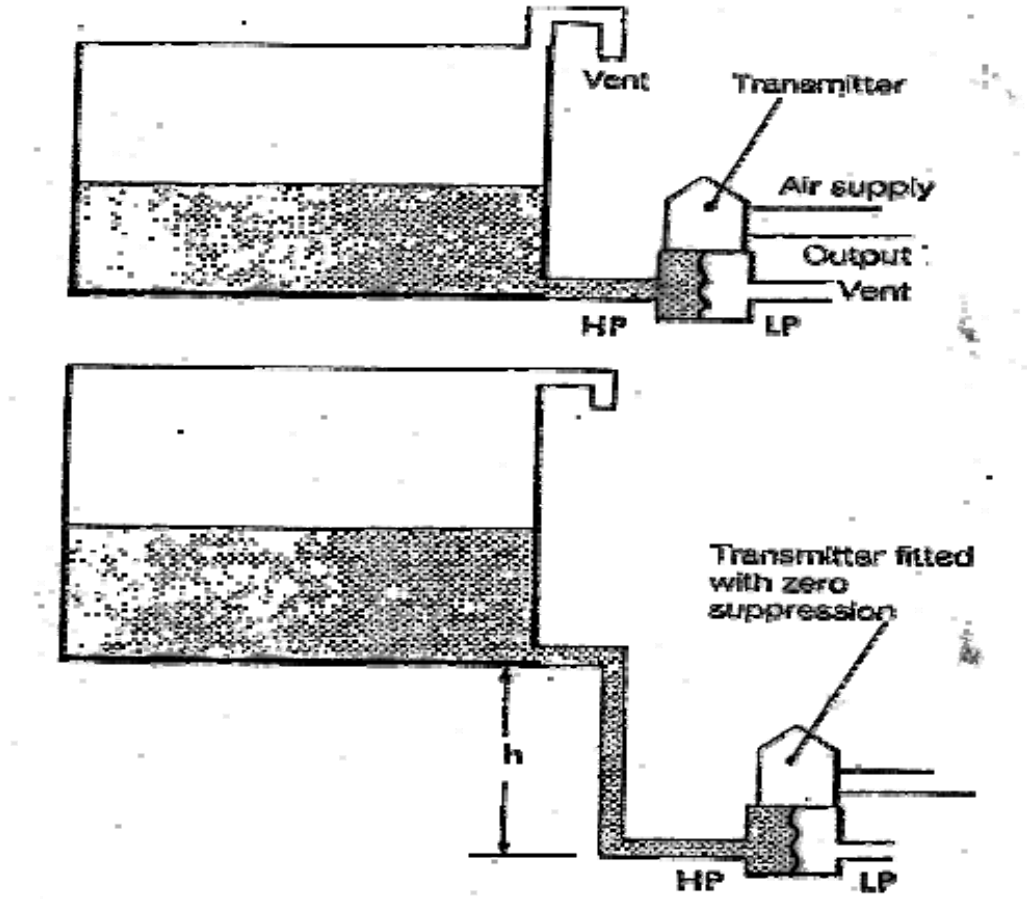


Figure 13. DP Transmitter installed for an Open Tank

Zero Suppression and Elevation

If the DP cell is mounted above or below the actual bottom of the vessel or in a closed vessel, then a **zero elevation** and **zero suppression** adjustments of the transmitter range will become necessary, as illustrated in figure 14.

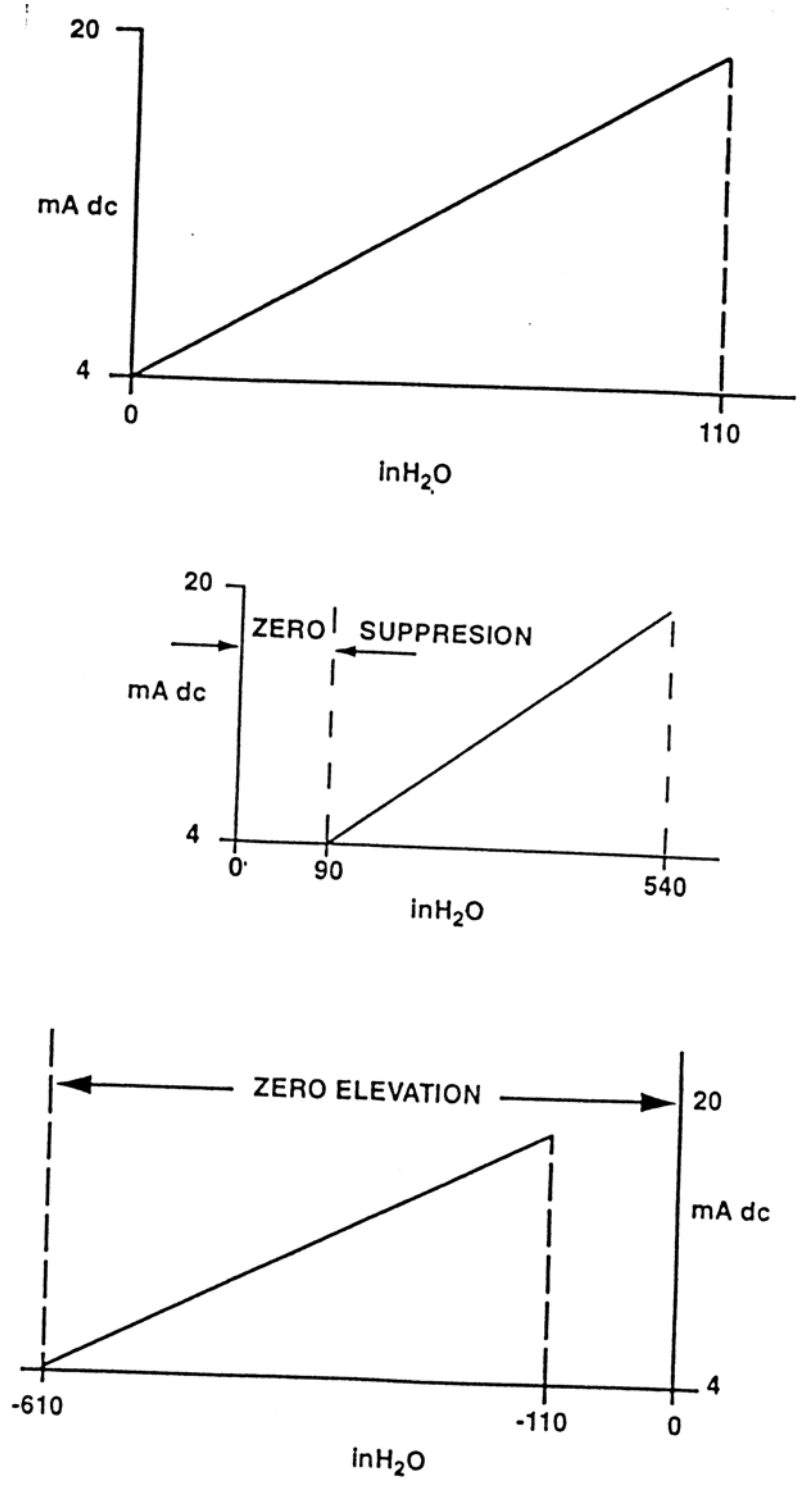


Figure 14. Zero Suppression & Zero Elevation

Closed Vessels

In closed vessels the pressure above the liquid will affect the pressure measured at the bottom. The pressure at the bottom of the vessel is equal to the height of the liquid multiplied by the specific gravity of the liquid plus the vessel pressure. To measure the true level, the vessel pressure must be subtracted from the measurement. This is achieved by making a pressure tap at the vessel and connecting this to the LP side of a DP transmitter. Vessel pressure is now equally applied to both sides of the transmitter resulting in the differential pressure proportional to the liquid height multiplied by the specific gravity.

However, these instruments are affected by changes in the process density and should only be used for liquids with fixed specific gravity or where errors due to varying specific gravity are acceptable.

Differential pressure devices require a constant head to be maintained on the external or reference leg. Two methods commonly available are:

- Dry leg
- Wet leg

Dry Leg

If the gas above the liquid does not condense the impulse piping to the low side of the transmitter will remain empty, see figure 15. If the DP transmitter is installed below the bottom of the tank then zero suppression must be made to offset the constant static head that present, otherwise there will be an an incorrect level reading. See figure 16.

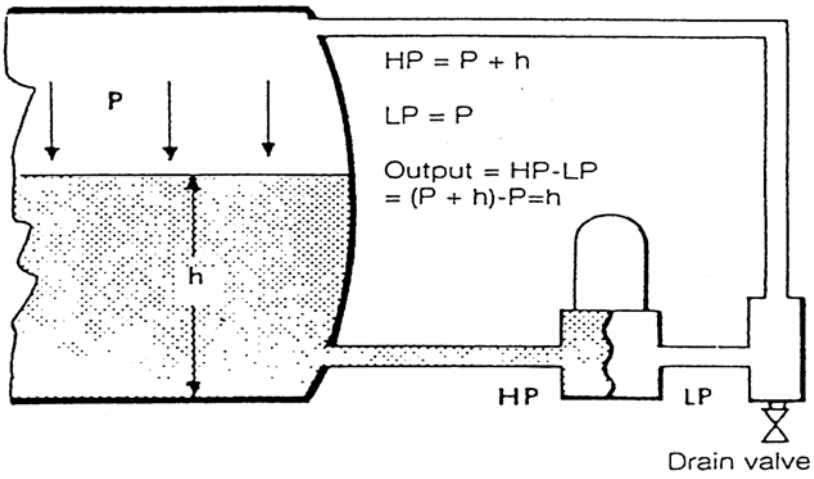


Figure 15. Dry leg, DP cell installed at the datum line

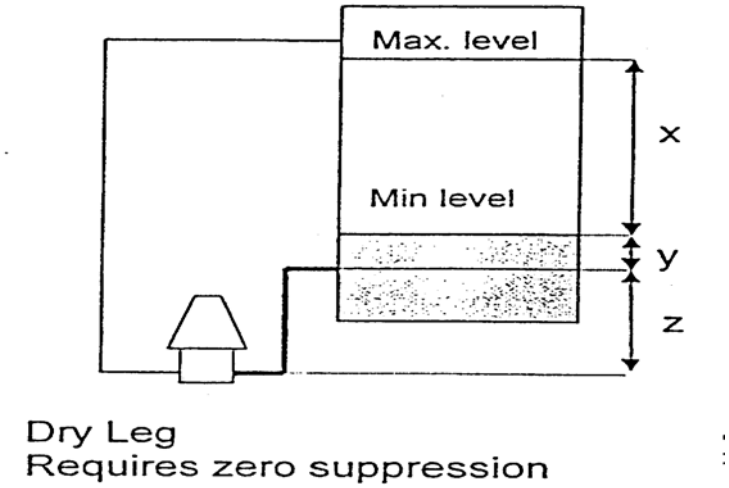


Figure 16. Dry leg, DP cell installed below the datum line

Calibration Formulas

Either open tank or closed-tank with dry-leg:

Calibrated range of the transmitter can be calculated as follow:

Span = (x) (Gl)

LRV at minimum level = (z) (Gs) + (y) (Gl)

URV at maximum level = (z) (Gs) + (x + y) (Gl)

Where:

Gl = Specific gravity of tank liquid

Gs = Specific gravity of seal liquid

x, y and z are shown In figure 16.

Example

An open tank system has a span of 80" (X) The HP connection side of a differential pressure transmitter has its datum 5" (y) below the minimum tank level. The DP cell itself is located 10" (z) below the datum level. The specific gravity of the tank liquid is 0.8 the specific gravity of the liquid in the connecting leg is 0.9. Calculate the required range of the transmitter.

Solution

Span= (80)(0.8) = 64 inches

LRV = (10)(0.9) + (5)(0.8) = 13 inches

URV = (10)(0.9) + (5 + 80)(0.8) = 77 inches

Calibrated range = 13 to 77 Inches head of water

Solve the following problem

An open tank system has span of 500" (x) the HP connection side of a differential pressure transmitter has its datum 100" (y) below the minimum tank level. The DP cell itself located at the datum level. The specific gravity of the tank liquid is 0.9; the specific gravity of the liquid in the connecting leg is o.9 Calculate the required range of the transmitter.

Solution

SPAN =

Suppression of HP head =

Range of transmitter required =

Wet Leg

If the gas above the liquid condenses in the piping the low side of the transmitter it will slowly fill up with liquid resulting in an incorrect level reading. To eliminate this potential error, the pipe is

purposely filled with a convenient reference fluid that possesses a higher specific gravity than, and is immiscible with, the liquid in the vessel A common liquid used for this function is common anti-freeze. In figure 17

Output = [P (Sg_{vapour}) + h (Sg_{liquid})] – [P (Sg_{vapour}) + Z (Sg_{leg})]

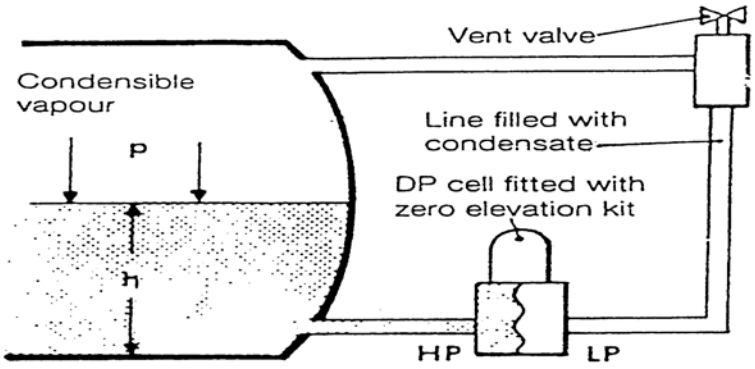


Figure 17. Wet leg, DP cell is installed at the datum line.

The reference fill fluid in the wet leg, will exert a head pressure on the low side of the transmitter requiring zero elevation. See figure 18.

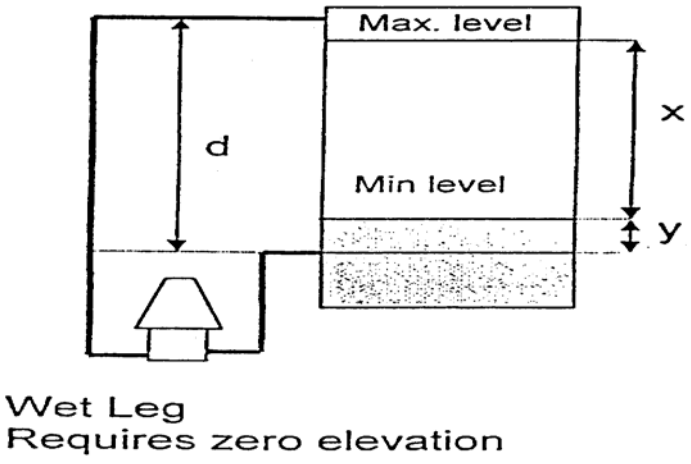


Figure 18. Wet leg, DP cell installed below the datum line.

Calibration Formulas

Closed tank with wet-leg

Calibrated range of the transmitter can be calculated as follow:

Span = (x)(Gl)

LRV at minimum level = (y)(Gl) - (d)(Gs)

URV at maximum level = (x + y)(Gl) - (d)(Gs)

Where:

Gl = Specific gravity of tank liquid

Gs = Specific gravity of seal liquid

x, y and d are shown in figure 18.

Example

A closed tank system has a span 70″ (x). The connection side of a differential pressure transmitter has its datum 20″ (y) below the minimum tank level. The distance between the HP and LP tapping points is 100″ (d) The specific gravity of the tank liquid is o.8; the specific gravity of the liquid the connecting leg is o.9 Calculate the required range of the transmitter.

Solution:

Span = (70)(0.8) = 56 inches

LRV = (20)(0.8) – (100)(0.9) = -74 inches

URV = (70 + 20)(0.8) - (100)(0.9) = -18 Inches

Calibrated range = -74 to -18 inches head of water

(Minus signs Indicate that the higher pressure is applied to the low-pressure side of the transmitter)

Boiler Discharge Vessel level Meter Installation

Figure 19 shows the DP cell installed to measure steam drum level.

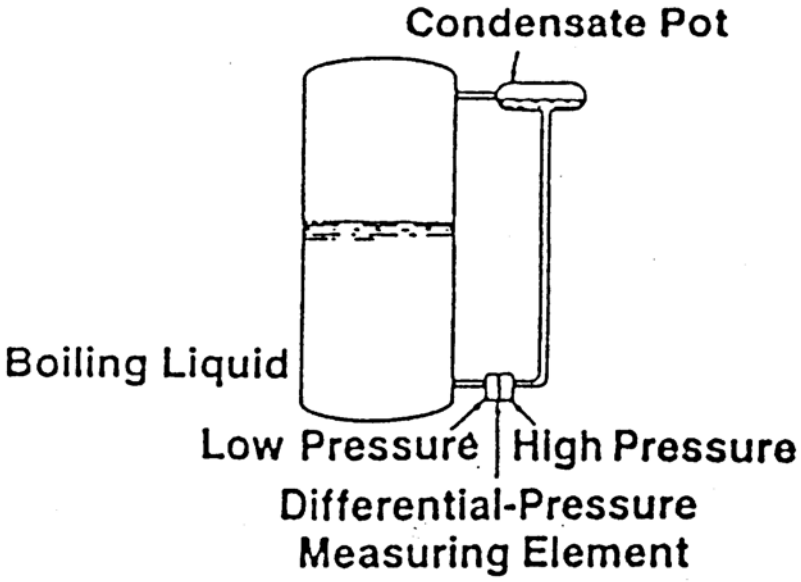


Figure 19

Bubble Tube (Purge) Systems

The bubble tube system continuously bubbles air or an inert purge gas through a tube that extends to nearly bottom of the vessel at low flow rate. As showing in figure 20, the back-pressure in the bubble tube will be a function of the hydrostatic pressure or head of the liquid in the vessel. The lowest point of the purge tube determines the zero point reading; therefore any liquid below it cannot be detected.

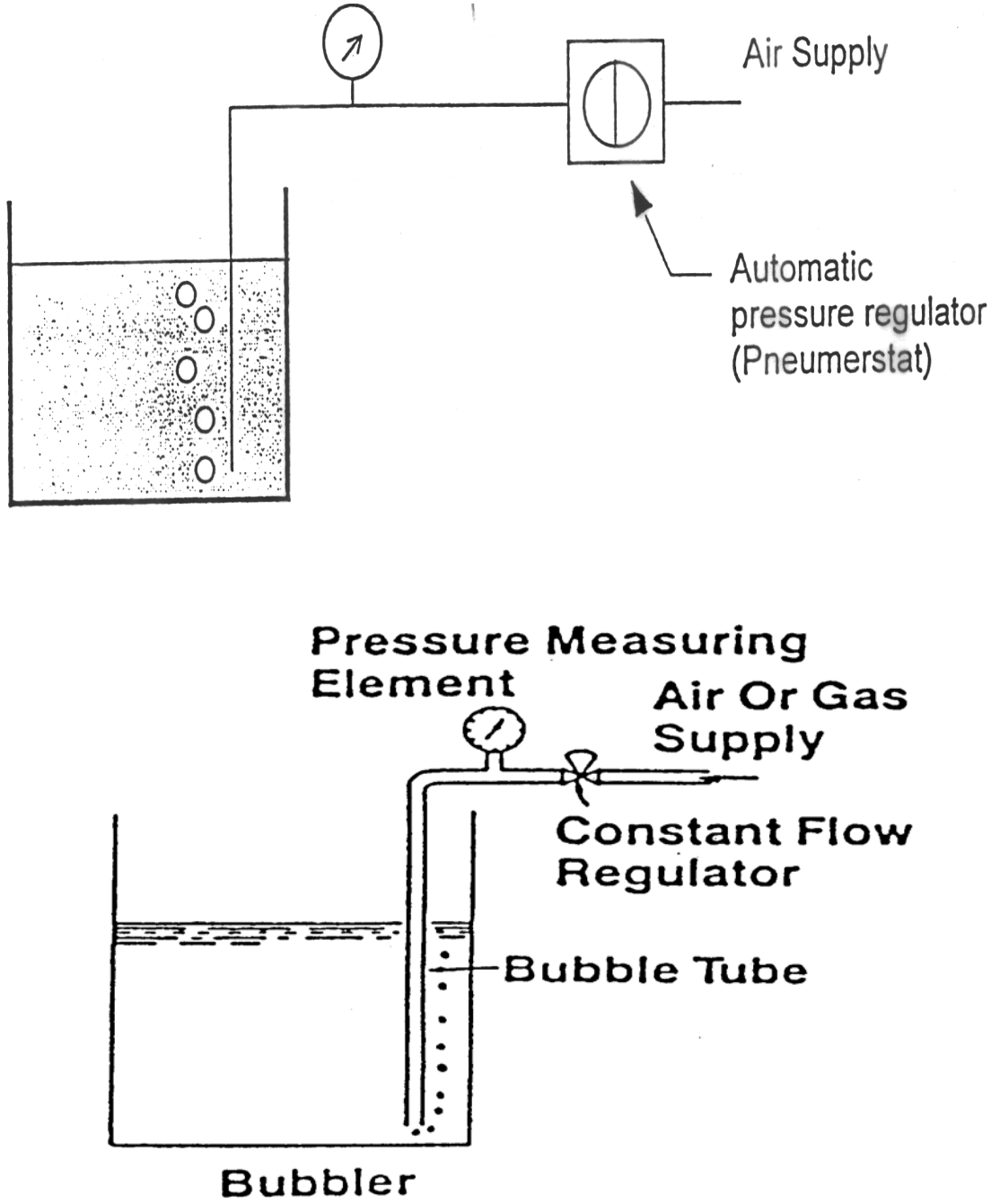


Figure 20. Bubble-tube (Purge) System

The bottom of the purge tube is notched to keep:

- The bubble size small
- Allow the bubbles to escape easily from the tube.
- Take care to minimise the back-pressure pulses.

A Clearance gap has to be maintained between the bottom of the vessel and the tip of the purge pipe so that sediment does not block the tube. A blocked tube will result in a false reading (reads maximum level).

This type of system is susceptible to freezing or blocking/plugging by process fluid. Care has to be taken to ensure that the purge gas does not cause a chemical reaction with the liquid in the vessel. Air **must not** be used where it is likely to cause a highly combustible mixture.

For this system to operate correctly there must be a constant airflow through the purge tube. Regulated pressure should be slightly higher than the maximum head pressure of liquid in the tank. The air pressure in the system will be equal to the hydrostatic head of the tank liquid at any point because any excess pressure will bubble out of the bottom of the tube. If the purge pressure is regulated at a value lower than this, then eventually a point will be reached where the bubbles will not escape from the tube leading to an incorrect measurement of the liquid level. Devices such as pneumerstats and constant differential pressure relays can be used to carry out the pressure adjustment automatically so that this problem will not occur.
Density variations of the liquid being measured will affect the reading.
Purge systems are particularly suited to measuring the level of:

- Corrosive liquids (brines)
- Viscous liquids
- Liquids containing entrained solids (slurry)

Displacement Devices

The displacement level transmitter is commonly used for continuous level measurement. It works on the buoyancy principle. As illustrated in figure 21, the displacer has a cylindrical shape therefore each increment of submersion in the

liquid; an equal increment of buoyancy change will result. This is a linear and proportional relationship.

When the weight of an object is heavier than an equal volume of the fluid into which it is submerged, full immersion results and the object never floats. Although the object (displacer) never floats on the liquid surface, it does assume a relative position in the liquid. As the liquid level moves up and down along the length of the displacer, the displacer undergoes a change in its weight caused by the buoyancy of the liquid. Buoyancy is explained by Archimedes' principle, which states that: "**the resultant pressure of a fluid on a body immersed in it acts vertically upward through the centre of gravity of the displaced fluid and is equal to the weight of the fluid displaced".**

The upward pressure acting on the area of the displacer creates the force called buoyancy. The buoyancy is of sufficient magnitude to cause the float (displacer) to be supported on the surface of a liquid or a float in float-actuated devices. But, in displacement level systems, the immersed body or displacer is supported by arms or springs that allow some small amount of vertical movement or displacement of the displacer due to buoyancy forces caused by the change in the liquid level. This buoyancy force can be measured to reflect the level variations.

When a body is fully or partially immersed in any liquid, it is reduced in weight by an amount equal to the weight of the volume of liquid displaced. A displacer arrangement is shown in Figure 21. The vessels shown are open to atmosphere, but the principle described applies to closed-tank measurement also.

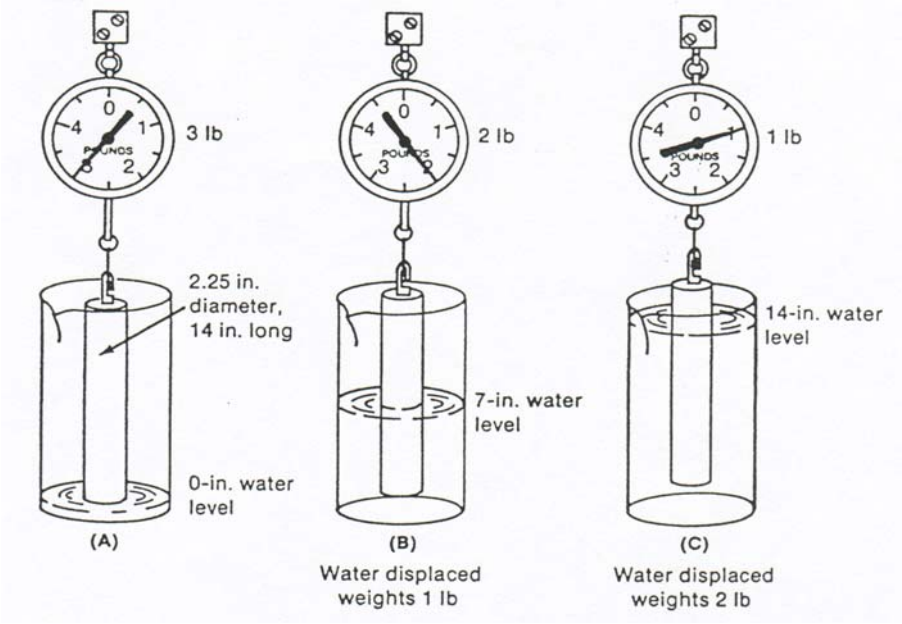


Figure 21 Displacement Level Measurement

In the vessel **A**, the displacer is suspended by a spring scale that shows the weight of the displacer in air. This would represent zero percent level in a measurement application. The full weight of the displacer is entirely supported by the spring and is shown to be 3 pounds.

In the vessel **B**, the water is at a level, in this case, represents 50 percent of the full measurement span. Note that the scale indicates a weight of 2 pounds. The loss in weight of the displacer (1 pound) is equal to the weight of the volume of water displaced.

In the vessel **C**, when the water level is increased by another 7 inches to a full-scale value of 14 inches, the net weight of the displacer is 1 pound. That represents a change of 2 pounds when the water level rises along the longitudinal axis of the displacer 14 inches. That is, when the water level changes from 0 to 100 percent (0 to 14 inches), the weight of the displacer changes from 3 pounds to 1 pound. As the weight of the displacer decreases, the net load on the spring scale decreases by an amount directly proportional to the increase in water level.

For the displacer in question, a 14-inch increase in level is equal to about 55 cubic inches of water displaced. This is the volume of the immersed portion of the displacer, which is determined by multiplying the cross-sectional area by the submerged length of the displacer.

Determining Suspended Weight for Dry Calibration:

To determine the total weight that must be suspended from the displacer rod to simulate a certain condition of fluid level or specific gravity, the following equation can be used:

$$Ws = Wd - [(0.0361)(V)(Sg)]$$

Ws = Total suspended weight in pounds (apparent weight).

Wd = Weight of displacer, dry, in pounds (determine by weighing displacer).

0.0361= Weight of one cubic inch of water, in pounds (specific gravity =1.0)

V = Volume of the displacer that would be submersed at the level required by the calibration Procedure (in cubic inches) **OR**,

$V = \Pi/4$ (displacer diameter)^2 * (length of displacer submerged)

Sg = Specific gravity of the process fluid at operating temperature.

For interface level measurement, the equation becomes

$Ws = Wd - [(0.0361)(V_L)(Sg_L) + (0.0361)(V_H)(Sg_H)$

Sg_L = Specific Gravity of the lighter fluid at operating temperature.

Sg_H= Specific gravity of the heavier fluid at operating temperature.

V_L = Volume of the displacer submersed by the lighter fluid, in cubic inches. OR

$V_L = \Pi/4$(displacer diameter) ^{2} * (length of the displacer submerged).

and

V_H = Volume of the displacer submerged by the heavier fluid, in cubic inches.

OR

$V_H = \Pi/4$ (displacer diameter)^2 * length of the displacer submerged).

Torque Tube

In this method a displacer body is connected to a torque tube which twists a specified amount for each increment of buoyancy change.

As shown in figure 22, the twisting force can drive a pointer, an indicator or be transferred to a pneumatic or electronic system. It is transferring the displacer movement from the inside of pressurised vessel to the readout mechanism, which is in atmospheric pressure. The torque tube rotates around 4° to 6° degrees angular to establish the 0-100% level readout.

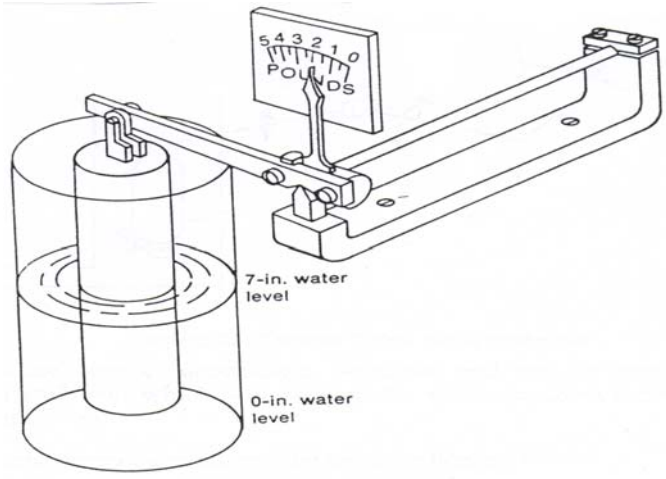


Figure 22. Torque Tube level principle (torsion spring)

Advantages

- These devices are simple and reliable.
- Instrument is accurate
- It can be mounted internally (inside a vessel) or externally in a still-well that will prevent the displacer from moving due to the process surface turbulence or agitation.

Applications

This type of measurement should only be used for liquids:

- With fixed specific gravity
- Where errors due to process variations are acceptable
- Where a change in process conditions will not create crystallisation or solids.

If the displacer is mounted in an external still-well then the block and drain valves should be installed for maintenance purpose. Trace heating, or insulation may by needed to maintain the temperature of the liquid in the well. Torque tube level system illustrated in Figure 23.

The piping arrangement should be designed to prevent the formation of sediment on the bottom of the float cage as eventually this can build up and affect the displacer movement. Coating build-up or dirt that clings to the displacer may affect the elements' buoyancy resulting in the accuracy of the measurement.

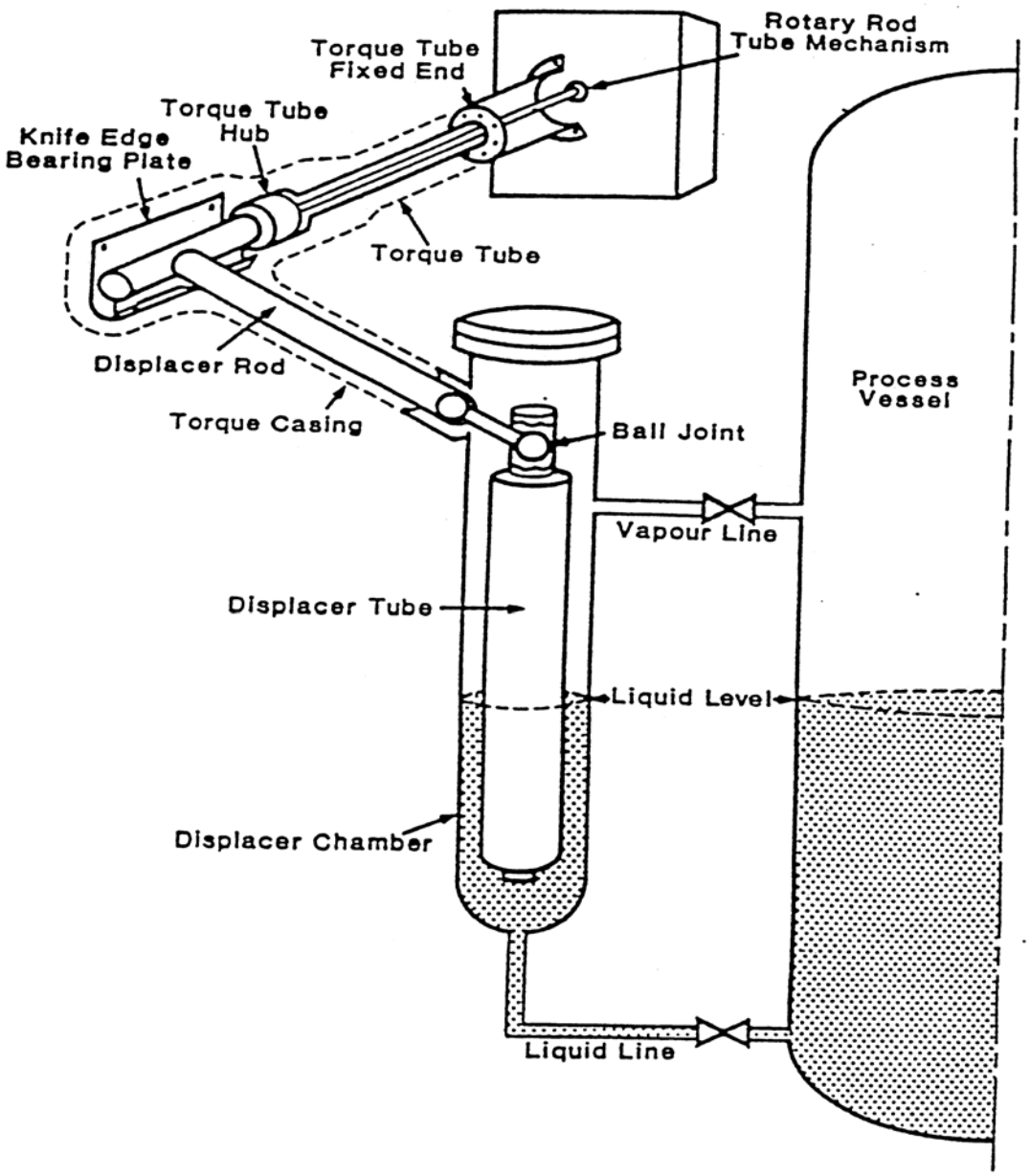


Figure 23. Torque-Tube Level System

Capacitance Probes

Capacitance probes are used when instruments that use specific gravity for sensing are not reliable and the difference in dielectric constant between the fluids is significant. Figure 24 showing the capacitance probes principle. Also figure 25 illustrates exploded view of level measurement by capacitance probe.

Heavy oil / water interfaces and emulsion are two of the most common examples. Electronic interfaces are available to use with the capacitance probes for detecting the interface (switching) by using horizontal mounting. Capacitance probes are installed on vertical mounting for continuous level measurement.

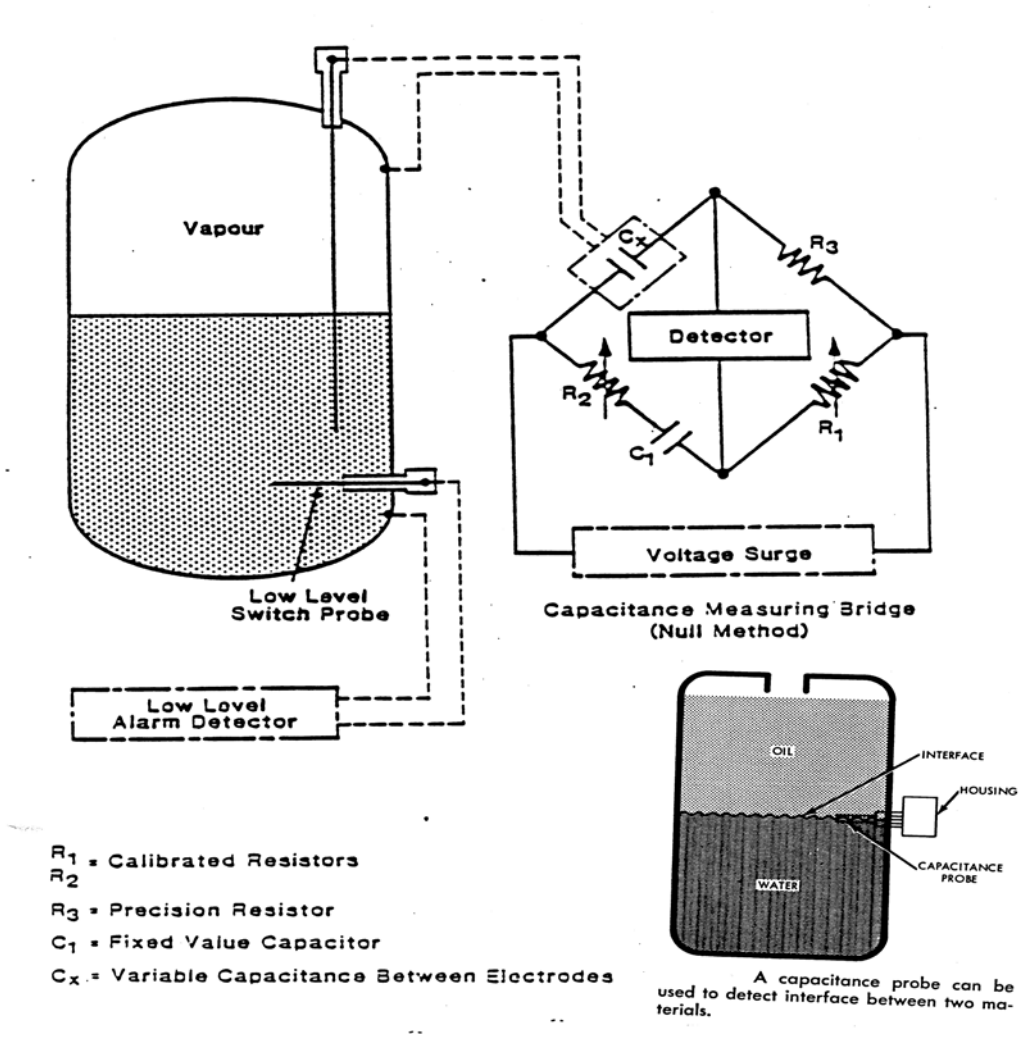


Figure 24. Capacitance Probes Principle of Operation

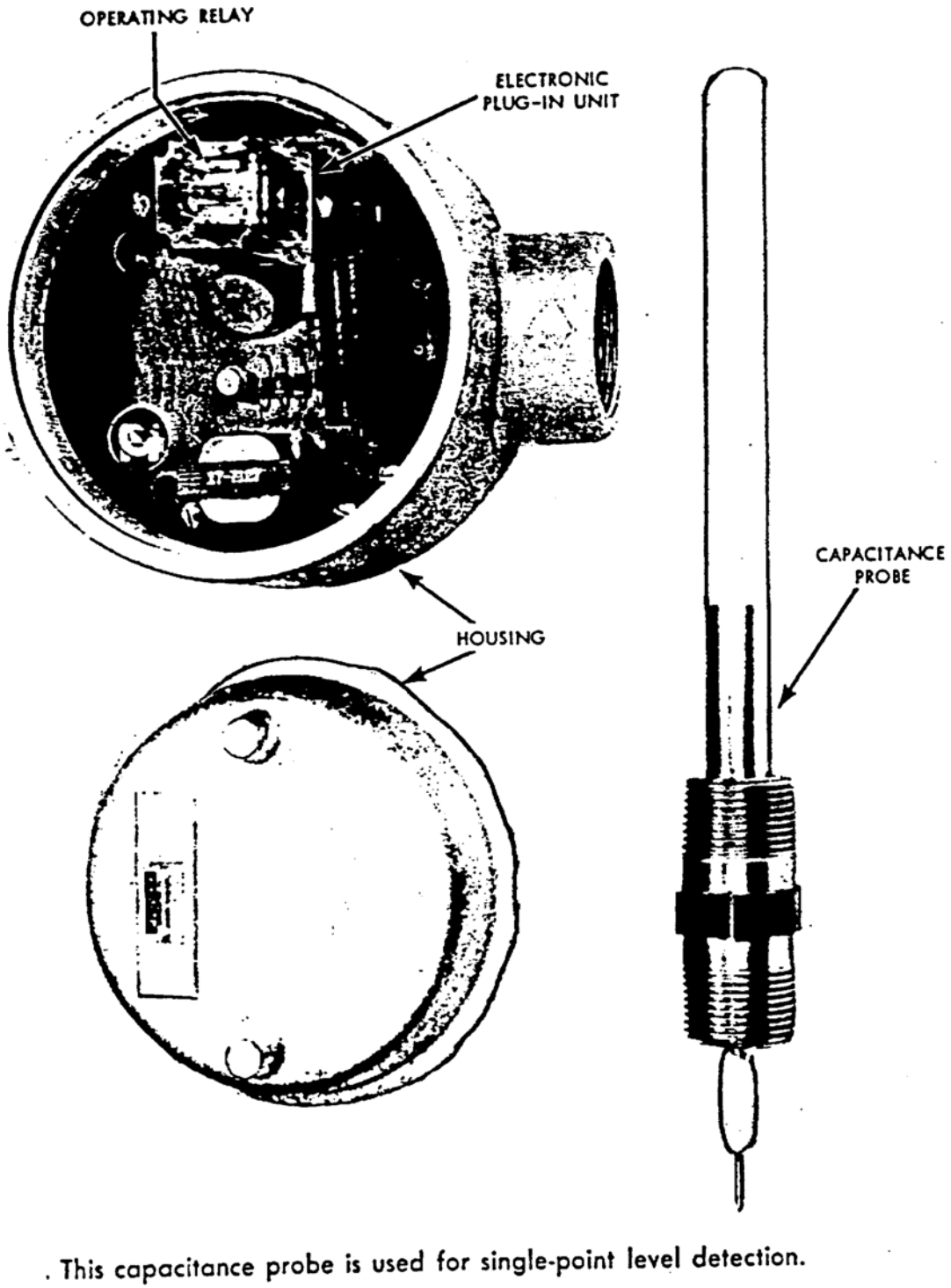


Figure 25. Capacitance Probes, Single-point level

Conductivity Level Sensors

Conductivity Level Sensors are typically used the detection of high or low Level, and are only suitable for use in conductive liquids such as water. When the fluid covers the probes, the measured resistance is low. Conversely, the resistance increases as the level drops below the probe.

Figure 26 showing the working principal of the conductivity level sensors.

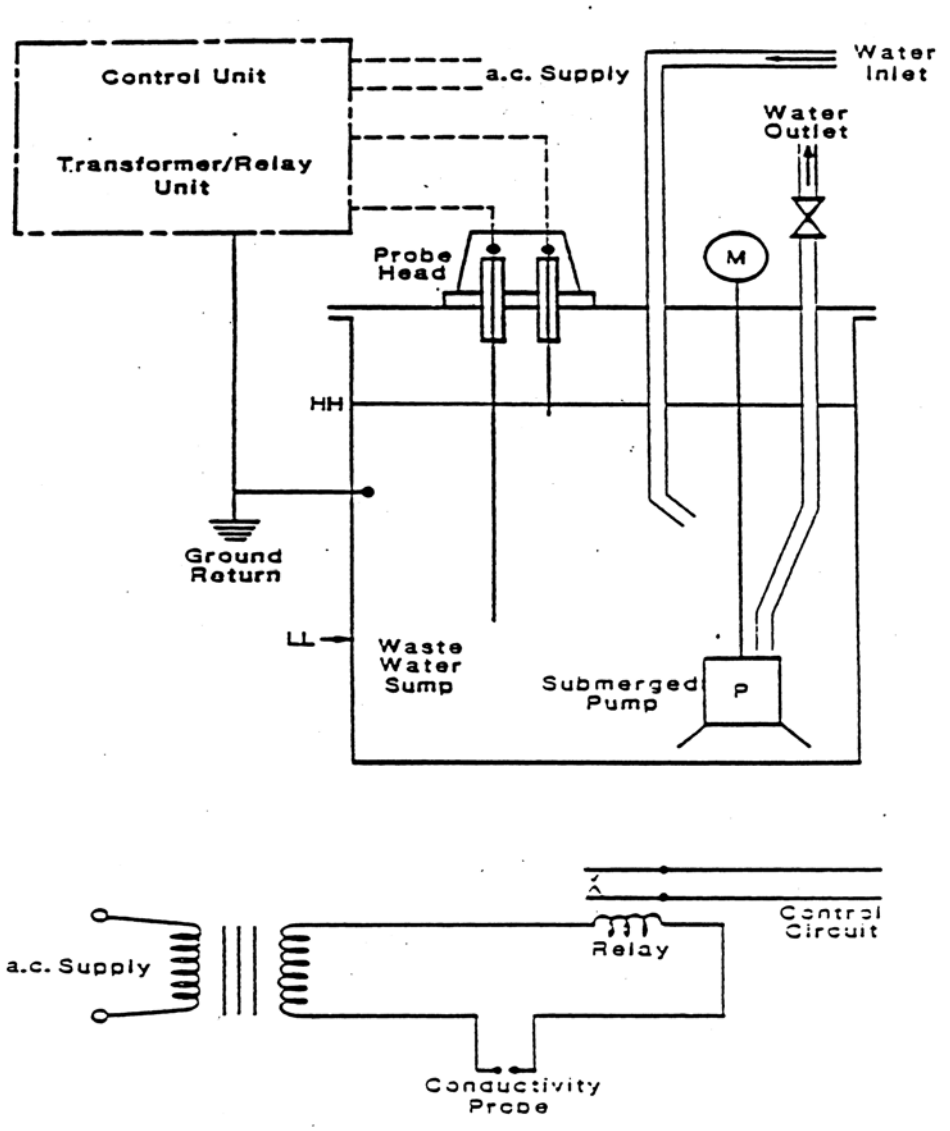


Figure 26. Conductivity level sensors working principle

Automatic Tank Gauge (ATG) Level System

These instruments are useful to measure the liquid level when the fluid stored at atmospheric pressure or slightly higher. A servo keeps constant tension on a tape attached to a float. The float follows guide wires so that tape is always vertical and the float stays at the surface of the liquid.

Figure 27 shows the parts of the ATG level gauge, which helps to understand the principle of operation. The ATG is basically a liquid level indicator but with some accessories added to the basic unit. It can work as an indicator plus level switch and/or level transmitter.

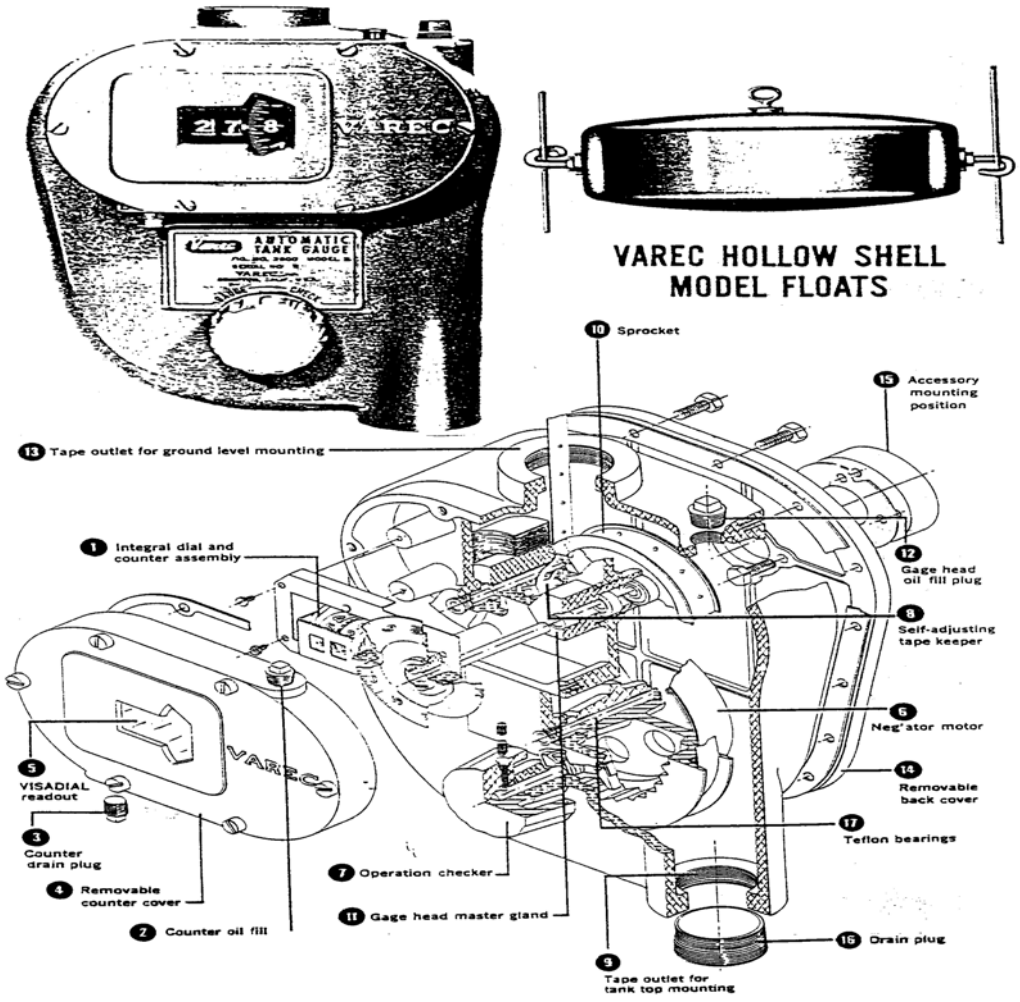


Figure 27. ATG Major Parts

Sonic and Ultrasonic Level Sensors

In applications when it is not acceptable for the level measuring instrument to come into contact with the process material, a sonic or ultrasonic device can by used These devices measure the distance from a reference point in the vessel to the level interface, using sonic or ultrasonic waves.

Sonic Sensors

As illustrated in figure 28, in sonic sensors, the unit uses the echo principle with a frequency in the audible range. After each pulse, the sensor detects the reflected echo. Note this will only work if the surface of the liquid is a good reflector and that the centre line of the transmitted beam is vertical.

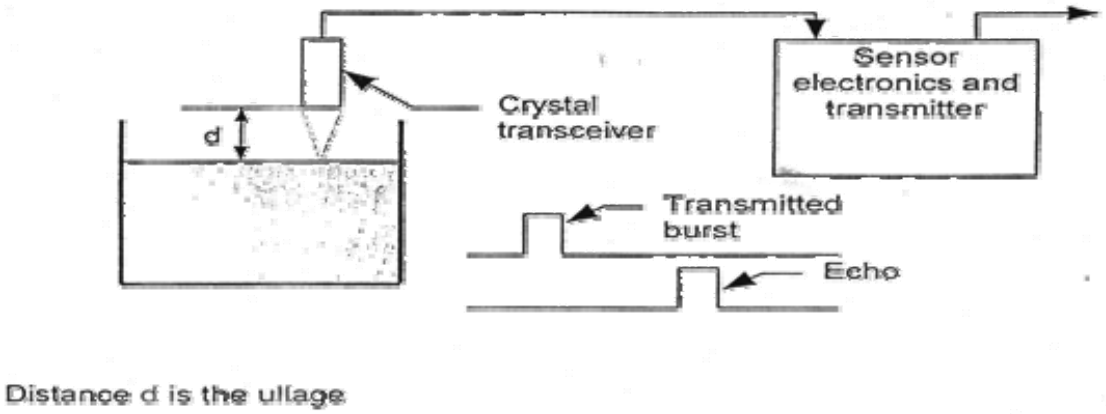
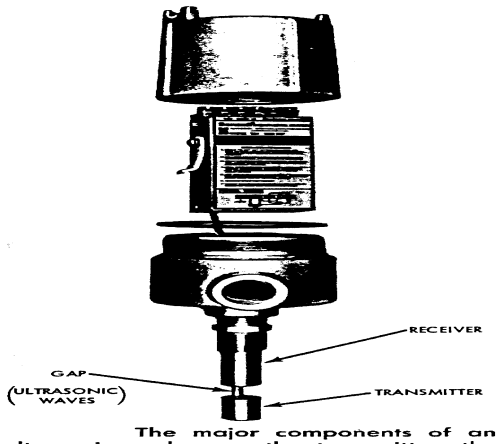


Figure 28. Ultrasonic level sensors principle



The major components of an ultrasonic probe are the transmitter, the receiver, and the gap.

Figure 29. Ultrasonic level transmitter

Ultrasonic Sensors

Figure 29 an exploded view of ultrasonic transmitter. A continuous measurement is made by measuring the elapsed time between the emission and the reception of the signal from a surface in a vessel for ullage measurement and between the surface of the liquid to the tank bottom for innage measurement. Selection of the method may be depending on type of liquid. Ultrasonic systems operate on the same principle only at a much higher frequency.

Figure 30 showing the ultrasonic level detection system.

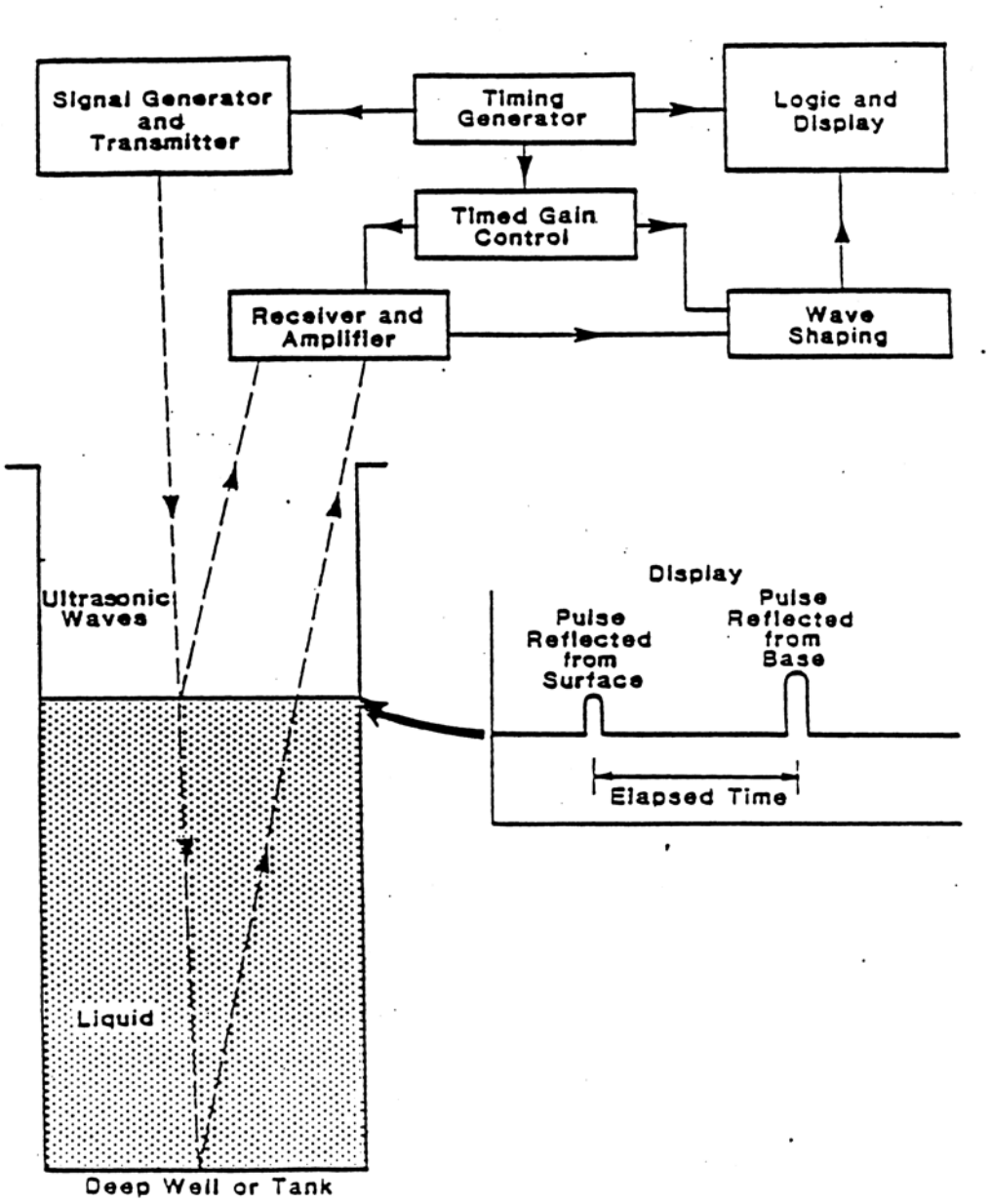


Figure 30. Ultrasonic Level Detection System

Problems can arise when a tank is emptied, as detecting the bottom of the tank will cause errors. Most ultrasonic equipment provides a loss of echo option.

In closed vessels with flat tops, it may be necessary to reduce the transmit repetition rate so that the echoes have enough time to die away before the next pulse is transmitted. Alternatively, introduce a blanking distance into the transmitter so that

it will not be able to see pulses from less than a certain distance One other option is the use of sound absorbing material installed on the underside of the tank top.

Advantage

- These devices are non-contacting, reliable and accurate.
- They can penetrate high humidity and may be used on dusty applications.
- They have no moving parts and are unaffected by changes in density, conductivity or composition.

Disadvantage

- Strong industrial noise or vibration at the operating frequency will affect the performance and tend to give false signals.
- A build-up of material on the probe will attenuate the signal; therefore the unit should not come into contact with the process fluid. Sonic ultrasonic devices cannot be used on foams because the foam absorbs the signal. The performance of these devices is dependent on the speed of sound in the vessel and obviously they cannot work in a vacuum. Various factors such as:
    - Vapour concentration
    - Process temperature
    - Relative humidity
    - The presence of another gas
- Can affect the speed of sound within a vessel resulting in the inaccuracy of the instrument. These errors can be minimised means of temperature compensation in the sensor head, which automatically varies the speed of sound used in its calculation.

Level Transmitters

A level transmitter is an instrument that converts the output of a Level sensor into either an analogue signal or a digital signal that can be transmitted to a remote location.

Generally, output of a level transmitter is analogue signal. Pneumatic level transmitter output is 3 -15 psig or 6-30 psig. Electronic level transmitter output is 4-20 mA. Pneumatic or electronic transmitter selection is made to the compatibility of indication, alarm and controlling system.

The signal from a transmitter may be transmitted to switches, controllers, or PLC. The transmitter output may be used for several different functions like controlling, alarm, shutdown, etc. The difference between a controller and a transmitter can be just a matter of semantics.

There are level transmitters with the dual head with a controller. In such instruments, the transmitter portion would be transmitting the level measurement to a remote location. The controller portion would be controlling the liquid level in the vessel through a final control element.

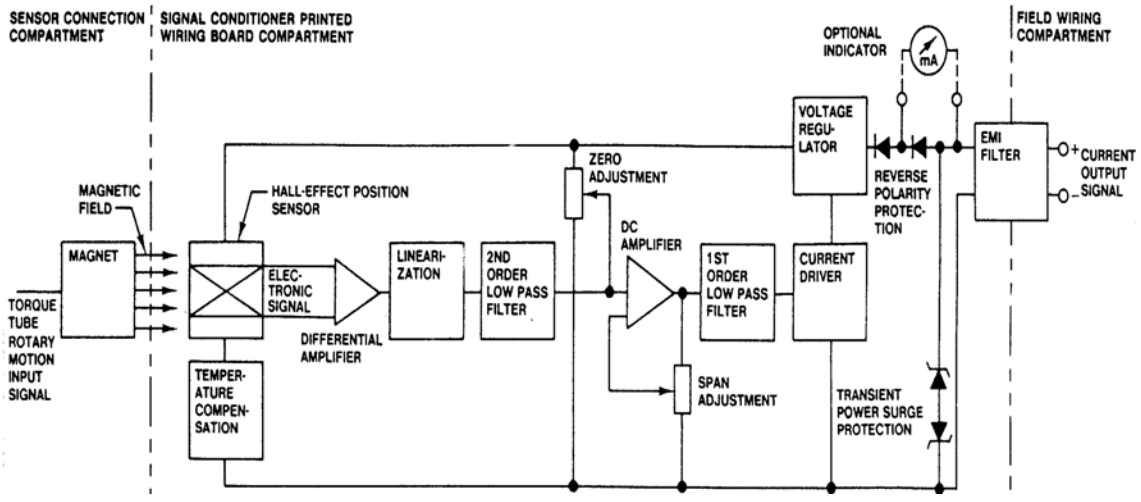
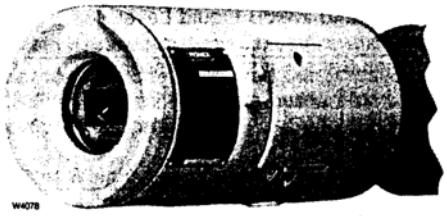
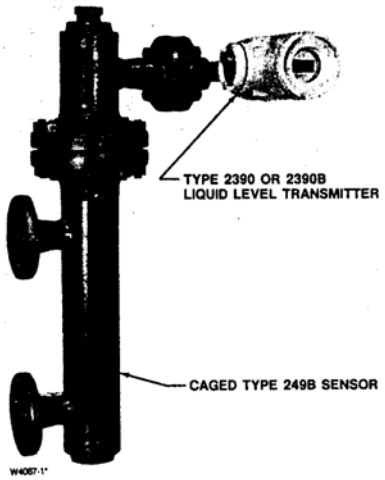


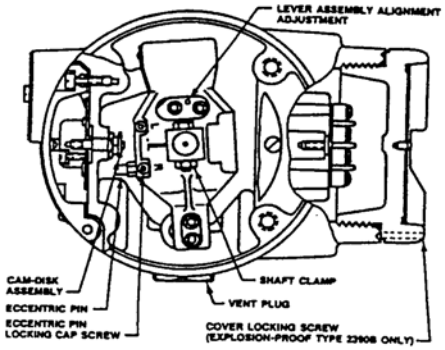
Figure 31. Block functional diagram of an electronic level transmitter 'Fisher Type".



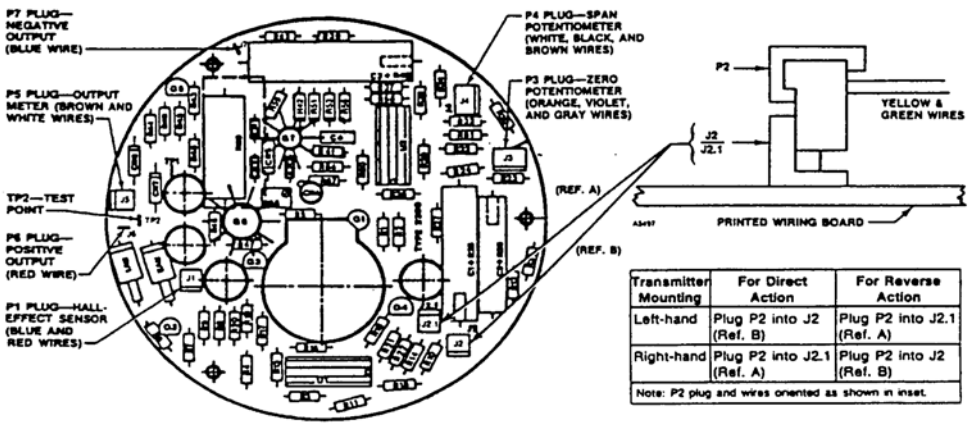
TYPE 2390 OR 2390B LIQUID LEVEL TRANSMITTER



Type 2390 or 2390B Liquid Level Transmitter and Typical Sensor



Sensor Connection Compartment



Transmitter Mounting	For Direct Action	For Reverse Action
Left-hand	Plug P2 into J2 (Ref. B)	Plug P2 into J2.1 (Ref. A)
Right-hand	Plug P2 into J2.1 (Ref. A)	Plug P2 into J2 (Ref. B)

Note: P2 plug and wires oriented as shown in inset.

Printed Wiring Board Connections

Figure 32. Visual view of electronic level transmitter and the printed circuit boards.

Level Troll

Level troll is a level measuring device. It consists of a measuring chamber, a displacer and a transmitter. Measuring chamber is built to with stand the process pressure and temperature. It is directly connected to the vessel or tank at the same elevation as the Level being measured.

Displacer is a hollow tube closed at both ends. It has a fixed volume and weight. It is hanging free on a hook inside the measuring chamber: The displacer loses weight as the process liquid level increase inside the measuring chamber. It is Archimedes principle that explains the loss of weight of the object due to immersion in the liquid is equal to the weight of the liquid of the same volume the object displaces. The loss of weight of the displacer due to buoyancy will vary with the liquid level. The maximum span of measuring liquid level is limited to the length of the displacer.

Level trolls are considered to be more reliable. The process liquid as the level changes always moves in or out of the measuring chamber, eliminating the possibility of line blockages due to stringent liquids.

A transmitter attached to the displacer senses the changes in buoyant forces and converts the changes in to a signal, which will be linear and proportional to the level. As required, a pneumatic or an electronic transmitter is used.

Even though span of measurement is fixed as the length of the displacer. Level trolls can be used for various liquids of different specific gravity.

In addition to the usual calibration adjustments a separate specific gravity adjustment is available on many models. A transmitter calibrated for water can be used for another liquid by simply changing the SG (specific gravity) adjustment dial, to the SG of the liquid level to be measured.

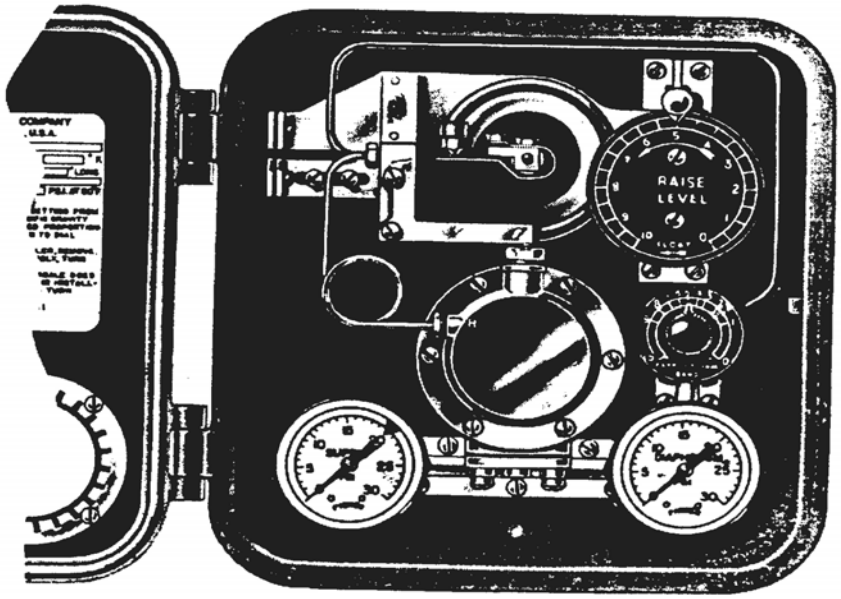


Figure 33. Exploded view of pneumatic level troll

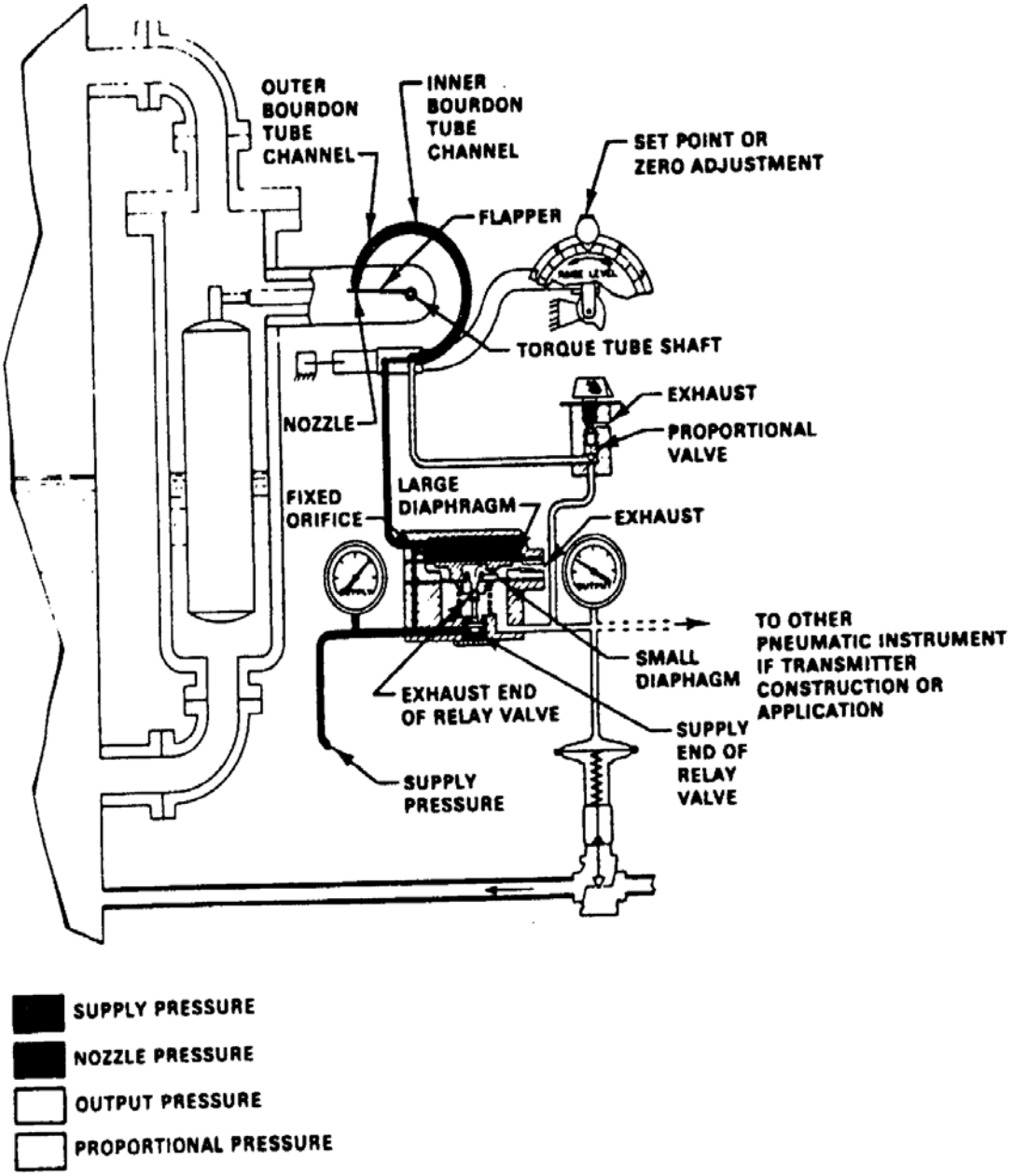


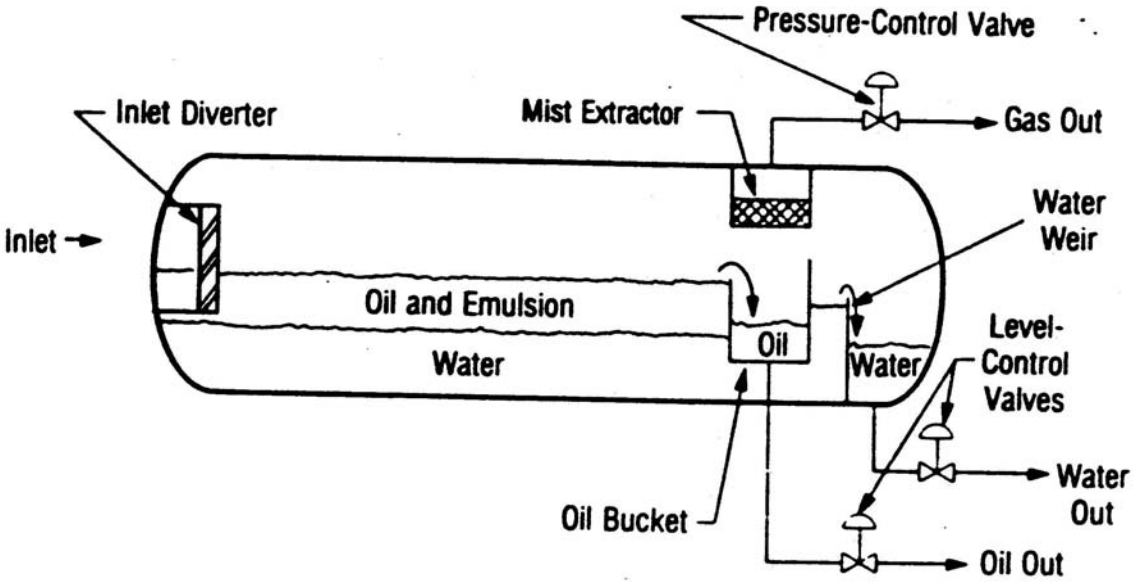
Figure 34. Working principle of pneumatic controller

Interface Level

If two or more immiscible liquids of different specific gravity are flown into a vessel/tank and are allowed sufficient time to settle, the higher SG (specific gravity) liquid settle down at the bottom of the vessel, over that the lower SG, over that the light and so on. The point of separation of settled liquids is called interface.

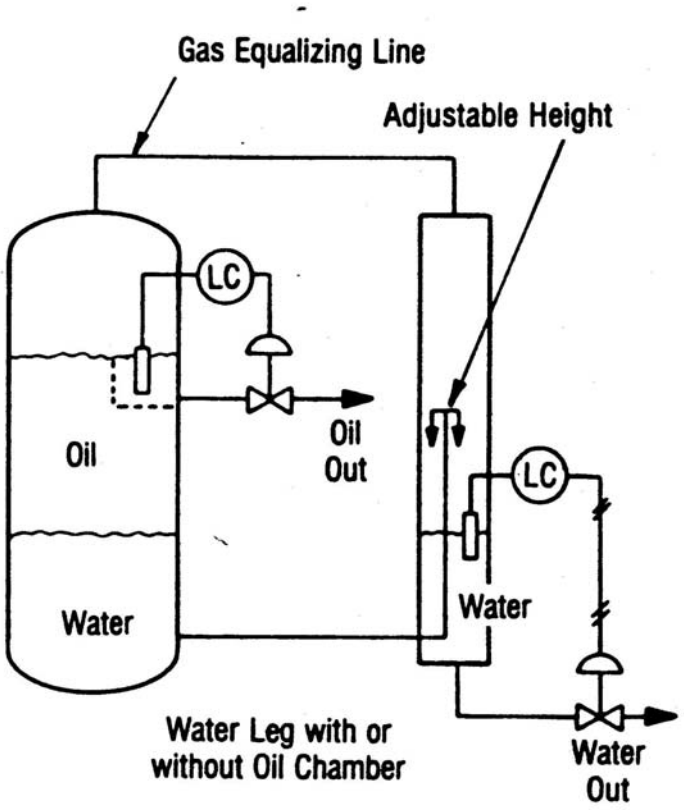
The height of the bottom liquid is interface Level. In the oil industry interface level measurement is very crucial to remove the unwanted water from the crude oil and gas condensate. Recovering of glycol from water…etc.

Differential pressure transmitters and Level-trolls are equally used for liquid interface measurement. The calibration procedure is slightly different on those instruments while using on interface application.



Simplified schematic of a typical, horizontal three-phase separator with a bucket-and-weir design.

Figure 35a.



Schematic of level control used on vertical three-phase separator.

Figure 35

Level Switches

Level switches are used to detect the liquid high and low levels. The level switch outputs are used for initiating the alarm and shutdown functions. The outputs are also used for On/Off controls, such as in the starting and stopping of pumps.

Switches are available in the "normally open or normally closed" position. 'Normally open' or 'Normally close' refers to the switch position without electrical power or pneumatic signal. Switches merely turn either an electronic or pneumatic signal on or off as required for the control schemes.

An electric switch should have the correct contacts for the application. There should be enough contacts for the circuits to be controlled. They should open on rising or falling level as required by the circuit. In " fail-safe: systems circuits are designed to alarm or shutdown when the contact opens. The electrical switch is usually single-pole, double throw or double-pole, double-throw.

The number of poles determines the number of separate circuits that can be controlled by the switch. Single-pole for one circuit and the double-pole for two circuits. The 'double throw' term means that a common terminal is connected to either of two other terminals, normally open or normally closed. The diagram labelled "SPDT" is the single-pole double-throw configuration.

With the level sensor in the normal position, the common terminal is connected to the normally closed terminal by a movable contact. (In the process health condition, the switch contact is close). When the level is increased above the set point, a plunger coupled to the movable contact moves the contact and breaks the contact between the common and normally closed terminals, and makes the contact between the common and normally open terminal.

A level switch may be used as a high- level sensor or a low-level sensor.

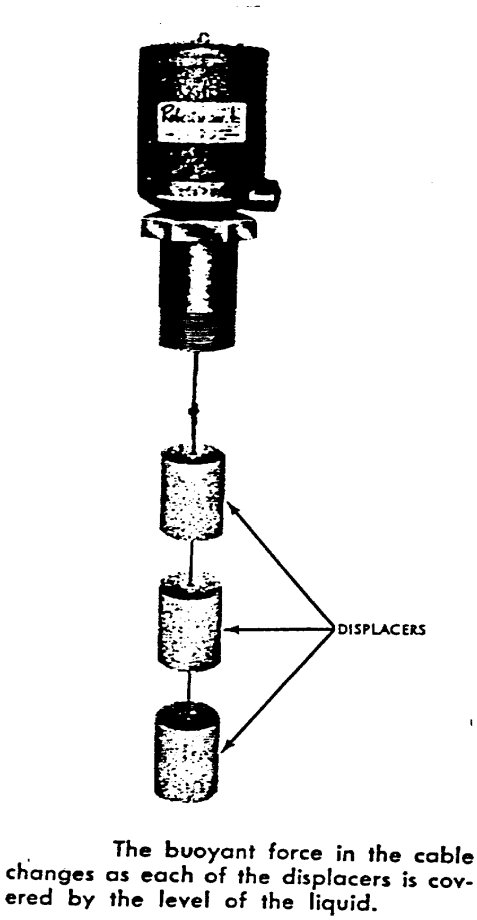


Figure 36. Electric level switch, multi-displacer type

High Level Alarm

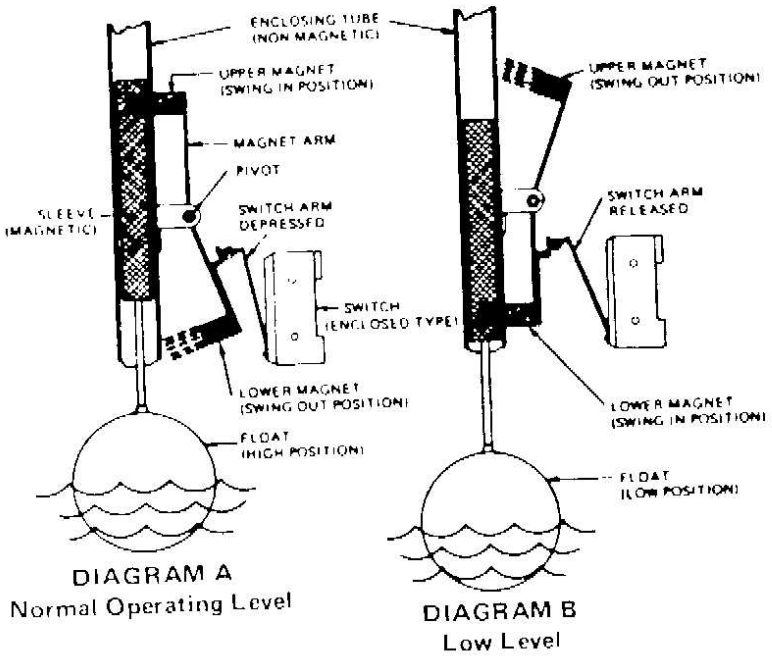
If the level switch is used for a high-level alarm, then the wiring are terminated on 'Common' and 'Normally close'.

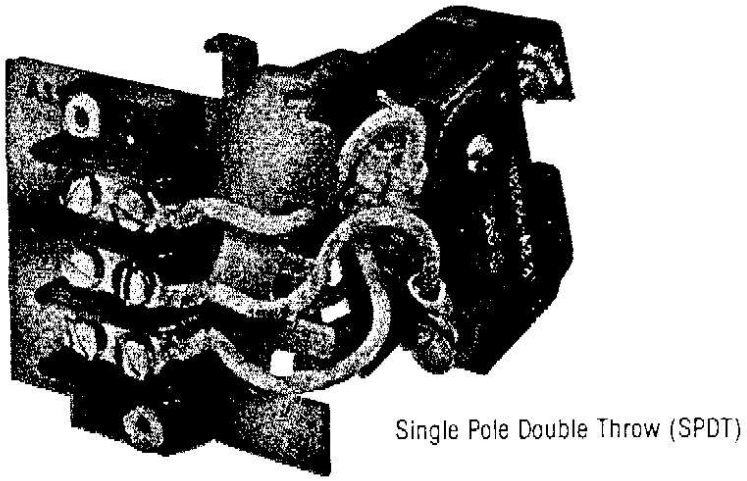
During the process healthy condition, the switch is not actuated and the switch contact remains close. If the process level goes above the set point (switch level) the switch contact breaks, resulting in a 'High level alarm'.

Low Level Alarm

If the level switch is used for a low-level alarm, then the wiring are terminated on 'Common' and 'Normally open.

During the process healthy condition, the switch is actuated and the switch contact remains close. If the process level goes below the set point (switch level) the switch contact breaks, resulting in a 'Low level alarm'.





Dry contact switch.

Figure 37. Electric level switch, float type.

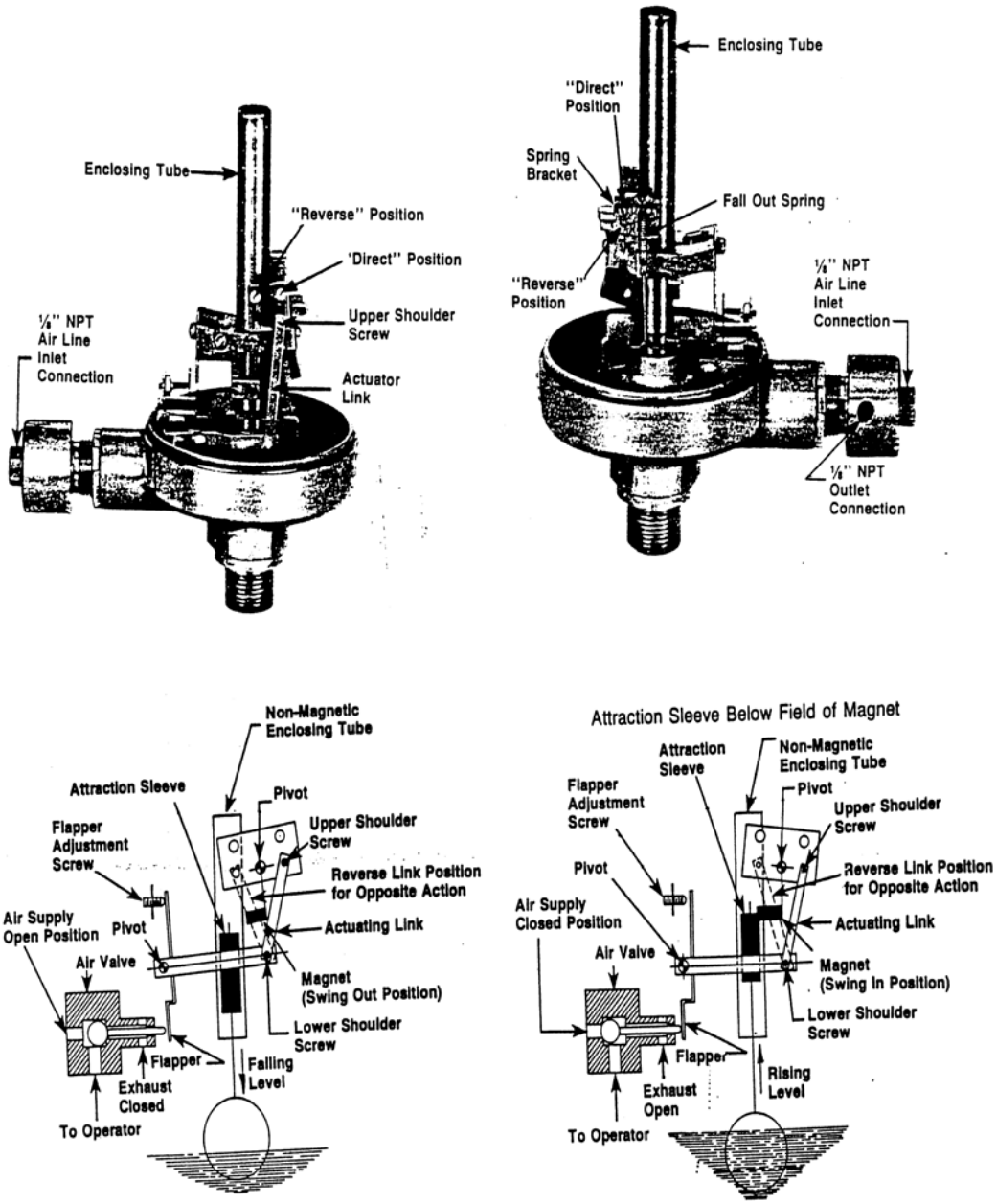


Figure 38. Pneumatic level switch, float type.

Troubleshooting Importance

Incorrect level measurement will result in many problems like:

1. Tank / vessel / separator over flowing
2. Loss of suction pressure to the transfer pumps…etc

On liquid level comparison, the zero reference point is very important. Usually a sight glass by the side of the vessel will cover a range greater than that of a level transmitter. The sight glass must be clean and checked for blockages of the impulse lines.

As the drains are connected to closed drain system, there is a possibility of pressure lock in the drain line leading to an interface level in the sight glass.
On interface level measurement, the lighter liquid must full in the vessel. I.e. above the transmitter measuring range. Other wise the total liquid head on high-pressure side of a DP transmitter will be less leading to erratic reading.

Emulsion is a status in liquids, where two or more liquids are somewhat in a homogeneous status. In an emulsion status, a clear interface is not visible. In crude oil and water, emulsion looks like crude oil but heavier than crude. So the transmitters measurement is uncertain for water.

During a field interface level calibration, care must be taken to make sure a proper interface is visible. Right dosage and type of chemical injections are done for quick settling of water in oil to get an interface status.

SGC HSE Regulation Related to:

Refer to HSE Regulation No. 7 "ISOLATIONS"

7.18 CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

4. Isolation of Hardware.

Isolation of control engineering hardware may be necessary to enable maintenance work to be done or permit removal of the hardware to effect repairs (either locally or remotely). Isolation of hardware can take several forms, for example isolation from:

a. Process plant.

b. Utilities (electric, pneumatic, hydraulic, cooling media etc).

c. Larger system of which the hardware is a subsystem or component.

5. Isolation from Process.

a. Isolation of instruments which are connected to or form a part of the process is usually achieved by valving. It is important that, where isolation of an instrument is required for maintenance purposes, correct venting/draining and valve closure procedures are adhered to.

b. Where instruments have local isolating valves in addition to the primary process isolating valves, the local valves may be used for some routine in-situ testing at the discretion of the Senior Control Engineer. If an instrument is to be removed from site, the process isolating valves must be used and any impulse pipe work must be drained or vented completely.

c. Where the process fluids are of a hazardous nature (eg toxic, flammable etc), particular care should be taken to ensure correct venting and draining, and also to clean or flush the instrument carefully, prior to effecting work or removal of the hardware from site for maintenance or repair. Gas testing may be required. On large items,

eg control valves, a certificate of cleanness is necessary prior to delivery to workshops.

d. On removal of a directly mounted instrument, from a process line containing hazardous fluids, eg pressure gauges etc, isolation by the primary isolation valve only is NOT acceptable. The valve outlet shall be blanked off, capped or plugged with a blank flange, solid screwed plug or cap, whichever is appropriate.

Refer to HSE Regulation No. 7 "Isolation"

7.18 CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

1. Preparation for Work.

When work is to be done on control engineering hardware, it is essential that the work-site is prepared correctly. The work permit system (Regulation No 06) provides the mechanism for ensuring that essential preparatory activity is documented and witnessed. It is important that the Control Engineering Person doing the work also has responsibility for effecting the task safely. This is particularly important in the control engineering discipline where the instrument may still be connected to the process, or may require to be serviced with power-on for fault finding.

2. Interaction.

Care must be taken to ensure that the work to be carried out on a specific item of instrumentation will not cause a hazard due to interaction with other protection systems or operational process controls.

3. Preparatory Work.

Prior to a work permit being issued all appropriate preparatory work at the site must be completed. The following items are some examples.

a. Removal of potential hazards from the area. Particular vigilance is needed for enclosed areas (refer to Regulation No 09 Confined Space Entry).

b. Gas testing the area for flammable, toxic or suffocating gases.

c. Construction of scaffolding to permit safe access.
d. Provision of additional fire-fighting apparatus.
e. Provision of necessary protective clothing.
f. Isolation of the control engineering hardware from the process.
g. In the case of equipment removal, isolation of the hardware from utilities (eg electricity, air supplies etc).

On completion of all necessary preparatory work (defined by the Senior Control Engineering Person and the Area Authority), a hot or cold work permit/entry permit will be issued signed by the appropriate responsible authorities.

7.18 CONTROL SYSTEMS PROCEDURES AND ISOLATIONS

6. Isolation from Electrical/Pneumatic Supplies.
   a. If practical, equipment must be made safe before any work is done on it. The operation of making the equipment safe must be done by a Competent Control Engineering Person. Care shall be taken when working on live equipment to ensure avoidance of contact with live electrical components (refer to Regulation No 19 Working with Electricity).
   b. Pneumatically operated equipment must be isolated before it is disconnected or removed for repair by closing the valve at the supply manifold for the individual instrument and venting through the drain/vent of the pressure regulators.
7. **Isolation from Utilities**
   a. Control engineering equipment may be connected to utilities (other than electrical associated with the hardware eg stream, cooling water, hydraulic fluid, chemicals, carrier gases (analysis) and air supplies. It is important that attention is given to rendering the utilities safe when the control engineering hardware is being serviced or removed.

b. Utilities should be isolated at the point of distribution to the control engineering equipment being removed (eg isolating valve at distribution head) and not solely at the hardware itself.

c. Where utility fluids are 'piped' to an instrument, the pipe work should be drained down or vented if the instrument is removed.

d. It is important that removal of a utility from a specific piece of hardware does not influence any other hardware to which the utility may also be connected (eg cooling water may have been series connected to more than one item of hardware).

TEMPERATURE MEASUREMENTS

TEMPERATURE MEASUREMENTS

Objectives

At completion of this module, the developee will have understanding of:

1. The purpose of measuring temperature in oil and gas facilities.
2. Temperature measurement scales and conversion equations.
3. Methods of temperature measurement.
4. Filled systems, principles, ranges and applications.
5. Bimetallic elements, principles, construction and function.
6. Thermocouples principle of operation.
7. Thermocouple types, extension cables, measurement ranges.
8. Thermocouples connection as thermopile, in parallel, in a switching circuit and in a differential circuit.
9. RTDs elements as temperature sensors.
10. Whetstone bridge circuit connection and function.
11. RTDs circuits and wiring connections.
12. Thermistors construction and function.
13. Comparison between thermocouples and RTDs.
14. How to use the pyrometer as a temperature measuring device.

Related Safety Regulations for Module I-4: Temperature Measurement

Juniors have to be familiarised with the following **SGC HSE** regulations, while studying this module:
Regulation No. 6: Work to permit system.
Regulation No. 7: Isolation
7.18 (1-10) control systems procedures and isolations.
Regulation No. 22: Hot and Odd Bolting.
Regulation No. 23: General Engineering Safety.
Regulation No. 27: General Services:
Safe use of hand tools and powered tools/equipment.

Temperature Measurement

There are changes in the physical or chemical state of most substances when they are heated or cooled. It is for this reason that temperature is one of the most important of the measured variables encountered in industrial processes.

Temperature is defined as the degree of hotness or coldness measured on a definite scale. Hotness and coldness are the result of molecular activity. As the molecules of a substance move faster, the temperature of that substance increases. Heat is a form of energy and is measured in calories or BTU's (British Thermal Units).

When two substances at different temperatures come into contact with each other, there is a flow of heat. The flow is away from the substance at a higher temperature toward the substance at a lower temperature. The flow of heat stops when both substances are at the same temperature.

Heat Transfer

The flow of heat is transferred in three ways: convection, conduction, and radiation.

Convection

Heat transferred by the actual movement of portions of a gas or liquid from one place to another is called convection. This movement is caused by changes in density due to rising temperature. For example, in a forced air heating system, the warm air entering the room through the supply duct is less dense, and therefore, lighter than the cooler air already in the room. As the warm air-cools, it drops and moves through the cool air return and back through the heating system. See Fig. 4-1. Another example of convection is a water heating system. The heavier cold water moves down, forcing the heated water up through the pipes of the system. Convection takes place only in fluids (either a liquid or a gas).

Conduction

When heat is applied to one part of a substance, it is transferred to all parts of the substance. The movement is from molecule to molecule. Gases and liquids are poor conductors. The flow of heat by conduction takes place most effectively in solids. See Fig. 4-2.

Radiation

Heat energy is transferred in the form of rays sent out by the heated substance as its molecules undergo internal change. See Fig. 4-3. Only energy is transferred. The direction of the flow of heat is from the radiating source. The radiant energy is then absorbed by a colder substance or object. Radiation takes place in any medium (gas, liquid, or solid), or in a vacuum.

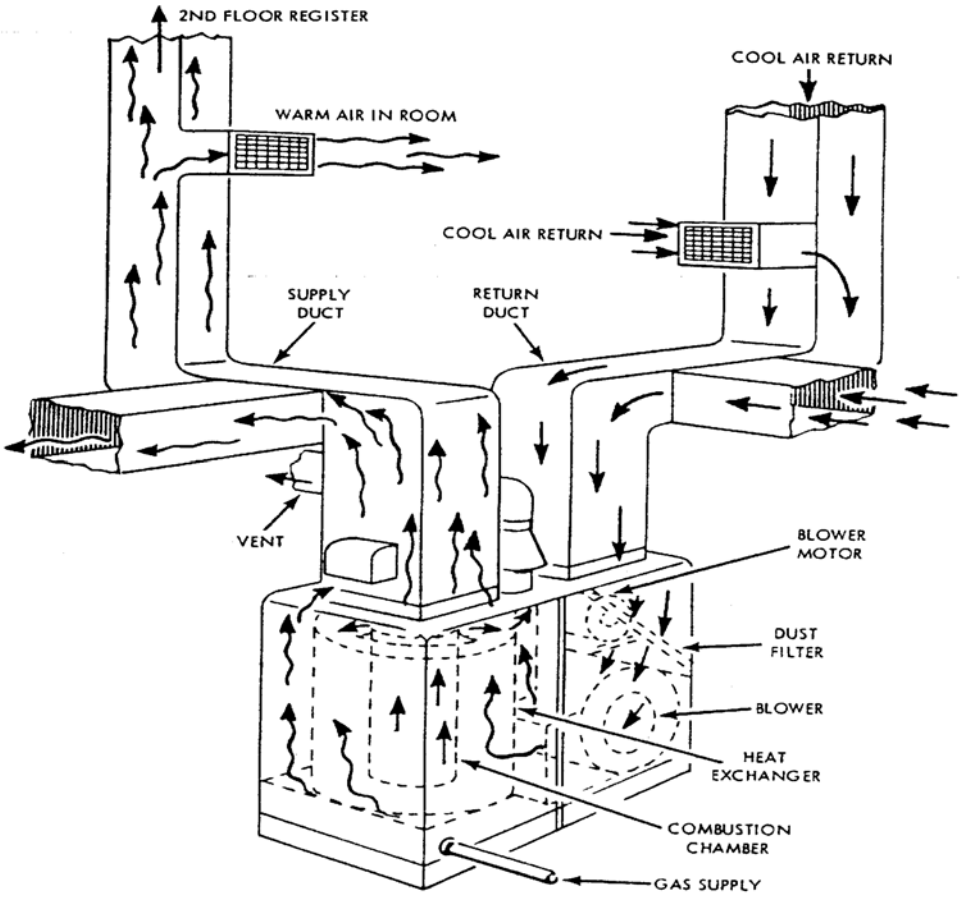


Figure 4.1

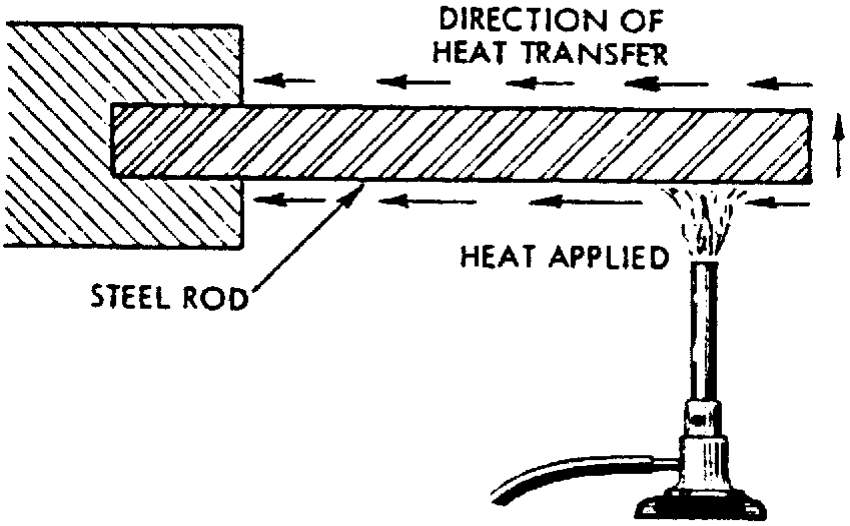


Figure 4.2

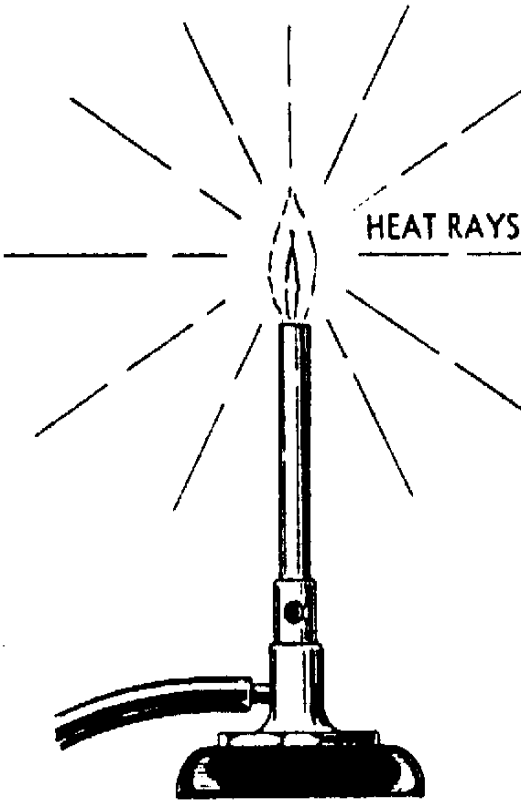


Figure 4.3

Temperature measurement is very important in industry. Which use equipment supply, remove and exchange heat energy in various processes. Critical factors such as process reaction rates, raw material usage, yield and quality can all affected the precision and frequency with which the temperature is measured. The measurement of temperature is also important for protection of the equipment, as uncontrolled high or low temperatures can cause structural deterioration of pipelines and vessels.

Temperature Scales

Heat is a from of stored energy. Temperature is the Measurement of intensity of heat. It is like measuring the pressure of a gas in cylinder, irrespective of the volume of the cylinder. Temperature measurement is very important in oil and processing industries the components in the crude oil and gas may vary in composition due to the variations in temperature while are treated in processing units.

The Machinery like pumps Compressors and equipment like the heating furnaces need to be monitored carefully on their heat generating parts in order to safe guard them from over heating there by preventing the damage of components and expensive break down. Temperature is expressed in degree. There are few temperature scales commonly used in Industrial measurement the centigrade the Fahrenheit and the Kelvin is most popular scale.

The centigrade scale zero starts at the point of pure water and divided into 100 graduations at the temperature of boiling point of pure water each division is known as a degree centigrade. The Fahrenheit scale zero starts below ice point It is divided into 10 equal graduations in between pure water ice point and boiling point. The ice point is 32°f and the boiling point is 212°F.

The absolute or the Kelvin scale zero reference starts from a point which is theoretically derived, where all the particles in the matter moving and seizes to a stand still It is 273.15 degrees below the ice point in centigrade scale. Hence, the ice point on a Kelvin scale is 273.15°k and the boiling point is 373. 15°k.

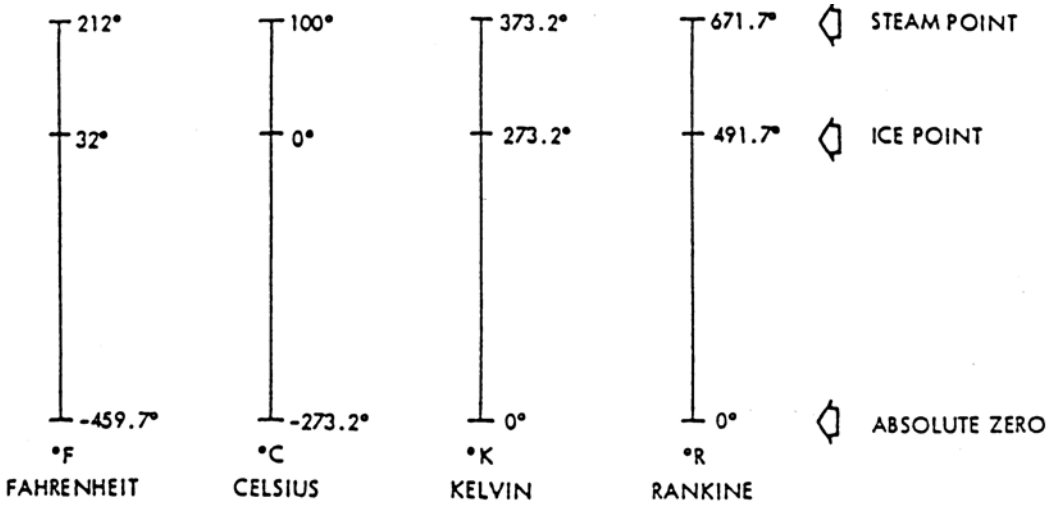


Figure 4.4: Absolute zero is the temperature at which the movement of molecules completely stops

Temperature value on given scale can be converted to express on other Scales:

$$^{\circ}C = (^{\circ}F-32) \times 5/9$$

$$^{\circ}F = (^{\circ}C \times 9/5) + 32$$

$$^{\circ}K = (^{\circ}C + 273.15)$$

Example: 1 Convert 100°C into Fahrenheit Scale?

$$^{\circ}F = ^{\circ}C \times 9/5 + 32$$

$$= (100 \times 9/5) + 32 = 212\ ^{\circ}F$$

Example: 2 Convert 122 °F into degrees centigrade?

$$^{\circ}C = (^{\circ}F - 32) \times 5/9$$

$$= (122 - 32) \times 5/9 = 50\ ^{\circ}C$$

Example: 3 Convert – 40 °C in degrees Fahrenheit?

$$^{\circ}F = (^{\circ}C \times 9/5) + 32$$

$$= (-40 \times 9/5) + 32 - 40$$

$$-40\ ^{\circ}C = -40\ ^{\circ}F$$

Temperature Measuring Sensors

1. Thermometers

Thermometers are used to measure temperature. But they vary greatly, depending on the requirements of the job they are intended for. In this section we discuss some of the more common types of thermometers, various principles and materials which go into their make-up, and how they are used.

a) Filled Thermometers

Filled system is a metallic assembly that consists of a bulb, capillary and a Bourdon tube assembly. Three types of metal bulb temperatures are in common use and they are categorised according the working fluid, Mercury, Liquid, Gas or Vapour.

The bulb capillary tube and Bourdon tube are completely filled with thermometric liquid and then sealed When the bulb is heated the liquid expands, moving the tip of the Bourdon tube. This movement is magnified and displayed on a local indicator.

Mercury-in-glass Thermometers, Mineral substances contract or expand a definite amount for each degree of temperature change. This is the principle of thermal expansion. When heat is applied to a mercury-in-glass thermometer, the mercury expands more than the glass bulb. This difference in expansion causes the mercury to rise in the small-bore (capillary) glass tube. Because the mercury rises uniformly with temperature, the tubing can be calibrated according to a temperature scale. The mercury-in-glass thermometer can be used for temperatures from -30°Fto +800°F.

Liquid-filled Thermometers; Mercury is not the only liquid used in glass thermometers. Other liquids, such as alcohol, are used to measure temperatures below the freezing point of mercury (-38.87°C or -37.96°F). The alcohol contains dye to enable the thermometer to be more easily read. Liquid-filled glass thermometers can be used for temperatures from -300°F to +600°F.

The scale can be etched directly on the glass tube or it can be engraved on a metal plate or tube to which the glass tube is attached. Floating glass thermometers are also available. Both the tube and the scale are enclosed in a glass envelope, which is weighted so that the thermometer floats in an upright position. See Fig. 4-5.

Gas-filled Thermometers; In this case a gas is used instead of thermometric fluid. The main advantage of this type of device is that in the event of leakage, there is no release of undesirable liquids.

Vapour pressure Thermometers; In this case the bulb is filled with volatile liquid which remains the bulb at all working temperatures. As the temperature rises vapour pressure increases, and it is this that moves the Bourdon tube to give a local indication.

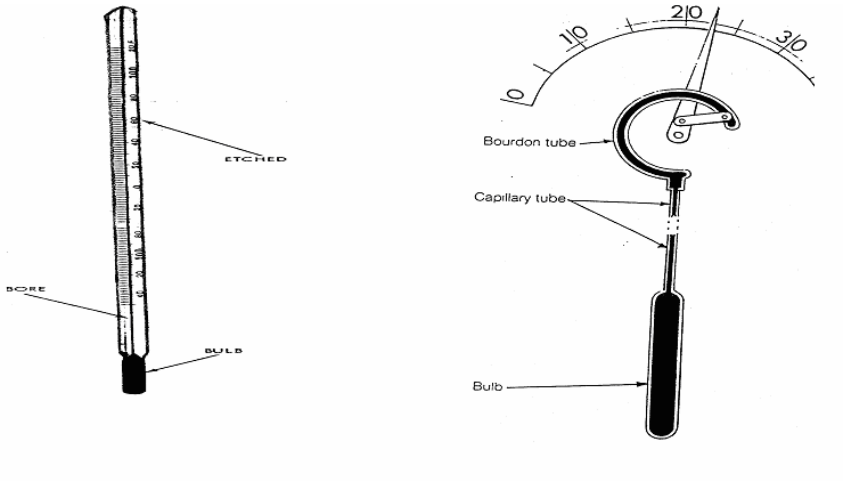
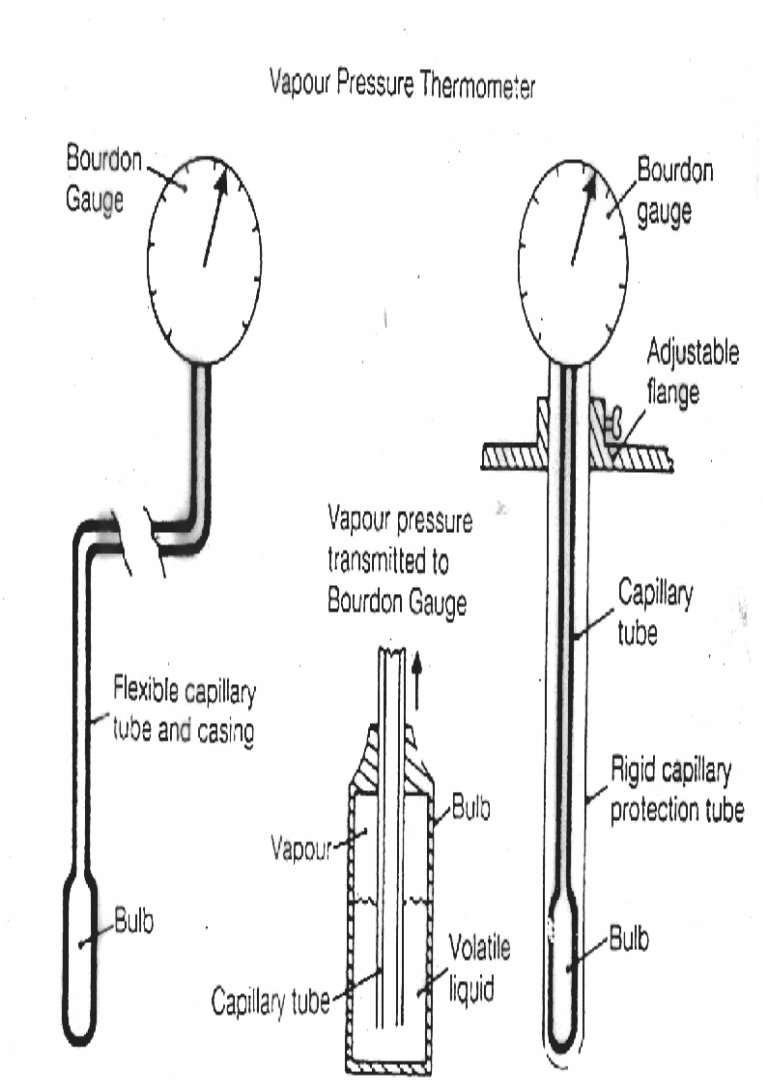


Figure 4.5 and 4.6

Operational Aspects

One of the major advantages of metal thermometer is the fact that the bulb may be placed at some distance from where the readings are taken. This advantage introduces the problem of changes in thermometer of the surroundings of the Bourdon tube and the capillary. In order to minimise this effect, liquid and gas filled thermometers must have some from of compensatory element included in their design.



Vapour pressure thermometers do not need compensation as the vapour pressure depends upon the liquid / vapour interface temperature, and the liquid surface is always in the bulb. Bimetallic thermometers are available in convenient range increments for measurements between – 80 °F (-50 °C) and 1000 °F (500 °C). A range should chosen so that the normal operating temperature is near the centre and both the high and low temperature of interest are covered.

Figure 4.7

They are not very susceptible to damage from over or under ranging. Dial calibrations are available in either Fahrenheit or Celsius or with both calibrations. An external adjustment screw is usually provided so that the thermometer can be calibrated at a single point, but there is usually no adjustment for span.

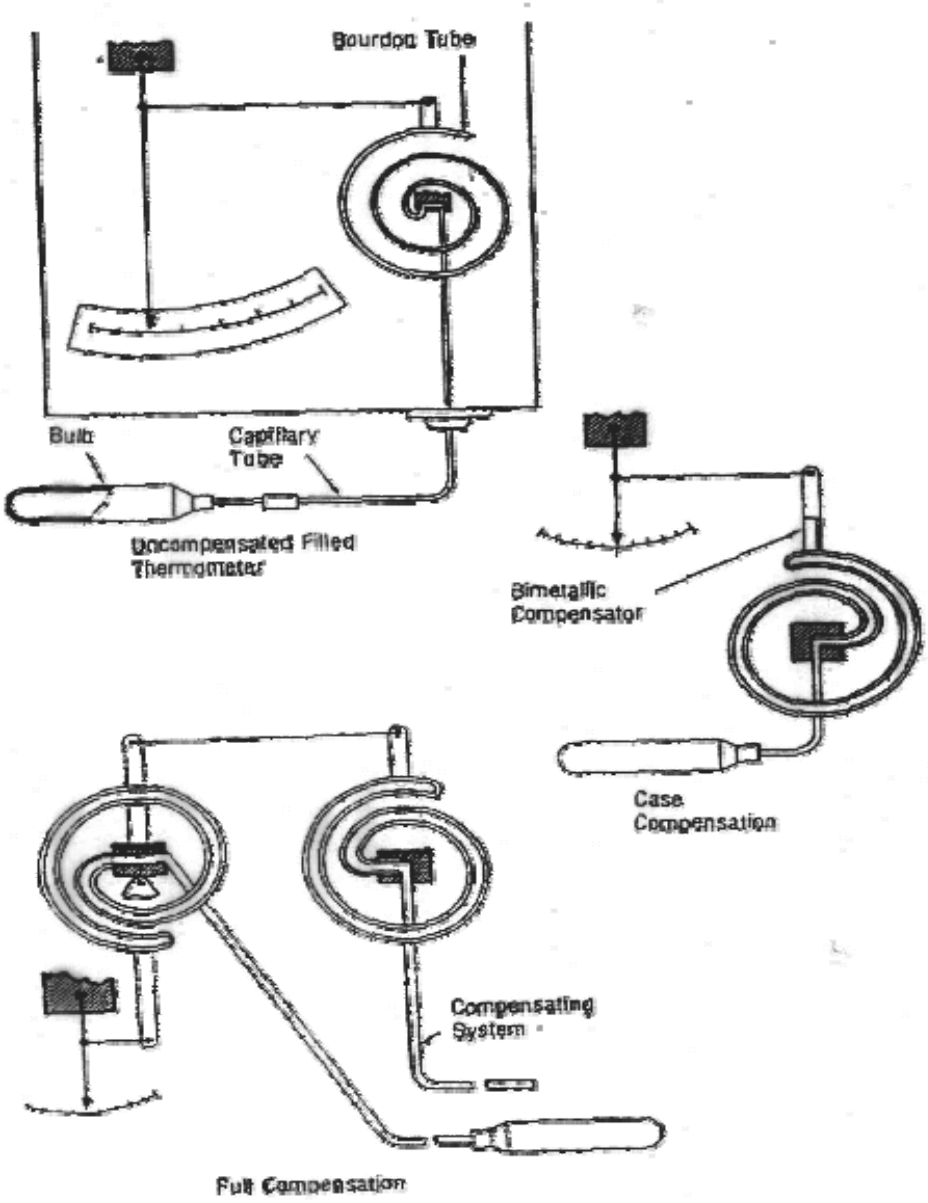
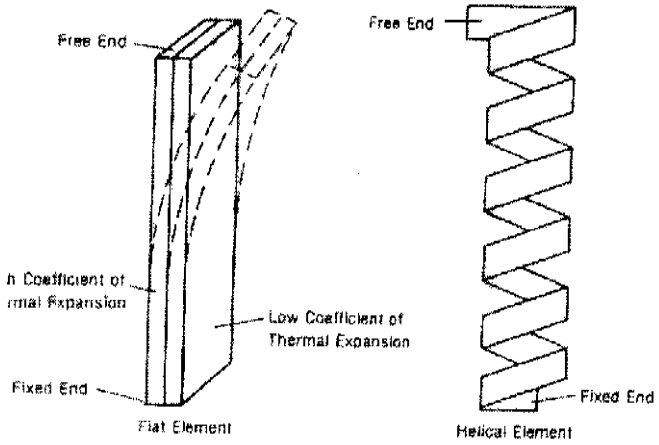


Figure 4.8

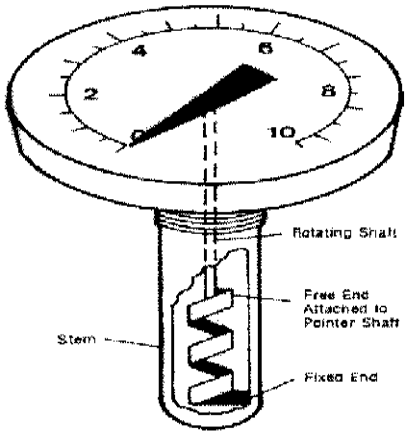
b) Bimetallic Thermometers

Most substances expand when the temperature increases and contract when the temperature decreases, but different substances expand and contract at different rates for a given material, the increase length per unit length per degree of temperature increase is called the coefficient of thermal expansion for that material. If two materials with different coefficients of thermal expansion are bonded together increase in temperature will cause the free end to bend toward the material with the lower coefficient of thermal expansion.

A bimetallic element can be formed spiral or helix to increase the amount of motion available for a given temperature change. The spiral form of bimetallic element is convenient for housing in a circular flat case and is typically used is dial thermometers that measure ambient temperature. The helical form is will suited for housing in a narrow tube (stem) for increase into a fluid directly or housing within a thermowell with a small bore.



Schematic Illustrating Concept of Bimetallic Temperature Elements



Schematic of a Typical Bimetallic Dial Thremometer

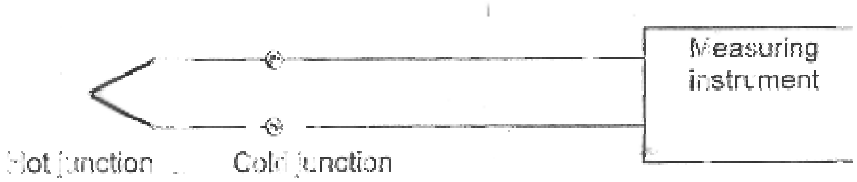
Figure 4.9

2. Thermocouples

Thermocouples are used in measuring wide range of temperatures from 250 °C to1400 °C. When any two dissimilar metals are joined together at both ends, and there is a temperature difference between the two ends, an e.m.f is produced. The

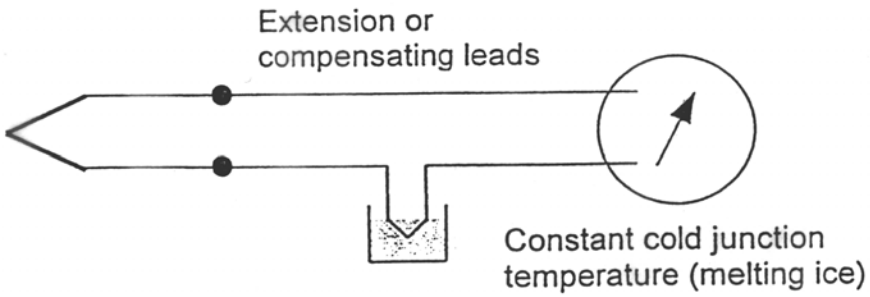
exposed to the temperature being measured is known as the HOT junction and the COLD junction consists of the measurement circuitry.

The terminal end of thermocouple will usually be close to the object whose temperatures is being measured and may itself be subjected to varying temperatures. It will therefore not provide a suitable stable temperature for cold junction measuring purposes.



Long thermocouple leads could be used to move the cold junction well away from any unstable temperature areas, but as thermocouple wires are manufactured to a close tolerance this would be expensive. Their use is therefore limited to the probe itself. To overcome the problem wires are used which have similar thermo-electric properties to the thermocouple materials over a limited temperature.

These are called compensating leads. Different may be used for the same thermocouple type and additional letters for example KCA or KCB distinguishes them. By having thermocouple connected directly compensating leads, the cold junction can moved to a location where only small changes in ambient temperature exist. These small changes in ambient temperature can be automatically compensated for.



The electrical limitations are that the junction, including any third metal, must be at the temperature to be measured the wires must be insulated from each other from the junction to be receiver, and if the junction is grounded there must be no other

ground. The only physical limitation is that the wires must be able to stand the environment to which they are subjected.

Types of Thermocouples

There are about a dozen commonly used thermocouples, which have been assigned a letter designation. By convention a slash mark is used to separate the material of each thermocouple wire and helps identify the polarity of the wires. The first wires has a positive polarity when the measuring junction is at a higher temperature than the reference junction.

Type J

Is the most common and expensive thermocouple is **Iron versus Constant**. Type J is usually furnished when no specific type is specified.

Type K

Is **Chromal versus Alumal** thermocouple. Type K offers better corrosion resistance and does not produce as much out as type J.

Type T

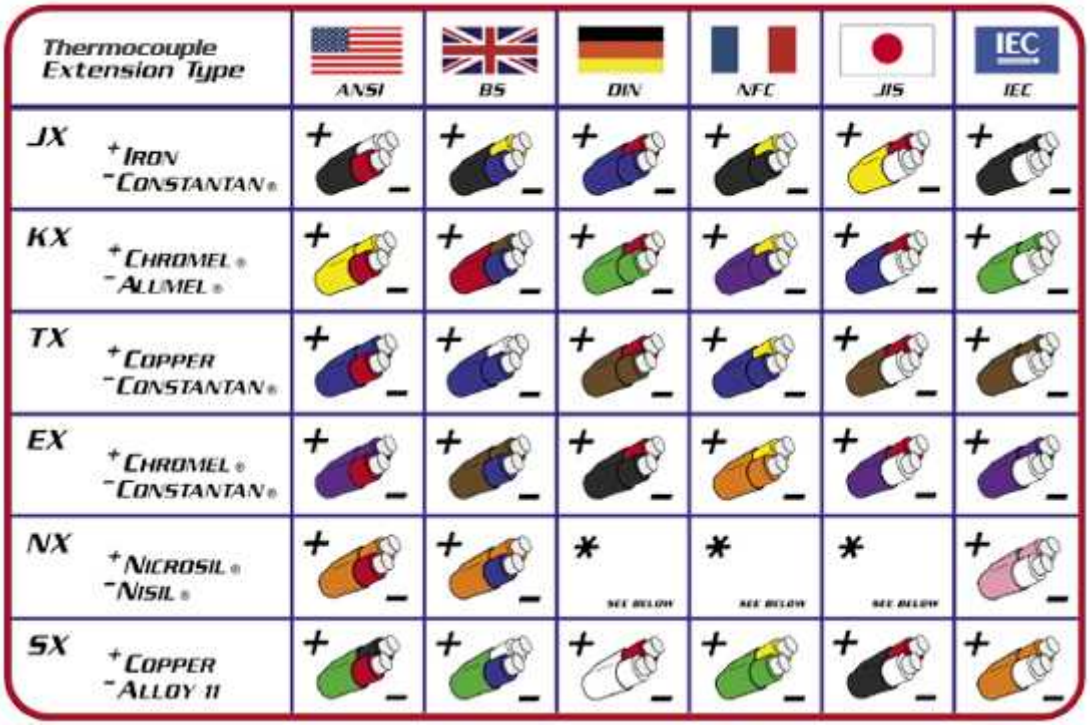
Is **Copper versus Constantine** thermocouple, is usually used when temperatures below zero are to be measured. The materials used in type T behave more predictably at low temperatures than those used for types J and K.

Type E

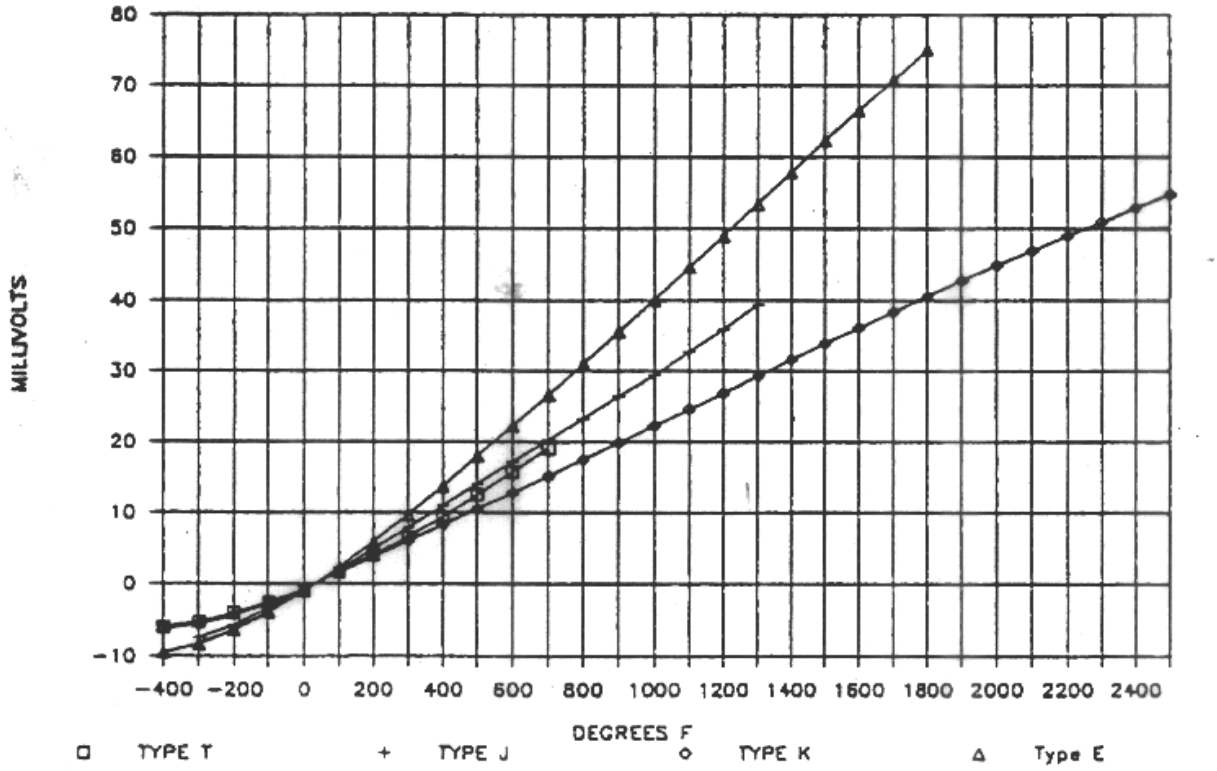
Is **Chromal versus Constantan** thermocouple**,** Provides the largest voltage changing per temperature change for standard thermocouples. An output of 40 millivolts at 1000ºF can be compared to 30 mv for type J and 22 mV for type K. Type E has more tendencies to change characteristics with time than type J, K and T.

These four types of thermocouples comprise the base metal thermocouples. Other thermocouple types, called the noble metal types are available for measurements where the base metal types are not auditable. They are made from expensive metals such as platinum, thodium, iridium and tungsten thus are more expensive. Also, they do not provide as much output as the base metal types.

These noble metal thermocouples are used in laboratories, for molten metals and other applications, but are rarely used in production facilities.



Type	Temperature Range	British coloursystem		Inrernational colour system	
		+ve	-ve	+ve	- ve
K	0° to 1100 °C	Brown	blue	Green	White
T	- 185° to 300 °C	White	blue	Brown	White
J	20° to 700 °C	Yellow	blue	Black	white
E	-150° to 1000 °C	White	blue	Orange white	



Output Versus Temperature Curves for the Four Types of Base Metal Thermocouples. (Types J, K, T and E)

Three Basic Types Of Thermocouple Assembly

Some thermocouple assemblies are manufactured so that the thermocouple makes electrical contact with the sheath (called **ground junction**) and some are manfactured where the thermocouple is electrically insulated from the sheath (called **ungrounded junction**) A third option is where the thermocouple extends slightly beyond the sheath (called **exposed junction**) exposed junction offer the fastest response, but are not used in oil and gas processing because they are subject to physical damage.

The exposed junction is often used for the measurement of static or flowing non-corrosive gas temperatures where the response time must be minimal. The junction extends beyond the protective metallic sheath to provide better response. The sheath insulation is sealed at the point of entry to prevent penetration of moisture or gas.

The ungrounded junction often is used for the measurement of static or flowing corrosive gas and liquid temperatures in critical electrical applications. The welded wire thermocouple is physically insulated from the thermocouple sheath by soft magnesium oxide (MgO) powder or equivalent.

The grounded junction often is used for the measurement of static or flowing corrosive gas and liquid temperatures and for high-pressure applications. The junction is welded to the protective sheath, providing faster response than an ungrounded junction does.

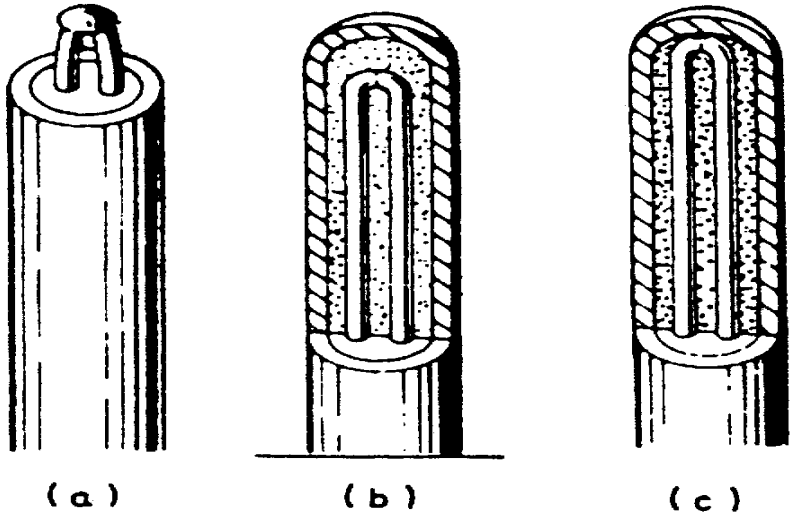


FIGURE 7 Thermocouple measuring junctions, (a) Exposed junction, (b) Ungrounded junction, (c) Grounded junction.

Extension cables

Extension cables are used in the same way as compensating cables but provide a greater accuracy. They are manufactured from the same material as the thermocouple being used.

Installation of Compensating Cables

The colour of the thermocouple and the compensating cable must be adhered to.

Note: If the correct compensating cable is used but the connections crossed at each end, the associated instrument will indicate an error equal to twice the temperature difference between the thermocouple head and the instrument environment.

Thermocople Circuit Flexibility

Normally one envisions the use of thermocouples one at a time for single temperature measurements. As shown below, thermocouples may be used in parallel, in series, and in switching and differential circuits.

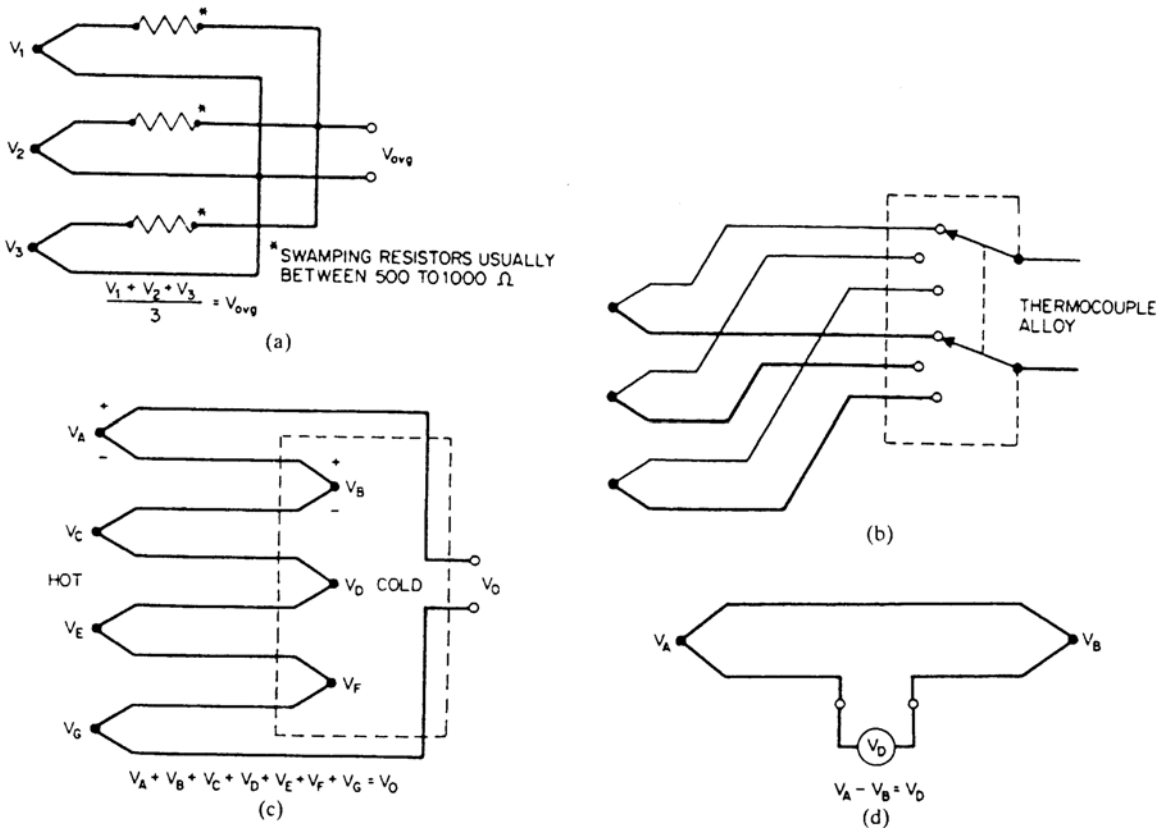


FIGURE 10. Use of thermocouples in multiples, (a) Thermocouples in parallel, (b) Thermocouples in switch circuit. Switch must be isothermal or made of the same thermocouple alloy material, (c) Thermocouples in series (thermopile). Note; Vb, Vd, and Vf are negative thermoelectric voltages compared with VA, VC, VE, and VG.

However, the alloys are also reversed, thus creating a net voltage, (d) Differential circuit. Note: Output voltage cannot be accurately cold-junction compensated because of the non-linearity of thermocouple EMF versus temperature. Approximations can be made if the absolute temperature is known.

Thermocouple reference tables

When using thermocouples, temperature reference tables can be used to convert the mV signal from the thermocouple into a temperature reading. All table values are referenced to a cold junction temperature of 0° C. if the reference junction of a thermocouple is not at 0°C, the tables can still be used by applying an appropriate Correction to compensate for the difference between the reference junction and 0 ^{o}C. The use of loop powerd head mounted transducers allow the transmission of a standard 4-20 mA signal . this avoids the necessity of compensating or extension cables and reduces the danger of electrical noise interference. More importantly the head mounted electronical will linearise the signal and automatically add cold junction compensation. Therefore the 4 – 20 mA signal is directly related to the actual temperature being measured.

Example:

The output of a type j thermocouple is 5.340 mV when the reference junction is at 20^{o}C. Calculate the measured temperature.

From the type J table 20^{o}C is 1.019 mV.

Therefore the actual output is 5.340 + 1.019 = 6.359 mV.

From the type J table. This gave a measured temperature of 120^{o}C.

Application Notes

Thermocouples can be constrcured either protected or exposed.

When protected they can be grounded which will give a faster response or ungrounded which are slower to respond out are electrically isolated and less

susceptible to electrical noise. Electeonic modules, can be used to detect thermocoupled failure by driving the indication fully upscale or downscale. This is usually know as upscale or dowriscale Burnout detection.

Nickel-Chromium/Nickel-Aluminum Thermocouple reference Table

K

International Thermocouple Reference Tables for **NICKEL-CHROMIUM/NICKEL-ALUMINIUM**

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
-270	-6458									
-260	-6441	-6444	-6446	-6448	-6450	-6452	-6453	-6455	-6456	-6457
-250	-6404	-6408	-6413	-6417	-6421	-6425	-6429	-6432	-6435	-6438
-240	-6344	-6351	-6358	-6364	-6371	-6377	-6382	-6388	-6394	-6399
-230	-6262	-6271	-6280	-6289	-6297	-6306	-6314	-6322	-6329	-6337
-220	-6158	-6170	-6181	-6192	-6202	-6213	-6223	-6233	-6243	-6253
-210	-6035	-6048	-6061	-6074	-6087	-6099	-6111	-6123	-6135	-6147
-200	-5891	-5907	-5922	-5936	-5951	-5965	-5980	-5994	-6007	-6021
-190	-5730	-5747	-5763	-5780	-5796	-5813	-5829	-5845	-5860	-5876
-180	-5550	-5569	-5587	-5606	-5624	-5642	-5660	-5678	-5695	-5712
-170	-5354	-5374	-5394	-5414	-5434	-5454	-5474	-5493	-5512	-5531
-160	-5141	-5163	-5185	-5207	-5228	-5249	-5271	-5292	-5313	-5333
-150	-4912	-4936	-4959	-4983	-5006	-5029	-5051	-5074	-5097	-5119
-140	-4669	-4694	-4719	-4743	-4768	-4792	-4817	-4841	-4865	-4889
-130	-4410	-4437	-4463	-4489	-4515	-4541	-4567	-4593	-4618	-4644
-120	-4138	-4166	-4193	-4221	-4248	-4276	-4303	-4330	-4357	-4384
-110	-3852	-3881	-3910	-3939	-3968	-3997	-4025	-4053	-4082	-4110
-100	-3553	-3584	-3614	-3644	-3674	-3704	-3734	-3764	-3793	-3823
-90	-3242	-3274	-3305	-3337	-3368	-3399	-3430	-3461	-3492	-3523
-80	-2920	-2953	-2985	-3018	-3050	-3082	-3115	-3147	-3179	-3211
-70	-2586	-2620	-2654	-2687	-2721	-2754	-2788	-2821	-2854	-2887
-60	-2243	-2277	-2312	-2347	-2381	-2416	-2450	-2484	-2518	-2552
-50	-1889	-1925	-1961	-1996	-2032	-2067	-2102	-2137	-2173	-2208
-40	-1527	-1563	-1600	-1636	-1673	-1709	-1745	-1781	-1817	-1853
-30	-1156	-1193	-1231	-1268	-1305	-1342	-1379	-1416	-1453	-1490
-20	-777	-816	-854	-892	-930	-968	-1005	-1043	-1081	-1118
-10	-392	-431	-469	-508	-547	-585	-624	-662	-701	-739
0	0	-39	-79	-118	-157	-197	-236	-275	-314	-353
0	0	39	79	119	158	198	238	277	317	357
10	397	437	477	517	557	597	637	677	718	758
20	798	838	879	919	960	1000	1041	1081	1122	1162
30	1203	1244	1285	1325	1366	1407	1448	1489	1529	1570
40	1611	1652	1693	1734	1776	1817	1858	1899	1940	1981
50	2022	2064	2105	2146	2188	2229	2270	2312	2353	2394
60	2436	2477	2519	2560	2601	2643	2684	2726	2767	2809
70	2850	2892	2933	2975	3016	3058	3100	3141	3183	3224
80	3266	3307	3349	3390	3432	3473	3515	3556	3598	3639
90	3681	3722	3764	3805	3847	3888	3930	3971	4012	4054
100	4095	4137	4178	4219	4261	4302	4343	4384	4426	4467
110	4508	4549	4590	4632	4673	4714	4755	4796	4837	4878
120	4919	4960	5001	5042	5083	5124	5164	5205	5246	5287
130	5327	5368	5409	5450	5490	5531	5571	5612	5652	5693
140	5733	5774	5814	5855	5895	5936	5976	6016	6057	6097
150	6137	6177	6218	6258	6298	6338	6378	6419	6459	6499
160	6539	6579	6619	6659	6699	6739	6779	6819	6859	6899
170	6939	6979	7019	7059	7099	7139	7179	7219	7259	7299
180	7338	7378	7418	7458	7498	7538	7578	7618	7658	7697
190	7737	7777	7817	7857	7897	7937	7977	8017	8057	8097
200	8137	8177	8216	8256	8296	8336	8376	8416	8456	8497
210	8537	8577	8617	8657	8697	8737	8777	8817	8857	8898
220	8938	8978	9018	9058	9099	9139	9179	9220	9260	9300
230	9341	9381	9421	9462	9502	9543	9583	9624	9664	9705
240	9745	9786	9826	9867	9907	9948	9989	10029	10070	10111
250	10151	10192	10233	10274	10315	10355	10396	10437	10478	10519
260	10560	10600	10641	10682	10723	10764	10805	10846	10887	10928
270	10969	11010	11051	11093	11134	11175	11216	11257	11298	11339
280	11381	11422	11463	11504	11546	11587	11628	11669	11711	11752
290	11793	11835	11876	11918	11959	12000	12042	12083	12125	12166

Absolute thermocouple e.m.f. in microvolts

K

International Thermocouple Reference Tables for **NICKEL-CHROMIUM/NICKEL-ALUMINIUM**

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
300	12207	12249	12290	12332	12373	12415	12456	12498	12539	12581
310	12623	12664	12706	12747	12789	12831	12872	12914	12955	12997
320	13039	13080	13122	13164	13205	13247	13289	13331	13372	13414
330	13456	13497	13539	13581	13623	13665	13706	13748	13790	13832
340	13874	13915	13957	13999	14041	14083	14125	14167	14208	14250
350	14292	14334	14376	14418	14460	14502	14544	14586	14628	14670
360	14712	14754	14796	14838	14880	14922	14964	15006	15048	15090
370	15132	15174	15216	15258	15300	15342	15384	15426	15468	15510
380	15552	15594	15636	15679	15721	15763	15805	15847	15889	15931
390	15974	16016	16058	16100	16142	16184	16227	16269	16311	16353
400	16395	16438	16480	16522	16564	16607	16649	16691	16733	16776
410	16818	16860	16902	16945	16987	17029	17072	17114	17156	17199
420	17241	17283	17326	17368	17410	17453	17495	17537	17580	17622
430	17664	17707	17749	17792	17834	17876	17919	17961	18004	18046
440	18088	18131	18173	18216	18258	18301	18343	18385	18428	18470
450	18513	18555	18598	18640	18683	18725	18768	18810	18853	18895
460	18938	18980	19023	19065	19108	19150	19193	19235	19278	19320
470	19363	19405	19448	19490	19533	19576	19618	19661	19703	19746
480	19788	19831	19873	19916	19959	20001	20044	20086	20129	20172
490	20214	20257	20299	20342	20385	20427	20470	20512	20555	20598
500	20640	20683	20725	20768	20811	20853	20896	20938	20981	21024
510	21066	21109	21152	21194	21237	21280	21322	21365	21407	21450
520	21493	21535	21578	21621	21663	21706	21749	21791	21834	21876
530	21919	21962	22004	22047	22090	22132	22175	22218	22260	22303
540	22346	22388	22431	22473	22516	22559	22601	22644	22687	22729
550	22772	22815	22857	22900	22942	22985	23028	23070	23113	23156
560	23198	23241	23284	23326	23369	23411	23454	23497	23539	23582
570	23624	23667	23710	23752	23795	23837	23880	23923	23965	24008
580	24050	24093	24136	24178	24221	24263	24306	24348	24391	24434
590	24476	24519	24561	24604	24646	24689	24731	24774	24817	24859
600	24902	24944	24987	25029	25072	25114	25157	25199	25242	25284
610	25327	25369	25412	25454	25497	25539	25582	25624	25666	25709
620	25751	25794	25836	25879	25921	25964	26006	26048	26091	26133
630	26176	26218	26260	26303	26345	26387	26430	26472	26515	26557
640	26599	26642	26684	26726	26769	26811	26853	26896	26938	26980
650	27022	27065	27107	27149	27192	27234	27276	27319	27361	27403
660	27445	27487	27529	27572	27614	27656	27698	27740	27783	27825
670	27867	27909	27951	27993	28035	28078	28120	28162	28204	28246
680	28288	28330	28372	28414	28456	28498	28540	28583	28625	28667
690	28709	28751	28793	28835	28877	28919	28961	29002	29044	29086
700	29128	29170	29212	29254	29296	29338	29380	29422	29464	29505
710	29547	29589	29631	29673	29715	29756	29798	29840	29882	29924
720	29965	30007	30049	30091	30132	30174	30216	30257	30299	30341
730	30383	30424	30466	30508	30549	30591	30632	30674	30716	30757
740	30799	30840	30882	30924	30965	31007	31048	31090	31131	31173
750	31214	31256	31297	31339	31380	31422	31463	31504	31546	31587
760	31629	31670	31712	31753	31794	31836	31877	31918	31960	32001
770	32042	32084	32125	32166	32207	32249	32290	32331	32372	32414
780	32455	32496	32537	32578	32619	32661	32702	32743	32784	32825
790	32866	32907	32948	32990	33031	33072	33113	33154	33195	33236
800	33277	33318	33359	33400	33441	33482	33523	33564	33604	33645
810	33686	33727	33768	33809	33850	33891	33931	33972	34013	34054
820	34095	34136	34176	34217	34258	34299	34339	34380	34421	34461
830	34502	34543	34583	34624	34665	34705	34746	34787	34827	34868
840	34909	34949	34990	35030	35071	35111	35152	35192	35233	35273
850	35314	35354	35395	35435	35476	35516	35557	35597	35637	35678
860	35718	35758	35799	35839	35880	35920	35960	36000	36041	36081
870	36121	36162	36202	36242	36282	36323	36363	36403	36443	36483
880	36524	36564	36604	36644	36684	36724	36764	36804	36844	36885
890	36925	36965	37005	37045	37085	37125	37165	37205	37245	37285

Absolute thermocouple e.m.f. in microvolts

K

International Thermocouple Reference Tables for **NICKEL-CHROMIUM/NICKEL-ALUMINIUM**

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
900	37325	37365	37405	37445	37484	37524	37564	37604	37644	37684
910	37724	37764	37803	37843	37883	37923	37963	38002	38042	38082
920	38122	38162	38201	38241	38281	38320	38360	38400	38439	38479
930	38519	38558	38598	38638	38677	38717	38756	38796	38836	38875
940	38915	38954	38994	39033	39073	39112	39152	39191	39231	39270
950	39310	39349	39388	39428	39467	39507	39546	39585	39625	39664
960	39703	39743	39782	39821	39861	39900	39939	39979	40018	40057
970	40096	40136	40175	40214	40253	40292	40332	40371	40410	40449
980	40488	40527	40566	40605	40645	40684	40723	40762	40801	40840
990	40879	40918	40957	40996	41035	41074	41113	41152	41191	41230
1000	41269	41308	41347	41385	41424	41463	41502	41541	41580	41619
1010	41657	41696	41735	41774	41813	41851	41890	41929	41968	42006
1020	42045	42084	42123	42161	42200	42239	42277	42316	42355	42393
1030	42432	42470	42509	42548	42586	42625	42663	42702	42740	42779
1040	42817	42856	42894	42933	42971	43010	43048	43087	43125	43164
1050	43202	43240	43279	43317	43356	43394	43432	43471	43509	43547
1060	43585	43624	43662	43700	43739	43777	43815	43853	43891	43930
1070	43968	44006	44044	44082	44121	44159	44197	44235	44273	44311
1080	44349	44387	44425	44463	44501	44539	44577	44615	44653	44691
1090	44729	44767	44805	44843	44881	44919	44957	44995	45033	45070
1100	45108	45146	45184	45222	45260	45297	45335	45373	45411	45448
1110	45486	45524	45561	45599	45637	45675	45712	45750	45787	45825
1120	45863	45900	45938	45975	46013	46051	46088	46126	46163	46201
1130	46238	46275	46313	46350	46388	46425	46463	46500	46537	46575
1140	46612	46649	46687	46724	46761	46799	46836	46873	46910	46948
1150	46985	47022	47059	47096	47134	47171	47208	47245	47282	47319
1160	47356	47393	47430	47468	47505	47542	47579	47616	47653	47689
1170	47726	47763	47800	47837	47874	47911	47948	47985	48021	48058
1180	48095	48132	48169	48205	48242	48279	48316	48352	48389	48426
1190	48462	48499	48536	48572	48609	48645	48682	48718	48755	48792
1200	48828	48865	48901	48937	48974	49010	49047	49083	49120	49156
1210	49192	49229	49265	49301	49338	49374	49410	49446	49483	49519
1220	49555	49591	49627	49663	49700	49736	49772	49808	49844	49880
1230	49916	49952	49988	50024	50060	50096	50132	50168	50204	50240
1240	50276	50311	50347	50383	50419	50455	50491	50526	50562	50598
1250	50633	50669	50705	50741	50776	50812	50847	50883	50919	50954
1260	50990	51025	51061	51096	51132	51167	51203	51238	51274	51309
1270	51344	51380	51415	51450	51486	51521	51556	51592	51627	51662
1280	51697	51733	51768	51803	51838	51873	51908	51943	51979	52014
1290	52049	52084	52119	52154	52189	52224	52259	52294	52329	52364
1300	52398	52433	52468	52503	52538	52573	52608	52642	52677	52712
1310	52747	52781	52816	52851	52886	52920	52955	52989	53024	53059
1320	53093	53128	53162	53197	53232	53266	53301	53335	53370	53404
1330	53439	53473	53507	53542	53576	53611	53645	53679	53714	53748
1340	53782	53817	53851	53885	53920	53954	53988	54022	54057	54091
1350	54125	54159	54193	54228	54262	54296	54330	54364	54398	54432
1360	54466	54501	54535	54569	54603	54637	54671	54705	54739	54773
1370	54807	54841	54875							

Absolute thermocouple e.m.f. in microvolts

Iron/Copper Nickel Thermocouple reference Table

J

International Thermocouple Reference Tables for **IRON/COPPER-NICKEL**

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
-210	-8096									
-200	-7890	-7912	-7934	-7955	-7976	-7996	-8017	-8037	-8057	-8076
-190	-7659	-7683	-7707	-7731	-7755	-7778	-7801	-7824	-7846	-7868
-180	-7402	-7429	-7455	-7482	-7508	-7533	-7559	-7584	-7609	-7634
-170	-7122	-7151	-7180	-7209	-7237	-7265	-7293	-7321	-7348	-7375
-160	-6821	-6852	-6883	-6914	-6944	-6974	-7004	-7034	-7064	-7093
-150	-6499	-6532	-6565	-6598	-6630	-6663	-6695	-6727	-6758	-6790
-140	-6159	-6194	-6228	-6263	-6297	-6331	-6365	-6399	-6433	-6466
-130	-5801	-5837	-5874	-5910	-5946	-5982	-6018	-6053	-6089	-6124
-120	-5426	-5464	-5502	-5540	-5578	-5615	-5653	-5690	-5727	-5764
-110	-5036	-5076	-5115	-5155	-5194	-5233	-5272	-5311	-5349	-5388
-100	-4632	-4673	-4714	-4755	-4795	-4836	-4876	-4916	-4956	-4996
-90	-4215	-4257	-4299	-4341	-4383	-4425	-4467	-4508	-4550	-4591
-80	-3785	-3829	-3872	-3915	-3958	-4001	-4044	-4087	-4130	-4172
-70	-3344	-3389	-3433	-3478	-3522	-3566	-3610	-3654	-3698	-3742
-60	-2892	-2938	-2984	-3029	-3074	-3120	-3165	-3210	-3255	-3299
-50	-2431	-2478	-2524	-2570	-2617	-2663	-2709	-2755	-2801	-2847
-40	-1960	-2008	-2055	-2102	-2150	-2197	-2244	-2291	-2338	-2384
-30	-1481	-1530	-1578	-1626	-1674	-1722	-1770	-1818	-1865	-1913
-20	-995	-1044	-1093	-1141	-1190	-1239	-1288	-1336	-1385	-1433
-10	-501	-550	-600	-650	-699	-748	-798	-847	-896	-945
0	0	-50	-101	-151	-201	-251	-301	-351	-401	-451
0	0	50	101	151	202	253	303	354	405	456
10	507	558	609	660	711	762	813	865	916	967
20	1019	1070	1122	1174	1225	1277	1329	1381	1432	1484
30	1536	1588	1640	1693	1745	1797	1849	1901	1954	2006
40	2058	2111	2163	2216	2268	2321	2374	2426	2479	2532
50	2585	2638	2691	2743	2796	2849	2902	2956	3009	3062
60	3115	3168	3221	3275	3328	3381	3435	3488	3542	3595
70	3649	3702	3756	3809	3863	3917	3971	4024	4078	4132
80	4186	4239	4293	4347	4401	4455	4509	4563	4617	4671
90	4725	4780	4834	4888	4942	4996	5050	5105	5159	5213
100	5268	5322	5376	5431	5485	5540	5594	5649	5703	5758
110	5812	5867	5921	5976	6031	6085	6140	6195	6249	6304
120	6359	6414	6468	6523	6578	6633	6688	6742	6797	6852
130	6907	6962	7017	7072	7127	7182	7237	7292	7347	7402
140	7457	7512	7567	7622	7677	7732	7787	7843	7898	7953
150	8008	8063	8118	8174	8229	8284	8339	8394	8450	8505
160	8560	8616	8671	8726	8781	8837	8892	8947	9003	9058
170	9113	9169	9224	9279	9335	9390	9446	9501	9556	9612
180	9667	9723	9778	9834	9889	9944	10000	10055	10111	10166
190	10222	10277	10333	10388	10444	10499	10555	10610	10666	10721
200	10777	10832	10888	10943	10999	11054	11110	11165	11221	11276
210	11332	11387	11443	11498	11554	11609	11665	11720	11776	11831
220	11887	11943	11998	12054	12109	12165	12220	12276	12331	12387
230	12442	12498	12553	12609	12664	12720	12776	12831	12887	12942
240	12998	13053	13109	13164	13220	13275	13331	13386	13442	13497

Absolute thermocouple e.m.f. in microvolts

J

International Thermocouple Reference Tables for IRON/COPPER-NICKEL

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
250	13553	13608	13664	13719	13775	13830	13886	13941	13997	14052
260	14108	14163	14219	14274	14330	14385	14441	14496	14552	14607
270	14663	14718	14774	14829	14885	14940	14995	15051	15106	15162
280	15217	15273	15328	15383	15439	15494	15550	15605	15661	15716
290	15771	15827	15882	15938	15993	16048	16104	16159	16214	16270
300	16325	16380	16436	16491	16547	16602	16657	16713	16768	16823
310	16879	16934	16989	17044	17100	17155	17210	17266	17321	17376
320	17432	17487	17542	17597	17653	17708	17763	17818	17874	17929
330	17984	18039	18095	18150	18205	18260	18316	18371	18426	18481
340	18537	18592	18647	18702	18757	18813	18868	18923	18978	19033
350	19089	19144	19199	19254	19309	19364	19420	19475	19530	19585
360	19640	19695	19751	19806	19861	19916	19971	20026	20081	20137
370	20192	20247	20302	20357	20412	20467	20523	20578	20633	20688
380	20743	20798	20853	20909	20964	21019	21074	21129	21184	21239
390	21295	21350	21405	21460	21515	21570	21625	21680	21736	21791
400	21846	21901	21956	22011	22066	22122	22177	22232	22287	22342
410	22397	22453	22508	22563	22618	22673	22728	22784	22839	22894
420	22949	23004	23060	23115	23170	23225	23280	23336	23391	23446
430	23501	23556	23612	23667	23722	23777	23833	23888	23943	23999
440	24054	24109	24164	24220	24275	24330	24386	24441	24496	24552
450	24607	24662	24718	24773	24829	24884	24939	24995	25050	25106
460	25161	25217	25272	25327	25383	25438	25494	25549	25605	25661
470	25716	25772	25827	25883	25938	25994	26050	26105	26161	26216
480	26272	26328	26383	26439	26495	26551	26606	26662	26718	26774
490	26829	26885	26941	26997	27053	27109	27165	27220	27276	27332
500	27388	27444	27500	27556	27612	27668	27724	27780	27836	27893
510	27949	28005	28061	28117	28173	28230	28286	28342	28398	28455
520	28511	28567	28624	28680	28736	28793	28849	28906	28962	29019
530	29075	29132	29188	29245	29301	29358	29415	29471	29528	29585
540	29642	29698	29755	29812	29869	29926	29983	30039	30096	30153
550	30210	30267	30324	30381	30439	30496	30553	30610	30667	30724
560	30782	30839	30896	30954	31011	31068	31126	31183	31241	31298
570	31356	31413	31471	31528	31586	31644	31702	31759	31817	31875
580	31933	31991	32048	32106	32164	32222	32280	32338	32396	32455
590	32513	32571	32629	32687	32746	32804	32862	32921	32979	33038
600	33096	33155	33213	33272	33330	33389	33448	33506	33565	33624
610	33683	33742	33800	33859	33918	33977	34036	34095	34155	34214
620	34273	34332	34391	34451	34510	34569	34629	34688	34748	34807
630	34867	34926	34986	35046	35105	35165	35225	35285	35344	35404
640	35464	35524	35584	35644	35704	35764	35825	35885	35945	36005
650	36066	36126	36186	36247	36307	36368	36428	36489	36549	36610
660	36671	36732	36792	36853	36914	36975	37036	37097	37158	37219
670	37280	37341	37402	37463	37525	37586	37647	37709	37770	37831
680	37893	37954	38016	38078	38139	38201	38262	38324	38386	38448
690	38510	38572	38633	38695	38757	38819	38882	38944	39006	39068
700	39130	39192	39255	39317	39379	39442	39504	39567	39629	39692
710	39754	39817	39880	39942	40005	40068	40131	40193	40256	40319
720	40382	40445	40508	40571	40634	40697	40760	40823	40886	40950
730	41013	41076	41139	41203	41266	41329	41393	41456	41520	41583
740	41647	41710	41774	41837	41901	41965	42028	42092	42156	42219

Absolute thermocouple e.m.f. in microvolts

J

International Thermocouple Reference Tables for IRON/COPPER-NICKEL

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
750	42283	42347	42411	42475	42538	42602	42666	42730	42794	42858
760	42922	42986	43050	43114	43178	43242	43306	43370	43435	43499
770	43563	43627	43692	43756	43820	43885	43949	44014	44078	44142
780	44207	44271	44336	44400	44465	44529	44594	44658	44723	44788
790	44852	44917	44981	45046	45111	45175	45240	45304	45369	45434
800	45498	45563	45627	45692	45757	45821	45886	45950	46015	46080
810	46144	46209	46273	46338	46403	46467	46532	46596	46661	46725
820	46790	46854	46919	46983	47047	47112	47176	47241	47305	47369
830	47434	47498	47562	47627	47691	47755	47819	47884	47948	48012
840	48076	48140	48204	48269	48333	48397	48461	48525	48589	48653
850	48716	48780	48844	48908	48972	49036	49099	49163	49227	49291
860	49354	49418	49481	49545	49608	49672	49735	49799	49862	49926
870	49989	50052	50116	50179	50242	50305	50369	50432	50495	50558
880	50621	50684	50747	50810	50873	50936	50998	51061	51124	51187
890	51249	51312	51375	51437	51500	51562	51625	51687	51750	51812
900	51875	51937	51999	52061	52124	52186	52248	52310	52372	52434
910	52496	52558	52620	52682	52744	52806	52868	52929	52991	53053
920	53115	53176	53238	53299	53361	53422	53484	53545	53607	53668
930	53729	53791	53852	53913	53974	54035	54096	54157	54219	54280
940	54341	54401	54462	54523	54584	54645	54706	54766	54827	54888
950	54948	55009	55070	55130	55191	55251	55312	55372	55432	55493
960	55553	55613	55674	55734	55794	55854	55914	55974	56035	56095
970	56155	56215	56275	56334	56394	56454	56514	56574	56634	56693
980	56753	56813	56873	56932	56992	57051	57111	57170	57230	57289
990	57349	57408	57468	57527	57586	57646	57705	57764	57824	57883
1000	57942	58001	58060	58120	58179	58238	58297	58356	58415	58474
1010	58533	58592	58651	58710	58769	58827	58886	58945	59004	59063
1020	59121	59180	59239	59298	59356	59415	59474	59532	59591	59650
1030	59708	59767	59825	59884	59942	60001	60059	60118	60176	60235
1040	60293	60351	60410	60468	60527	60585	60643	60702	60760	60818
1050	60876	60935	60993	61051	61109	61168	61226	61284	61342	61400
1060	61459	61517	61575	61633	61691	61749	61807	61865	61923	61981
1070	62039	62097	62156	62214	62272	62330	62388	62446	62504	62562
1080	62619	62677	62735	62793	62851	62909	62967	63025	63083	63141
1090	63199	63257	63314	63372	63430	63488	63546	63604	63662	63719
1100	63777	63835	63893	63951	64009	64066	64124	64182	64240	64298
1110	64355	64413	64471	64529	64586	64644	64702	64760	64817	64875
1120	64933	64991	65048	65106	65164	65222	65279	65337	65395	65453
1130	65510	65568	65626	65683	65741	65799	65856	65914	65972	66029
1140	66087	66145	66202	66260	66318	66375	66433	66491	66548	66606
1150	66664	66721	66779	66836	66894	66952	67009	67067	67124	67182
1160	67240	67297	67355	67412	67470	67527	67585	67643	67700	67758
1170	67815	67873	67930	67988	68045	68103	68160	68217	68275	68332
1180	68390	68447	68505	68562	68619	68677	68734	68792	68849	68906
1190	68964	69021	69078	69135	69193	69250	69307	69364	69422	69479
1200	69536									

Absolute thermocouple e.m.f. in microvolts

Copper/Copper Nickel Thermocouple reference Table

T

International Thermocouple Reference Tables for **COPPER/COPPER-NICKEL**

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
-270	-6258									
-260	-6232	-6236	-6239	-6242	-6245	-6248	-6251	-6253	-6255	-6256
-250	-6181	-6187	-6193	-6198	-6204	-6209	-6214	-6219	-6224	-6228
-240	-6105	-6114	-6122	-6130	-6138	-6146	-6153	-6160	-6167	-6174
-230	-6007	-6018	-6028	-6039	-6049	-6059	-6068	-6078	-6087	-6096
-220	-5889	-5901	-5914	-5926	-5938	-5950	-5962	-5973	-5985	-5996
-210	-5753	-5767	-5782	-5795	-5809	-5823	-5836	-5850	-5863	-5876
-200	-5603	-5619	-5634	-5650	-5665	-5680	-5695	-5710	-5724	-5739
-190	-5439	-5456	-5473	-5489	-5506	-5522	-5539	-5555	-5571	-5587
-180	-5261	-5279	-5297	-5315	-5333	-5351	-5369	-5387	-5404	-5421
-170	-5069	-5089	-5109	-5128	-5147	-5167	-5186	-5205	-5223	-5242
-160	-4865	-4886	-4907	-4928	-4948	-4969	-4989	-5010	-5030	-5050
-150	-4648	-4670	-4693	-4715	-4737	-4758	-4780	-4801	-4823	-4844
-140	-4419	-4442	-4466	-4489	-4512	-4535	-4558	-4581	-4603	-4626
-130	-4177	-4202	-4226	-4251	-4275	-4299	-4323	-4347	-4371	-4395
-120	-3923	-3949	-3974	-4000	-4026	-4051	-4077	-4102	-4127	-4152
-110	-3656	-3684	-3711	-3737	-3764	-3791	-3818	-3844	-3870	-3897
-100	-3378	-3407	-3435	-3463	-3491	-3519	-3547	-3574	-3602	-3629
-90	-3089	-3118	-3147	-3177	-3206	-3235	-3264	-3293	-3321	-3350
-80	-2788	-2818	-2849	-2879	-2909	-2939	-2970	-2999	-3029	-3059
-70	-2475	-2507	-2539	-2570	-2602	-2633	-2664	-2695	-2726	-2757
-60	-2152	-2185	-2218	-2250	-2283	-2315	-2348	-2380	-2412	-2444
-50	-1819	-1853	-1886	-1920	-1953	-1987	-2020	-2053	-2087	-2120
-40	-1475	-1510	-1544	-1579	-1614	-1648	-1682	-1717	-1751	-1785
-30	-1121	-1157	-1192	-1228	-1263	-1299	-1334	-1370	-1405	-1440
-20	-757	-794	-830	-867	-903	-940	-976	-1013	-1049	-1085
-10	-383	-421	-458	-496	-534	-571	-608	-646	-683	-720
0	0	-39	-77	-116	-154	-193	-231	-269	-307	-345
0	0	39	78	117	156	195	234	273	312	351
10	391	430	470	510	549	589	629	669	709	749
20	789	830	870	911	951	992	1032	1073	1114	1155
30	1196	1237	1279	1320	1361	1403	1444	1486	1528	1569
40	1611	1653	1695	1738	1780	1822	1865	1907	1950	1992

Absolute thermocouple e.m.f. in microvolts

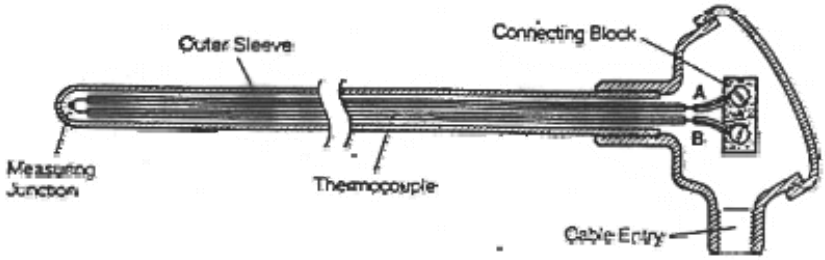
T

International Thermocouple Reference Tables for COPPER/COPPER-NICKEL

Reference junction at 0°C

°C	0	1	2	3	4	5	6	7	8	9
50	2035	2078	2121	2164	2207	2250	2294	2337	2380	2424
60	2467	2511	2555	2599	2643	2687	2731	2775	2819	2864
70	2908	2953	2997	3042	3087	3131	3176	3221	3266	3312
80	3357	3402	3447	3493	3538	3584	3630	3676	3721	3767
90	3813	3859	3906	3952	3998	4044	4091	4137	4184	4231
100	4277	4324	4371	4418	4465	4512	4559	4607	4654	4701
110	4749	4796	4844	4891	4939	4987	5035	5083	5131	5179
120	5227	5275	5324	5372	5420	5469	5517	5566	5615	5663
130	5712	5761	5810	5859	5908	5957	6007	6056	6105	6155
140	6204	6254	6303	6353	6403	6452	6502	6552	6602	6652
150	6702	6753	6803	6853	6903	6954	7004	7055	7106	7156
160	7207	7258	7309	7360	7411	7462	7513	7564	7615	7666
170	7718	7769	7821	7872	7924	7975	8027	8079	8131	8183
180	8235	8287	8339	8391	8443	8495	8548	8600	8652	8705
190	8757	8810	8863	8915	8968	9021	9074	9127	9180	9233
200	9286	9339	9392	9446	9499	9553	9606	9659	9713	9767
210	9820	9874	9928	9982	10036	10090	10144	10198	10252	10306
220	10360	10414	10469	10523	10578	10632	10687	10741	10796	10851
230	10905	10960	11015	11070	11125	11180	11235	11290	11345	11401
240	11456	11511	11566	11622	11677	11733	11788	11844	11900	11956
250	12011	12067	12123	12179	12235	12291	12347	12403	12459	12515
260	12572	12628	12684	12741	12797	12854	12910	12967	13024	13080
270	13137	13194	13251	13307	13364	13421	13478	13535	13592	13650
280	13707	13764	13821	13879	13936	13993	14051	14108	14166	14223
290	14281	14339	14396	14454	14512	14570	14628	14686	14744	14802
300	14860	14918	14976	15034	15092	15151	15209	15267	15326	15384
310	15443	15501	15560	15619	15677	15736	15795	15853	15912	15971
320	16030	16089	16148	16207	16266	16325	16384	16444	16503	16562
330	16621	16681	16740	16800	16859	16919	16978	17038	17097	17157
340	17217	17277	17336	17396	17456	17516	17576	17636	17696	17756
350	17816	17877	17937	17997	18057	18118	18178	18238	18299	18359
360	18420	18480	18541	18602	18662	18723	18784	18845	18905	18966
370	19027	19088	19149	19210	19271	19332	19393	19455	19516	19577
380	19638	19699	19761	19822	19883	19945	20006	20068	20129	20191
390	20252	20314	20376	20437	20499	20560	20622	20684	20746	20807
400	20869									

Absolute thermocouple e.m.f. in microvolts

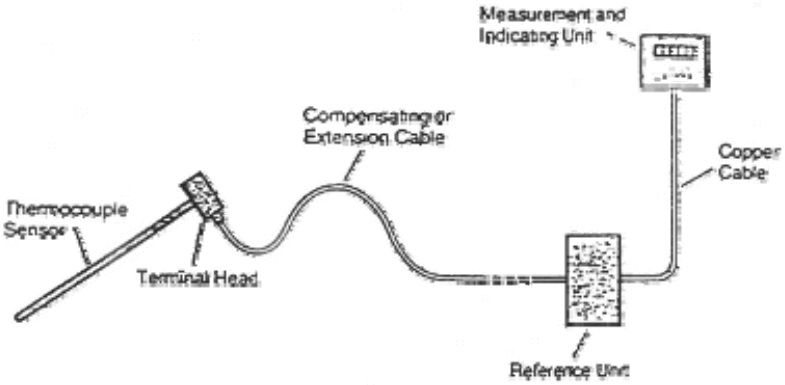


Types of measuring junction configuration

INSULATED: Hot junction insulated from sheath.
Gives floating output with typical insulation resistance in excess of 100 megohms.

GROUNDED: Hot junction welded to sheath tip giving earthed output and faster response to temperature changes.

EXPOSED. Fastest response mainly for the measurement of air temperature in ducts and continuous ovens. Restricted to a maximum operating temperature of 500°C.



3. Resistance Temperature Detectors

The resistance of a conductor usually increase as the temperature increase .if the properties of that conductor are known, the temperature can be calculated from the measured resistance. A resistance temperature can be calculated from the measured resistance temperature detector (RTD) is a conductor of known characteristics constructed for insertion into the medium for temperature measurement.

Any conductor can be used to construct an RTD, but a few have been identified as having more described characterstics than others. The characteristics which are desired include.

1. **Stability:** in the temperature range to be measured. The material must not melt, correde, embattle or change electrical characteristics when subjected to the environment in which it will operate.
2. **Linearity**: The resistance change with temperature should be as liner as possible over the rang of interst to simplify readout.
3. **High resistively**: Less material is needed to manufactor an RTD with a specified resistance when the matrial has a high characteristic resistively.
4. **Workability:** The material must be suitable for configuring for insertion into the media.

The materials which have been identified as having acceptable characteristics are: Copper, Nickel, tungsten and platinum.

Copper has good linearity, workability , and is able up to 250° F (120°C), but has low resistively, thus either a long conductor or one with a very small crose-sectional area is required for a reasonable resistance.

Nickel and nickel alloys have high resistively, good stability and good workability, but have poor linearity.

Tungsten is brittle and difficult to work with.

Platinum has been accepted as the materialwhich best fist all the criteria and has been generally accepted for industrial measurement between –300 and 1200° F (-150 and 650 °C). The effect of resistances inherent in the lead wires of the RTD circute on the temperature measurement can be minimised by increasing the resistance of the sensor; however, the sensor will also be increased. RTDS are commercially available with resistances from 50 to 1000 ohms at 32°F (0°C) and increase resistance 0.385 ohms for every °C of temperature rise.

This is called the European (E) standard and is in accordance with the DIN (Deutsche Instilut fuer Normung) 43760 Standard .

Chemically pure plantium has a rise of . 392 ohms per °C for a 100 ohm RTD in accordance with the american (A) standard.

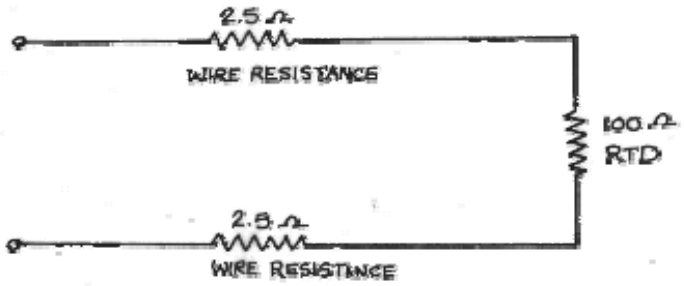
The European standard is dominant, Even in the United States the American standard Is seldom used.

When the resistance of the RTD is found by measurement, the temperature can be calculated:

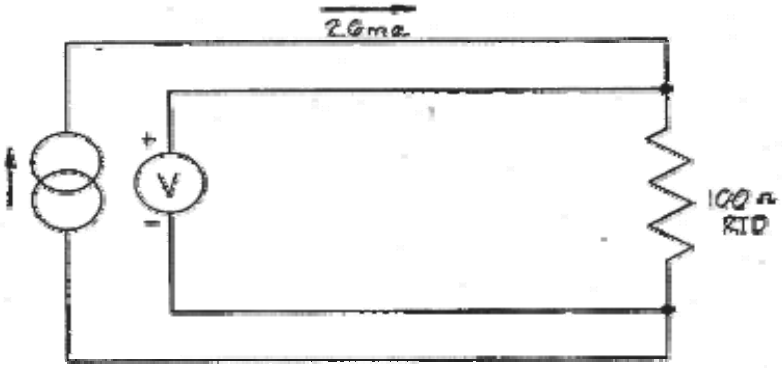
°C = (Ohms reading – 100) / 0.385

the accuracy of this calculation is determined primarily by the accuracy of the reading. Modem instruments can measure resistance very accurately and the temperature can be determined precisely if the resistance of the connecting circuit is insignificant or is known. Unfortunately , this resistances usually not negligible or known for most parcticl circuits. The wire that's usually used (16 AWG standed copper) has a resistance of approximately 4 ohms per 100 feet (305 m). If it is assumed that the RTD is connected to the instrument by a 625- foot cable as shown in figure , the total resistance will be 5 ohms lanrge than the RTD resistance, which will cause a 23,4 °F (13°C) error/ furthermore, copper wire has a temperature coefficient of about 0.0039 ohms /°C/so the reading will very about a degree for every 20° change in ambient temperature these errors can be compensated for by

measuring the resistance of every loop and keeping track of the ambient temperature, but fortunately there are better methods.



Schematic Two-Wire RTD Circuit



Schematic of four-wire RTD Circuit

The most accurate method for connecting the RTD is with four wires as shown in Figure 8. a constant current source forces a known current through the RTD, which for this discussion will be assumed to be 2.6 milliamperes. The resistance of the wires conducting this current does not need to be known.

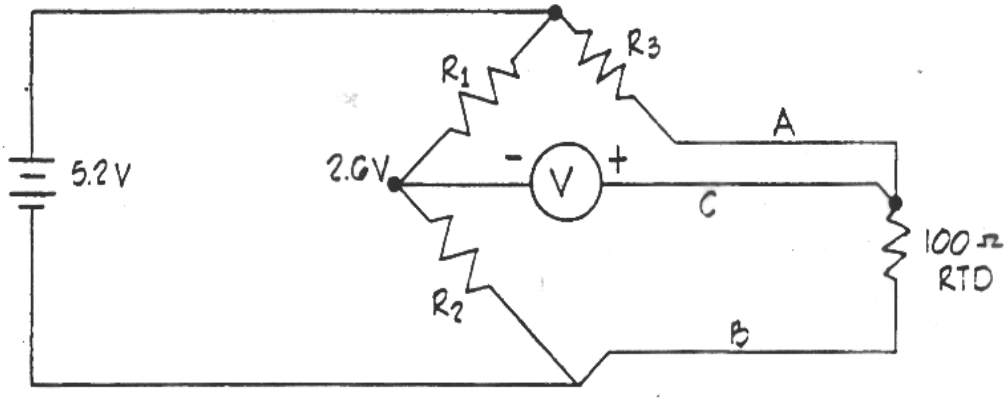
By ohm's law, the voltage across the RTD will be this current multiplied by the resistance of the RTD. This voltage is measured by a high-impedance voltmeter on the other set of wires. The voltmeter will read 260 millivolts when the temperature of the RTD is 32°F (0°C) and its resistance is 100 ohms.

For every 1.8 °F (1°C) of temperature rise, the resistance of the RTD will increase 0.385 ohms, which when multiplied by the 2.6 milliamperes flowing increases the

voltage b 1 millivolt and the temperature in °C can be read by subtracting 260 from the reading.

The resistance of the leads is not important, Even in relation to each other as long as the current can be maintanined and the resistances of the leads are small compared to the resistance of the voltmeter.

Use of 4-wire RTD circuits is usually limited to laboratories and situations where very high accuracy is desired because less expensive 3-wire circuite almost always provide the needed accuracy.



Schematic Of Three-Wire RTD Circuit With A Balanced Bridge

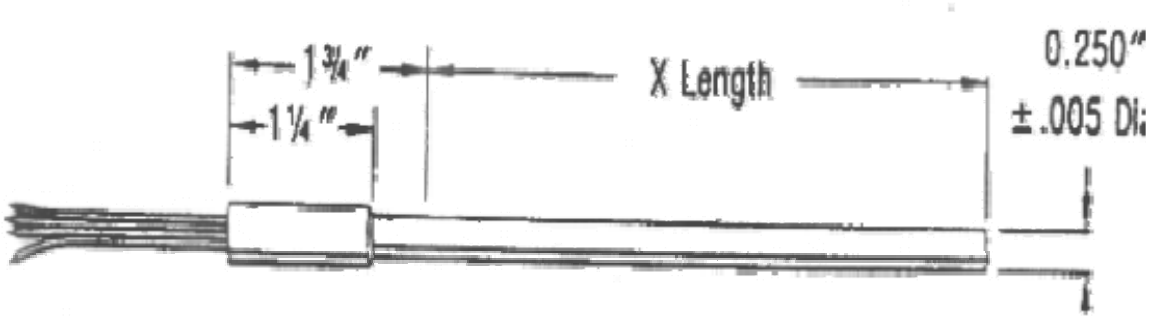
A compromise connection method for RTD that uses three wires and a balanced bridge circuit is shown above. For this circuit, R1 and R2 are selected to be the same resistance so that the voltage at the negative terminal of the voltmeter is one half o the supply voltage. R3 is selected to be the same resistance as the RTD at the base temperature, 100 ohms if 0°C is used as the base. For this circuit , it is important that wire a and wire b have the same resistance . the usual practice is to run the three wires as a shielded raid, thus they will all be the same length and the same resistance Within manufacturing tolerance.

At the base condition, the positive terminal will also see one-half of the supply voltage and the reading will be zero. If 5.2 volts is used to power the bridge, the

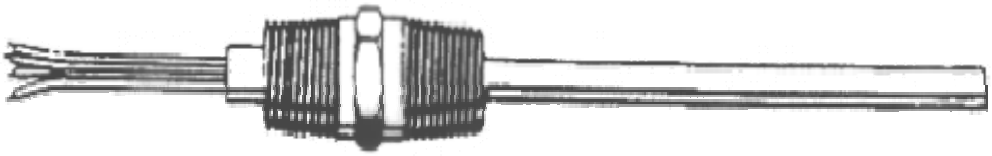
voltage will be 2.6 at each terminal of the voltmeter at the base temperature. When the temperature of the DTD is raised one degree C, the voltage reading will increase to one millivolt. The symmetry will be upset as the reading moves away from the base temperature and the one millivolt per degree will not continue to be exact, but various schemes of completion are available to give an acceptable reading.

The proceeding paragraphs are intended to explain the basis of two, three and four wire RTD connections. The selection of resistors and compensation schemes are left to the manufacturer of the instrument, but the facilities engineer selects which of the connection methods to use. The three wire method is the proper selection for virtually all production facility applications.

Resistance temperature detectors (RTDS) are the most frequently used electronic temperature sensors for production facilities. The industry has standardised on RTDS that are calibrated to Din standard 43760 which is also known as the European standard RTDS which meet this standard measure 100 ohms at 0°C, are made of platinum and exhibit a resistance increase of 0.385 ohms per °C temperature increase. Another stadard, called the American Standard, is available but is not in wide use, even in the United States . typical RTDS Are shown below.



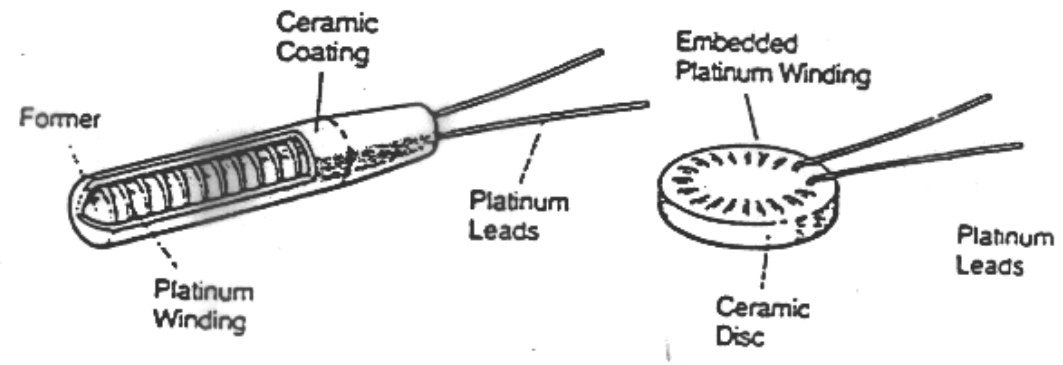
Capsule Sensor for Compression Fitting

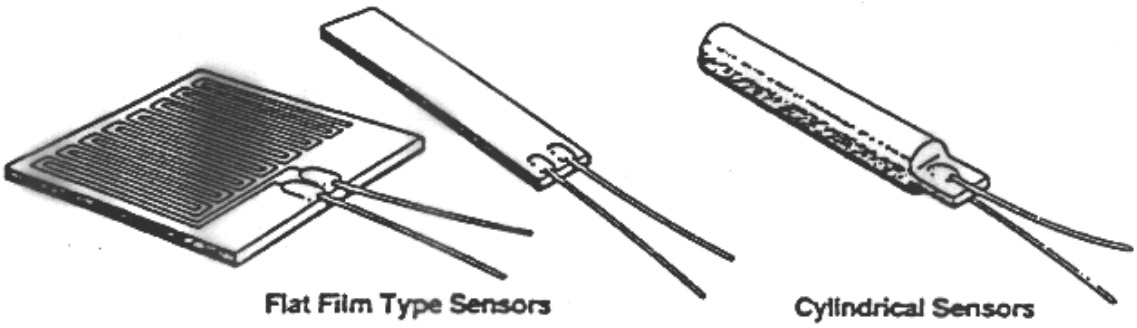


Sheath Silver Soldered to Fitting

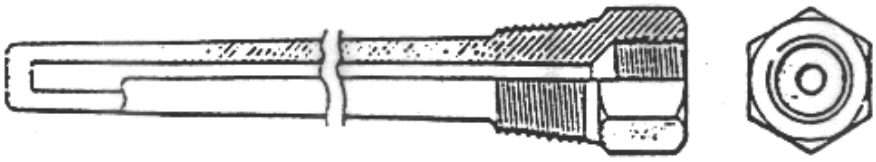
Spring Loaded Sensor

Various Methods for attaching an RTD Or Themocouple Sheath to a Thermowell fitting

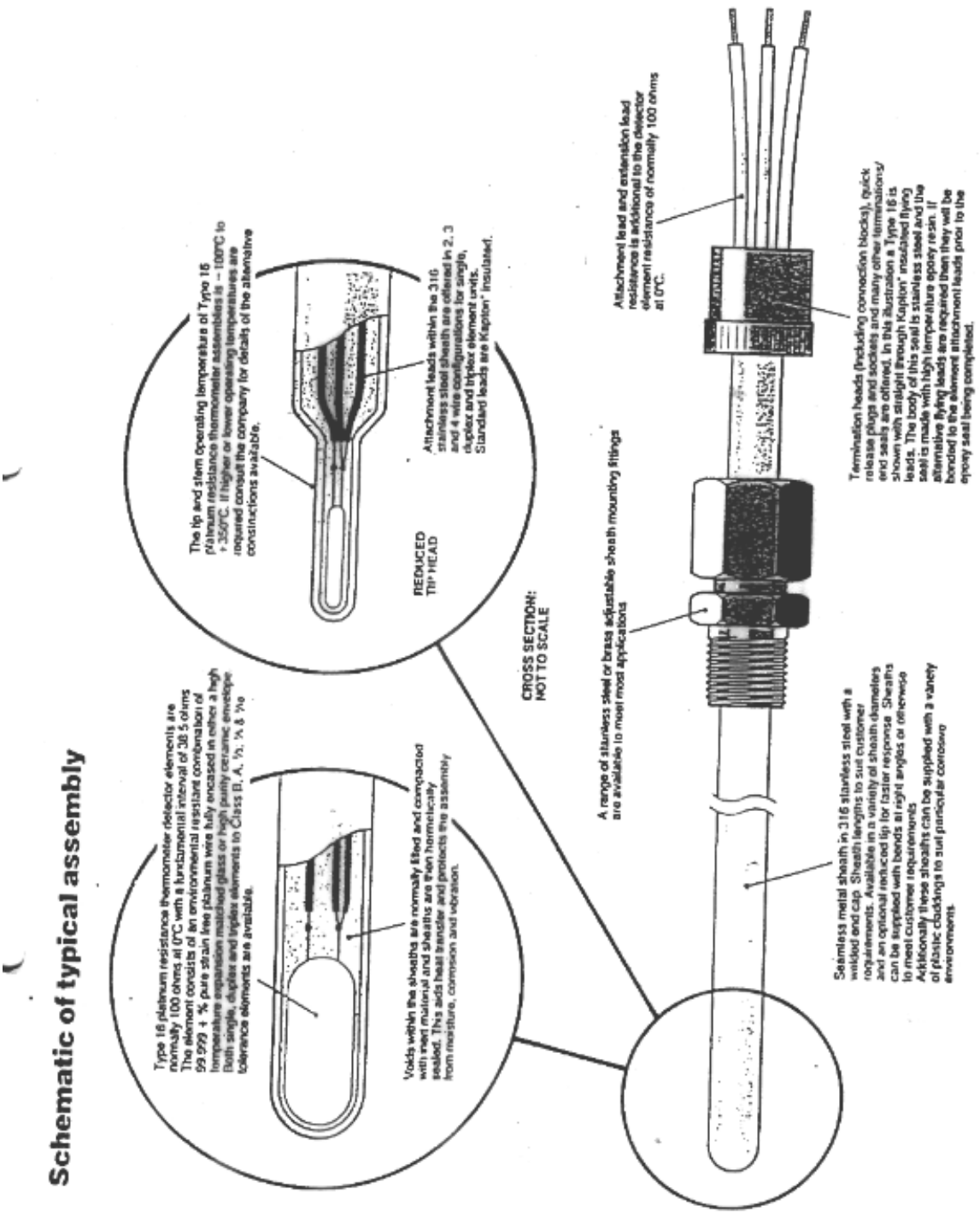




Resistance Sensor Protection Tube



Thermowell


Schematic of typical assembly
Type 18 platinum resistance thermometer detector elements are normally 100 ohms at 0°C with a fundamental interval of 38.5 ohms. The element consists of an environmental resistant combination of 99.999 + % pure strain free platinum wire fully encased in either a high temperature expansion matched glass or high purity ceramic envelope. Both single, duplex and triplex elements to Class B, A, ½, ⅓ & ⅙ tolerance elements are available.
Voids within the sheaths are normally filled and compacted with inert material and sheaths are then hermetically sealed. This aids heat transfer and protects the assembly from moisture, corrosion and vibration.
CROSS SECTION: NOT TO SCALE
REDUCED TIP HEAD
The tip and stem operating temperature of Type 18 Platinum resistance thermometer assemblies is – 100°C to +350°C. If higher or lower operating temperatures are required consult the company for details of the alternative constructions available.
Attachment leads within the 316 stainless steel sheath are offered in 2, 3 and 4 wire configurations for single, duplex and triplex element units. Standard leads are Kapton® insulated.
Attachment lead and extension lead resistance is additional to the detector element resistance of normally 100 ohms at 0°C.
Termination heads (including connection blocks), quick release plugs and sockets and many other terminations/ end seals are offered. In this illustration a Type 18 is shown with straight through Kapton® insulated flying leads. The body of this seal is stainless steel and the seal is made with high temperature epoxy resin. If alternative flying leads are required then they will be bonded to the element attachment leads prior to the epoxy seal being completed.
A range of stainless steel or brass adjustable sheath mounting fittings are available to meet most applications
Seamless metal sheath in 316 stainless steel with a welded end cap. Sheath lengths to suit customer requirements. Available in a variety of sheath diameters and an optional reduced tip for faster response. Sheaths can be supplied with bends at right angles or otherwise to meet customer requirements.
Additionally these sheaths can be supplied with a variety of plastic claddings to suit particular corrosive environments.

Pt100 – Resistance Vs Temperature table

Resistance vs Temperature Relationship over the range −200°C to +850°C for Platinum Resistance Thermometer Detector Element

Ohms BS 1904:1964 (Superseded 1984)	Temp-erature	Ohms IEC 751:1983 (BS 1904:1984, DIN43760:1980)				
0	°C	0	2	4	6	8
18.56	—200	18.49				
22.83	—190	22.80	21.94	21.08	20.22	19.36
27.11	—180	27.08	26.23	25.37	24.52	23.66
31.34	—170	31.32	30.47	29.63	28.78	27.93
35.54	—160	35.53	34.69	33.85	33.01	32.16
39.72	—150	39.71	38.88	38.04	37.21	36.37
43.88	—140	43.87	43.04	42.21	41.38	40.55
48.01	—130	48.00	47.18	46.35	45.52	44.70
52.12	—120	52.11	51.29	50.47	49.64	48.82
56.21	—110	56.19	55.38	54.56	53.74	52.92
60.28	—100	60.25	59.44	58.63	57.82	57.00
64.32	—90	64.30	63.49	62.68	61.87	61.06
68.34	—80	68.33	67.52	66.72	65.91	65.11
72.35	—70	72.33	71.53	70.73	69.93	69.13
76.34	—60	76.33	75.53	74.73	73.93	73.13
80.32	—50	80.31	79.51	78.72	77.92	77.13
84.29	—40	84.27	83.48	82.69	81.89	81.10
88.23	—30	88.22	87.43	86.64	85.85	85.06
92.16	—20	92.16	91.37	90.59	89.80	89.01
96.09	—10	96.09	95.30	94.52	93.73	92.95
100.00	—0	100.00	99.22	98.44	97.65	96.87
100.00	0	100.00	100.78	101.56	102.34	103.12
103.90	10	103.90	104.68	105.46	106.24	107.02
107.79	20	107.79	108.57	109.35	110.12	110.90
111.67	30	111.67	112.45	113.22	113.99	114.77
115.54	40	115.54	116.31	117.08	117.85	118.62
119.40	50	119.40	120.16	120.93	121.70	122.47
123.24	60	123.24	124.01	124.77	125.54	126.31
127.07	70	127.07	127.84	128.60	129.37	130.13
130.89	80	130.89	131.86	132.42	133.18	133.94
134.70	90	134.70	135.46	136.22	136.98	137.74
138.50	100	138.50	139.26	140.02	140.77	141.53
142.29	110	142.29	143.04	143.80	144.55	145.31
146.06	120	146.06	146.81	147.57	148.32	149.07
149.82	130	149.82	150.57	151.33	152.08	152.83
153.57	140	153.58	154.32	155.07	155.82	156.57
157.31	150	157.31	158.06	158.81	159.55	160.30
161.04	160	161.04	161.79	162.53	163.27	164.02
164.76	170	164.76	165.50	166.24	166.98	167.72
168.46	180	168.46	169.20	169.94	170.68	171.42
172.16	190	172.16	172.90	173.63	174.37	175.10
175.83	200	175.84	176.57	177.31	178.04	178.78
179.50	210	179.51	180.24	180.97	181.71	182.44
183.16	220	183.17	183.90	184.63	185.36	186.09
186.82	230	186.82	187.54	188.27	189.00	189.72
190.45	240	190.45	191.18	191.90	192.63	193.35
194.07	250	194.07	194.80	195.52	196.24	196.96
197.69	260	197.69	198.41	199.13	199.85	200.57
201.29	270	201.29	202.01	202.72	203.44	204.16
204.88	280	204.88	205.59	206.31	207.02	207.74
208.46	290	208.45	209.17	209.88	210.59	211.31

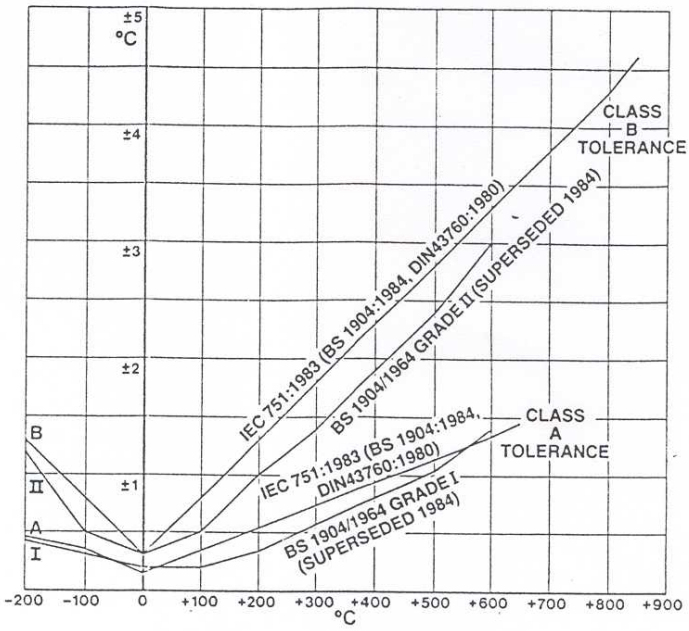
Nominal Resistance: 100 ohms @ 0°C
Fundamental Interval: 38.5Ω
For measuring circuit schematics see page 6, section 3.3

Resistance vs Temperature Relationship over the range −200°C to +850°C for Platinum Resistance Thermometer Detector Element

Ohms BS 1904:1964 (Superseded 1984)	Temp-erature	Ohms IEC 751:1983 (BS 1904:1984, DIN43760:1980)				
0	°C	0	2	4	6	8
212.02	300	212.02	212.73	213.44	214.15	214.86
215.58	310	215.57	216.28	216.99	217.70	218.41
219.12	320	219.12	219.82	220.53	221.24	221.94
222.26	330	222.65	223.35	224.06	224.76	225.46
226.18	340	226.17	226.87	227.57	228.27	228.97
229.69	350	229.67	230.37	231.07	231.77	232.47
233.19	360	233.17	233.87	234.56	235.26	235.96
236.68	370	236.65	237.35	238.04	238.74	239.43
240.16	380	240.13	240.82	241.51	242.20	242.90
243.61	390	243.59	244.28	244.97	245.66	246.35
247.08	400	247.04	247.73	248.41	249.10	249.79
250.52	410	250.48	251.16	251.85	252.53	253.22
253.95	420	253.90	254.59	255.27	255.95	256.63
257.37	430	257.32	258.00	258.68	259.36	260.04
260.77	440	260.72	261.40	262.08	262.76	263.43
264.17	450	264.11	264.79	265.47	266.14	266.82
267.56	460	267.49	268.17	268.84	269.51	270.19
270.94	470	270.86	271.53	272.20	272.88	273.55
274.29	480	274.22	274.89	275.56	276.23	276.89
277.64	490	277.56	278.23	278.90	279.56	280.23
280.98	500	280.90	281.56	282.23	282.89	283.55
284.31	510	284.22	284.88	285.54	286.21	286.87
287.63	520	287.53	288.19	288.85	289.51	290.17
290.93	530	290.83	291.49	292.14	292.80	293.46
294.22	540	294.11	294.77	295.43	296.08	296.74
297.50	550	297.39	298.04	298.70	299.35	300.00
300.76	560	300.65	301.31	310.96	302.61	303.26
304.02	570	303.91	304.56	305.20	305.85	306.50
307.27	580	307.15	307.79	308.44	309.09	309.73
310.51	590	310.38	311.02	311.66	312.31	312.95
313.72	600	313.59	314.24	314.88	315.52	316.16
316.93	610	316.80	317.44	318.08	318.72	319.36
320.12	620	319.99	320.63	321.27	321.91	322.54
323.31	630	323.18	323.81	324.45	325.08	325.72
326.5	640	326.35	326.98	327.61	328.25	328.88
329.6	650	329.51	330.14	330.77	331.40	332.03
332.8	660	332.66	333.28	333.91	334.54	335.17
335.9	670	335.79	336.42	337.04	337.67	338.29
339.1	680	338.92	339.54	340.16	340.79	341.41
342.2	690	342.03	342.65	343.27	343.89	344.51
345.4	700	345.13	345.75	346.37	346.99	347.60
348.4	710	348.22	348.84	349.45	350.07	350.69
351.5	720	351.30	351.91	352.53	353.14	353.75
354.5	730	354.37	354.98	355.59	356.20	356.81
357.7	740	357.42	358.03	358.64	359.25	359.86
360.6	750	360.47	361.06	361.68	362.29	362.89
363.6	760	363.50	364.10	364.71	365.31	365.91
366.6	770	366.52	367.12	367.72	368.32	368.93
369.7	780	369.53	370.13	370.73	371.33	371.93
372.7	790	372.52	373.12	373.72	374.32	374.91
375.6	800	375.51	376.10	376.70	377.29	377.89
378.7	810	378.48	379.08	379.67	380.26	380.85
381.6	820	381.45	382.04	382.63	383.22	383.81
384.6	830	384.40	384.98	385.57	386.16	386.75
387.5	840	387.34	387.92	388.51	389.09	389.68
390.4	850	390.26				

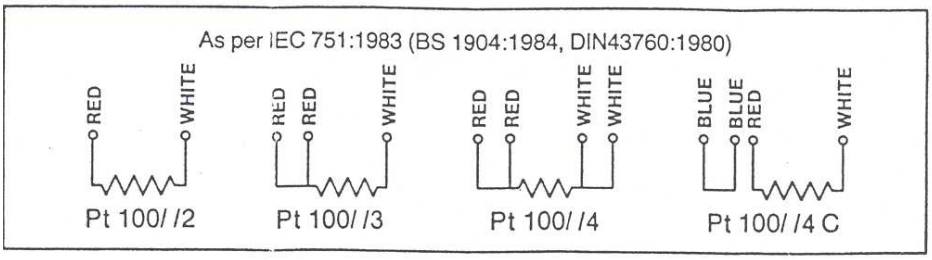
Resistance vs Temperature Relationship over the range −200°C to +850°C for Platinum Resistance Thermometer Detector Element

Tolerance values for 100 ohm elements, 38·5 ohm fundamental interval



Temperature °C	Tolerance IEC 751:1983 (BS 1904:1984, DIN43760:1980)			
	Class A		Class B	
	± °C	± Ohms	± °C	± Ohms
−200	0.55	0.24	1.3	0.56
−100	0.35	0.14	0.8	0.32
0	0.15	0.06	0.3	0.12
100	0.35	0.13	0.8	0.30
200	0.55	0.20	1.3	0.48
300	0.75	0.27	1.8	0.64
400	0.95	0.33	2.3	0.79
500	1.15	0.38	2.8	0.93
600	1.35	0.43	3.3	1.06
650	1.45	0.46	3.6	1.13
700	—	—	3.8	1.17
800	—	—	4.3	1.28
850	—	—	4.6	1.34

Typical Connection Schematics for 2, 3 & 4 wire lead configuration for Resistance Thermometer Detectors



RTD/Thermocouple Comparison

	RTD	Thermocouple
Advantage	• Potentially the most accurate method. • Simple installation. • Needs only copper cables for long runs.	• Wide temperature range. • Versatile. • Simple application, just the tip needs to take up the required temperature.
Disadvantage	• Needs energising current. • Sensor types limited.	• Needs a temperature reference. • Needs extension cables for long runs.

Thermocouples measure the temperature at their tip only and are therefore faster to respond than RTDs.

RTDs are the most stable and the most accurate at moderate temperatures.

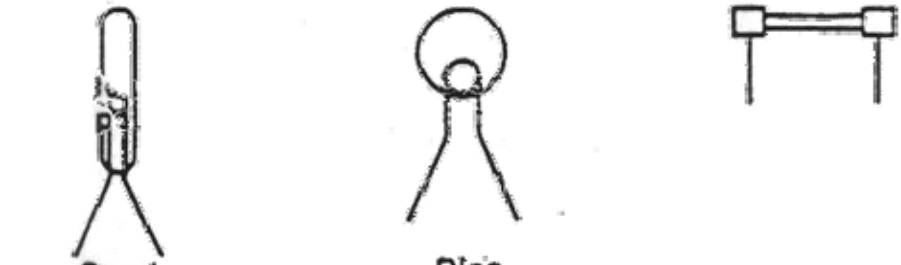
RTDs are less susceptible to electrical noise.

RTDs are relatively expensive compared to thermocouples and have a slow response since the whole device averages the temperature over the element.

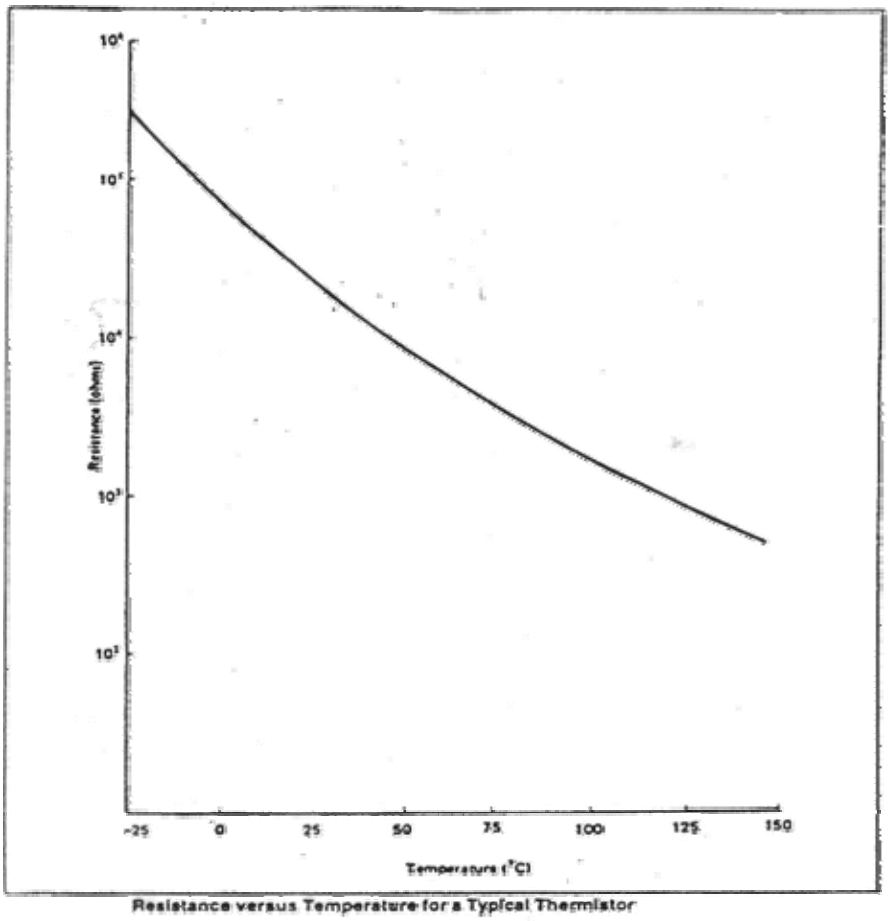
Thermistors

Thermistors are resistance temperature elements made from a semiconductor material and basically do the same job as an RTD. These elements generally have a negative Temperature coefficient (NTC) but positive temperature coefficients are also available over a limited range.

The advantage of a Thermistor is it is highly sensitive to temperature changes making them useful in temperature trip alarms. Unfortunately they posses highly non-liner resistive properties which restrict their useful range.



Commercial Thermistor Sensors



Resistance versus Temperature for a Typical Thermistor

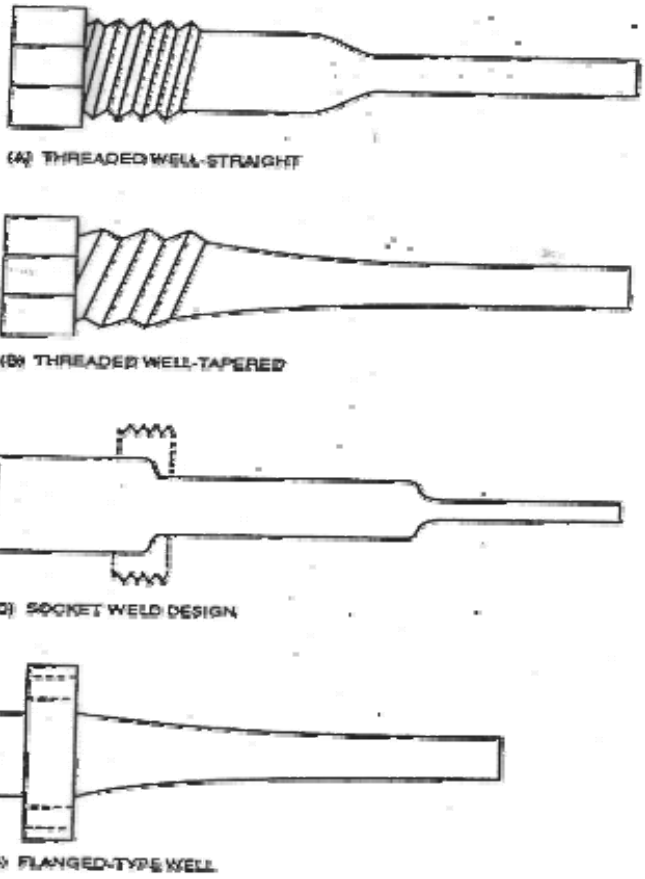
Thermowells

Thermowells are used to protect the detector and so that the detector can be changed without interrupting the process. One downside of using a thermowell os the time delay it introduces into the measurement system due to thermal lag.
Thermowlls should be installed where a good representative sample of the process fluid temperature can be measured.

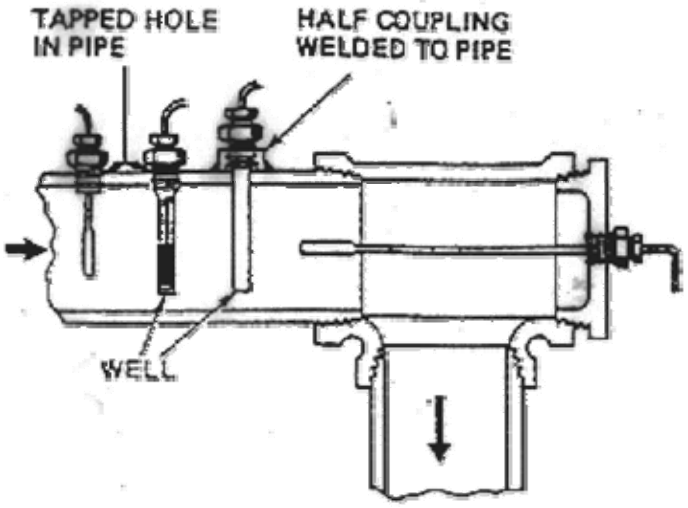
The optimum immersion length of a thermowell depends on the application

- If the well is installed perpendiculat to the line, the tip of the well should be between one half and one third of the pipe diameter.

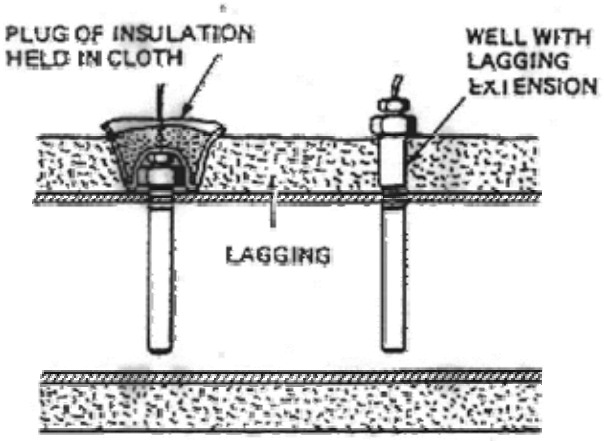
If the well is installed in an elbow, the tip should point towards the flow.
The speed of response of a sensor in a thermowell will be slower than that of an unprotected buib. Keeping the clearance between bulb and pocket down to an absolute minimum and filling the space with oil or glycol (antifreeze) can reduce this effect.

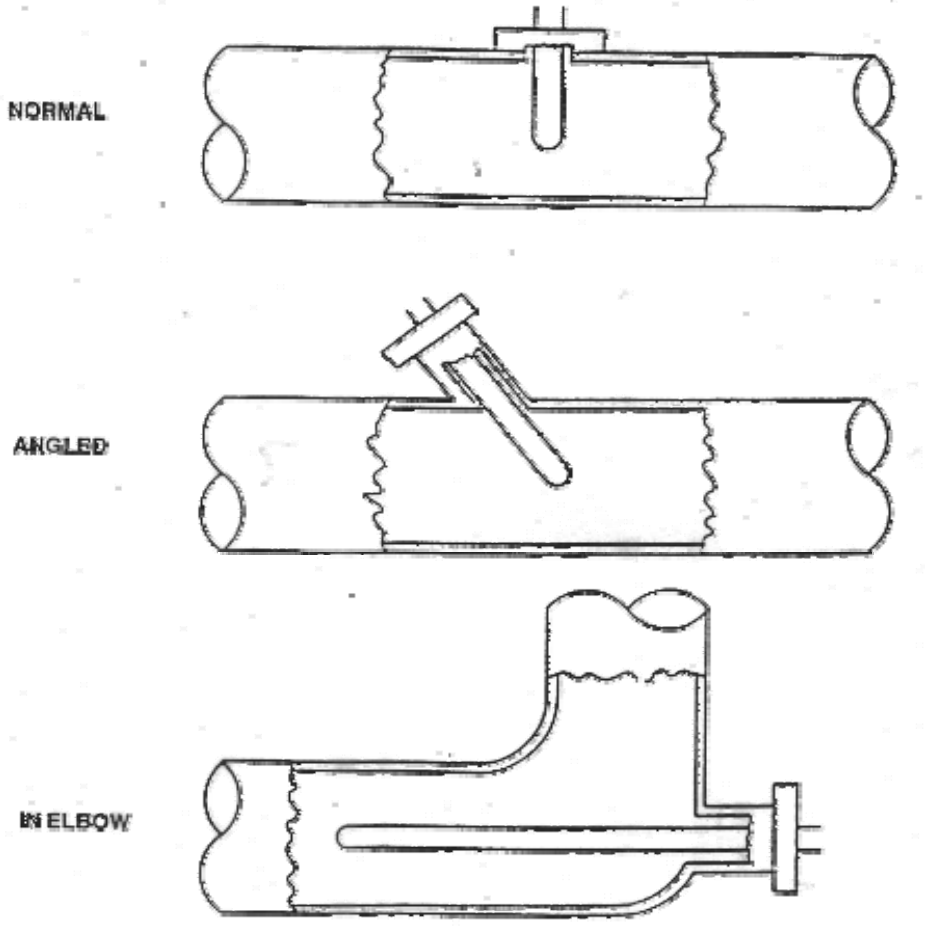


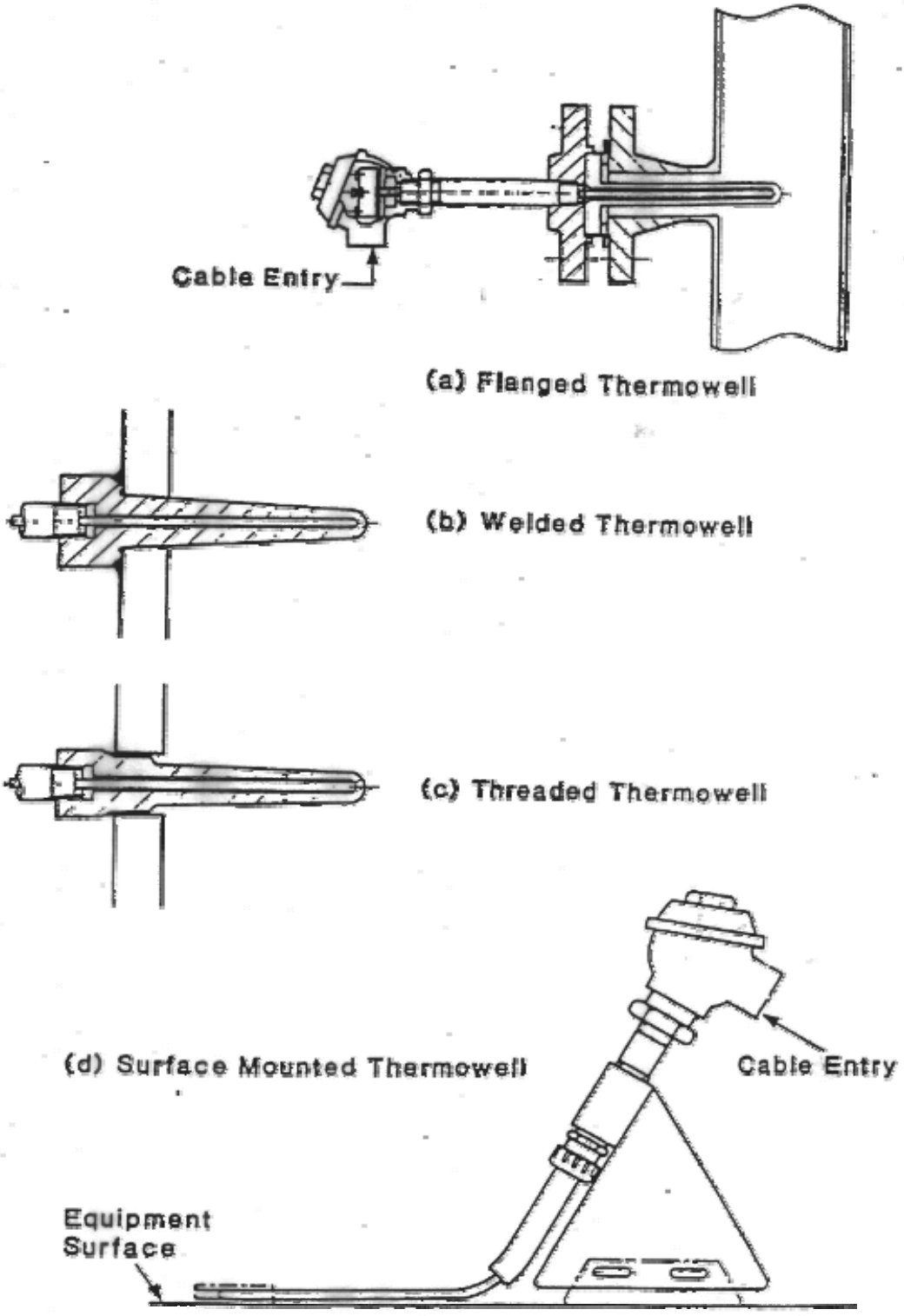
IN PIPE LINE, USING BUSHING OR WELL

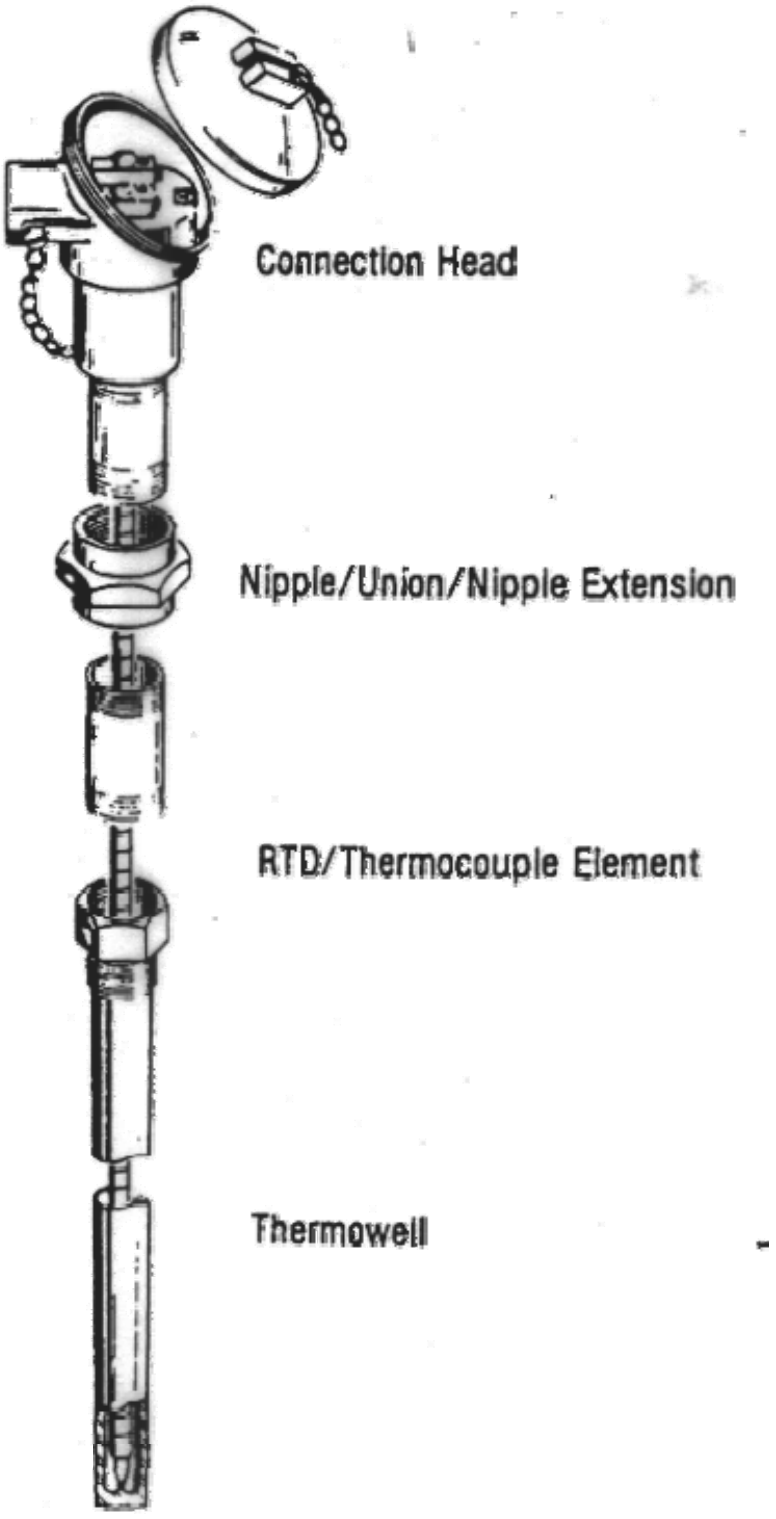


IN LAGGED (INSULATED) PIPE LINE






Cable Entry
(a) Flanged Thermowell
(b) Welded Thermowell
(c) Threaded Thermowell
(d) Surface Mounted Thermowell
Cable Entry
Equipment
Surface



A Nipple-Union-Nipple Extension Assembly for Installing an RTD or Thermocouple Element into a Thermowell

4. Pyrometer

These are non-contact measurement systems and are especially suited for measurement on poor heat conductors such as:

- Ceramics
- Plastics…etc

For example a contact probe can only measure temperature accurately if it is the same temperature as the object/process being measured. With poor heat conductors this would be impossible or the response times would be too long.

Pyrometers are also useful for measuring temperatures of:

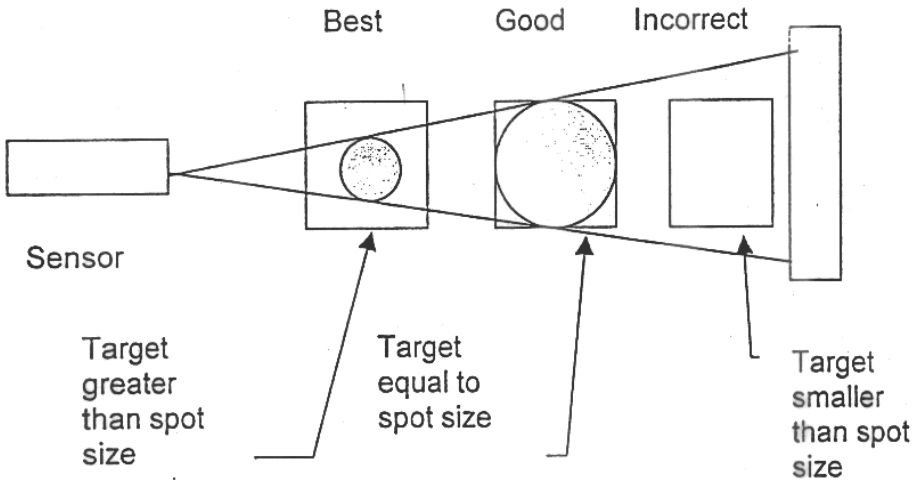
- Moving parts
- Parts that cannot be touched or are out of reach
- Live parts
- Very small items

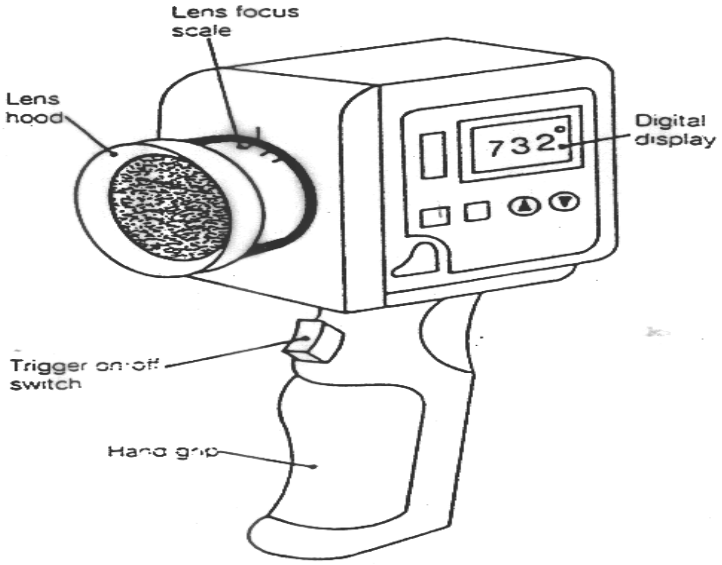
Operational Aspects

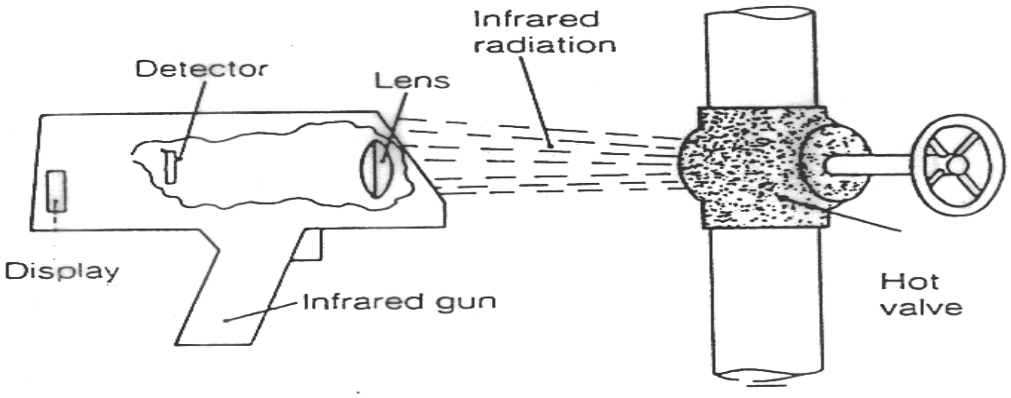
Some of the points that should be considered when using a pyrometer are:

- What is the temperature range of the process?
- What is the size of the target?
- How close to the target can the instrument be installed?
- Does the target fill the field of view?
- What is the target material?
- How fast is the process moving?
- What is the ambient temperature?
- Are the ambient conditions contaminated with dust, smoke or steam?

For accurate temperature when using a pyrometer, the target should be larger than the instruments field of view or spot size. If the spot size is larger than the target, the energy emitted from the background or other surrounding objects will also be measured.


Best
Good
Incorrect
Sensor
Target greater than spot size
Target equal to spot size
Target smaller than spot size


Lens focus scale
Lens hood
Digital display
732°
Trigger on-off switch
Hand grip


Infrared radiation
Detector
Lens
Display
Infrared gun
Hot valve

Burnout Protection

As the temperature sensors are continuously subjected to process temperature, there is a likely chance of failure due to excess temperature or mechanical damage. In either event, the RTD or thermocouple circuit will be open, discontinuing the electrical path. This is called burnout. It is a facility provided within the receiving instruments like the recorders, indicating, controllers, to respond for such loss of input signals.

Burnout Protection up Scale

When the input to the instrument is disconnected the instrument shows a range maximum value.

Burnout Protection down Scale

When the input signal wiring to the instrument is disconnected. The instrument shows scale minimum value. The associated circuits with the above instrument like switches-alarm; shutdown logic fail safe option is selected in Burnout Protection.

Temperature Transmitters

Temperature transmitters are used when it is necessary to convert the signal from a temperature sensor to one of the standard signals for transmission over a long distance or interface with other instruments.

The transmitter output (signal) is usually 4 to 20 mA. for electronic transmission and 3 to 15 psig (20 to 100 kpa) for pneumatic transmitter. Other signals can be used if required by the receiver, but these are most common and should be used if possible. It is also possible to bring a temperature measurement into a control room without using a transmitter A thermocouple RTD can be wired directly to an instrument in the control room and this is acceptable practice.

Temperature transmitters for new installations are predominantly electronic with 4 to 20 mA output. Inputs to the transmitters are from thermocouples or RTDS. These transmitters can be mounted in the field and on the thermo-well or in the field on a support and connected to the sensor by a cable.

Temperature transmitter mounted in the field must be protected from the elements by an appropriate housing. A weatherproof (INEMA 4) housing is adequate for m most applications, even in Division 2 hazardous area because there are no arching contacts in a typical temperature transmitter. An explosion-proof (NEMA 7) housing is required for Division 1 area unless the installation is certified intrinsically safe. The energy level required in temperature transmitters is such that they can be used in intrinsically safe installations if isolated from the power supply and receiver by approved barriers and approved by an agency recognized in the country where installed.

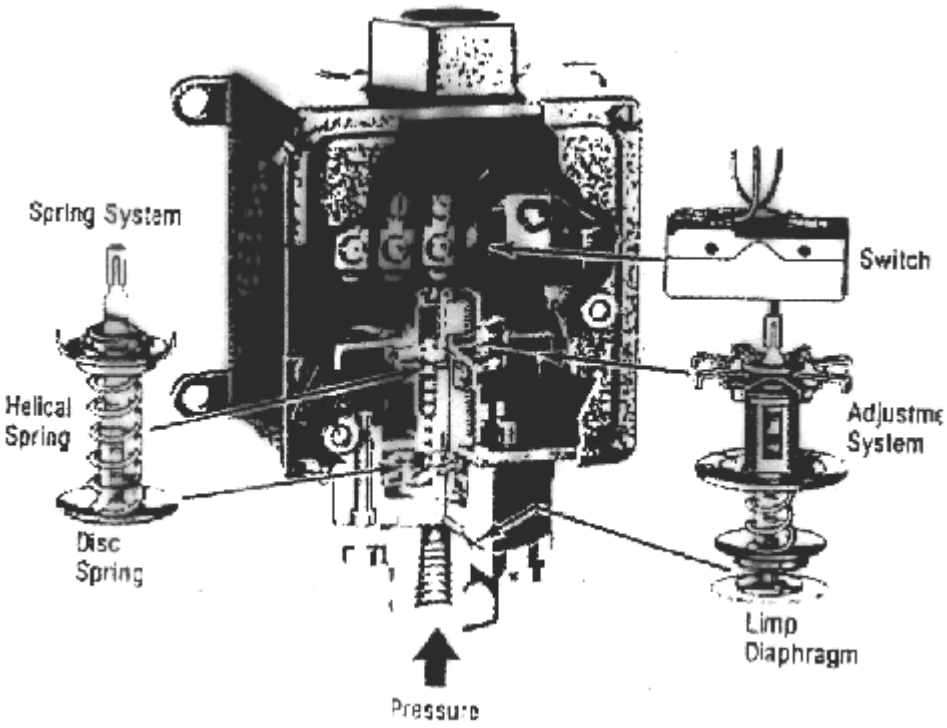
Electrical Temperature Switches

An electric temperature switch is a device, which causes a contact to open or close with a change in temperature. Most switches can be used as either high temperature or low temperature sensors, depending on how they are calibrated and electrically connected.

Mechanically operated temperature switches are used more frequently in production facilities most mechanically operated temperature switches use a vapor–filled system or a liquid–filled system to operate pressure switch. Gas-filled systems generally do not develop enough power for switch use.

Filled system switches are available for both local and remote mounting. The local mounting type has the bulb rigidly attached to the switch mechanism and housing. The assembly has a threaded connection so that it cab be screwed into and be supported by a thermowell. The remote mounting type has the bulb connection to

the switch mechanism by a capillary tube from 6 feet (2 meters) to 25 feet (8 meters) or more longs. The local mounting type is less expensive to purchase and install, while the remote mounting type provides isolation of the switch from process vibration and more convenient access. The switch cannot be separated from the bulb in the field for either these design.



SGC HSE Regulation Related to:

Refer to HSE Regulation No. 7 "Isolation"

7.18 Control Systems Procedures and Isolations

1. Preparation for Work

When work is to be done on control engineering hardware, it is essential that the work-site is prepared correctly. The work permit system (Regulation No 06) provides the mechanism for ensuring that essential preparatory activity is documented and witnessed. It is important that the Control Engineering Person doing the work also has responsibility for effecting the task safely. This is particularly important in the control engineering discipline where the instrument may still be connected to the process, or may require to be serviced with power-on for fault finding.

2. Interaction

Care must be taken to ensure that the work to be carried out on a specific item of instrumentation will not cause a hazard due to interaction with other protection systems or operational process controls.

3. Preparatory Work

Prior to a work permit being issued all appropriate preparatory work at the site must be completed. The following items are some examples.

a. Removal of potential hazards from the area. Particular vigilance is needed for enclosed areas (refer to Regulation No 09 Confined Space Entry).
b. **Gas** testing the area for flammable, toxic or suffocating gases.
c. Construction of scaffolding to permit safe access.
d. Provision of additional firefighting apparatus.
e. Provision of necessary protective clothing.
f. Isolation of the control engineering hardware from the process.

g. In the case of equipment removal, isolation of the hardware from utilities (eg electricity, air supplies etc).

On completion of all necessary preparatory work (defined by the Senior Control Engineering Person and the Area Authority), a hot or cold work permit/entry permit will be issued signed by the appropriate responsible authorities.

4. Isolation of Hardware

Isolation of control engineering hardware may be necessary to enable maintenance work to be done or permit removal of the hardware to effect repairs (either locally or remotely). Isolation of hardware can take several forms, for example isolation from:

a. Process plant.
b. Utilities (electric, pneumatic, hydraulic, cooling media etc).
c. Larger system of which the hardware is a subsystem or component.

5. Isolation from Process

a. Isolation of instruments, which are connected to or form a part of the process is usually achieved by valving. It is important that, where isolation of an instrument is required for maintenance purposes, correct venting/draining and valve closure procedures are adhered to.
b. Where instruments have local isolating valves in addition to the primary process isolating valves, the local valves may be used for some routine in-situ testing at the discretion of the Senior Control Engineer. If an instrument is to be removed from site, the process isolating valves must be used and any impulse pipework must be drained or vented completely.
c. Where the process fluids are of a hazardous nature (eg toxic, flammable etc), particular care should be taken to ensure correct venting and draining, and also to clean or flush the instrument carefully, prior to

effecting work or removal of the hardware from site for maintenance or repair. Gas testing may be required. On large items, eg control valves, a certificate of cleanness is necessary prior to delivery to workshops.

d. On removal of a directly mounted instrument, from a process line containing hazardous fluids, eg pressure gauges etc, isolation by the primary isolation valve only is NOT acceptable. The valve outlet shall be blanked off, capped or plugged with a blank flange, solid screwed plug or cap, whichever is appropriate.

6. Isolation from Electrical/Pneumatic Supplies

a. If practical, equipment must be made safe before any work is done on it. The operation of making the equipment safe must be done by a Competent Control Engineering Person. Care shall be taken when working on live equipment to ensure avoidance of contact with live electrical components (refer to Regulation No 19 Working with Electricity).

b. Pneumatically operated equipment must be isolated before it is disconnected or removed for repair by closing the valve at the supply manifold for the individual instrument and venting through the drain/vent of the pressure regulators.

7. Isolation from Utilities

a. Control engineering equipment may be connected to utilities (other than electrical associated with the hardware eg stream, cooling water, hydraulic fluid, chemicals, carrier gases (analysis) and air supplies. It is important that attention is given to rendering the utilities safe when the control engineering hardware is being serviced or removed.

b. Utilities should be isolated at the point of distribution to the control engineering equipment being removed (e.g isolating valve at distribution head) and not solely at the hardware itself.

c. Where utility fluids are 'piped' to an instrument, the pipe work should be drained down or vented if the instrument is removed.

d. It is important that removal of a utility from a specific piece of hardware does not influence any other hardware to which the utility may also be connected (eg cooling water may have been series connected to more than one item of hardware).

8. Use of Tools and Test Equipment

a. Tools and test equipment must be suitable for use in the work area. They should be checked before and after use and all calibration equipment should itself be calibrated periodically, at intervals determined by the Senior Control Engineering Person.

b. Use of tools and test equipment are subject to the Work Permit System (refer to Regulation No 06). It is particularly relevant to ensure that electrical tools and test equipment comply with the area safety classification of the work place. This may be achieved either by certification or using the Permit to Work.

9. Workshop Practice.

It is essential that good workshop practice is adhered to at all times, for example:

a. Machinery will not be operated without guards or suitable personal protection.

b. All workshop equipment will be maintained in good working order.

c. Good housekeeping is essential to safety.

d. All necessary spares, cleaning materials, tools and test equipment should be available and must be correctly maintained, tested (where appropriate) and stored.

10. Changes and Modifications

Changes to alarm settings and transmitter ranges must be approved by the operating authorities and documented. No such change will be made unless the Passport Work Order has been completed. Changes to trip settings should only be carried out under the authority of a PMR endorsed by the ED. This is mandatory at all times. Similarly, no modification will be undertaken unless the proposal has been through the established PMR and appropriate authorisation. Thus, before work starts, the job must be approved and finance made available.

19.7.4 Precautions on Low Voltage Systems

1. The consequences of shock, or serious burns, from short circuits associated with low voltage systems (50 - 1000V ac/120 - 1500V dc between conductors, or 50 - 600V ac/120 - 900V dc between conductor and earth) can be serious and often fatal. Whenever possible therefore, work on low voltage equipment and cables shall be carried out after they are proved **DEAD** by use of an approved instrument and where appropriate **EARTHED** using an Electrical Isolation Certificate (refer to Paragraph 19.8).
2. If it is not possible to make **DEAD,** to prove **DEAD** and where appropriate **EARTH** low voltage systems, work on them shall be carried out as if they were **LIVE** using a Sanction For Test Certificate (refer to Paragraph 19.9).

19.7.5 Precautions on Extra Low Voltage Systems

1. Control and telecommunications plant operating at extra low voltage (< 50V ac/120V dc between electrical conductors or to earth) shall not be worked on without an Electrical Isolation Permit being issued. This is necessary to prevent the possibility of sparks in a hazardous area (refer to Paragraph 19.8).
2. Battery systems with high stored energy can be dangerous to personnel and therefore precautions should be taken when working with such systems. In particular flooded cells requiring electrolyte replacement are hazardous. Where these types of cells exist, a local procedure should be produced for work on battery systems.

MODULE 3:

CONTROLLERS AND CONTROL THEORY

CONTROLLERS AND CONTROL THEORY

Objectives

At completion of this module, the developee will have understanding of:

1. Basic objectives of the process control system.
2. Control loop components and functions.
3. Basic requirements of an open control loop.
4. Basic requirement of a closed control loop.
5. Semi-automatic control and full automatic control.
6. Control modes.
7. ON-OFF mode advantages and disadvantages.
8. Proportional only control function and reaction curve.
9. Proportional band setting effects on control loop.
10. Direct action and reverse action control.
11. Advantages and disadvantages of proportional control mode.
12. Purpose of using Integral action in conjunction with proportional mode.
13. Integral mode reaction curve at different settings.
14. Advantages and disadvantages of integral control mode.
15. Purpose of using derivative action in conjunction with proportional + Integral actions in a control loop.
16. Derivative mode reaction curve at different values.
17. P + I + D Controller behaviour at different values.
18. Finding the optimum controller settings using the empirical method.
19. Finding the optimum controller settings using the ultimate method.
20. Different controller reaction curves in different modes.
21. Cascade control loops.
22. Ratio control loops.
23. Override Control.

Related Safety Regulations for Module 3: CONTROLLERS & CONTROL THEORY

Juniors have to be familiarised with the following SGC HSE regulations, while studying this module:

Regulation No. 6: Work to permit system.

Regulation No. 7: Isolation

7.18 (1-10) control systems procedures and isolations

" Regulation No. 19/20: Working with Electricity,

19.7.4: Precautions on low voltage systems.

19.7.5: Precautions on Extra low voltage systems.

The Objectives of Process Control

The basic objectives of any process control system are:

- Closely monitor the condition of the process
- Maintain the process in a safe and stable condition
- Compensate for changes in the process conditions and maintain production to a given specification
- Increase profitability

This section examines the basic principles behind the development of process control and its applications in the oil industry.

Manual Verses Automatic Systems:

In any installation there will be a series of process quantities, such as level, flow and pressure that will need to be maintained at a pre-defined or target value. A difference from these values could result in a dangerous condition arising, or result in a loss of revenue due to a loss in production.

If the task of maintaining a process at a desired level were left in the hands of a human operator, several disadvantages would quickly become apparent.

Fatigue: Once the operator becomes tired and bored with the task, the level of interest and accuracy drops.

Reaction time: Depending on the process involved, the reaction of the human operator may not be fast enough to maintain the accuracy required.

Limited power: Humans are relatively weak and cannot operate heavy equipment without using some form of mechanical advantage, i.e. a gearbox, levers etc.

Safety: Many products found in processes can be hazardous to humans, in that it may be toxic, carcinogenic or at a very high or low temperature.

Accuracy: When the shift changes over, what may be accurate enough for the day shift operator may not be good enough for the night shift operator, or visa versa.

Cost: If you had to pay a skilled operator to sit and constantly monitor and control a process then the running cost of the process would be extremely high.

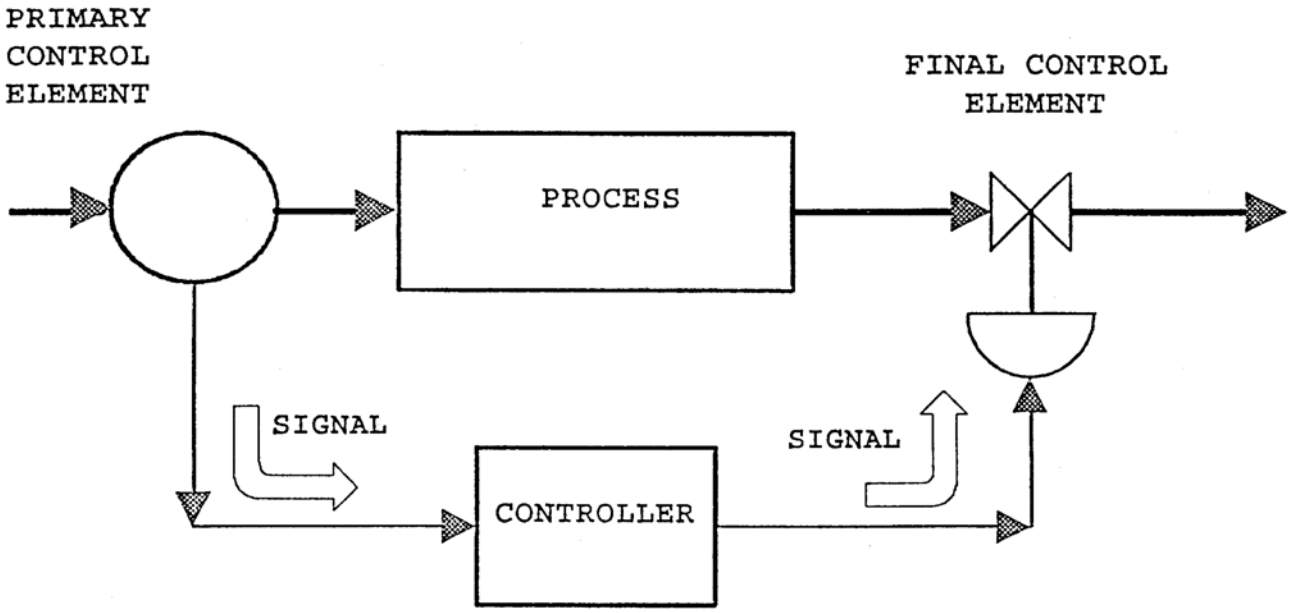
It is because of these limitations and problems that the demand for precise control by automatic control systems has arisen.

Automatic Control Systems Produce

- A more consistent product
- Release skilled operators for other productive work
- Reduce the physical effort required, lessening fatigue and boredom
- Decrease the physical workload on an operator
- Improve safety and working conditions

Once an automatic control system has been installed and commissioned, it should be able to maintain a pre-set operating condition over an extended period of time without any operator involvement.

Elements of a Control Loop



Elements of Control Loop

Primary Element

This is the first instrument in the control loop. It is usually in physical contact with the process and sense changes in the process variable. Examples of Primary Elements are thermocouples; orifice plates; pressure transmitters.

Transmitter

If the signal from the primary element has not been converted then it needs to be standardised before it can be used by standard controllers. The transmitter takes the signal from the primary element and gives a standard proportional output. There are pneumatic and electronic transmitters. Some common output signals are 4 to 20mA; 10 to 50mA; 1 to 5V; 3 to 15 psi and 0.2 to 1 bar.

Controller

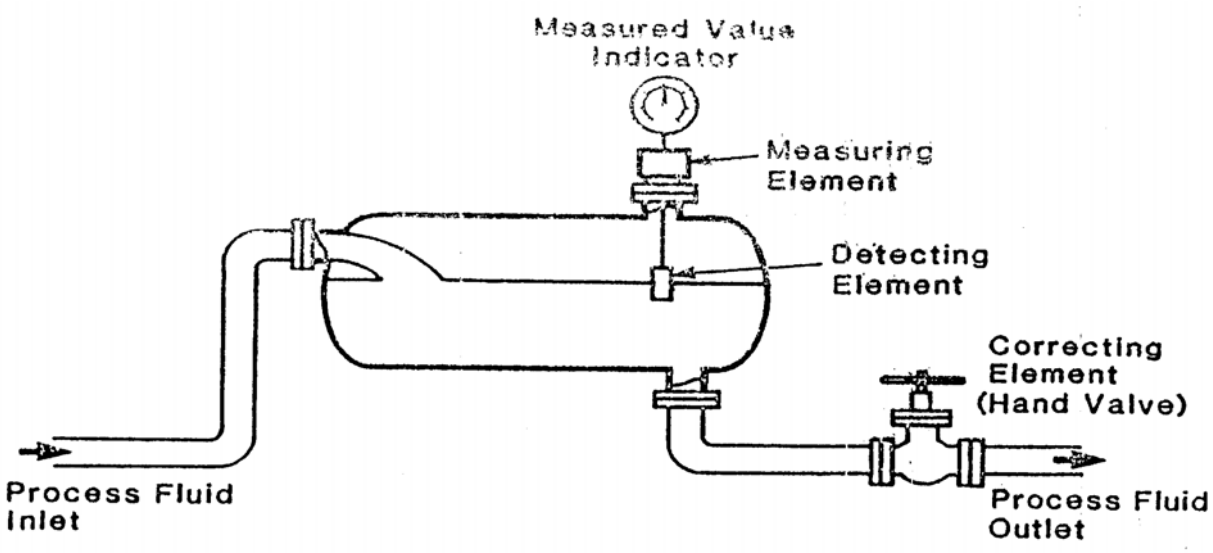
Controllers are the 'brains' of the control loop. The controller **receives** a signal from the transmitter and then **compares** this value with the set point and **computes** the amount of output signal needed to remove the difference between the measurement and the set point (the offset).

Final Control Element

This is the correction device in the control loop. It is usually a valve but can be a heater or motor. It gets the signal from the controller and alters its output accordingly. The final control element manipulates the manipulated medium.

Classification of Control Systems

Open Loop Control



Open Loop Control System

In an open loop system the controller has no information or feedback about the current condition of the process. Therefore the controller is unaware of the effect of its output.

An example is that of water flowing into a tank with an outlet at the bottom. As the tank fills up then the flow of water will increase (due to the higher pressure). A point will be reached where the flow out of the tank will be the same as the flow into the tank and the level will not change.

The Basic Requirements of an Open Loop System are:

- A process
- A measuring element
- A correcting element

With open loop control, the controller's output is fixed regardless of changes in demand or process conditions. It is unusual to find this form of control, because of the lack of feedback. Here is a plant being operated in the open loop mode. Note that although there is a measurement of the level in the vessel no action will be taken as a result of any change in the measured condition, i.e. it only provides information. This is because the position of the hand valve is fixed and would have to be manually adjusted to compensate for any error. The presence of an operator would then "close" the loop.

Summary of Open Loop Control:

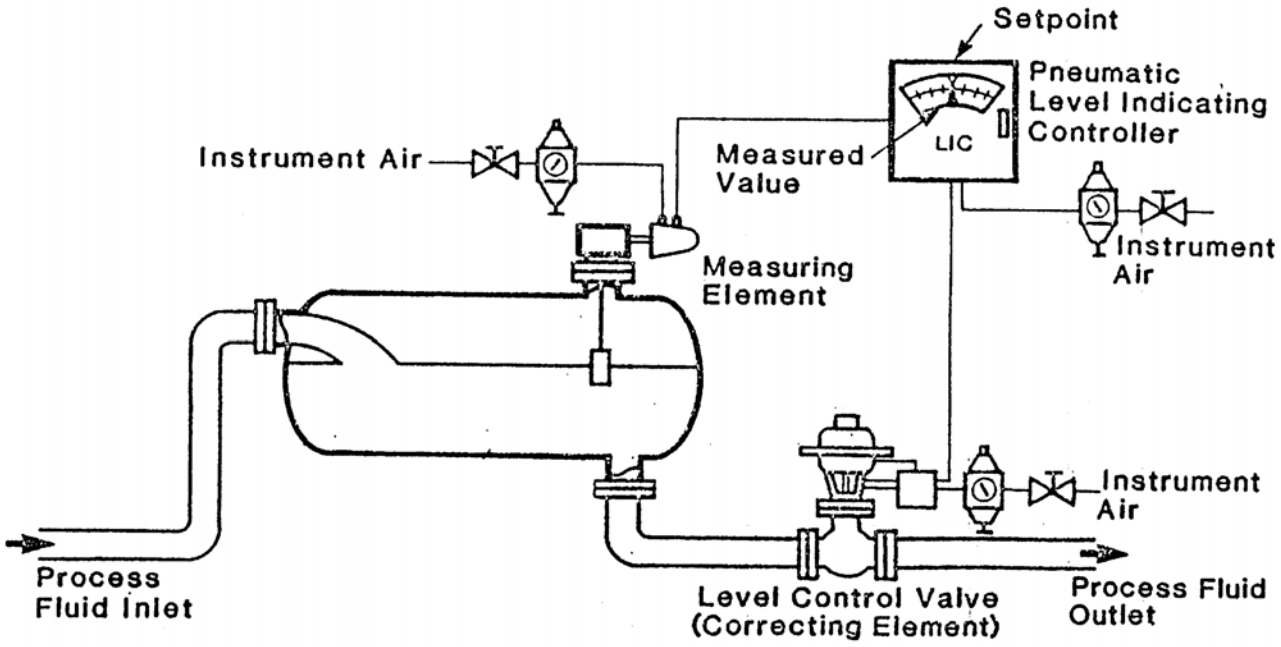
1. Open loop control has no information or feedback about the measured value.
2. The position of the correcting element is fixed.
3. It is unable to compensate for any disturbances in the process.

Closed Control Loop

In a closed loop control system the output of the measuring element is fed into the loop controller where it is compared with the set point. An error signal is generated when the measured value is not equal to the set point. Subsequently, the controller adjusts the position of the control valve until the measured value fed into the controller is equal to the set point.

The Measured Value (MV) signal is fed back to the controller after adjustment of the control valve (correcting element) by the controller. The controller continuously compares this feedback (MV) signal with the Set Point (SP) and readjusts the control valve to maintain MV = SP. Thus closed loop control is often referred to as feedback control.

The preceding illustration shows a practical representation of a pneumatic closed loop control system where the process level is measured by a displacer level transmitter which transmits the level measured value (process variable) to the loop controller. In the controller the measured value is compared with the manually adjusted set point (desired value). If the set point is not equal to the measured value, then a deviation exists. Depending upon the magnitude and direction of the deviation, the controller will make the necessary adjustment to its output in order to move the control valve position.

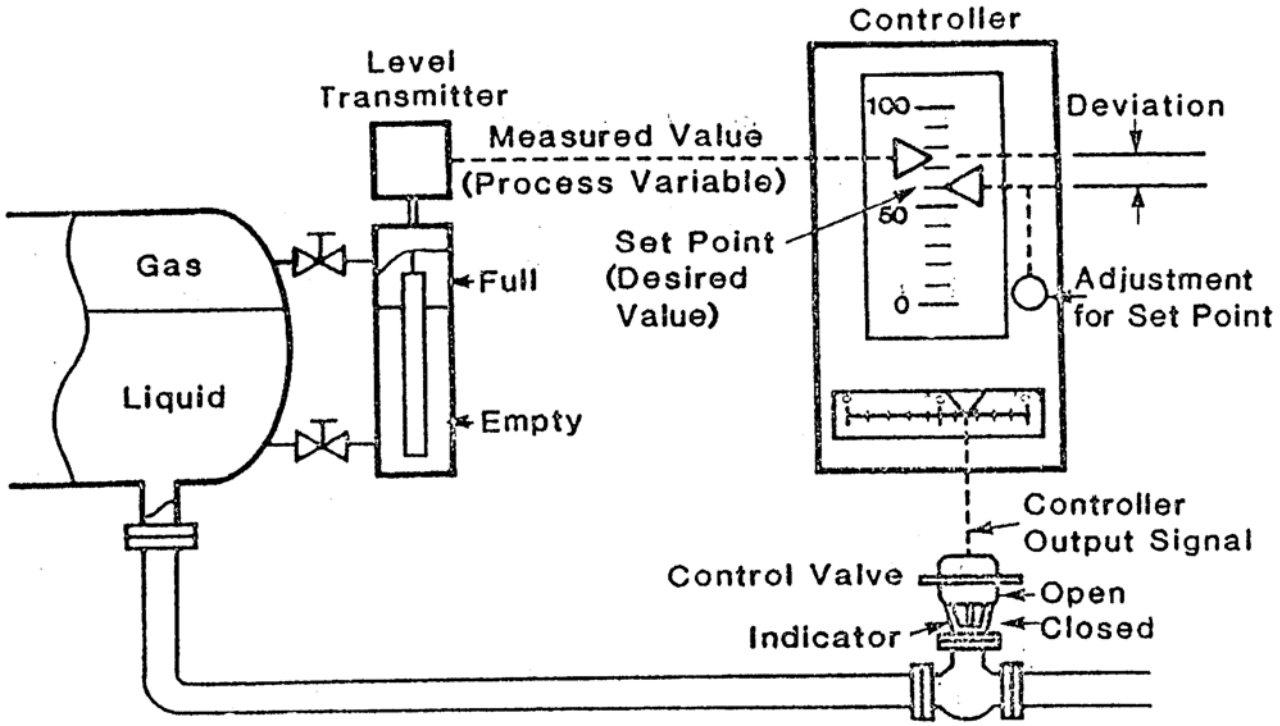


Pneumatic Closed Loop Control System

In the accompanying illustration the vessel level is above set point. Therefore, the control valve must be moved towards the open position, so that the vessel discharges faster. The level will begin to drop, provided the flow of fluid into the vessel remains constant. The

measured value will now fall towards the set point and the control loop will attempt to stabilise.

The response of the process, the speed at which stable conditions can be achieved and the amount of deviation between the measured value and set point are important conditions, which affect the performance of any process. It is these constraints which must be acknowledged when considering automatic control.



Example of a Closed Loop Control System

Summary of Closed Loop Control:

1. Closed loop control has information and feedback about the measured value.
2. The position of the correcting element is variable.
3. It is able to compensate for any disturbances in the process.

Modes of Process Control System

A System Falls into one of the Following Categories:

1. Two Steps; On-Off
2. One term; Proportional only
3. Two term; Proportional plus Integral
4. Three term; Proportional plus Integral plus Derivative

The more controller terms, the more expensive the controller and the tighter the control of the process, although all three terms are not always required.

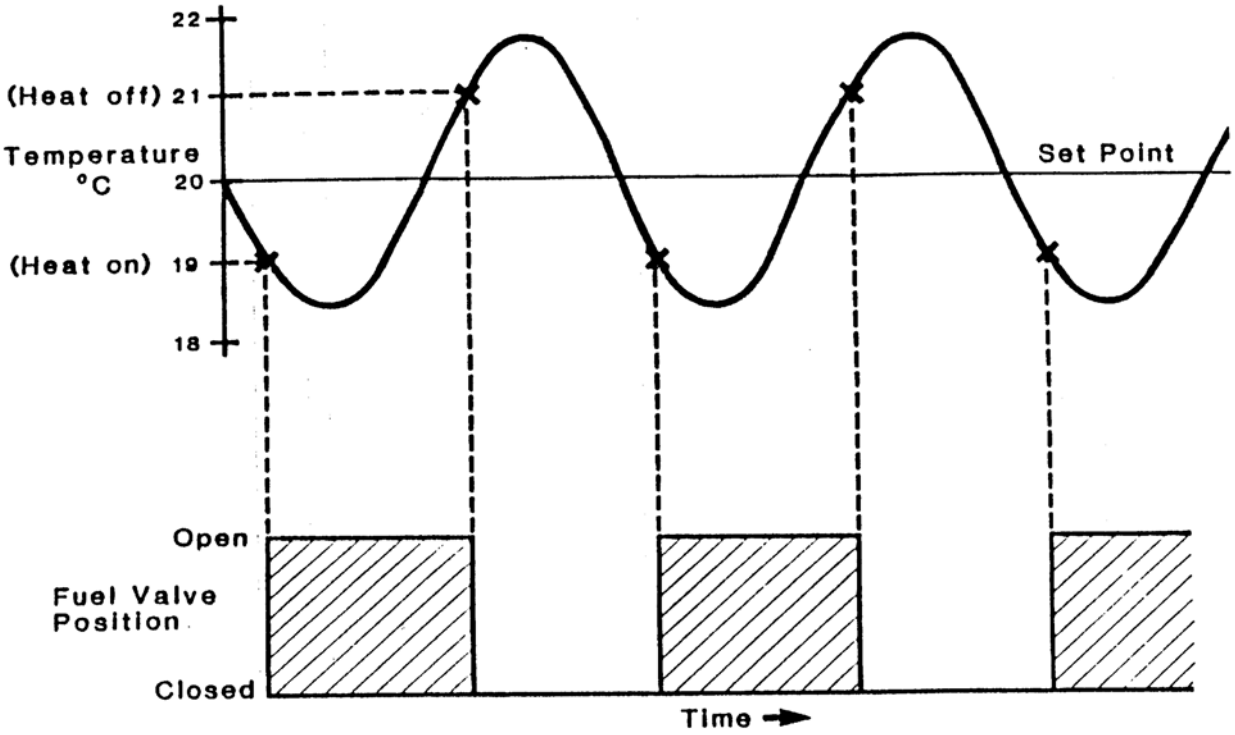
The purpose and the effect of each of the terms is considered below:

Two Steps: On-Off

Two-step is the simplest of all the control modes. The output from the controller is either on or off with the controller's output changing from one extreme to the other regardless of the size of the error. This leads to a very cyclic control system. For effective control with a two-step system the demand for energy must be very much larger than the supply of energy.

In this way the size of the oscillations can be minimised. It is very unlikely that this control mode would be found in normal process operations.

On some plants the controlled variable must be kept very close to the set point to ensure product quality. Complex control loops must be used on these plants. On other plants it may be acceptable to have a large difference between the measurement and the set point (offset). On these plants simple control loops may be used.



Characteristics of Two Steps Action

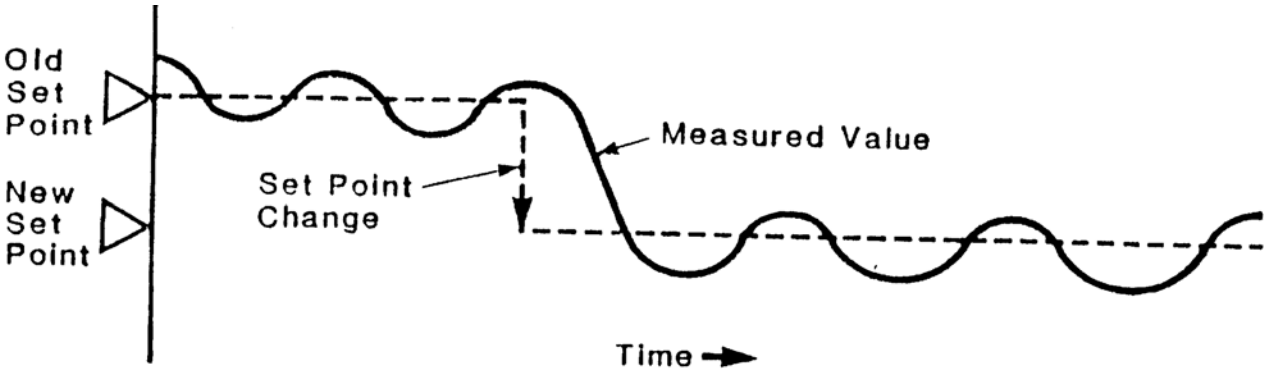
Two-position control can only be in one of two positions, either 0% or 100%. A switch is an example of On/Off control.

Advantages:

On/Off control makes "trouble shooting" very easy and requires only basic types of instruments.

Disadvantages:

1) The process oscillates.
2) The final control element (usually a control valve) is always opening and closing. This causes excessive wear.
3) There is no fixed operating point.



Two - Step Control

Proportional Action

With proportional control action, the correcting element is adjusted In proportion to the change in the measured value from the set point. The largest movement is made to the correcting element when the deviation between measured value and set point is greatest. Usually, the set point and measured value are equal when the output is midway of the controller output signal range.

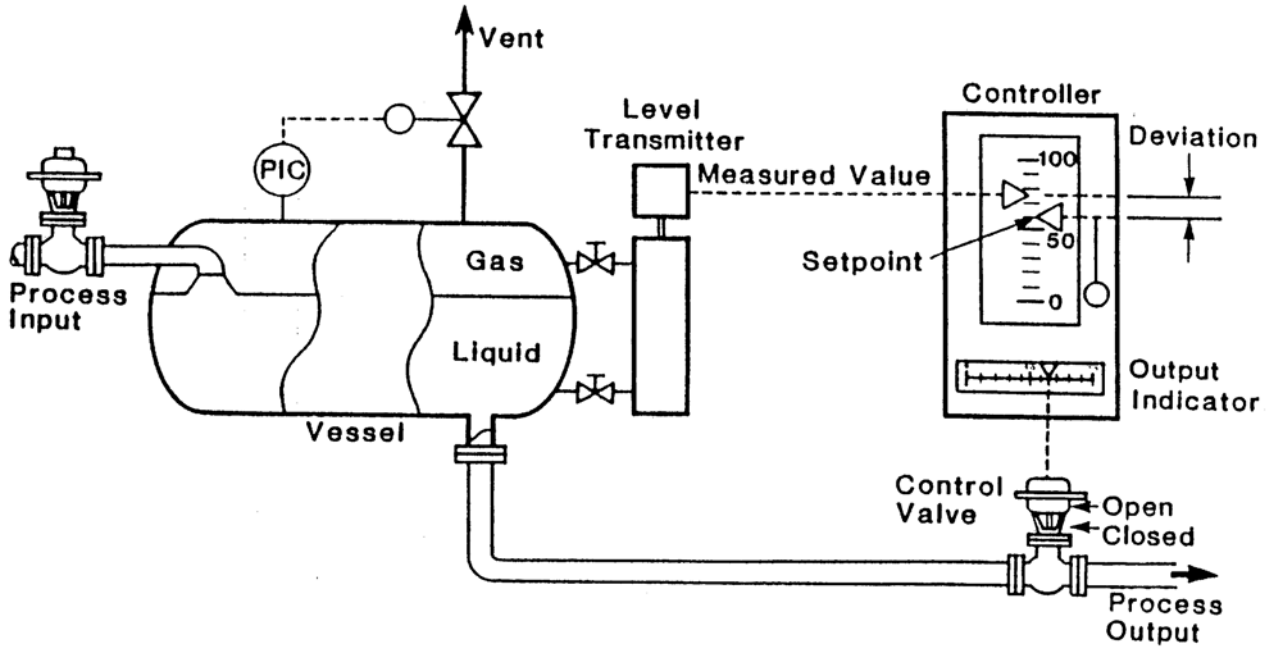
In the accompanying diagram, the set point is shown at 60%, the measured value at 75% and the output at 65%. If the measured value were to drop to 60%, that is, equal to the SP, the output would stabilise at the designed 50%. By repositioning the set point to 50% the measured value falls to 50%, the output would again be 50%.

Assuming that the level transmitter, controller and control valve are all operating correctly and have been recently calibrated, when set point and measured valve are equal and the system is in stable condition, the valve will be 50% open. The valve would have been sized during design to maintain the stable condition under a set of known conditions.
The process throughput, the fluid condition, the vessel, operating pressure and the back-pressure from the downstream process can all affect the throughput of the control valve. From .the diagram, it can be seen that the process input 1s equal to the process output and steady state conditions have been achieved with a level stabilised at 75%, but with a SP of 60%.

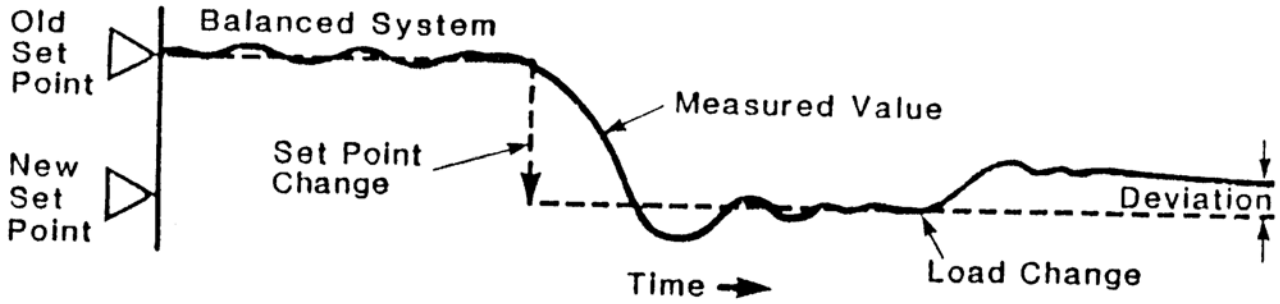
Under these conditions, the control valve would need to be 65% open; the magnitude of deviation is used to reposition the valve from its normal 50% open position. Deviation from other changes in operating conditions, particularly load changes, would also open or close the valve to achieve the new stable level.

The process load can be changed in the following ways to remove the deviation:

- Reduce the process input to the vessel allowing the level to drop so that a stable level is achieved at 60%'when the valve is 50% open.
- Increase the operating pressure of the vessel. This creates a higher differential pressure across the control valve, causing the fluid to flow from the vessel at an increased rate. This allows the level to' fall so that» a stable level is achieved at 60% when the valve is 50% open.
- Reduce the back pressure from the downstream process, creating a higher differential pressure across the control valve. This also causes the fluid to flow from the vessel at an increased rate.
- Increase the capacity of the contro1 valve to allow more process fluid to flow through the valve so that at 50% open a 60% level in the vessel is achieved.
- Any combination of the above conditions will a1 so remove the deviation. Over compensation may cause the measured value to move below the set point, causing a deviation in the opposite direction.



Offset in a Proportional Control Loop



Proportional Control

Proportional Band (PB)

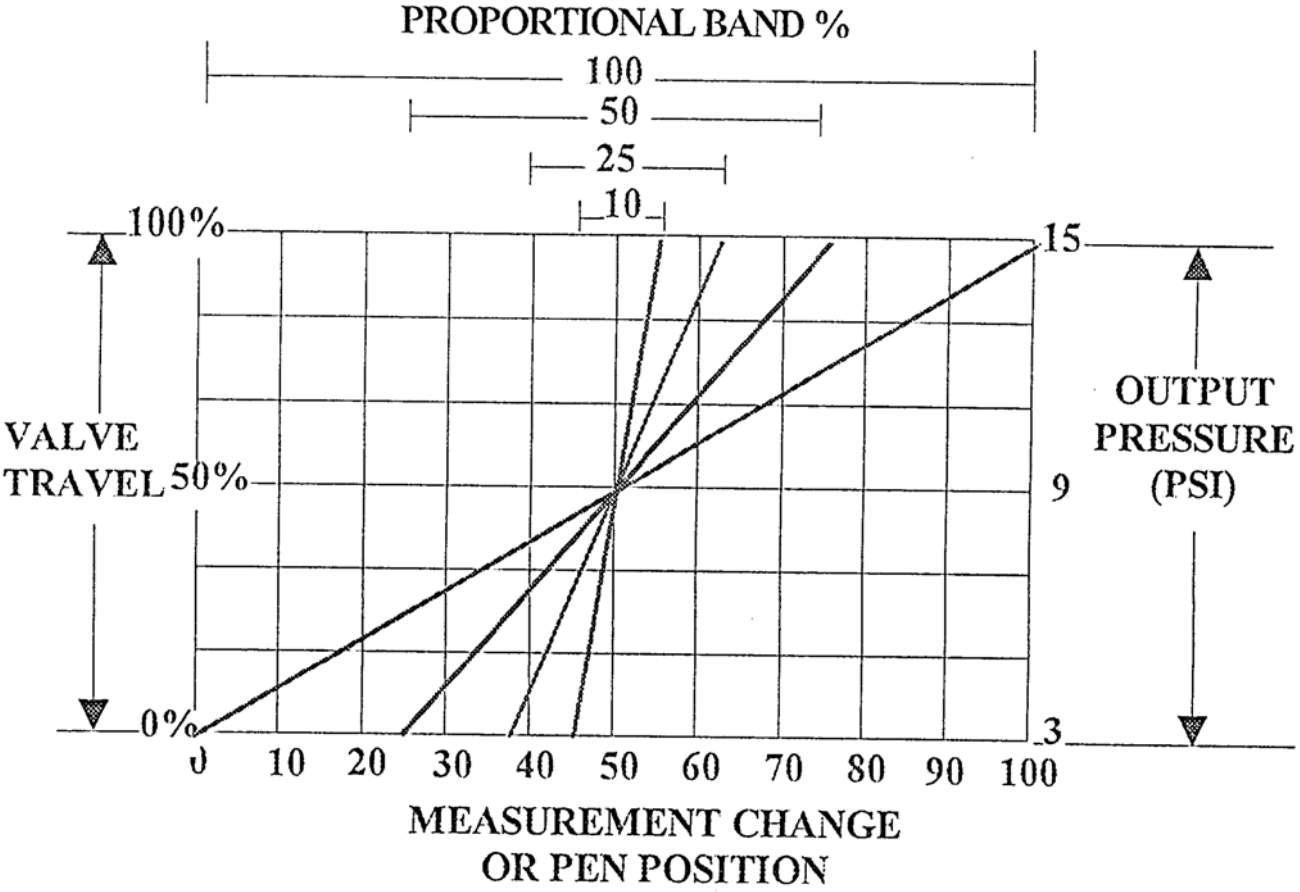
The simplest and most common form of control action to be found on a controller is proportional. With this form of control the output from the controller is directly proportional to the input error signal, i.e. the larger the input error the larger the output response from the controller.

The actual size of the output depends on another factor, the controller's proportional band or gain. (The controller's sensitivity)

A controller's proportional band is defined as the range of input values that will result in the controller's output sending the correcting element from one extreme to another. The proportional band (PB) is normally quoted as a percentage i.e. 10%, 50% 100% etc. or in terms of how wide or narrow it is.

The accompanying diagram shows that, with the set point at 50%, the measured value would have to move, to 100% to open the valve fully and drop to 0% to shut the valve. In other words, it required a deviation of 50% above and below the set point, which is 100% of the span of measurement over which the controller can operate, to give full valve movement.

For a 50% PB setting only a 25% deviation above and below set point gives full valve movement. For a 25% PB setting a deviation of 12.5% either side of the set point is required for full valve movement, which is not practicable. Similarly, for a 10% PB setting a deviation of 5% to achieve full valve movement is also impracticable.

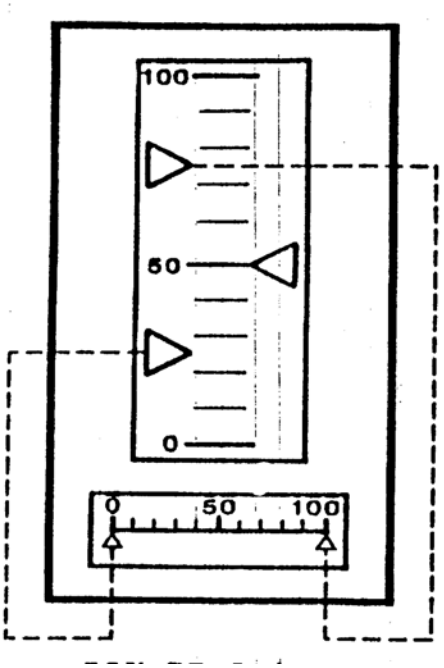


Proportional Band

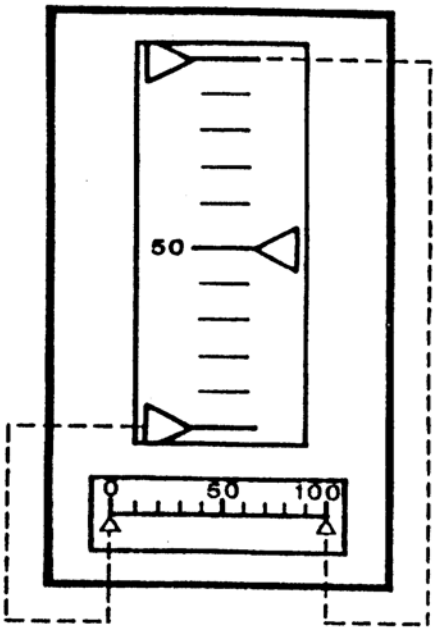
Deviation and Output Changes for Various PB Settings

The following diagram shows the flow sets of conditions in a graphical representation of how they would appear on the controller. It can be seen that the lower the PB, the higher the amount of valve movement for a given deviation. This can be related to gain or sensitivity.

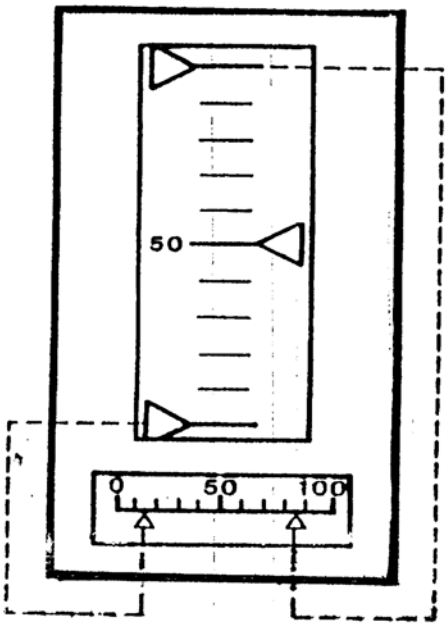
Some manufactures do no use the term PB, but use the term GAIN instead. Gain is just the inverse of PB multiplied by 100 or gain = 100/PB, so a PB of 100% would, have a gain of 1 and a PB of 10% would have a gain of 10.



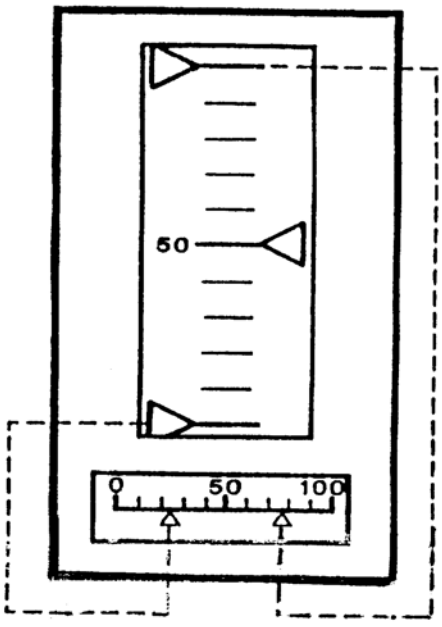
50 % PB Gain = 2



100 % PB Gain =1



150% PB Gain = 0.67



200 % PB Gain = 0.5

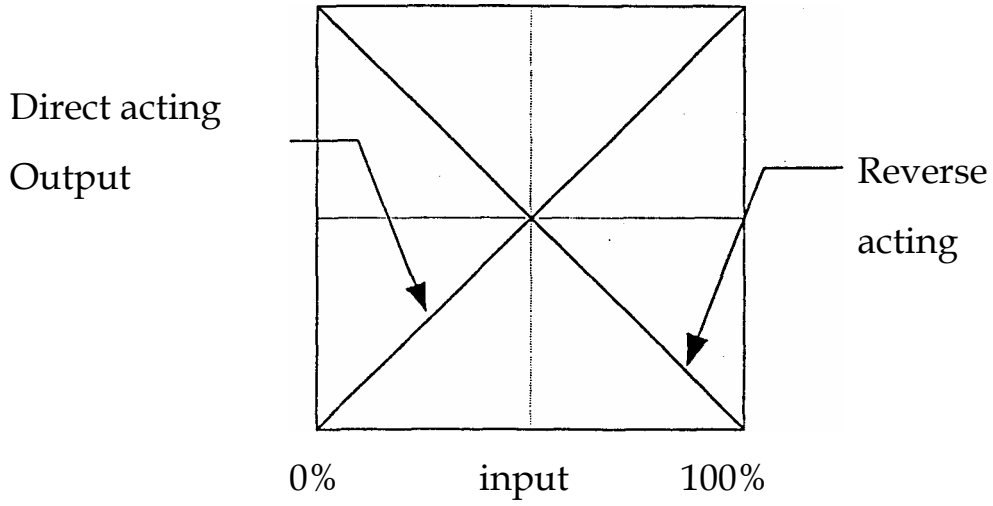
The vertical scale shows the measured value and the horizontal scale shows the controller's output. From these scales it is easy to see how the proportional band relates the size of the input error to the resulting controller output. Note that each controller has its set point at 50%.

Wide proportional bands produce slow response to changes in input but give a quick settling time. Narrow bands produce quick response but longer settling time.

Proportional control provides good process stability, but it suffers from OFFSET when the process is subject to sustained load changes.

Offset is the difference between the actual process value and the desired value. The size of the offset will be dependent on the size of the proportional band and the load. Unfortunately if the proportional band is narrowed too much the process will become unstable and constantly oscillate. Because of this, proportional band alone cannot be used to eliminate offset. Proportional only control is usually used where offset can be tolerated.

A controller's output is either direct acting or reverse acting.



A direct acting controller's output increases as the input signal increases, whereas a reverse acting controller's output decreases as the input signal increases.

When a controlled variable has to be kept at the set point proportional control is used. If the controlled variable moves away from the set point by a small amount the controller output will change by a small amount. The amount the controller output changes is proportional to the size of the error and the proportional band.

If the measurement and set point are both at the same value the controller output is 50%. If the measurement changes the output will change.

It can be seen that historically this proportionality between deviation and valve position was called proportional band, PB. However, due to the growth of analytical techniques for process control the term 'gain' (K) is now more commonly used.

The proportional mode of control can be described mathematically as:

V = K (E) + M

Where

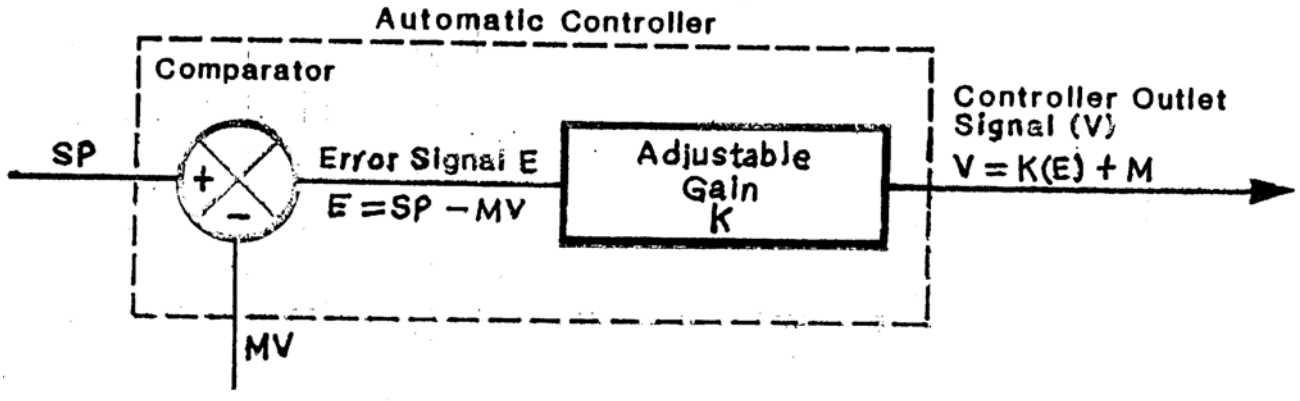
V = controller output signal to correcting unit,

K = adjustable gain,

E = magnitude of error signal,

M = constant which is the position of the valve when there is no deviation, that is, SP = MV and E = 0.

This can be shown diagrammatically as in the following diagram and gain settings can be shown graphically as in the following diagram.



Proportional Controller Block Diagram

Summary of Proportional Control

Proportional Band

The proportional band is the percentage change in the controller input divided by the percentage change in the controller output multiplied by 100%. If a controller has a proportional band of, say, 20% that means that there will be a full output change for a change of just 20% of the input (10% either side of the set point). With a controller, the lower the proportional band is the more sensitive is the controller.

Gain

The gain of a controller is output change divided by input change. When the proportional band of a controller is very low (and so the gain is very high) the controller is very sensitive and acts like an ON/OFF controller. With a controller, the higher the gain the more sensitive the controller is.

Proportional Control

- Stable control
- Suffers from offset due to load changes.

Narrow PB%

Fast to respond,

Large overshoot,

Long settling time,

Small offset

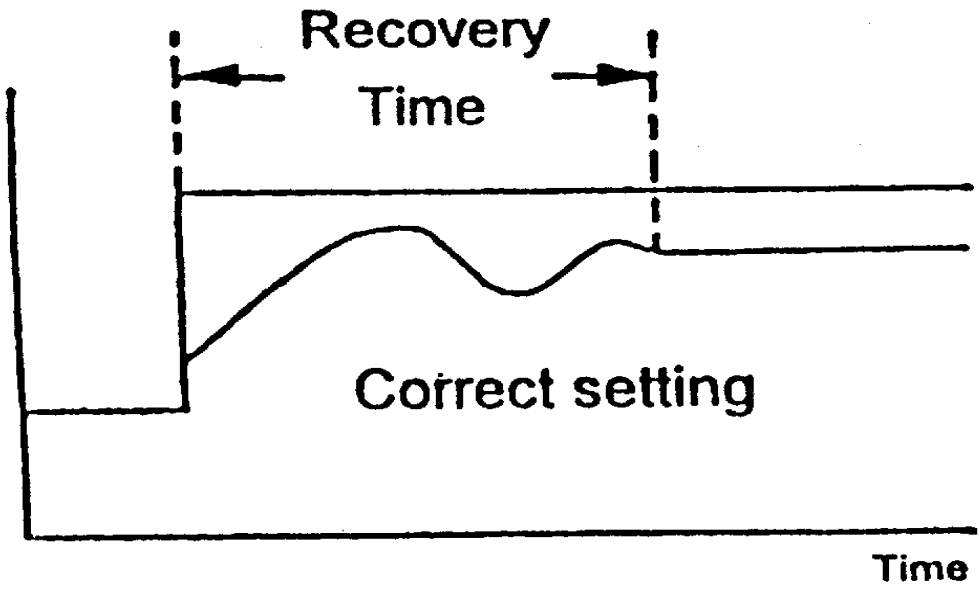
Wide PB%

Slow to respond,

Quick to settle,

Large offset

- Used in process where load changes are small and the offset can be tolerated.



Optimum Setting of P Control

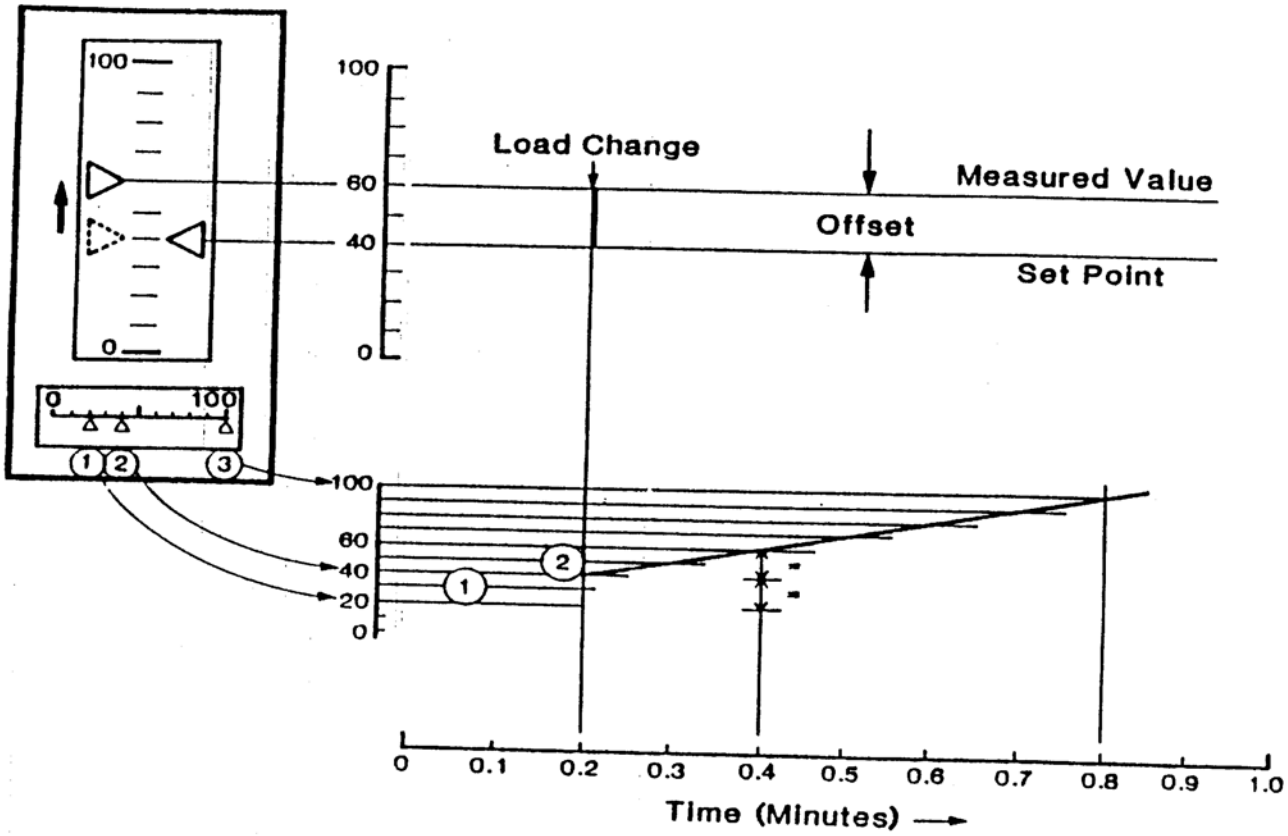
Integral Action

Proportional control suffers from offset, and the use of proportional action alone cannot eliminate this. Integral action is used in conjunction with proportional action to eliminate this problem. So long as an offset occurs integral action will keep the valve moving until the offset is reduced to zero.

Integral action is more commonly know as RESET; this comes from its action of resetting the error between the actual value and the desired value to zero.

The amount of integral action present is measured in minutes per repeat or repeats per minute depending on the controller manufacturer. In terms of minutes per repeat the smaller the number the more integral action present and the greater the effect; the larger the number the less integral action present. In terms of repeats per minute the smaller the number the smaller effect the integral action has and the larger the number the larger the effect.

An example of integral action in a proportional and integral controller is shown in the following diagram, here if the process is operating under steady state conditions at a set point of, say, 40% at time T = 0 minutes, the output of the controller is at 20%. In a proportional only controller the output would be 50% when the measured value is equal to set point, but this is not necessarily the case in a proportional plus reset controller.



Example of Proportional Plus Integral Action Control

At the time T = 0.2 minutes a sudden load change occurs which Causes the measured value to rise 20% above set point to 60%. Proportional action increases the output 20% to 40%, which indicates a PB of 100% or a gain of 1.

If the offset is maintained after this output change because the increased output cannot cause the measured variable to drop, the controller output will begin to increase in a ramp fashion.

The time it takes to ramp the controller output up to a value equal to the effect of the initial proportional action is called the integral action time. The previous illustration shows that the initial proportional action is a 20% increase in output. This action is repeated by integral action in 0.4 - 0.2 = 0.2 minutes to move the output from 40% to 60%, so for this example, integral action time = 0.2 minutes per repeat.

Integral action is sometimes quoted as 'reset time" or "repeats per minute* which is the reciprocal of minutes per repeat.

Reset time = 1/0.2 minutes per repeat,

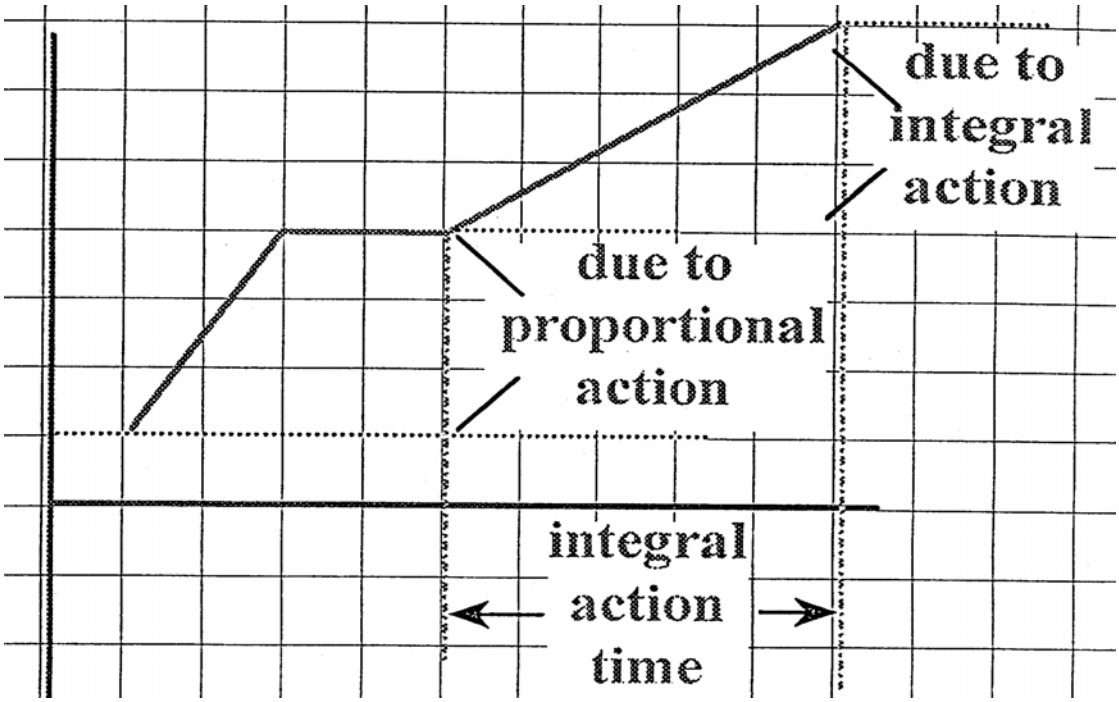
Reset time = 5 repeats per minute.

As can also be seen from the previous illustration, the valve ramps from 40% open to 100% open in 0.6 minutes (0.8 - 0.2) which is three repeats of 20% opening in 0.2 minutes per repeat. After the output has achieved 100% It continues to increase beyond the value necessary to open the valve fully. This is called integral saturation or reset wind-up.

Typical reset rates which are adjustable in modern controllers' range from 0.01 minutes to 1.0 minutes per repeat.

It should be noted that, although the explanation is for a set of conditions which require the valve to open with the measured value above set point, the reverse set of conditions can occur where the valve is required to close, when the measured value is below set point.

Referring to the mathematical equation for proportional only control, V = K (E) + M, the addition of integral action adjusts the M term of the equation, that is, it continues to adjust the valve position (M) all the time that a deviation between SP and MV exists.



Integral Action

The addition of integral action into a proportional controller has the advantage of eliminating the offset due to load changes, but unfortunately makes the process less stable and take longer to settle down.

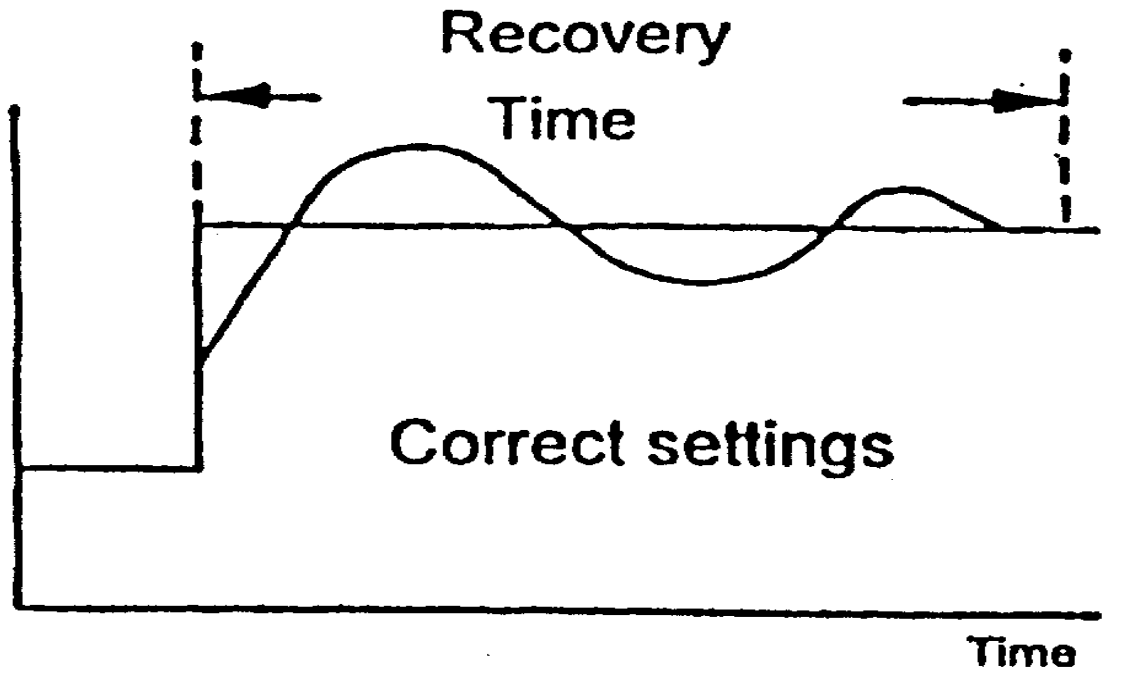
A common problem caused by integral action is called integral saturation or wind-up. During the time a process is shut down the integral action will keep trying to move the valve to correct for the error between its set point and the actual process value. When the process is started up it will take time for the process controller to gain control of the valve again. This time delay could result in damage to the plant or shutdown due to the plant safety devices cutting in. Normally a process such as this would be brought up on manual control and then switched over to automatic.

To prevent saturation from occurring controllers are fitted with integral de-saturation or anti wind-up devices. De-saturation relays prevent the controller's output from falling below 0.2 bar and rising above 1 bar.

Proportional plus integral control is usually used when offset cannot be tolerated but the long settling time is not a problem, such as in the hot water system.

Summary of integral action (Reset)

- Measured in minutes per repeat or repeats per minute depending on the manufacturer.
- Eliminates offset
- Makes the process less stable and take longer to settle down.
- Can suffer from integral saturation or wind-up on batch processes.
- Used when offset must be eliminated automatically and integral saturation due to a sustained offset is not a problem.



Integral Action

Derivative Action

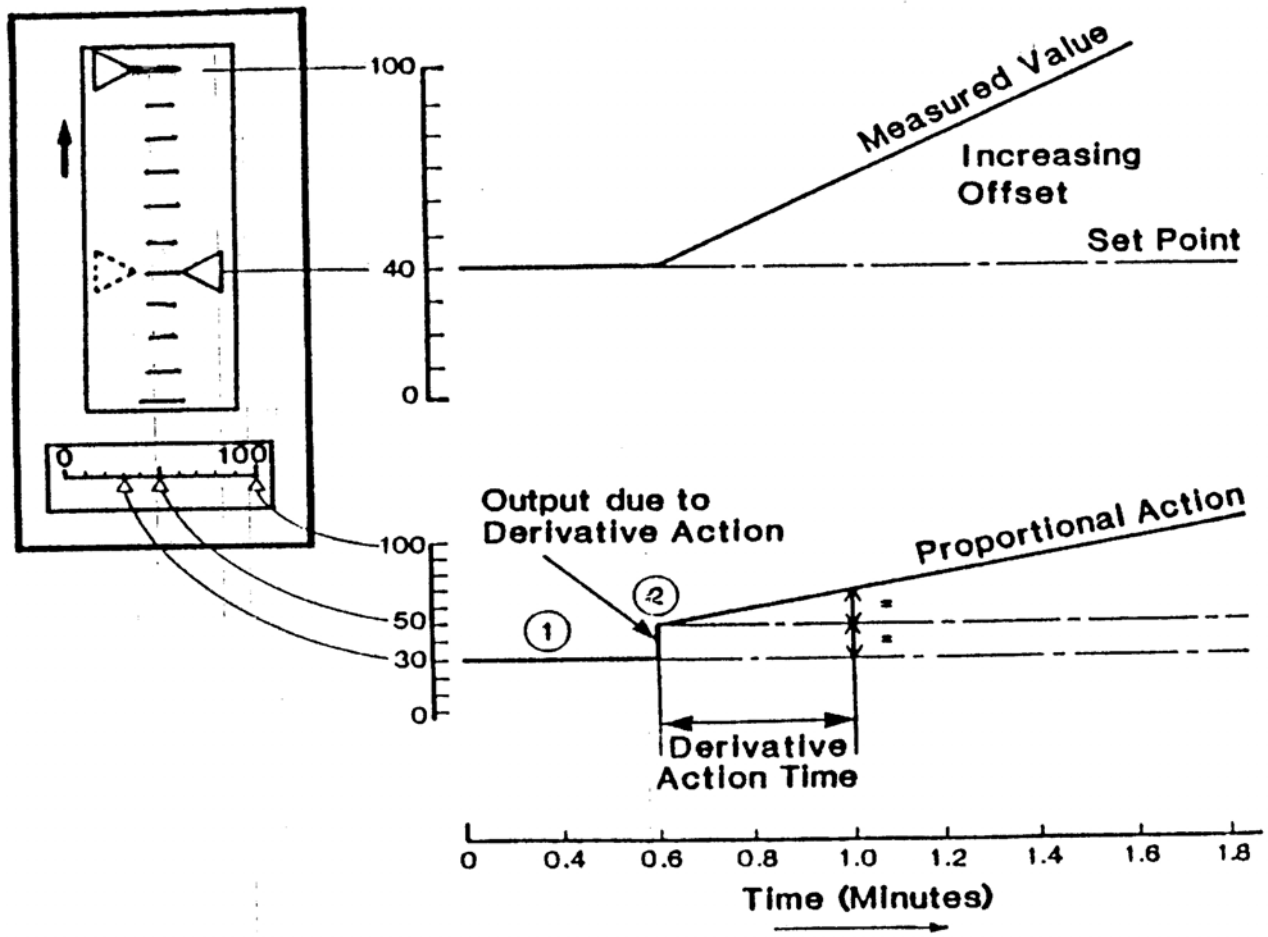
On a large and sudden load change the proportional action tends to cause a large overshoot and undershoot on the process variable, with a consequent increase in recovery time before the process stabilises. To reduce the amplitude of the swing and decrease the recovery time, derivative action can be used.

Derivative action is always incorporated in controllers having proportional action and cannot be used on its own. It is used in conjunction with proportional action to reduce the settling time of process.

Derivative action is commonly known as RATE control because it works on the rate of change in the process variable. It is sometimes known as pre-act because of its attempt to "anticipate "the control output required.

If the controller output is moved an amount which depends on the rate at which the deviation is changing, the recovery time will be reduced, as will the amplitude of the oscillations. This is known as derivative action. When the deviation is constant, no matter how large it is, derivative action will not alter the position of the correcting unit. It is mostly used where plants are large, for example, for temperature and pressure control, but is not usually necessary for flow control.

Derivative action time is defined as the time interval, in minutes, in which the output change due to proportional action is equal to the output change due to derivative action, so long as the deviation is changing at a constant rate. In terms of minutes the larger the number the greater the derivative action present and greater effect the smaller the number the less derivative action present.



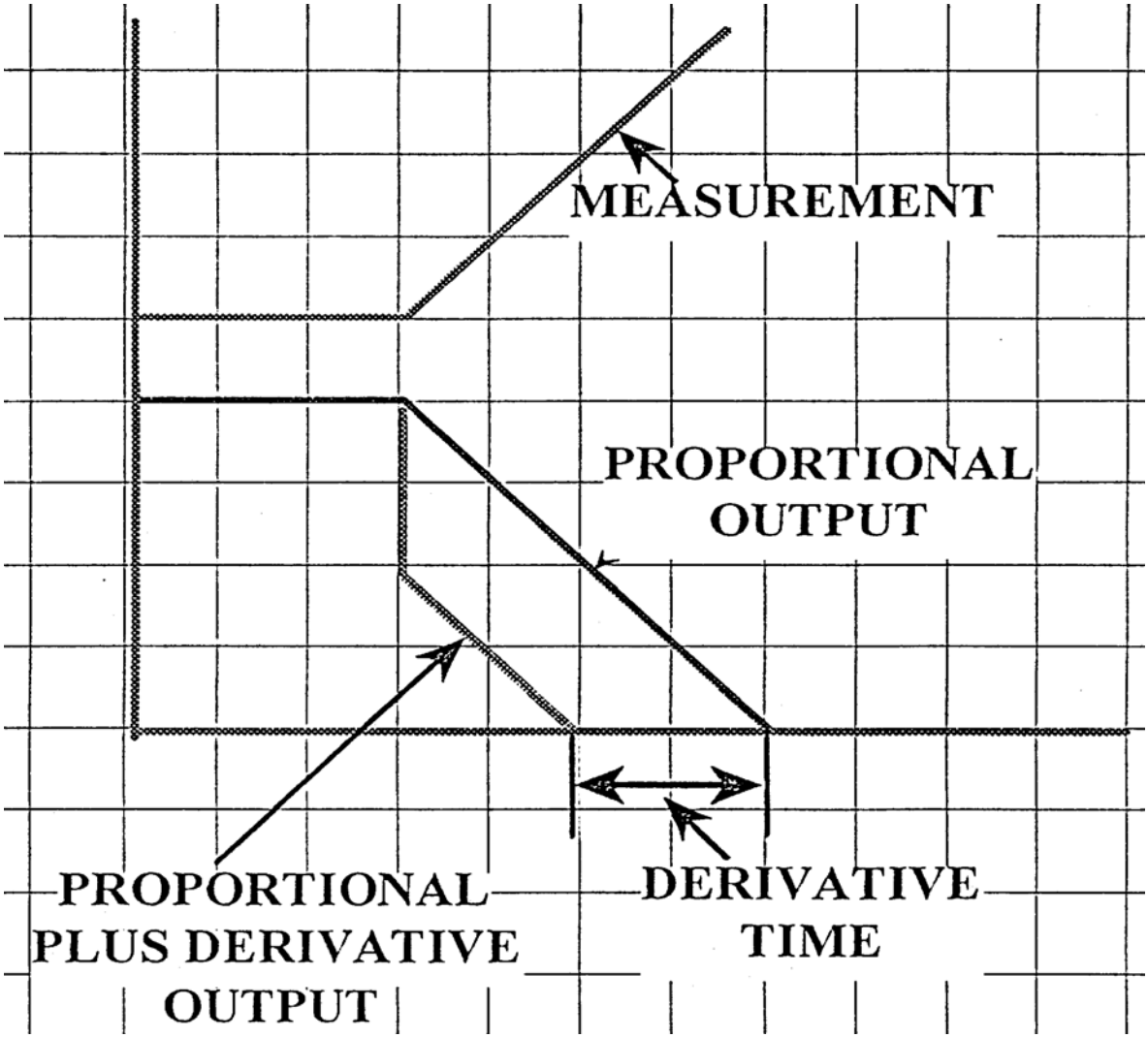
The illustration shows the effect of derivative action when a constant rate of change of offset is considered (the derivative time is 0.4 minutes (1. - 0.6)). When the set point is equal to the measured value the output remains constant.

Once the rate at which the measured value is increasing from the set point is determined, then derivative action acts to increase the controller output, in this case, from 30% to 50%. The output then increases due to proportional action.

The additional correction exists only while the error is changing, it disappears when the error stops changing even though there may still be a large value of error signal.

Derivative action has no effect on the offset in a proportional only controller and therefore it is unusual to find a proportional plus derivative controller.

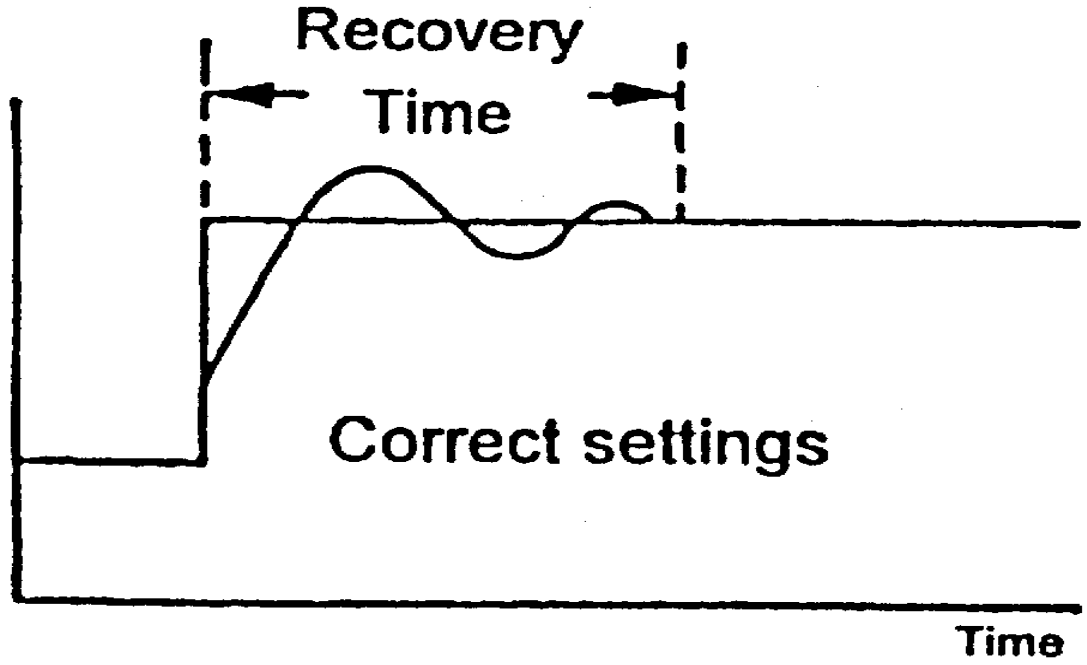
The introduction of derivative action makes the controller more sensitive and can tend to cause instability especially on "noisy" process signals such as flow.



Derivative Action

Summary of derivative action (RATE)

- Measured in minutes per repeat or minutes
- Has no effect on offset
- Reduces the process settling time
- Cannot be used on noisy signals



-Derivative Action-

Control Mode Comparison

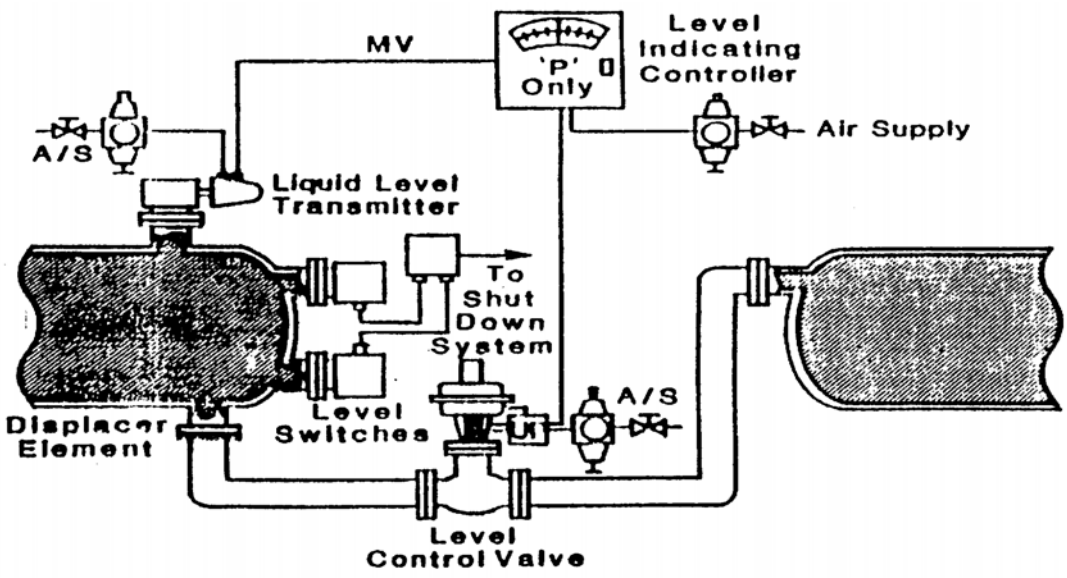
Mode	Advantages	Disadvantages
On-Off	Simple, inexpensive	Constant cycling
Proportional	Does not add lag	Almost always has offset
Integral	Eliminates offset	Adds time lag to system
Derivative	Speeds up response	Responds to noise

Use of one, two and three term controllers

- **Proportional only**

Proportional only controllers have their place in industrial control where the effect of offset is not important, but these situations are few and far between. Proportional only action is used in applications where:

- Process load changes are small,
- Offset can be tolerated.

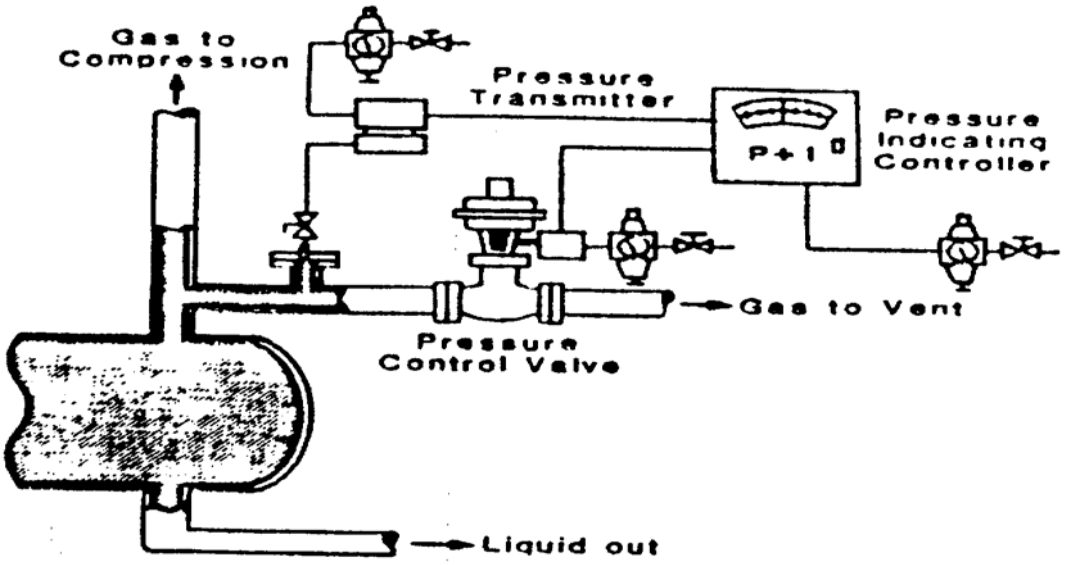


Example – Liquid Level Control (P Only Control)

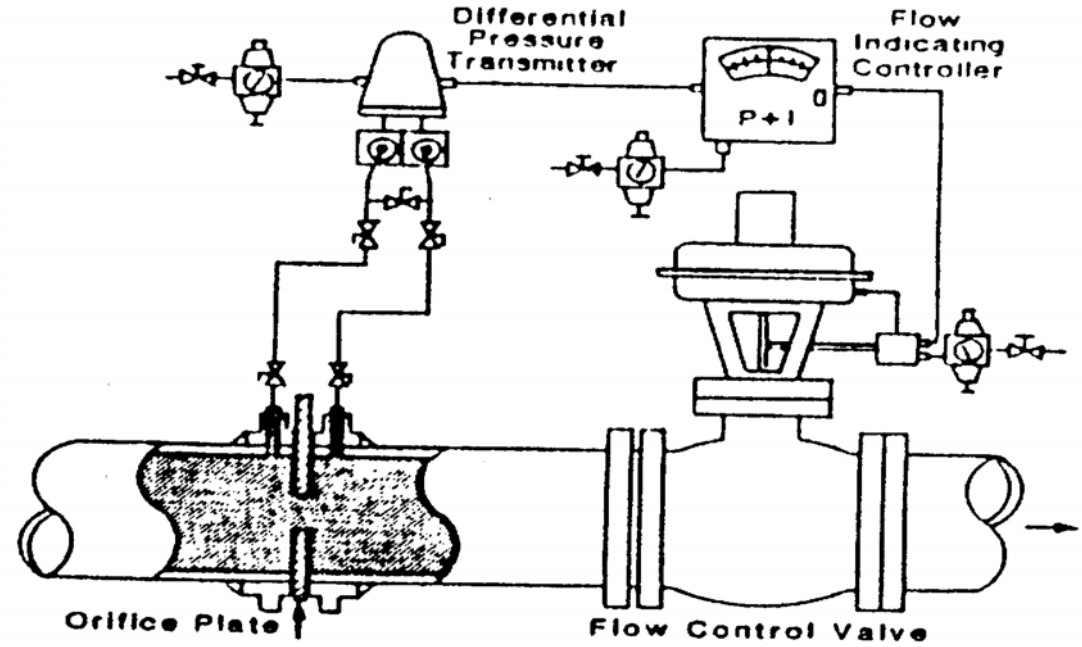
- **PI controllers**

PI controllers are more commonly used because of the control system performance requirements of no offset, an unfortunate by-product of the addition of integral action is an increased settling time. Integral action is typically used in applications where:

- Offset must be automatically eliminated,
- Integral saturation due to a sustained offset is not objectionable.



Example – Pressure Control (P + I Control)



Example – Flow Control (P + I Control)

- **PD controllers**

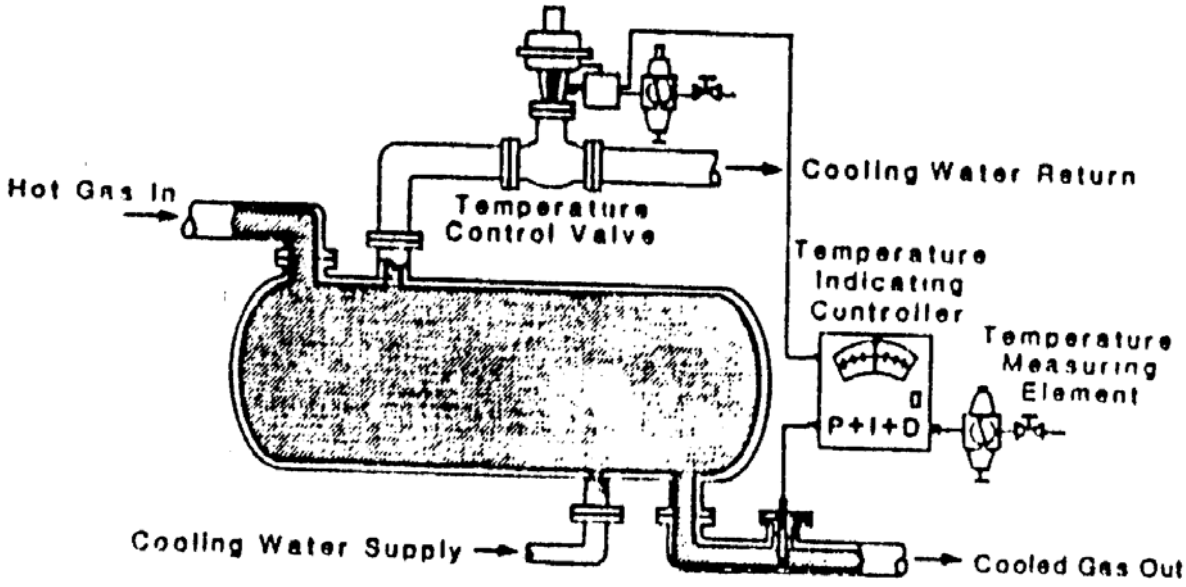
PD controllers are very rarely found due to the offset problem of proportional controllers and the susceptibility of derivative action to noisy flow signals.

- **PID controllers**

PID controllers are only required where tight control over the process variable, such as in temperature systems, is essential. This control mode will give you the best in terms of stability, settling time and removal of offset, but three term controllers are notoriously difficult to set up correctly. Derivative action is used in applications where:

- Large transfer lags or distance/velocity lags are present,
- The amount of deviation caused by plant load changes is required to be minimised.

With traditional pneumatic controllers you only added the control terms you needed because of the relative expense of adding each term. With electronic and digital based controllers this is no longer a problem as they come "free" with the controller.

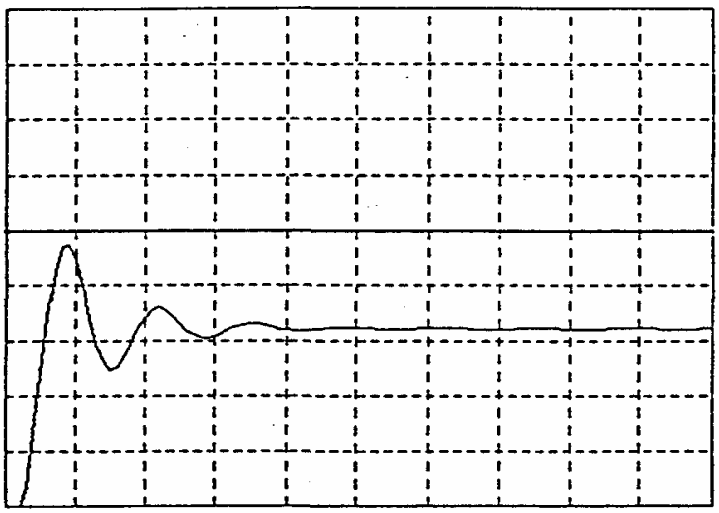


Example – Temperature Control (P + I + D Control)

Typical Controller Reaction Curves

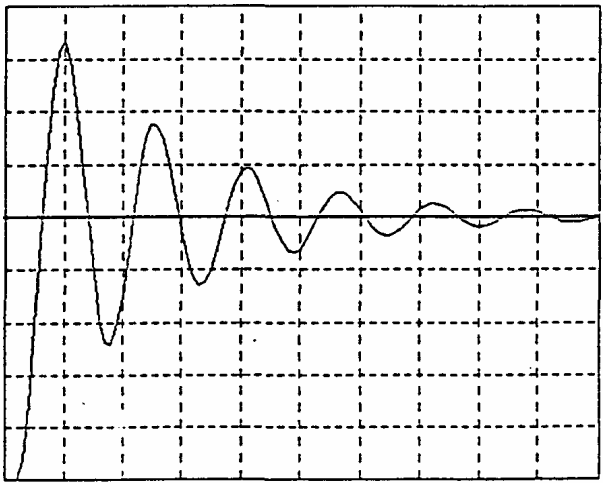
The following graphs, help to describe the response of closed loop two and three term control systems. These graphs show the typical effect of offset, overshoot and settling time that can affect the performance of a process.

a) Proportional Only Response



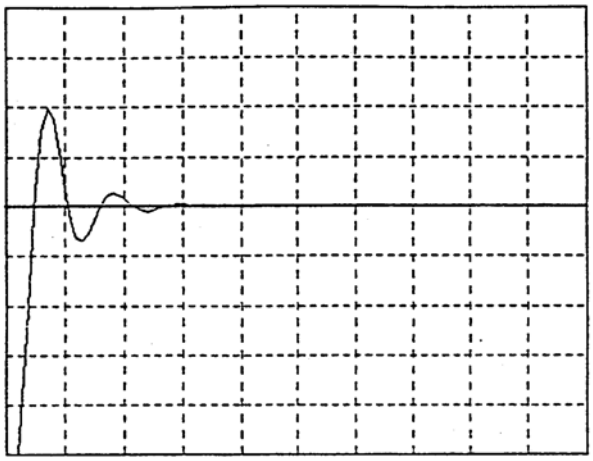
Proportional only response

b) Proportional + Integral Response



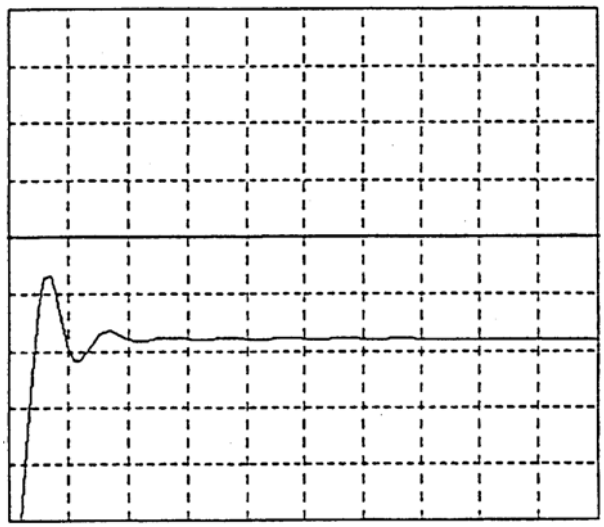
P + I response

c) Proportional + Integral + Derivative Response

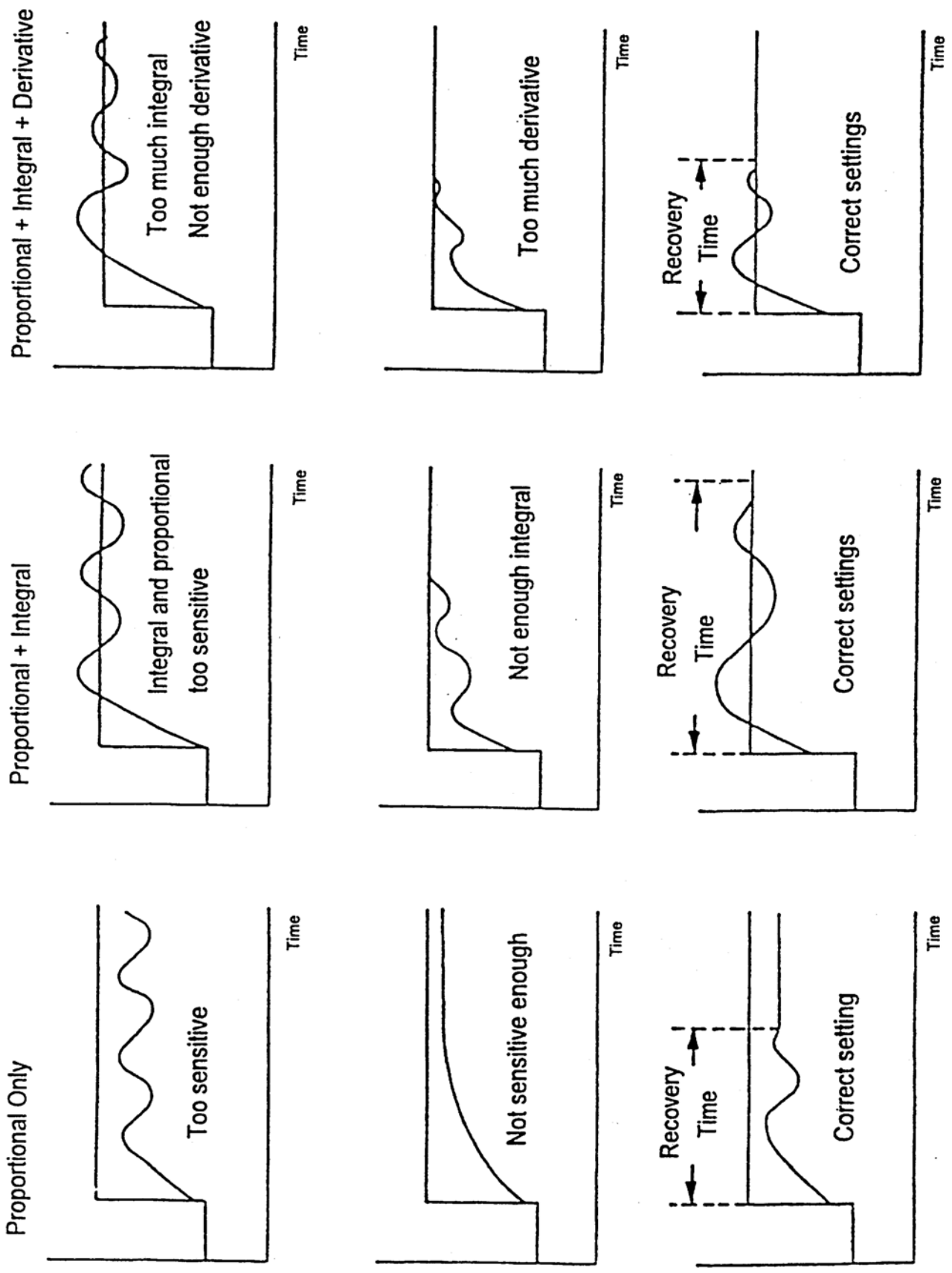


P + I + D response

d) Proportional + Derivative Response



P + D response


Proportional + Integral + Derivative
Too much integral
Not enough derivative
Time
Too much derivative
Time
Recovery
Time
Correct settings
Time
Proportional + Integral
Integral and proportional
too sensitive
Time
Not enough integral
Time
Recovery
Time
Correct settings
Time
Proportional Only
Too sensitive
Time
Not sensitive enough
Time
Recovery
Time
Correct setting
Time

AUTOMATIC CONTROLLER ADJUSTMENTS

The accompanying diagram shows a typical rack mounted controller, which fits flush to the control panel face at the bezel and gives access to the following controls.

- Set point adjustment, which allows the operator to select the required operating point for the process when the controller is in automatic mode.
- Auto/manual selector switch. When in the manual position the controller output becomes independent of the measured value and set point, that is, the controller 'operates in open loop.
- Output adjustment which allows the position of the final control element to be controlled by the operator when the controller is in manual mode so that the correcting element can be moved from fully closed to fully open and can be held at any position in between.

Bumpless Transfer

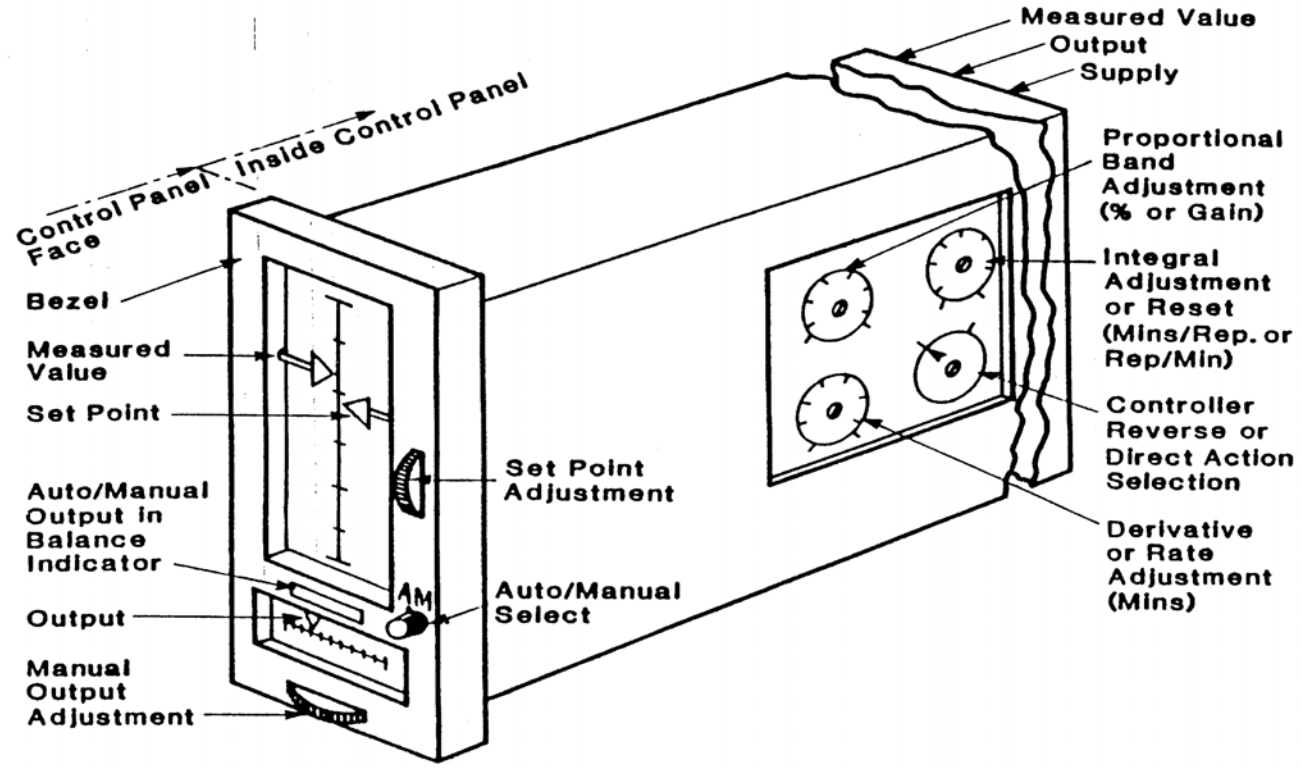
When switching a controller from auto to manual or vice versa, care must be taken that the output signal does not move sharply when the auto/manual switch is operated. This may cause a severe disturbance in the process, which may result in damage or shutdown.

Auto to Manual

If the controller is operating in auto under steady process conditions, the measured value will be equal to set point and the output signal will be at some value to maintain the measured value.

If it is now required to switch to manual mode, the following procedure is usual.

- Adjust manual output until the balance indicator shows that the manually adjusted output pressure is equal to the output pressure generated by the auto mechanism. The balance indicator mechanism varies according to the manufacturer of the controller, but all indicate by a flag or some similar device when the two output pressures are equal.



- Once the balance position has been found, It Is safe to switch from auto to manual without any process bump. The manual output adjustment now has control of the output to the final control element.

- When switching from manual to auto, the set point should Initially be moved towards the measured value to see if an output balance can be found. It is usual to find balance where there is an offset between set point and measured value. When the balance point has been found, it is then safe to switch to auto and slowly reposition the set point to the desired operating condition.

In some controllers it is possible for the auto to manual output systems to track each other so that the operator may switch from auto to manual and vice versa without finding the point of balance. This method of switching is usually called procedure-less bump-less transfer.

Controller Tuning

If the controller is withdrawn from the control panel face, further adjustments are available which are used to tune the controller to the process.

When a control loop is commissioned, the controller settings are adjusted to correspond to those, which have been specified during the design of the control system. If a large section of process is to be commissioned; possibly a mathematical model of the process will have been developed from which the optimum controller settings can be calculated for efficient and stable operation. It is these values which are set into the controller before start-up and, if calculated correctly, no further adjustment will be required.

In some cases it will be necessary to tune a controller without having the benefit of knowing what the settings should be. It must always be remembered that the adjustments cover a very wide range of sensitivity and response. If adjusted haphazardly, the process may shut down and damage to equipment and lost production may occur.

The task of controller tuning is usually left to an instrument technician with experience in the cause and effect of process reaction and controller adjustments.

There are many trial and error methods of controller tuning which do not involve mathematical analysis and should be demonstrated by an experienced person, otherwise shutdowns may occur.

The first adjustment, which would normally be made, would be to set forward or reverse action as required. A forward acting controller has increasing output in response to an increasing measured variable. A reverse acting controller has decreasing output in response to an increasing measured variable.

PB at Optimum Value

Controller optimisations can then be carried out as follows. For any particular control system there is a value of the proportional band, which will produce the best controller performance. Increasing the proportional band above this value will result in greater deviations of the controlled condition from the desired value owing to disturbances in the process.

Decreasing the Proportional band below the critical value will increase the tendency for the process to hunt, and disturbances will cause prolonged oscillation of the controlled condition about the control point. Indeed, if made too narrow, the system becomes unstable and instead of the oscillations dying out they will increase in amplitude.

Trained observation of the chart record, following a plant disturbance, thus provides a method of initially adjusting a controller's settings to the process. Process disturbances are easily simulated by moving the set point away from the desired value and returning it to its original position.

Empirical Tuning Method

Proportional Only Controller

- With transfer switch at manua1, set PB at maximum or at safe high value, usually 200% PB.
- Move transfer switch to auto and make changes in set point. The time required for the disturbance to settle may then be noted.
- Continue reducing band-width to half its previous value until the oscillation do not die away, But continue to be perceptible.
- Now increase the band-width to twice its value. This gives the required stability, that is, the minimum stabilising time and minimum offset.

Proportional Plus Integral Action

- Set the Integral Action Time (IAT) to maximum.
- Adjust the proportional band as for a proportional controller.
- Decrease the IAT in steps, each step being such that line IAT 1s halved at each adjustment. Below some critical value, depending upon the lag characteristics of the process, hunting will occur. This hunting Indicates that the IAT has been reduced too far.
- Now increase the time to approximately twice this value to restore the desired stability.

Proportional Plus Derivative Action

- Adjust the Derivative Action Time (DAT) to its minimum value.
- Adjust the proportional band as described for proportional controller, but do not increase the band when hunting occurs.
- Increase the DAT (that is, double each setting) so that; the hunting caused by the narrow band is eliminated.
- Continue to narrow the band and again increase the DAT until the hunting is eliminated.
- Repeat previous step until further increase of the derivative action time fails to eliminate the hunting introduced by the reduction of the proportional band, or tends to increase it. This establishes the optimum value of the DAT and the hunting should be eliminated by increasing the width of the proportional band slightly.

Proportional Plus Integral Plus Derivative Action

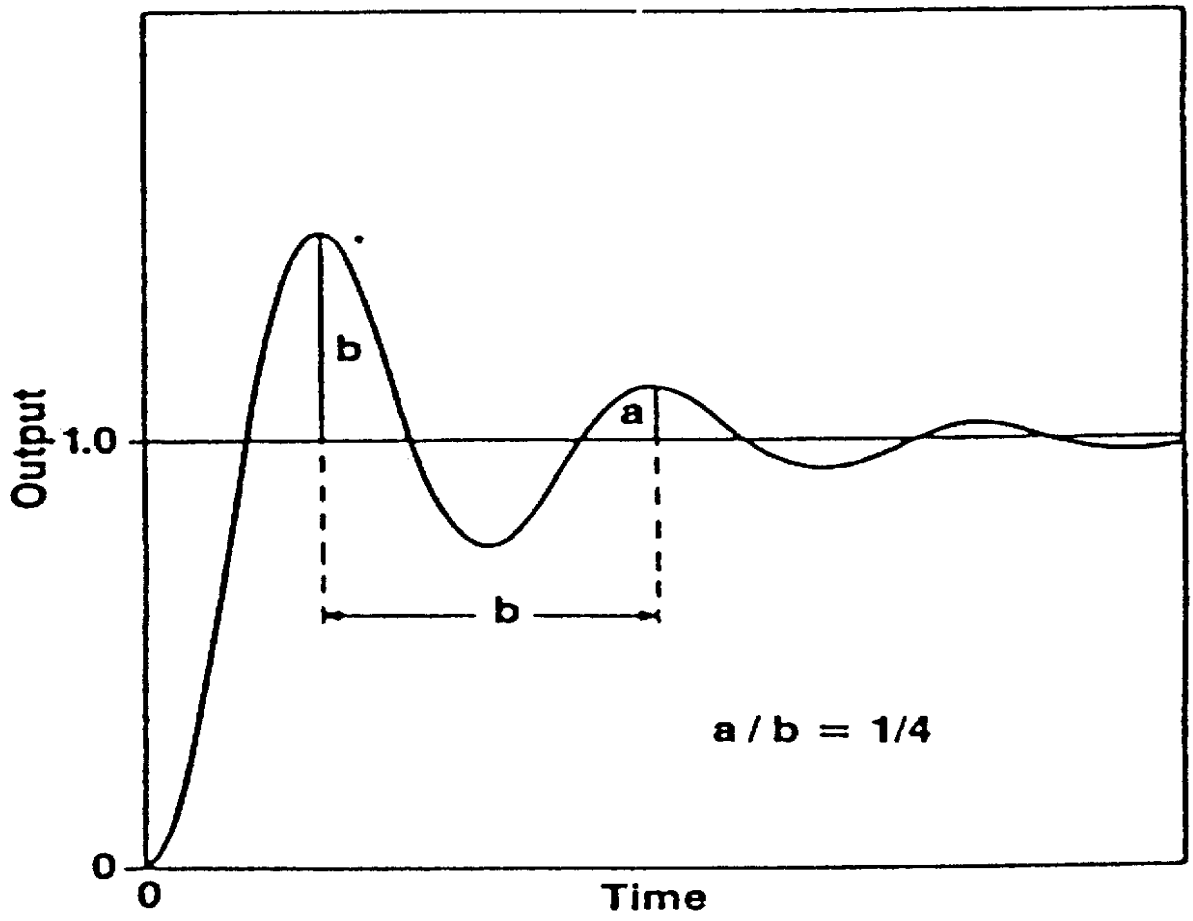
- Set IAT to a maximum.
- Set DAT to a minimum.
- Adjust the proportional band as for a P + D controller.
- Adjust derivative using same procedure as for above, P + D.
- Adjust integral to a related value of the final derivative setting.

A three-term controller is therefore adjusted as for a P + D controller and the integral value simply related to the derivative value.

In many cases, the setting procedure may be shortened by omitting settings, which are outside the probable range.

The process should then respond to set point or load changes, where integral action removes offset and the second overshoot of set point is approximately 1/4 the amplitude of the first. This is commonly referred to as the 1/4 decay method and is generally agreed to be the optimum controller setting for a P + I controller.

The above method is only used when no other controller setting data is available and must be practiced with care.



(b) Response Curve For 1/4 Decay Ratio

Optimum Settings (Ultimate Method)

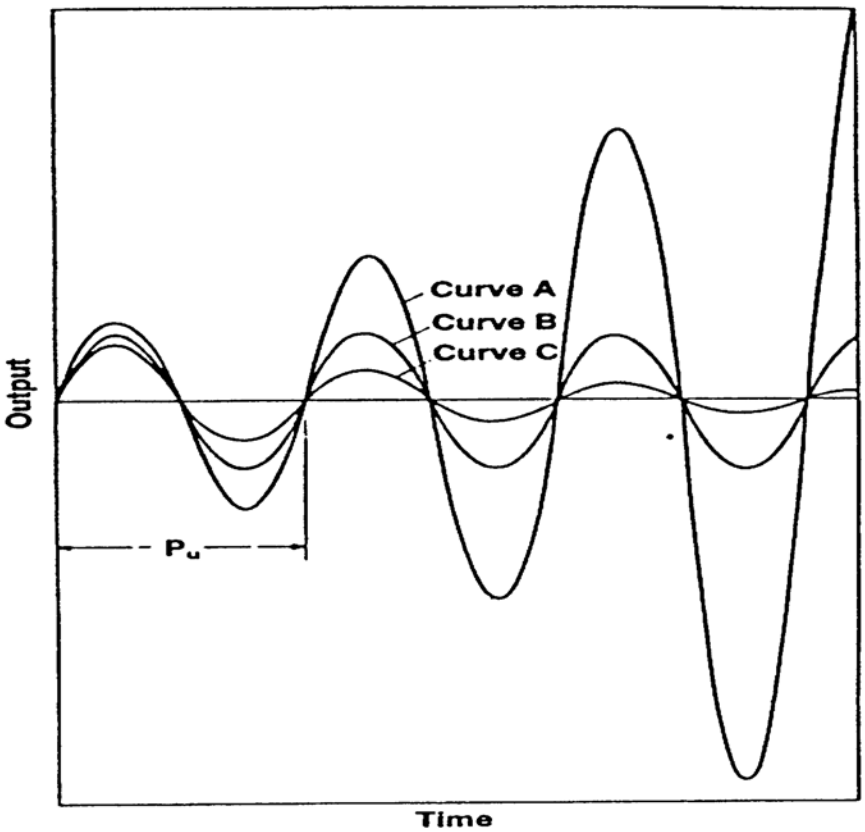
The closed loop or ultimate method involves finding the point where the system becomes unstable and using this as a basis to calculate the optimum settings.

The following steps may be used to determine ultimate PB and period:

1. Switch the controller to automatic.
2. Turn off all integral and derivative action.
3. Set the proportional band to high value and reduce this value to the point where the system becomes unstable and oscillates with constant amplitude. Sometimes a small step change is required to force the system into its unstable mode. The

below figure showing typical response obtained when determining ultimate proportional band and ultimate period time.

4. The proportional band that required causing continuous oscillation is the ultimate value Bu.
5. The ultimate periodic time is Pu.
6. From these two values the optimum setting can be calculated.



Curve A: unstable.
Curve B: continuous cycling.
Curve C: stable.

- **For proportional action only**

PB% = 2 Bu %

- **Proportional + Integral**

PB% = 2.2 Bu %

Integral action time = Pu / 1.2 minutes/repeat

- **Proportional + Integral + Derivative**

PB%=1.67Bu

Integral action time = Pu / 2 minutes/repeat

Derivative action = Pu / 8 minutes

Example 1

When a step change was applied to a closed loop system with a PB% of 14% sustained oscillation of the output was observed. The time between two adjacent peaks was measured as 1.2 minutes. Calculate the optimum setting for a P only & P+I+D system.

[(28%) & (23.38%, 0.6 min/rep, 0.15 min)].

Example 2

A closed loop control system is found to oscillate when the proportional band is reduced to 23%. A trace on a chart gives a measurement of 6 mm between adjacent peaks. If the chart speed of the recorder is 10mm per minute, calculate the optimum controller settings for P+I+D control.

(38.41%, 0.3 min/rep, 0.08 min)

Typical Controller Settings

Process	Gain	PB(%)	Integral		Derivative
			T_i (sec)	min/rpt.	T_d (sec)
Flow	0.6-0.8	167-125	3.0-1.8	0.05-0.03	0.0
Pressure	5.0	20.0	120-60	2.0-1.0	0.0
Temp.	1.0-2.0	100-50	120-30	2.0-0.5	6.0-12
Level	0.8-1.2	125-83	600-300	10.0-5.0	0.6-1.2

TYPES OF CONTROL LAGS

Control lags can be described as the time that elapses between a change in the process variable from the desired value until the process variable returns to the desired value.

The total time lag of a control loop is the addition of the following lags: -

- Measurement lags,
- Process lags,
- Transfer lags,
- Distance/velocity lags,
- Controller and correcting unit lags.

Measurement, process and transfer lags are also known as capacity and resistance lags.

Capacity refers to the parts of the process and equipment, which store energy; in the following diagram, for example, the wall of the heating coil and the oil in the tank

can store heat energy. This energy storing property gives the heating coil and oil the ability to show any change in temperature required by the controller.

Resistance refers to the parts of the process and equipment that resist transfer of energy between the heating coil and the oil. The walls of the heating coils and the insulating effect of the layers of fluid and oil on either side of the heating coil resist the transfer of energy from the heating media to the oil.

Measurement Lags

Measurement lag is the time it takes the measuring device to give a signal which accurately represents the process variable; for example, a temperature measuring device must be in thermal equilibrium with the process before it can give an accurate reading of the process temperature. If the detecting element is the bulb of mercury and steel thermometer, and the process temperature that the bulb is measuring rises, the bulb and the mercury will require a definite amount of heat from the process to reach thermal equilibrium.

The speed of the bulb reaction depends upon the following:

- Thermal conductivity of the bulb material and the medium which surrounds the bulb. When the bulb is fitted in a thermowell, it is better to fill the space between the thermowell and the bulb with a liquid, which will conduct the heat better than air; this will cut down measurement lag.

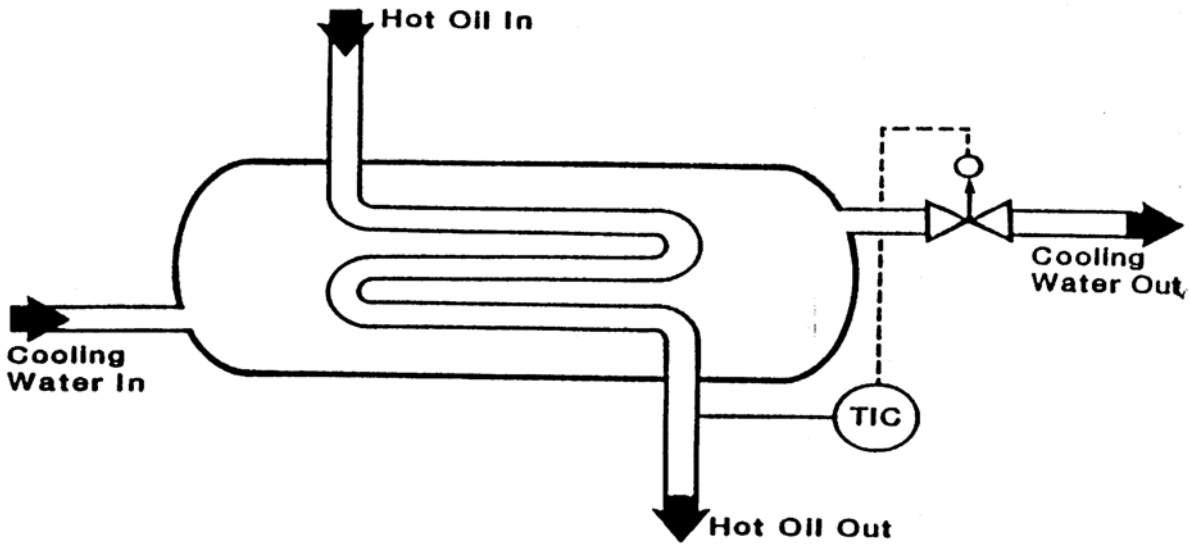
- Heat capacity of the process, which comes into contact with the bulb or thermowell, the process velocity, the density of the process, and the specific heat of the fluid. The bulb or thermowell should have a smooth external finish as the process will tend to cling to a rough surface, reducing the rate of heat transfer to the bulb.

- The ratio of surface area to the mass of the bulb, as the greater the surface area, the less amount to be heated by conduction.

- The measurement lag of this example may be regarded as the resistance to the transfer of heat to the bulb.

Process Lags

Process lags are best explained by considering the examples of a simple process. The simple process illustrated, shows oil which, is being cooled by being passed through a coil in a tank through which cooling water passes. The rate of decrease in the temperature of the oil will depend upon the thermal capacity of the tank and its contents.

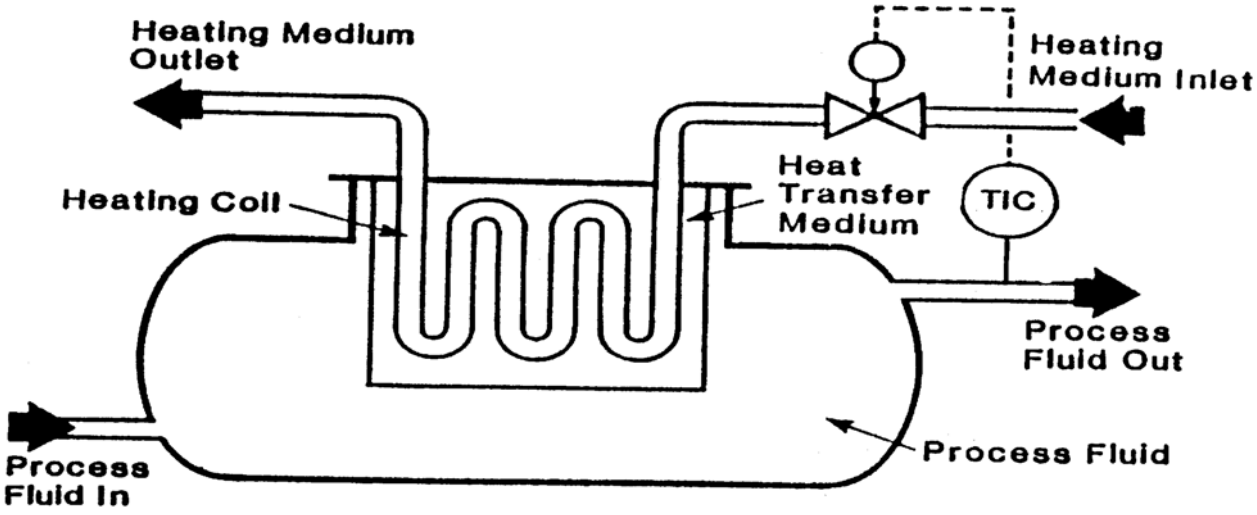


Process Lags

Transfer Lags

If the process is indirectly heated, as shown in the following diagram, the system will have two capacity lags and a resistance lag between the heating medium and the process. The heating medium must transfer heat energy to the heating liquid and the heating liquid must then transfer heat energy to the process. There will be a

definite time lag before the temperature of the process begins to rise. Owing to the time taken for the heat to 'transfer' to the outer tank, the process material will reach its maximum rate of temperature rise more slowly; the process is said to have a transfer lag.



Transfer Lags

Distance and Velocity Lags

Distance and velocity lags are also known as 'dead time'. This is the time required for a change to travel from one point of the process to another. If the temperature of the cold oil (as shown in transfer lag diagram) decreases, some time will elapse before the colder oil will reach the thermometer; this is known as dead time.

The various lags discussed can cause unstable control of the process. However, these lags can be anticipated and reduced if, at the design stage, the correct measuring element, transmitter, controller and final control element are chosen. It is also necessary that at the installation stage these Instruments are fitted correctly and at the commissioning stage they are calibrated and set up correctly to meet process demands. During production these instruments should be maintained to a high

standard. If these conditions are met, there should be a high standard of process automatic control.

Controller and Correcting Unit Lags

Lags in the controller and correcting unit will also affect the quality of control achieved. As in the measuring unit and the process, these lags are due mainly to resistance and capacity and they may be treated in similar fashion to measurement and, process lags.

MULTIPLE LOOP CONTROL

Multiple control loops are considered to exist when two or more input signals jointly affect the action of the control system. The following are termed as multiple control loops: -

- Cascade Control,
- Ratio Control,
- Override Control

Cascade Control

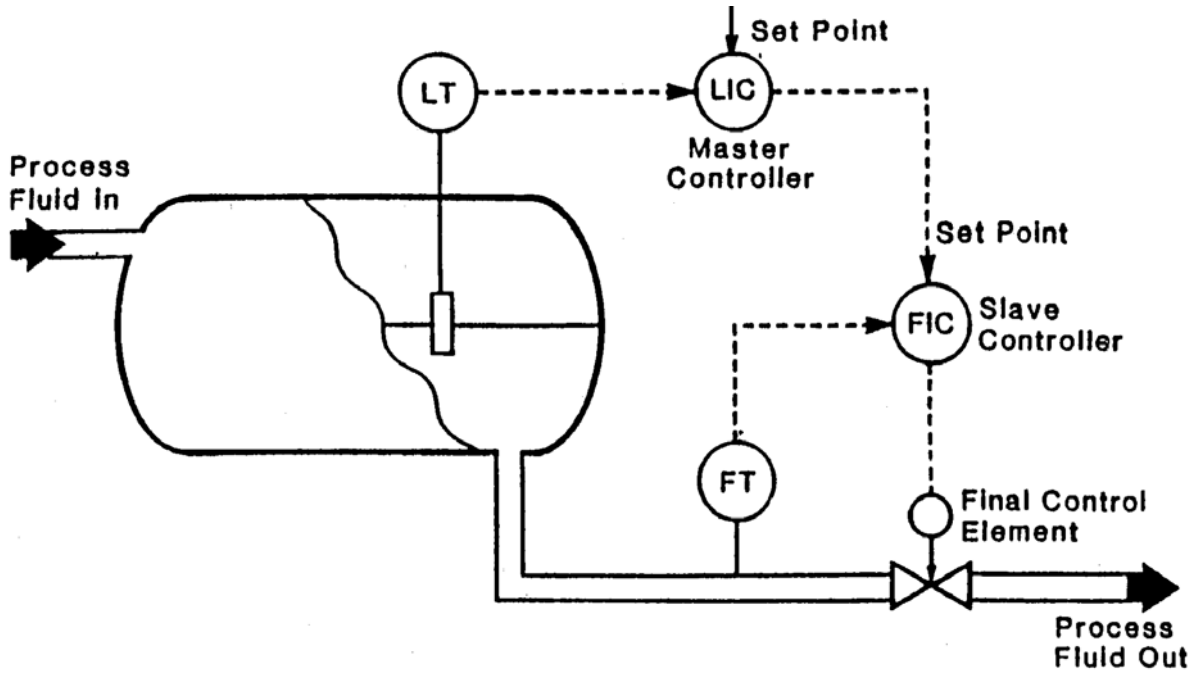
In cascade control the output from one controller "called the master" is the set point for another controller "commonly referred to as the slave". The master will have an independent plant measurement. Only the slave controller has an output to the final control element.

The advantages of cascade control are:

1. Variations of the process variable measurement by the master controller are corrected by the slave control systems.
2. Speed of response of the master control loop is increased.

3. Slave controller permits an exact manipulation of the flow of mass or energy by the master (to maintain the process variable, measured by the master controller within the normal operating limits)

Disadvantage: However, cascade control is more costly. Thus, it is normally used when highly accurate control is required and where random process disturbances are expected.



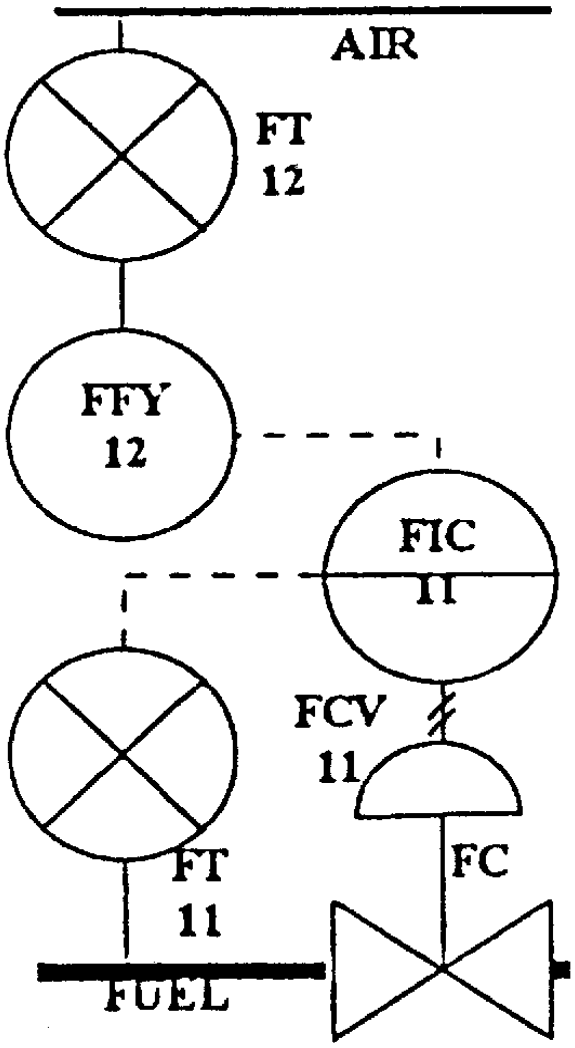
Cascade Control

Ratio Control

Ratio control is where a predetermined ratio is maintained between two or more variables. Each controller has its own output to separate final control elements.

The set points of both controllers are set from a master primary signal. The set points on the controllers can be adjusted by altering the required ratio between the two process variables. Ratio control is commonly used on the air/fuel flow for combustion chambers or gas turbines.

Advantage It follows that the advantage of ratio control is the ability to maintain a consistent ratio of two or more process variables, rather than relying on two independent controls.



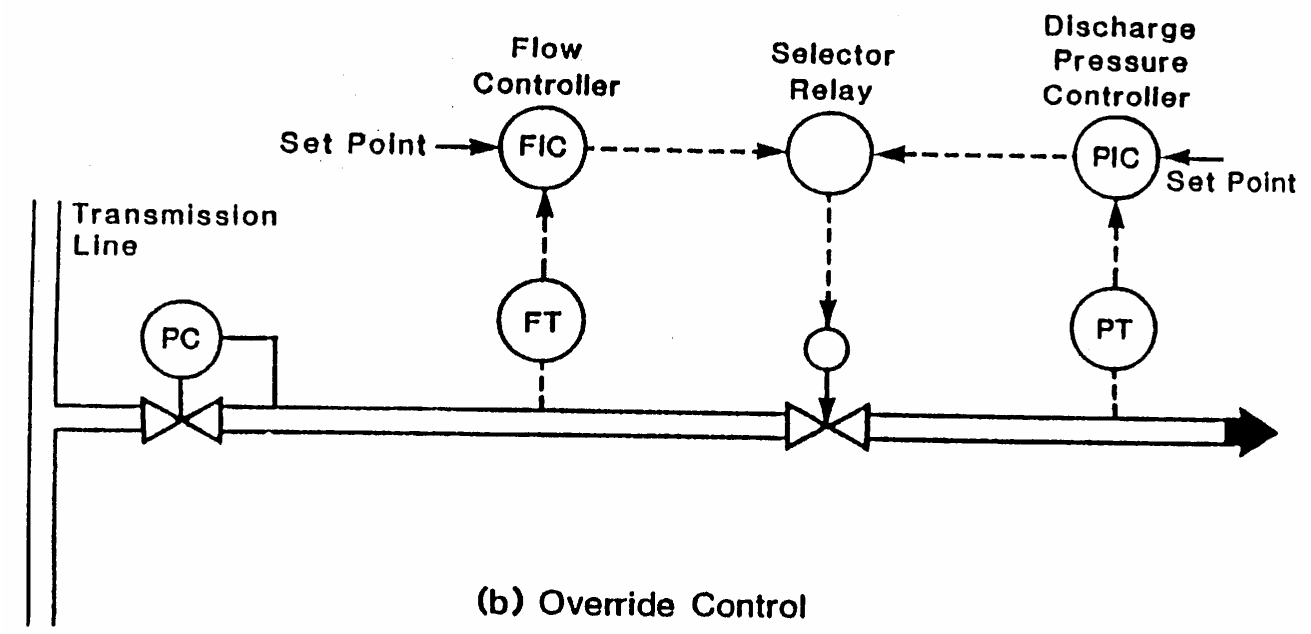
This is a ratio control system with two flow transmitters (FT 11 and FT 12). The ratio station FFY 12 gives an electrical output to the flow-indicating controller FIC 11, which controls the ratio of fuel to air. This controller operates a pneumatic control valve (FCV 11) The FC next to the valve indicates that the valve will close if the air supply fails. This system is often used on furnace systems.

Override Control

In override control either the highest or lowest signal from two or more input signals is automatically selected by the selector relay.

An example of override control is shown in diagram (b) for flow and pressure control of a gas distribution system. Normally, the distribution valve is controlled by the discharge pressure controller, but under high demand conditions the control is transferred to the flow controller.

This system limits the maximum flow rate and the maximum pressure in a distribution system.



(b) Override Control

SGC HSE Regulation Related to:

1. Refer to HSE Regulation No. 6 "Permit To Work System" to:

- Issue a cold work permit before starting the job.

2. Refer to HSE Regulation No. 7 "Isolation"

7.18 Control Systems Procedures and Isolations

1. Preparation for Work

When work is to be done on control engineering hardware, it is essential that the work site is prepared correctly. The work permit system (Regulation No 06) provides the mechanism for ensuring that essential preparatory activity is documented and witnessed. It is important that the Control Engineering Person doing the work also has responsibility for effecting the task safely. This is particularly important in the control engineering discipline where the instrument may still be connected to the process, or may require to be serviced with power-on for fault finding.

2. Interaction

Care must be taken to ensure that the work to be carried out on a specific item of instrumentation will not cause a hazard due to interaction with other protection systems or operational process controls.

3. Preparatory Work

Prior to a work permit being issued all appropriate preparatory work at the site must be completed. The following items are some examples.

a. Removal of potential hazards from the area. Particular vigilance is needed for enclosed areas (refer to Regulation No 09 Confined Space Entry).
b. Gas testing the area for flammable, toxic or suffocating gases.
c. Construction of scaffolding to permit safe access.
d. Provision of additional fire-fighting apparatus.
e. Provision of necessary protective clothing.
f. Isolation of the control engineering hardware from the process.
g. In the case of equipment removal, isolation of the hardware from utilities (eg electricity, air supplies etc).

On completion of all necessary preparatory work (defined by the Senior Control Engineering Person and the Area Authority), a hot or cold work permit/entry permit will be issued signed by the appropriate responsible authorities.

4. Isolation of Hardware

Isolation of control engineering hardware may be necessary to enable maintenance work to be done or permit removal of the hardware to effect repairs (either locally or remotely). Isolation of hardware can take several forms, for example isolation from:

a. Process plant.
b. Utilities (electric, pneumatic, hydraulic, cooling media etc)
c. Larger system of which the hardware is a subsystem or component

5. Isolation from Process

a. Isolation of instruments, which are connected to or form a part of the process, is usually achieved by valving. It is important that, where isolation of an instrument is required for maintenance purposes, correct venting/draining and valve closure procedures are adhered to.
b. Where instruments have local isolating valves in addition to the primary process isolating valves, the local valves may be used for some routine in-situ testing at the discretion of the Senior Control Engineer. If an instrument is to be removed from site, the process isolating valves must

be used and any impulse pipe-work must be drained or vented completely.

c. Where the process fluids are of a hazardous nature (e.g. toxic, flammable etc), particular care should be taken to ensure correct venting and draining, and also to clean or flush the instrument carefully, prior to effecting work or removal of the hardware from site for maintenance or repair. Gas testing may be required. On large items, eg control valves, a certificate of cleanness is necessary prior to delivery to workshops.

d. On removal of a directly mounted instrument, from a process line containing hazardous fluids, eg pressure gauges etc, isolation by the primary isolation valve only is NOT acceptable. The valve outlet shall be blanked off, capped or plugged with a blank flange, solid screwed plug or cap, whichever is appropriate.

6. Isolation from Electrical/Pneumatic Supplies

a. If practical, equipment must be made safe before any work is done on it. The operation of making the equipment safe must be done by a Competent Control Engineering Person. Care shall be taken when working on live equipment to ensure avoidance of contact with live electrical components (refer to Regulation No 19 Working with Electricity).

b. Pneumatically operated equipment must be isolated before it is disconnected or removed for repair by closing the valve at the supply manifold for the individual instrument and venting through the drain/vent of the pressure regulators.

7. Isolation from Utilities

a. Control engineering equipment may be connected to utilities (other than electrical associated with the hardware e.g. steam, cooling water, hydraulic fluid, chemicals, carrier gases (analysis) and air supplies. It is important

that attention is given to rendering the utilities safe when the control engineering hardware is being serviced or removed.

b. Utilities should be isolated at the point of distribution to the control engineering equipment being removed (e.g. isolating valve at distribution head) and not solely at the hardware itself.

c. Where utility fluids are 'piped' to an instrument, the pipe work should be drained down or vented if the instrument is removed.

d. It is important that removal of a utility from a specific piece of hardware does not influence any other hardware to which the utility may also be connected (eg cooling water may have been series connected to more than one item of hardware).

8. Use of Tools and Test Equipment

a. Tools and test equipment must be suitable for use in the work area. They should be checked before and after use and all calibration equipment should itself be calibrated periodically, at intervals determined by the Senior Control Engineering Person.

b. Use of tools and test equipment are subject to the Work Permit System (refer to Regulation No 06). It is particularly relevant to ensure that electrical tools and test equipment comply with the area safety classification of the work place. This may be achieved either by certification or using the Permit to Work.

9. Workshop Practice

It is essential that good workshop practice is adhered to at all times, for example:

a. Machinery will not be operated without guards or suitable personal protection.

b. All workshop equipment will be maintained in good working order.

c. Good housekeeping is essential to safety.

d. All necessary spares, cleaning materials, tools and test equipment should be available and must be correctly maintained, tested (where appropriate) and stored.

10. Changes and Modifications

Changes to alarm settings and transmitter ranges must be approved by the operating authorities and documented. No such change will be made unless the Passport Work Order has been completed. Changes to trip settings should only be carried out under the authority of a PMR endorsed by the ED. This is mandatory at all times. Similarly, no modification will be undertaken unless the proposal has been through the established PMR and appropriate authorisation. Thus, before work starts, the job must be approved and finance made available.

3. Refer to HSE Regulation No. 19

19.7.4 Precautions on Low Voltage Systems

1. The consequences of shock, or serious burns, from short circuits associated with low voltage systems (50 - 1000V ac/120 - 1500V dc between conductors, or 50 - 600V ac/120 - 900V dc between conductor and earth) can be serious and often fatal. Whenever possible therefore, work on low voltage equipment and cables shall be carried out after they are proved **DEAD** by use of an approved instrument and where appropriate **EARTHED** using an Electrical Isolation Certificate (refer to Paragraph 19.8).

2. If it is not possible to make **DEAD,** to prove **DEAD** and where appropriate **EARTH** low voltage systems, work on them shall be carried out as if they were **LIVE** using a Sanction For Test Certificate (refer to Paragraph 19.9).

19.7.5 Precautions on Extra Low Voltage Systems

1. Control and telecommunications plant operating at extra low voltage (< 50V ac/120V dc between electrical conductors or to earth) shall not be worked on without an Electrical Isolation Permit being issued. This is necessary to prevent the possibility of sparks in a hazardous area (refer to Paragraph 19.8).

2. Battery systems with high stored energy can be dangerous to personnel and therefore precautions should be taken when working with such systems. In particular flooded cells requiring electrolyte replacement are hazardous. Where these types of cells exist, a local procedure should be produced for work on battery systems.

FINAL CONTROL ELEMENTS

FINAL CONTROL ELEMENTS

OBJECTIVES

At completion of this module, the developee will have understanding of:

1- Introduction "Purpose of final control element"

2- Control valve major components

3- Main types of control valves

4- Control valve body definition and components

5- Valve flow characteristics

6- Globe valve components that controls its characteristics

7- Use of each characteristic

8- Valve shut off capability

9- Control valve selection criteria

10- Definition of control valve sizing coefficient

11- Definition of Flashing & Cavition phenomena

12- Brief about control valve noise

13- Actuator types and modes of actuation

14- Actuators sizing considerations (general)

15- Actuators for sliding stem and rotary shaft valves

16- Valve positioners and their benefits

17- Basic principle of positioner mechanism

17.1 Motion balance

17.2 Force balance

18- Types of positioners

18.1 Pneumatic input type

18.2 Current input type

19- I/P converters (Current to pneumatic converter)

19.1 Electromagnetic type

19.2 Solid State type

20- Limit Switches

21- other accessories:

- Volume Booster, Trip Valve

22- Position Transmitters

23- Hand Wheels and Manual Operators

24- Control Valve Installation and Maintenance

Related Safety Regulations for Module 3: Final Control Elements

Juniors have to be familiarised with the following SGC HSE regulations, while studying this module:

Regulation No. 5: Risk assessment

Regulation No. 6: Work to permit system

Regulation No. 7: Isolation

7.18 (1-10) control systems procedures and isolations

Regulation No. 8: Breaking containment

Regulation No. 23: General Engineering Safety

Regulation No. 27: General services

Safe use of hand tools and powered tools/equipment

Introduction

In process systems, the final control element is normally a pneumatically actuated control valve, which is used to regulate the flow of a fluid. It provides the necessary power to translate the controller's output to the process. Pneumatics is used because of the original popularity of pneumatic control systems and the comparatively low operating pressures used, also for safe operation in the oil & gas facilities.

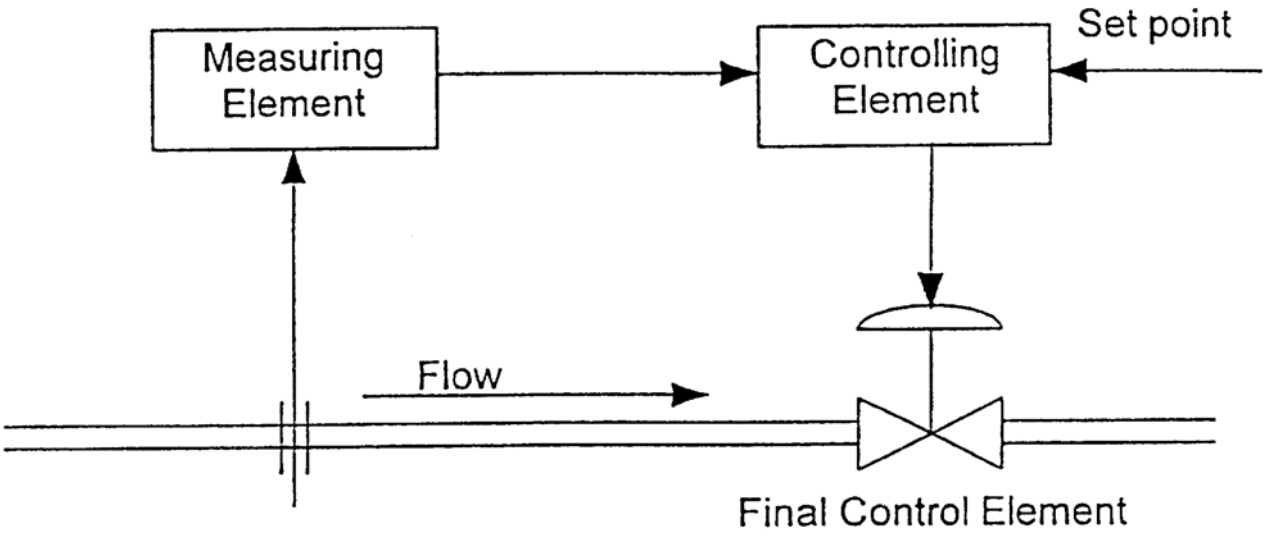


Figure 1. Final Control Element in a Control Loop.

As shown in figure 1, in the basic components of a control loop, the control valve is subject to the harshest conditions. A control valve is also the most expensive item and the most prone to incorrect selection.

Major Parts of the Control Valve:

The major parts of any control valves are:

1. The actuator, and
2. The valve body assembly

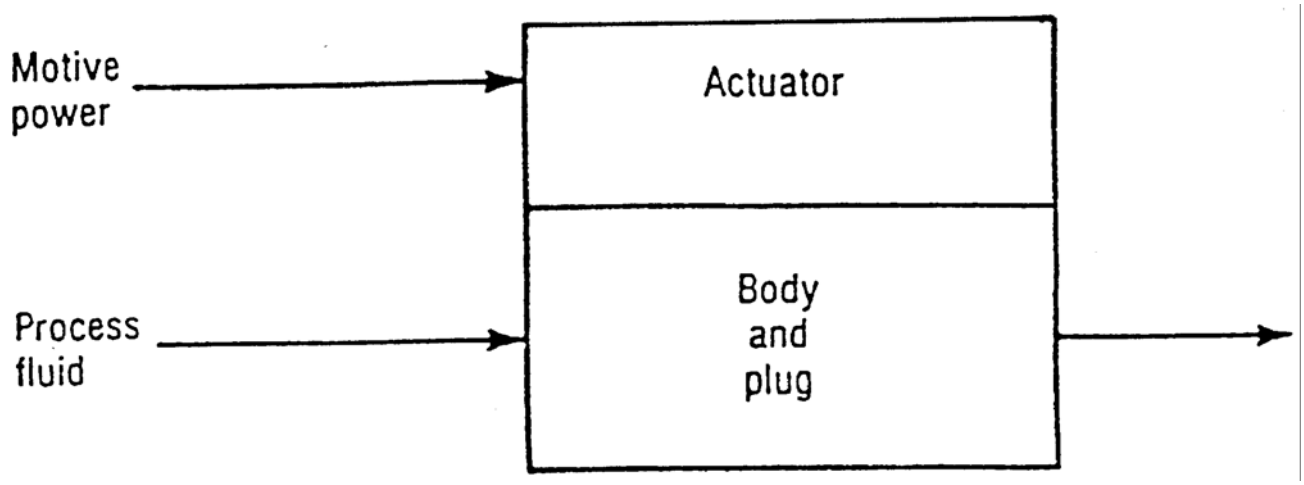


Figure 2. Major Parts of the Control Valve

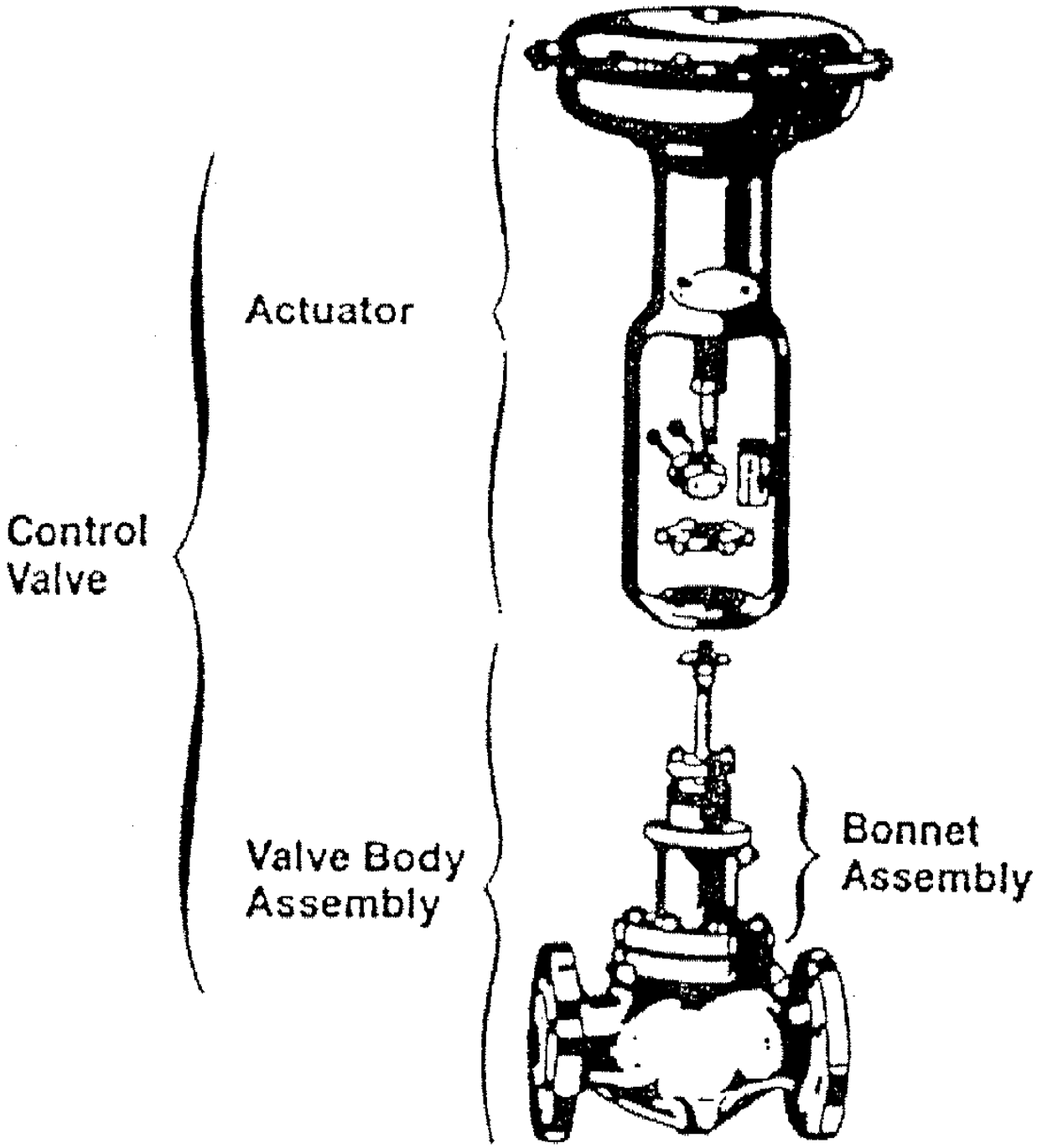


Figure 3. Major Parts of the Control Valve

There are also several types of body designs, flow characteristics, actuator types and trim designs.

Figure 4. Control Valve Terminology

Functional block Diagram of the control valve:

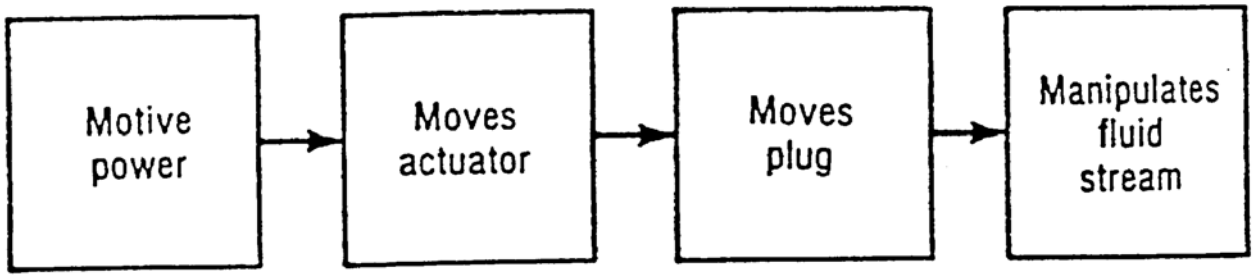


Figure 5. Functional Block of the control valve

In most cases a control valve is expected to respond to a control signal to keep a process variable steady.

Main Types of Control Valves:

Control valves can be classified based on body design as follow:

1) Sliding Stem Control Valves

1.1- Globe Bodies

Globe valves are the most common type in use today. They may single port, double port and three-way. Split body and angle valves are classified as special type globe valves

a) Single Port

Single port valves are simple in construction, frequently used in sizes 2 inches and below, provides tight shutoff, but may have high unbalanced forces on the plug requiring large actuators. These valves can be constructed to have the valve plug move into or out of the port with increasing actuator-loading pressure. Figure 6 shows a typical design for a single port body unbalanced design.

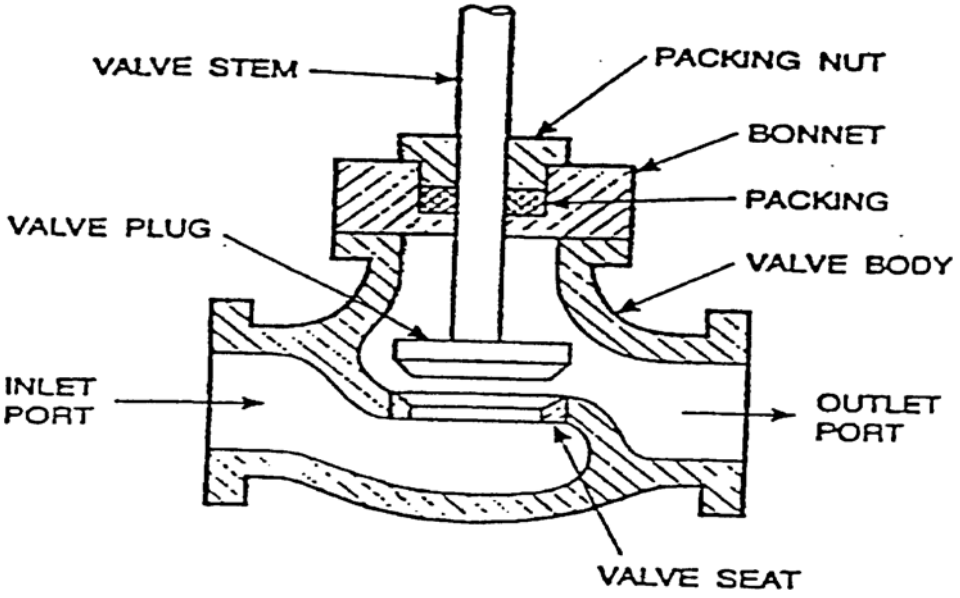


Figure 6. Single port - globe body with plug to move into the seat with increasing signal Pressure.

b) Double Port

Double port valves balance the forces acting on single port valves (figure 7). They have higher flow capacities and require smaller stem forces compared to the same size single port valve. They are frequently specified for sizes larger than 2 inches but should not be used when leakage is unacceptable. Reversible plug design is available to open or close the valve with increasing loading pressure.

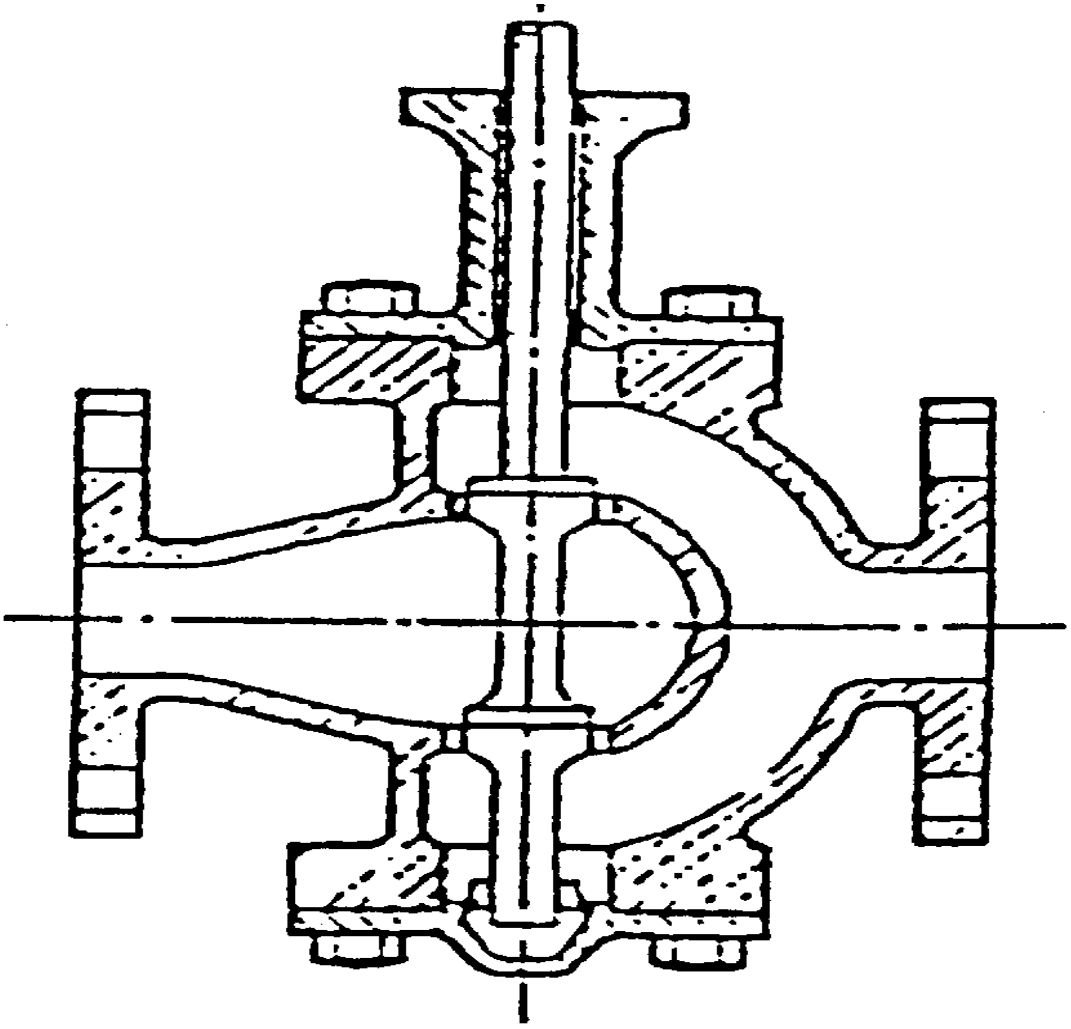


Figure 7. Double port globe body provides fewer imbalances of plug forces.

c) Three- Way

Three-way valves are designed to **blend (mix)** or to **divert (split)** flowing streams. Total flow is proportioned only, not controlled, in either service. Most three-way valves have the characteristic of unbalanced forces on the valve plug and require large operators. They are usually installed with the flow tending to open the valve plug discs to prevent "slamming" of the valve plug.

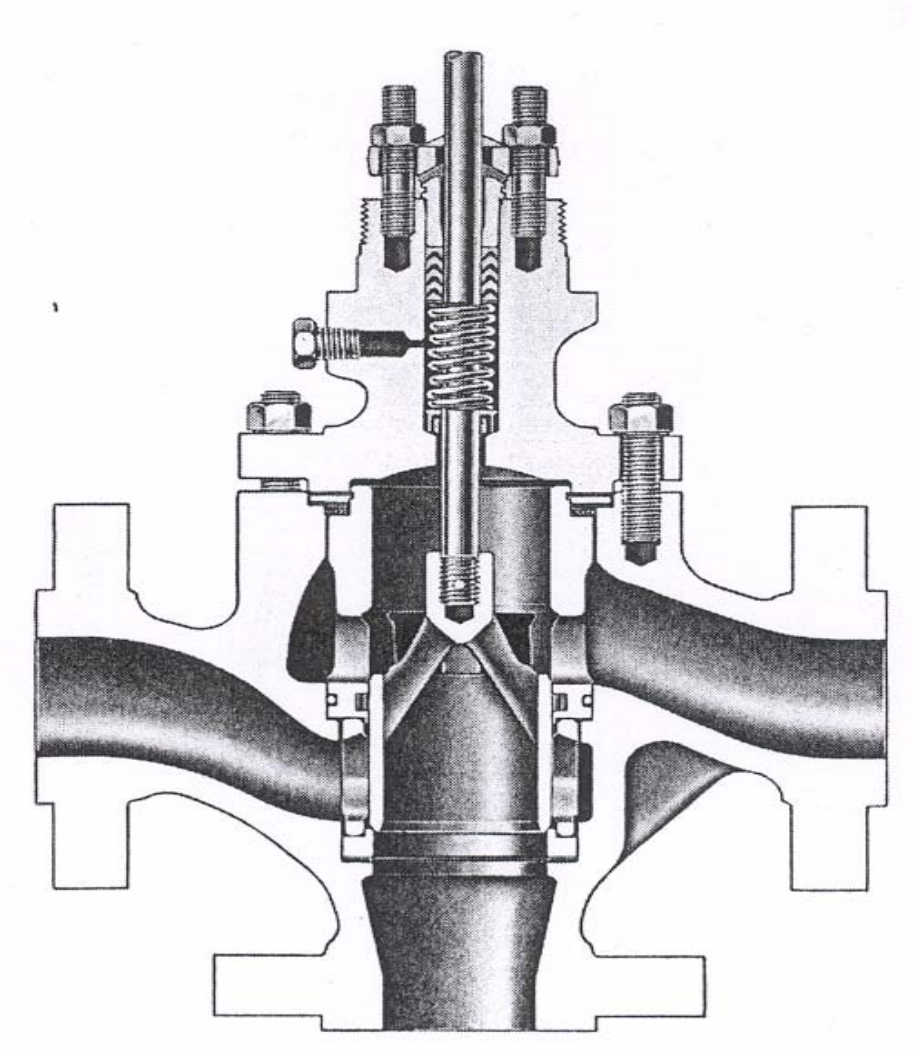


Figure 8. Three-way valve have three connections for converging (mixing) or diverting (spliting) operation.

d) Angle Valves

Angle valves nearly always single ported are often used where space is at a limited. They are easily removed from the line and can handle sludge and erosive materials.

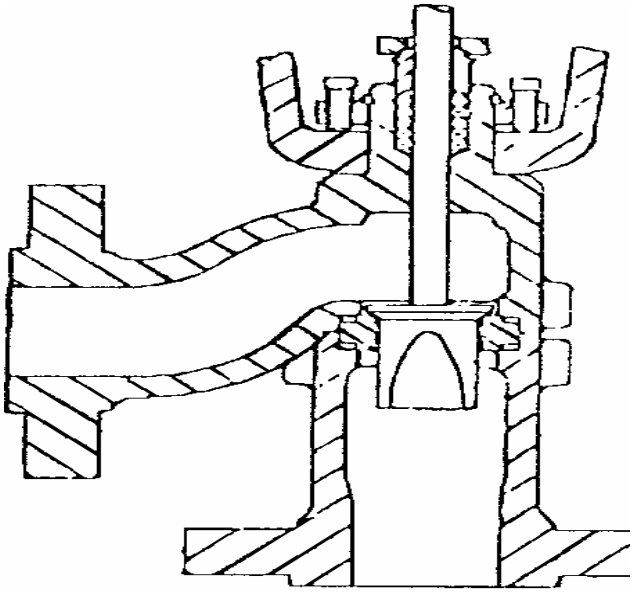


Figure 9. Angle valve with split body construction is easily removed and reinstalled.

1.2- Diaphragm Valve

The diaphragm valve consists basically of a body, bonnet and flexible diaphragm (figure 10). It is more often referred to as a Saunders-Type valve. Closure is made by using a flexible dome-like diaphragm against a weir.

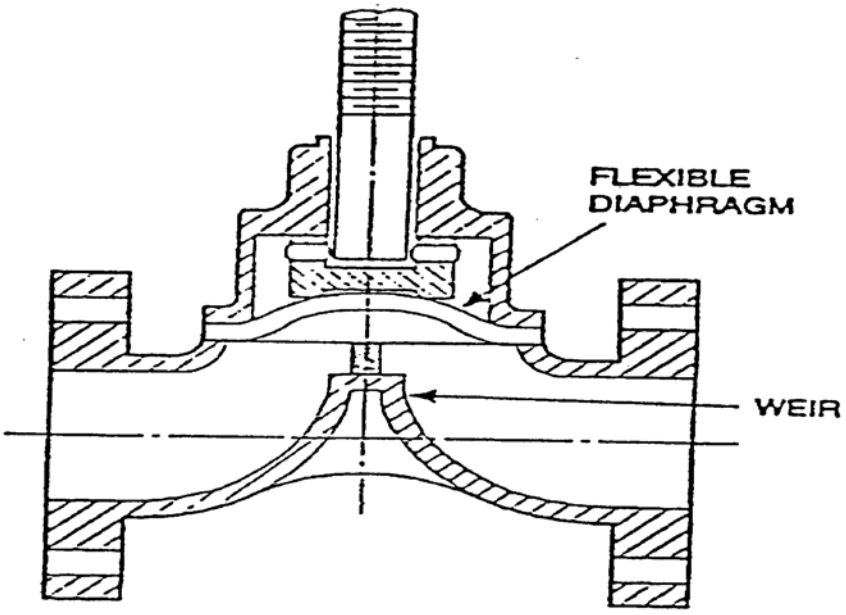


Figure 10. Diaphragm/Saunders' Valve

This type is suited for slurries and viscous fluids the diaphragm valve has high capacity, its cost is relatively low. The diaphragm seals the working parts of the valve from the process fluid and is the only wearing part of the valve. The Saunders-Type valve exhibits relatively poor control characteristics and has a low turndown ratio.

2) Rotary Stem (Shaft) Control Valves.

a) Full Ball and Vee notch Ball valves

Figure 11 shows ball valve design made for hard-to-handle fluids such as paper stock polymer slurries, heavy crud and other fluids with entrained solids. These high-recovery (low-pressure loss) valves have good control characteristic and high rangeabilities. Full Ball valves are used mainly for S/D and isolation, but not for control. Vee notch ball valves are used for control.

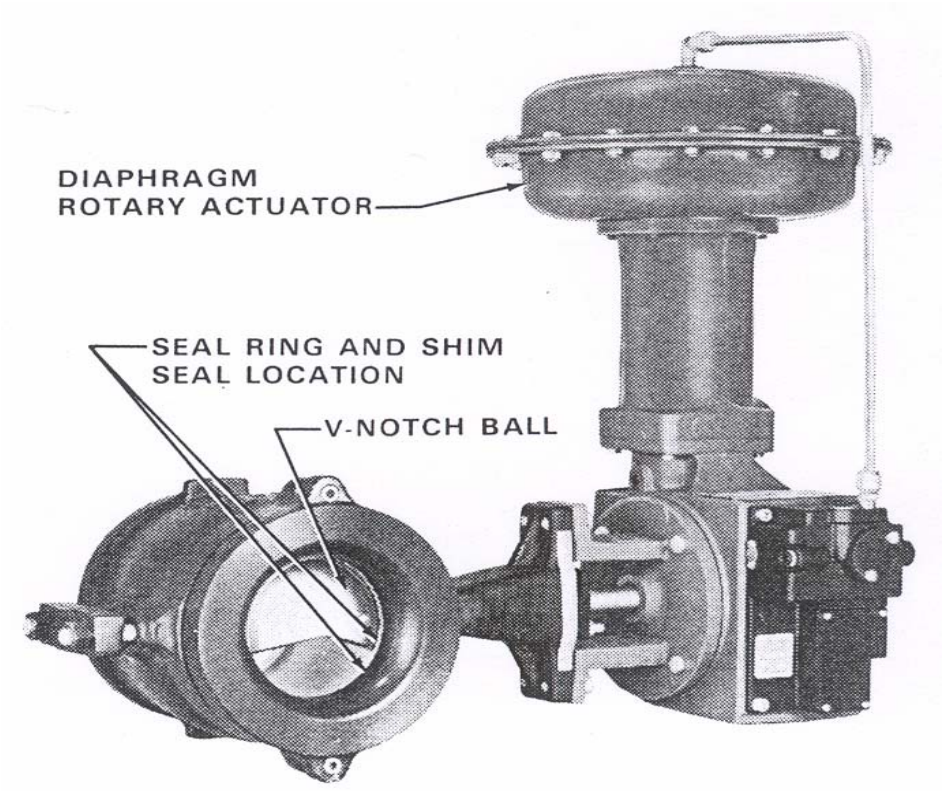


Figure 11. Partial ball (Vee notch ball) body design for hard to handle fluids as paper stock, heavy crude and polymer slurries.

b) Eccentric Rotating Plug (desk)

The "Camflex" valve is a rotating plug valve that has a centre of rotation eccentric to the centreline of the seat (figure 12) When the plug rotates to close the valve port, the plug face moves into the seat with a cam-like motion. Design is such that little or no rubbing action occurs after contact is made between plug and seat and the stem elastically deforms to give a tight shutoff. Some valve designs permit installation of reduced-trim seats without replacement of the plug. Valve flow characteristics are between equal percentage and linear buts are nearly linear. It can be used for hard to handle fluids. It has good tightness class and can be used at relative high pressure.

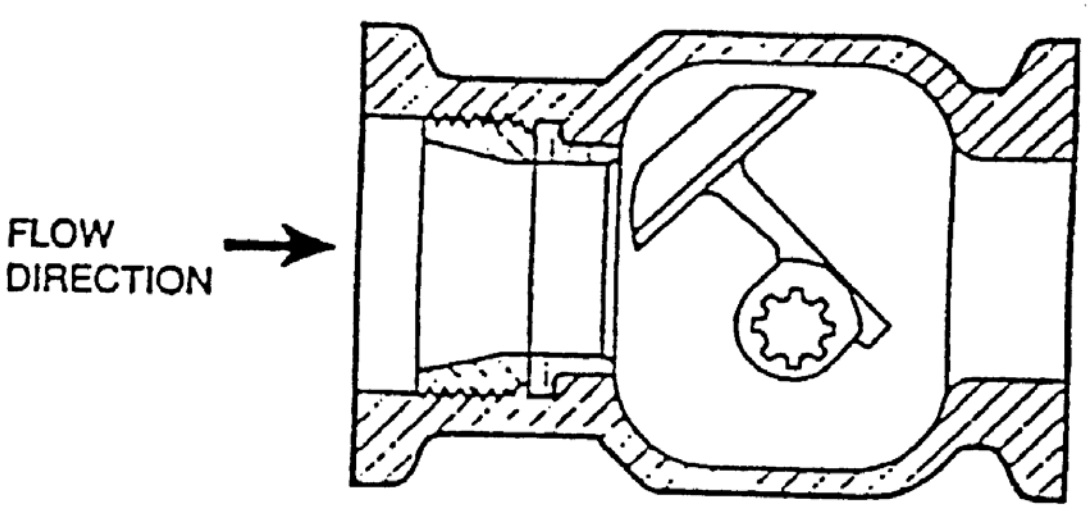


Figure 12. The Camflex valve has a centre of rotation eccentric to the centre line of the seat.

c) Butterfly

A butterfly valve consists of a shaft-supported vane or disc capable of rotating within a cylindrical body. In early industry use butterfly valves were specified primarily for low-pressure drop applications at low static pressures where control was not critical and where high leakage rates could be tolerated. In the last few years butterfly designs have been

upgraded for high-pressure drops high static pressures and tight shutoff. Tight shutoff is accomplished through use of soft composition seats for seating the metal vanes (figure 13) Butterfly valves are economical especially in larger sizes because of their simple design and high capacity. They require a minimum space for installation and often reduce pumping costs because of their low-pressure drop characteristic.

One of the disadvantages of the butterfly valve is the high operating torque requirement due to fluid How through the valve. Butterfly valves commonly have been used for throttling control between 10° and 60° openings because torque conditions cause instability beyond this range. Its tightness class is relatively low.

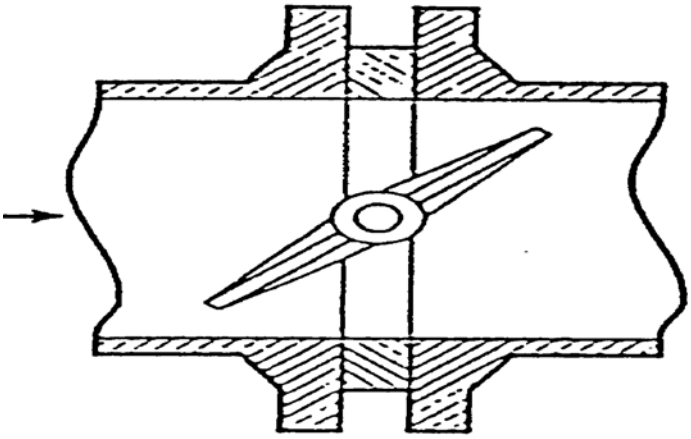
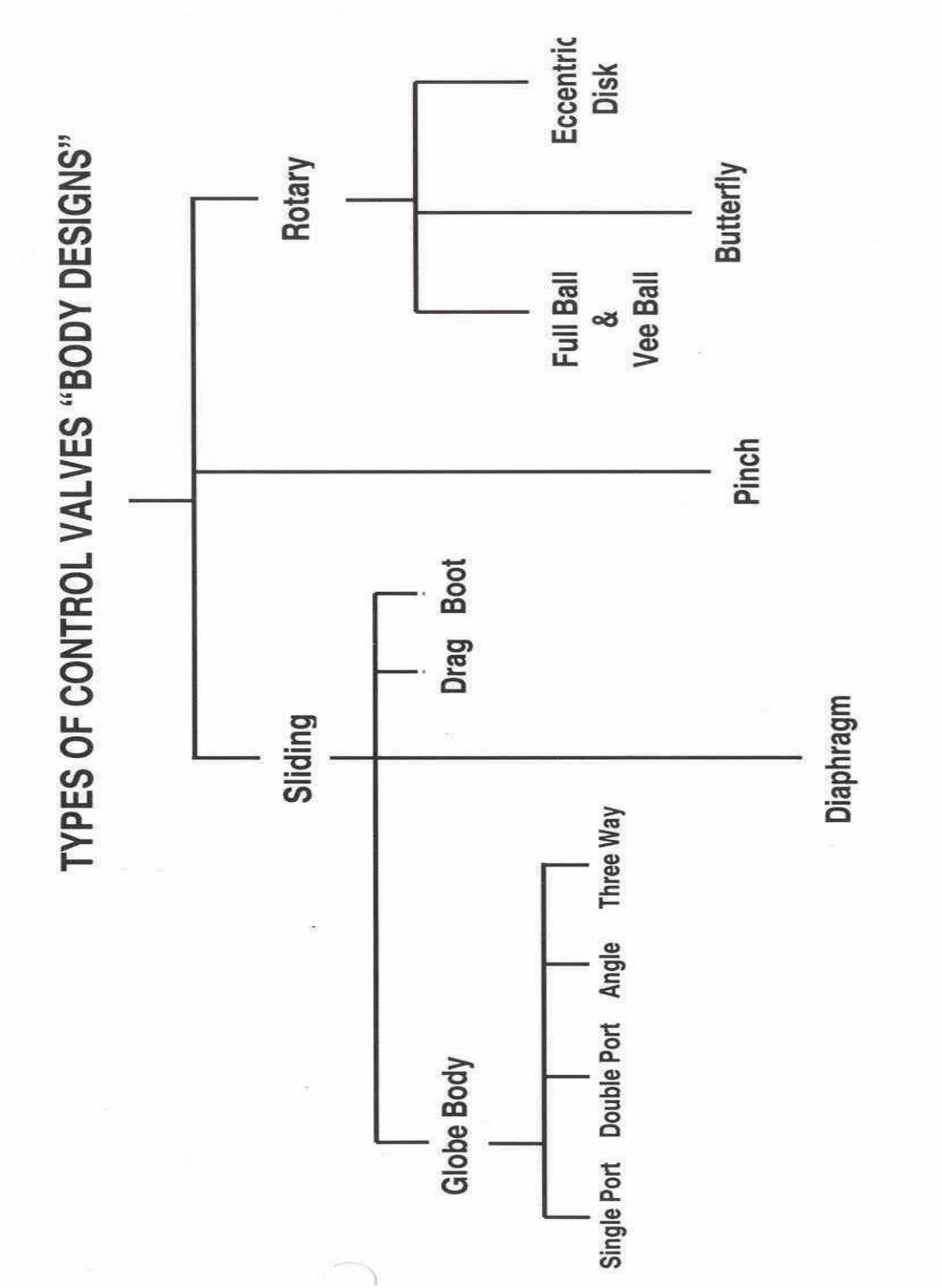


Figure 13. Butterfly valve with rubber lining (soft seat) for tight shutoff characteristic


TYPES OF CONTROL VALVES "BODY DESIGNS"
Rotary
Eccentric Disk
Butterfly
Full Ball & Vee Ball
Pinch
Sliding
Drag Boot
Diaphragm
Globe Body
Three Way
Angle
Double Port
Single Port

Control Valve Body definition and components

As shown in figure 14, valve body is a housing for internal parts that having inlet and outlet flow connections. Among the most common valve body constructions are:

a) Single-ported valve bodies having one port and one valve plug,
b) Double-ported valve bodies having two ports and two valve plugs on the same stem,
c) Two-way valve bodies having two flow connections, one inlet and one outlet,
d) Three-way valve bodies having three flow connections, two of which may be inlets with one outlet (for converging or mixing flows), or one inlet and two outlets (for diverging or diverting flows).

(The term Valve Body, or even just Body, frequently is used in referring to the valve body together with its bonnet assembly and included trim parts. More properly, this group of components should be called the Valve Body Assembly)

Valve Body Assembly

An assembly of a body, bonnet assembly, bottom flange (if used), and trim elements. The trim includes the valve plug, which opens, closes, or partially obstructs one or more ports.

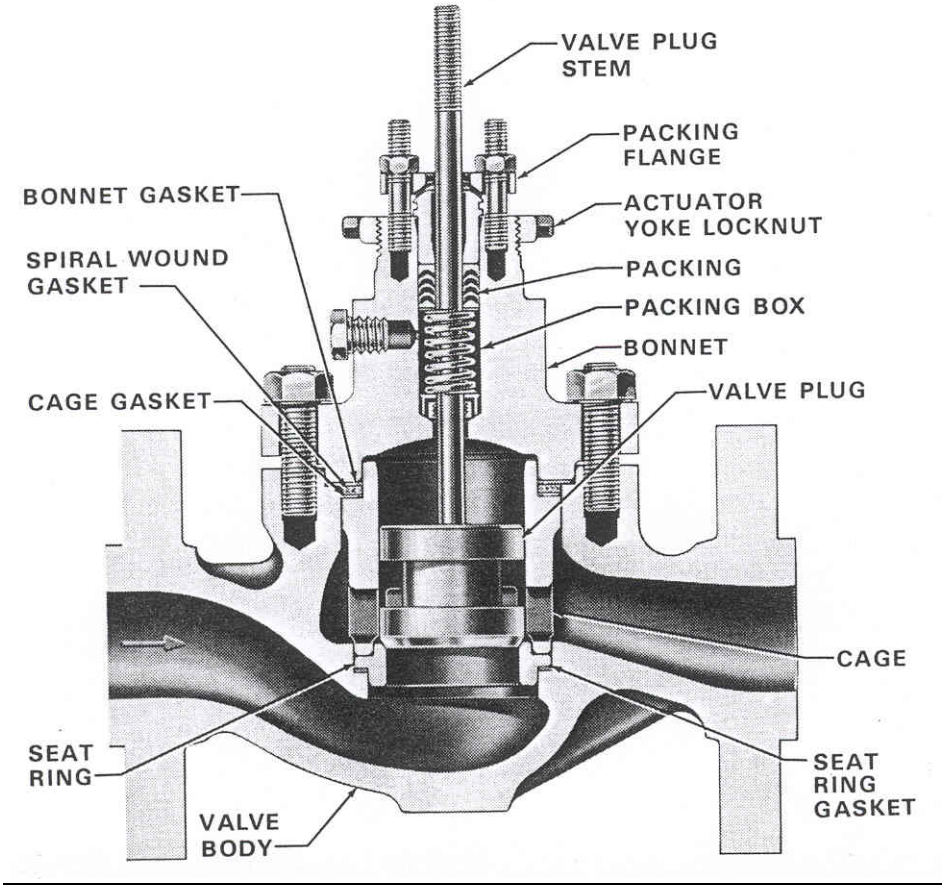
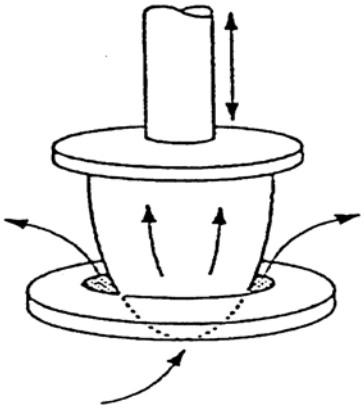


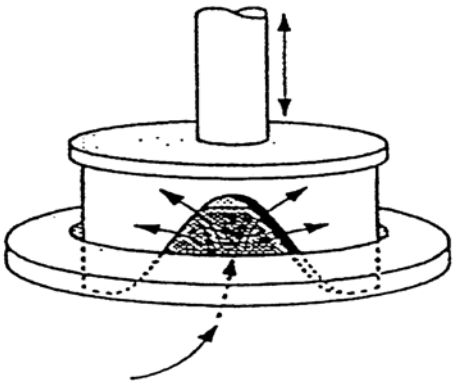
Figure 14. Valve Body Assembly.

Valve Trim:

Valves control the rate of flow by introducing a pressure drop across the valve trim. (In a globe valve, the valve trim would be typically include valve plug, seat ring, cage, stem and



Contoured plug



'V' - port plug

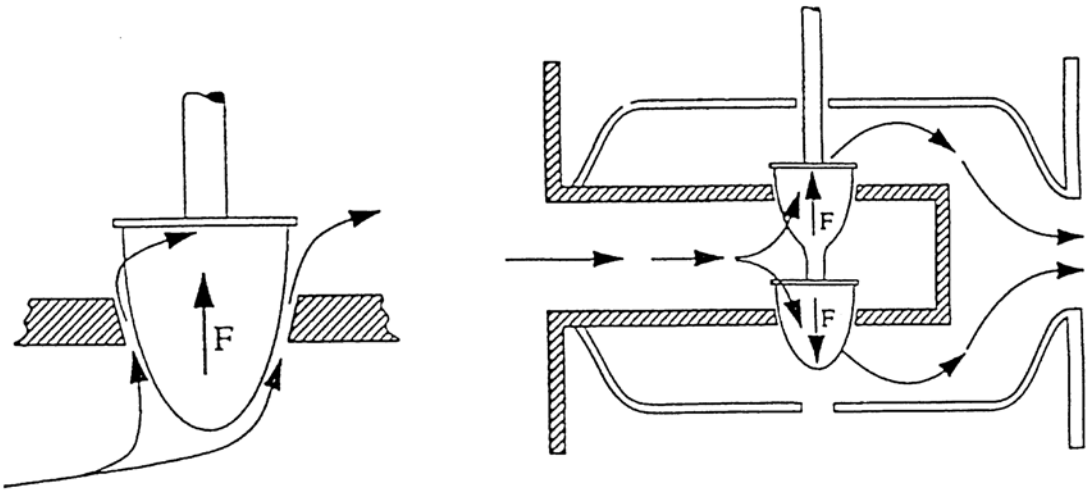
stem pin. These are usually sold as matched sets, which have been ground to a precise fit in the fully closed position). Figure 15 illustrates some of valve plugs and seats.

Figure 15. Valve Trim.

Bonnet Assembly:

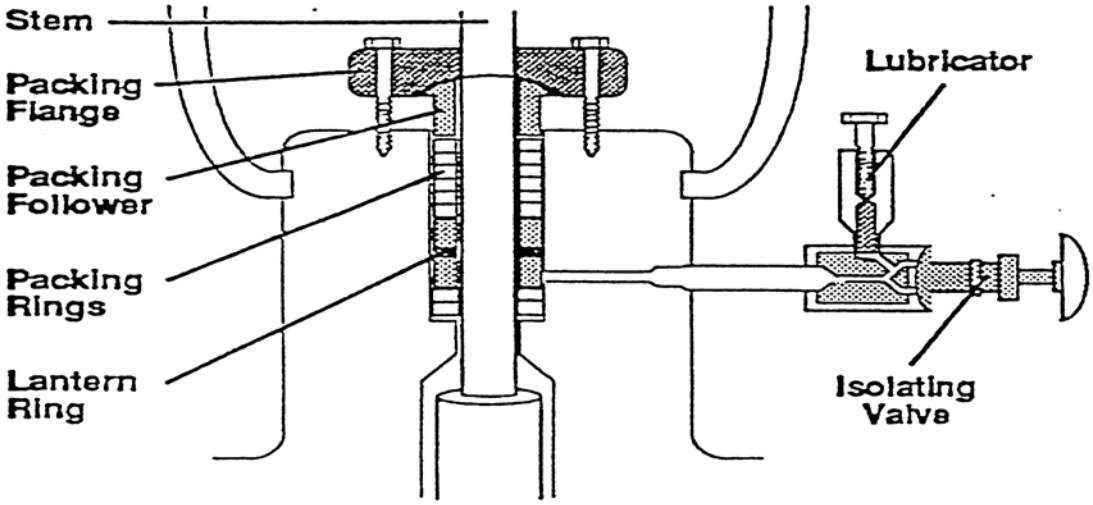
Bonnet Assembly: An assembly including the part through which a valve plug-stem moves and a means for sealing against leakage along the stem. It usually provides a means for mounting the actuator.

Packing Box Assembly:

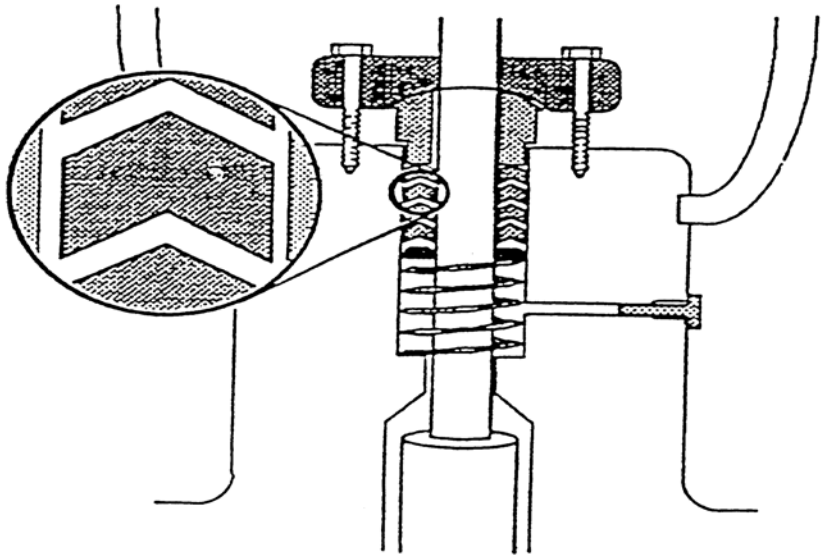


Single seat Double Seat

The part of the bonnet assembly used to seal against leakage around the valve plug stem. Included in the complete packing box assembly are various combinations of some or all of the following component parts: Packing, Packing Follower, Packing Nut, Lantern Ring, Packing Spring, Packing Flange, Packing Flange Studs or Bolts, Packing Flange Nuts, Packing Ring, Packing Wiper Ring, Felt Wiper Ring. Figure 16 shows Packing box assembly.



Bolted Packing Box Assembly



V-Shape Teflon Packing

Figure 16. Packing box assembly.

Valve Flow Characteristics

Valve flow characteristic was defined as the relationship that exists between valve flow and valve position. Almost any kind of characteristic can be obtained by proper shaping of the seat and plug. The purpose of characterising is to provide control loop stability over the expected range of operating conditions.

Flow characteristics fall into three major types (figure 17), quick opening, linear and equal percentage.

Many variations of these types occur because of inherent valve design or because changes are engineered into the plug and seat design. The three major types are discussed below as well as some of their modifications.

Quick Opening

A quick opening characteristic provides for a maximum change in flow rate at low stem travel while maintaining a linear relationship through most of the stem travel. In Figure 17 about 90 percent of valve capacity is obtained at 30 percent valve opening and a straight-line relationship exists to that point.
Quick opening valve plugs are used primarily for on-off service or in self-actuated control valves or in regulators. They are also suitable for systems with constant pressure drops where linear characteristics are needed.

Linear

A valve with a linear flow characteristic produces flow directly proportional to the valve lift. Fifty percent of valve lift produces 50% of valve flow etc. This proportional relationship produces a constant slope so that each incremental change in valve plug position produces a like incremental change in valve flow if the pressure drop is constant. Linear valve plugs are commonly specified for liquid level control and for control applications requiring constant gain.

Equal Percentage

An equal percentage flow characteristic is one in which equal increments of stem travel produce equal percentage changes in existing flow. For example when the flow is small the change in flow (for an incremental change) is small; when the flow is large the change in flow (for an incremental change) is large. The change is always proportional to the quantity flowing before the change. Equal percentage valve plugs are used on pressure control applications where only a small percentage of the system drop is available for the control valve. It can be used for flow control.

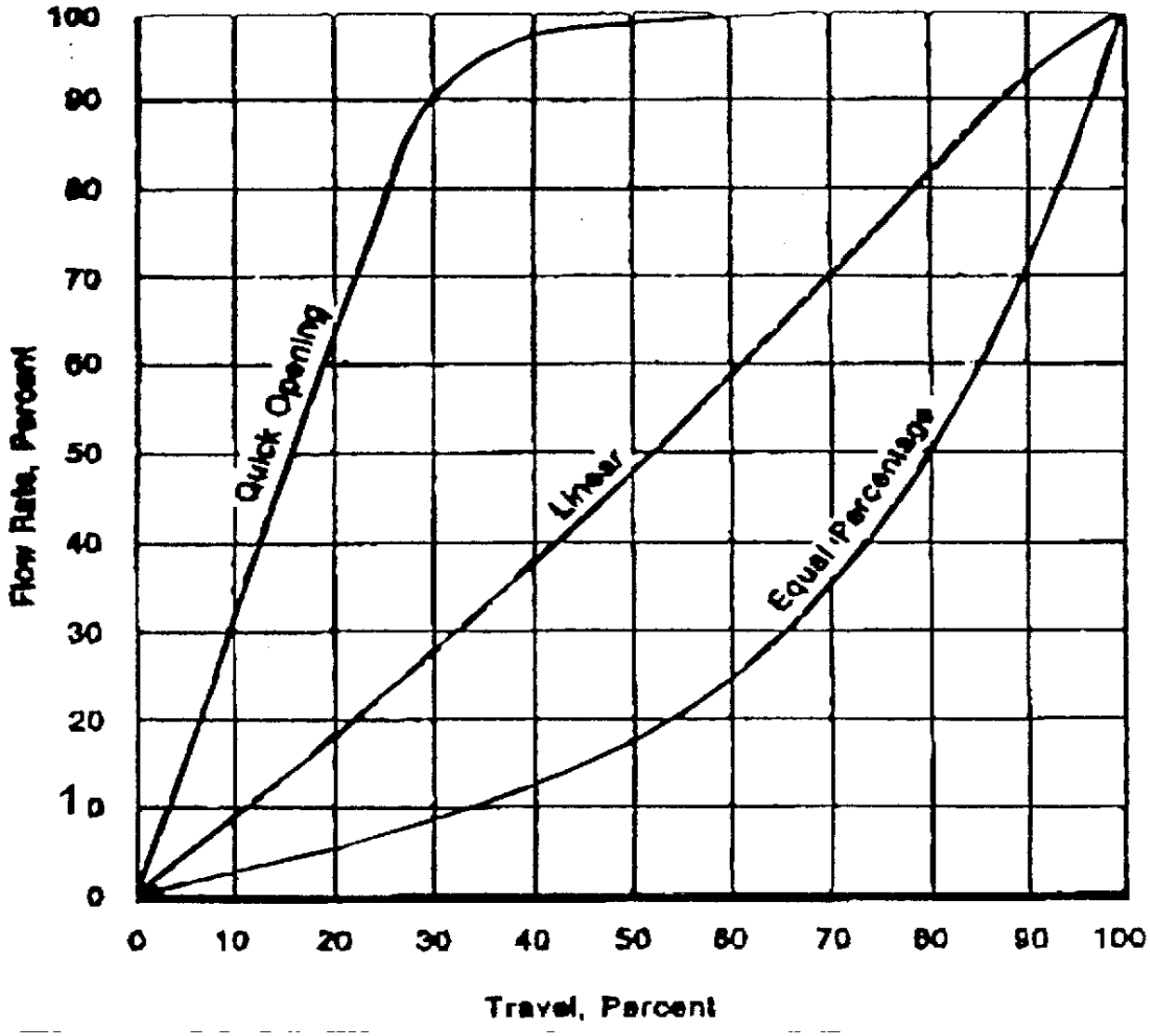


Figure 17. Percentage Flow Characteristics

Trim Design and Components of Globe Valves

The shaping of plugs, seats and cages to obtain the desired flow characteristic would logically be a function of trim design. This section however covers other design concepts relating to valve trim that affect not only the characteristic curve but also how the valve responds to problems such as erosion, cavitation, vibration, high pressure drop, noise and other similar problems. The term trim applies to the parts of a valve (except the body housing) that come into contact with the flowing fluid another term often used is wetted parts.

Plugs

Primarily valve plug shapes or patterns determine valve flow characteristics. Figure 18 shows some typical plugs for linear trim single port valve, equal percentage trim and quick opening characteristic valves.

Seats

The seat or seat ring is that portion of the valve trim or body that the plug contacts for closure. The seat ring may be screwed or welded to the body.

Metal-to-metal contact between plugs and seats is standard practice. They can be machined accurately enough to prevent high leakage rates. However, when tight shut-off is required, soft seats made of Teflon, hard rubber or other resilient composition materials, are used to provide the necessary tight closure. The resilient part may be an insert in the seat.

Cages

Cage is a hollow cylindrical trim element that is a guide to align the movement of a valve plug with a seat ring in the valve body. The walls of the cage contain openings, which usually determine the flow characteristic of the control valve. As shown in figure 19.

Figure 18. Typical examples of linear plug, equal percentage plug and quick opening plug with type guiding shown.

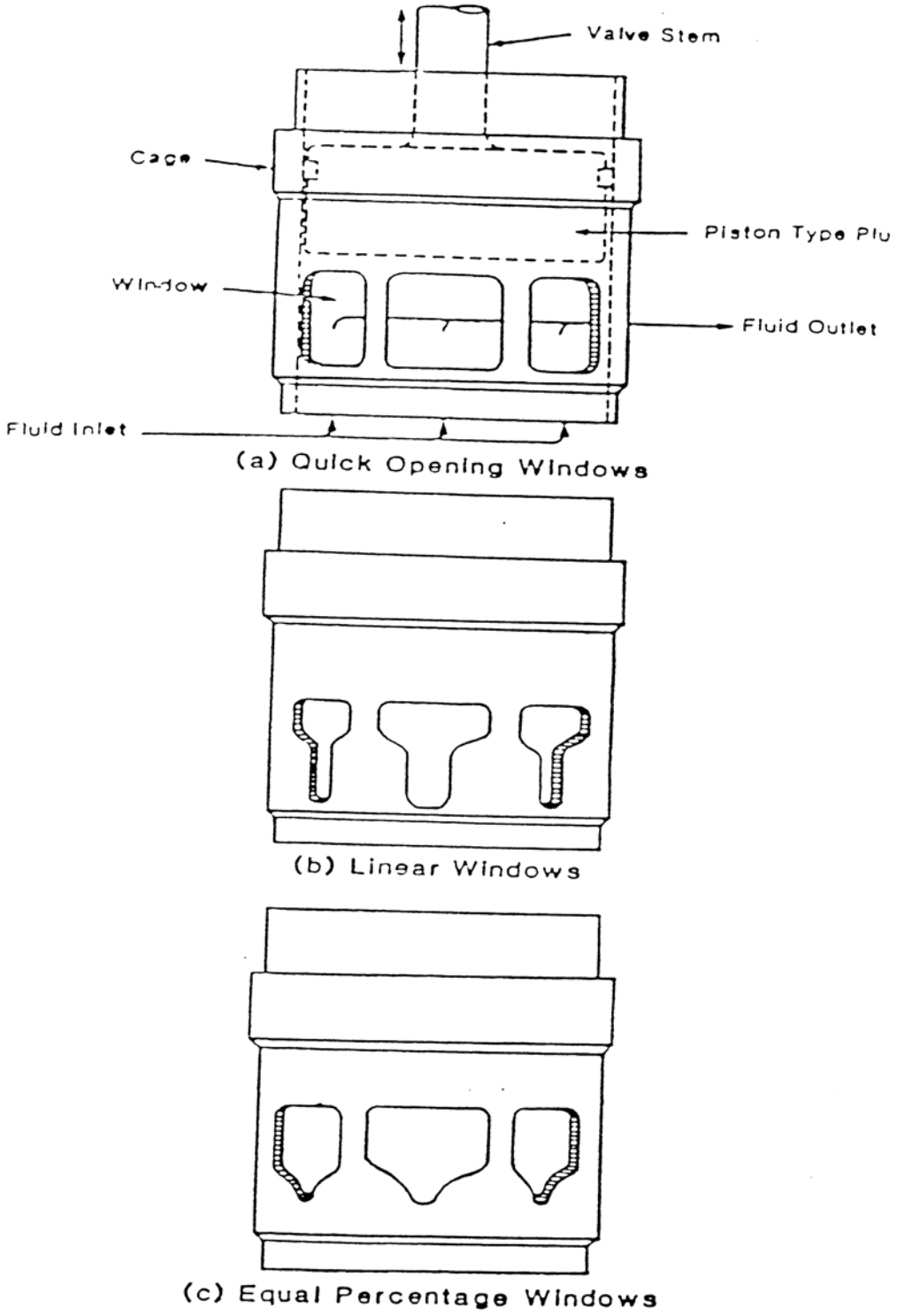


Figure 19. Flow Characterised Cage Windows

Use of each characteristic

Normally the choice of valve characteristic required by the loop is established by carrying out a dynamic analysis of the control loop but there are rules of thumb that can be applied to general situations.

- Linear trims are used in situations, such as level control, where the pressure drop across the valve is constant.
- Equal percentage trims are best used in flow situations where the pressure drop across the valve will vary as the flow goes from its minimum value to its maximum value. This is especially true on pumped systems (pressure & flow control).
- Quick opening valves are useful in by-pass or re-cycle lines where a basic on-off control of flow is required.

Valve shutoff

There are six classes of valve leakage

- Class I no test
- Class II 0.5% of rated valve capacity
- Class III 0.1% of rated valve capacity
- Class IV 0.01 % of rated valve capacity
- Class V 0.0005 mL/min of water per inch of port diameter per psi differential
- Class VI bubble tight (1 to 45 bubbles per minute for port sizes 1″ to 8″ diameter).

If tight shutoff is required, it is good practice to provide a tight shutoff isolation valve in series with the throttling valve. The soft seat in a throttling valve will need to be frequently replaced if it used to carry out tight shutoff.

Control Valve Selection Criteria

Normally a valve is designed to handle its maximum flow when it is at 75% open. Making the valve too big or too small would be detrimental to the operation of the valve and the loop. Valves should not operate below the 10% open position or above the 90% open position. The choice of control valve will depend upon the application, i.e. flow control, ESD etc. The main factors to take into account to select a valve for service are:

1. The valves' ability to regulate the flow.
2. The pressure loss/recovery when the valve fully open.
3. The shut-off leakage when the valve fully closed.
4. Suitable flow characteristics to match the process
5. Fail safe mode
6. Proper choice of valve body type and accessories.
7. Correct installation

For instance a globe valve gives good flow regulation, has poor pressure recovery at high flow rates and does not give tight shut-off, whereas a ball valve has poor flow regulation characteristics, low pressure loss at high flow rates and has the advantage of tight shut-off. Space is another factor that can come into the consideration.

To determine the control valve size, the process data are required to find out the valve-sizing coefficient by relevant calculations.

The process data required for control valve selections are:

1. Type of fluid to be controlled.
2. Flowing pressure (max., min. and normal),
3. Flowing temperature (max., min. and normal),
4. The differential pressure across the valve (Max. and Min.),
5. The maximum and minimum flows and the degree of shutoff required,

6. Fluid viscosity,
7. Fluid specific gravity,
8. Inlet and outlet pipe size and schedule,
9. Maximum permissible noise level.

Valve Sizing Coefficent (CV): The following is the definition of the valve sizing cofficient which is to be calculated in view of the above factors and then the control valve could be selected from any manufacturer product guid.

"**Valve flow coefficent (cv) is defined as the number of US gallons of water at 60 F that will flow through the valve in one minute when pressure differential across the valve is 1 psi".**

Flashing & Cavitation

All valves have a throttling action that causes a reduction in pressure. If the pressure increases again too rapidly, gas bubbles, entrained in the fluid implode, causing rapid erosion of the valve plug and seat surfaces. This process is known as cavitation.
Refer to course attachment No. 1 for more details about flashing and cavitation

Control Valve Noise

Control Valves have long been recognised as a major source of excessive noise levels inherent to many fluid process and transmission systems.

There are different sources of noise in the control valves and therefore different methods are used for noise abatement in order to have better process operation, less effect on the valve and equipment parts and better environment for personnel.
Refer to attached No. 2, for more information about the causes of the control valve noise and applicable solutions.

Actuators

The actuator provides the power to vary the orifice area of the valve in response to a signal received.

Control valve actuators may be operated pneumatically electrically, hydraulically, manually or by a combination of electrical, pneumatic and hydraulic forces. Pneumatic operation is the most widely used method. The forces actuators must overcome are the unbalanced forces caused by the pressure drop across the valve, friction between and weight of moving parts and stem unbalance. The actuators have mainly two actuating modes which are air to close and air to open(see fig. 21).

Types of Actuators

A) Pneumatic Actuators

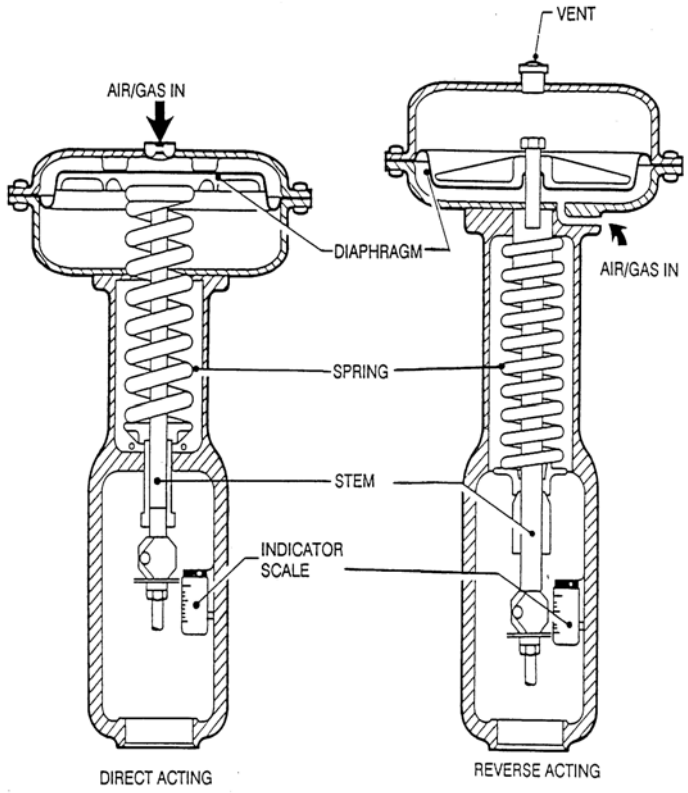
Pneumatic operators may be classified into two basic types:

1) Spring type (diaphragm or piston actuator), and
2) Spring-less (piston actuator).

1) Spring Type Actuators.

a) Diaphragm Type

Diaphragm type actuator is the most frequently used type. These actuators may be direct acting or reverse acting. As shown in figure 20, a direct acting operator is designed so that air pressure (usually 3-15 psi) on the top of the diaphragm moves the stem downward, closing the valve. This action is termed fail-open (air-to-close). This force opposed by compression of the spring, and loss of the operating medium (usually air) allows the compressed spring to open the valve.



Air to lower actuator Air to left actuator

Figure 20. Spring type diaphragm actuators

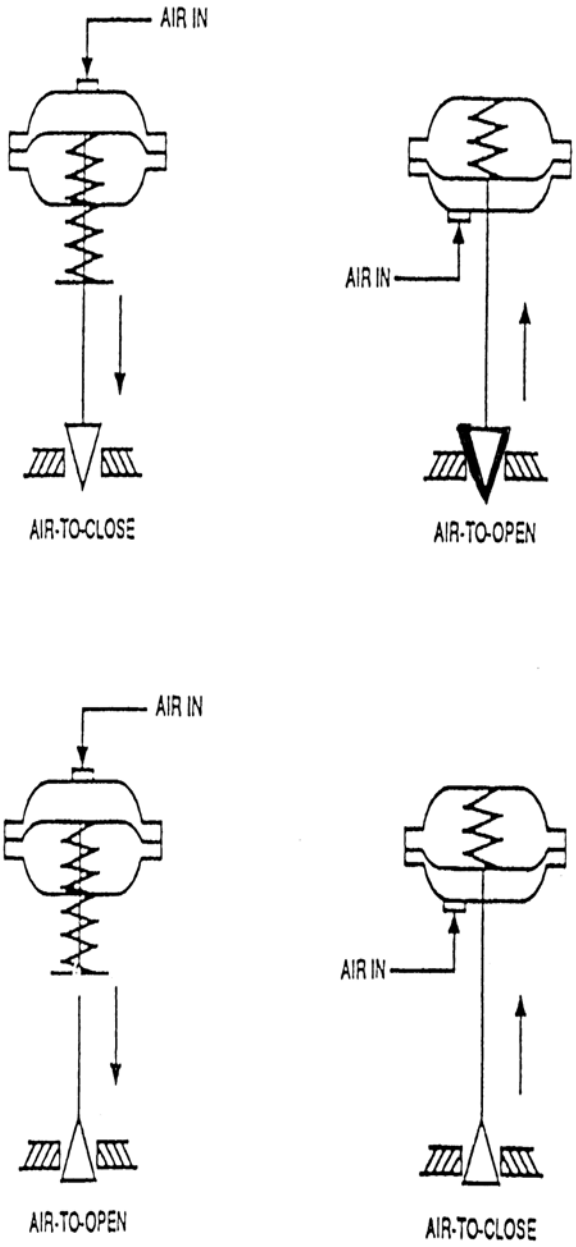


Figure 21. Valve failure mode with different valve/actuator set-ups.

b) Piston Type

The air piston provides high torque or force and has a fast stroking speed. It provides a high power to weight ratio, has few moving parts and an excellent dynamic response. It can handle high differential pressures and provides high shutoff capability. It is also easily adapted to services where high ambient temperature is involved. Figure 22 illustrates a single acting piston type actuator.

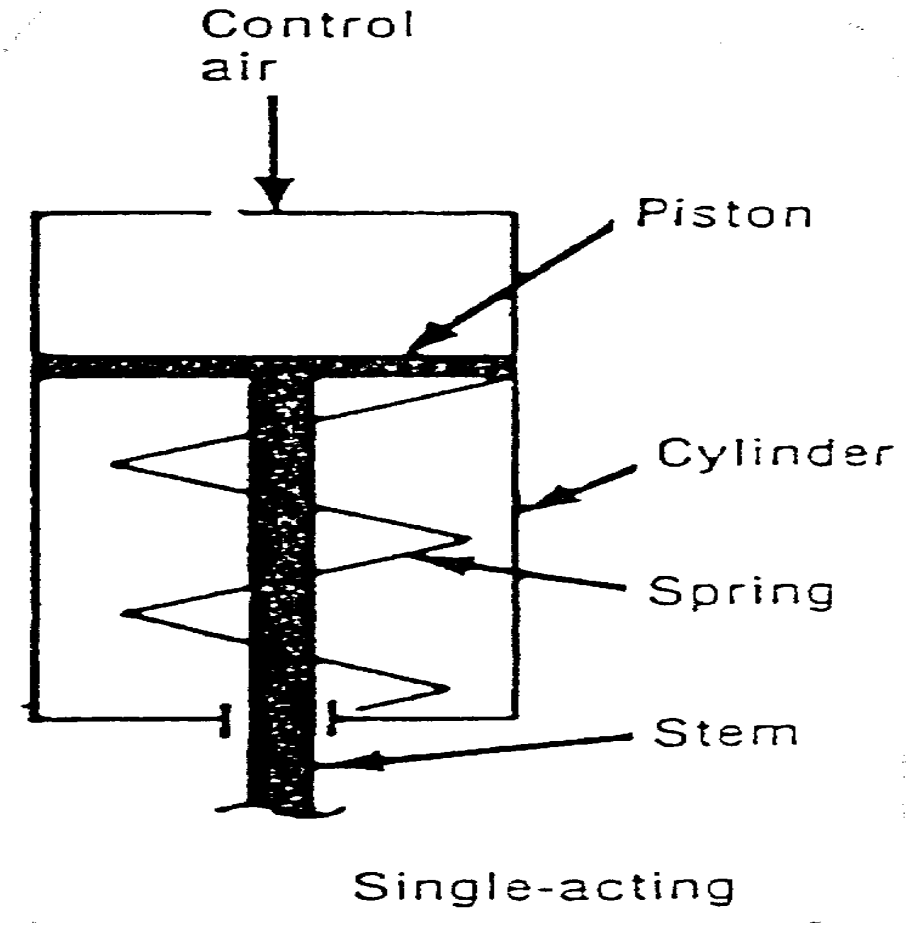


Figure 22. Single acting Piston Type Actuator.

2) Spring-less (Piston type).

Spring-less operators include pneumatic cylinder or piston operators. Cylinder or piston operators are increasing in usage because of the need for increased power and fast action. Increased power results from their ability to use higher-pressure supply air. These operators sometimes include built-in valve positioners.

Figure 23, shows how the piston is forced upward by a constant pressure from a reducing regulator, adjustable to suit the stem load. The chamber above the piston is dynamically loaded. An increase in instrument air pressure increases the pressure in the chamber above

the piston, moving it downward. This extends the range spring until the positioner forces are brought back into balance, at which point the positioner stabilises the pressure on top of the piston to hold the new position. A decrease in instrument air pressure reverses the procedure. Higher supply pressures provide greater power and faster stroking speeds. To provide fail-closed or fail-open modes, cylinder operators can be furnished with spring-return features. For fail-safe operation on electric power loss and for air supply loss, bottled gas with appropriate regulators and trip valves is sometimes employed.

Valves are sometimes required to maintain the position they were in when supply pressure or signal is lost. Such a state is known as "fail-last position," and can be accomplished by trapping the last signal pressures within the cylinder or piston assembly using a special trip relay.

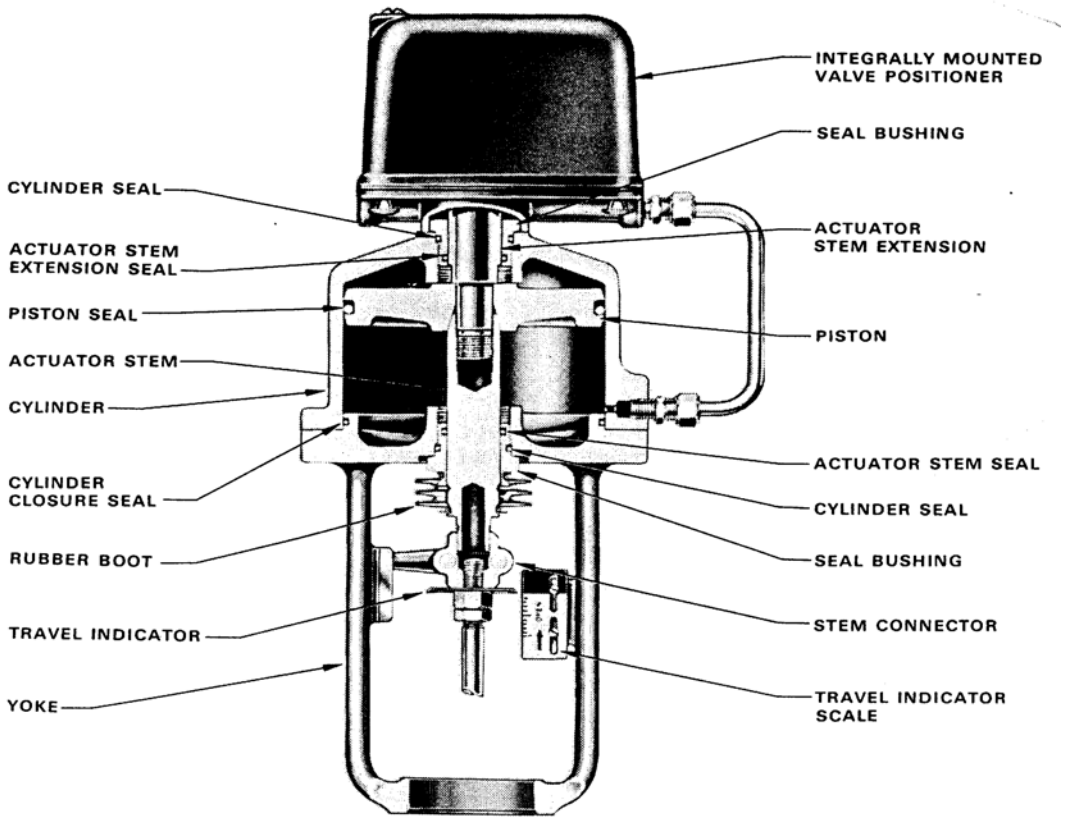


Figure 23. Typical Double Acting Piston Actuator.

B) Electro-Hydraulic Actuators

- Require only electrical power to the motor and an electrical input signal from the controller.

- Ideal for isolated locations where pneumatic supply pressure is not available but where precise control of valve plug position is needed.

- Units are normally reversible by making minor adjustments and are usually self-contained, including motor, pump, and double acting hydraulically operated piston within a weatherproof or explosion proof casing.

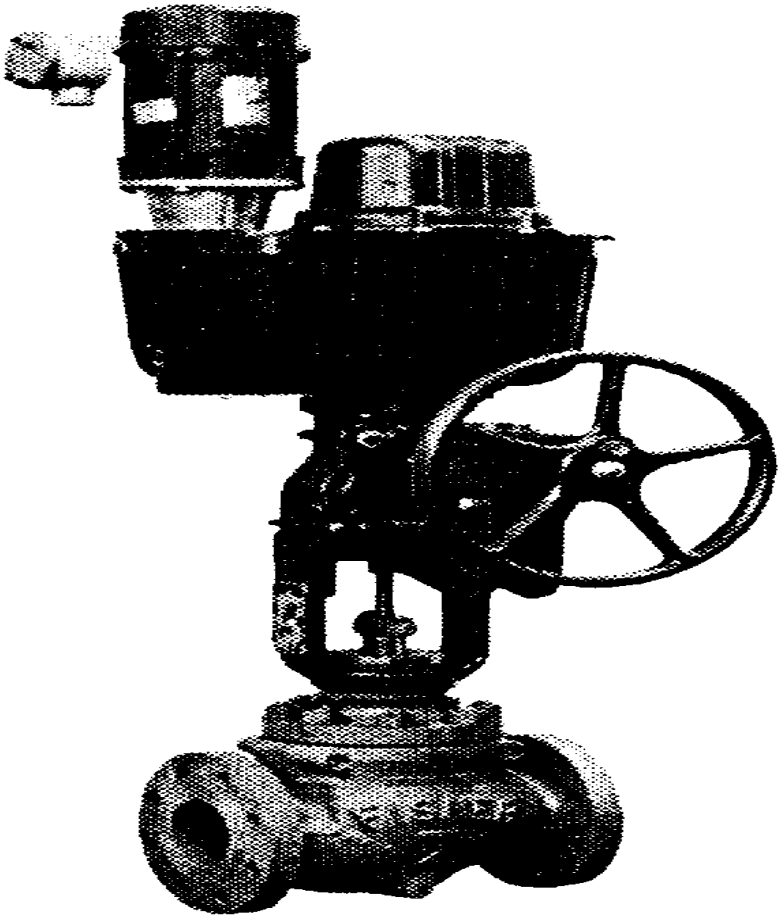


Figure 24. Control Valve with Double-Acting Electro-Hydraulic Actuator and Hand-wheel.

C) Electrical Actuators

Electric operators with proportional or infinite positioning control have limited use in the process industries. Their primary use has been in remote areas, such as tank farms and pipeline stations, where no convenient air supply is available. Slow operating speeds, maintenance problems in hazardous areas and economics have prevented wide acceptance for throttling applications. However, several companies have offered electrically powered units. Figure 25 shows an electrically operated butterfly valve, which can be supplied with an automatic amplifier-relay control package for use with a remote command potentiometer. The remote potentiometer is part of a Wheatstone bridge arrangement with the feedback potentiometer in the actuator. Changes in the command potentiometer cause the actuator to reposition to a pint where bridge balance is re-established.

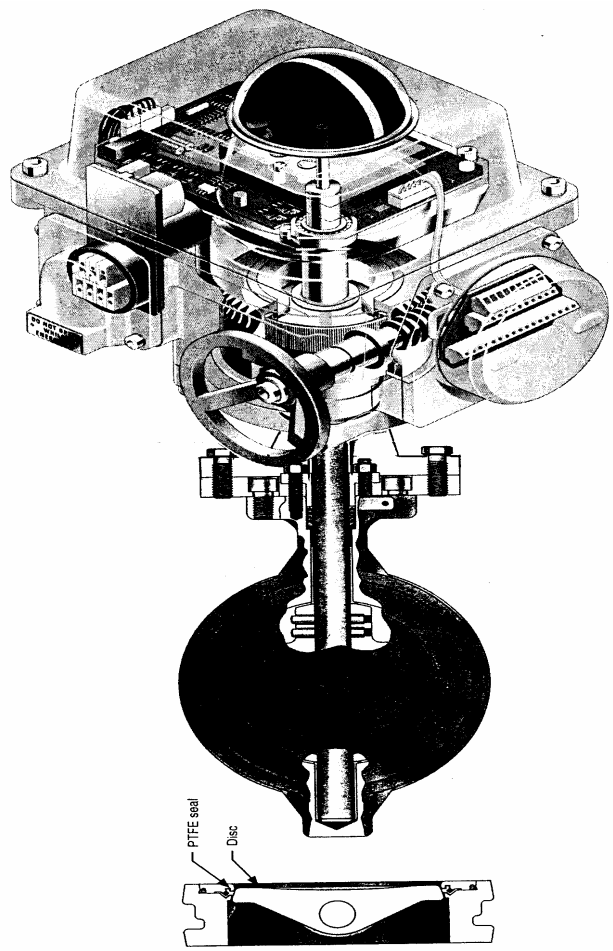


Figure 25. Butterfly valve with electric actuator for remote positioning control.

Solenoid Actuators

Solenoid actuator is an electromagnetic device, which moves its plunger/valve plug when electrical power is applied on its coil.

These are only used on small control systems where on-off control is required. Mostly they are found in the form of three way valves on the signal lines from the controller to the valve for ESD use. On removal of the power the valve will disconnect the controller from the valve and vent the air in the valve to atmosphere. Figure 26 illustrates a solenoid valve.

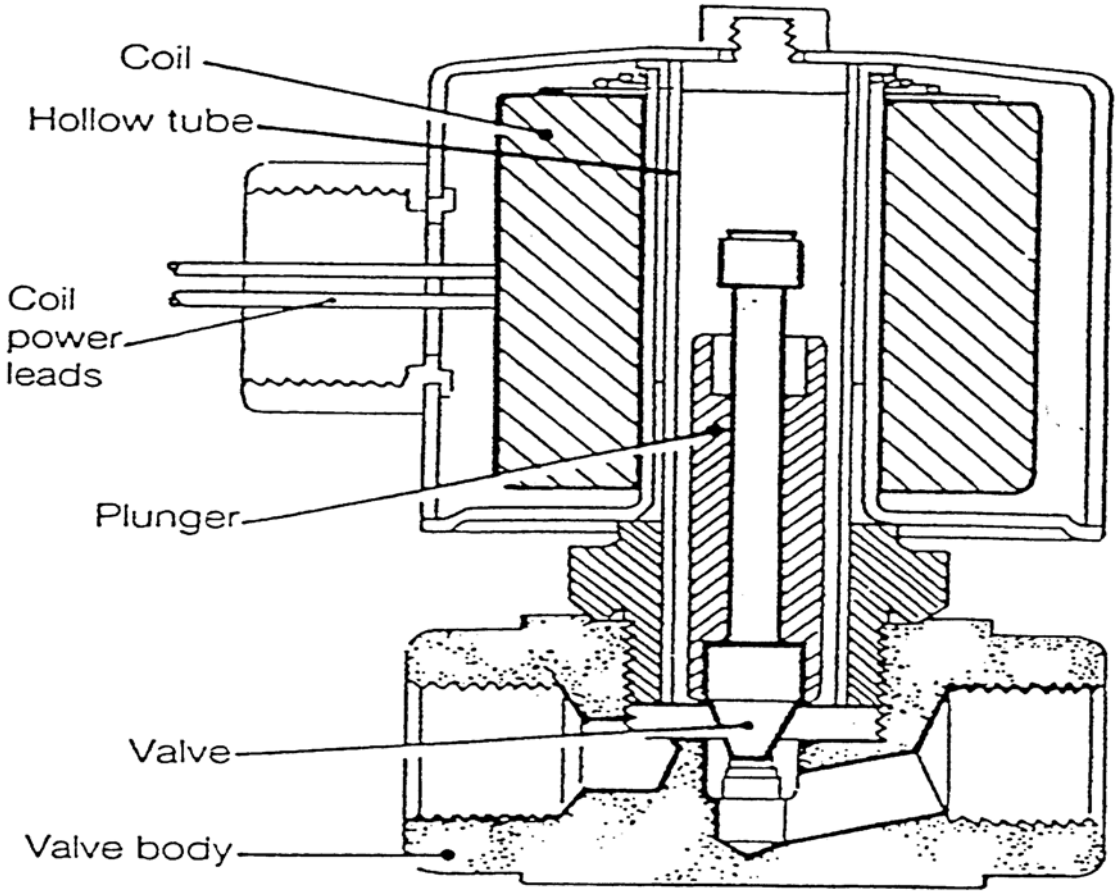
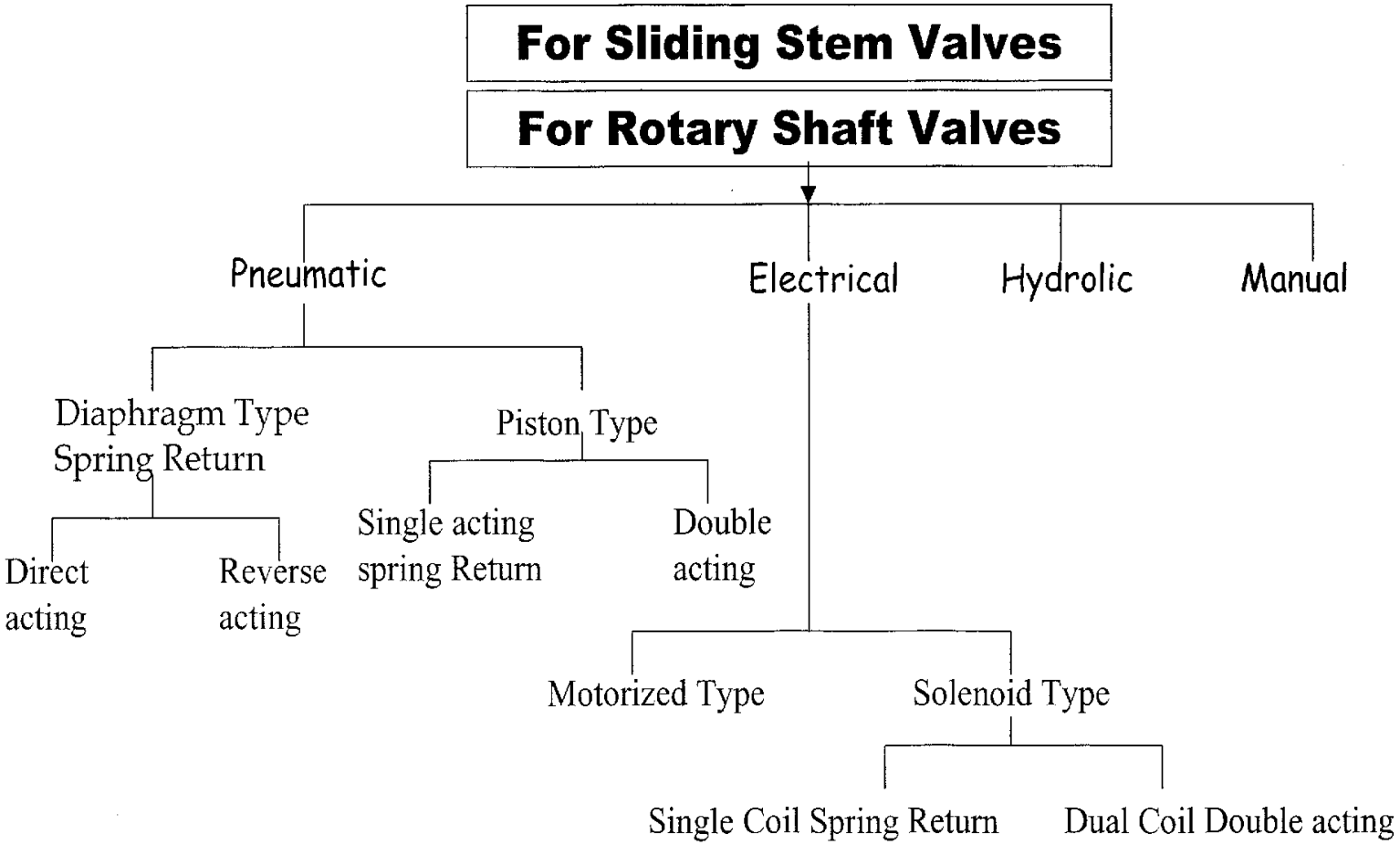


Figure 26. Solenoid Valve.

Types of Control Valve actuators



Actuator Sizing Considerations

The actuator must be proper-sized since the use of too large an actuator adds unnecessary expense and increased response time to a control valve, while use of an undersized actuator might make it impossible to open the valve or close it completely. However, selection of an optimum-sized actuator for a given control valve application is a subject of greater scope than can be completely detailed here. Consequently, the information furnished is summarised to provide basic knowledge of the factors that must be considered.

Actuators for Sliding-Stem Valves

After a valve has been selected to meet given service conditions, the valve must be matched with an appropriate actuator to achieve maximum efficiency. The actuator must provide sufficient force to stroke the valve plug to the fully closed position with sufficient seat loading to meet the required leak class criteria. With spring-return actuators, the spring selected must be sized to properly oppose the force provided by the air supply pressure.

Sizing an actuator involves solving a problem in static. The forces, and the direction in which each force acts, depend upon actuator design and flow direction through the valve. The free body diagram illustrates the forces involved in achieving static equilibrium. The figure depicts a direct-acting (push-down-to-close) valve body where the flow tends to open the valve plug. The actuator is a reverse-acting spring-and-diaphragm construction that closes the valve in case of supply pressure failure.

To stroke the valve to the fully closed position, the actuator must provide enough force to overcome friction forces and to overcome the unbalance force due to the flow through the valve. The actuator force available is the product of the air supply pressure and the area against which that pressure is applied (i.e., the diaphragm area or piston area). Packing friction varies with stem size, packing material(s), and packing arrangement. Specific friction forces must be obtained from the packing manufacturer or the actuator manufacturer. Other friction forces, such as friction due to metal piston rings, depend on valve design and must be obtained from the valve manufacturer.

The unbalance force is the product of the force of the flowing medium and the area against which that force is applied (either the total port area or, in the case of a "balanced" construction, .the specific unbalance area obtained from the valve manufacturer's specifications).

Unbalance Force = $DP_{shutoff}$ X Unbalance Area

To meet required leakage criteria, the actuator must provide some seating force beyond that force required to stroke the valve to the fully closed position. The specific force required depends upon valve style and size. Seat load is usually expressed in pounds per lineal inch of port circumference. For a given leak class, valve designs with large ports usually require greater seating load than is required for valves with smaller ports. The seat load is the product of the port circumference and the pounds-per-lineal-inch force recommended by the valve manufacturer.

Seat Load = Port circumference (inches) x Recommended seating force (pounds per lineal inch)

The actuator force available must be greater than the sum of the forces, which the actuator force must oppose to achieve static equilibrium. For a spring-opposed diaphragm actuator or a spring-return piston actuator, the spring force (spring rate X travel) must be considered in the equilibrium calculations.

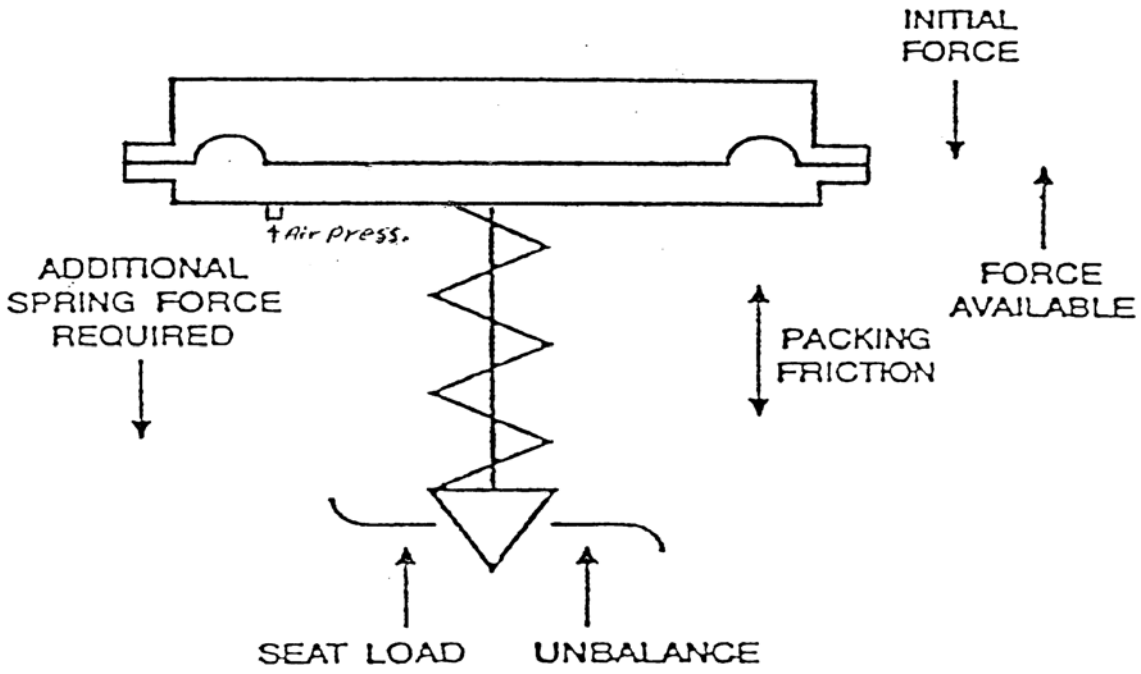


Figure 27. Free Body Diagram/or Reverse-Acting, Spring-Opposed Diaphragm Actuator on Flow-Tends-To-Open Valve Body

Actuators for Rotary-Shaft Valves

The actuator selected must be capable of providing adequate torque output to overcome the dynamic torque forces on the disc or ball of the valve under flowing conditions. The actuator must also be capable of exceeding the "breakout" torque requirements of the disc or ball at shutoff, in order to initiate rotation of the rotary valve shaft. Breakout torque requirement determination begins by multiplying the actual pressure drop across the closed valve times a tested breakout torque/pressure drop relationship provided by the valve manufacturer. Another factor is added that includes tested or predicted breakout torque for the body when it is not pressurised. Total calculated valve breakout torque must be less than the maximum allowable breakout torque limit of the actuator size being considered, as published by the actuator manufacturer. By the same token, total calculated valve dynamic torque must not exceed the maximum allowable actuator dynamic torque limits published by the actuator manufacturer. Dynamic torque requirements are calculated by multiplying the pressure drop that produces critical (gas) or choked (liquid) flow times a

pre-calculated effective pressure drop coefficient for the size and style of valve being considered.

CONTROL VALVES ACCESSORIES

1) Valve Positioners

It receives the controller output, compare it with the actual valve position then give an output to the actuator to put the valve in the required position accurately. In theory, a control valve should respond quickly to small output changes from the controller. However, if the controller output pressure is very small, the actuator may not be able to develop sufficient force to position the valve correctly and/or fast enough for good control. This failure could be caused by stiction between the stem and the valve packing, unbalance of the valve plug due to the hydrostatic forces of the process fluid or hysteresis within the valve itself.
This creates two main problems.

- It takes a greater force to initially move the valve in any direction, so causing a dead spot.

- Once the valve is moving, the initial force applied will cause the valve to accelerate and this in turn could cause overshoot and instability in the process.

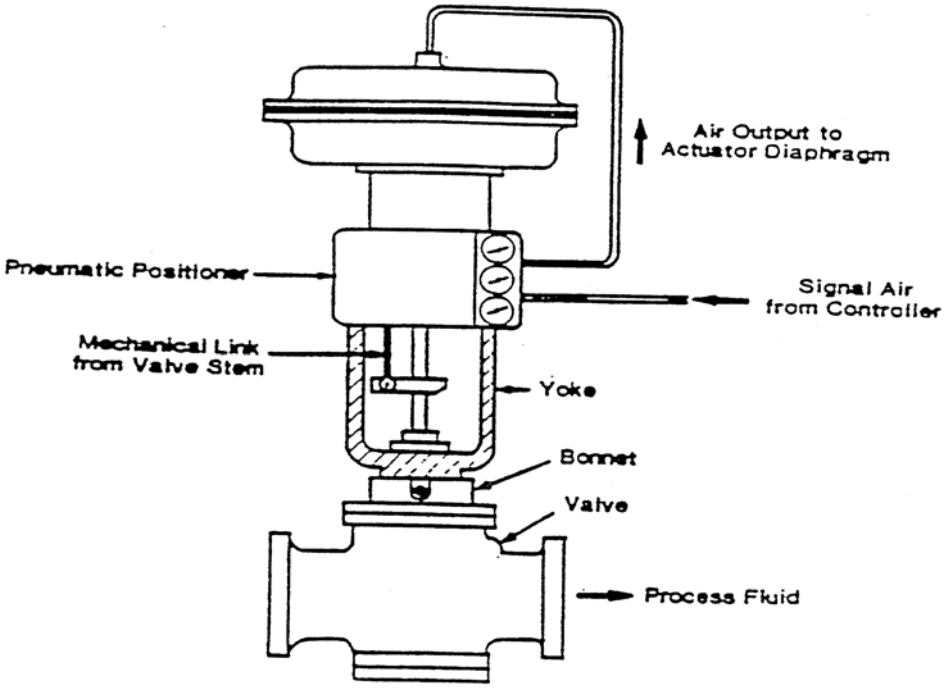
Therefore instead of sending the signal from the controller directly to the valve diaphragm, the signal is passed to a slave controller, with its own air supply, known as a positioner. A positioner is fitted to a valve in such a way that it can monitor the valve position and adjust its output signal until the valve is at the position required, i.e. the inclusion of a positioner provides closed loop control of the valve position.

Positioners can provide the following benefits

- Accurate positioning of the valve stem
- Ability to change the valve characteristics
- Split the operating range of two or more valves
- Increase the speed of response
- Reverse the action of a valve

Types of Positioners and Methods of Mounting:

There are many valve positioners available, but two basic design approaches have been made: motion balance and force balance. Positioners usually are mounted on the side of diaphragm actuators and on the top of piston actuators (for both sliding stem, Figure 1& 2, and rotary actuators, as shown in figure3). They are mechanically connected to the valve



stem or piston so that the stem piston can be compared with the piston dictated by the controller. Typical designs are shown and described.

Figure 1. Pneumatic Positioner is mounted on yoke of valve actuator (Sliding Stem).

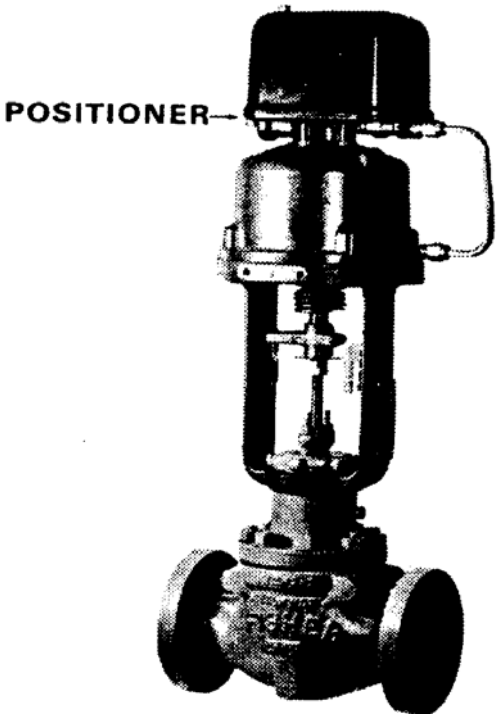
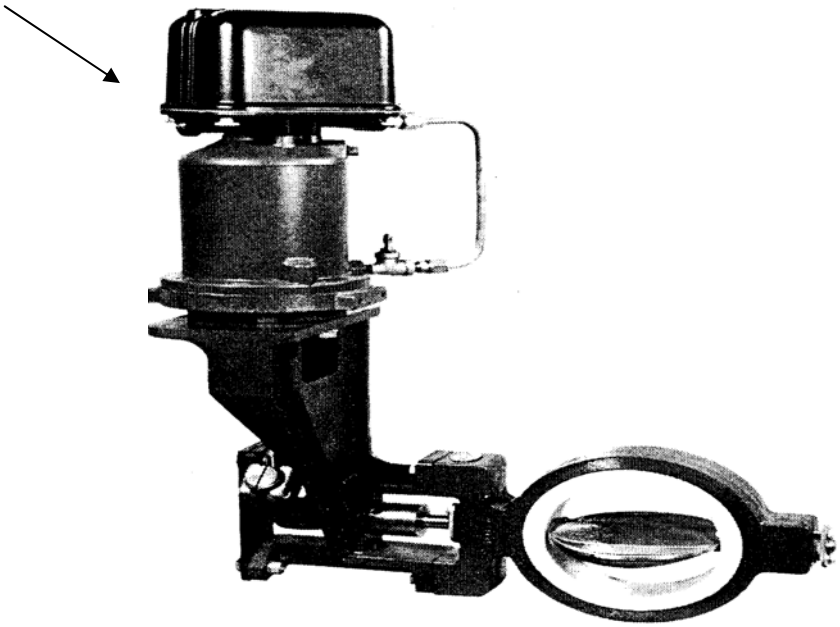


Figure 2. Pneumatic Positioner mounted on top of piston actuator (sliding Stem)



Pneumatic Positioner

Figure 3. Pneumatic positioner mounted on top of piston actuator (Rotary Shaft Control Valve)

Basic Principles of the Valve Positioners:

There are two main principles on which the valve positioners are designed:

a) Motion balance.

b) Force balance.

The following drawings are showing these two basic principles:

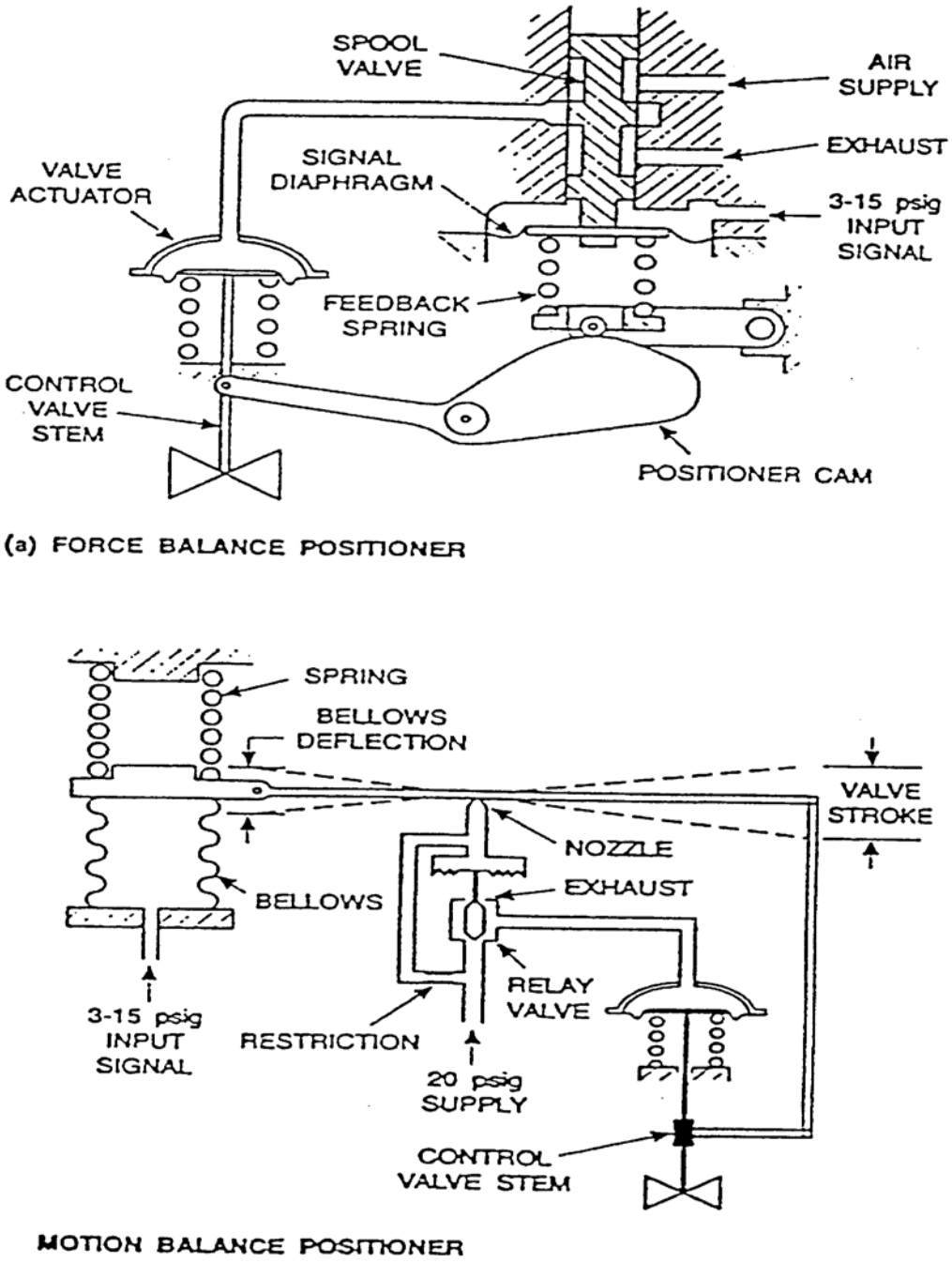


Figure 4. Motion Balance and Force Balance Positioners

Pneumatic input Positioner: (Typical example for Motion Balance type)

Figure 5A, shows a schematic of a motion balance positioner and its physical appearance is seen in Figure 5B. It consists basically of (a) a bellows to receive the instrument signal, (b) a beam fixed to the bellows at one end and, through linkage, to the valve stem at the other end and (c) a relay whose nozzle forms a flapper-nozzle arrangement with the beam. As the bellows moves in response to a changed instrument signal, the flapper-nozzle arrangement moves, either admitting air to, or bleeding air from the diaphragm until the valve stem position corresponds to the instrument air signal. At this point the positioner is once again in equilibrium with the changed instrument signal.

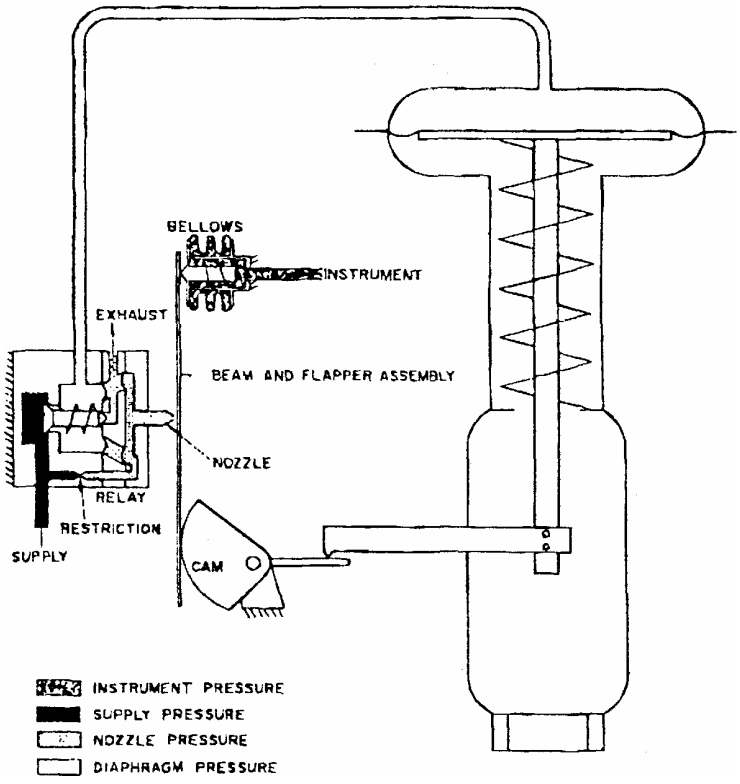


Figure 5A. Schematic of the motion balance pneumatic positioner reveals how it operates.

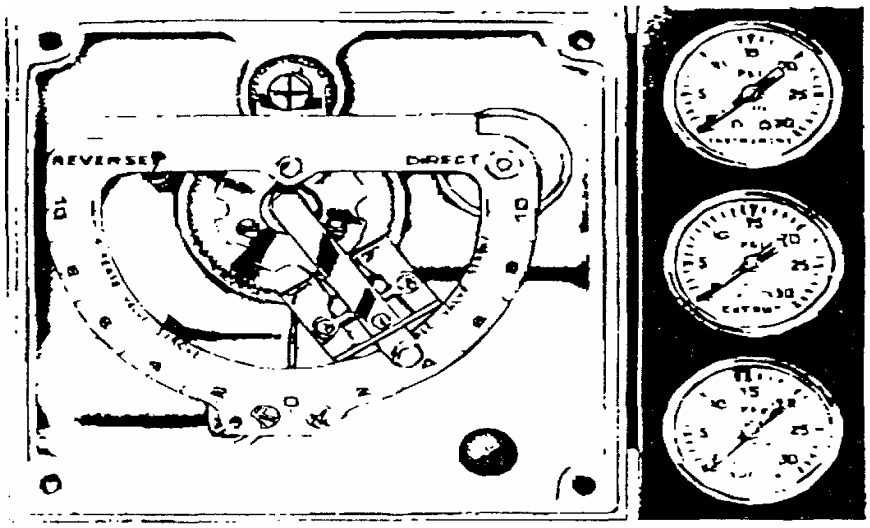


Figure 5B. Motion Balance Positioner with outside cover removed

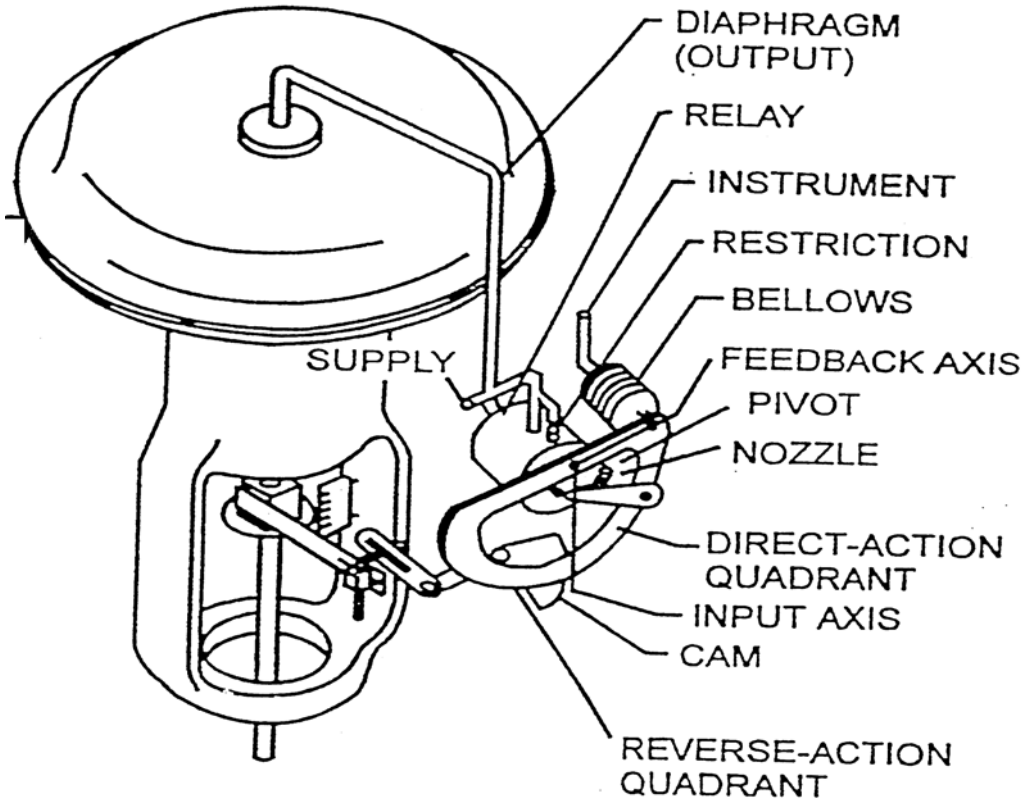
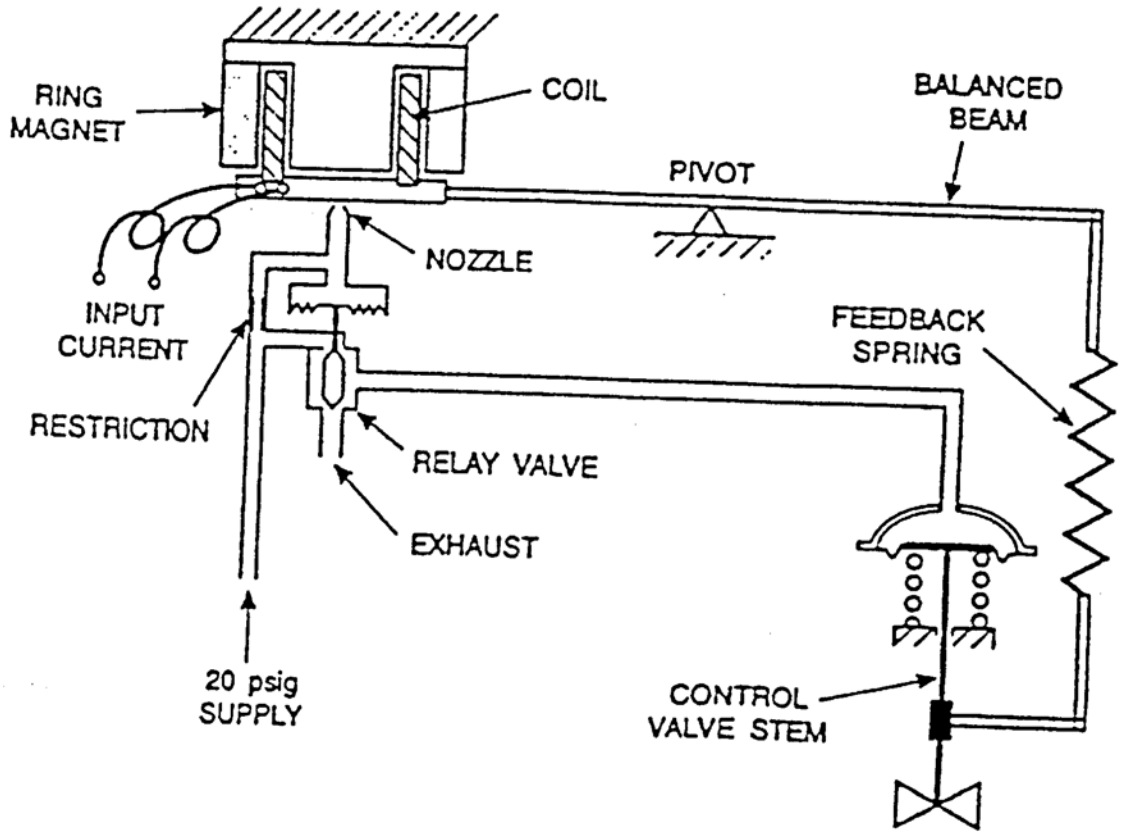


Figure 6. Motion balance positioner hooked up to a diaphragm actuator.

Electro Pneumatic Positioner:

Use of electronic control loops with air-operated valves has led to the development of Electro-pneumatic positioners, a combination of an Electronic-to-air transducer and a positioner figure 7. This is a force balance device and is supplied as direct or reverse acting.

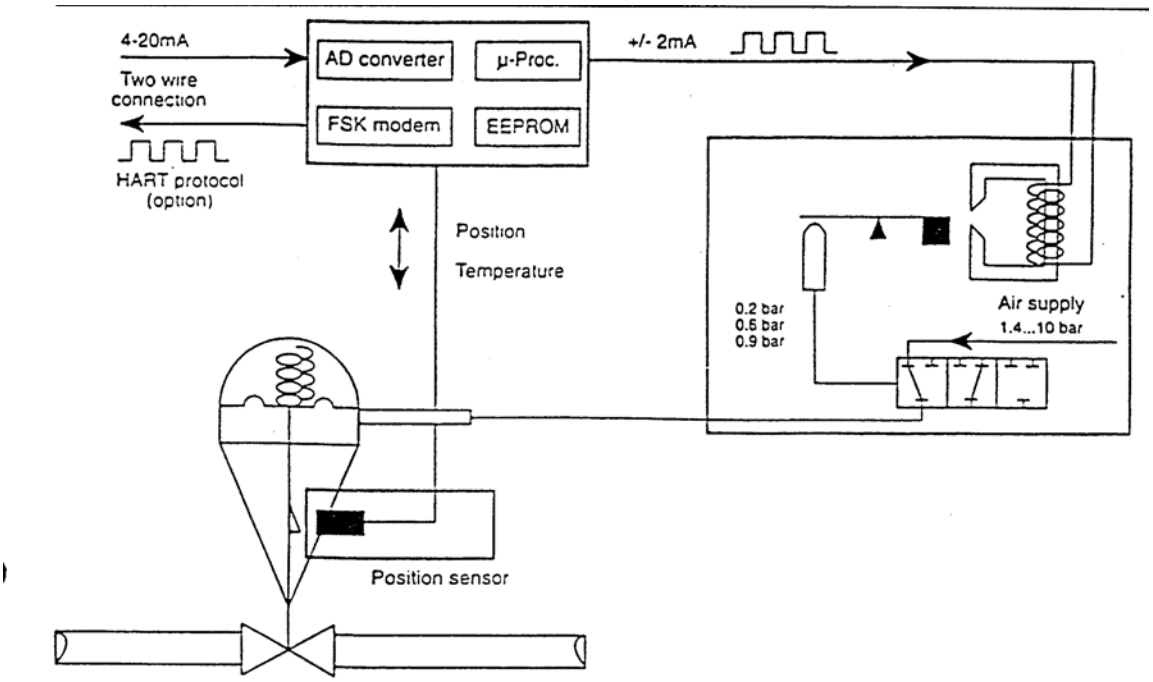
With direct action (increasing electrical input signal increases the air output pressure), an increase of the input signal causes the coil to produce a force on the beam, moving the flapper to cover the nozzle. The increase in nozzle back pressure causes the relay plug to close the exhaust in the diaphragm block and open the inlet, increasing positioner output pressure to the control valve actuator. The resultant valve stem motion extends the spring (through linkage) until the spring-force is balanced by the coil force. As they equalise, the nozzle back pressure decreases, allowing the relay plug to close the inlet and open the



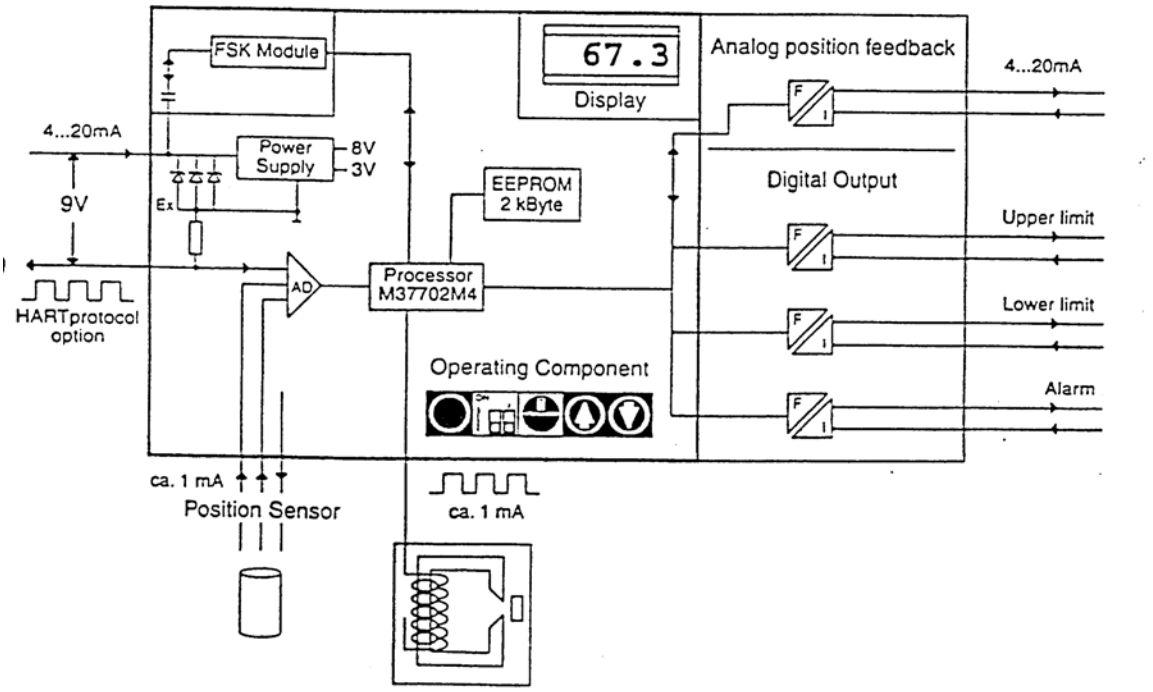
exhaust. The system is in equilibrium, and the positioner output is stabilised at a value

Figure 7. Electro-Pneumatic Positioner.

Smart Positioners:



Smart Positioner Schematic Diagram.



Smart Positioner Electrical Operation

Figure 8. Smart Positioner Schematic.

Use of the Valve Positioners in Split-Range Control Loops

Typical loop as an example:

As illustrated in figure 9, the purpose of using this split-rage loop is to provide the two consumers with fuel gas if gas supply pressure is high enough. With supply pressure decreasing, the control loop reduces the opening of the low priority consumer control valve to save the gas for the high priority consumer.

The positioner of PV-A is calibrated to operate its control valve from 0 to 100% travel at the lower half of the input signal range (4 – 12 mA), while the positioner of PV-B is calibrated to operate its control valve from 0 to 100% travel at the upper-half of the input signal range (12 – 20 mA)

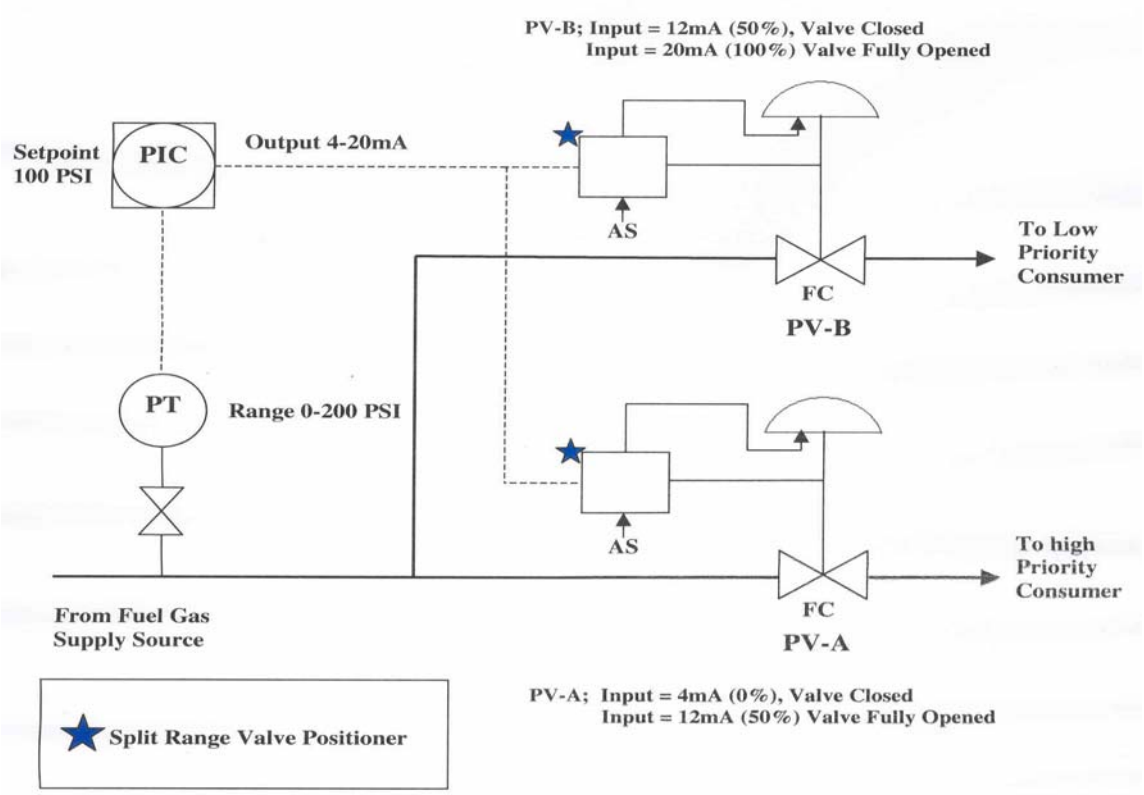


Figure 9. Sample Split Range Loop

2) Current to Pneumatic Converter (I/P)

a) Electromagnetic type

Principle of operation

High-pressure air enters the inlet and passes through the valve to the lower side of the control diaphragm assembly and also to the outlet port. The output pressure is controlled by the servo assembly position, which is moved by the air pressure applied to either side. Initially a spring applies a force to the control diaphragm allowing the valve to open slightly; as the output pressure rises the diaphragm will lift closing the valve.

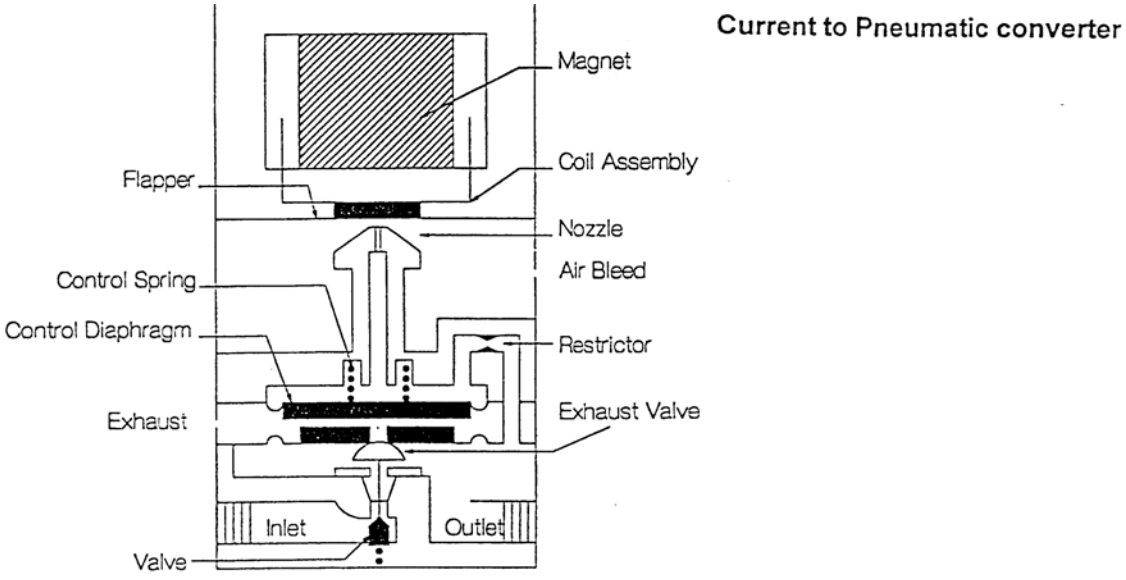


Figure 10. Current to Pneumatic Converter

Note:

In the absence of an electrical signal, there will be a small output pressure (leakage) determined by the spring. This is required for the correct operation of this type of device.

A permanent magnet applies a magnetic field to the coil assembly that is free to move in the gap. The coil assembly includes a flat plate that forms a flapper nozzle structure. The

output pressure is applied to the lower side of the nozzle through a restrictor. When a current is applied to the coil an Electro-magnetic field is produced which opposes that of the permanent magnet. This results in the flapper moving towards the nozzle and increasing the pressure in the space above the diaphragm, and opening the valve. This increases the output pressure until the pressures on each side of the diaphragm is the same.

An integral volume flow booster provides adequate flow capacity to give fast response for the majority of applications.

Due to the mechanical nature of this device it is sensitive to changes in position and must be calibrated and utilised in a vertical position.

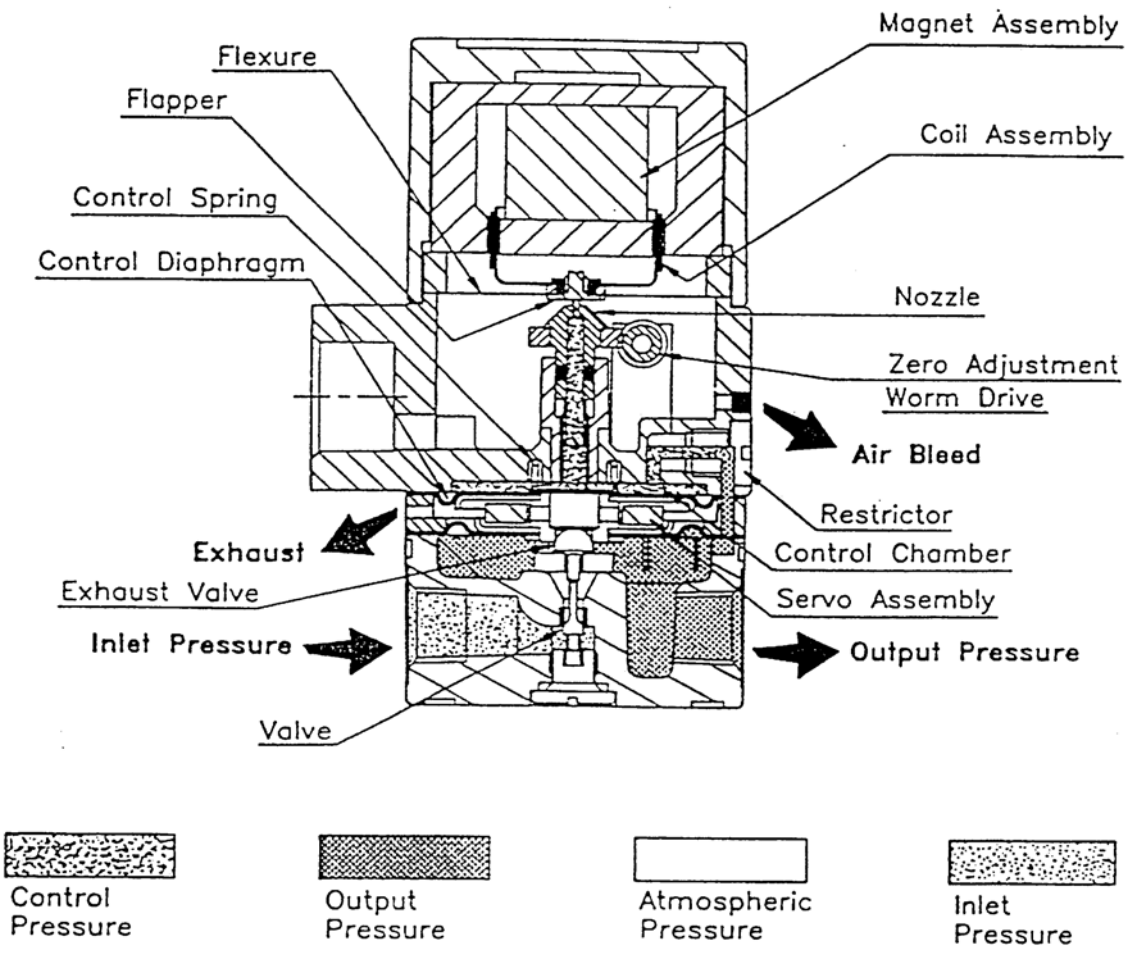


Figure 11. Cross-section of a typical Current-to-Pneumatic Converter

This device uses the 4-20 mA loop signal as its power source and requires only a connection to a regulated air supply. A small but constant air-flow bleed is maintained through the instrument via the restrictor valve and the nozzle; this is essential for the correct instrument operation.

Vibration

In many process plants and industrial installations, vibration is a common problem and these vibrations can cause problems with both the performance and longevity of instrument equipment.

Vibration is the repeated harmonic motion often encountered as a consequence of the operation of rotating machinery. Most installations have such machinery in the form of air compressors, fans, mixer motors etc. This gives rise to areas of the plant that suffer from an almost constant background vibration effect,

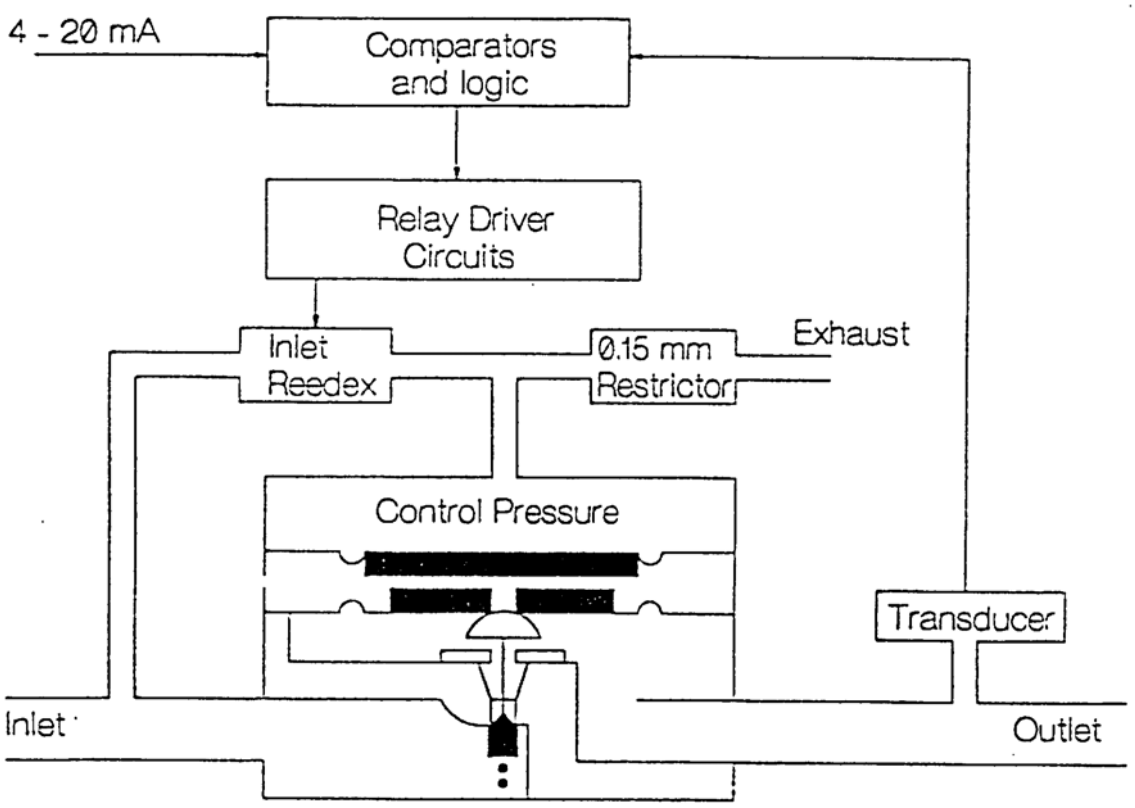
Vibration and I/P converters

Traditional I/P converters using the flapper/nozzle system suffer from the effects of vibration in the form of output instability. One way to protect the 1/P from the vibration problems is to locate it away from the valve that it is controlling, which leads to a more difficult plant design and increased installation complexity. Another way to overcome this problem is to use a solid state based I/P.

b) Solid state type I/P converters

These come in one of two formats

- Fail safe
- Fail freeze



Fail Safe

Figure 12. Fail Safe Solid State I/P

Control of the outlet pressure is achieved by variation of the control pressure. The steady state of the diaphragm assembly is such that the inlet valve and the relief valve are both closed reducing the overall air consumption. Increasing the control pressure moves the diaphragm assembly down opening the inlet valve. Supply air flows to the outlet; and the outlet pressure starts to increase. This increases the force on the bottom of the diaphragm assembly closing the inlet valve when a steady state pressure has been established.

Reducing the control pressure causes the diaphragm to rise and open the relief valve allowing the outlet pressure to decrease. When the pressure balance is achieved the relief valve is closed.

Air is constantly being bled from the control volume allowing a steady fall in the control pressure. Pressure control is achieved by the use of a Reedex high speed, precision, solenoid valve, which operates in a similar way to an electrical reed relay. Inside the Reedex the reed has a small orifice which is normally closed by a seal. Deflection of the reed causes the orifice to open. The reedex valve is opened for a few milliseconds by means of a variable pulse width 10 Hz signal to allow the supply pressure to enter the control volume and increase its pressure.

In the steady state condition the air supplied through the Reedex balances that lost through the bleed, maintaining a constant pressure in the control volume and the outlet port.
An electronic transducer constantly monitors the outlet pressure. This is then compared with the signal current to produce an error signal. If the outlet pressure falls or the signal current rises then the width of the pulses sent to the Reedex valve increases causing the control pressure to rise. As the outlet pressure rises the width of the pulses decrease until a new steady state is achieved.

If the outlet pressure rises or the signal current falls the width of the pulse sent to the Reedex decreases allowing the control pressure to fall and open the relief valve. As the outlet pressure falls the Reedex pulse width gradually increases until a new steady state is achieved.
On a signal failure the Reedex valve is unable to open and the pressure falls to a low value due to the exhaust bleed, ensuring a fail-safe operation.

Fail Freeze

In the fail freeze option pressure control. is achieved by the use of two Reedex solenoid valves, one inlet Reedex and one exhaust Reedex. In the steady state both Reedex valves are closed maintaining a constant pressures in the control volume.

If the outlet pressure changes or the signal current alters then the width of the pulse sent to the appropriate Reedex valve changes causing the control pressure to rise or fall as required by either supplying or exhausting air to/from the control volume.
When the signal current fails neither Reedex valve is able to open and the pressure remains at the last set point, ensuring a fail-freeze operation.

General

Because of the solid state nature of this device it has a high immunity to vibration problems and can be mounted in any orientation.

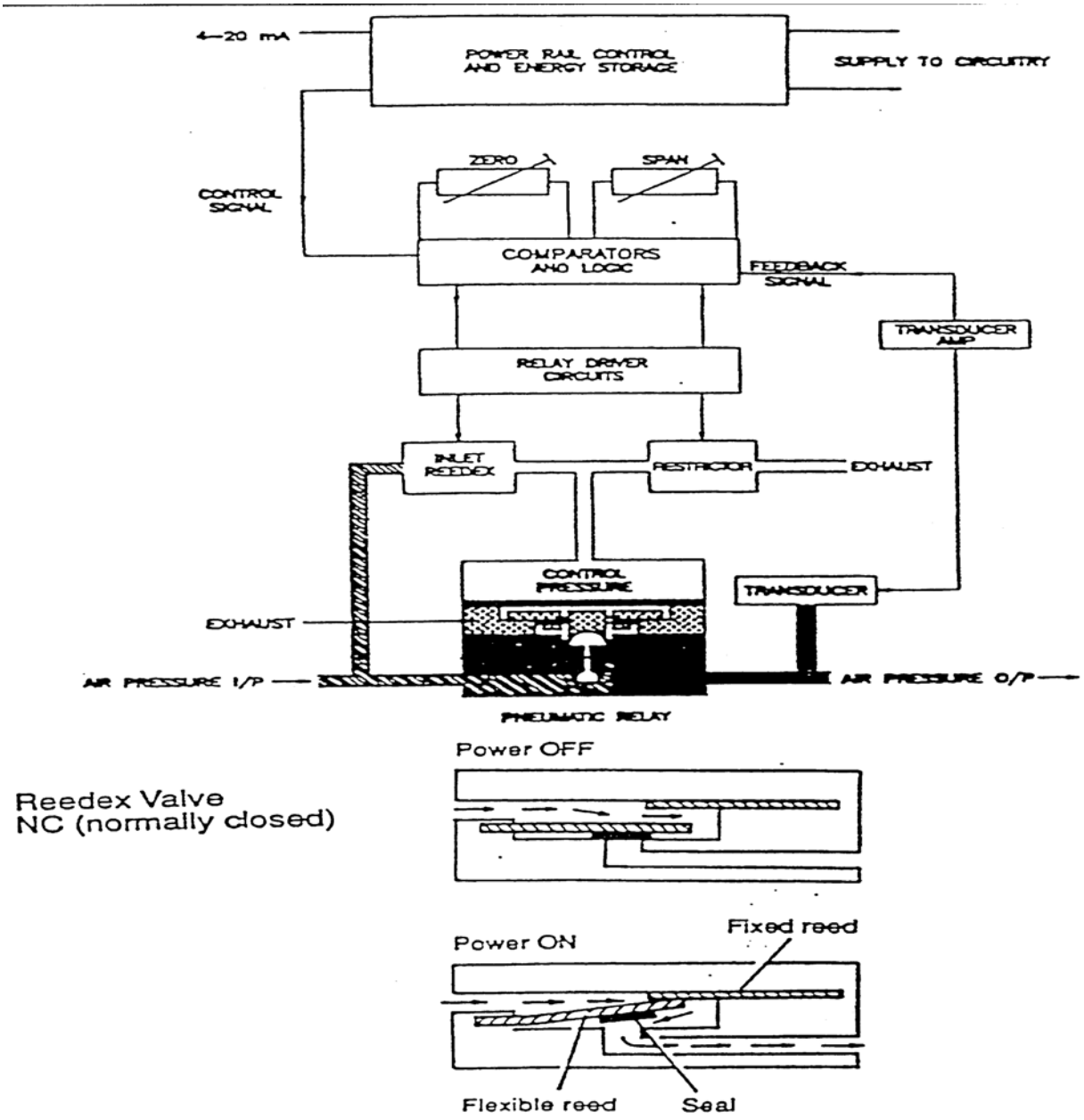


Figure 12. Fail Freeze Solid State I/P Converter Schematic and Reedex Valve Details

3) Limit Switches

Limit switches are used to operate signal lights, small solenoid valves, electric relays, or alarms when the control valve position reaches a predetermined point. These switches can be fully adjustable units with multiple switches or stand alone switches. The cam- operated type shown is available with from two to six individual switches operated by movement of

the valve stem. The switches are to be housed in an assembly that mounts on the side of the actuator. Each switch is individually adjustable and can be supplied for either alternating current or direct current systems. The styles of valve mounted limit switches are available in different designs to suit all types of control valves(sliding stem, rotary shaft. etc) and the area at which the control valve is installed.

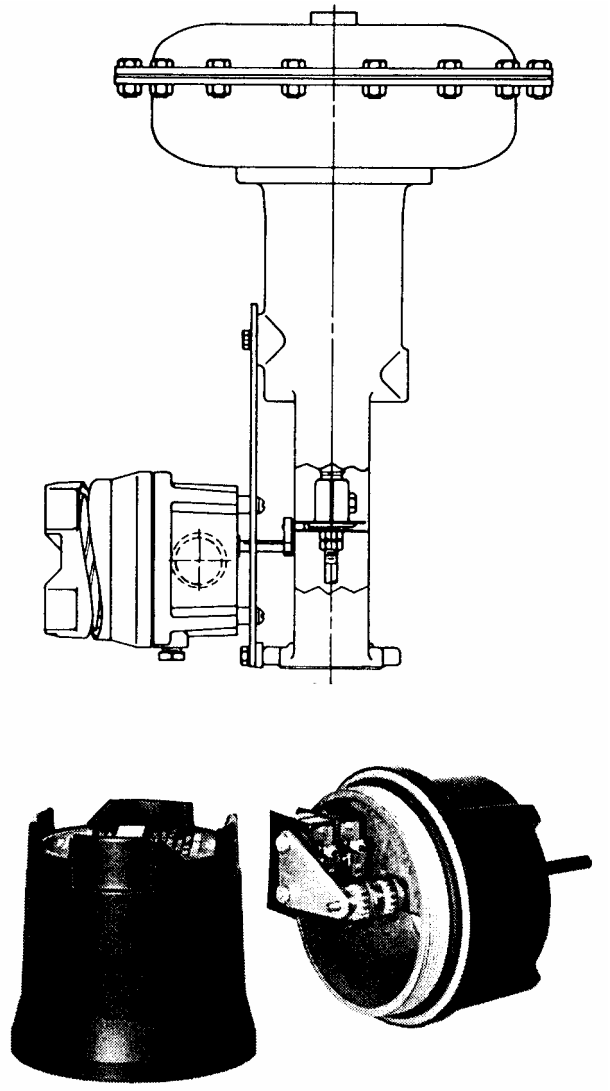


Figure 13. Cam-Operated Limit Switches

4) Position Transmitter

Electronic position transmitters are available that send either analogue or digital electronic output signals to control-room devices. The instrument senses the position of the valve and provides a discrete or proportional output signal. Electrical position switches are often included in these transmitters (Fig. 14).

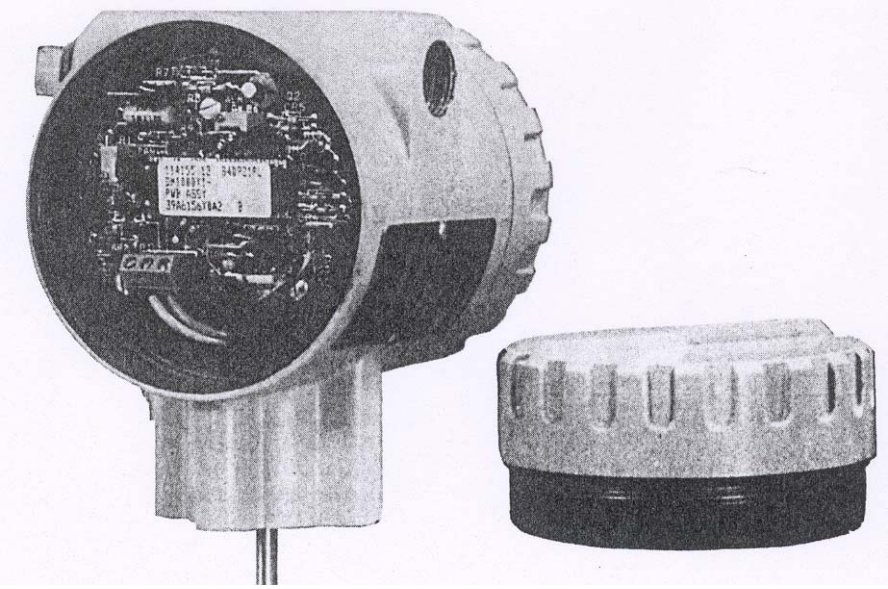


FIGURE 14. Stem position transmitters provide discrete or analogue output of valve position for use by control-room instrumentation.

4) Handwheels and Manual operators

A variety of actuator accessories are available which allow for manual override in the event of signal failure or lack of signal previous start-up. Nearly all actuator styles have available either gear-style or screw-style manual override wheels. In many cases, in addition to providing override capability, these hand wheels can used as adjustable position or travel stops. Figures below are showing different installations of hand wheels on different types of actuators.

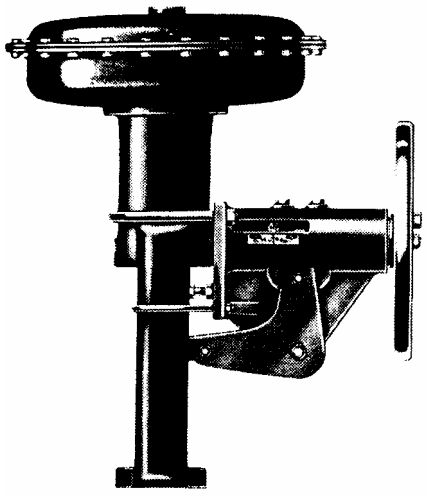


Figure 15. Side-Mounted Handwheel for Diaphragm Actuators

Used with either direct-acting or reverse-acting actuators, this unit acts as an adjustable travel stop to limit either full opening or full closing of the valve or to position the valve manually.

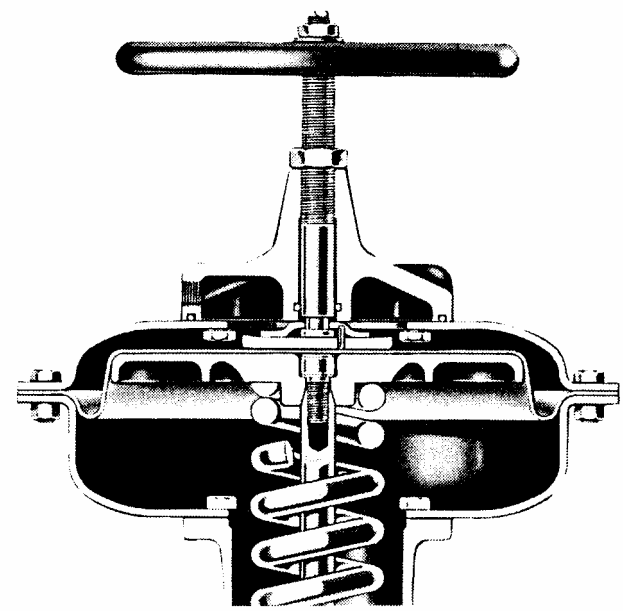


Figure 16. Top Mounted Handwheel for Direct-Acting Diaphragm Actuator

This unit can be used as an adjustable travel stop to limit travel in the upward direction or to manually close push- down-to-close valves.

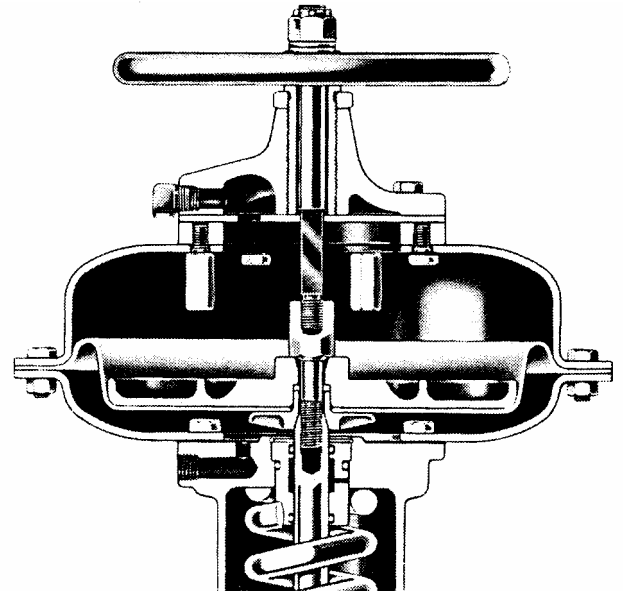


Figure 17. Top Mounted Handwheel for Reverse-Acting Diaphragm Actuator

This unit can be used as an adjustable travel stop to limit travel in the down- ward direction or to manually close push-down-to-open valves.

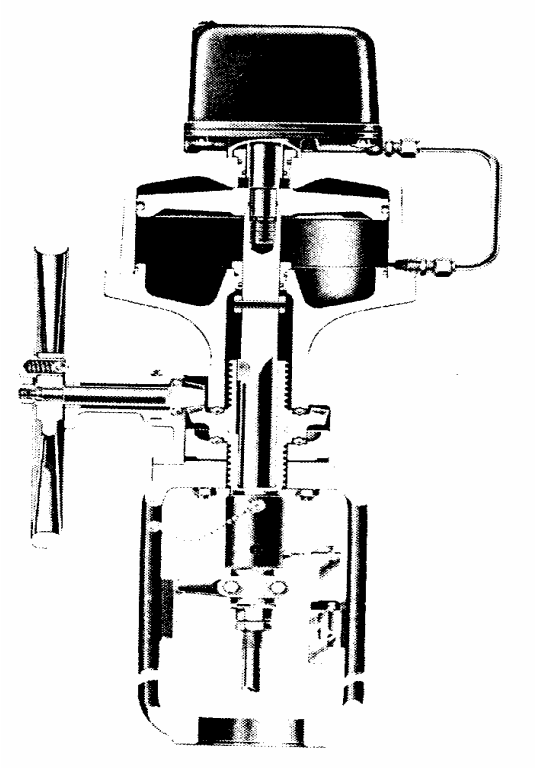


Figure 18. Side-Mounted Handwheel for Piston Actuators

Used to manually position the valve, this unit also acts as an adjustable travel stop to limit either full opening or full closing of the valve.

6) Other accessories

6.1- Volume Booster:

The volume booster is normally used in control valve actuators having no positioner to increase the stroking speed. These pneumatic devices have a separate supply pressure and deliver a higher volume output signal to move actuators rapidly to their desired position.

6.2- Trip Relay:

Pressure sensing trip valves are sometimes used for control applications where a specific actuator action is required when supply pressure fails or falls below a specific value. When supply pressure falls below the preadjusted trip point, the trip valve causes the actuator to **fail up** or **lock in last position** or **fail down.** When supply pressure rises above the trip point, the valve automatically resets, allowing the system to return to normal operation.

Auxiliary power to provide for actuator action in case of trip is provided by pneumatic volume tank.

CONTROL VALVES INSTALLATION & MAINTENANCE

Control Valves Installation and Maintenance

1) Control Valve Installation

Installation considerations (brief):

1- use a recommended piping arrangement:

- Proper isolation valves.
- ample room is allowed above, below, right & left of the control valve to permit easy removal, reinstallation and maintenance work on any part of it
- Proper alignment with piping system.
- Proper clearance between flanges for gaskets.
- Pressure taps up stream and downstream of the valve are useful for several checks and measurement purposes. Such taps should be installed in straight runs of pipe for accurate results.
- Tubing of *VA* OD or 3/8 OD to the actuator should be at minimum length and min. number of fittings and elbows in order to reducing system lag. if long distance is involved, a valve positioner and/or booster relay should be used on the control valve.

2- Be sure the pipeline is clean before installation of the control valve.

3- Inspect the control valve prior to installation and ensure no damage in any part of it and no foreign materials are inside the valve body.

4- Review the valve test certificate or test it prior to installation.

5- Install the control valve in vertical position to avoid mechanical stress on the actuator yoke. If it is necessary to be mounted at any other position, make relevant support for the actuator.
6- follow the criss-cross method in bolting the valve

Correct sizing and selection procedures, proper installation techniques, and periodic preventive maintenance are all factors that can lengthen control valve service life. Most valve manufacturers furnish detailed installation and operation instructions with each valve. These instruction sheets normally outline specific installation and maintenance procedures which apply to the particular valve described. Naturally, the specific instructions should be read by the purchaser prior to valve installation and closely followed during installation and operation. The suggestions furnished below are general in nature and should not take precedence over the valve manufacturer's detailed instructions for a particular valve.

Use a Recommended Piping Arrangement

The Instrument Society of America has published a Recommended Practice, ISA RP-4.2, on Standard Control Valve Manifold Designs to promote uniform control valve installations. Following one of the recommended practices, such arrangement will be of benefit in the event that piping components have to be replaced due to changing service requirements..

Be Sure the Pipeline is Clean

Foreign material in the pipeline could damage the seating surface of the valve, or even obstruct the movement of the valve plug or disc so that the valve could not shut off properly. To help reduce the possibility of a dangerous situation occurring, all pipelines should be blown out with air prior to valve installation. Make sure pipe scale, metal chips, welding slag, and other foreign materials are removed. Also, if the valve has screwed end connections, a good grade of pipe sealant compound should be applied to the male pipeline threads only.

Do not use sealant on the female threads in the valve body because excess compound on the female threads would be forced into the valve body. This could cause sticking of the valve plug or accumulation of dirt, which would prevent good valve shutoff.
Inspect the Control Valve Before Installation

While valve manufacturers take steps to prevent shipment damage, such damage is possible and should be discovered and reported before the valve is installed.

DO NOT INSTALL A CONTROL VALVE KNOWN TO HAVE BEEN DAMAGED IN SHIPMENT. Before installation check for and remove all shipping stops and protective plastic plugs or gasket surface covers. Check inside the valve body to make sure no foreign objects are present.

Use Good Piping Practice

Most control valves can be installed in any position. However, the most common method is with the actuator vertical and above the valve body. If horizontal actuator mounting is necessary, consider the possibility of providing additional vertical support for the actuator. Be sure the body is installed so that fluid flow will be in the direction indicated by the flow arrow on the body.

Be sure ample room is allowed above and/or below the valve installation to permit easy removal of the actuator or valve plug for inspection and maintenance procedures. Clearance distances are normally available from the valve manufacturer in the form of certified dimension drawings. For flanged valve bodies, be sure the flanges are properly aligned to provide uniform contact on the gasket surfaces. Snug up the bolts gently in establishing proper flange alignment and then finish tightening them in a criss-cross pattern as depicted in Figure 1. This will avoid uneven gasket loading and will help in preventing leaks, as well as avoiding the possibility of damaging, or even breaking, the flange itself. This precaution is particularly important when connecting flanges of different materials, such as would be the case when a cast iron body is bolted between steel pipeline flanges.

Pressure taps installed upstream and downstream of the control valve are useful for checking flow capacity or pressure drop. Such taps should be located in straight runs of pipe, away from elbows, reducers, or expanders, to minimise inaccuracies resulting from fluid turbulence.

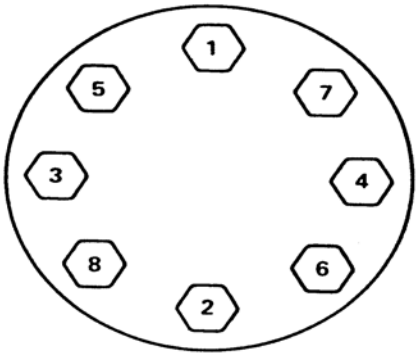


Figure Recommended Bolt Tightening Sequence for Uniform Gasket Loading

Use 1/4-inch or 3/8-inch tubing or pipe from the pressure connection on the actuator to the controller. Try to keep this distance relatively short and try to minimise the number of fittings and elbows in order to reduce system time lag. If long distances are involved, a valve positioner or a booster should be used on the control valve.

2) Control Valve Maintenance

Maintenance considerations (brief):

Do periodical checks on the control valve for any abnormal conditions such like (its called routine maintenance).

a) Ensure valve/actuator free movement, if hard movement is observed, follow the relevant procedures for repair.
b) Ensure no air leaks from the actuator, tubing, positioner ...etc; if there is, follow the relevant procedures for tightening.
c) Ensure no fluid leaks from the packing bonnet flange, valve flanges (inlet/outlet flanges), if there is any leak follow relevant procedures for tightening.

d) Ensure that valve position is matching the received control signal (if not, do the calibration procedure).
e) Ensure no leaks through the plug and seat ring when the valve is fully closed. If a leak is detected, follow the relevant procedures to take out the control valve to the work shop and do the repair as per the manufacturer instructions

Do a leak test (with the valve under the test pressure) after repair to verify no further leaks through the trim.

In order to perform even routine maintenance procedures on a control valve, it is important that the maintenance man have a thorough understanding of the fundamental construction and operation of the valve. Without this knowledge, the equipment could be damaged inadvertently, or could cause injury to the maintenance man and others in the area. Most valve manufacturers provide suggested safety measures in their detailed instruction and operation manuals. Usually, a sectional drawing of the equipment is also furnished to help in understanding the operation of the equipment as well as to provide identification of component parts.
In all major types of control valves, the actuator provides force to position a movable valve plug, disc, or ball in relation to a stationary seat ring or sealing surface. The moveable member should respond freely to changes in actuator loading pressure. If proper operation is not being received, service is indicated. Before any maintenance procedures are started, be sure that all line pressure is shut off and released from the valve body and also that all pressure to the actuator is shut off and captive pressure gradually relieved. Failure to take adequate precautions could create a situation that would damage the equipment or injure personnel.

Often corporate maintenance policy or existing codes require preventive maintenance on a regular schedule. Usually such programs include inspection for damage of all major valve components and replacement of all gaskets, 0-ring seals, diaphragms, and other elastomer parts. Following is a series of commonly performed maintenance procedures and some

general instructions for performing each procedure. The reader is reminded that specific, detailed maintenance procedure instructions are normally furnished with control valve equipment and should be carefully followed.

Replacing Actuator Diaphragm

After isolating the valve from all pressure, relieve all spring compression in the main spring, if possible. (On some spring and diaphragm actuators for use on rotary-shaft valve bodies. spring compression is not externally adjustable. Initial spring compression is set at the factory and does not need to be released in order to change the diaphragm.) Remove the upper diaphragm case.

On direct-acting actuators, the diaphragm can be lifted out and replaced with a new one. On reverse-acting actuators, the diaphragm head assembly must be dismantled to change the diaphragm.

Most pneumatic spring-and-diaphragm actuators utilise a moulded diaphragm for control valve service. The moulded diaphragm facilitates installation, provides a relatively uniform effective area throughout the valve's travel range, and permits greater travel than could be possible if a flat-sheet diaphragm were used. If a flat-sheet diaphragm is used in an emergency repair situation, it should be replaced with a moulded diaphragm as soon as possible.

When re-assembling the diaphragm case, tighten the cap screws around the perimeter of the case firmly and evenly to prevent leakage.

Replacing Stem Packing

Bonnet packing, which provides the pressure seal around the stem of a globe-style or angle-style valve body, may need to be replaced if leakage develops around the stem, or if the

valve is completely disassembled for other maintenance or inspection. Before starting to remove packing nuts, make sure there is no pressure in the valve body.

If the packing is of the split ring variety, it can be removed (with considerable difficulty) without removing the actuator by digging it out of the packing box with a narrow, sharp tool. This is not recommended, because the wall of the packing box or the stem could easily be scratched, thereby causing leakage when the new packing was installed.

Don't try to blow out the old packing rings by applying pressure to the lubricator hole in the bonnet. This can be dangerous and frequently doesn't work very well anyway. (Many packing arrangements have about half of the rings below the lubricator opening.)
The approved method is to:

1. Separate the valve stem and actuator stem connection.
2. Remove the actuator from the valve body.
3. Remove the bonnet and pull out the valve plug and stem.
4. Insert a rod (preferably slightly larger than the stem) through the bottom of the packing box and push or drive the old packing out the top of the bonnet. (Don't use the valve plug stem because the threads could be damaged in the process.)
5. Clean the packing box. Inspect the stem for scratches or imperfections that could damage new packing.
6. Check the valve plug, seat ring, and trim parts as appropriate.
7. Re-assemble the valve body and put the bonnet in position.
8. Tighten body/bonnet bolting in sequence similar to that described for flanges on page 49.
9. Slide new packing parts over the stem in proper sequence, being careful that the stem threads do not damage the packing rings.
10. Install the packing follower, flange, and packing nuts.
11. For spring-loaded TFE V-ring packing, tighten the packing nuts as far as they will go. For other varieties, tighten in service only enough to prevent leakage.

12. Replace and tighten the actuator onto the body. Position and tighten the stem connector to provide desired valve plug travel.

Replacing Threaded Seat Rings

Many conventional sliding-stem control valves use threaded-in seat rings. Severe service conditions can cause damage to the seating surface of the seat ring(s) so that the valve does not shut off satisfactorily. In that event, replacement of the seat ring(s) will be necessary.

Before trying to remove the seat ring(s), check to see if the ring has been tack-welded to the valve body. If so, cut away the weld and apply penetrating oil to the seat ring threads before trying to remove the ring. The following procedure for seat ring removal assumes that a seat ring puller, such as that shown in Figure 2, is being used. If a puller is not available, a lathe or boring mill may be used to remove the ring(s).

1. Place the proper size seat lug bar across the seat ring so that the bar contacts the seat lugs as shown.

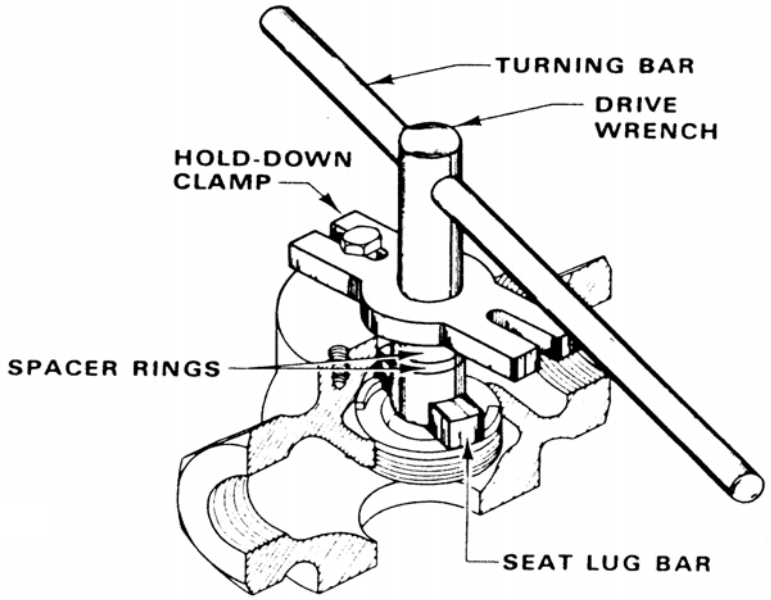


Figure 5-4. Seat Ring Puller

2. Insert drive wrench and place enough spacer rings over the wrench so that the hold-down clamp will rest about 1/4-inch above the body flange. Slip hold-down clamp

onto drive wrench and secure the clamp to the body with two cap screws (or hex nuts for steel bodies) from the bonnet. Do not tighten cap screws or nuts.

3. Use turning bar to unscrew the seat ring. Stuck seat rings may require additional force on the turning bar. Slip a 3- to 5-foot length of pipe over one end of the turning bar, and while applying a steady force, hit the other end of the bar with a heavy hammer to break the ring loose. In addition, a large pipe wrench can be used on the drive wrench near the hold-down clamp.
4. After the seat ring is loose, alternately unscrew the flange bolts (or nuts) on the hold-down clamp and continue to unscrew seat ring.
5. Before installing new ring(s), thoroughly clean threads in the body port(s). Apply pipe compound to the threads of the new seat ring(s).

Note

On double-port bodies, one of the seat rings is smaller than the other. On direct-acting valves (push-down-to-close action), install the smaller ring in the body port farther from the bonnet before installing the larger ring. On reverse-acting valves (pushdown-to-open action), install the smaller ring in the body port closer to the bonnet before installing the larger ring.

Screw the ring(s) into the body. Use the seat ring puller, lathe, or boring mill to tighten seat rings in the body. Remove all excess pipe compound after tightening. The seat ring can be spot welded in place to ensure that it does not loosen.

6. Reassemble the valve.

Grinding Metal Seats

A certain amount of leakage should be expected with metal-to-metal seating in any globe-style valve body. If the leakage becomes excessive, however, the condition of the seating surfaces of the valve plug and seat ring can be improved by grinding. Large nicks should be machined out rather than ground out. Many grinding compounds are available commercially. Use one of good quality or make your own with a mixture of 600-grit silicon

carbide compound and solidified vegetable oil. White lead should be applied to the seat to prevent excessive cutting or tearing during grinding. In cage-style constructions the bonnet or bottom flange must be bolted to the body with the gaskets in place during the grinding procedure to position the cage and seat ring properly and to help align the valve plug with the seat ring. A simple grinding tool can be made from a piece of strap iron locked to the valve plug stem with nuts.

On double-port bodies, the top ring normally grinds faster than the bottom ring. Under these conditions, continue to use grinding compound and white lead on the bottom ring, but use only a polishing compound (rotten-stone and oil) on the top ring. If either of the ports continues to leak, use more grinding compound on the seat ring that is not leaking and polishing compound on the other ring. This procedure grinds down the seat ring that is not leaking until both seats touch at the same time. Never leave one seat ring dry while grinding the other.

After grinding, remove bonnet or bottom flange, clean seating surfaces, and test for shutoff. Repeat grinding procedure if leakage is still excessive.

Lubricating Control Valve Packing

A lubricator or lubricator/isolating valve (as shown in Figure 3) is required for semi-metallic packing and is recommended for graphite asbestos and TFE-impregnated asbestos packing. The lubricator or lubricator/ isolating valve combination should be installed on the side of the valve bonnet, replacing the pipe plug used with packing types not requiring lubrication. Use Dow Corning X-2 lubricant or equivalent for standard service up to 450°F (232°C) and Hooker Chemical Corporation Fluorolube lubricant or equivalent for chemical service up to 300°F (149°C).

With, lubricator, isolating valve, and pipe nipple (if used) completely filled with lubricant and installed on bonnet, open isolating valve (if used) and rotate lubricator bolt a full turn clockwise to force lubricant into the packing box. Close the isolating valve after lubrication.

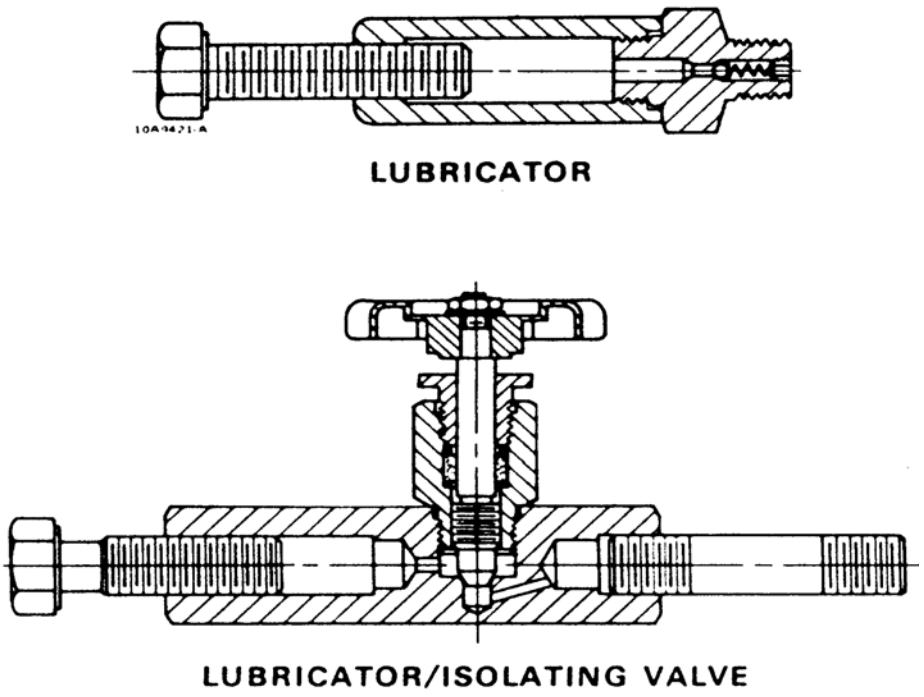


Figure 5-5. Lubricator and Lubricator/ Isolating Valve

Adjusting Travel and Connecting Stem

Sliding-Stem Control Valves Part names used throughout the following section is shown in Figure 4. The procedure is appropriate for sliding-stem valves with either spring-and-diaphragm or piston actuators. When performing the travel adjustment procedure, be careful to avoid damaging the valve plug stem. Scratches on the stem can lead to packing leakage. If the unit includes a bellows seal bonnet, the stem must not be rotated or the bellows will be damaged. On all other units, the stem may be rotated for minor travel adjustment, but the valve plug should not be in contact with the seat ring during rotation of the stem.

1. Assemble the body and mount the actuator. Screw the stem lock nuts onto the valve plug stem and set the travel indicator disc on the lock nuts with the "cupped" portion downward. Leave enough threads exposed above the disc for the stem connector.
2. Be sure the actuator stem is in the position that equates with the "closed" valve plug position—fully "down" for push-down-to-close valve styles; fully "up" for push-down-to-open valve styles. To achieve this condition, will often be necessary to pressure loading the actuator to properly position the stem.
3. Move the valve plug to the "closed" position, contacting the seat ring.
4. Change actuator-loading pressure in order to move the actuator stem 1/8-inch. Install the stem connector, clamping the actuator stem to the valve plug stem.
5. Cycle the actuator to check availability of desired total travel and that the valves plug seats before the actuator contacts the upper travel stop. Minor adjustments in total travel can be made, if necessary, by loosening the stem connector slightly, tightening the lock nuts together, and screwing the stem either into or out of the stem connector by means of a wrench on the lock nuts. If overall travel increase is desired, the increase must be less than the 1/8-inch the actuator rod was moved in step 4 above or the valve will not shut off.

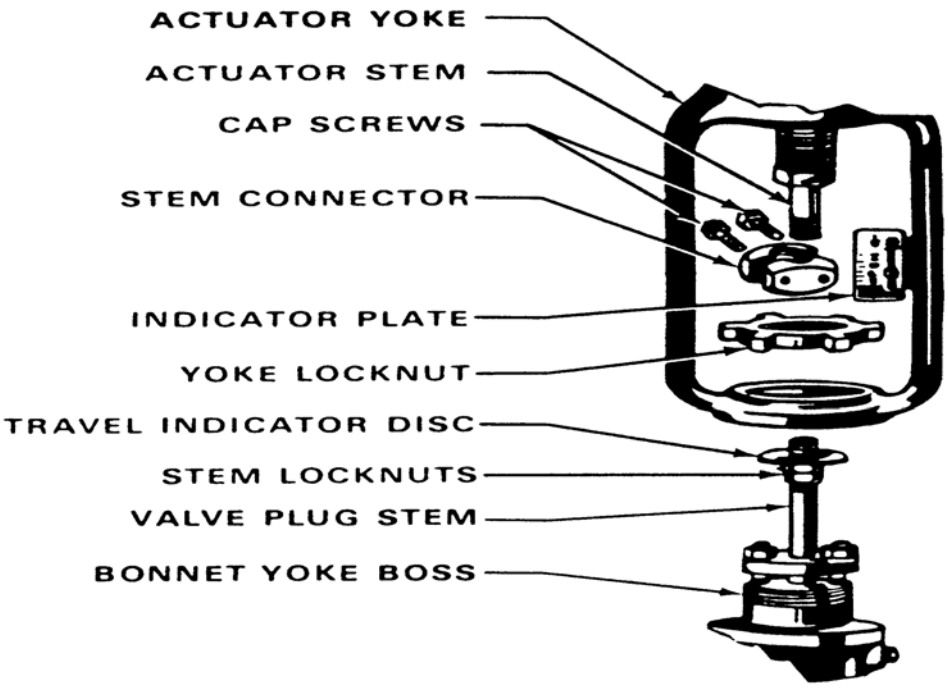
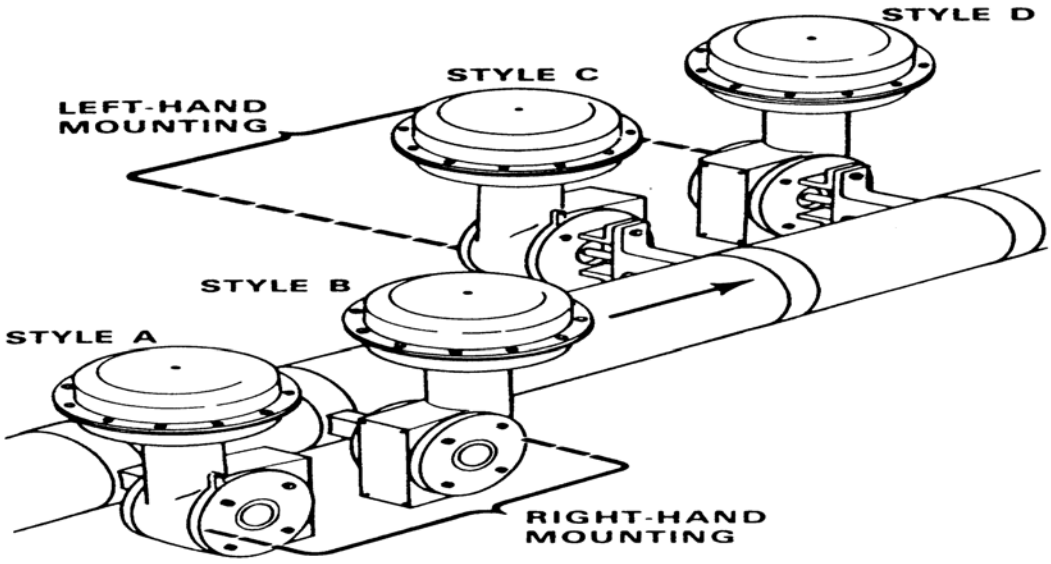


Figure 5-6. Exploded View of Stem Connection Components

6. If the total travel is adequate, tighten the stem connector securely, lock the travel indicator disc against the connector with the lock-nuts, and adjust the indicator plate on the yoke to show valve plug position.
7. Provide a gauge to measure the pressure to the actuator. Make a final adjustment on the actuator or its positioner to set the starting point of valve travel and to obtain full travel for the desired instrument range.

Rotary-Shaft Control Valves

As shown in Figures 5 and 6, there are a variety of actuator mounting styles and positions possible with rotary-shaft control valve bodies. Specific adjustment procedures vary depending on whether desired valve action is push-down-to-close or pushdown-to-open. The connecting linkage between the actuator and the valve body normally includes a lever, which is attached to the valve shaft by means of a key and keyway slot or by mating multiple cut splines on the lever and shaft. A rod end bearing and turnbuckle usually connect the lever to the actuator stem



Figur 5-7. Mounting Styles for Actuators on Rotary-Shaft Control Valve Bodies

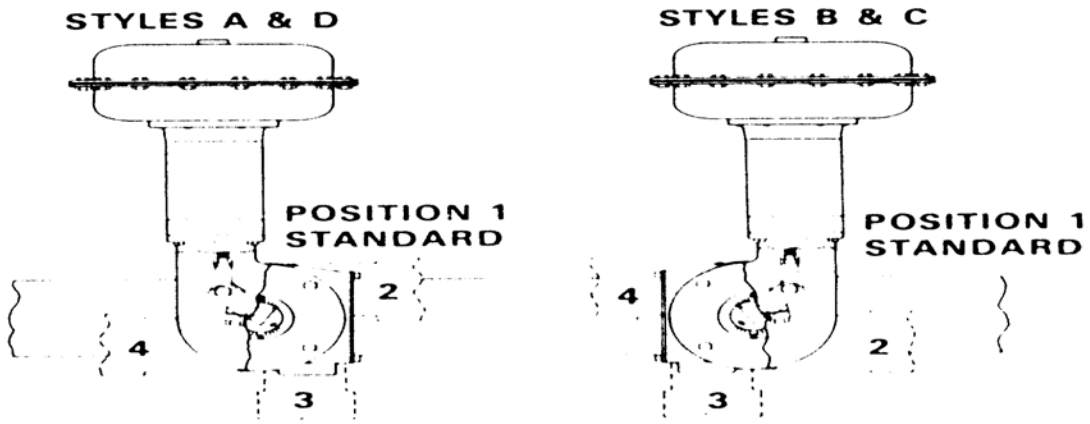
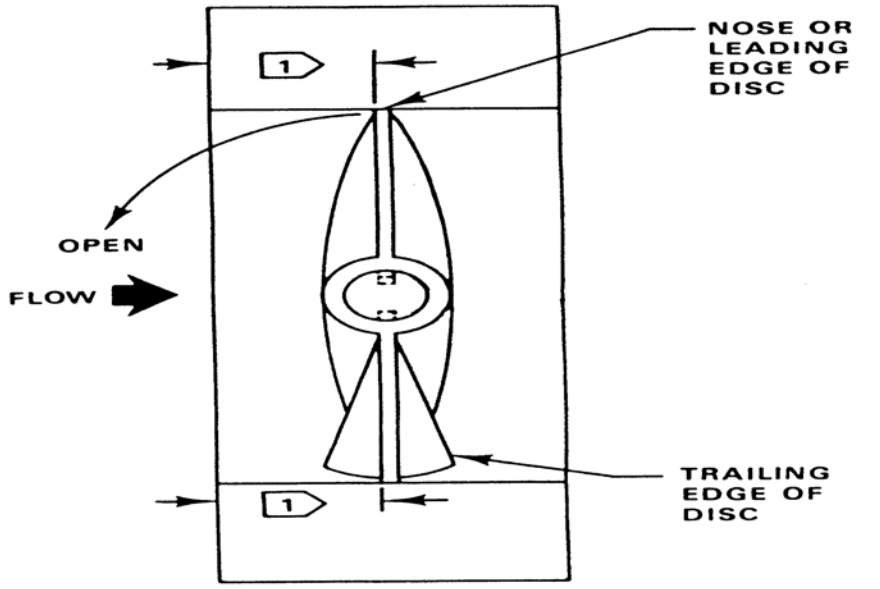


Figure 5-8. Mounting Positions for Actuators on Rotary-Shaft Control Valve Bodies



1 EQUAL MEASUREMENTS BETWEEN VALVE FACE AND DISC EDGE AT TOP AND BOTTOM ENSURE FULLY CLOSED DISC.

Figure 5-9. Check for Proper Disc Closure

The valve shaft and disc or V-notched ball is stamped with indicating marks to show proper orientation for mating splines. Similar indicating marks are used to show shaft and lever orientation. Fine adjustment is accomplished by lengthening or shortening the turnbuckle to achieve full disc or V-notch ball closure at 0° indicated rotation.

For disc-style rotary valves, fine travel adjustment should be performed with the valve body out of the pipeline so that measurements can be made as suggested in Figure 7. Refer to the manufacturer's instruction manuals for specific adjustment details for the body and actuator being used.

Valve Terminology

- **Balanced valve**: a body design in which the same pressure acts on both sides of the valve plug.

- **Bonnet assembly**: an assembly including the part through which a valve plug stem moves and a means for sealing against leakage along the stem.

- **Cavitate**: the formation of voids or cavities in a valve resulting from increased fluid velocity through the restricted areas of the valve. It occurs in liquids when the valve operates near the vapour pressure of the liquid.

- **Characteristic**: relation between flow through the valve and percent travel as the travel is varied from zero to 100%.

- **Corrosion**: the reactions between materials of the valve and the fluids handled which cause valve deterioration.

- **Cv**: flow coefficient, the capacity of a valve. It is defined as the number of gallons per minute of water at room temperature which will pass through a given flow restriction with a pressure drop of one psi.

- **Dead Band**: the amount the diaphragm pressure can be varied without moving the valve plug.

- **Diaphragm Actuator**: an actuator that uses diaphragm assembly for moving the actuator stem.

- **Equal Percentage flow characteristics**: flow characteristic, which for equal increments of rated travel will give equal percentage changes of the existing flow.

- **Erosion:** wearing action on valve trim and body. It is common in steam service and where high pressure drops occur.

- **Extension bonnet:** a bonnet with an extension between the packing box assembly and bonnet flange.

- **Guide bushing:** a bushing in a bonnet, bottom flange or body to align the movement of a valve plug with a seat ring.

- **Leakage:** quantity of fluid passing through an assembled valve when the valve is fully closed.

- **Linear flow characteristic:** flow characteristic, which can be represented by a straight line on a graph of flow versus percent, rated travel.

- **Modulate:** continually move the valve between the closed and full open positions.

- **Normally closed:** applying to a normally closed control valve assembly one, which closes when the actuator pressure is reduced to atmospheric.

- **Normally open:** applying to a normally open control valve assembly one, which opens when the actuator pressure is reduced to atmospheric.

- **DP:** the pressure drop across a valve the condition must be specified For example: DP for sizing; DP at normal flow; DP at valve closure; etc.

- **Packing box assembly:** the part of the bonnet assembly used to seal against leakage around the valve plug stem.

- **Plug:** a moveable part, which provides a variable restriction in a port.
- **Seat:** that bit of a seat ring or valve body, which with which the valve plug contacts for closure.
- **Seat ring:** a separate piece inserted in a valve body to form a valve body port.

- **Stem:** a rod extending through the bonnet assembly to permit positioning the valve plug.

- **Trim:** the parts (except the body) of a valve, which come into contact with the flowing fluid.

- **Valve body:** a housing for internal valve parts having inlet and outlet flow connections.

- **Valve plug guide:** that portion of a valve plug, which aligns its movement in either a seat ring, bonnet, bottom flange any two of these.

- **Yoke:** a structure by which the diaphragm case assembly is supported rigidly on the bonnet assembly.

SGC HSE Regulation Related to:

HSE Regulation No. 5 "RISK ASSESSMENT"

To identify all hazards and to set the required precautions A sanction for the procedures may be required if the isolation methods necessitate so.

HSE Regulation No. 6 "Permit To Work System"

To issue a cold work permit before starting the job

HSE Regulation No. 7 "ISOLATIONS"

To ensure the most safe method of isolation in place, and mechanical isolation certificate issued before start repairing the valve.

To ensure Isolating valve(s) locked and tagged other isolation devices like plugs assessed

7.18 Control Systems Procedures and Isolations

1. Preparation for Work

When work is to be done on control engineering hardware, it is essential that the worksite is prepared correctly. The work permit system (Regulation No 06) provides the mechanism for ensuring that essential preparatory activity is documented and witnessed. It is important that the Control Engineering Person doing the work also has responsibility for effecting the task safely. This is particularly important in the control engineering discipline where the instrument may still be connected to the process, or may require to be serviced with power-on for fault finding.

2. Interaction

Care must be taken to ensure that the work to be carried out on a specific item of instrumentation will not cause a hazard due to interaction with other protection systems or operational process controls.

3. Preparatory Work

Prior to a work permit being issued all appropriate preparatory work at the site must be completed. The following items are some examples.

a. Removal of potential hazards from the area. Particular vigilance is needed for enclosed areas (refer to Regulation No 09 Confined Space Entry).
b. Gas testing the area for flammable, toxic or suffocating gases.
c. Construction of scaffolding to permit safe access.
d. Provision of additional fire fighting apparatus.
e. Provision of necessary protective clothing-
f. Isolation of the control engineering hardware from the process.
g. In the case of equipment removal, isolation of the hardware from utilities (e.g. electricity, air supplies etc).

On completion of all necessary preparatory work (defined by the Senior Control Engineering Person and the Area Authority), a hot or cold work permit/entry permit will be issued signed by the appropriate responsible authorities.

4. Isolation of Hardware

Isolation of control engineering hardware may be necessary to enable maintenance work to be done or permit removal of the hardware to effect repairs (either locally or remotely). Isolation of hardware can take several forms, for example isolation from:

a. Process plant.
b. Utilities (electric, pneumatic, hydraulic, cooling media etc).
c. Larger system of which the hardware is a subsystem or component.

5. Isolation from Process

a. Isolation of instruments, which are connected to or form a part of the process, is usually achieved by valving. It is important that, where isolation of an instrument is required for maintenance purposes, correct venting/draining and valve closure procedures are adhered to.

b. Where instruments have local isolating valves in addition to the primary process isolating valves, the local valves may be used for some routine in-situ testing at the discretion of the Senior Control Engineer. If an instrument is to be removed from site, the process isolating valves must be used and any impulse pipe-work must be drained or vented completely.

c. Where the process fluids are of a hazardous nature (e.g. toxic, flammable etc), particular care should be taken to ensure correct venting and draining, and also to clean or flush the instrument carefully, prior to effecting work or removal of the hardware from site for maintenance or repair. Gas testing may be required. On large items, e.g. control valves, a certificate of cleanness, is necessary prior to delivery to workshops.

d. On removal of a directly mounted instrument, from a process line containing hazardous fluids, e.g. pressure gauges etc, isolation by the primary isolation valve only is NOT acceptable. The valve outlet shall be blanked off, capped or plugged with a blank flange, solid screwed plug or cap, whichever is appropriate.

6. Isolation from Electrical/Pneumatic Supplies

a. If practical, equipment must be made safe before any work is done on it. A Competent Control Engineering Person must do the operation of making the equipment safe. Care shall be taken when working on live equipment to ensure avoidance of contact with live electrical components (refer to Regulation NO.19 Working with Electricity).

b. Pneumatically operated equipment must be isolated before it is disconnected or removed for repair by closing the valve at the supply manifold for the individual instrument and venting through the drain/vent of the pressure regulators.

7. Isolation from Utilities

a. Control engineering equipment may be connected to utilities (other than electrical associated with the hardware e.g. streams, cooling water, hydraulic fluid, chemicals, carrier gases (analysis) and air supplies. It is important that attention is given to rendering the utilities safe when the control engineering hardware is being serviced or removed.

b. Utilities should be isolated at the point of distribution to the control engineering equipment being removed (e.g. isolating valve at distribution head) and not solely at the hardware itself.

c. Where utility fluids are 'piped' to an instrument, the pipe-work should be drained down or vented if the instrument is removed.

d. It is important that removal of a utility from a specific piece of hardware does not influence any other hardware to which the utility may also be connected (e.g. cooling water may have been series connected to more than one item of hardware).

8. Use of Tools and Test Equipment

a. Tools and test equipment must be suitable for use in the work area. They should be checked before and after use and all calibration equipment should itself be calibrated periodically, at intervals determined by the Senior Control Engineering Person.

b. Use of tools and test equipment are subject to the Work Permit System (refer to Regulation No 06). It is particularly relevant to ensure that electrical tools and test equipment comply with the area safety classification of the work place. This may be achieved either by certification or using the Permit to Work.

9. Workshop Practice

It is essential that good workshop practice be adhered to at alt times, for example:

a. Machinery will not be operated without guards or suitable personal protection.

b. All workshop equipment will be maintained in good working order.

c. Good housekeeping is essential to safety.

d. All necessary spares, cleaning materials, tools and test equipment should be available and must be correctly maintained, tested (where appropriate) and stored.

10. Changes and Modifications

Changes to alarm settings and transmitter ranges must be approved by the operating authorities and documented. No such change will be made unless the Passport Work Order

has been completed. Changes to trip settings should only be carried out under the authority of a PMR endorsed by the ED. This is mandatory at all times.

Similarly, no modification will be undertaken unless the proposal has been through the established PMR and appropriate authorisation. Thus, before work starts, the job must be approved and finance made available.

11. Procedure for Adjusting Limit Switches of MOVs within the Process Plant

Conditions of related work:

a. This procedure is considered as an explanation and does not constitute deviations from the SGC Safety Regulations and Standard Procedures.
b. This procedure is limited to work on the Limit Switch Compartment of any MOV.
c. The MOV is provided with a local 415V isolator. (If this is not available, electrical isolation by the Electrical Department must be made.)
d. A continuous gas-monitoring device is to be provided on site.
e. A competent Control Supervisor/ Foreman/Technician shall perform Work.

Work Procedures:

a. Obtain a hot work permit from the Area Authority and observe all of the conditions.
b. Switch off the local 415V isolator using a padlock (keys are to be kept with the performer).
c. Check that there is no power supply available by operating the local open/close switches.
d. Adjust the valve open and close, limit switches as necessary.
e. Switch on the supply to the valve actuator from the local switch and test the valve operation.
f. Repeat step b to e until satisfactory adjustment of the limit switches is made.
g. Hand over the valve to the Operating Authority by signing off the work permit.

HSE Regulation No. 8 "Breaking Containment"

For emptying containment, cleaning and gas freeing methods removal of sludge (pyrophoric scales), trapped oil or vapour general ventilation.

HSE Regulation No. 23 "General Engineering Safety" concerning the Safety Relief Valves

23.20 Safety Relief Valves

1. The use of safety valves in series is forbidden, because of the consequent reduction in the rate of discharge.
2. The fitting of an intervening stop valve between a vessel and its relief valve or protecting device, should only be resorted to after careful consideration of the possibilities of maloperation of the stop valves.
3. Where stop valves are so fitted, they must be so arranged that they can be locked in the open or closed position. Permission must be obtained from the Area Authority before any lock is removed and the stop valve operated.
4. The keys for the locks of any stop valves, on the upstream side of relief valves, will be held in the custody of the Area Authority.
5. 5- if it is necessary to remove or isolate a safety valve from any pressure vessel in service, the sanction of the Area Authority must first be obtained.
6. Venting of relief valves should be arranged to avoid the release of toxic or hazardous chemicals into areas that could affect personnel.
7. The Area Authority should maintain a register of all safety valves on the plant, giving the PSVs current status, e.g. locked open, locked closed, on standby, to maintenance, etc.

HSE Regulation No. 27 " General Services"

- To safe use of hand tools and electric/or pneumatic powered tools and equipment,
- Precautions for Crane/lifting appliances and gears and precautions for industrial powered trucks/hydraulic work platforms,
- Manual handling and lifting
- Scaffolding to do the repair Job

MODULE 4:

ADVANCED CONTROL METHODS

ADVANCED CONTROL METHODS

Introduction

This module will cover Process Automatic (Advanced Control Methods) and Transmission Systems (Comparison of Electronic & Pneumatic Systems)

Objectives:

Participants will learn about the basics of process control theory. They will also get acquainted with the advanced methods, techniques and applications of advanced process control. Participants will also learn how to evaluate the system performance and automated tuning. As well as, gaining basic knowledge and techniques practically used in process plant measurements.

Feed forward control, decoupling methods, relative gain, and process modifications are techniques for resolving the difficulties that arise in some process-control loops. Though simple feedback loops are dominant in controlling a typical process plant, about 10 to 20 % of the control loops are more complicated typical examples include cascade, ratio, and auto-selector and override controls. The most difficult of these (about 5%) will require the use of advanced techniques.

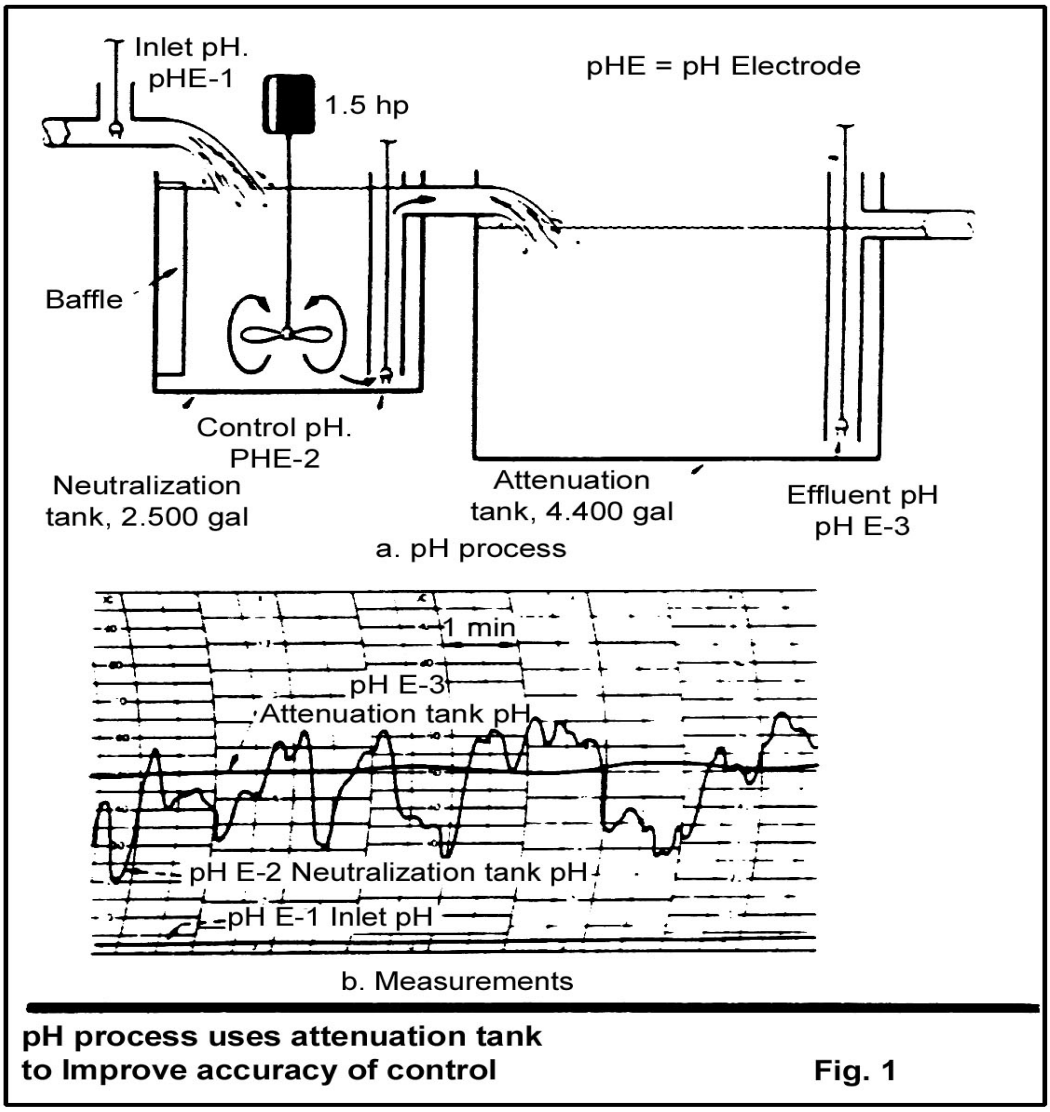
In this article, we will discuss the advanced techniques of feed forward control and decoupling. Succeeding articles will cover additional procedures. Also, the problem of interaction in multivariable process will be illustrated by using the concept of relative gain, and by comparing techniques for decoupling.

Better results can often be obtained by modifying the process design. This should not be overlooked as a means for achieving improved control. In this article, process modification is considered the most important method and will be discussed first.

Modifying the Process Design

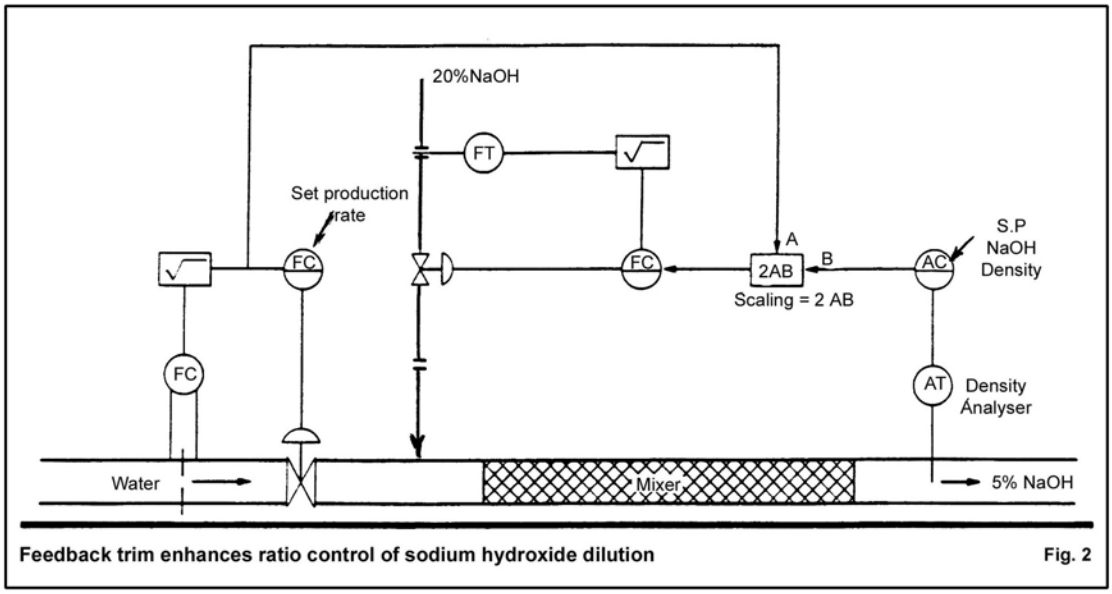
The classic example for achieving advanced control through design modifications is the pH process. Here attention must be paid to details such as sizing mixers with sufficient horsepower, and optimizing impeller speed to minimize dead time. In addition, the pH electrode must be placed to allow fast response without excessive signal noise, and the control valve must be located at the point of reagent addition to avoid downstream piping and unwanted draining and dripping after the valve is shut. In single-vessel pH processes even all this attention to detail will not be enough to meet the accuracy requirements. Which sometimes exceed one part in a million .When simultaneous upsets occur. Hence a design solution must be considered. For example. A second tank could be placed downstream from the first for attenuation, and a pH recording measurement could then be made on the effluent from the attenuation tank (fig.1). Let us suppose that the pH neutralization tank (fig.1) is oscillating at high frequency (30s. peak-to-peak) with amplitude that exceeds the required control range. The attenuation tank acts as a capacitor does in filtering a noisy electrical signal, and will significantly reduce the amplitude of the pH oscillations from the control tank, producing an effluent that is within the required range. The attenuation tank is a low-pass filter- attenuating high frequencies and passing low ones (an electrical capacitor attenuates oscillations in voltage .Similarly, an attenuation tank smoothes out oscillations of ions in solution.) it is necessary that the dead time in the pH-control tank be minimized in order to keep the frequency high and the attenuation-tank volume and cost to a minimum. A tendency exists to blame the controller for poor control. Small improvement can be made by modifying the pH controller: e.g.. a properly tuned nonlinear controller will help to keep the amplitude of pH oscillations to a minimum and the frequency as high as the process

will allow. In the final analysis, modifying the controller will have only a minor effect on performance, while modifying the process will have a major one.



pH process uses attenuation tank to Improve accuracy of control **Fig. 1**

PROCESS AUTOMATION



Feedback trim enhances ratio control of sodium hydroxide dilution **Fig. 2**

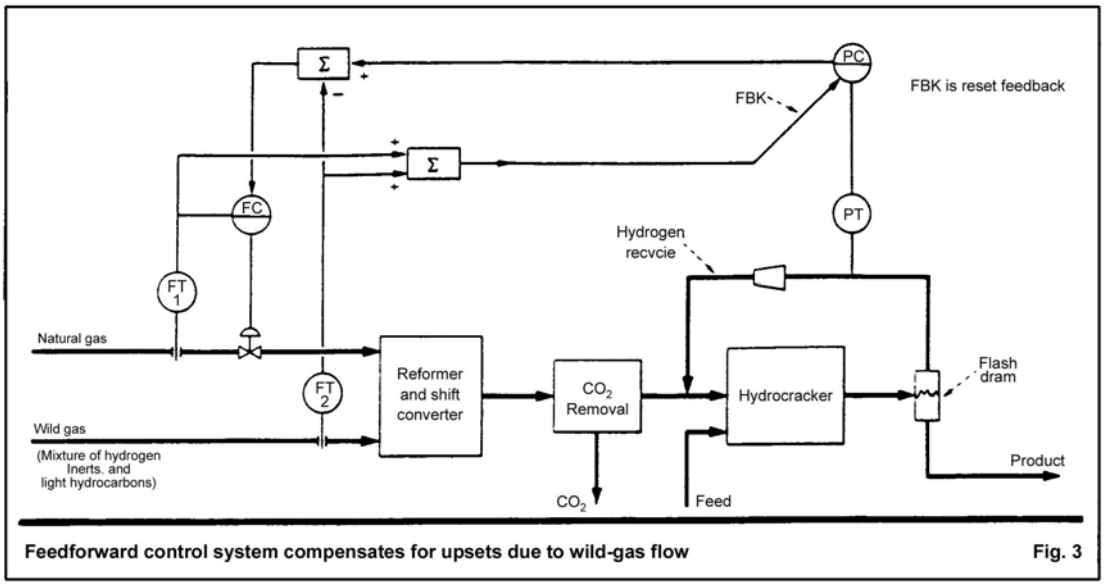
Feedwords Control Techniques

In principle if an upset can be measured en route to a controlled variable, feed forward techniques could be applied in a manner that would allow corrective action. This exactly cancels the upset and maintains the trolled variable constant in practice perfect correction is seldom achieved with feed forward control because accurate feed forward compensation can be very complex on all but the simplest systems. Combining feed forward control with feedback trim can be an effective way to achieving improved regulatory control. High accuracy in the steady state can be obtained with a feedback controller (If it has an integral or resets mode). In addition, good response to upsets can be achieved by using feed forward techniques.) See part 2 of this section. . p148.) Clearly, Conditions causing upsets should be eliminated or reduced when this is more cost-effective than adding feed forward control. When feedback controls cannot be arranged to respond fast enough to catch the upset. Feed forward controls can be an effective answer

Hardware and Software Constraints

Feed forward techniques were first applied to control boiler-drum level (three element control) but it was not until the pneumatic multiplier/divider was introduced that they began to gain acceptance for other processes. In the late 1960s electronic analog computers (based on the operational amplifier) provided highly accurate (up to 1 part in 100,000) simulation and control capabilities this ability to solve equations accurately and continuously was ideal for feed forward control. Today analog computers in the control industry have been separated into modules to perform specific functions such as addition multiplication .Division integration filtering (lag) differentiation (lead or derivative). Characterization ramping. And signal selection. The pure time-delay (dead time) algorithm needed for realistic simulation and for dynamic compensation of feed forward control systems cannot be implemented practically in analog controls .however, this algorithm can be easily implemented in digital processors. Along with all the functions previously available in analog hardware in addition .Digital processors can easily handle complicated systems of equations .Gate logic sequencing and iterative calculations. With present- day analog and digital computers, application of feed forward controls is limited only by the availability of suitable load measurements or sufficiently accurate process models.

ADVANCED CONTROL METHODS



Feedforward control system compensates for upsets due to wild-gas flow **Fig. 3**

Ratio Control

Ratio control is an effective form of feed forward control, fig,2 shows a commonly used ratio-control system which continuously adds 20% NaOH to a varying flow of water to produce 5% NaOH .If the water flowrate were to change. The set point to the caustic flow-controller would be increased or decreased proportionately maintaining a constant ratio of caustic flow to water flow, in this way; the upset has been compensated for before the composition (density) has been affected. Feed forward action substantially reduces the amount of feedback correction required for upsets in the water flowrate. The multiplier is scaled for twice the product of the A and B function to obtain a feedback controller output of 0.5 (i.e. midscale). This allows the feedback trim to adjust the ratio equally well up or down from the normal value. The "2AB" rule of thumb is acceptable when the flowmeter sizing for both water and caustic flows is consistent with respect to orifice overranging in other words both flow measurements are normally of the same fraction of the full-scale range. Feedback trim can be introduced with a summer, adding to or subtracting from the feed forward calculation. The choice of using a summer or a multiplier for feedback trim is mostly a matter of minimizing feedback corrections. Preferably, both flow measurements have been linearized (i.e.. square-root extractor for differential-pressure transmitter). Ratio control can work if both signals are "flow squared" without square-root extraction. The ratio of caustic flow "squared" to water flow "squared" will be maintained. This is an accurate and acceptable implementation of a mathematical model. Mixing squared and linear signals does not fit the mathematical model for blending, and would not produce accurate results. Feed forward controls are based on a model of the process. In the ratio-control example, the model seems intuitive actually, the model for the blending is based on two simultaneous equations: the overall and the caustic-materials balances or:

$$F_t = F_X + F_c \qquad (1)$$

$$F_t x_t = F_c \qquad (2)$$

Solving for the desire value of the manipulated variable, F_c the required caustic-flow set point can be calculated from the measured value of water flow, F_w, and the desired dilution concentration. X_v or:

$$F_c = \frac{F_x}{[(x_c / x_t) - 1]} = KF_w \qquad (3)$$

For constant concentrations, the flow of caustic is directly proportional to the flow of water. Unlike open loop ratio control, the ratio value, K is not calculated but is determined by the feedback controller output. The output limits of the controller can be used to restrict the adjustable range of K by setting them for minimum and maximum ratios.

Feedforward Reactor Control

Fig. 3 shoes a feed forward control system for a refinery reformer. This is reactor manufacturing hydrogen for a downstream hydrocracker. The hydrogen pressure in the hydrocracker system is an indication of hydrogen inventory. if conversion at the hydrocracker is increased, more hydrogen is consumed and the pressure will fall to maintain constant pressure, the controller increase the

Nomenclature

C_t Controlled variable i

F_A Flow of stream A , lb/h

F_B Flow of stream b, lb/h

F_C Flow of 20 % caustic, lb/h

F_L Hydrogen consumption rate (fig 3) standard ft^3h

F_t Total flow, lb/h

F_x Water flow, lb/h

F_1 Hydrogen flow at FT-1 (fig 3), standard ft^3/h

F_2 Hydrogen flow at FT-2 (fig 3), standard ft^3/h

G_l Gain L, dimensionless

G_2 Gain 2, dimensionless

K Constant

K_1 Ratio of valve sizes dimensionaless

M_j Manipulated variable j

X Ratio of flows v_2/v_1

X Number of carbon atoms in alkane

X_c Caustic concentration, weight %

X_t Dilution concentration of caustic, weight %

Y Total flow (v_1+ v_2), lb/h

λ_η Relative, gain, dimensionless

Moles of Hydrogen Produced Per moles of Feed

Component	Component moles	H_2 produced, moles
H_2	x_a	x_a
Inerts	x_b	o
CH_4	x_c	$4x_c$
C_2H_6	x_d	$7x_d$
C_3H_8	x_e	$10x_e$
$C_xH_{(2x+2)}$	x	3x+1
	———	
	$\sum$ = 100	

Natural gas to the reformer in order to produce more hydrogen .This restores the inventory of hydrogen at the hydrocracker.Hydrogen consumption at the hydrocracker changes slowly, and pressure control would be good except for an uncontrolled feed to the reformer. This uncontrolled feed (called "wild gas") is a mixture of hydrogen, inert, and light hydrocarbons. When the wild-gas flow changes suddenly, the pressure in the hydrocracker system will be upset. The feedforward system should reduce natural-gas flow whenever wild-gas flow is increased by an

amount that will not upset the hydrocracker. Since wild gas has a composition that is different from natural gas has a composition that is different from natural gas, it does not produce the same volume of hydrogen. The overall reaction in the reformer and the shift converter for methane is :

$$CH_4 + 2H_2o \longrightarrow CO_2 + 4H_2 \qquad (4)$$

Notice that four moles of hydrogen are produced for each mole of methane. The overall reaction for an arbitrav alkane $C_x H_{2x-2x}$ is:

$$C_1H_{2-2} + 2_xH_2O \longrightarrow {}_xCO_2 + (3x+1)H_2 \qquad (5)$$

For light hydrocarbons (where $x \geq 2$) fed to the reformer. Moles of hydrogen per mole of feed are more than four. The Hydrogen-producing power of the wild gas and natural gas feeds must be compared on a volumetric basis because they are metered as gas (using an orifice-plate flowmeter) the accompanying table lists the moles of hydrogen produced for each component in the feed.

The natural gas contains some ethane along with methane. And will produce 4.2 volumes of hydrogen per volume of feed the wild gas contains some hydrogen for each volume fed to the reformer in addition to the ideal gas law.100% reaction conversion is assumed and losses oh hydrogen is neglected.

The scaling of the summer can be obtained from a material balance of hydrogen in the hydrocrcker system, if good pressure control is maintained, the overall hydrogen balance should be:0= accumulation = inflow – outflow. Specifically:

$$0 = (F_1 + | F_2) - F_1 \qquad (6)$$

The identification of transmitter ranges and the use of the previously identified production factors will yield a scaled equation for the summer:

$$F'_i = g_1F'_1 - g_2F'_2 \qquad (7)$$

Where the prime mark (′) indicate scaled valued and $g_1$and g_2 are gain terms determined by scaling if the pressure controller is in automatic mod and is controlling well its output could be determined by:

$$F'_1 = \frac{F'_i}{g_L} + \frac{g_2 F'_2}{g_L} \qquad (8)$$

The calculation for F'_L in Eq.(8) should be used by the pressure controller for its external integral (reset) feed-back connection to help prevent windup if the natural gas flow controller cannot follow its setpoint. The process operator greatly appreciates this feature particularly when the pressure controller is put in track mode, as it should be if the flow controller were to be put on local setpoint or on manual .when the track bit (0=no tracking, 1=tracking) of the pressure controller is set, the controller output is immediately kept equal to the integral feedback value. Thus, the output of the pressure controller is back-calculated to precisely the correct value to keep the external setpoint of the flow controller matched to its measurement. To to put the system on control the operator merely puts the flow controller in automatic with remote set-point. Since the operator is relieved of adjusting the pressure-controller output to line up the external set-point and measurement of the flow controller, the possibility of bumping the process is reduced

Two-variable Feedforward Control

Feedforward techniques can be used to compensate for simultaneous upsets let us consider the control system in Fig, 4 the feed composition and flowrate to a single continuous distillation column are variable. This can cause excessive impurities to appesr in the bottoms products. The temperature controller cannot be tuned fast enough to catch these upsets.

Feedforward control is used in order to keep the bottom temperature steady. The temperature of the two component mixture in the tower-bottoms sump (column pressure is controlled) has a direct relationship to the impurity concentration. Even with Multicomponent mixtures, a sufficient correlation often exists if the feed, F is a liquid at its bubble point when its flow is increased the level on the feed tray will rise, spilling over the down comer weirs onto the tray below.

The feed is analyzed for the fraction of lights. $_Z$ The first multiplier simply calculates F_z which is equal to the flow of the lighter component entering the tower. The flow rate of steam should be nearly proportional to F_z because an increased flow of lights must be vaporized and removed as distillate. This proportionality however will not exist for all columns because variations in feed composition may affect reflux ratio or a substantial amount of steam may be required merely because the feed is sub cooled.

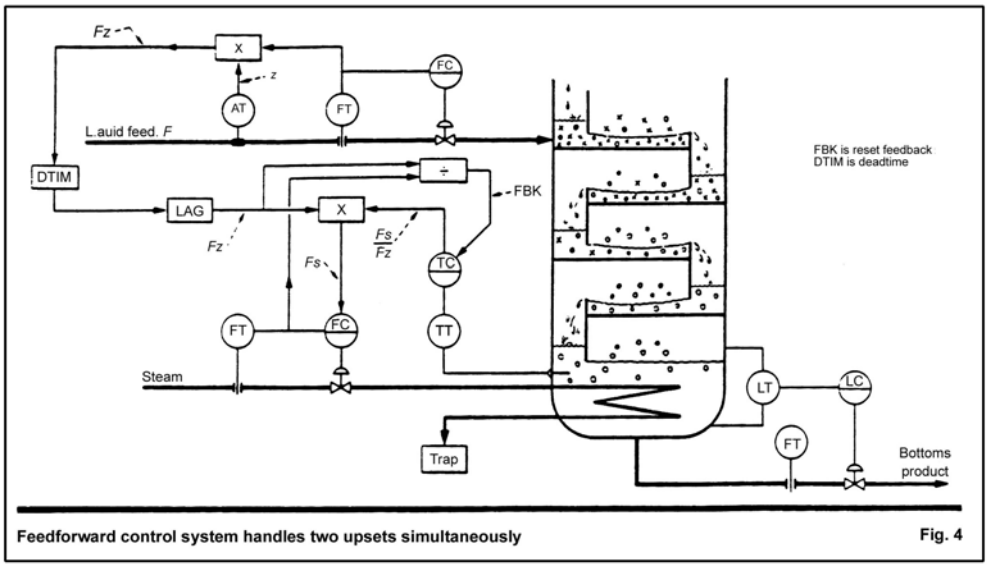
The second multiplier simply allows the temperature controller to adjust the ratio. The feedback trim from the temperature controller compensates for possible error in the measurement model and calculation.

The most difficult part of this system involves the dynamic compensation if the flow of feed suddenly increases; the effect at the bottom of the column is delayed by the length of time it takes for the increased liquid to cascade down the column.

For most columns, it will takes roughly 10 s/tray before the upset begins to affect the bottoms temperature the deadtime and lag settings for the feed forward dynamic compensation should be based on actual column testing. The deadtime is primarily a function of mixing on the tray and transport delay in the downcomer liquid level. The lag time of each tray is approximately equal to the actual volume of liquid on the tray divided by the liquid flowrate.

The easiest and fastest way to get an estimate for dead time is to increase the feed flow about 10% with the steam flow held constant, while at the same time recording feed and bottoms flow rates. This procedure has been successfully used to accurately set the dynamic compensation time constants, which change with flow rate composition and tray level. These may require periodic readjustment

ADVANCED CONTROL METHODS



Feedforward control system handles two upsets simultaneously Fig. 4

Multivariable Control Techniques

Process interactions arise from interconnected networks of mechanical, fluid or electrical components. In some cases, the interactions are intentional: in others they arise as an unavoidable consequence of the process design. For example, if a large steam user suddenly starts up, it will decrease the pressure in the steam header possibly causing upsets to other steam users on the same header these interactions occur as result of the piping network.

When a user or a supplier moves a control valve, all others users and suppliers are affected usually, the header pressure is controlled by manipulating the source of steam – typically by adjusting firing rate at a boiler. If this pressure controller is fast

compared with those of the other users or suppliers, the pressure in the header can be maintained, thus minimizing interaction.

Such interactions can be decoupled explicitly or implicitly. Explicit decouples use a process model (often including dynamics) in order for each controller to influence other interacting controllers in such a way that any changes in output reduce or eliminate the propagation of upsets to the other interacting controllers. Implicit decoupling involves rearranging and/or tuning the controllers in ways that makes individual loops inherently less interactive. the major problem with interaction in multivariable process is the lack of identification of the extent and mechanism of interaction.

Relative Gain

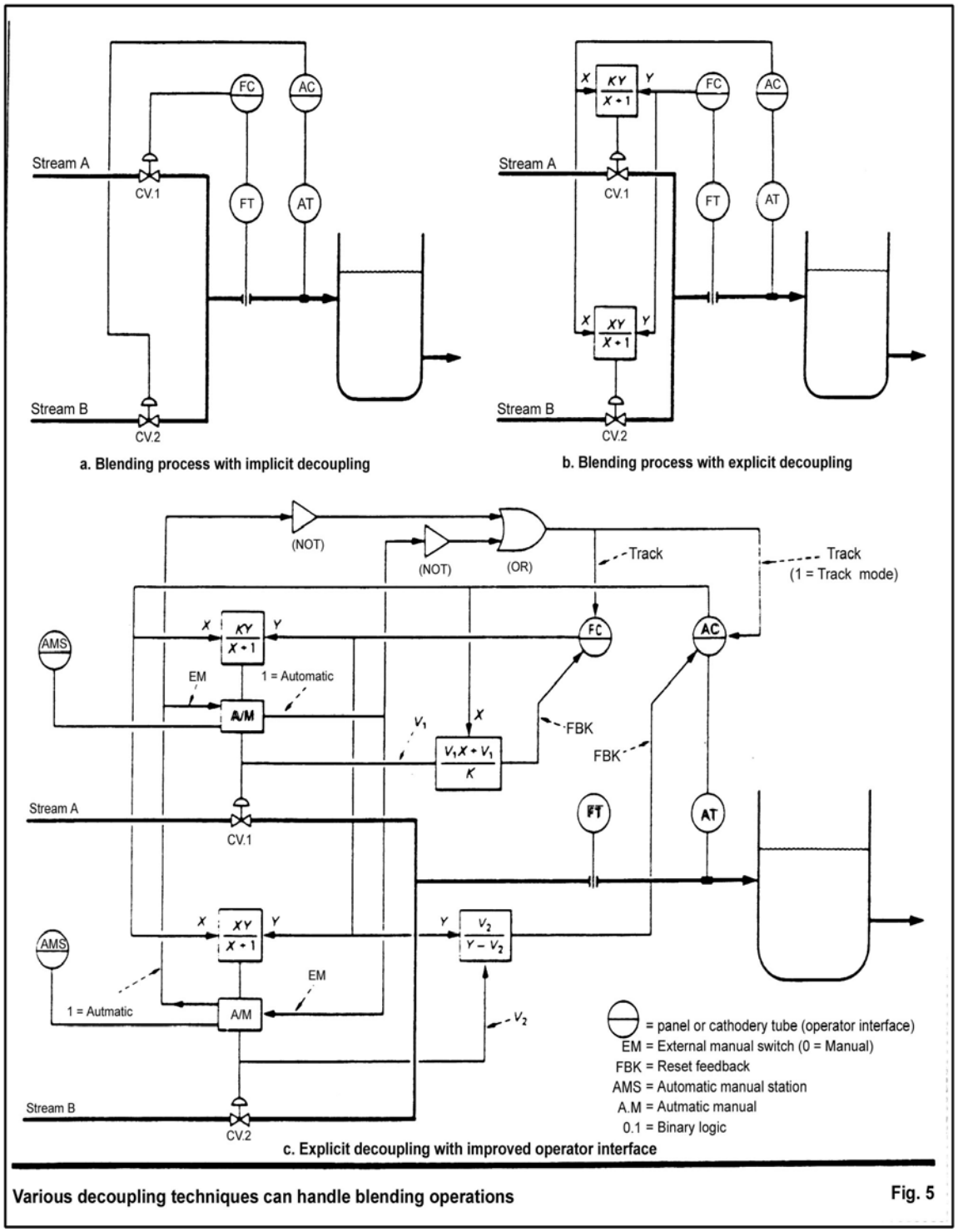
To analyze loops for interaction, shins key (2) employs a techniques developed by Bristol (4) called relative gain it has been increasingly used to guide control-system arrangement for distillation columns and is applicable to a wide range of interaction problems. It is successful because it quantifies the specific amount of interaction and can be used for any control lop, relative gain is defined as:

$$\lambda_{\eta} = \frac{\left.\frac{\partial C_1}{\partial m_1}\right|_{m=k}}{\left.\frac{\partial C_1}{\partial m_j}\right|_{C=k}} \qquad (9)$$

Where c = controller I, λ_{η} = relative gain for controller I with valve j.

The relative gain, which is ratio of two gains, can be used to determine if controller I should be connected to valve j. the numerator in Eq (9) is change in the controller measurement with a change in the valve position. With m = k (this is all other valve positions in the plant are fixed).

PROCESS AUTOMATION



Various decoupling techniques can handle blending operations

Fig. 5

The denomination is the change in the controller measurement with a change in the valve position with C = k (that is all other controllers in the plant moving their valves as needed in order to maintain their measurements at stepoint).

The relative gain can be obtained by field testing .In practice. It has usually been calculated from simple material balance equations ideally, moving the valve should affect the measurements to be controlled equally, whether the other controllers are in automatic or manual. In this ideal case, numerator equals denominator.
Relative gain is equal to 1.0 and valve 2 will control measurement without interaction from any other controller in the plant.

Let us consider the blending process in fig 5a to determine which valve should be used to control the total flow calculated the relative gain for flow that is controlled by valve CV-1. If valve 1 is opened and valve 2 is fixed. The flow will increase .Indicating that the numerator of Eq. (9) is a positive number. If the composition controller is in automatic when valve 1 is opened the flow will increase. However, the composition controller will open valve 2, increasing the flow further in order to keep the composition at setpoint. Thus the denominator is greater than the numerator and the relative gain is somewhere between zero and one.

To determine the specific relative gain .Calculated the analysis is slightly easier if valve position is proportional to individual flows .I.e. a linear installed valve characteristic. The material balance is:

$$F_T = F_A - F_B \qquad (10)$$

The Controlled Variable .C is the total flow. F_T. and the manipulated variable M. is the flow F_A through valve 1. therefore .with valve 2 held fixed . Calculated the numerator as:

$$\left.\frac{\partial c_1}{\partial m_1}\right|_{m=k} = \left.\frac{\partial f_t}{\partial f_A}\right|_{F_B=K} = \frac{\partial (F_A + F_B)}{\partial F_A} = 1.0 \qquad (11)$$

To calculate the denominator, the flow F_B is no longer constant but must be adjusted to allow the composition controller to hold its stepoint .Cleary whenever F_A is changed. F_B must also change if composition is to be held constant. If perfect composition control is to be realized, then the ratio of the two flows must be constant. i.e.:

$$F_B/F_A = K \qquad (12)$$

Keeping in mind that F_B is now a function of F_A:

$$\left.\frac{\partial C_1}{\partial m_1}\right|_{c=k} = \left.\frac{\partial F_T}{\partial FA}\right|_{\frac{F_B}{F_A}=K} = \left.\frac{\partial (F_A + F_B)}{\partial F_A}\right|_{\frac{F_B}{F_A}=K} = 1$$

$$\left.\frac{\partial C_T}{\partial m_1}\right|_{c=k} = \frac{\partial (F_A + F_B)}{\partial F_A} = 1+K \qquad (13)$$

The relative gain of the flow controller connected to valve 1 is the ratio of the numerator and denominator calculated for Eq.(13). Or:

$$\lambda_\eta = 1/(1+k)$$

Here it is apparent that the relative gain is not a constant but is a variable, dependent on the value of k.

If F_B Is Equal to F_A .Then k is equal to 1.0 the flows will be equal at only one particular setpoint for the composition controller if feed compositions are constant .For this particular setpoint .The relative gain is 0.5, which is worst interaction possible for the blending example. A relative gain of 1.0 which occurs when the numerator is equal to the denominator is ideal: without interaction this ideal case will occur when none of the other controllers in the plant has any effect on the prospective control loop, the flow controller in our example will have a relative gain of 1.0 when the flow through valve 1 is much larger than flow through valve 2.

In this analysis it has been assumed that no other valves in the plant affect the total flow. Other loops can interact if they cause variation in feed composition or in valve pressured-drops .these effects must be considered in evaluating the denominator for

the relative gain. If they are neglected .The relative gain will not reflect these potentially troublesome interactions.

If the relative gain for a controller connected to either one of two valves is equal to 0.5. Interaction is maximum the controller connected to either valve will work equally well or equally bad .If one pairing gives a relative gain of 0.8. the opposite pairing would give gain of 0.2. The 0.8 combination would be the preferred controller arrangement. But some interaction would still exist.

For two valves and two controllers .The relative–gain array 2x2 matrix .Here the sum of the columns and the sum of the rows is equal to 1.0 .For a few processes. The relative gain is actually negative. The general matrix is:

	M_1	M_2	
C_1	λ_{11}	$1-\lambda_{11}$	(14)
C_2	$1-\lambda_{11}$	λ_{11}	

This property of Eq. (14) helps to simplify calculations because only one relative gain. λ_{11}. In Eq (14) needs to be evaluated in order to determine the interaction of two loops.

Implicit Decoupling

It would seem logical that a decoupler be required to handle two interacting loops with relative gains of 0.5 however, its use may not be justified even in the case of maximum interaction. If the blending process does not have upsets or setpoint changes .Both the flow and composition controllers would eventually settle out and an extra decoupling device would not be needed to achieve stability .Also the consequences of an occasional upset or setpoint change might not be sufficient to justify a decoupler.

Even if the interaction mechanism is well understood .The vast majority of the interacting loops do not have decouplers because a large number of less-costly and easier alternatives often exist.

When a controllers have been assigned to particular valves. Tuning can have a dramatic effect on the amount of interaction. The blending example of the fig.5a shows implicit decoupling: controller tuning can have a decisive impact on its effectiveness.

Typically, it is more important to keep composition under control. If the composition measurement is fast-responding (density conductivity or infrared analysis) it would be possible to tune the composition controller tightly. This would be done using techniques discussed earlier in the series expect that the flow controller should be in manual while the composition controller is tuned. (See part 3 of this section on process automation .p.155.) The flow controller should be tuned sluggishly (wider or larger proportional band). And the integral (reset) time should be longer.

Tuned in this manner .The flow controller takes the brunt of an upset, while composition control remains nearly as good as could be achieved in a non-interacting process.

If the composition measurement is flow (e.g. a process chromatograph) or a fast measurement is used having a long time delay associated with the sampling system. The composition controller will be slow with a long period in this case, the flow controller can be tuned tightly. The composition control will be slow, but it would be nearly as slow without interaction.

Generally, if two interacting loops oscillate at different frequencies, their controllers should be tuned to further separate these frequencies-thus minimizing interaction. Two interacting loops oscillating at the same frequency can always be separated somewhat by tuning furthermore .frequency separation can also be accomplished by changing the deadtimes or lages in the process. Sometimes deading can be reduced inexpensive by moving the measurement location, changing the sampling system for the measurement .Or moving the control valve. If surge tanks are used to smooth flows or compositions. Then the associated lag can be used to stabilize one or both of the interacting loops.

Another way to help reduce interaction is to prevent upsets from reaching the process. Feedforward control could be used to catch these upsets. Also, some upsets can be scheduled or can be reduced by changing operating procedures at the origin of the upset.

In the home feedforward scheduling and changing operating procedures are used to reduce interaction .for example. The shower water becomes scalding hot whenever cold water is used at the kitchen sink .to overcome such an interaction. The temperature loop can be tuned faster .the use of the sink can be delaved until after the faster. The use of the sink can be delaved until after the shower .the piping to the sink can be restricted the thermostat on the water heater can be tuned down, or a temperature controlling shower valve installed .all of these techniques .expect the last. Are examples of implicit decoupling.

Explicit Decoupling

Implicit decoupling techniques such as previously described are not always practical or effective way of removing troublesome interaction. The temperature controlling shower valve is an effective decoupler with independent adjustment of flow and temperature .many building codes require them for new construction. The decoupling shower valve is effective because it (1) works, (2) is economical to build and install, and (3) is comprehensible to the operator .these three requirements are prerequisites for a successful explicit decoupler.

The explicit decoupler shown in fig .5b is designed to allow the flow controller to adjust the total flow, Y, and the composition controller to adjust the ratio of flows. X, linear installed valve characteristics are assumed, and the dynamics can be neglected if both valves have the same time constant and are very close to the blending junction. The factor K in the fig .5b has been added to equalize valve size.

The flow controller output Y is proportional to the total flow.$V_1 - V_2$:

$$V_1 + V_2 = \frac{Y}{X+1} + \frac{XY}{X+1} = \frac{Y(1+X)}{X+1} = Y \qquad (15)$$

The Composition controller output X is proportional to the ratio of flowrates. V_2/V_1:

$$\frac{V_2}{V_1} = \frac{\frac{XY}{X+1}}{\frac{Y}{X+1}} = X \qquad (16)$$

This Coupling Scheme May Meet The First Two Requirement of an effective decoupler, but the operator interface may be confusing. if a standard controller display is used. the controller output would not be representative of the valve position and a separate indicator display would have to be used.

If direct manual control of the valves is required .An automatic/manual (auto/manual) device could be inserted between the control valves and the decoupler.

The operator could use the auto/manual station in manual mode to stroke the valves to any desired position.

However, upon return to automatic mode, a complicated balancing procedure would be needed to avoid bumping the valve position .if this trial and error balancing procedure must be done often. The operator would probably find the decoupler difficult to use.

Fig.5c shows the decouple system with an improved operator interface and automatic balancing. If an auto manual station is put in manual, a logic signal switches the other auto /manual station into manual, and puts both of the feedback controllers into track this allows the operator to stroke one valve with the other valve fixed.

Upon return to automatic mode, bumbles transfer can be achieved because the controller reset feedback is back calculated. With both controllers in the track mode .A positive feedback loop is established and the possibility of an unstable feedback calculation exists. A first order lag with a setting of perhaps 0.1 min could be inserted downstream of either decoupler (see fig .5c) to ensure stability.

The cost and complexity of the explicit decoupler is often increased in order to provide an acceptable operator interface. The operator need not know how to start. Operate, and shut it off .If the operator is comfortable with the decoupler interface and if the decoupler works .then the operator will view it as a way to achieve improved operation.

MODULE 5:

DISTRIBUTED CONTROL SYSTEMS (DCS)

DISTRIBUTED CONTROL SYSTEMS (DCS)

Introduction & and Historical Background

The history of man's attempts to control industrial processes through automatic means is a long one (see Refs. l.l and 1.5), starting with such early developments as Cornelis Drebbel's furnace thermostat (1620) and James Watt's centrifugal governor for steam engines (1788). However, the major advances in integrated control system architectures, as compared to individual controllers, have taken place over the last fifty years. This section reviews several key developments during these years to provide the rationale for the recent emergence of the distributed control system architecture. The references at the end of the chapter provide additional historical detail.

Control systems have developed from the 1930s to the present day in response to two intertwined influences: user needs and technological advances. One factor that has influenced the control needs of the user is the continual growth in the size and complexity of industrial processes over the past fifty years. Also, the costs of raw materials and energy required to process these materials have increased substantially in this time. Finally, the labor costs involved in plant startup, operation, and maintenance have grown substantially. These influences have motivated the owners and operators of industrial processes to place a greater amount of emphasis on automation and on efforts to optimize their operations. In response to these user needs, the suppliers of industrial controls have been motivated to develop totally integrated plant management systems that are more than the combination of individual control, monitoring, and data logging systems.

Fortunately, the explosive advances in technology that have taken place over the past fifty years have provided the capabilities needed for the evolution of such plant management systems. For example, the development of transistors, integrated analog circuits, and solid-state relays resulted in a growth in capability and an increase in reliability of electronic control systems that enabled them to largely replace pneumatic control systems. Similarly, the development of digital technology in the form of improved large-scale integrated logic circuits, microprocessors, semiconductor memories, and cathode-ray tube (CRT) displays has led to even more impressive improvements in digital control system capabilities. These improvements have allowed control systems based on digital technology to replace electronic analog systems in many applications. References 1.2, 1.3, 1.4, and 1.6 trace the history of many of these technological developments.

The lines of technological development can be divided into two separate streams, as illustrated in Figure 1.1. The upper stream with its two branches is the more traditional one, and includes the evolution of analog controllers and other discrete devices such as relay logic and motor controllers. The second stream is a more recent one that includes the use of large-scale digital computers and their mini and micro descendants in industrial process control. These streams have merged into the current mainstream of distributed digital control systems. The dates of several key milestones in this evolutionary process are shown in Table 1.1 to illustrate the pace of these advances.

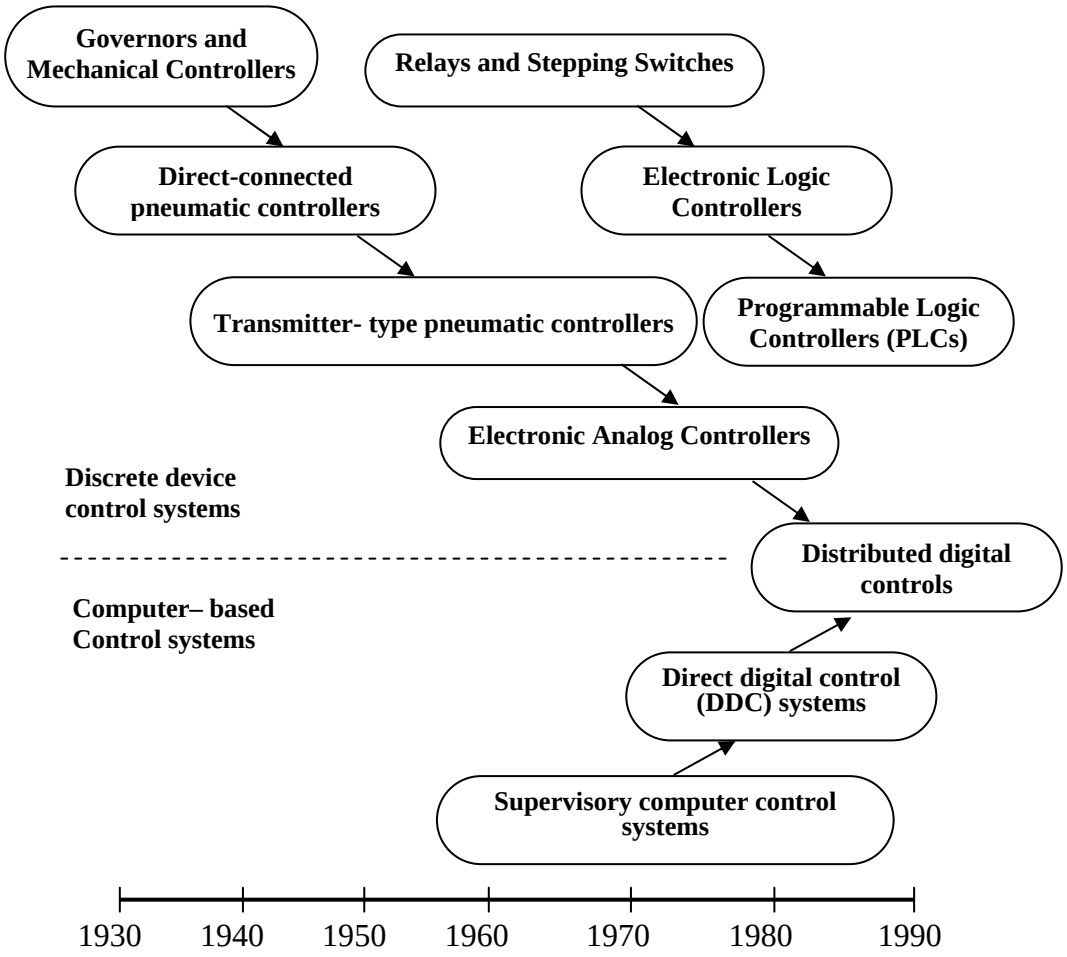


Table 1.1. Key Milestones in Control System Evolution

1934 Direct-connected pneumatic controls dominate market.

1938 Transmitter-type pneumatic control systems emerge, making centralized control rooms possible.

1958 First computer monitoring in electric utility.

1959 First supervisory computer in refinery.

1960 First solid-state electronic controllers on market. 1963 First direct digital control (DDC) system installed. 1970 First programmable logic controllers (PLCs) on market. 1970 Sales of electronic controllers surpass pneumatic.

1975 First distributed digital control system on market.

Traditional Control System Developments

The concept of distributed control systems is not a new one. In fact, the early discrete device control systems listed in Figure 1.1 were distributed around the plant. Individual control devices such as governors and mechanical controllers were located at the process equipment to be controlled. Local readouts of set points and control outputs were available, and a means for changing the control mode from manual to automatic (or vice versa) usually was provided. It was up to the operator (actually, several operators) to coordinate the control of the many devices that made up the total process.

They did this by roaming around the plant and making corrections to the control devices as needed and using the "Hey, Joe!" method of communications to integrate plant operations. This was a feasible approach to the control of early industrial processes because the plants were small geographically and the processes were not too large or complex. The same architecture was copied when direct-connected pneumatic controllers were developed in the late 1920s. These controllers provided more flexibility in selection and adjustment of the control algorithms, but all of the elements of the control loop (sensor, controller, operator interface, and output actuator) were still located in the field. There was no mechanism for communication between controllers other than that provided by each operator to other operators in the plant using visual and vocal means.

This situation changed of necessity in the late 1930s due to the growth in size and complexity of the processes to be controlled. It became more and more difficult to run a plant using the isolated-loop control architecture described above. The emphasis on improving overall plant operations led to a movement towards centralized control and equipment rooms. This was made possible by the development of transmitter-type pneumatic systems. In this architecture, measurements made at the process were converted to pneumatic signals at standard

levels, which were then transmitted to the central location. The required control signals were computed at this location, and then transmitted back to the actuating devices at the process. The great advantage of this architecture was that all of the process information was available to the operator at the central location. Thus, the operator was able to make better control decisions and operate the plant with a greater degree of safety and economic return.

The centralized control structure described above is still the dominant one in plants operating today. In the late 1950s and early 1960s, the technology used to implement this architecture started to shift from pneumatics to electronics. One of the key objectives of this shift was replacing the long runs of tubing used in pneumatic systems with the wires used in electronic ones. This change reduced the cost of installing the control systems and also eliminated the time lag inherent in pneumatic systems. Both of these advantages became more significant as plant sizes increased. Another consequence of the centralized control architecture was the development of the split controller structure. In this type of controller, the operator display section of the controller is panel mounted in the control room and the computing section is located in a separate rack in an adjoining equipment room. The split controller structure is especially appropriate for complex, interactive control systems (e.g., for boiler controls) in which the number of computing elements greatly exceeds the number of operator display elements.

Both pneumatic and electronic versions of the centralized control architecture still exist and are being sold, delivered, and operated today. In fact, it was not until 1970 that the sales of electronic controllers exceeded the sales of pneumatic controllers in the industrial process control marketplace (1.4).

The discussion to this point has focused on continuous, or analog, control devices, in which both the inputs and outputs to the controllers vary continuously over a selected range (e.g., 1-5 volts or 3-15 psi). Similar developments have taken place in the realm of sequential logic control devices, in which the inputs and outputs to the controllers take on only one of two discrete states (e.g., On/Off, 0/24 volts). These devices generally are used in controlling certain types of pumps, motors, or valves in a process. They also are used in safety override systems that operate in parallel to and back up the continuous systems described above. The original versions of these logic systems were implemented using simple electronic devices such as relays and stepping switches. Later, the development of solid-state electronic modules allowed logic systems to be implemented using the same level of technology as the corresponding electronic analog controllers.

In the early 1970s, a sophisticated device known as the programmable logic controller (PLC) was developed to implement sequential logic systems. This device is significant because it was one of the first special-purpose, computer-based devices that could be used by someone who was not a computer specialist. It was designed to be programmed by a user who was familiar with relay logic diagrams but was not necessarily a computer programmer. This approach to control system configuration was inspired by early efforts of process computer specialists to develop a process-oriented control language. However, it was more successful than most of these efforts in eliminating the user's dependence on a priesthood of computer specialists in running a process control system.

All of the versions of sequential logic systems described above have been implemented in direct-connected distributed architectures as well as in centralized ones. In each case, the logic controller has been associated directly with the corresponding unit of process equipment, with little or no communication between it and other logic controllers. It was not until the late 1970s that PLCs and computers started to be connected together in integrated systems for factory automation. For

the purposes of this book, the PLC and networks of PLCs are considered to be special cases of the general distributed control system architecture described later in this chapter.

Computer-based Control System Developments

In addition to the evolution of the traditional types of-control systems described above, a more recent (and equally important) evolution of computer-based process control systems has been taking place, as shown in the lower part of Figure I.I. The first application of computers to industrial processes was in the areas of plant monitoring and supervisory control.

In September 1958, the first industrial computer system for plant monitoring was installed at an electric utility power generating station (1.3). This innovation provided an automatic data acquisition capability not available before, and freed the operator from much drudgery by automatically logging plant operating conditions on a periodic basis. Shortly thereafter (in 1959 and 1960), supervisory computer control systems were installed in a refinery and in a chemical plant (1.4). In these applications, analog controllers were still the primary means of control. The computer used the available input data to calculate control set points that corresponded to the most efficient plant operating conditions. These set points then were sent to the analog controllers, which performed the actual closed-loop control. The ability of supervisory control computers to perform economic optimization as well as to acquire, display, and log plant data provided the operator with a powerful tool for significantly improving plant operations.

The next step in the evolution of computer process control was the use of the computer in the primary control loop itself, in a mode usually known as direct digital control, or DDC. In this approach, process measurements are read by the computer directly, the computer calculates the proper control outputs, then sends the outputs directly to the actuation devices. The first DDC system was installed in

1963 in a petrochemical plant (1.3). For security, a backup analog control system was provided to ensure that the process could be run automatically in the event of a computer failure. This proved to be a wise precaution, because this early DDC installation (as well as many others) was plagued with computer hardware reliability problems. Despite these problems, it demonstrated many of the advantages digital control has over analog control: tuning parameters and set points do not drift, complex control algorithms can be implemented to improve plant operation, and control loop tuning parameters can be set adaptively to track changing operating conditions.

Resulting System Architectures

As a result of the developments described above, two industrial control system architectures came to dominate the scene by the end of the 1970s. While there are many variations, typical examples of these architectures are shown in Figures 1.2 and 1.3. The first architecture is a hybrid one, making use of a combination of discrete control hardware and computer hardware in a central location to implement the required control functions. In this approach, first level or local control of the plant unit operations is implemented by using discrete analog and sequential logic controllers (or PLCs). Panel board instrumentation connected to these controllers is used for operator interfacing and is located in the central control room area. A supervisory computer and associated data acquisition system are used to implement the plant management functions, including operating point optimization, alarming, data logging, and historical data storage and retrieval. The computer also is used to drive its own operator interface, usually consisting of one or more video display units (VDUs). A substantial amount of interfacing hardware is required to tie the analog and sequential control equipment to each other as well as to the supervisory computer. The other dominant architecture, shown in Figure 1.3, is one in which all system functions are implemented in high-performance computer hardware in a central location. In general, redundant computers are required so that the failure of a single computer does not shut the whole process down. Operator interfacing for

plant management functions is provided using computer-driven VDUs, just as in the hybrid control system architecture described above. Operator interfacing for first-level continuous and sequential closed-loop control also may be implemented using VDUs. Optionally, the computers can be interfaced to standard panel board instrumentation so that the operator in charge of first-level control can use a more familiar set of control and display hardware.

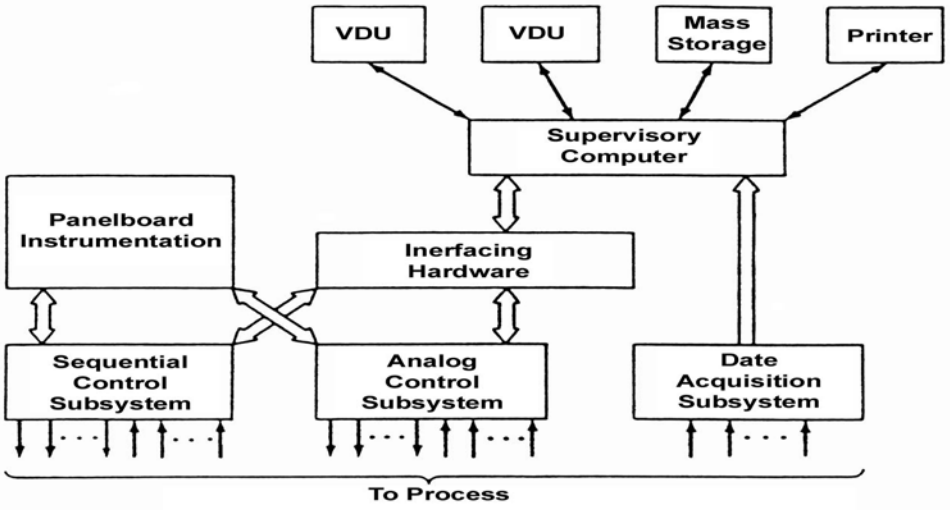


Figure 1.2. Hybrid System Architecture.

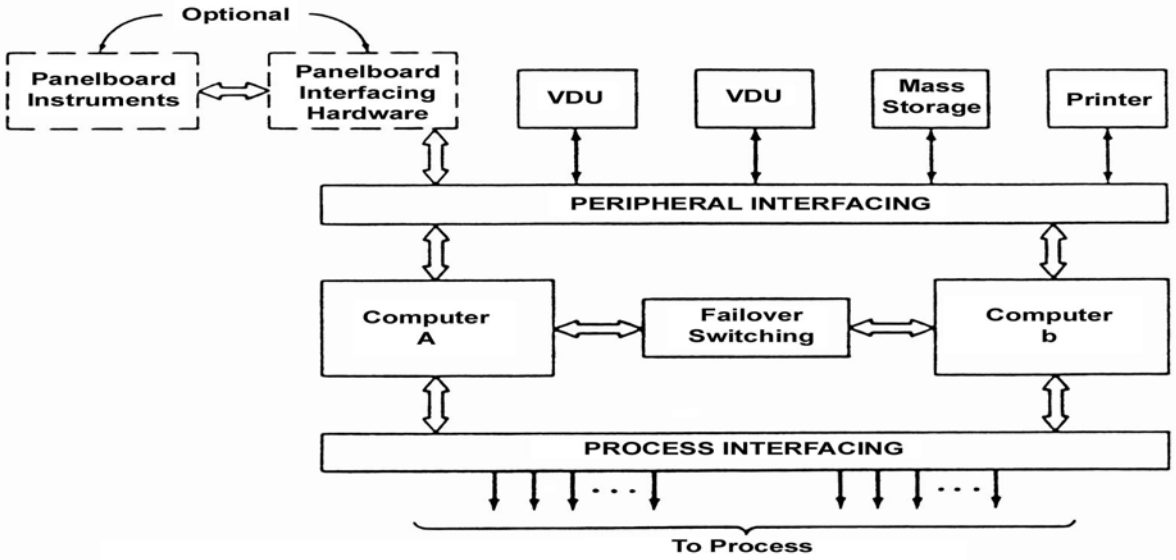


Figure 1.3. Central Computer System Architecture.

Note that both of the above systems use computers. The main difference between the two systems is the location of the implementation of the first-level continuous and sequential logic control functions. By the late 1970s, the hybrid system became by far the more prevalent approach in industrial control practice. The chemical and petroleum process industries heavily favored this approach, perhaps as a result of their disappointing experiences using early versions of direct digital control systems. In contrast, the use of large centralized computer systems to implement almost all plant control functions was limited primarily to the electric utility industry.

Emergence of the Distributed Control System Architecture

While the central computer and hybrid system architectures provide significant advantages over earlier ones, they also suffer from a number of disadvantages. The biggest disadvantage of the centralized computer control architecture is that the central processing unit (CPU) represents a single point of failure that can shut down the entire process if it is lost. Since early industrial computer hardware was notoriously unreliable, two approaches were developed and have been used to attack the reliability problem: either a complete analog control system is used to back up the computer system, or another computer is used as a "hot standby" to take over if the primary control computer fails. Either approach results in a system significantly more expensive than an analog control system that performs a comparable set of functions.

Another problem with these computer-based systems has been that the software required to implement all of the functions is extremely complex, and requires a priesthood of computer experts to develop the system, start it up, and keep it running. This is the natural result of an architecture in which a single CPU is required to perform a variety of functions in real time: input scanning; database updating; control algorithm computation; logging, long-term storage and retrieval of data; and man-machine interfacing (among others). Finally, the centralized system is

limited in its capability to accommodate change and expansion. Once the loading on the computer approaches its limit, it becomes very difficult to add on to the system without a significant decrease in performance or increase in cost.

The hybrid system architecture of Figure 1.2 also has its deficiencies. One of the worst is simply that it is composed of many different subsystems, often manufactured by different vendors. Just interfacing the subsystems to one another is a significant challenge, given the variety of different signal levels and conventions that exists in each. Starting them up and making them work as an integrated whole is no less difficult a task. The hybrid approach also is functionally limited compared to the central computer-based system. The benefits of digital control outlined in Section 1.1.2 are lost, since the closed-loop control is done by discrete analog and sequential devices. Also, the speed and accuracy of plant performance computations suffer due to the limitations of the analog input equipment and the problems in accessing the database, which is no longer centralized as in the computer implementation approach.

Because of these problems, it became clear to both users and system designers that a new architectural approach was needed. Control system engineers had been sketching out concepts of distributed systems composed of digital control and communication elements since the middle 1960s. Unfortunately, the technology to implement these concepts in a cost-effective manner was not available at that time. It was not until the microprocessor was introduced in 1971 that the distributed system architecture became practical. Supporting technology also became available during the early 1970s: inexpensive solid-state memories were developed to replace magnetic core memories; integrated circuit chips to implement standard communication protocols were introduced; display system technology flourished with the emergence of light-emitting diode (LED) and color CRT displays; in the software area, structured design techniques, modular software packages, and new on-line diagnostic concepts were developed.

The result of this fortunate confluence of user needs and technological developments was the introduction of a large number of distributed digital control system product lines by vendors in the late 1970s and early 1980s. (See Refs. 1.7-1.12 for tutorial information on distributed control systems and 1.13-1.26 for a review of the development of these systems). While each system has a unique structure and specialized features, the architectures of most of these systems can be described in the context of the generalized one shown in Figure 1.4. The devices in this architecture are grouped into three categories: those that interface directly to the process to be controlled or monitored, those that perform high-level human interfacing and computing functions, and those that provide the means of communication between the other devices. A brief definition of each device is given below, and this terminology is used throughout the book:

1. **Local Control Unit (LCU)—**The smallest collection of hardware in the system that can do closed-loop control. The LCU interfaces directly to the process.

2. **Low-level Human Interface (LLHI)**—A device that allows the operator or instrument engineer to interact with the local control unit (e.g., to change set points, control modes, control configurations, or tuning parameters) using a direct connection. LLHIs can also interface directly to the process. Operator-oriented hardware at this level is called a low-level operator interface; instrument engineer-oriented hardware is called a low-level engineering interface.

3. **Data Input/output Unit (DI/OU)—**A device that interfaces to the process solely for the purpose of acquiring or outputting data. It performs no control functions.

4. **High-level Human Interface (HLHI)—**A collection of hardware that performs functions similar to the LLHI but with increased capability and user

friendliness. It interfaces to other devices only over the shared communication facilities. Operator-oriented hardware at this level is called a high-level operator interface; instrument engineer-oriented hardware is called a high-level engineering interface.

5. **High-level Computing Device (HLCD)—** A collection of microprocessor-based hardware that performs plant management functions traditionally performed by a plant computer. It interfaces to other devices only over the shared communication facilities.

6. **Computer Interface Device (CID)—** A collection of hardware that allows an external general-purpose computer to interact with other devices in the distributed control system using the shared communication facilities.

7. **Shared Communication Facilities—** One or more levels of communication hardware and associated software that allow the sharing of data among all devices in the distributed system. Shared communication facilities do not include dedicated communication channels between specific devices or between hardware elements within a device.

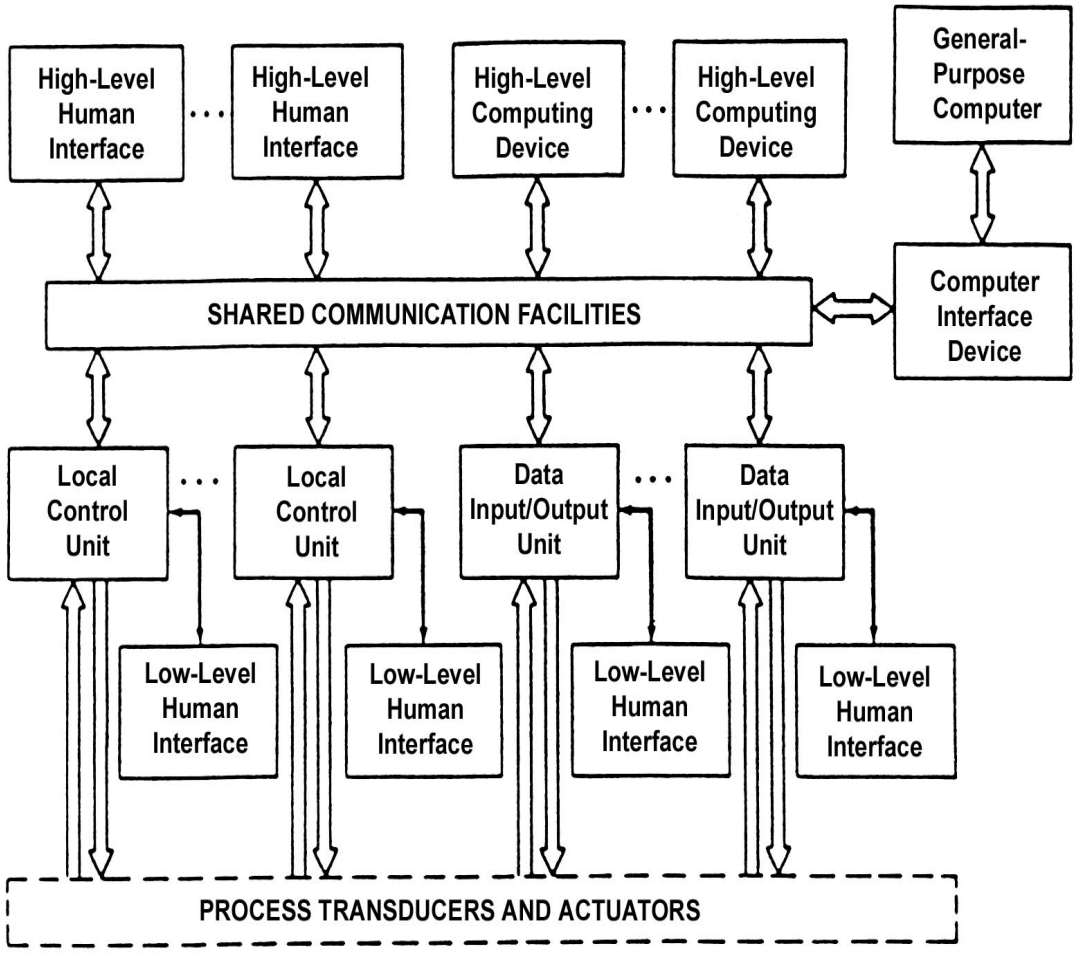


Figure 1.4 Generalized Distributed Control System Architecture.

Not included in the architecture in Figure 1.4 but of vital importance to the design of a distributed control system are the packaging and electrical power systems.

Detailed descriptions of the seven distributed control system elements mentioned above and a discussion of the major issues involved in selecting, using, and designing these elements form the bulk of the remaining chapters in this book, as follows:

1. **Local Control Unit**—Architectural and hardware issues are discussed in Chapter 2; software and language issues are covered in Chapter 3; and security design issues are discussed in Chapter 4.

2. **Low-level Human Interface**—The low-level operator interface is discussed in Chapter 6; and the low-level engineering interface is covered in Chapter 7.

3. **Data Input/output Unit**—Many design issues overlap with local control unit discussions and are found in Chapters 2 and 3; specific process input/output design issues are covered in Chapter 4.

4. **High-level Human Interface**—The high-level operator interface is discussed in Chapter 6, while Chapter 7 covers engineering interface.

5. **High-level Computing Device**—Discussed in Chapter 8.

6. **Computer Interface Device**—Discussed in Chapter 8.

7. **Packaging and Power Systems**—Discussed in Chapter 8.

COMPARISON WITH PREVIOUS ARCHITECTURES

One of the main objectives in the development of distributed control systems has been to maintain the best features of the central computer control and hybrid architectures described in the previous section. Most importantly, the new systems have been structured to combine the power and flexibility of digital control with the user-oriented familiarity of the traditional analog and sequential control systems. A summary of some of the key features of distributed control systems compared to previous ones is given in Table 1.2, and additional information on the architectural advantages and disadvantages of distributed systems is provided in references (1.27-1.36). The following discussion of these features expands upon the table:

1. **Scalability and Expandability**—Refers to the ease with which a system can be sized for a spectrum of applications, ranging from small to large, and the ease with which elements can be added to the system after initial installation. The hybrid system is quite modular, so it ranks high on both counts; the same holds for the distributed system architecture. On the other hand, the central computer architecture is designed for only a small range of applications. It is not cost-effective for applications much smaller than its design size and it cannot be expanded easily once its memory and performance limits are reached.

2. **Control Capability**—Refers to the power and flexibility of the control algorithms that can be implemented by the system. The capability of the hybrid architecture is limited by the functions available in the hardware modules that make up the system. To add a function involves both adding hardware and rewiring the control system. On the other hand, central computer and distributed architectures both provide the full advantages of digital control: drift less set points and tuning parameters, availability of complex control algorithms, ability to change algorithms without changing hardware, remote and adaptive tuning capabilities, and many others.

3. **Operator Interfacing Capability**—Refers to the capability of the hardware provided to aid the operator in performing plant monitoring and control functions. The operator interface in the hybrid system consists of conventional panel board instrumentation for normal control and monitoring functions and a separate video display unit

Table 1.2 Comparison of Architectures

FEATURE	HYBRID ARCHITECTURE	CENTRAL COMPUER ARCHITECTURE	Distributed Architecture
1- scalability and expandability	Good due to modularity	Poor- very limited range of system size	Good due to modularity
2- control capability	Limited by analog and sequential control hardware	Full digital control capability	Full digital control capability
3- Operator interfacing capability	Limited by panel board instrumentation	Digital hardware provides significant improvement for large systems	Digital hardware provides improvement for full range system sizes
4- Integration of system functions	Poor due to variety of products	All functions performed by central computer	Functions integrated in a family of products
5- Significance of single-point failure	Low due to modularity	High	Low due to modularity
6- Installation costs	High due to discrete wiring and large volume of equipment	Medium-saves control room and equipment room space but uses discrete wiring	Low- savings in both wiring costs and equipment space
7- Maintainability	Poor-many module types, few diagnostics	Medium – requires highly trained computer maintenance personnel	Excellent – automatic diagnostics and module replacement

(VDU) for Supervisory Control. In the central computer and distributed architectures, VDUs generally are used as the primary operator interface for both the normal and supervisory control functions. The VDUs provide significant benefits to the operator: reduction in time needed to access control stations, flexibility of station grouping, graphics displays that mimic the process layout, and others. Since the VDUs in the distributed system are driven by microprocessors rather than by a large computer, they can be applied in a cost-effective way to small systems as well as large ones.

4. **Integration of System Functions**—Refers to the degree with which the various functional subsystems are designed to work with one another in an integrated fashion. A high degree of integration minimizes user problems in procuring, interfacing, starting up and maintaining the system. Since the hybrid system is composed of a variety of individual product lines, it is usually poorly integrated. The central computer architecture is well integrated because all of the functions are performed by the same hardware. The distributed system lies somewhere in between, depending on how well the products that make it up are designed to work together. (There are both good and bad examples of system integration out on the market today.)

5. **Significance of Single-Point Failure**—Refers to the sensitivity of the system's performance to a failure of any of its elements. In the central computer architecture, the failure of a hardware element in the computer can cause the system to stop performing completely unless a backup computer is used. Therefore, this system is very sensitive to single-point failures. On the other hand, both the hybrid and distributed architectures are relatively insensitive to single-point failures due to the modularity of their structure.

6. **Installation Costs**—Refers to the cost of system wiring and the cost of control room and equipment room space needed to house the system. The installation costs of the hybrid system are high: much custom wiring is needed for internal system interconnections; long wiring runs are needed from sensors to control cabinets; much control room space is required to house the panelboard instrumentation; and the large volume of control modules needed to implement the system take up a lot of equipment room space. The central computer architecture cuts down on this cost by eliminating the module inter-connection wiring and by using VDUs to replace much of the panel-board instrumentation. The distributed system reduces costs further by using a communication system to replace the sensor wiring runs and by reducing required equipment room space through the use of space-efficient microprocessor-based modules.

7. **Maintainability**—Refers to the ease with which a system can be kept running after installation. Low maintainability implies high maintenance costs, including the cost of spares, costs of process downtime while repairs are being made, and personnel training costs. The hybrid system is particularly poor in this area because of the large number of spare modules required, the lack of failure diagnostics in the system, and the personnel training required to cover the diverse subsystems. The central computer architecture is somewhat better: the range of module types is reduced and a certain number of failure diagnostics are provided. However, relatively sophisticated personnel are required to maintain the complex computer hardware and software. On the other hand, the maintainability of the distributed system architecture is excellent. Since there are only a few general-purpose control modules in the system, spare parts and personnel training requirements are minimal. Automatic on-line diagnostics are available to isolate failures to the module level, and module replacements can be made without disrupting a major portion of the process.

It can be seen from the table that the distributed control system architecture provides the user with many benefits over the hybrid and central computer architectures. The comparison is not all one-sided, however; as with any new venture, moving from a conventional analog control system to a distributed one requires the user to deal with a number of potential difficulties and changes in operation. One of the most obvious changes is that a microprocessor-based control system represents a new technology that plant personnel must learn. A certain amount of retraining of operating, instrument and maintenance people is required to ensure the success of any first installation of a distributed control system in a plant. Operating procedures will change; the operators will be spending a greater percentage of their time monitoring the process from the control room than patrolling the plant. When in the control room, they will be running the process from a video display unit instead of from panelboard instrumentation. During the early introduction of VDUs to the control room, the switch was expected to be traumatic for the operators. However, the transition turned out to be relatively painless; whether this was due to an underestimation of the adaptability of humans or to the pertaining effects of video games and home computers is not clear.

The new distributed systems offer the user a tremendous amount of flexibility in choice of control algorithms and location of equipment in the plant. While this is an advantage in most ways, it also requires that the user plan the installation carefully so that the control system is partitioned properly and that there is appropriate space and protection in the remote locations for the control hardware. These decisions must be well documented so that the installation and startup process proceeds smoothly.

When partitioning the control strategy, the user must be aware of the consequences of the various processing and communication delays that are inherent in a distributed control system. While the rapid advances in digital system hardware are

fast making these delays negligible in most situations, the user must be aware of the needs of his or her particular application. If the control system is distributed geographically as well as functionally, the user must make sure that in the remote locations the installed hardware can survive the environment and the proper backup hardware is provided to accommodate any equipment failures.

The above comparisons and design considerations only begin to cover the issues involved in evaluating and designing distributed digital control systems. More detailed comparisons and design discussions on specific technical issues will be provided later in the book. References 1.37-1.44 contain additional information on selecting and evaluating distributed control systems.

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COMMUNICATION FACILITIES

COMMUNICATION FACILITIES

Introduction

In conventional nondistributed control systems, the connections that allow communication between the various system elements are configured on a per-job basis. Figure 5.1 shows an example of this for the case of a hybrid system, such as that described in Chapter 1 (see Figure 1.2 and Section 1.1). This system consists of a combination of continuous controllers, sequential controllers, data acquisition hardware, panelboard instrumentation, and a computer system. The controllers communicate with each other by means of point-to-point wiring, usually within the control cabinets. This custom wiring reflects the particular control system configuration selected. The controllers are connected to the corresponding panelboard instrumentation and to the computer system by means of prefabricated cables. The computer obtains information from the data acquisition modules using similar hard wiring or cabling that is specific to the particular module configuration implemented.

This approach to interconnecting system elements has proven to be expensive to design and check out, difficult to change, burdensome to document, and subject to errors (see References 5.1-5.2). It becomes even more cumbersome if the system elements are distributed geographically around the plant. The first step taken to improve this situation was to introduce the concept of distributed multiplexing in the early 1970s (see References 5.3-5.4). This concept was first used in the process control industry to implement large-scale data acquisition systems, which at that time had grown to several thousand inputs in size. To reduce the cost of wiring, remote multiplexers located near the sensors in the field were used to convert the inputs to digital form and transmit them back to the data acquisition computer over a shared communication system.

When distributed control systems were introduced in the late 1970s, the use of digital communications was extended to control-oriented systems as well. The communication system began to be viewed as a facility that the various elements and devices in the distributed network share, as the "black box" representation in Figure 5.2 shows. Replacing dedicated point-to-point wiring and cabling with this communications facility provides a considerable number of benefits to the user:

1. The cost of plant wiring is reduced significantly (see References 5.5-5.7 for analyses), since thousands of wires are replaced by the few cables or buses used to implement the shared communication system.

2. The flexibility of making changes increases, since it is the software or firmware configurations in the system elements that define the data interconnection paths and not hard wiring.

3. It takes less time to implement a large system, since the wiring labor is nearly eliminated, configuration errors are reduced, and less time is required to check out the interconnections.

4. The control system is more reliable due to the significant reduction in physical connections in the system (a major source of failures).

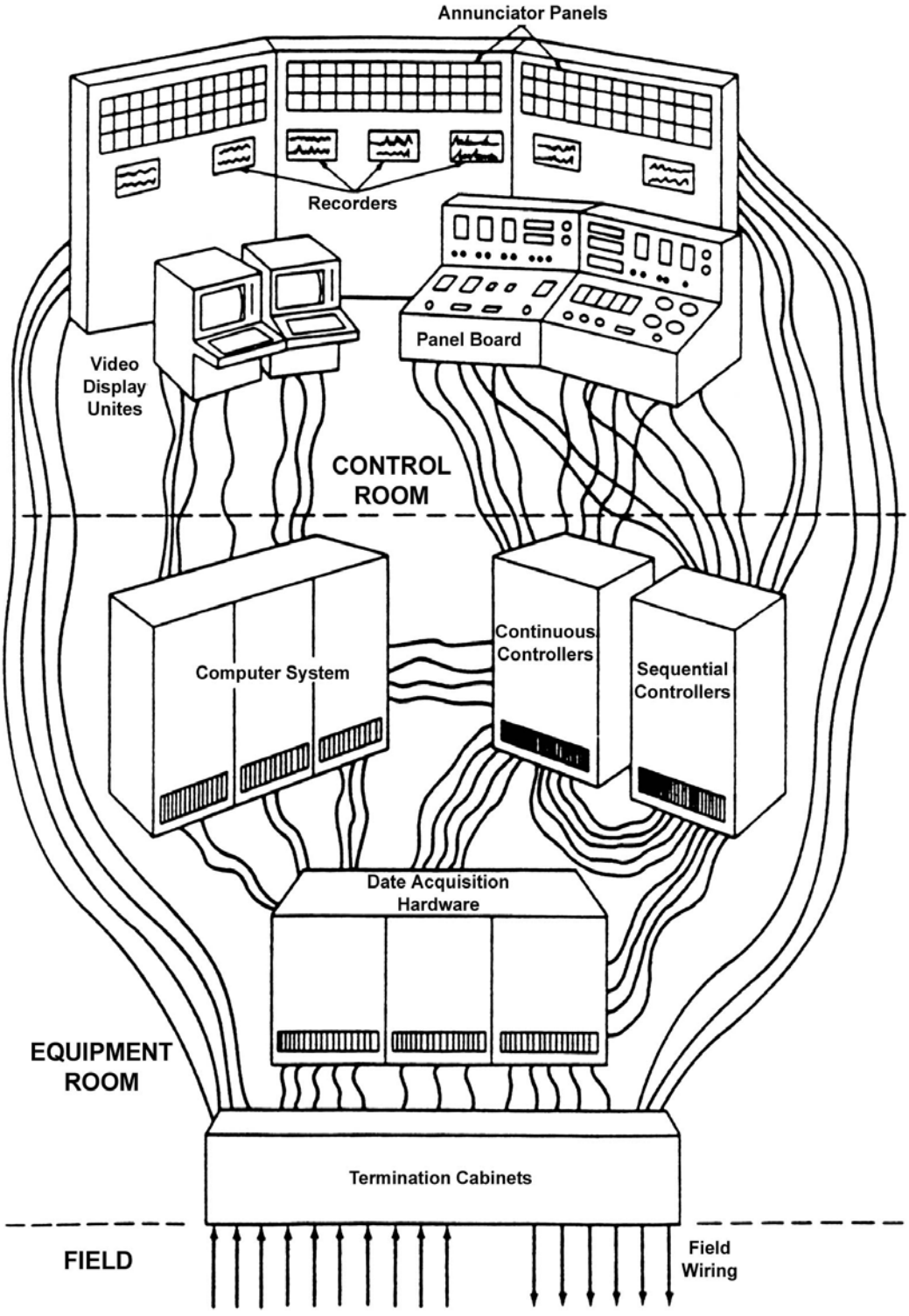

Figure 5.1 Conventional Point-to-Point Wiring

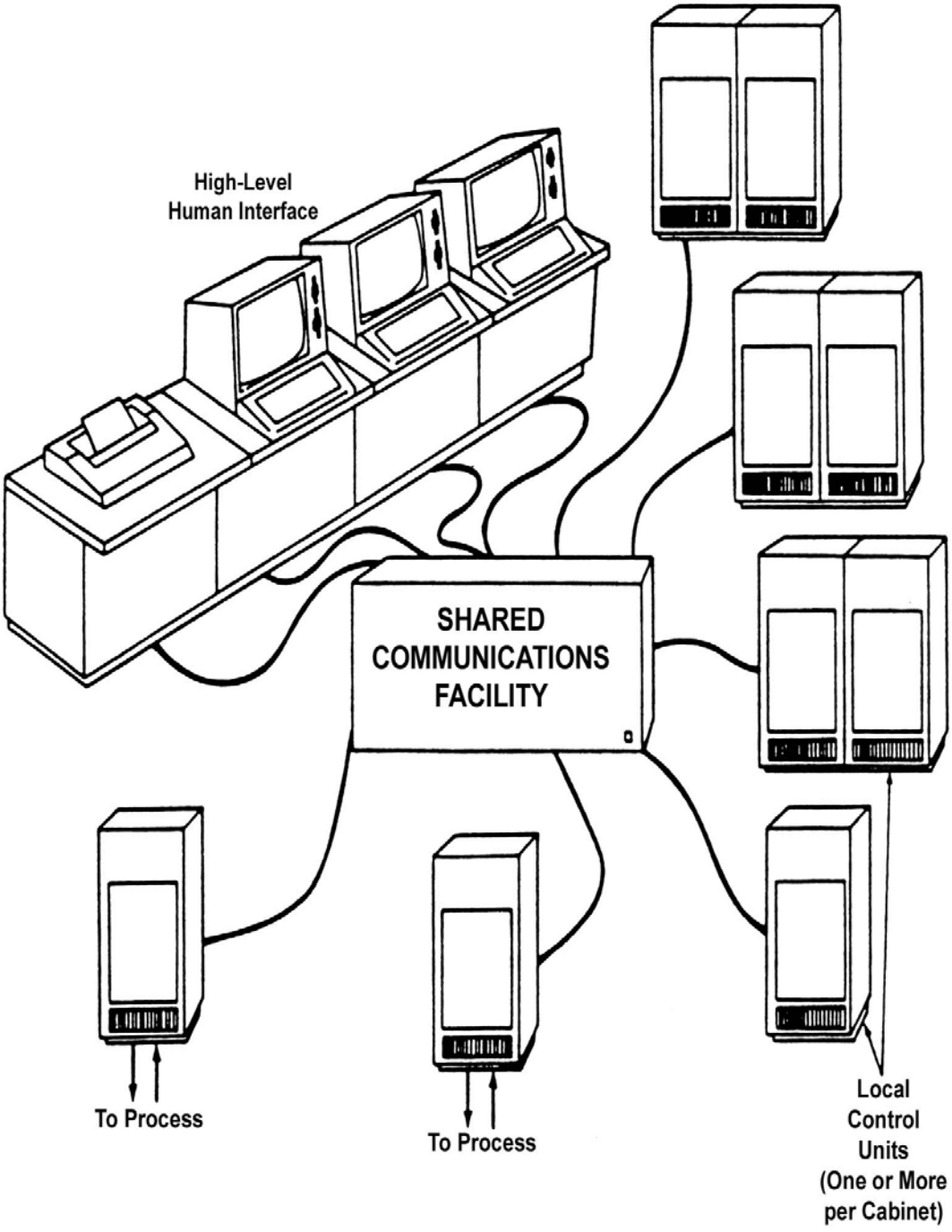

Figure 5.2. Communication Facility as a "Black Box"

However, replacing hard wiring with the shared communications network of a distributed control system also raises a number of questions for the user. In a conventional system, communications between system elements travel at the speed of light, essentially with zero delay. Also, since the hard-wired communication channels between elements are dedicated, there is no danger of overloading a channel. In the case of a shared communication system, the user must be able to judge whether the response time and capacity of the shared system (in addition to many other performance factors) are adequate for the application. This can be a difficult problem for the user, since there are significant differences and few standards among the various communication systems available on the market today.

The purpose of this chapter is to identify the key issues in evaluating and designing a shared communication system used in distributed control. Because of the vast scope of this subject, I have not attempted an exhaustive treatment of these issues. Rather, the discussion concentrates on functional requirements, major design tradeoffs, and critical features to consider. References 5.8-5.11 provide the reader with tutorial information on digital communication systems, while other references at the end of the chapter provide additional details on particular issues.

Section 5.2 lists the various functions implemented by the communication system and discusses the corresponding requirements on the performance of these functions. This section also summarizes the alternative design approaches to consider in evaluating a particular communications system. **Section 5.3** covers architectural issues and S**ection 5.4** deals with protocol issues. **Section 5.5** discusses several other issues that are relatively independent of architectural considerations. Finally, **Section 5.6** summarizes the status of recent attempts to develop standards for communication networks in distributed control systems.

COMMUNICATION SYSTEM REQUIREMENTS

As Section 5.1 just implied, the shared communications facility in a distributed control system must at least duplicate the functions previously implemented by the hard-wired connections in a conventional control system. In the context of the distributed control architecture shown in Figure 5.2. these wire-replacement functions include the following:

1. Transmission of control variables between local control units in the system. This is a requirement for all applications in which the control strategy requires multiple interacting controllers. To minimize delays and maximize security of transmission, the LCUs should be able to communicate directly with one another and not through an intermediary.
2. Transmission of process variables, control variables, and alarm status information from the LCUs to the high-level human interfaces and to the low-level human interfaces in the system (i.e., operator and engineer consoles and panelboard instrumentation).
3. Communication of set-point commands, operating modes, and control variables from the high-level computing devices and human interface devices to the LCUs for the purpose of supervisory control.

In addition to these wire-replacement functions, the shared communications facility also may implement functions more closely related to the distributed control architecture:

4. Downloading of control system configurations, tuning parameters, and user programs from the high-level human interfaces to the LCUs.
5. Transmission of information from the data input/output units to the high-level computing devices for purposes of data acquisition or transfer.
6. Transfer of large blocks of data (e.g., console displays, historical trends and

logs, or large data bases), programs, or control configurations from one high-level computing device or human interface to another.

7. Synchronization of real time among all of the elements in the distributed control system.

The shared facility can also implement other communication functions, such as transferring voice and video images. However, the current state of technology usually makes it more cost-effective to implement these functions using a separate, dedicated communications medium.

Once the set of functions to be implemented in the shared communications facility is established, the next step is to specify the performance requirements that the system must meet and the features it must include. Often these are highly application-dependent. However, the discussion in the following paragraphs may help the user or designer identify the key parameters to consider.

Maximum Size of the System. This specification includes two parameters: the geographical distances that the communication system must cover, and the maximum number of devices allowed within the system (where a device can be any one of the elements in Figure 5.2). In some communication systems, a third parameter, the maximum distance between devices, is also important. Commercially available systems often extend over several miles of plant area and can handle several hundred devices.

Maximum Delay Time through the System. As mentioned previously, the delay time across a hard-wired connection is essentially zero on the time scale at which industrial processes operate, since the signals travel at the speed of light. (The actual delay depends on the distance traveled— about I nanosecond delay per foot.) In the case of a shared communication system, however, there always are some message

delays due to a combination of factors: it takes time to get access to the shared network, to propagate the message, and to process the message at both the sending and receiving ends. The maximum acceptable delay time depends on both the communication function and the particular industrial process. If the shared communication system is used only to monitor temperature signals generated by thermocouples, for example, a delay of several seconds usually would not be significant. On the other hand, if the communication link is part of a fast-acting flow control loop, a delay of a few tenths of a second could introduce instabilities into the loop.

A simple experiment (either actual or hypothetical) for evaluating these communication delays is to introduce a sine wave signal source into an analog input at one end of the distributed system and observe the response at an analog output at the other end, as Figure 5.3 shows. One can hardwire a strip chart recorder to both the input and the output signals to measure the delay and distortion of the input signal as a function of the frequency of the sine wave input. Of course, the results of this experiment are affected by the necessary sampling and digitization of the input and output signals; but the experiment does provide a direct end-to-end check on the effectiveness of the communication-and-control system as a wire replacement medium.

Interactions between LCU Architecture and the Communications Facility. There is a significant interaction between the architecture of the LCUs and the shared communications facility in terms of the latter's required performance. One example of this type of interaction is related to the first wire-replacement function listed previously, that of transmitting control variables between LCUs. Figure 5.4 illustrates this interaction. In this figure, each circle represents a single control function, such as a PID controller or a computational block. The lines connecting the circles represent the required transfer of an internal variable from one control function to another. Now, suppose that the LCUs are designed in such a way that

each can implement a maximum of nine control functions. In this case, the control logic shown in the figure would have to be partitioned along the solid lines shown, and a total of 12 internal variables would have to be transmitted from one LCU to another across the solid boundaries, using the shared communications facility. On the other hand, if the LCU is designed to implement only four control units, the control logic would be partitioned along the dotted lines shown in the figure. This would result in a total of 24 variables that would have to be transmitted from one LCU to another across the dotted lines, doubling the throughput requirements on the communication facility.

Although this is an artificial example, it illustrates that distributed control systems employing relatively small LCUs usually require a higher rate of communications between elements than those employing large LCUs. In practice, a control system should be partitioned to minimize the need for communications between LCUs, whatever the size of the LCU; however, the trend of interaction from larger to smaller LCUs is clear from the example. Similar relationships between LCU size and required data rates exist with respect to other types of communications, such as transmission of control loop information (e.g., set points and process variables) between LCUs and human interface devices.

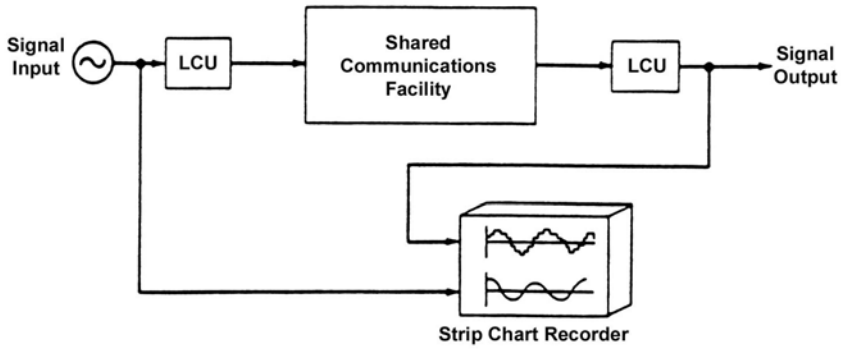

Figure 5.3. Conceptual Time Delary Experiment

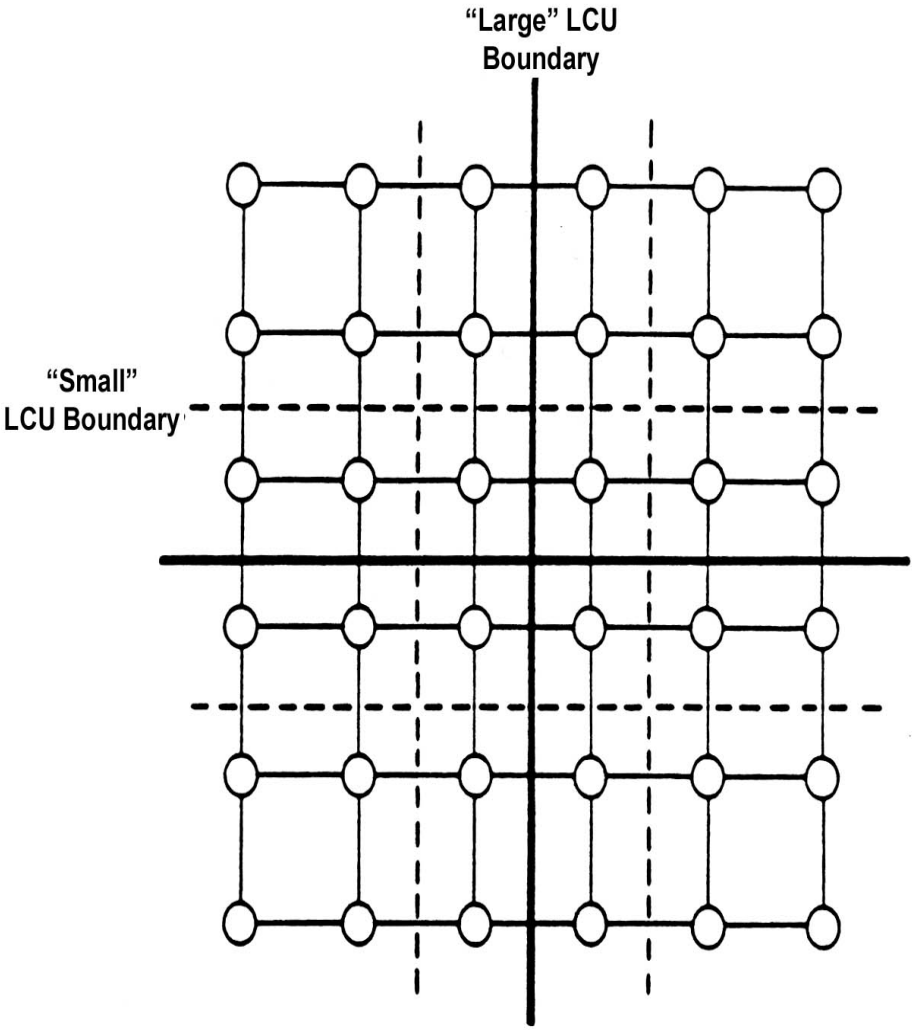

Figure 5.4. Interaction Between LCU Archltecture and Communications Facility

Rate of Undetected Errors Occurring in the System. Every communication link, including a direct hard-wired connection, is subject to errors due to electrical noise in the environment. One advantage of a shared digital communication system is that it can detect these errors at the receiving end and either correct them or request a retransmission of the message. The number of raw errors in a communication system is a function of the noise level and the rate of message transmissions. Most of these errors are detected by the communication system itself; they are not significant in the system's performance except that they increase the number of message delays because garbled information requires retransmission of the data. The significant parameter is the rate of undetected errors, since these can cause problems in controlling and monitoring an industrial process. Most industrial communication systems are designed for no more than one undetected error every 100 years. Section 5.5.2 describes some of the design approaches used to achieve this level of security.

Sensitivity to Traffic Loading. All shared communication systems are designed to operate satisfactorily under light loading conditions (i.e., when message traffic on the network is light). The critical test of a shared network is how it behaves under heavy traffic conditions, such as during a major plant upset, when many critical variables are changing rapidly. The message delay time and undetected error rate of the network must not degrade in any significant way during these conditions. One can evaluate the effect of increased loading on a particular network using the same conceptual experiment as shown in Figure 5.3. Adding more input/output signal pairs will increase the traffic loading on the communication network, and one can then evaluate this effect on the delay between any source-destination pair as a function of that loading level. The relationship between loading and the effective communication delays for a particular network involves the network's topology, the physical communication medium, and the message protocols. This chapter will discuss these issues later.

System Scalability. Ideally, the communication facilities should be designed to be cost-effective for a small monitoring or control application but also expandable to larger applications without requiring a major restructuring. This approach provides the user with the most flexibility in configuring the distributed control system. This scalability requirement, together with the requirement on maximum geographical size, often leads to a multilevel architecture of the communications facilities. Section 5.3 will discuss this in more detail.

System Fault Tolerance. It is clear from Figure 5.2 that the shared communications facility is the "spinal cord" of a distributed control system. As such, it must be designed in such a way that the failure of any one of its components will not affect its performance. This requirement for fault tolerance leads to the use of failsafe and redundant architectures in the design of its elements.

Interfacing Requirements. In most commercially available distributed control systems supplied by a particular vendor, the communication facility is designed to interface only with elements supplied by the same vendor. However, it is important that the vendor provide a mechanism (sometimes called a gateway) that allows connection to other elements using a generally accepted interface standard such as the RS-232C, RS-422, or IEEE 488 standards (described in more detail later in the chapter). This minimizes problems when the user must interface the distributed control system with "smart" instruments, sensors, or computing devices that adhere to these standards.

Ease of Application and Maintenance. To maximize its ease of application, the communications facility should be designed in such a way that the user can view it as a simple "black box" to which elements of the distributed control system can be connected. As much as possible, any operations for setting up, starting up, or restarting the communications facility should be simple, automatic, or eliminated. There should be

application tools that assist the user in configuring the system and automatically checking for potential overloading conditions.

This philosophy also should extend to the area of maintenance. The facility should have self-diagnostic capabilities that detect and announce internal failures. The facility should be modular so that relatively unskilled plant personnel can make repairs quickly. It should be possible to replace modules while the rest of the system is powered and operating: a complete system shutdown should not be necessary.

Of course, all these features are important in designing other elements in the distributed system. However, they are especially important in the case of the communications facility since this is the part of the system with which users are usually least familiar and comfortable.

Environmental Specifications. Environmental specifications are likely to be much more stringent for the communication facility than for the other elements of the distributed control system, since the former is the least likely to be enclosed in a protective physical environment. The discussion on packaging and power in Chapter 8 lists a typical set of such specifications.

ARCHITECTURAL ISSUES

Until now, we have viewed the communications facility from the outside in as a black box (Figure 5.2) having certain external characteristics and performance capabilities. This viewpoint is adequate at the first stage of system evaluation and design, during which the main concerns have to do with the communication system's basic scope (How long can the cables extend? How many terminals can the communication system support?) and its overall performance (What is its speed? What kind of delays can be expected?). However, in later stages of evaluation, one has to look at the internals of the communication system to review its architecture and detailed performance characteristics. This is the only way one can understand

the system's limitations, strong points, and modes of potential failure. This section will review some of the architectural alternatives available in structuring a communications facility and discuss their advantages and disadvantages.

Channel Structure

The first decision to make in evaluating or designing a communications facility is whether to choose a parallel or serial link as the communication medium. In the parallel approach, multiple conductors (wires or fiber optic links) carry a combination of data and handshaking signals (the latter control the flow of data between the nodes in the system). The serial approach uses only a single coaxial cable, fiber optic link, or pair of wires. Given the same communication capacity on each link, it is clear that the parallel approach provides a higher message throughput rate than does the serial approach. Also, the existence of separate handshaking lines to control the transfer of data between sender and receiver simplifies the coordination of the communication process. However, the parallel approach requires more circuitry and interconnection hardware at each interface to support the multiple channels, resulting in a higher cost of electronics per node. Also, the timing of the data in the multiple channels can become skewed (i.e., arrive at different times) if the distance between nodes becomes large. As a result, usually only applications requiring high data transfers over relatively short distances (examples are local computer buses and the IEEE 488 instrumentation interface standard) use the parallel approach. Most communication subsystems used in distributed process control use the serial channel approach, especially in the long-distance plant wide communication subsystem (see References 5.12 and 5.13 for examples).

For similar reasons of cost and complexity, frequency multiplexing of multiple communication channels over a single physical link is seldom if ever used in commercial distributed control systems (except in some military applications). Usually, so-called base band signaling is used to transmit a single digital signal over a single physical channel. In this type of signaling, information is transmitted

through a change in a voltage or a current level rather than a change in the amplitude, phase, or frequency of a sine wave.

Levels of Sub networks

The next issue to settle in evaluating or designing a communications facility for distributed control is whether a single network is sufficient for interconnecting all of the elements in the system, or whether multiple subnetworks are necessary; if the latter, then how many subnetworks are required? In this context, a subnetwork is defined to be a self-contained communication system that:

1. Has its own address structure (that is, a numbering system that uniquely identifies each drop on the subnetwork);
2. Allows communications among elements connected to it using a specific protocol (i.e., data interchange convention);
3. Allows communications between elements directly connected to it and elements in other subnetworks through an interface device that "translates" the message addresses and protocols of the two subnetworks.

Usually, there is a time penalty involved in communicating between subnetworks because the interface device mentioned in (3) adds a message delay greater than the time delay experienced within the subnetwork.

The decision to use subnetworks and if so, how to structure them for a particular application, depends on a number of factors, including:

1. The number and types of system elements to be connected;
2. The geographical distribution of the elements;
3. The communication traffic patterns generated by the elements.

For example, a data acquisition and control application in a small laboratory may involve only a few system elements located in the same geographical area. A single subnetwork may easily handle the amount of message traffic these elements generate. In this case, partitioning the communication system into multiple subnetworks would unnecessarily increase the cost and complexity of the system.

However, in a distributed control system application involving plantwide process control and data acquisition, this is usually not the situation. In this case, there are usually a large number of system elements that must be interconnected over a widespread geographical area. These elements often generate large volumes of message traffic; however, the traffic usually follows certain natural patterns of activity in the plant, such as:

1. Between controllers within a cabinet or in a given plant area;
2. Between high-level devices within the central control room and equipment room area;
3. Between the various plant areas and the central control room area.

In this situation, it often makes sense to partition the communication system into subnetworks that follow the natural patterns of message traffic, while providing a mechanism for the subnetworks to intercommunicate as needed.

Figure 5.5 shows one possible partitioning structure. In this case, several high-level operator interfaces and computing elements located in the central control room area must communicate with each other at moderate levels of message traffic. These elements must also be able to communicate with data acquisition and control elements located near the process units to be controlled. In this example, these latter elements are LCUs (e.g., single-loop controllers) that must communicate with each other at high rates within each process area. The natural communication system partitioning that results from these requirements has three levels:

1. A local bus or subnetwork in each cabinet allows the individual controllers to intercommunicate without interfering with message traffic in other cabinets;
2. A local subnetwork in the central control room area allows the high-level devices to intercommunicate;
3. A plant wide communication system interconnects the control room elements with the distributed elements in the process areas.

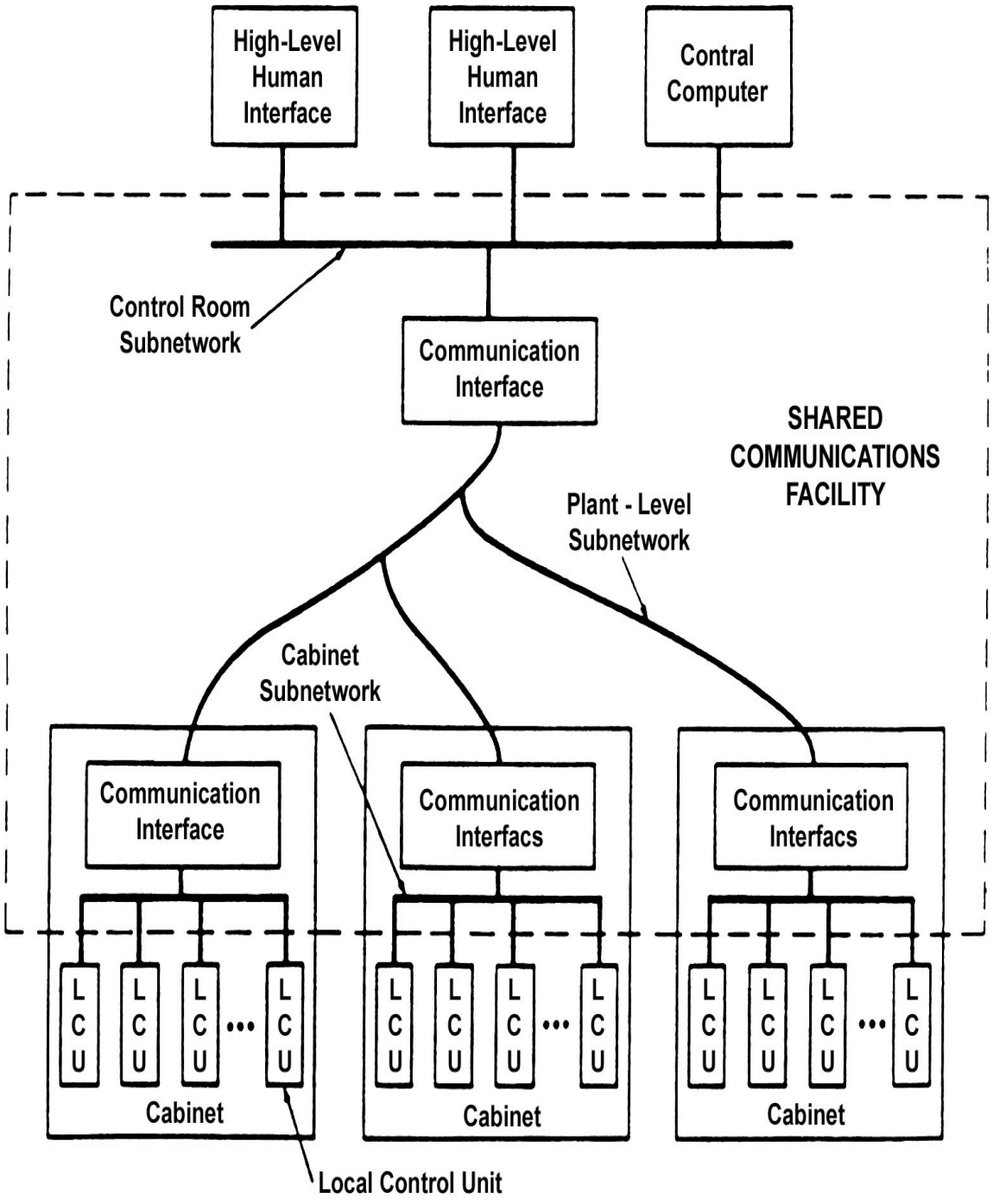

Figure 5.5. Communication System Partitioning - Example 1

This partitioning may not be appropriate if the communication requirements and controller structure change, even slightly. For example, suppose that the required communication rates between the high-level elements increase significantly (e.g., to allow for rapid dumps of large databases from one element to another). Also, assume that larger, multiloop controllers are used instead of the single-loop controllers in the previous example. Figure 5.6 illustrates the communication system partitioning that may be appropriate in this case. It consists of the following elements:

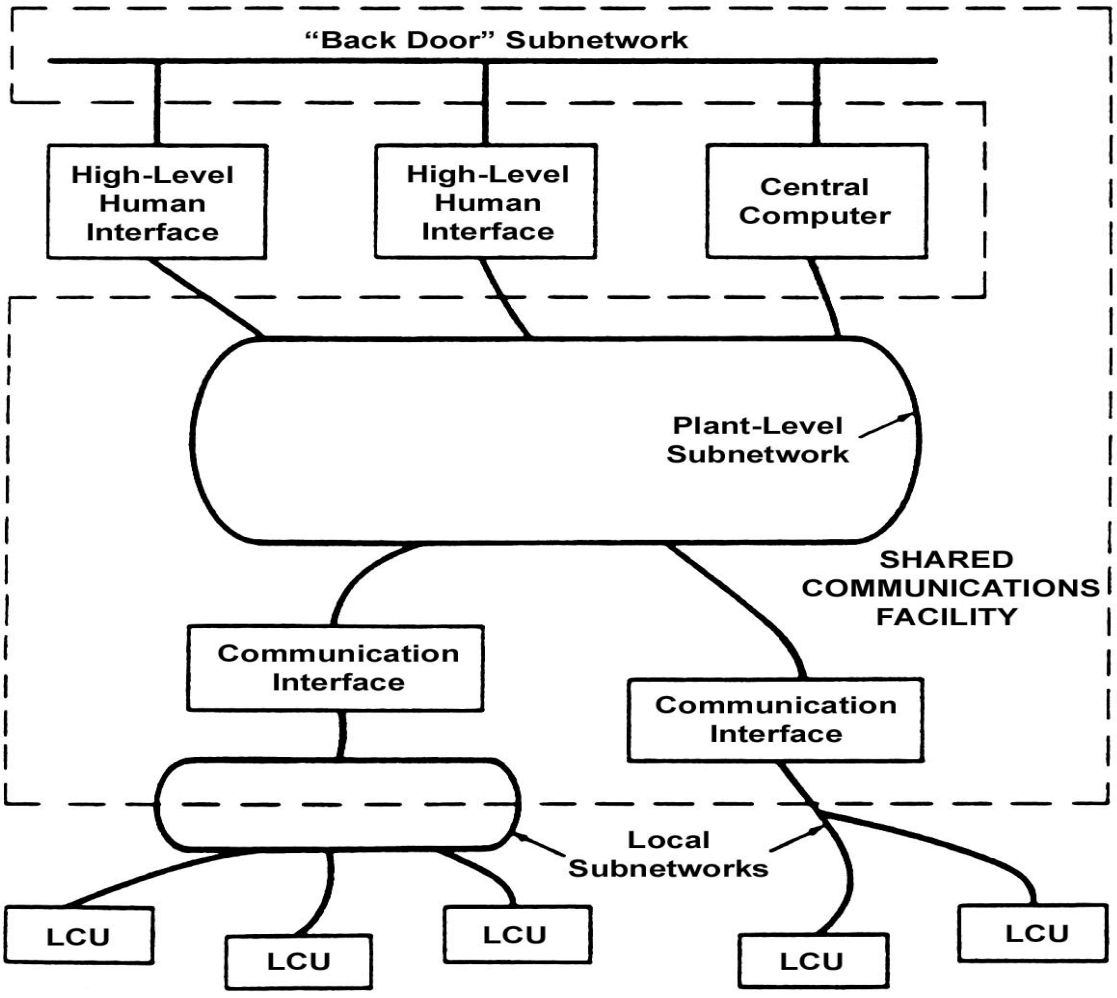

Figure 5.6. Communication System Partitioning - Example 2

1. A local subnetwork that allows the controllers in a given process area to intercommunicate;
2. A plantwide communication system that connects the high-level elements with the local subnetworks;
3. A "back door" subnetwork that allows rapid data transfers between high-level elements to take place without interfering with the process area traffic.

Subnetworks have advantages or disadvantages, depending on the situation. In general, providing multiple levels of subnetworks improves the flexibility of the communication system structure: only the lowest level of subnetwork (presumably the least expensive one) need be used in simple applications, while the higher levels can be added if needed. One can configure very large communication system structures with such a multilevel approach. On the other hand, the multilevel approach suffers from a number of potential disadvantages: (1) message delays through a large number of interfaces between subnetworks can be significant: (2) as more hardware is put in the communication chain between elements, the probability of a failure or error goes up; and (3) the addition of product types increases the complexity and maintenance problems in the system.

Network Topologies

Once the user or designer has established the necessary overall architecture of the communication system (including the use of subnetworks), the next step in the evaluation process is to select the topology of each subnetwork. Topology refers to the structure of the physical connections among the elements in a subnetwork.

Rose (5.14) and others (5.15-5.18) have analyzed a number of topologies that have been considered for use in a distributed control system. Figure 5.7 illustrates the most popular ones—star, bus, mesh, and ring configurations. In this figure, the outer six boxes in each diagram represent the system elements to be interconnected; the

boxes within the dotted lines represent the devices that make up the communication subnetwork.

The star topology has the advantage of being the simplest (and therefore likely to be the least expensive) of the four. In this approach, a single "intelligent" switching device routes the messages from one system element to another. However, a failure of this device would cause the entire subnetwork to stop functioning. Adding a redundant switching device to improve reliability increases the complexity and cost of the star topology considerably. As a result, this approach is rarely used in commercial distributed control systems.

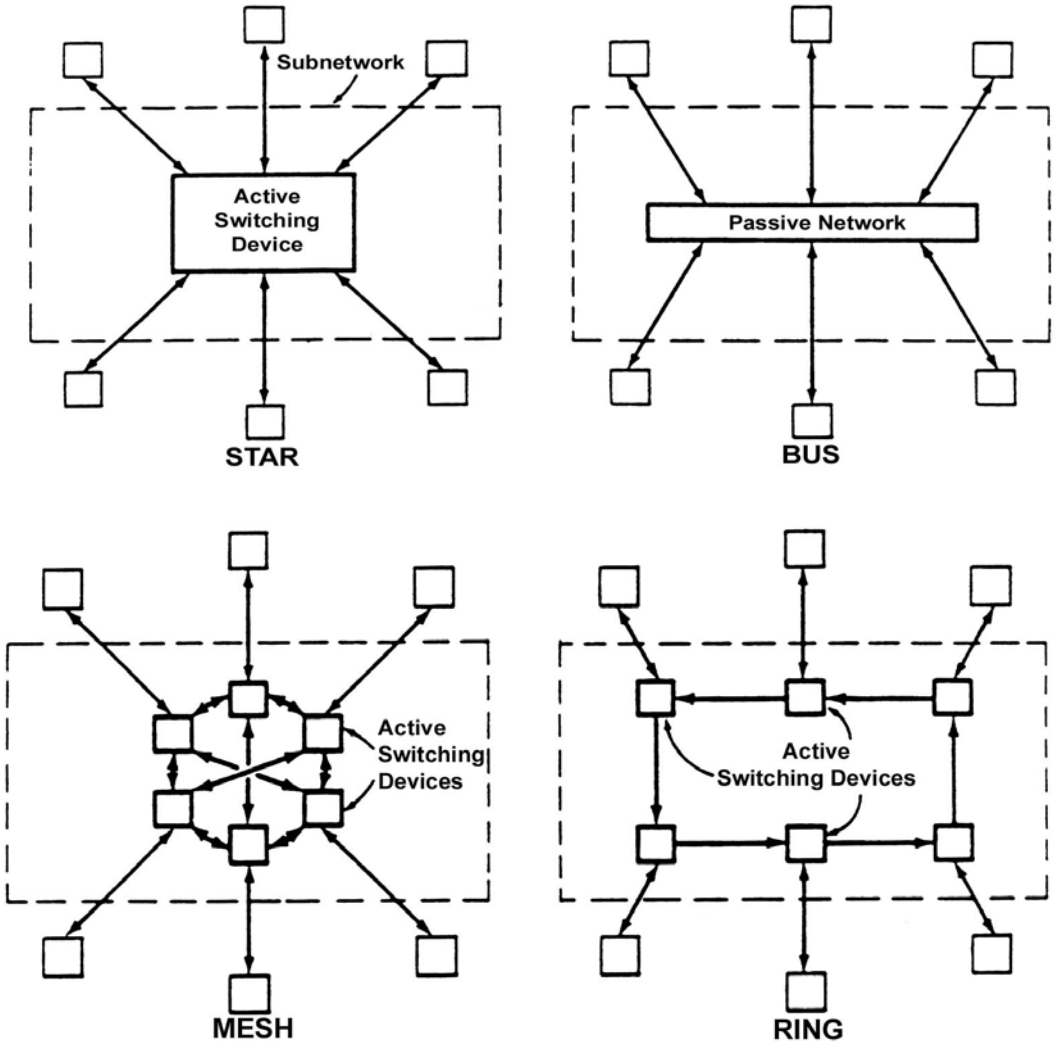

Figure 5.7. Communication Subnetwork Topologies

The bus topology is similar to the star topology in that all of the messages in the subnetwork must pass through a common piece of hardware. However, in this case the hardware is a passive network of connections instead of an active switching device. Each element that wishes to communicate over the network obtains control of it in turn (through mechanisms to be discussed later) and transmits its messages directly to the receiving elements. Since the network is passive, introducing redundant connections improves the reliability of this topology without overcomplicating the system. However, in the bus topology the system is vulnerable to a device failure that incapacitates the bus, keeping other devices from gaining control of the bus and communicating on it. For this reason, each device on the bus is designed to be failsafe to the maximum extent possible (i.e.. to fail only in a mode that disconnects it from the bus).

The mesh topology attempts to overcome some of the disadvantages of the star topology by introducing multiple switching devices that provide redundancy in active hardware as well as alternative message pathways between communicating elements. This results in a very flexible communication structure that has been used successfully in applications requiring that level of security. However, this approach is complex and expensive; it also results in significant delays as each switching device stores and forwards the messages. As a result, industrial distributed control systems generally have not adapted this topology.

The ring, or loop, topology is a special case of the mesh topology that provides connections only between adjacent switching devices. To simplify the system further, messages usually are permitted to travel in only one direction around the ring from origin to destination. Since no message-routing decisions are necessary in this approach, the switching device can be very simple and inexpensive. For this reason, one can add a redundant ring to increase the reliability of the subnetwork without significantly increasing the cost or complexity of the system. The major

potential failure mode in this topology is in a switching device, which would block message traffic around the ring. Therefore, redundant rings with automatic failsafe bypass capabilities usually are used to ensure that a single device failure does not cause a total loop breakdown.

Because of their relative cost-effectiveness and insensitivity to failures, the bus and the ring are the topologies most favored in commercially available distributed control systems. Often the total communication system uses more than one of these types among its various subnetworks.

Protocol Issues

The previous section introduced the architectural concepts of network topologies and subnetworks. These are physical characteristics of communication systems that determine the pathways over which messages can travel. The operations that must take place to accomplish the safe and accurate routing of each message along these pathways from origin to destination also must be defined for a particular communication system. The rules or conventions that govern the transmission of data in the system are usually referred to as protocols. Selecting protocols is as critical as (or more critical than) selecting the physical architecture for determining the performance and security of the communication system. This section defines the types of protocols community used in distributed control systems and briefly describes several examples of the most popular protocols.

Protocol Reference Model

As implied above in the previous paragraph, many types of communication protocols have been developed over the years (see Refs. 5.19-5.21 for tutorial information on protocols). Some of these have been implemented in software or firmware in the communication processors; others have been standardized to the extent that they have been implemented in hardware (e.g., in special memory or communication processor chips).

To attempt to put some order into the discussion of these protocols, the International Standards Organization (ISO) has developed a reference model for protocols used in communication networks, formally named the Reference Model for Open Systems Interconnection (ISO/OSI). Here, open refers to communication systems that have provisions for interfacing to other nonproprietary systems using established interface standards. It will be helpful to refer to this model in the course of discussions in the following paragraphs, so a brief summary of the model follows. (See References 5.22 and 5.23 for more details.)

The ISO model categorizes the various protocols into seven "layers," each of which can be involved in transmitting a message from one system element to another using the communications facility. Suppose, for example, that one LCU in the system (call it LCU A) is executing a control algorithm that requires the current value of a process variable in another LCU (call it LCU B). In this situation, LCU A obtains that information from LCU B over the communication system. All seven layers of protocol may be involved during this process of message transmission and reception. Figure 5.8 illustrates this process. Each layer provides an orderly interface to the next higher layer, thus implementing a logical connection from LCU A down to the communication hardware and then back up to LCUB.

The various layers provide the following services (described in simplified form):

I. **Physical layer**—This layer defines the electrical and mechanical characteristics of the interface between the physical communication medium and the driver and receiver electronics. These characteristics include voltage levels, channel structure (parallel or serial transmission), and the signaling or modulation technique used by the hardware to transmit the data.

2. **Data Link Layer**—The communication network hardware is shared among a number of system elements. One function of this layer is to determine which element has control of the hardware at any given time. The other function is to structure the transmission of messages from one element to another at the bit level; that is, this level defines the formatting of the bits and bytes in the message itself so that the arrangement makes sense to both the sender and the receiver. The level also defines the error detection and error correction techniques used and sets up the conventions for defining the start and stop of each message.

3. **Network Layer**— Within a network having multiple pathways between elements, this protocol layer handles the routing of messages from one element to another. In a communication system consisting of multiple subnetworks, this layer handles the translation of addresses and routing of information from one subnetwork to another. If the communication system consists of a single subnetwork having only single pathways between elements, this layer is not required in the communication system protocol structure.

4. **Transport Layer**—The transport layer is the mechanism in each communicating element ensuring that end-to-end message transmission has been accomplished properly. The services provided by the transport protocol layer include acknowledging messages, detecting end-to-end message errors and retransmitting

the messages, prioritizing messages, and transferring messages to multiple receivers.

5. **Session Layer**—This level of protocol schedules the starting and stopping of communication activity between two elements in the system. It may also specify the "quality" of transport service required if multiple levels of service are available.

6. **Presentation Layer**—This layer translates the message formats in the communication system into the information formats required by the next higher layer, the application layer. The presentation layer allows the application layer to properly interpret the data sent over the communication system and, conversely, it puts the information to be transmitted into the proper message format.

7. **Application Layer**—This layer is not strictly part of the communication protocol structure; rather, it is the part of the application software or firmware that calls up the communication services at the lower layers. In a high level language program, it might be a statement such as READ/COM or INPUT/COM that requests information from another system element over the communications facility. In a function block logic structure, it would be an input block that requests a certain process variable to be read from another system element over the communications facility.

These definitions are somewhat abstract and difficult to appreciate fully without referring to concrete examples. The following paragraphs provide some of these examples, and they will illustrate the convenience of the ISO layer structure as a method of organizing a discussion of the functions of a communication system

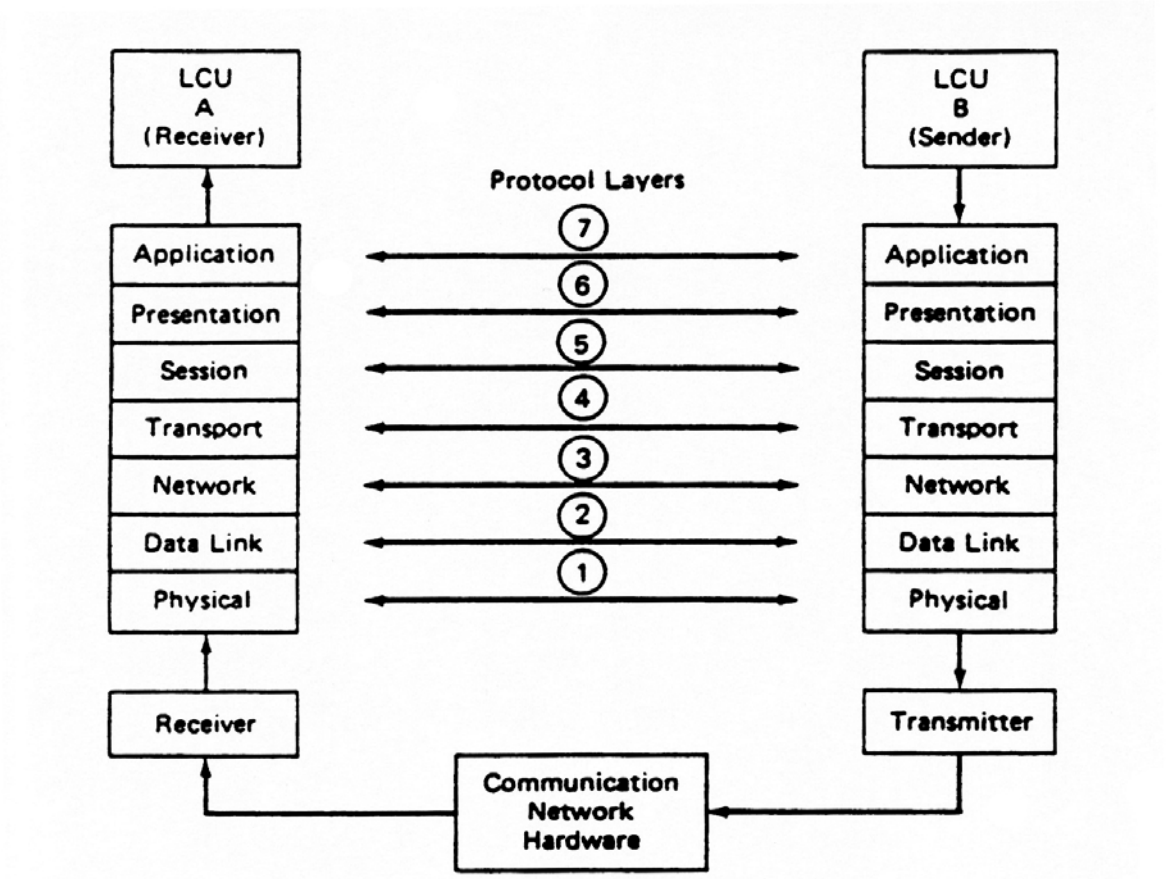

Figure 5.8. ISO Protocol Reference Model

Physical Layer Protocols

The purpose of the physical layer is primarily to ensure that the communication system electronics for driver and receiver interface properly with the communication medium itself. Specifications at the physical layer might include:

1. Type of connector and number and functions of connector pins;
2. The method of encoding a digital signal on the communication medium (e.g., the voltage levels and pattern that defines a 1 or a 0 on the signal wire or coaxial cable—see Reference 5.24 for a summary of the most common encoding schemes);
3. Definitions of hardware functions that deal with the control of access to the communication medium (such as defining handshaking control lines or detecting a busy signal line).

From the examples in this list, it is clear that defining communication system functions at the level of the physical layer is closely tied to selecting the communication medium (e.g., twisted pair wire, coaxial cable, or fiber optics) and the channel structure (e.g., serial or parallel transmission as discussed in the previous section).

A number of standard interface specifications have been developed at the level of the physical layer, including the following:

1. **RS-232C**—As described in References 5.25 and 5.26, this interface standard defines the electrical characteristics of the interface between a data terminal or computer and a 25-conductor communication cable. The standard covers allowable voltage levels, number of pins, and pin assignments of the connector. It also defines the maximum recommended transmission speed (19,200 bits/second) and distance (50 feet) for the type of voltage-based signaling approach specified. Reference 5.27 describes the speed-distance tradeoffs that can be made to extend the standard while adhering to the RS-232C specifications. The RS-232C standard was written primarily for a communication link having only a single transmitter and a single receiver.

2. **RS-449**—As described in References 5.28 and 5.29, this interface standard is similar in scope to RS-232C, but specifies a different method of voltage signaling using a "differential" or "balanced" voltage signaling technique. It references two other standards (RS-422A and RS-423) that provide specifics about the voltage driving and receiving approaches allowed. The signaling technique specified in the RS-449 standard is an improvement over RS-232C in several respects. It provides greater immunity to electrical noise, and it permits faster data transmission rates over longer distances (e.g., 250.000 bits/second over 1,000 feet). It also allows multiple receivers to be connected to the same communication link.

3. **RS-485**—As described in Reference 5.30, this standard goes beyond the RS-232C and RS-449 standards in that it includes the handshaking lines needed to support multiple transmitters and receivers on the same communication link (up to 32 transmitters and/or receiver stations per link). This structure allows multiple devices to intercommunicate at a low cost and at speeds and distances on the same order of magnitude as the RS—449 standard.

It should be pointed out that none of the standards listed above is helpful in defining the formats or meanings of the messages transmitted across the serial communication link; only the higher layers of protocol perform this defining function (described below).

Data Link Layer Protocols

As stated in the definition above, the data link layer of protocol encompasses two functions: (I) controlling access to the shared communication medium and (2) structuring the format of messages in the network. This section describes the two functions and gives examples of their implementation.

Since a communication network (or subnetwork) consists of a single communication medium with many potential users, the data link protocol must provide the rules for arbitrating how the common hardware is used. There is a wide variety of approaches to implementing this function (see Reference 5.12, for example), some of which are used only with certain network topologies. Table 5.1 lists some of the more common network access protocols, along with their key characteristics. They are defined as follows:

1. **Time Division Multiplex Access (TDMA)**—This approach is used in bus-type network topologies. A bus master transmits a time clock signal to each of the nodes in the network, each of which has a preas-signed time slot during which it is allowed to transmit messages. In some implementations, the assignment of time slots is dynamic instead of static. While this is a simple approach, it does

not allow nodes to get rapid access to the network, nor does it handle bursty message traffic (where the messages come in spurts, or bursts) very efficiently. Also, it requires the use of a bus master, which can be a single point of failure unless it is made redundant (which increases cost and complexity).

2. **Polling**—This approach can be used in either bus or ring networks. Like TDMA, it requires that a network "master" be used to implement the protocol. In this approach, the master polls each of the nodes in the network in sequence and asks it whether it has any messages to transmit. If the reply is affirmative, the node is granted access to the network for a fixed length of time. If not, the master moves on to the next node. Since time is not reserved for nodes that do not need to transmit, this protocol is more efficient than TDMA. However, it suffers from the same disadvantages as TDMA: slow access to the network and need for a redundant master for reliability. The polling approach has been used extensively in computer-based digital control systems and in certain proprietary distributed control systems.

3. **Token Passing**—This method can be used in either bus or ring networks. In this protocol, a token is a special message that circulates from each node in the network to the next in a prescribed sequence. A node transmits a message containing information only when it has the token. An advantage of this approach over the previous protocols is that it requires no network master. The access allocation method is predictable and deterministic, and it can be used in both large and small distributed networks. The main disadvantage of this approach is the potential danger that a token may get "dropped" (lost) or that two nodes may think they have the token at the same time. Reliable recovery strategies must be implemented to minimize the chance of these errors causing a problem in the network communication function. Token passing is one of the access protocols defined by the IEEE 802 local area network (LAN) standard (described in more detail in Section 5.6).

4. **Carrier Sense/Multiple access with Collision Detection (CSMA/CD)**— This approach is used in bus networks. It is analogous to a party-line telephone network, in which a caller (in this case a node or device in the network) listens on the line until it is no longer busy. Then the device initiates the call (i.e., the message transmission), while listening at all times for any other device trying to use the line. If another device starts to send a message at the same time, both devices detect this and back off a random length of time before trying again. This approach has a number of advantages. It is simple and inexpensive to implement, it does not require a network master, and it provides nodes or devices with fast access to the network. Its efficiency decreases in geographically large networks, since the larger signal propagation times require a longer wait before the device is sure that no other device is trying to use the network. Also, it is not possible to define an absolute maximum time it can take to gain access to the network, since the access process is not as predictable as in other access protocols (the token-passing approach, for example). However, queuing analyses and simulations can provide excellent information on the behavior of a CSMA/CD network, so predicting its performance generally is not a problem (see Reference 5.32. for example). The CSMA/CD protocol is used in the Ethernet proprietary communication system, and is specified in the IEEE 802 local area network standard.

5. **Ring Expansion**—This approach is applicable only to ring networks. In this technique, a node wishing to transmit a message monitors the message stream passing through it. When it detects a lull in the message traffic, it inserts its own message, while at the same time buffering (and later retransmitting) any incoming message. In effect, this method "expands" the ring by one message until the original message or an acknowledgment returns back to the original sender. This protocol is very useful in ring networks, since it does not require a network master; it also permits multiple nodes to transmit messages simultaneously (thereby increasing the effective bandwidth of the communication system). This

approach is used in the serial version of the CAMAC (computer-automated measurement and control) system and in certain proprietary communication networks.

Table 5.1 Network Access Protocols

NETWORK ACCESS PROTOCOL	NETWORK TYPE	ADVANTAGES	DISADVANTAGES
Time division/multiplex access (TDMA)	Bus	Simple structure	• Not very efficient for normal (bursty) mes-sage traffic • Redundant bus master required to maintain master clock
Polling	Bus or ring	• Simple structure • More efficient than TDMA • Deterministic allocation of access	• Redundant network master required • Slow access to the net-work
Token passing	Bus or ring	• Deterministic allocation of access • No master required • Can be used in large bus network topolo-gies	• Must have recovery strategies for a dropped token
Carrier sense/Multiple ac-cess With collision Detection	Bus	• No master required • Simple implementation • Stable performance at high message traffic levels	• Efficiency decreases in long-distance net-works • Access time to network is

(CSMA/CD)			probabilistic. not deterministic
Ring expansion	Ring	• No master required • Supports multiple simultaneous message transmissions	• Usable only on ring network

Once the control of the communication medium has been established by one of the mechanisms just described, data can be sent from one node to another in the form of a sequence of bits. It is the data link layer of protocol that defines the format in which the bits are arranged to form an intelligible message. It also defines the details of the message transmission and reception operations (including error detection and correction). Most commercial communication systems used in distributed control implement this level using one of a number of protocols that have become standards in the communication industry. Some of the more popular ones are:

1. **BIS YNC (Binary Synchronous Communications Protocol)**—Character-oriented protocol developed by International Business Machines (IBM).
2. **DDCMP (Digital Data Communications Message Protocol)**—Character-oriented protocol developed by the Digital Equipment Corporation (DEC).
3. **SDLC (Synchronous Data Link Control)**—Bit-oriented protocol developed by IBM.
4. **HDLC (High-level Data Link Control)**—Bit-oriented protocol standard defined by the Consultative Committee for International Telephony and Telegraphy (CCITT).
5. **ADCCP (Advanced Data Communications Control Procedures)**—Bit-oriented protocol standard defined by the American National Standards Institute (ANSI).

The first two protocols are similar in that they define node-to-node messages on the basis of characters or bytes (eight-bit units of data). In contrast, the last three protocols are bit-oriented; that is, the messages are broken up into frames in which the individual message bits have significance. The second group of protocols has largely supplanted the first in current communication systems because of their superior performance and efficient use of the communication medium. The protocols defined in the second group also have been implemented in off-the-shelf chips available from the semiconductor manufacturers, thus simplifying their usage in commercial control systems. References 5.8, 5.9, 5.11, and 5.31 discuss the characteristics and relative merits of these protocols in detail.

Network Layer Protocols

The network layer of protocol handles the details of message routing in communication networks with multiple source-to-destination routes (for example, the ARPANET packet-switching network). The network layer protocol also implements any address translations required in transmitting a message from one subnetwork to another within the same overall communication network. As References 5.9 and 5.29 describe, certain standard protocols (such as CCIT X.25) have emerged to support networking in communication systems. However, due to their cost and complexity, networks allowing alternative routings (such as the mesh topology shown in Figure 5.7) are rare in industrial control systems. Bus and ring topologies with redundant links between nodes are common; however, this type of redundancy generally does not include any options on message routing within the network. It is used only to allow the communication system to continue operating if a cable breaks or a failure disables one of the primary links.

Because of this, most industrial systems need a network layer of protocol primarily to manage the interfaces between subnetworks. The subnetwork interface protocol is usually specific to the particular vendor's proprietary communication system.

Industrial systems also need the network layer to implement gateways, which are interfaces between the proprietary system and external communication links. The network layer accomplishes this by translating the proprietary message structure into one that conforms to one of the lower-level protocol standards described earlier (e.g., RS-232C or SDLC).

Transport and Session Layer Protocols

In communication systems designed for industrial control, the transport and session layers are often combined for simplicity. These protocols define the method for initiating data transfer, the messages required to accomplish the transfer, and the method for concluding the transfer.

In a distributed control system, each node or element in the system is intelligent (i.e., has a microprocessor and acts independently) and performs a particular function in the overall system context. To perform this function, each node requires input information, some of which is generated within the node and the rest obtained from other nodes in the system. One can view the shared communications facility as the mechanism that updates the database in each node with the required information from the other nodes. The updating process is carried out at the level of the session and transport layers of protocol. In industrial control systems, one of the following three methods is used most often to accomplish this updating:

1. **Polling**—The node requiring the information periodically polls the other nodes, usually at a rate consistent with the expected rate of change of the variables in question. The exchange takes place on a request-response basis, with the returned data providing acknowledgment that the polled node received the request and understood it properly.
2. **Broadcasting**—In this approach, the node containing the desired information broadcasts this information to all other nodes in the system, whether the other nodes have requested it or not. In some systems, the receivers acknowledge these broadcast messages; in others, they do not.

3. **Exception Reporting**—in this approach, the node containing a particular item of information maintains a "subscriber's list" of the nodes needing that information. When that information changes by a specified amount, the subscribers are furnished with the updated value. Usually, the receiving nodes acknowledge this update.

Reference 5.33 describes in detail the characteristics and relative performance of these three implementations of the transport-session layer of protocol. The polling approach is the protocol most commonly used in distributed control systems, particularly those employing computers or network masters to run the communication network. However, this approach is relatively inefficient in its use of communication system bandwidth, since it uses many of the request-response messages to update unchanging information. Also, it responds slowly to changing data. The broadcast method is better in both regards, especially if a pure broadcast approach is used without acknowledgment of updates. However, this latter approach suffers from a potential problem in data security, since the data sender has no assurance that the data user received the update correctly. The exception-reporting technique has proved to be very responsive and efficient in initiating data transfers in distributed control systems (see Reference 5.15). Often a pure exception-reporting approach is augmented with other rules to ensure that:

1. Exception reports on the same point are not generated more often than necessary, which would tend to flood the network with redundant data;
2. At least one exception report on each point is generated within a selected time interval, even if the point has not changed beyond the exception band. This ensures that the users of that information have available to them a current value of the point.

Higher-Level Protocols

The higher levels of protocols in the ISO model, the presentation and application layers, perform the housekeeping operations required to interface the lower-level protocols with the users of the shared communications facility. Therefore, it is usually not necessary for the user to consider these protocols as separate from the distributed control functions performed through the communication system. (Section 5.2 gave a partial list of these functions)

One system feature that the higher levels of protocols could implement is to differentiate between classes of messages and designate their corresponding priorities in the communication system. For example, a designer may choose to subdivide the types of messages to be transmitted over the shared communications facility into the following priority levels:

1. Messages performing time synchronization functions;
2. Process trip signals and safety permissive;
3. Process variable alarms;
4. Operator set-point and mode change commands;
5. Process variable transmissions;
6. Configuration and tuning commands;
7. Logging and long-term data storage information.

Given this priority scheme, the higher layer of protocols could be used to sort out these messages and to feed the highest priority messages to the lower protocol levels first. While this would appear to be a desirable goal, the complexities and costs involved in implementing such a priority technique in practice are formidable. To date, the suppliers of commercially available distributed control systems have not incorporated a priority message structure in their communication systems. Rather, they have depended on keeping the message traffic in their systems low enough compared to capacity so that messages at all priority levels can get through without undue delays.

Other Issues in Evaluating Communication Systems

The previous two sections discussed issues dealing with communication system architectures and network protocols. This section reviews a number of other issues to consider in evaluating a communication system. Most are relatively independent of architecture and protocol issues; where there is an interrelationship, it will be identified.

Selecting a Communication Medium

One basic decision to make in evaluating a communication system approach is selecting the physical medium used in conveying information through the system. There are many factors to take into account when choosing the best medium for a particular application, including speed and distance of data transmission, topology of the network, and target costs. While there are many options, the media most often selected for use in industrial control systems are the following:

1. **Twisted Pair Cable**—A twisted pair of wires usually surrounded by an external shield to minimize noise susceptibility and a rugged jacket for protection against the environment.
2. **Coaxial Cable**—An inner metal conductor surrounded by insulation and a rigid or semi rigid outer conductor, enclosed in a protective jacket.
3. **Fiber Optic Cable**—An inner core surrounded by an outer cladding, both layers constructed of glass or plastic material. Fiber optic cables conduct light waves instead of electrical signals. The cable usually includes an internal mechanical member to increase pulling strength, and all elements are encased in a rugged jacket.

Selected industrial applications have employed the technique of transmitting information through free space using infrared light or radio frequencies, but this approach has not been very widespread for a variety of reasons (including problems of cost, security, and noise).

Table 5.2 summarizes the major differences in characteristics of twisted pair cable, coaxial cable, and fiber optic cable, and Figure 5.9 compares typical transmission speed and distance capabilities of the three media.

Cables using a shielded twisted pair of conductors as the communication medium have been used for many years in industrial applications to convey analog signal information and point-to-point digital information. These cables, and their associated connectors and supporting electronics, are low in cost and have multiple sources due to standardization efforts that have taken place over the years. They are easy to install and maintain, and service people can handle and repair them with minimum special training. Twisted pair cable is used most often to implement ring architecture networks, especially the high-speed, long-distance networks characteristic of plant-wide communication systems. These cables are quite suitable for use in rugged environments, particularly if they are built to high-quality standards. As Figure 5.9 shows, point-to-point links using twisted pair cable generally do not operate at transmission speeds over five megabits per second at distances over a few kilometers. Of course, decreasing the distance between nodes in the system allows messages to be sent at higher transmission speeds.

Widespread use of coaxial cable in cable television (CATV) networks has led to a standardization of components and corresponding reduction in costs. As a result, coaxial cable has become quite prevalent in industrial communications systems. If applied properly, this type of medium has a number of advantages over twisted pair cable. First, it can implement a bus network (as well as a ring network) as long as the system uses the appropriate "tee" connectors and bus termination components. Also, it has an advantage in the area of noise immunity, resulting in a potential increase in communication system performance. As Figure 5.9 shows, a communication system using coaxial cable is capable of operating at speeds and distances greater than twisted pair-based systems by a factor of 10 or more. However, more complex and

expensive electronics are needed to achieve this higher potential level of performance. As a result, most coaxial-based industrial systems operate at points on the speed-distance graph that are well within the maximum performance limits (i.e.. closer to the limits for twisted pair cable). This approach results in communication system costs that are not much higher than those for twisted pair-based systems.

Table 5.2 Characteristics of Communication Media

MEDIUM / FEATURE	TWISTED PAIR CABLE	COAXIAL CABLE	FIBER OPTIC CABLE
Relative cost of cable	Low	Higher than twisted pair	Multimode fiber cable comparable with twisted pair
Cost of connectors and supporting electronics	Low due to standardization	Low due to CATV standardization	Relatively high–offset by high performance
Noise immunity	Good if external shield used	Very good	Excellent-not susceptible to and does not generate electromagnetic interference
Standardization of components	High – with multiple sources	Promoted by CATV influences	Very little standardization or second sourcing
Ease of installation	Simple due to two-wire connection	Can be complicated when rigid cable type is used	Simple because of light weight and small size
Field repair	Requires simple solder repair only	Requires special splice fixture	Requires special skills and fixturing
Network types supported	Primarily ring networks	Either bus or ring networks	Almost solely ring networks
Suitability for rugged environments	Good, with reasonable cable construction	Good, but must protect aluminum conductor from water or corrosive environment	Excellent – can survive high temperatures and other extreme environments

In the area of installation and repair, coaxial cable is somewhat more difficult to handle than twisted pair cable. One type of coaxial cable (broadband cable) has a rigid outer conductor that makes it difficult to install in cable trays or conduit. Coaxial cable requires special connectors, whereas a simple two-wire lug connection is possible with twisted pair cable. Field repair of coaxial cable is more complicated, requiring special fixturing tools and splicing components to perform the operation. Coaxial cable whose outer conductor is made of aluminum requires special care to protect it from water or corrosive elements in underground installations.

The state of the art of fiber optic cable and connector technology has advanced significantly during the 1980s in the areas of performance improvements, standardization of components, and ease of field installation and repair (see References 5.34-5.39). As Figure 5.9 indicates, there is no question that the potential performance of fiber optic communication systems far exceeds that of systems based on coaxial or twisted pair cable. This is primarily due to the fact that fiber optics is not susceptible to the electromagnetic interference and electrical losses that limit the performance of the other two approaches. As in the case of coaxial cable, however, current industrial communication systems employing fiber optics operate well within the possible outside limits of the speed-distance range to keep the costs of the drivers and receivers relatively low and compatible with other elements in the distributed control system. Some systems use fiber optic cable only in selected portions of the total network to take advantage of its high immunity to electrical noise.

Despite recent progress, there still are a number of significant issues to consider when evaluating fiber optic technology for use in industrial applications. First, at its current stage of development, only ring networks can use this medium effectively. (See Reference 5.40.) While short-distance fiber optic bus systems have been implemented and run in specialized situations, cost-effective and reliable industrial-

grade bus systems are still being perfected. This is primarily due to the fact that a fiber optic bus system requires the use of special components (light splitters, couplers, and switches) that are still evolving and that require careful selection and installation to be effective. (See References 5.35, 5.38, and 5.42 for additional information in this area.) This development is proceeding rapidly, however, and it is likely that long-distance fiber optic bus systems will emerge eventually.

Another obstacle to the widespread use of this technology is the slow development of standards for fiber optic cables and connectors. At present, the technology is developing much more rapidly than the standardization of its components. The Electronic Industries Association has produced a generic specification for fiber optic connectors (RS-475, see Reference 5.41). While this and related documents on nomenclature and testing are helpful, a standard that would promote second-sourcing and mixed use of cables, connectors, and other components from more than one vendor does not yet exist. As a result, the costs of these components are still relatively high compared to the costs of components used in twisted pair or coaxial systems (although the cost of the fiber optic cable itself is becoming comparable to that of electrical cable in low- or medium-speed applications).

The desirability of fiber optics from the point of view of field installation and repair is still mixed. Because of its light weight and small size, one can easily install fiber optic cable in conduit or cable trays. Since it is immune to electromagnetic interference and cannot conduct electrical energy, it can be run near power wiring and in hazardous plant areas without any special precautions. Fiber optic cable also provides electrical isolation from ground, thus eliminating concerns about ground loops during installation. Recent developments in materials and packaging have made fiber optic cable very suitable for use in high-temperature and other rugged environments. However, although the field repair of fiber optic cable is possible, it requires special fixturing and trained personnel to be accomplished reliably on a routine basis. Also, cost-effective field equipment suitable for locating cable breaks

and connector failures is still under development. As in other problem areas, that of field repair is expected to improve dramatically as fiber optic technology matures.

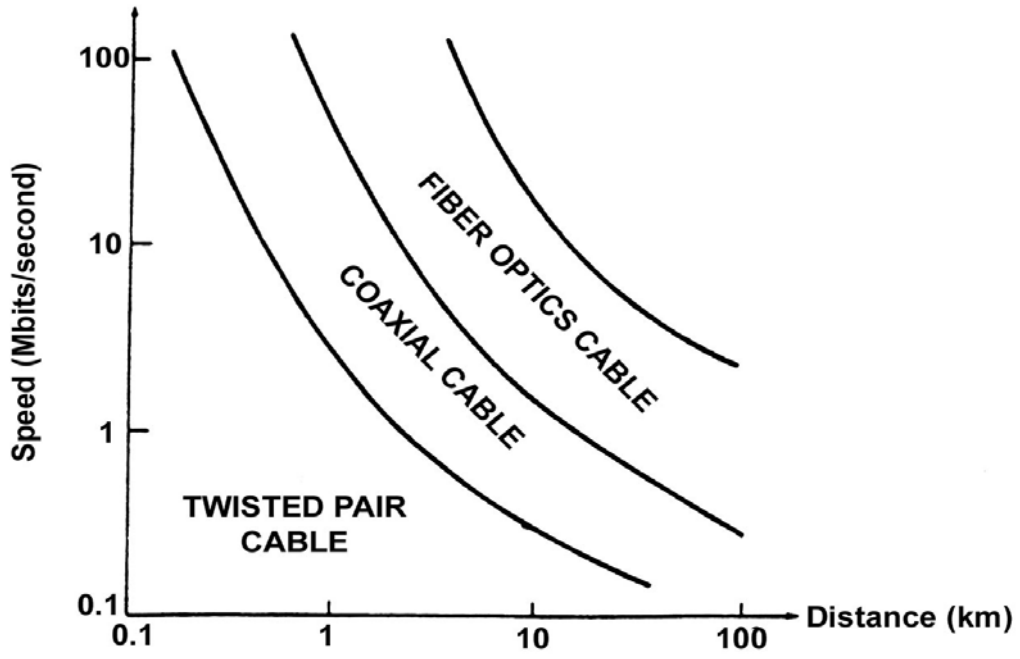

Figure 5.9. Typical Speed-Distance Tradeoffs

Message Security

As stated in the beginning of the chapter, the communication facility in a distributed control system is shared among many system elements, instead of being dedicated as in the case of an analog control system. As a result, the security of message transmission is a key issue in evaluating alternative communication techniques. There is always a concern that component failures or electrical noise will cut off or degrade the accuracy of information flowing from one system element to another. Section 5.3 noted that one way to avoid a cutoff of information flow is to ensure that the architecture of the communication system does not include any potential single points of failure.

However, additional safeguards are necessary to minimize the chance that message errors due to electrical noise will propagate through the system without detection. Many commercial distributed control systems provide four levels of protection:

1. Handshaking (i.e., acknowledgment of message receipt) between system elements involved in the transfer of information. This ensures that the information has arrived safely. Handshaking is used especially in transmitting critical information such as control system variables.
2. Ability to detect the vast majority of errors in messages caused by external noise.
3. Provision of message retransmission or some other scheme that allows an orderly recovery if a message is lost or contains an error.
4. Inclusion of "reasonableness checks" in the application layer of network protocol to catch any message errors not detected in the lower layers.

The first level of security (handshaking) occurs automatically if the session and transport layers of network protocol use the polling or exception-reporting protocols (Section 5.4). In the polling approach, the node sending a message requesting information is automatically notified that the request arrived safely when it receives the requested information back from the other nodes. If the information is not received, the polling node assumes that an error has occurred and asks for it again. In the exception-reporting approach, the report message usually is followed by an acknowledgment message from the receiving node that informs the sender that it received the information properly. If the broadcast approach is used, a separate mechanism for validating the transfer of information needs to be included to ensure that errors were not introduced in the broadcast messages. For example, this mechanism can consist of a separate set of acknowledgment messages sent by the receiving nodes.

One can best understand the second and third levels of message security by reviewing a typical error handling process, as illustrated in Figure 5.10. In any distributed system, there is a base load of messages that the nodes generate for transmission to other nodes using the shared communications facility. The transmission mechanisms in the nodes handle these messages and send them through the transmission medium. In the process of transmission, the noise environment corrupts some portion of these messages. At the receiving end, an error detection mechanism decides whether each incoming message is good (error-free) or bad (contains an error). If a message is determined to be bad, the receiving node makes the sending node aware of the error through the acknowledgment process previously described, and the sending node then retransmits the message. As a result, the total message traffic in the communication facility is the sum of the normal message traffic and the retransmitted message traffic. In this scheme, an increase in noise from the environment causes a corresponding increase in total message traffic; however, the error detection mechanism maintains message security. If this mechanism were perfect, there would be no undetected bad messages received. In practice, however, the mechanism can never be perfect, and a certain number of bad messages get through.

One can see from Figure 5.10 that the error rate of undetected bad messages is a function of the external noise environment, the characteristics of the transmission medium (type of cable, connectors, and driver and receiver electronics), and the effectiveness of the error detection mechanism. Usually, the error detection mechanism is implemented at the data link layer of network protocol by means of cyclic redundancy check (CRC) codes built into the message formats (see References 5.9 and 5.43 for detailed descriptions). The CRC code is inserted into the message at the source and checked at the destination for consistency with the transmitted information. In most industrial communication systems, these codes are designed to produce an undetected error rate of less than one bad message per 100 years of operation under an assumed noise environment. (Of course, the error rate in any

particular installation is a function of the actual, not the assumed, noise environment that exists.)

To pick up any errors that get through the data link layer of protection, most industrial control systems perform checks on the reasonableness of data acquired over a shared communications facility. For example, if an analog input is outside the expected range or changes faster than is physically possible, then the application logic making use of the input marks the input "bad" and takes appropriate action. In the case of a process variable input to a control loop, this action may consist of putting the loop into manual mode. If the input is used in a calculation, the application logic may substitute a default value or ask the operator to enter a value manually.

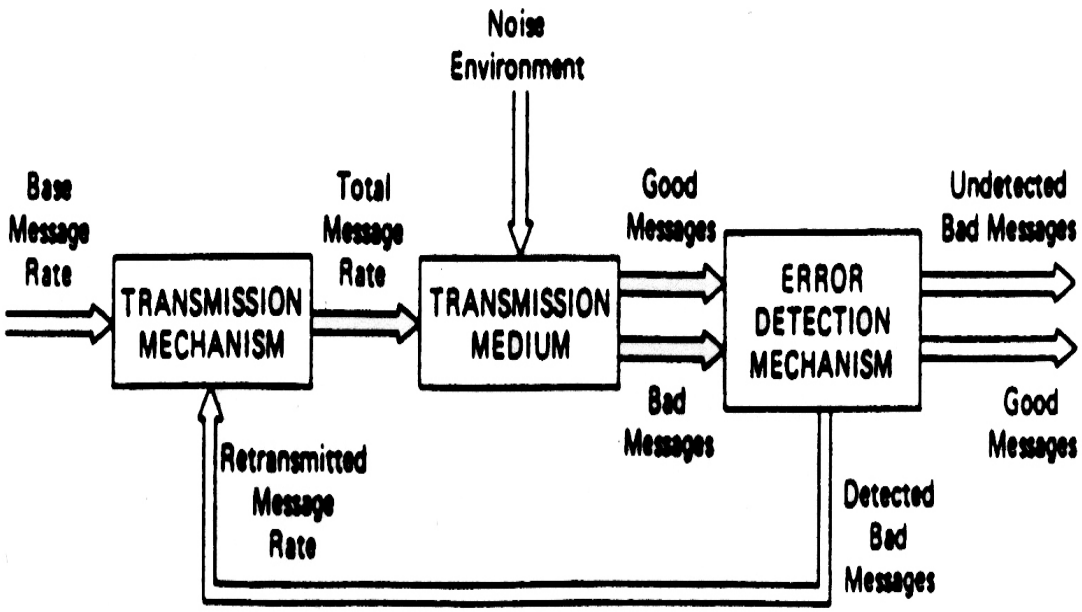

Figure 5.10. Error Handling Process

Efficiency of Bandwidth Usage

One often-used measure of the performance of a communications facility is the raw bit rate of data transmission between nodes. By this measure. a communication system that uses a 2 Mbit/sec, data rate is assumed to carry twice as much information as one running at 1 Mbit/sec. Unfortunately, the situation is not quite that simple. The efficiency of usage of the information bandwidth provided by 1 Mbit/sec, data links (for example) varies dramatically from one system to another. Some of the factors that can have an impact on this efficiency include the following:

I. **Topology of the Communication Network**—In ring networks, multiple messages can travel on the ring simultaneously if certain physical layer network protocols are used. As a result, this type of network can support more message transfers per second than a bus network having the same raw bit transmission rate.

2. **Message Formats**—A communication system that supports variable-length message formats can tailor the message size to the type of information to be sent (e.g., an analog input data message would be longer than a message reporting contact status). Therefore, a system with variable-length messages is more efficient than one in which the message length is fixed at the maximum, worst-case size.

3. **Repertoire of Message Types**—A communication system that supports "packed" messages (e.g., contact status information in groups of eight contacts at a time) is more efficient than a system requiring one message per variable. Similarly, systems that can send a single message to multiple destinations are more efficient than those requiring a separate message for each destination.

4. **Transport and Session Layer Protocol Used**—As pointed out previously, the exception-reporting protocol is much more efficient than the polling or

broadcast protocols, since the exception protocol initiates messages only when information is changing. One can consider transport layer protocols requiring acknowledgment of messages to be less efficient than those that do not; however, the use of acknowledgments significantly increases message security.

5. **Retransmission Rate Due to Message Errors**—If the data rate selected is too high for the noise environment, the transmission medium, or both, there will be a high rate of detected message errors. The resulting load of data retransmission messages can have a significant impact on the true information throughput rate of the system.

It is not possible in this space to discuss all of the factors that can affect the efficiency of bandwidth usage in a shared communications facility. However, the user or designer of such a facility must be aware of these factors and evaluate the alternative communication system approaches with them in mind.

DCS CONFIGURATION GUIDELINES

DCS CONFIGURATION GUIDELINES

Introduction

The following section is intended to give guidelines for the configuration of the Distributed Control System.

This section specifies the mandatory requirements to be incorpcrc during configuration of the system.

The following documents will require approval by the Purchaser during the configuration phase.

- Point to Group allocations for each console (operating and alarm).
- Custom Displays/Graphic Displays.
- Management Reports.

Prior to the above documents the following shall be submitted for Purchaser's approval. These shall be provided at an early stage o: engineering in order that configuration and engineering precedes an acceptable manner:

- DCS description and layout in full details.
- Operator interface philosophy.
- Display philosophy for normal plant case.
- Display philosophy for emergency plant case.
- Display philosophy for start-up plant case.
- Display philosophy for shutdown plant case.
- Group layout philosophy.
- Graphic layout and live area philosophy.
- Logging philosophy.

Configuration shall use standard software and hardware, any spec items require approval by the Purchaser.

All program source codes shall be fully commented on. All source codes and configuration listings shall be provided to the Purchase: at Project completion.

System Configuration

The process of system configuration shall be through a fill-in th-3 blanks configuration process the configuration software shall be user friendly and provide help functional for each of the template fields, fields that have a choice of several fixed entries shall be provided with a scrolling function to show the available choices without the need of typing in choice. Default valves shall be provided for all data fields where this is practical.

Configuration changes shall only be way of the system configuration. It shall not be possible to modify the system configuration without having the changes automatically recorded in the master image of the global data base.

The system shall be capable of on-line loop configuration changes without shutting off the controller or placing the controller in configuration mode. The loop being changed is the only one that shall be affected.

SOFTWARE

GENERAL

The seller will be required to make available to the client all new releases of software till site final acceptance (release and performance band).

The seller will be responsible to provide all required software to perform the functions described in this specification.

The ability to configure or modify configurations shall be under key-lock or password protection.

The ability to upload a control loop configuration must be possible. This is to include all configuration information as well as tuning constants and engineering ranges. The same information shall be readily downloadable.
Up loaded configurations shall be easily modified and an in the same application Engineering input format as the original configuration before down loading took place.

The DCS data base, graphic, displays, trend functions, control loops and interface functions shall be configurable while the DC: is functioning.

System configuration shall be possible of the engineering software (personalities).
The system shall utilize the latest version of the series microprocessor running at least 20 MHz and each controller shall have a minimum of 80k bytes memory for user configuration.

The system shall have a package that dynamically and Automatically tunes proportional, integral, derivative (PID) control loops, a real-tinie (on line) statistical process central and horizon predictive control is a model- based controller, us primarily in non linear process.
The system shall able to provide secure, supervisory control for process loops that require more functionality, advanced calculations to write complex expressions for calculating process / unit performance and optimization, and interface between device within the system and custom graphics displays.
The seller should mention his operating system and it should be. based on industry standards such as UNIX, SQL, X window, ... a

PROGRAMMABILITY

The system shall have free programming capability features for BASIC, FORTRAN, PASCAL, "C" or other high level languages shall be provided.

The language shall be comprehensive enough to perform the following functions at a minimum:

- Batch control.
- Product scheduling.
- Sequential operation.
- Supervisory control.
- Processing monitoring.
- Automatic start up / shutdown control.
- Process modeling.
- Computations.
- Emergency processing.
- Store program data and initial execution of program (bath) reports.
- Operator message generation.
- I/O interface drivers.

The language shall have provision for unit relating programs u. can run or symmetrical process units while providing unit specific tag data during runtime.

The generation and editing of program source code, including recipes, shall be performed on any of the control system's console monitors, control program execution, and trouble shoot programs in a tuntime mode from any console monitor without the necessity of the user creating any of the interactive program displays.

The system utilities shall be provided to allow source code print out, tag cross. Reference print out, and program backup.

Sequences / Logic Function / Batch Control

The system shall have provision to develop sequences. The system shall be of a type capable of being programmed directly from logic flow charts using an operator - oriented language.

It shall have the following minimum features:

- Programmable from logic diagrams.
- Internal (software) timers, flags and counters.
- Continuous monitoring of program execution.

The system shall have the capability to perform logic functions to support integrated operations and control. Pre-defined software blocks shall give the possibility of opening and closing on/off valves, starting and stopping motors, pumps . . .etc.

System Data Base

The system shall have a global data base that is distributed among the system nodes during routine. During system operatic/., each node will have its data base only, but must also be able tc access data from any where in the system. The control system software shall provide the following data base management functions:

- Allow direct tag/attribute access anywhere in the system without any knowledge of the tag's physical address.
- Data security (protection from unauthorized access and modifications).
- Data integrity (to insure correctness of data).
- Real - time access in a distributed environment.
- Data base generation, back-up, fail over and recovery.
- Query capability (search and retrieval).

Control Algorithms

The system shall have the following software functions for the control algorithms such as:-

- Regulatory control module continuous control
- Input monitoring
- PID control
- PID control with dead band

- Two-position on/off control
- Pulse duration on/off control (time proportioning)
- Manual control with input monitoring auto/man switch
- Ratio setting with auto/man transfer switch
- Automatic signal selector, average switch position
- Program set function (generate function of time)
- Signal switch
- totalizing/integration
- first order lead/lag unit
- Rate limit
- Dead time
- Moving/cumulative average
- Line-segment function for non-linear signal
- Square root extraction, average switch position
- Addition/subtraction
- Multiplication/division, common and natural logarithmic
- Scaling
- Dynamic compensation (impulse, lead-lag, dead time, ..etc)
- Compensation and conversion (characterization, pulse counter accumulator, high/low clamp, rate of change clamp,..etc)
- Boolean: AND,OR,NAND,NOR,XOR,XNOR
- Logic: switch, compare, bi-directional delay, on-off. .etc
- PID with wind-up protection
- OID feed forward
- Cascade control (the tracking of CAS loops shall be made automatically so that the balance less and bumbles .
- Operation can be achieved at any time without making the configuration for signal tracking)
- Discrete control (handing discrete points such as digital input, digital output, timer, logic gate,...etc)

Data Acquisition

Date acquisition system shall interface process for both analogue and digital signals. The DCS shall have the capability of converting various input signals into types which are direct us? to the system.

The data may be from various sources which include, but are net limited to the following devices:-

- from the controllers via the input cards
- from the operator input devices, keyboards,...etc
- via communication links with other devices (when required)
- from digital input cards (acquisition)

Data base concurrency shall be maintained such that all configuration and tuning changes to the run time data base sh.'.l". be automatically saved to the run time master copy of the global data base.

Process Inputs

The I/O modules shall be able to handle a wide variety of signals including analog and digital. The interface cards may support multiple inputs or outputs of a similar type and shall be capable of powering 2-wire 4:20 mA loops. Moreover the system shall be able to connect also the signals using three wire system (i.e. RTD, flow meter signals, etc...).

The following inputs/outputs shall be covered:

- Analog inputs : 4-20 - mAdC.
- Analog inputs : Thermocouples type K-T-J-R-E.
- Analog inputs : Platinum RTDS 100 ohm - 0 C
- Analog inputs : 1-5 Volt DC.
- Pulse inputs : 0:6000Hz.
- Digital inputs : Potential free contacts.

- Analog outputs : 4:20 mA DC.
- Analog outputs : 1-5 VDC.
- Digital outputs: Relays.
- Digital outputs : Transistor contact.
- Serial links with other micro process - based systems.

All of the I/O cards must be provided with galvanic or optical isolation. the discrete outputs shall operate latched relays, momentary relays or solenoid valves.
Interposing relays shall be provided if the contact rating does noc meet the requirements.
The minimum acceptable technical requirements are :

- Contact rating 3.0 Amp at 110VAC or 24 VDC with resistive load.
- Contact inputs will select dry contact field on switch. Status changes with power supplied from the DCS system.

System Redundancy and Security

System Redundancy

Key components of the DCS system shall be fully redundant so that neither a hardware nor a software failure will result in the loose of the features supplied by the components.

The size of a controller shall be selected based on maximizing the number of implemented input/output points so as to minimize the number of ysed controllers and consequently the nui-oer of redundant controllers.
For critical loops assumed 20% of the total number of loops, The I/O modules redundancy (a 100% redundancy is required)

System Reliability

A system reliability analysis shall be provided for the total system quoted. Failure of a DCS computing module shall not be considered when calculating the reliability of the DCS, except when normal operation of the DCS is affected by these devices.

The analysis shall include Mean Time between Failure and Availability values of every module employed in the system. A Mean Time To Repair of a failed item can be set at 8 hours for all self revealing failures. For unrevealing. faults a test interval of 168 hours shall be used to establish fractional dead times. Any loss of a signal channel or a failure in processing a signal shall be considered as a failure. Using these premises the total system Mean Time between Failure and the Availability figures shall be established.

To provide a high system Availability it can be assumed that every failure would be rectified by circuit board replacement, sub-assembly, keyboard, VDU unit, disc drive or power supply replacement. To assess spare part requirements for such maintenance, Mean Time between Failure figures shall be provided for all such components employed in the system quoted.

Four weeks before the final hardware freeze date a new reliability analysis shall be issued for approval of the Purchaser, to represent the intended final hardware configuration of the DCS.

All system components shall have inbuilt self-diagnostic facilities and failure shall be indicated to the operator via the VDU' s.
Furthermore the system shall be so designed, engineered and backed up that a total failure of the complete system can no: occur under any foreseeable circumstances e.g. single process failure, power and cable.

Diagnostics

The system shall include comprehensive on-line /off-line diagnostics.

The on-line and off-line diagnostic programs and a malfunctions check program, shall consists of a peripheral exercises program that will alarm the operator upon the detection of a fault. This shall be a unique and separate system from plant alarms. The self diagnostics shall detect a fault or malfunction in, but not be limited to the following:

- Data highways and communication lines.
- Power supplies.
- All cards (I/O's controllers, CPU'S ...ets.)
- Off-line shall be programs which can be loaded when required to test in detail all functions of the system and diagnose fault type and location.

Electrical noise / Radio frequency Immunity

No assessment of electrical noise levels on the plant is available. Electrical noise is presented to some degree. Both as RFI and no power supplies to the equipment. Sources induce . heavy electrical machinery, HVAC equipment and static invention.

Equipment to be suitable for operating at co. DEG.C higher than the stated figure in order to allow a temperature rise due to the self reating effect of equipment within the cabinets.

The system shall be totally immune from UHF and VHF radio interference.

Operational / Maintenance Security

The seller must explicitly point out in writing any single point: of failure which will affect the control operator interface, communications or any other function required for the continued control of process. Any devices which can not be replaced while the system is running must be explicitly identified and the consequences of tuning without that device must be fully explained in the proposal. For processors that can

be replaced while the system is running, the necessary steps to bring the processor to full functional operation shall be described.

All key boards will be secure against interference by means of a key lock, passwords, or equivalent method.

The system shall be designed so that any failure can be eliminated as quickly as possible. The repair policy shall be based on replacing printed circuit boards or subsystems.

SELECTION OF THE BEST DCS FOR A SPECIFIC APPLICATION

1- EVALUATION AND SELECTION OF THE BEST DISTRIBUTED CONTROL SYSTEM (DCS) FOR A SPECIFIC APPLICATION

ABSTRACT

The first DCS was announced in the year 1975, and since that date various vendors introduced their own DCS's on the market. The DCS is basically a type (most recent) of computer based control systems. For this respect, the historical background and a brief description is given to highlight the pros & cons of the different types of commercially available computer based control systems, namely: The supervisory digital control system (SDC), the direct digital control system (DDC), and the distributed control system (DCS) which is the subject of this paper.

Once a DCS type of system is selected for a specific application, the next step should be to define the detailed evaluation procedure to be followed for selecting the best DCS out of the various DCS's to be offered from vendors for this application.

The objective of this paper is to set a specific, yet flexible, technique for the evaluation and selection of the best DCS for the specific application. This technique, which is based on the matrix chart concept, shall explore in detail the factors governing the evaluation and selection procedure, namely: the technical, the price and the delivery factors, where two new concepts for considering the effect of price and delivery were introduced.

Introduction & and Historical Background

Since the late 50's, the trend to use computer based control systems in plant monitoring and control became increasingly attractive due to the benefits achieved out of introducing the computer power to the control system (history, trending, dynamic graphics, alarming capabilities, optimization, ... etc). The evolution of industrial computer based control systems is illustrated in FIG.l. The **FIRST STEP** was the application of computers to industrial processes in the areas of plant monitoring and supervisory control. Such control systems were known as supervisory digital control systems, or SDC.

The **SECOND STEP** in the evolution was the use of the computer in the primary control loop itself, in a mode usually known as direct digital control, or DDC. The **THIRD STEP** in the evolution was the introduction of a new system architecture to overcome the problems encountered with SDC & DDC systems. Such architecture is known as distributed control system, or DCS. Other steps in the evolution of computer based control systems are surely yet to come. In fact, the DCS is a control concept of today-not tomorrow. There is no telling where this rapidly growing field will lead, however, DCS will certainly be around for a long time. To properly understand the evolution of industrial computer based control systems which led to the development of DCS, one needs to look more closely at the above mentioned three types of systems:

Supervisory Digital Control (SDC)

In SDC, the computer talks directly to single loop analogue controllers. The controllers then communicate with the actual process. Fig.2 shows a typical, generalized SDC configuration. Any simulators or optimizers (in the form of computer programs), current trending, historical trending and alarming functions, run or reside in the computer memory. All such functions continually use the CPU

of the computer, thus seriously impacting operating speed and efficiency, memory requirements and system costs. In fact, SDC is actually a form of DDC, the functions performed by the computer are essentially the same, the major difference being that the computer sends signals to, and receives signals from, analogue controllers instead of field instruments (such as process choices". The project plant needs are: The plant technical requirements, project schedule and DCS project budget price. This method shall also facilitate revealing the technical points which would lead to a decision that a system is not accepted technically, thus ceasing to proceed with it's evaluation process.

PROCEDURE

Selection of the Committee Members who are to Develop the Procedure

This method basically involves selecting an appropriate group of people who are involved in the project, and then having them assess in a uniform, coordinated way how competing DCS's stack up against various factors that have been weighted to reflect their importance for the plant. The committee to develop the matrix should, preferably, consist of the following members:

- ***Operating/Manufacturing Company***
  * Two (2) experienced personnel from operations: They will act as representatives for the people who will be using the system
  * One (1) experienced instrument maintenance engineer: He will act as a representative for the people who will service and maintain the system.
  * One (1) instrument engineer from the project management team: He knows about the project requirements and schedule limitations.

- ***Engineering Contractor***
  * The process lead engineer: He knows about the process complexity and operational/safety requirements.

* The instrument lead engineer: The prime responsibility for preparation of the technical specifications and tender documents, as well as the preparation of the technical evaluation report is laid on the instrument lead engineers of the engineering company. Consequently he shall act as the coordinator for the whole committee.

The committee should develop a matrix of all the important factors for the DCS selection-BEFORE SELECTING THE DCS VENDORS TO BE INVITED FOR BIDDING. If they wait until after selecting the DCS vendors to be invited for bidding, it may bias .the weighting factors in favor of one of the vendors. Each item in the matrix must be well defined via the listing of all relevant governing factors that affect the assessment of it's weight and corresponding rating, hence the score for that item. To follow out the different steps of the procedure, reference should be made to the MATRIX CHART at the end of this paper.

The First Step in the Procedure is

to develop the matrix chart. The matrix chart shall consist of three basic sections (or basic factors), namely: the technical factor, the price factor, and the delivery factor. For the technical factor, the matrix should include (but not be limited to-because each company will likely have other/additional needs, which should be included on the matrix) the following items:

1- Deviations/Exceptions from Tender Requirements

A "YES" or "NO" should be indicated. In case of "YES", reference should be made to relevant sections in project technical evaluation report for details of the deviations/exceptions with vendor explanation and purchaser's opinion. The tender requirements shall be categorized as follows:

* Technical specifications
* Technical drawings
* Vendor documentation during bidding phase and after purchase award

* Inspection and Tests
* Spare Parts
* Training

2- System Architecture

Complete definition of each offered item shall be given via the listing of all factors affecting that item, and indicating the relevant description of each of these factors in the MATRIX CHART. The following are the main items of a typical DCS with their relevant factors:

2.1 Operator Station

- Quantity
- Manufacturer / Model No.
- Screen size
- Screen resolution
- Maximum No of colors on screen
- Operating system
- Windows feature (X-windows, NT windows ...etc)
- Data refresh rate
- Speed to call up another display (during plant normal/upset conditions)
- RAM capacity
- Processor manufacturer/model No.
- Processor size (number of bits)
- Processor speed
- Keyboards
  * Quantity
  * Manufacturer/model No.
  * No. of keys per keyboard
- Page selector & alarm panel
  * Quantity

* Manufacturer/model No.
* Number of buttons per board

2.2 *Storage Media*

- Fixed Media (Hard Disk HD/Dieital Audio Tape DAT)
  * Quantity/Location
  * Manufacturer / Model No.
  * Capacity
  * Access Time
- Removable Media (streaming Tape/ Floppy Diskette)
  * Quantity / Location
  * Manufacturer / Model No.
  * Capacity
  * Access Time

2.3 *Controller*

* Quantity
* Manufacturer / Model No. Redundancy
* Processor Size (number of bits)
* Processor speed
* RAM capacity
* Battery back-up for RAM
* Controller sizing calculation submitted by vendor, based on:
- Specified maximum number of loops/controller
- Specified execution periods for each type of input/output Specified free loading capability

2.4 *Input/Output Modules*

* Quantity (from each type)
* Manufacturer / Model No. (of each type)
* Number of channels (per each type)
* Redundancy
* Installed spares

2.5 *DCS Communication Network*

* Cable type
* Cable manufacturer / Model No.
* Maximum extendable length
* Maximum data block size
* Minimum data block size
* Communication PROTOCOL
* Compatibility with ISO/OSI 7 layers model
* Compatibility of network topology with IEEE 802.X
* BAUD rate
* Maximum loading of communication network (during plant upset conditions)

2.6 *DCS Cabinets*

* Quantities
* Manufacturer / Model No.
* Weather protection class (ventilation, dustfilters, ...etc)
* Types of I.S. barriers
* Quantities of each type of I.S. barriers (including installed spares)
* Space foreseen for future expansion (uninstalled spares)

2.7 *Power Supplies*

* Quantity/Location
* Manufacturer / Model No.
* Redundancy
* Voltage/Frequency of input supply (by purchaser)

2.8 *Printers (Lodging/Graphic)*

For each type of printer, the following factors shall be considered :

* Quantity.
* Manufacturer / Model No.
* Operating speed (characters per second CPS)
* Resolution (dot per inch DPI)

* Number of colors
* Power supply (voltage/frequency)
* Method of connection to DCS:
  - Directly to a specific OS
  - To DHW via a special gateway (software assignment to any OS)

3. Operator Interface

The factors clarifying the power of the operator interface, some of which are listed below, should be presented in detail by each of the vendors bidding on the project during a clarification meeting where the vendor shall use any of his convenient resources (DEMO equipment, video tapes, overhead projectors,...etc) to clarify his DCS operator interface capabilities to the committee members.

* How easy and convenient is it to navigate between graphics & displays?
* How well and convenient does the system present alarms to the operator (alarm sequence, grouping, resolution, ...etc)?
* How quickly can operator change displays (during normal plant conditions and during plant upset conditions)?
* How many windows can be simultaneously (yet conveniently) displayed?
* To which extent is the system capable to distinguish between events (operator actions), process alarms and DCS diagnostic alarms? Will the system historize such information in separate files?

How easy can the operator recall for display each type of alarms or the events?

* How easy and convenient (e.g how many buttons involved) is it to perform key functions like changing the mode of the controller or adjusting a given set point?

4. Configuration Ease

The factors clarifying the configuration ease, some of which are listed below, should be presented in detail by each of the vendors bidding on the project during a clarification meeting where the vendor shall use any of his convenient resources (DEMO equipment, video tapes, overhead projectors,...etc) to clarify

how easy it is to configure his DCS to the committee members.

* How complex is it to do the configuration (building of a dynamic graphic, a simple control loop, a complex control loop, logic/sequencing function, ... etc)
* What kind of access control is there to the configuration (key or pass word or both)?
* How well is on-line configuration done? How hard is it to make a range change on an input?
* Can off-line configuration be done on a personal computer? If so, how is the transfer done to the controllers?

5. Software and Functional Capability

The factors clarifying the software and functional capability, some of which are listed below, shall be clarified out of vendor offer and/or during clarification meeting.

* What is the DCS maturity? I.E, How long has this product been on the market?
* How are system software revisions made? I.E, is it via read-only-memory (ROM) changes or software down loads to RAM?
* What algorithms are available? can the system perform automatic and dynamic on-line tuning of PID control loops?
* To what extent can the system perform sequential logic? where will this software reside (in controller or in a network device)?
* Can the system perform advanced control strategies such as predictive control? Where will this software reside (in controller or in a network device)?

6. Maintenance Ease

The factors clarifying the maintenance ease, some of which are listed below, shall be clarified out of vendor offer and/or during clarification meeting.

* Does maintenance require board changes or chip changes?

* Are light-emitting-diode (LED) indicators provided on all of the boards in the system?
* Does on-line diagnostic programs run continuously?
* To what extent is the system capable of reporting diagnostics (unit fault, or graphic(s) showing system detailed architectural view to indicate fault location?
* Are mean-time-to-repair MTTR statistics available?
* What is the status of system parts availability and the cost of spare parts?
* Is an instrument-technician training program available? and what are the costs and locations of such training programs?
* How fast could vendor personnel become available to trouble shoot the system at the plant?

7. Vendor Capability

The factors clarifying the vendor capability, some of which are listed below, shall be clarified out of vendor offer and/or during clarification meeting.

* How long has this vendor been in the DCS business?
* What is it's international market share?
* How many of vendor DCS's are currently operating existing plants in your country?
* What is the full scale of training courses available (operational, engineering/configuration, technician maintenance)? and where could these courses be received?

The Second Step in The Procedure

Is to give a weighting to each item on the matrix except price and delivery. This weighting number should range from 1 to 10. With 10 being most favorable. You can have more than one item with the same weight. For example, if you think that "software & functional capability" is as important as configuration ease and that they are very important, then give them both a 10 weighting factor.

This number should be agreed by all committee members: it is likely that some compromises will be needed to achieve a consensus. After all, what is most important to operations is not necessarily most important to engineering. Remember, however, that this system must meet the need of all.

The definition of the terms shown on the matrix chart relevant to the technical factors are described as follows:

$$W_T = \sum_{i=1}^{n} W_i$$

where : W_T is the total weight W_i is the weight of the i^{th} item

$$S_T = \sum_{i=1}^{n} S_i$$

where : S_T is the total score (due to technical factors only) S_i is the score of the i^{th} item

$$S_i = W_i \times R_i$$

where: R_i is the rating of the i^{th} item

$$S_T = \sum_{i=1}^{n} W_i \times R_i$$

W_{TPCT} = 100, where W_{TPCT} is the total weight in percent (made equal to 100, i.e.

$$W_{TPCT} = W_T \times \frac{100}{W_T}$$

$$S_{TPCT} = S_T \times \frac{100}{W_T}$$

Where S_{TPCT} is the % of total score (due to technical factors only).

To be able to proceed with the formulation of the procedure, we have to define the criteria for identifying the best DCS out of the technically accepted DCS's, in the following manner:

The most favorable technically accepted DCS's shall have their scores lying in the top most 10% range of scores ST (i.e difference between the top most score and the lower most score is 10%). Any DCS with a score STPCT which is lesser by more than 10% from the top most score shall be discarded, unless the number of bidders lying within the top 10% score range is less than 3 (or less than the number required by the company/country regulation), in this case the DCS with the next highest score shall be qualified, and this procedure shall continue, up till the required number of bidders is reached.

The Third Step in The Procedure

is to consider the impact of system price and delivery on the selection of the best DCS. The criteria for considering the DCS's within the top 10% range of the technical scores which was developed in the second step of the procedure shall be used as the basis for the formulation of the criteria to determine the impact of the price and delivery.

Criteria for Price Impact

The price impact will generally follow the and serves to give in one single picture the whole basis for the evaluation processfor selecting the one DCS that is the best for the concerned project.

It is worth noted that a spread sheet software package like LOTUS 123 or EXCEL or similar could be used to construct the matrix chart, thus automatically solving all shown equations.

Conclusion

This paper has presented a specific, yet flexible, technique for the evaluation and selection of the best DCS for a specific application. The technique utilized the MATRIX CHART WITH WEIGHTED FACTORS CONCEIT.

TWO NEW CONCEPTS were introduced to allow consideration of the impact of price and delivery on the DCS score (which is normally calculated based on technical

factors only), so as to allow for a final (overall) DCS score that should directly reveal the one DCS that best suits the concerned application.

The mathematical foundation and derivation for the relationship between technical score price impact (P,), delivery impact (D,) and final score Sfpcr was shown. The step-by-step calculating equations were indicated on the MATRIX CHART in a manner suitable for implementation with a spread sheet software package like the LOTUS 123, EXCEL or similar, where all equations could be solved automatically.

Acknowledgements

The Author wish to thank all colleagues who helped preparation of the shown material.

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Theory of Operation for The Programmable Logic Controller (PLC) and Sequential Control

"Today, with microprocessors and more sophisticated hierarchical programming concepts, it is theoretically possible, though not always cost-effective, to replace or parallel humans in many continuous-flow and batch-flow operations. For industries requiring a lot of assembly steps, like automobile making, however, even theoretical total automation seems a long way off".

Thus, it was originally in the automotive assembly line, faced with costly scrapping of controls due to changes during model changeovers, that the first PLCs were installed in 1969 as an electronic replacement of electro-mechanical relay controls. Here the PLC presented the best compromise of existing relay ladder schematic techniques and expanding solid-state technology. It eliminated the need of costly rewiring of relay controls, reduced downtime, increased flexibility, considerably reduced space requirements, and presented a more efficient system.

A Programmable Logic Controller, according to NEMA standards, is a digitally operating electronic apparatus that uses a programmable memory for the internal storage of instructions that implement specific functions such as logic (interlocks, alarms & sequencing), timing, counting, and arithmetic, to control machines and processes. The detailed description and operation of PLC's and their relation to standard relay control schemes will be explained in the following chapters.

In the 1980s a new breed of PLCs has become available. This new breed, in addition to logic, timing, counting and arithmetic functions, has the capabilities of performing advanced process control and process monitoring functions.

Initially, due to development costs and relatively high costs of custom packaged chips and semiconductors, the installed cost of PLCs compared to equivalent relay systems was considerably higher. But, due to rapid developments in the semiconductor industry, development of solid-state memories, and LSI (Large Scale Integrated) chips that are now available at extremely low costs, PLC prices have shown a steady decline (See Figure 1.1), in 1978, the initial cost of a programmable controller installation became the same as the installed cost of an equivalent relay control system. As labor and maintenance costs continue to rise in industry, the cost of PLCs shows a downward trend through the 1980s. With the present trend, PLCs will continue to gain wider acceptance in industry - including the process industry.

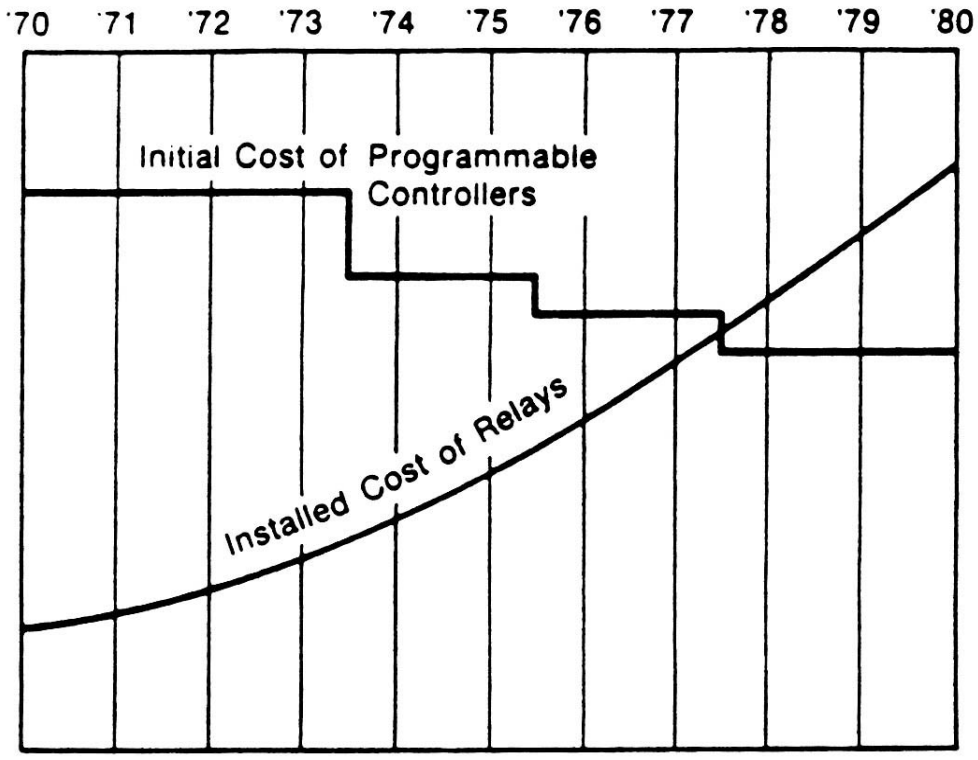


Figure : ! PLC prices have shown a steady decline, and now compare favorably with relay systems.

Relay system costs are greatly affected initially by installation labor costs. Later, trouble-shooting and rewiring as required by changes in the control scheme affected the cost. Solid-state PLCs will get less expensive as they get more compact and are widely accepted.

Let us look at the technical characteristics of each of the relay systems and the PLC and compare them.

Relays and their associated contacts are "hardwired", point to point, in accordance with the circuitry shown on the relevant ladder diagram. It is usually very difficult to make changes in the field, especially where additional relays are required, or contacts are to be reversed (from normally open to normally closed).
Programmable controllers are usually considered where speed and reliability are most important. For example, the average electromechanical relay will operate in 6-8 ms, whereas microprocessors require only 2-3ms.

Programmable-controller logic is generated using fixed software routines programmed to conform to the interlock logic required. Keyboards are furnished with the controller, and programming may be done using ladder diagrams. The complete system consists of input buffers, logic modules, and output buffers.
The input buffers condition the signal from the external field contacts (e.g., thermocouples) to the logic module. Output buffers condition the signal from the logic modules to the final controller (e.g. solenoid valves). Logic modules are built using solid-state components on printed-circuit cards. They generate the logic functions (OR, AND) which are equivalent to series and parallel contact configurations in a relay matrix.

Logic functions (OR, AND) operate in a binary mode (0,1); thus, undesired external pulses may affect the status. Since the state of the logic is determined by a pulse (change in contact status), the effects of contact-bounce when the external switches

are activated must be minimized. This can be eliminated by ensuring that a pulse must be held for a minimum time before its status is recognized by the logic circuity. Input/output modules may be either the isolated or nonisolated type. Isolated modules require a separate external power supply, each with its own fuses and circuit breakers, to drive the input or output components. Nonisolated modules use a common bus - thus all components are powered from the same source.

Solid-state circuity will fail safe when deenergized, in the same way that relays do. Thus, in the event the power should fail, the logic outputs will return to Logic 0. Outputs modules that are used to drive a.c. components use triacs as solid-state switches, to open and close circuits electronically. These units may fail "shorted", i.e., with the contacts remaining closed. If this is so, when the logic calls for them to turn off and deenergize the final element, they may fail to do so. To prevent this, it may be necessary to monitor the output of the module and use the internal logic to disconnect the external power, thereby effectively deenergizing the final control element.

Reliability is measured by the Mean Time Between Failures (MTBF). It has been determined that, beyond the infant mortality (early failures), the MTBF of solid-state circuits surpasses that of electromechanical relays. To reduce the early failures, the units are "burned in" for a period of time before delivery. The relay's life span (ca. 20,000 cycles) is affected by its frequency.

Solid-state circuits require much less maintenance than that required for relays. Component failure is minimal as there are no mechanical parts and the components are conservatively designed. Heat dissipation is provided and overvoltage protection is built in. On the other hand, relays require more maintenance, since coils fail, contacts oxidize, and springs lose their tension. In systems that are static for long periods, relays can become inoperable. Therefore, it may be necessary to use redundant relays to increase reliability.

BASIC CONCEPTS OF THE PLC

A programmable controller, as previously defined, is essentially meant to replace relays, timers, and sequencers in traditional relay control systems, and is designed for installation and operation in industrial and process plants.
PLC's are primarily used for logic control, although in the last few years, a new breed of PLC's acquired the additional capabilities of performing analogue process control and process monitoring functions.

In this course, the basic function of the PLC, which is "logic control", shall be considered.

One example of a process which requires logic sequence control is an ION exchange package used in a water treatment plant. One of the four typical ION exchange trains comprising the package is shown in Fig. 2.1. In any one train, if any of the following conditions occurs, the train will undergo a regeneration cycle.

1. Low totalized through-put flow.
2. High conductivity outlet from anion exchanger.
3. High silica outlet from anion exchanger.

Regeneration Cycle Main Steps:

Step 1 Stop water feed by closing XI then Yl then Z2.

Step 2 Back wash cation exchanger by opening X3 and X2.

Step 3 Stop backwash of cation exchanger by closing X2.

Step 4 Start backwash of anion exchanger by opening Y3 and Y2.

Step 5 Stop backwash of anion exchanger by closing Y2, Y3, X3.

Step 6 Start regenerants addition simultaneously to cation & anoin exchangers and allow the spent regenerant to drain (for a specified period of time), by opening X4, Y4, X5, Y5.

Step 7 Stop regenerant addition by closing X4, Y4

Step 8 After a specified time delay, start rinsing cation exchanger for a specific duration by opening XI.

Step 9 Stop rinse water going to drain by closing X5

Step 10 Start rinsing anion exchanger with rinse water from cation exchanger by opening Yl.

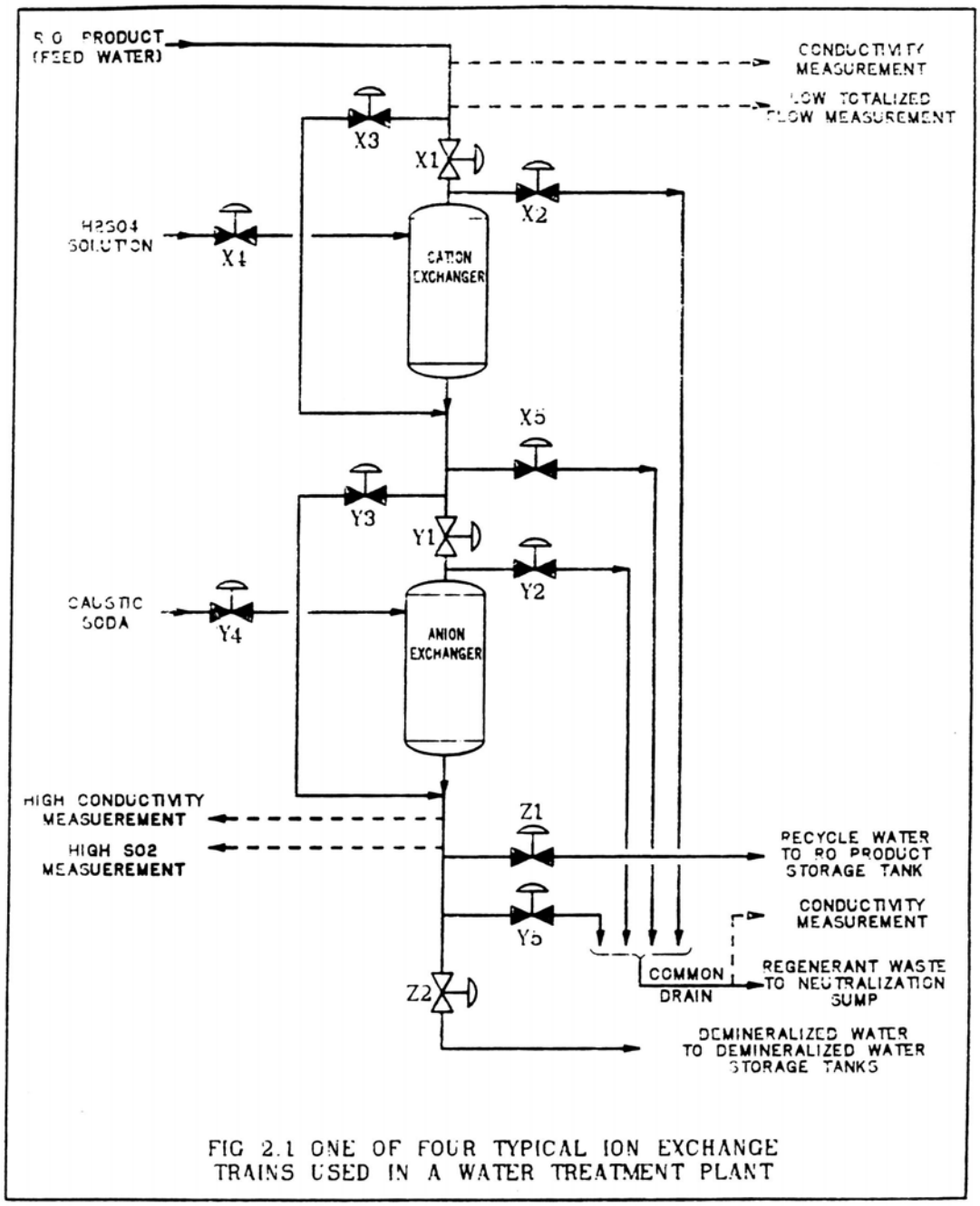


FIG 2.1 ONE OF FOUR TYPICAL ION EXCHANGE TRAINS USED IN A WATER TREATMENT PLANT

Step 11 Continue rinsing both exchangers with rinse water going to neutralization sump till such time the conductivity of the feed water equals that of the water going to sump. Then close Y5 and open Zl to recycle rinse water back to feed water storage tank for a certain period of time.

Step 12 Put the train (one of four typical trains) which has just undergone the regeneration cycle to service if only two other trains were operating, or put the train on standby if all the other three trains were operating so that at any time, 3 out of 4 trains are operating and one is standby.

Step 13 Before lining up the standby train to service, rinse the train again to satisfy the conditions in 11.

The above sequence and conditions could be transformed into a PLC program which could take several methods of presentation, one of which is the PLC ladder diagram which is a common method of presentation by all PLC manufacturers.

Typical PLC system architecture is shown in Figure 2.2. Models by Texas Instruments, Gould-Modicon, Allen-Bradley, Struthers-Dunn, Tenor Co., General Electric, Square D, Cutler-Hammer, Restbury, Siemens, Foxborow, Westinghouse, and other manufacturers are available. The models have various levels of capabilities and complexities, and are continuously updated and improved to meet industry requirements. Figure 2.2 shows the CPU, memory, power supply, input/output section, and programming device. These main blocks and their functions, which are basically the same in all available PLCs, are explained in the following sections.

The Central Processing Unit (CPU)

The CPU is the heart of a PLC, computer, minicomputer, or microcomputer because it receives instructions from the memory and generates commands to the output modules. Input commands, device status, and instructions are converted to logic signals, "1" for input present, and "0" for output signal in positive logic. These logic

signals are then processed by the CPU. As in traditional ladder diagrams where the NO/NC contacts of the field devices activate relays and timers, PLCs process logic signals and activate output TRIACS that can be normally, energized or de-energized. In a relay control system, NO/NC contacts from relays are available for use in the control scheme. Similarly in a PLC, internal and output coils have NO/NC contacts that can be used in the logic scheme. In contrast to hard-wired relay control systems, no wiring is needed for implementing the control logic in a PLC. All sequence control logic is internal to the PLC, and is processed by the CPU as explained.

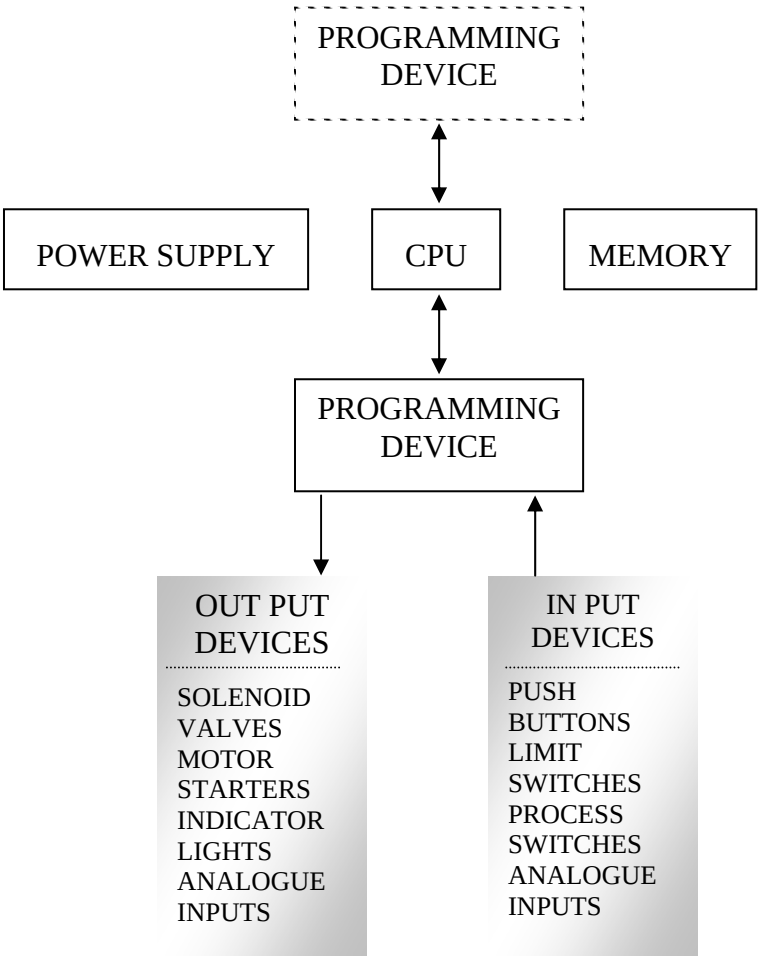


Figure 2.2 PLC Systems Have Basic Architecture

That Are Usually Similar

Available PLCs are based on various microprocessor chips which are preprogrammed with a main "executive program." The executive program enables the CPU to understand input command instructions and status signals, and provides logic processing capability. These capabilities (solving simple "and/or" logic, timing, sequencing, addition, subtraction, multiplication, division, and counting) vary from different models and manufacturers and often affect the hardware price. A careful study of the requirements of a control application should be made to decide the features currently needed and ones that may be needed in the future.

MEMORY

Memory in a PLC is where the central program is stored. The CPU utilizes program instructions stored in memory to tell itself to scan certain inputs and then to generate output commands. Memory capacities vary, and generally store 256, 512, 1024(IK), 2K or 4K words, depending on word size. Some models may have higher capacities. The memory size furnished in the PLC varies with the size of the control functions to be performed, and should be carefully selected only after evaluating present and future needs.

Various PLCs have different limitations on the number of horizontal and vertical contacts that can be programmed into each step. This affects net memory usage for a given ladder schematic (refer to the example under "Programming"). Also, the number of words of memory used per contact varies from model to model (Modicon uses two words per contact, whereas Square "D" uses one word per contact), and should be checked before selecting a particular memory size. Some PLCs use part of the advertised memory for the "executive program", which thereby reduces the available memory for the control program.

An important concept to understand in the operation of a PLC is "Memory Scan." A typical multi-node format network used by the Modicon Model 484 is shown in

Figure 2.3. Some PLCs can be programmed only one horizontal rung at a time. Their maximum number of vertical elements is also limited. In PLCs such as Texas Instruments' Model 5 TI, there is no limitation on the number of series or parallel elements that can be used in a logic line or step.

As a typical example (Figure 2.3) the controller will solve each network of interconnected logic elements (NO and NC contacts, timers, counters, etc.) in their numerical sequence - the order in which they were programmed. The first network is scanned from the time that power is applied, first from top left to bottom left, and then continuing to the next vertical column to the right. Within a network, the logic elements are solved during the scan, then the coils are appropriately energized or de-energized to complete the scan. Since the scanning rate is very fast (4 milliseconds for 250 words, to 20 milliseconds for 4K memory for a Modicon 484) , it appears that all logic is solved simultaneously. The result (change in coil state, numerical values, etc.) of each network scan is then available to all subsequent networks. Thus all inputs and outputs are updated once per scan. The time from solving any individual network on one scan until that network is again solved on the next scan is defined as the scan time of the memory. It depends upon the complexity of the programmed logic and memory size. For memories with a longer scan time, a fast close/open input signal could possibly be missed by the PLC scan. In such a case, a push button would have to be held in longer. Though this seldom presents a problem in the average control systems, it should be considered.

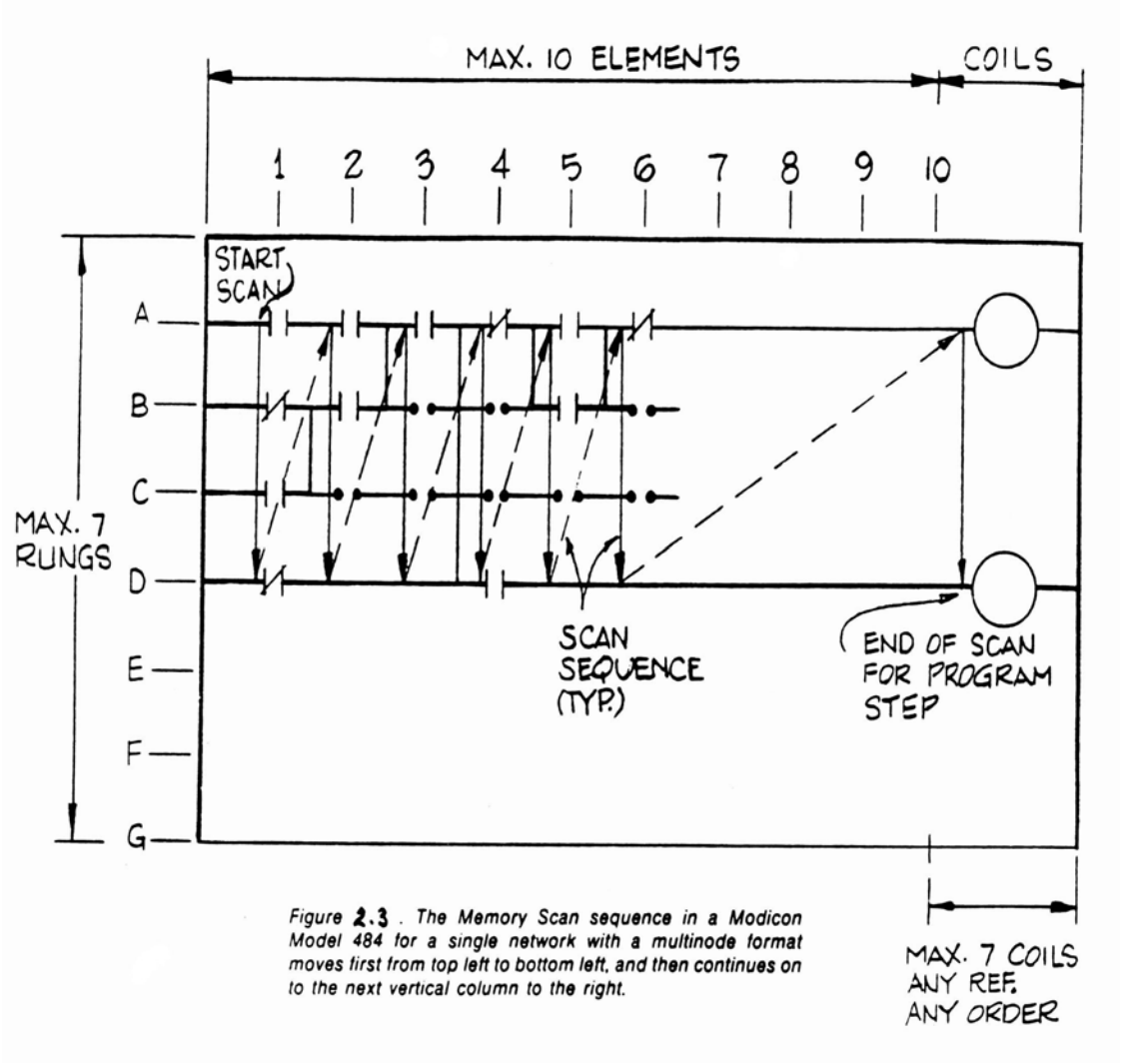


Figure 2.3 . The Memory Scan sequence in a Modicon Model 484 for a single network with a multinode format moves first from top left to bottom left, and then continues on to the next vertical column to the right.

Different forms of memory are available. The trend is towards using solid-state CMOS memories. Memory types include:

1. Torodial Core Read/Write Memory: extremely flexible and easiest to reprogram, but is susceptible to voltage transients. In each scan, the program memory is read, stored in a register, and then rewritten in the original location.
2. Erasable Programmable Read-only Memory (EPROM): offers high noise immunity, and, for reprogramming, the original program can be erased by exposure to ultraviolet light. Portable erasing equipment is now available. These memories are also known as LEROM (Light Erasable ROM).

3. Random Access Memories (RAM): volatile, require battery backup in case of power loss. Generally, a nickel-cadmium cell(s) is provided on the memory boards to ensure that the program is not lost. Some models offer a combination of RAM/PROM (Programmable ROM), in which the program is first developed on the RAM and then transferred to the PROM.
4. Electrically Alterable ROM (EAROM): nonvolatile, so battery back up is not required. Electrical alteration is possible via the "programmer," which can be either hand-held or a CRT type, where the program steps are displayed. PLCs with EAROM are more desirable because it is possible to add new program steps or revise the existing program in the processor while it is operating in the RUN mode. Gould, in their MODICON PLCs, and Square "D", in their SY/MAX-20 models, offer this feature.

Various models offer "scratch pad," or "trial" read/write memories in addition to the main memory. These enable the programmer to make changes, add to or delete from the program, debug the program, and then transfer it to the main memory.
An additional feature is "memory protect." A key interlock is provided to prevent unauthorized tempering with the stored program.

POWER SUPPLY

The power supply is an integral part of the PLC and is generally mounted -in the mainframe enclosure. Line power specified is converted to the appropriate DC voltages required by the solid-state circuitry and memory.

For volatile memories that require constant power to retain the stored program, DC cells are provided to ensure retention of the memory in case of main power failure. The power supply is designed to operate both the CPU and the basic number of inputs and outputs. For expanded input/outputs, an optional heavy-duty power supply usually has to be specified.

The power supply generally is designed for a controlled "power-down" sequence in case line power is lost. In this case, the CPU stops solving logic and retains the status of all coils, inputs, outputs, and registers. The outputs are all turned off. This eliminates the possibility of failure in an undetermined mode. In the "power-up" sequence, the stored inputs, outputs, status of coils and registers are checked, and then the memory-scan sequence is started.

Input/Output Section

One of the main characteristics that has made PLCs extremely attractive is that the input/output modules are designed to interface directly with industrial equipment. Input modules are generally available for interface with a wide variety of signal levels; for example, 120 VAC, 24VDC, 48VDC, 4-20 maDC, 5 VDC (TTL). Most manufacturers offer optically isolated inputs, which permit mixing of discrete and analog inputs, and prevent transients on the field wiring from affecting the internal logic. Input cards (modules) for each type of input signal are of plug-in construction and can usually be inserted or removed without a system shutdown. Most manufacturers now offer a status indicator light for individual inputs on the card.

Output modules are also available in the same wide variety of voltage ranges as are input modules. Each output is optically isolated and fused, and is available with output status indication. Field devices such as small motor contactors, valves, solenoids, and lights can be directly operated from the output modules. In some models the input/output section is directly connected to the mainframe, while in other it can be remotely located if the CPU is kept in a central location.

There is a basic difference between a PLC and a standard relay control system with regard to input/outputs. A section of a relay ladder diagram is shown in Figure 2.4. Its equivalent PLC input/ output diagram is shown in Figure 2.5.

Note that all field switches are wired to input points identified by the PLC numbers 1001, 1002, etc. A closed-field contact, such as PSH-351, essentially energizes an internal PLC relay 1001, and all internal NO contacts referenced to 1001 in the logic will close (NC contacts of PLC relay 1001 will open). If PSH-351 opens, the PLC relay 1001 will deenergize and all references to NO contacts in the logic program will open (NC reference contacts will close). The logic operations during the scan are done on the internally programmed reference contacts shown in Figure 2.5.B. If motor starting conditions are satisfied, output coil 0001 energizes and seals in. This causes the output TRIAC labeled 0001 to energize, which in turn energizes the motor contractor MC and starts pump P-101. The alarm output 0016 is wired to a 24 VDC output module as shown in Figure 2.5.A.

Programming Devices

The programmer for a PLC is the device (usually, an external unit) that transforms the control scheme into useful PLC logic. The logic program is then stored in memory, where it is made available to the CPU for logic operations.

Various kinds of programming devices are available from PLC manufacturers. These range from a CRT Programming Panel, a hand-held calculator-like device, a thumbwheel input system, a cassette tape loader, or a hookup to a central computer or programmer through a telephone interface. For simplicity and compatibility with existing relay ladder schemes, most programming devices use either standard relay symbols for NO/NC contacts, timers, counters, etc., or use Boolean terminology (AND, OR, NOT, etc.) Thus, there is no need to learn a sophisticated programming language or to redraw standard ladder diagrams in special format to program a PLC.

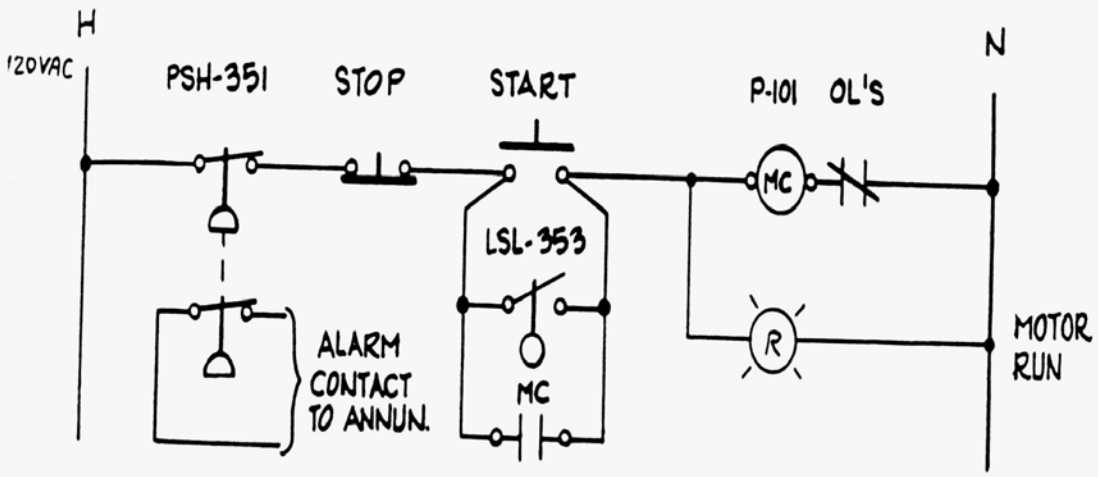
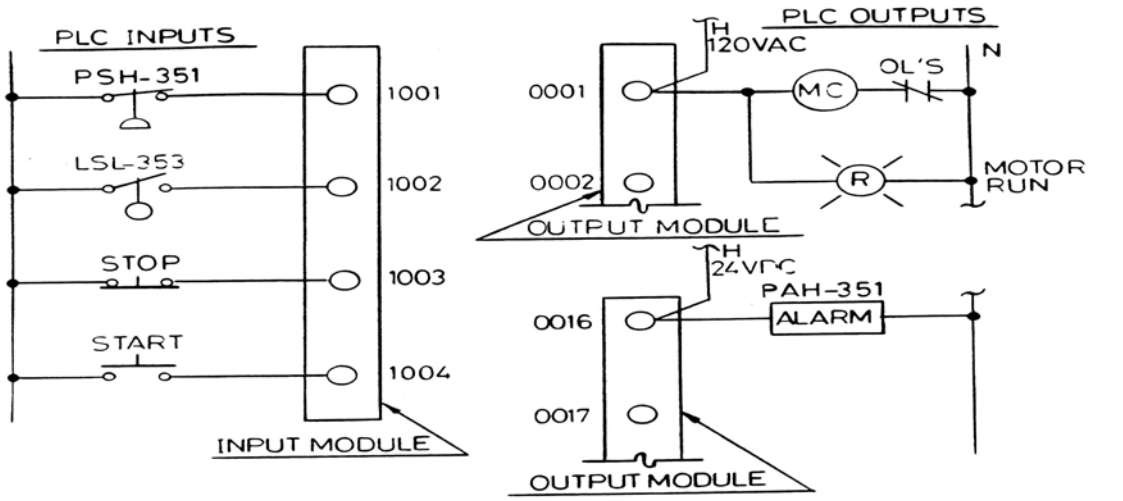
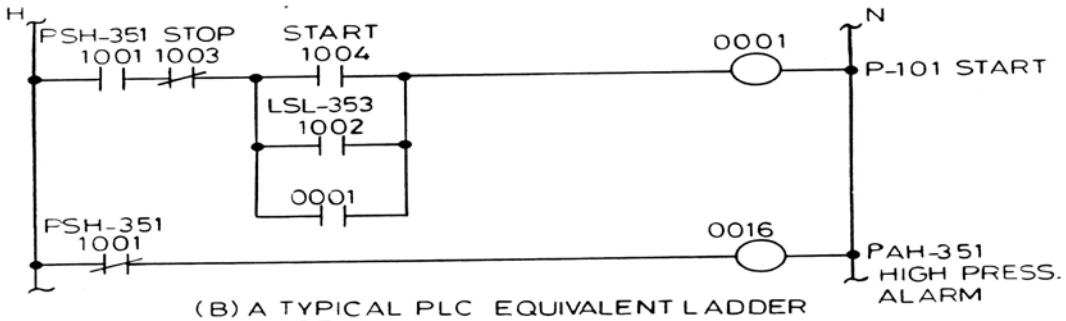


Figure 2.4 A typical relay system ladder diagram.



(A) TYPICAL PLC WIRING TO INPUT/ OUTPUT MODULES



(B) A TYPICAL PLC EQUIVALENT LADDER DIAGRAM

FIG. 2.5 DIAGRAMS SHOWING TYPICAL PLC WIRING TO INPUT/OUTPUT MOUDULES AND EQUIVALENT LADDER DIAGRAM FOR THE SYSTEM SHOWN IN FIG. 2.4

In. addition to ease and flexibility of programming, most programming devices have a power flow light that enables the user to modify or troubleshoot the system. In CRT-based devices, visual display of the logic ensures the programmer of the accuracy of the punched-in program. Read/write or "scratch pad" memories are offered in some systems. These allow the programmer to add, delete, or modify the program before entering it in the main memory.

Some manufacturers offer, as a service to their customers, factory-written programs loaded into the PLC via a telephone interface. Programs can also be stored on cassette tapes and loaded into the PLC through a special interface unit.

One typical example of a programming panel is shown in Figure 2.6.A. Logic is entered line by line. The programming panel is plugged into the service port of the programmable controller. A new program is manually entered into the memory by pushbuttons marked with the same symbols as the four relay contact types. The line number and the individual contact reference number are dialed on a thumbwheel; one of the pushbuttons marked with the element position (A, B, C, d, output) is then pushed; then one of the pushbuttons, marked with the logic symbol, is then selected and pushed. The programming panel contains other types of logic functions, including time delay or elapsed time and event counting for control of a logic line output. The logic program lines are designed with seal circuits for momentary contact inputs or with latch relay circuits for retentive output action in case of power failure.

The programming panel is a valuable tool to check the functional operation of programs. It is used to isolate the logic program from field wiring and to test the output by a simple pushbutton procedure. Individual field inputs or outputs are easily checked. The simulation testing of logic sequences is initiated by dialing a line output or input contact. Then, the logic operation is disabled and the output condition is turned on and off manually, by pushbutton. By this method, an output

signal determines whether field wiring is correct. When the final operator is activated, the external wiring is proved. The logic program is reactivated for the simulation of input contact conditions to establish whether or not the logic design and operation are correct. If a change is desired, it can be made quickly, by pushbutton, without removing a single wire or relay. The design checkout is fast and complete.

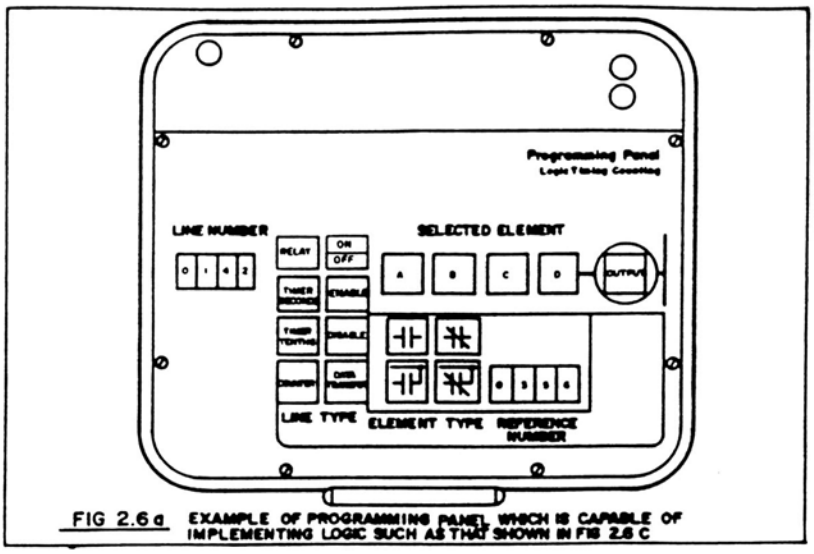


FIG 2.6 a EXAMPLE OF PROGRAMMING PANEL WHICH IS CAPABLE OF IMPLEMENTING LOGIC SUCH AS THAT SHOWN IN FIG 2.6 C

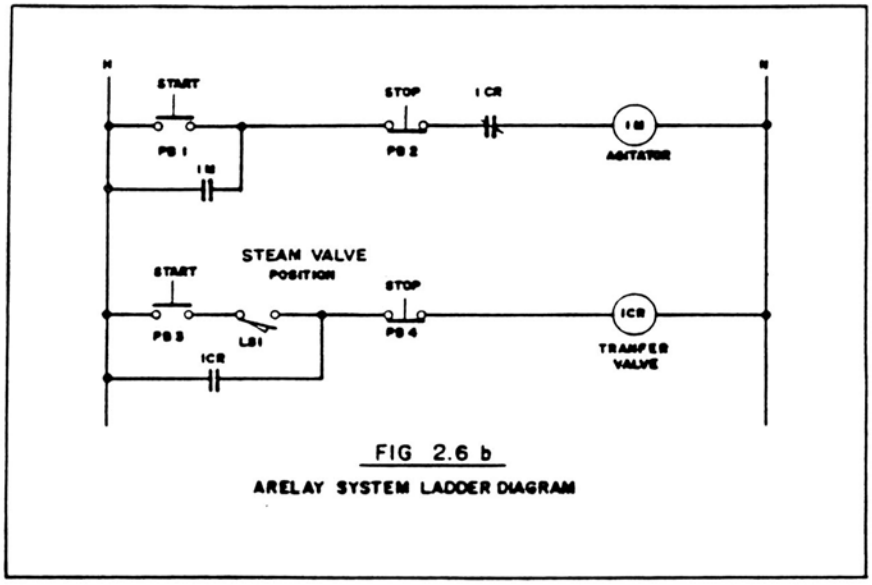


FIG 2.6 b
ARELAY SYSTEM LADDER DIAGRAM

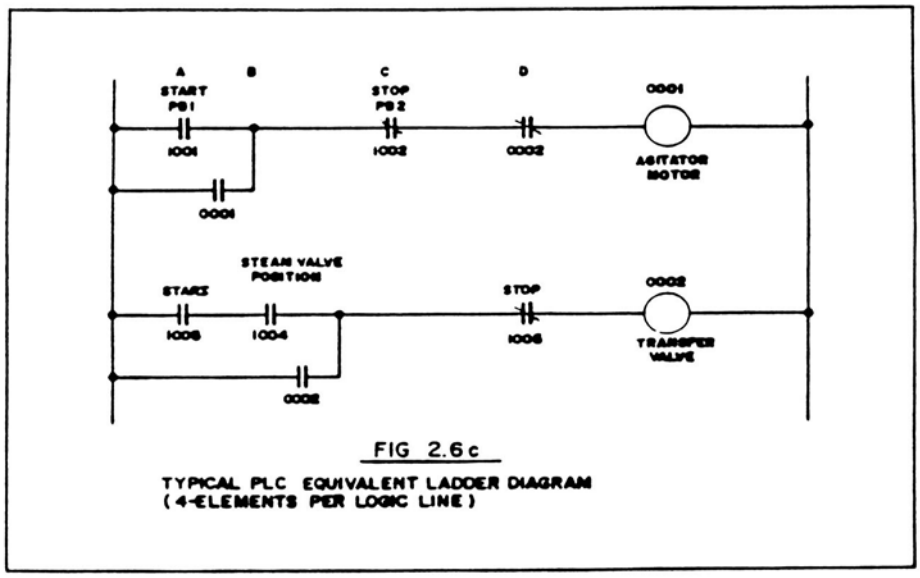


FIG 2.6c

TYPICAL PLC EQUIVALENT LADDER DIAGRAM
(4-ELEMENTS PER LOGIC LINE)

Programming

Programming a PLC can be simple and straightforward when approached in a systematic manner. As an example, convert a control scheme in relay ladder diagram form to a PLC ladder schematic for a Gould MODICON 484. A partial schematic for an oil heater burner ignition system is represented in Figure 2.7. The addresses of the input/output relays are assigned in a way similar to that shown in Figure 2.4 and 2.5. It is essentially a matter of converting the relay ladder control logic to PLC logic, using NO/NC contacts with the appropriate addresses to coincide with the input/output assignments. This is achieved by:

1. Draw a relay ladder schematic of the control scheme.
2. Assign an input address to each field input device (pressure switches, temperature switches, pushbuttons, selector switches, etc.). These field contacts are wired to the Input Modules, per their respective address assignments. Generally, most PLCs allow any number of references (NO/NC contacts) to the input or output coils. Where an output is not required, internal "coils" are used. They are referenced to NO/NC contacts (e.g., coil No. 0258 in Figure 2.8).

3. Assign output relay coil numbers, where required, for solenoid valves, motors, indicator lights, alarms, etc. The output coil assignments and module types must be compatible with the system control voltage.
4. Then, enter the program into the processor using the CRT Programming Panel (Figure 2.8 shows how the screen display will look. It represents the PLC ladder schematic for Figure 2.7.

Once the input assignments have been made, the field wiring can be hooked-up, and control logic changes carried out by addressing the particular program step. Some PLCs allow for on-line changes. Also, outputs can be locked into an energized/de-energized position, the program change made, and the output re-enabled.

Each type of programming device requires a different approach to actual program loading. A sample program done on a Texas Instruments 5 TI 2000 Series R/W programming device is shown in Figure 2.11. Programming can be done either directly from a ladder diagram or a Boolean logic diagram. The same program, done on a Modicon P180 CRT programming device, is shown in Figure 2.12. Even though the control functions are the same, a different amount of memory is used in each case.

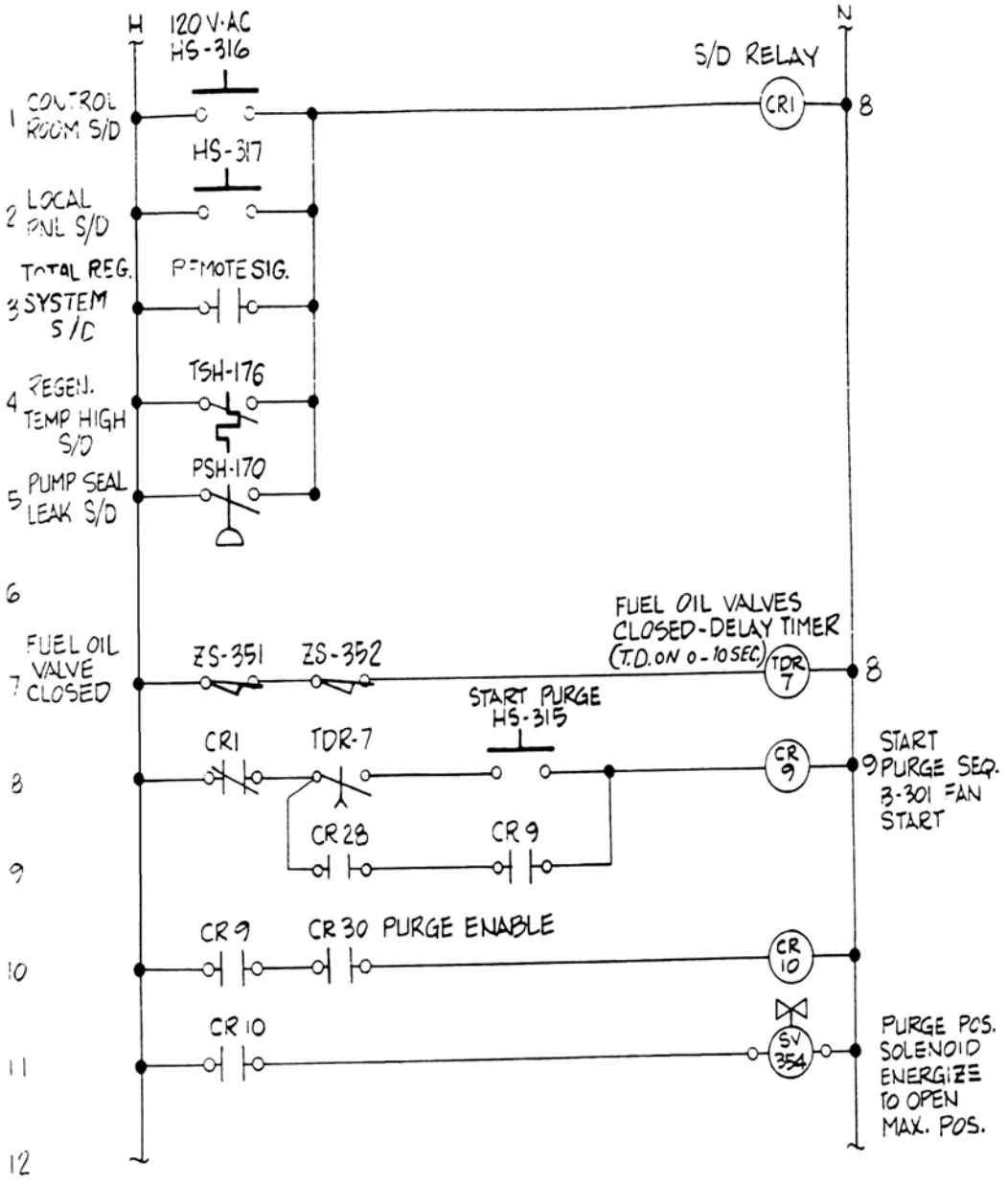


Figure 2.7 A partial schematic of an oil heater burner ignition system.

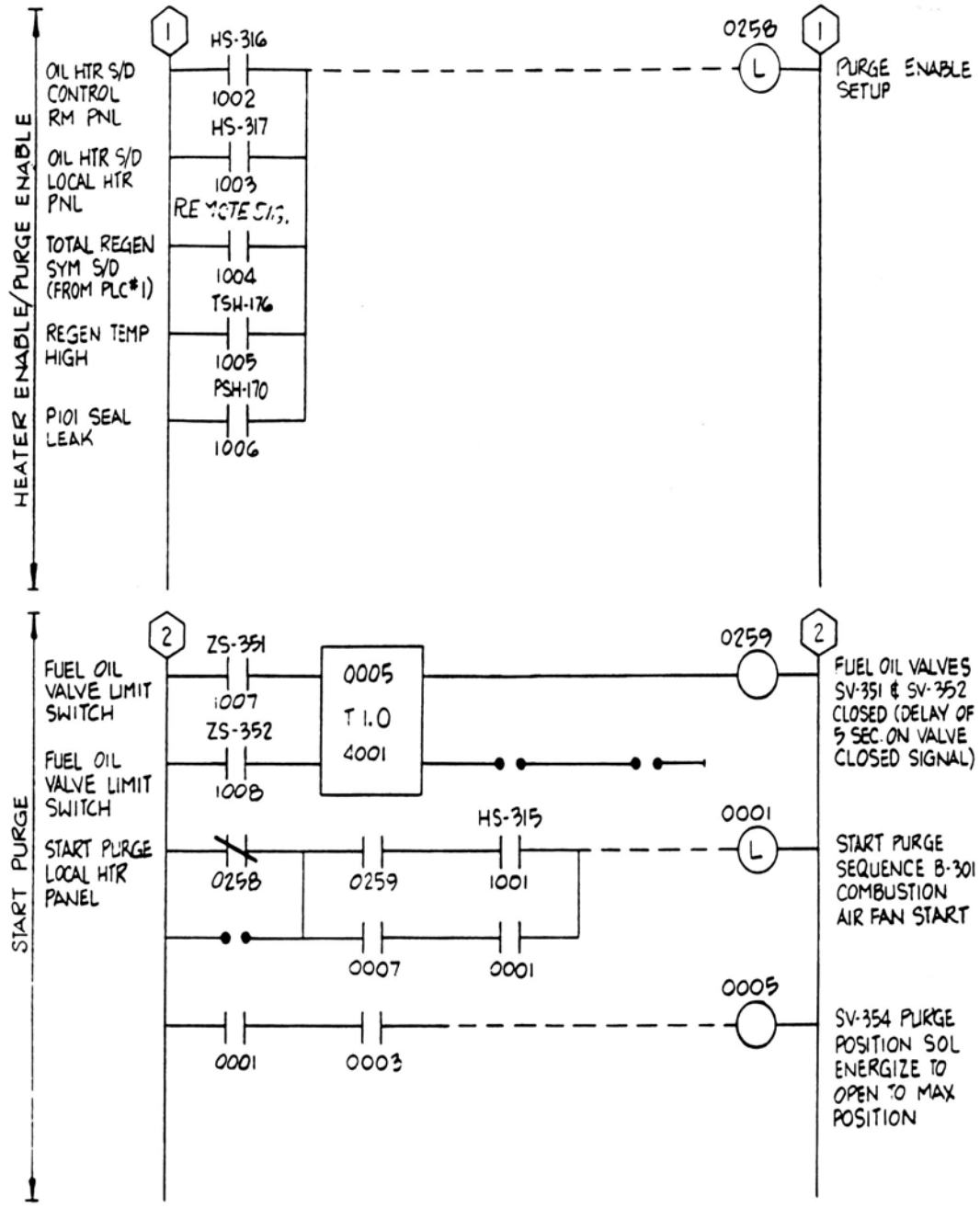


Figure 2.8 A PLC schematic of the oil heater schematic shown in Figure 2.7

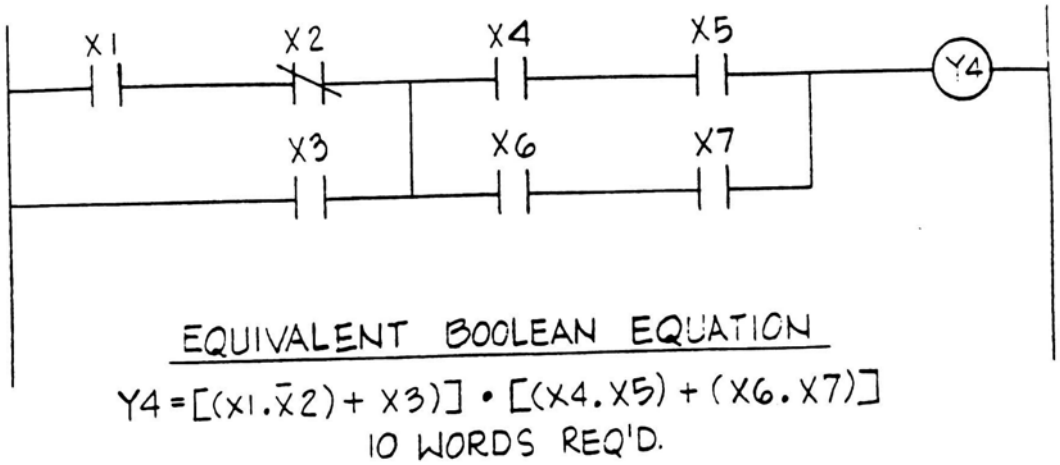


Figure 2.11. A program done on a Texas Instruments 5TI2000 Programmer.

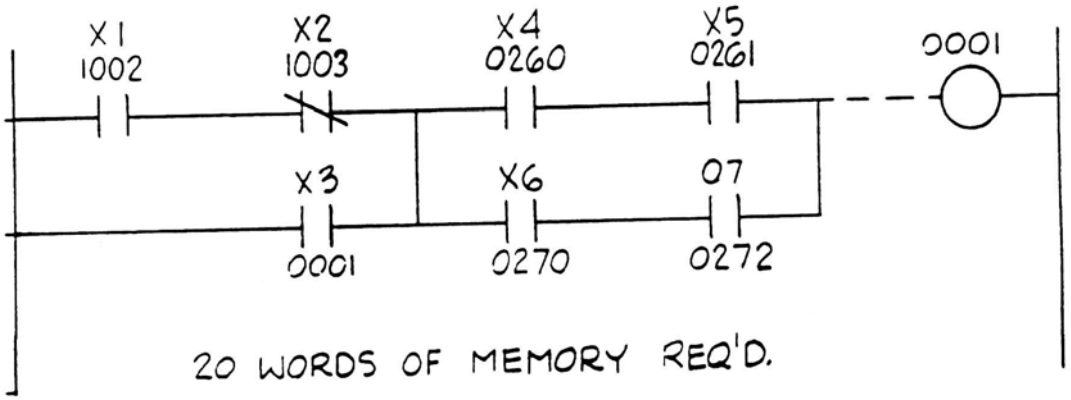


Figure 2.12. The logic of Figure 14.46 done on a Modicon P180 CRT Programmer.

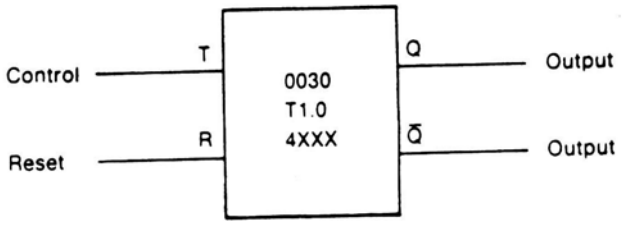


Figure 2.13. A schematic representation of a time delay on timer.

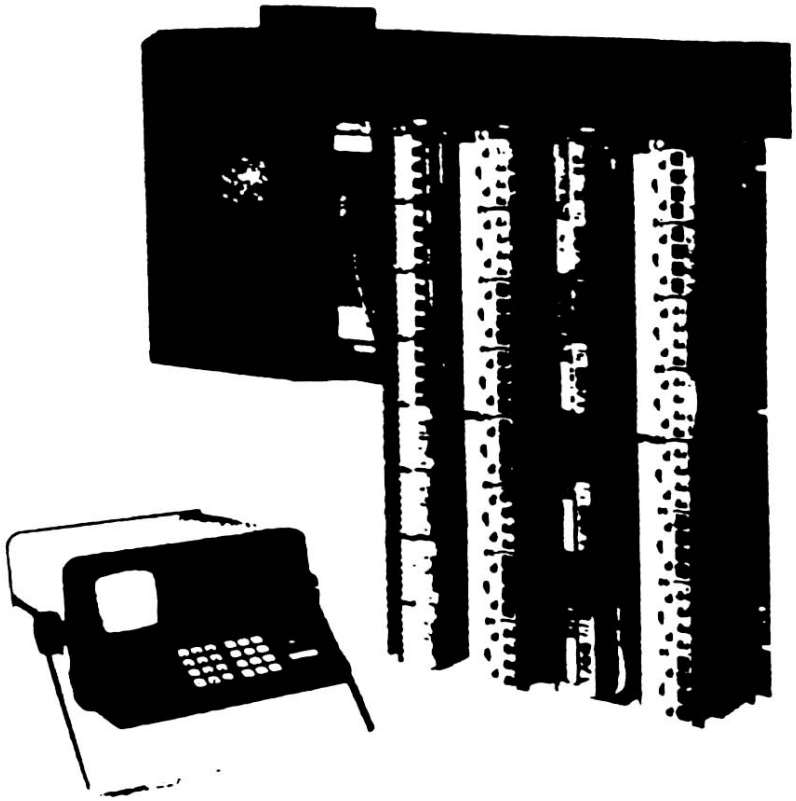


Figure 2.9. A Modicon Model 484 Programmable Controller system. (Courtesy of Gould, Inc.)

Figure 2.10. A Texas Instruments Model 5TI Programmable Controller system (Courtesy of Texas Instruments, Inc.)

Timers and counters can also be programmed easily. PLCs have a crystal-controlled clock signal that drives the timers. Maximum timing periods vary (999 seconds for a Modicon 484, and 54.6 minutes for a 5 TI). However, by cascading timers (counters), any desirable timing period (count) can be obtained. One form of a timer is shown in Figure 2.13.

Timer or counter functions are addressable from the programming panel. In Figure 2.13, 0030 is the timing period of 30 seconds. Tl.O denotes steps of 1 second each, and 4xxx gives the register address in the CPU. The timing function will start when a signal is given at T and when a logic "1" is present at the reset terminal. When the set

period has elapsed, the Q output is energized. When the reset signal at terminal R turns to "0", the timer resets. When R = "1", the timer is enabled. Q signal is the inverse (opposite) of the Q signal. Counters operate in much the same way as timers except they increment the current count by one only when the energizing signal changes to "1"

In addition to the programming features, newer models of PLCs offer mathematical computing abilities (math packages) and enhanced instruction sets. These provide the ability to add, subtract, multiply, and divide, as well as other features. This enables the calculation of new values for process variables and set-points, comparison of field data with reference data stored in memory, and generation of statistical information for display or print out.
Only in the last few years did some PLCs acquire the additional capability of performing analogue process control functions. But as was mentioned earlier in this chapter, this course will be dedicated to the basic function of the PLC which is "logic control".

In the field of programmable logic controllers, the software is playing an increasingly important role, especially where more complex automation tasks are involved. In view of the fact that overall automation costs are being determined more and more by the software costs (see Figure 2.14), the economical development of software and the use of standard programs are especially important.

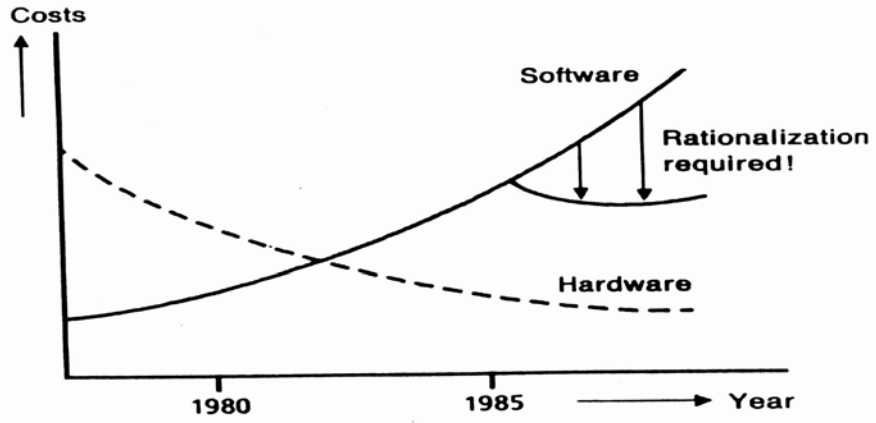


Fig. 2.14 Trends in software and hardware costs

Application of Programmable Logic Controllers

Some examples of actual PLC application in industry are listed, with a brief description of each control system.

I. On Exchange Package Control: Four typical ion exchange trains are used in a water treatment plant for producing demineralized water. One PLC is used for controlling the sequence of operation of the four trains. This example was explained in Chapter

2. Boiler control: A separate PLC is used for each of four boilers in a chemical plant to control the process of purging, pilot light-off, flame safety checks, all interlock and safety shutdown checks, main burner light-off, temperature control, and valve switching for converting from natural gas to fuel oil. The PLC is programmed to be an energy management system for maximum efficiency and safety.

Compressor station control: A compressor station with multiple compressors is controlled by a PLC, which handles start-up and shutdown sequences and all safety interlocks.

Ethylene drying facility: In an ethylene drying facility, where moist gas is first

removed from salt domes and then dried and pumped into the main pipeline, two PLCs are used for controlling the entire operation. One PLC is used to control heater combustion controls and shutdown system. The second PLC is used to control the drying and regeneration cycles in the dryer units. Both PLCs act as "slaves", and are tied into a "master" PLC located in a Central Operations Control Room. The "master" PLC has control over shutdown sequences, and also monitors some of the critical process variables.

Automatic welding: PLC have been successfully applied to the control of automatic welding machines in the automotive industry. The use of aluminum in automobile bodies for weight reduction created a load distribution (AC power) problem because welding aluminum consumes more current per weld than welding steel. To eliminate this, controllers time-share automatic welding machines on a priority scheme that utilizes the data handling and arithmetic capability of the PLCs.

Coal fluidizing process: A PLC installed on a fluidized bed to determine the amount of energy generated from a given amount of coal. A mixture of crushed coal and limestone is blown through jets over a heated bed. Burning rates and temperatures are monitored. The PLC controls the sequencing of the valves, and takes the place of a relay control system. The analog capabilities of the PLC enable jet valves to be controlled by the control system that is doing the sequencing. Control devices are also monitored on a CRT.

Material handling: In a storage/retrieval system controlled by a PLC, parts are loaded and carried through the system in the totes (bins). The controller keeps track of the totes. An operator's console allows parts to be rapidly loaded or unloaded. A printer provides inventory printout, such as storage lane number, parts assigned to each lane, and the quantity of parts in a lane.

These examples show the variety of applications for PLCs in some phases of industry. Arithetic and analog operations capacities are being used in conjunction with normal sequence control, thus making PLCs attractive and cost effective.

Logic System Design

The design of Interlock and Alarms

The need for safe interlocks and alarms is greater than ever. We will examine this need, and detail the different types of failure that interlocks might have to handle. We will than look at the failsafe concept, seven principles of interlock design, and various types of interlocks and alarms. Finally, we will examine the role of power distribution and the human factor in interlock design.

The Need

Today's chemical plants have more complex processes, are larger, and are operated closer to their safety limits than in the past. As a result, these plants are more likely to become unsafe, thus having a potential for causing greater damage, jeopardizing personnel, and resulting in costly shutdowns.

In providing plant, process and personnel protection, the design of plant safety systems, with their associated alarms, is extremely important. These systems must function so that permissible conditions exist before start-up, and the overall operation is safe when abnormal or dangerous conditions arise. In addition, safety systems and alarms serve to minimize personnel operating errors in emergency situations.

An interlock system consists of inputs (e.g., pushbuttons, limit switches, process switches and other external contacts) and outputs (e.g., final actuators such as solenoids) that are related and interconnected to perform a defined function, such as a startup or shutdown, through a logical sequence of events as determined by certain

hardware (such as relay contact arrangements in series, parallel, or in combinations of both) and/or programmed software.

The purpose of the interlock systems is to automatically and/or manually cause a predictable set of operations when process limits are exceeded, mechanical equipment fails, power is lost, or components fail either individually or in combination. The system should operate so as to render the plant safe.

Interlocks and safety systems will not prevent harmful process upsets or catastrophic damage, but will reduce the risk of such occurences to an acceptable level. Since an element of risk is involved and "acceptable level" must be defined, we will now deal with probabilities of occurence, random or undetermined variables, and disturbances outside the system. These factors all contribute to the unreliability of safe operating conditions.

We can be sure that sooner or later any or all of the following will happen: processes will not stay within safe limits (e.g., flammability limits will be exceeded, toxic emissions and decompositions will occur); equipment will fail (e.g., a compressor will malfunction); performance will decay (e.g., heat exchangers will become fouled); utilities will be interrupted; and process control and interlock safety systems will be unreliable. Thus we must make sure that, if all else fails, the plant attains the safest mode of operation, or shuts down as a last resort.

Analysis of Failure

The reliability of system is inversely proportional to the rate at which failure occurs. One of the difficulties in determining system reliability is defining failure. The nature of the failure depends on the effect it has on the system and the output. For example, if a power supply fails and this causes an unacceptable upset in the output, the system has failed. However, if a component or subsystem fails, but the output is not affected or responds as intended, then the system has not failed.

Failure occurs in different ways:

1. **Infantile Failures** - These take place during the early part of the component's life and are usually due to a manufacturing or design defect. Such failures can be detected and eliminated by inspection and burn-in.
2. **Casual Failures** - These appear during the working life of the component. They are distributed according to the laws of probability.
3. Wear Failures - These are due to the progressive aging and deterioration of the components, and determine the useful life of the system. Once this level of failure is reached, there is, considerable increase in the failure rate, making the system entirely uneconomical.

   The above failures relate to a single component. However, when a component is integrated into a system, its failure will affect the integrity of the entire system. This may result in marginal or catastrophic failure:
4. **Marginal Failures** - When these occur, the resulting variations in the mechanical or electrical characteristics do not materially affect the operation of the system.
5. **Catastrophic Failures** These may result from complete breakage, short circuits, open circuits, misoperation, poor maintenance, etc., which may cause plant shutdown, equipment damage, or personnel injury.

Various methods have been devised, such as Fault-Tree Analysis and Hazard Analysis, to determine events that might occur singularly or in combination to cause a potentially unsafe condition. A probability is assigned to each potential failure, and the cumulative effect is evaluated to determine the ultimate risk. An overall analysis of possible failures will statistically determine the possibility of occurence, but will not lead to the most reliable, safest system, because these analyses usually include the most likely failures, not the least likely ones.

"Least likely failures" obviously occur less frequently, and usually reside outside of the system, e.g., plant blackout due to lighting, tripping devices improperly set or bypassed, accidental trip due to maintenance, misoperation of the process, failure to correct or heed early-warning alarms, operator misjudment, etc. These and similar events are less predictable, but can be just as devastating as any others.

Principles of Design

Interlock design should follow these principles:

1. **Every system should fail to its lowest energy state, or to a state away from its critical operating limit.** Each process, or portion thereof, should be analyzed to determine the prime source of energy for operation. The source may be: steam to reboilers; cooling water to heat exchangers; the reaction itself, if it is exothermic; a liquid/vapor mixture, if it is flammable; or any other source of energy that may drive the process to its limits. Decreasing the amount of energy reduces the risk of exceeding equipment design limits or at least minimizes the potential damage if these limits are exceeded. In other words, "remove the fuel from the fire".

2. **Interlock and safety systems should be independent of all other plant and process controls.** It is dangerous to couple interlock systems with plant and process controls. Interlocks are usually designed to override the process control. Therefore, if they are dependent on one another (through common power supplies or transmitter signal outputs), and the process control system fails, the interlock system will also fail. Thus, it is essential that the safety system stand alone, with all redundancy backup it requires to enhance reliability.

3. **Process-control signal failures should drive the final actuator (control valve, pump motor or whatever)to the failsafe direction.** The failsafe direction of the final actuator is determined from process considerations, to remove or limit the-amount of energy in the process (see Point 1 above). The direction of control signals should be consistent, from the transmitter to the final operator, so that if any one component fails, resulting in the loss of signal, the control system will cause the process to fail safe. Insofar as possible, loss of process control should cause the system to fail in the same direction as the interlock.

4. **All electrical components that make up an interlock circuit should be powered from the same power supply or individual circuit.** Thus, when power failure occurs, all components will be deenergized in the failsafe condition. Some interlock systems require backup power (such as batteries) to prevent actuation on loss of power. Circuits that require such power should be separated from those sevred by conventional power supplies.

5. **Interlock circuits and their components should be designed to actuate the final operator in the direction required to cause the process to fail safe upon loss of power.** Thus, on alarm or trip, relay and solenoid coils should be deenergized, and initiating contacts should open. This will ensure that the system will fail safe should there be a loss of power or should any component fail.

6. **Annunciators, alarms, indicating lights and electrical instruments, except those associated with interlocks, should be powered independently from the interlock system.** Each service having a shutdown interlock should be provided with two alarms: an early-warning or precursor alarm that will be activated before the interlock comes into effect (e.g., a flashing display "High pressure"); and an adjustable-trip alarm that can be set to go off when the interlock is activated.
7. **Interlock and trip systems for each section of the plant and its related equipment should be designed so that failure of one system will not affect others.** Interaction between individual interlock systems is usually through the process. For example, a compressor that furnishes process air may trip due to a malfunction. The loss of this air may cause the process to reach its explosive limit, requiring that the feed be shut off. An interlock system on the feed must therefore not be adversely affected by the failure of the compressor system. The interlocks should be designed from a holistic point of view, to ensure that interlock failure will not jeopardize related process systems.

As with all criteria, the above should serve as guidlines. Each circuit or system should be considered separately and in the whole picture, so that its failure will not affect plant safety.

Interlock Analysis

The action taken by interlocks is determined from consideration of the interdependence of the process, equipment, utilities and controls. Interlock requirements must be analyzed to determine not only their necessity but also the subsequent effects of their actions and failures.

1. **Process Failures.** The analysis of process failures depends on the causes for the upsets (e.g., process variables out of limits) and the actions taken (e.g., stopping feeds, reducing energy input, introducing quench mediums, stopping mechanical equipment). When the causes and effects have been ascertained, the following should be done:
   a) **Protect Mechanical Equipment**. Such items should be shutdown in an orderly manner to prevent damage.
   b) Prevent the restart of the plant or equipment until the unsafe conditions have been cleared.
   c) Reset automatic process controllers to their startup setpoints to prevent driving control valves to their limits, which might cause unsafe process conditions.
   d) Prevent personnel from overriding safety and interlock controls, which can result in unsafe startup or shutdown.
2. **Utility Failures.** Possible loss of any of the utilities, whether local or plantwide, must be considered, where their loss cannot be tolerated, or where they are necessary to activate critical interlocks, standby utilities must be provided. These may be supplied by instrument air bottles, uninterruptible power supplies, standby pumps, emergency cooling water, etc.
3. **Interlock Component Failures.** It should be assumed for design purposes that all interlock components will malfunction at one time or another. Therefore, the need for redundancy must be evaluated, particularly in critical installations, where even a partial failure might render an interlock system ineffective. In particular, the designer should:
   a) Provide redundant interlock circuits with contacts wired so that either circuit will initiate a trip (contacts in series).
   b) Monitor dissimilar but related alarm conditions in the same system. For example, monitor low coolant flow and high temperature, rather than having two flow alarms, to preclude common-mode failure of similar switches and potential loss of both interlocks.

c) Provide redundant initiating contacts with a means for switching without causing a trip, so that they may be serviced. Suitable indication should be provided to denote which of them is active.

Fail Safe Design

The development of interlock logic and process control loops must be consistent with a frame of reference, so that a given sequence of events follows the same direction. The reference is established in relation to the failsafe mode of: the process, the final control component (e.g., valve, pump), the process control loop, and the alarm contact.

Process control loops must be designed to satisfy two criteria: to correct the process when it deviates from a setpoint, and to prevent the process from going out of control if any component fails. Every component, from transmitter to final control element, should be specified so that its failure will not cause an unsafe condition. Component selection and signal direction must satisfy both the above criteria for each control loop.

It is quite possible (and acceptable) to have contradictory signals from controller and interlock. For example, a control signal may open a valve for control, whereas a trip may close the valve. In an emergency, the interlock will override to ensure all-round safety.

Transmitters - As a general rule, it is assumed that an increase in the measured variable (say, pressure) will increase the transmitter output signal. A transmitter failure, therefore, would imply a low measurement. If a high process measurement (e.g., high pressure) is at the limit, then the failure of the transmitter would aggravate the dangerous situation. Redundant instruments should then be used; one for control, the other (preferably reverse-acting) connected to alarms or interlocks, so that the control signal and interlocks can be directed accordingly.

A possible exception is a temperature transmitter, using a thermocouple. This instrument can be specified to fail upscale or downscale if the thermocouple burns out. Thus, the output signal can be designed to either increase or decrease, depending on the failsafe direction required.

Temperature transmitters using "filled" thermal elements should not be used for critical temperature measurements. Failure of the thermal element will indicate a low temperature, even though the temperature may be rising.

Controllers - Controller output is based on the valve action required to control the process. This output can be selected to either increase or decrease with respect to transmitter input. The direction is predicated on the valve action required to control the process. Loss of power supply (air or electric) will inherently cause a low signal output, causing the valve to fail safe. Therefore, the controller output should be consistent with control and failsafe requirements.

Final Control Devices - Final control devices are used to control the process by regulating flow. These devices may be control valves, pumps, compressors, etc. They may fail by stopping, shutting off the flow, or going out of control and thereby increasing the flow beyond its maximum rate.

Control valve actuators are usually of the spring-and-diaphragm type. They are pneumatically actuated and can be specified to fail either open or closed in the event of air failure. The action the valve should take depends on its function, i.e., whether it is on control and/or trip. Where different actions are required, positioner, solenoid valves and signal-reverse relays may be used to reverse the signal, depending on its function.

The final control component should operate to put the process in a safe condition if either the process control signal fails or the interlock is tripped on power failure. If this is not possible because of process control requirements, or power is required to energize the interlock, then backup air supply and/or power should be provided.

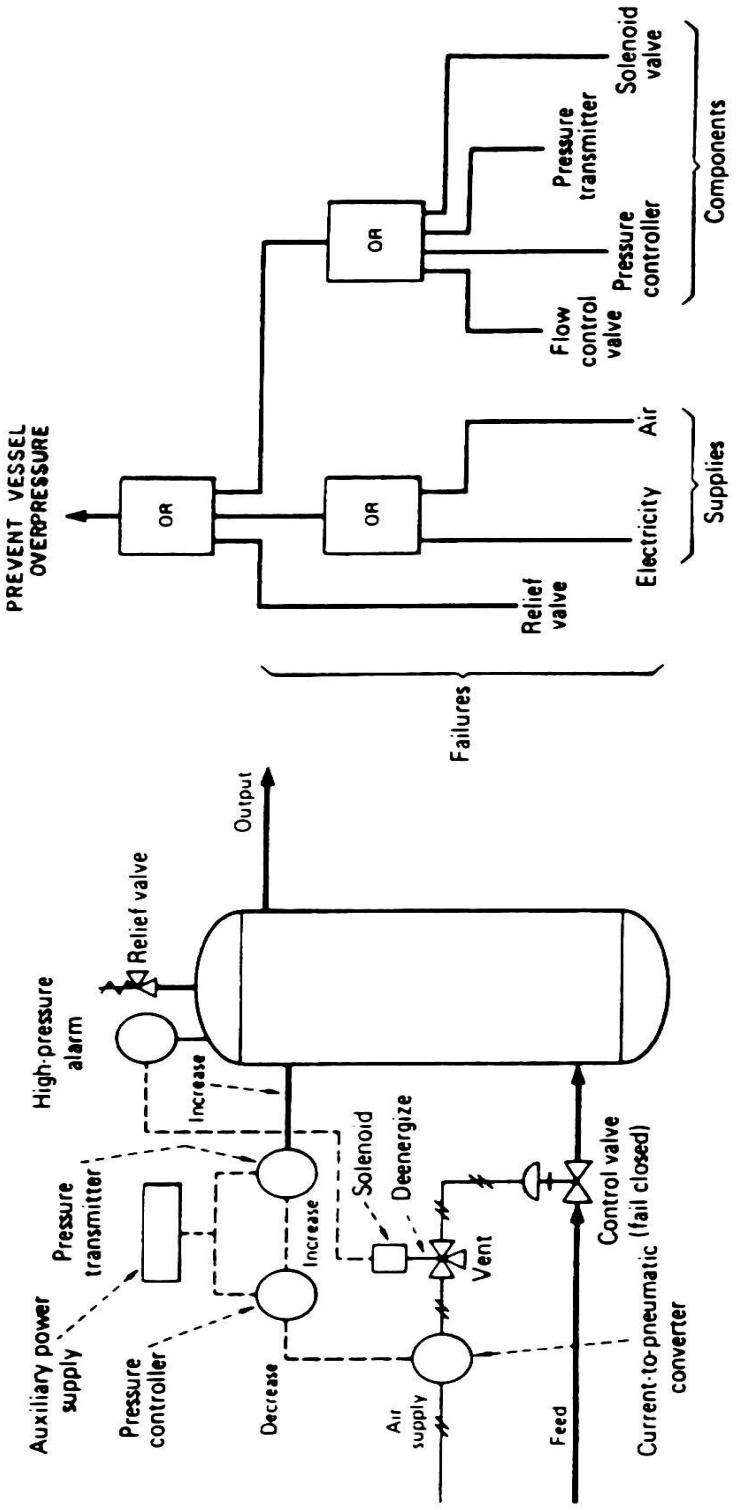
In general, each control loop should be designed from a failsafe point of view with regard to the process. This does not necessarily have to be done in a formal manner, using sophisticated fault-tree analysis, but just with a constant awareness and appreciation of conditions that might arise - with proper consideration of cause and effect.

Example of Failsafe Design

The pressure-control loop in Fig. 4.la consists of a pressure transmitter, controller, and pressure-control valve. A pressure increase will create an increase in the pressure transmitter signal, which is translated by the pressure controller into a signal decrease. This is converted by the current-to-pneumatic converter into a decrease in air supply, which positions the control valve to reduce the pressure in the vessel, until the controller setpoint is reached.

A high-pressure alarm switch is provided to override the controller, in the event of control loop failure. A high-pressure condition opens the switch to deenergize the solenoid and close the control valve.
The system is designed to fail safe, so that failure of any component within the loop will cause the pressure control valve to close, preventing overpressure in the vessel:

1. The valve is chosen to close on air failure so that, if all else fails, at least the process will not exceed its limit of high pressure.



(A) PROCESS & INSTRUMENT SYSTEM (B) FAULT-TREE ANALYSIS

FIG. 4.1 FAIL SAFE DESIGN OF A PRESSURE CONTROL LOOP

2. The pressure-controller output is selected to decrease as the transmitter signal rises above the setpoint, thereby closing the valve to reduce the pressure.
3. Failsafe analysis of the control loop is such that if power supply to the pressure-control valve or pressure-controller or pressure override fails, the valve will close. Also if the pressure-control valve itself fails, it will close by virtue of its its spring action.
4. The pressure-alarm switch contact opens on high pressure, deenergizing the solenoid, and venting the valve diaphragm actuator to close the valve. Failure of interlock power supply will also cause the system to fail safe.

Thus, failure of any component, power supply, or loss of control signal will cause the system to fail safe.

Types of Interlocks

Interlocks must be designed in two directions - starting up and shutting down. Conventionally, the basic frame of reference assumes that the process is shutdown (the lowest energy state), with interlocks unpowered. Circuits are then designed using the convention of positive logic (contact closed, light on; contact open, light off), with initiating and relay contacts shown in their normal (deenergized) state. The circuits are designed to operate while the process is starting up.

The complementary logic is developed when the process is assumed to be running (in its highest energy state). Interlock circuits are then designed to operate while the system is shutting down. Interlock systems must be designed from both (complementary) points of view.

Interlock circuits are usually arranged in three parts:

1. Input - Consisting of field switches, panel switches, pushbuttons, selector switches.
2. Logic - Relay contact arrangement or programmable-controller programs that establish the relationship between the inputs and outputs.

3. Output - Actuating devices, solenoid valves, motor starters, indicating lights, and alarms.

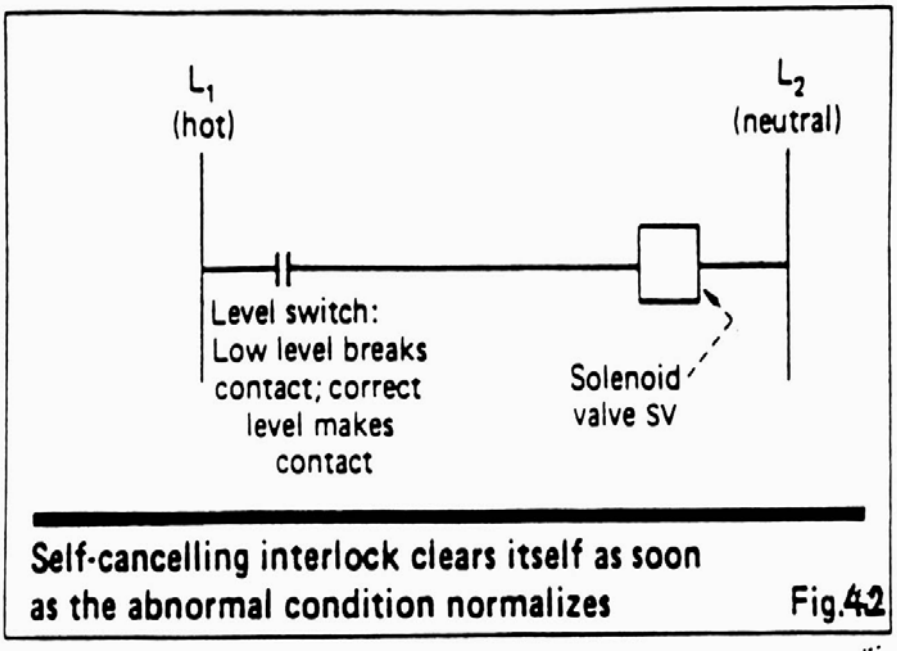
The ladder diagrams in Fig. 4.2 - 4.7 are presented to illustrate some of the most common interlocks. Such interlocks are arranged singly or in combination to fulfill process requirements.

Self Canceling Interlock - This circuit clears itself as soon as the abnormal condition returns to normal. For example, in Fig 4.2 a solenoid valve in a level-control circuit is energized through a level switch connecting the "hot" line, LI, to the neutral line, L2. Suppose that the energized solenoid maintains the fail-open inlet valve closed. A low level opens the contact on the level switch. The solenoid is deenergized and the flow valve opens. The low level corrects itself and the level-switch contact is made again. The solenoid is reenergized and the inlet is closed.

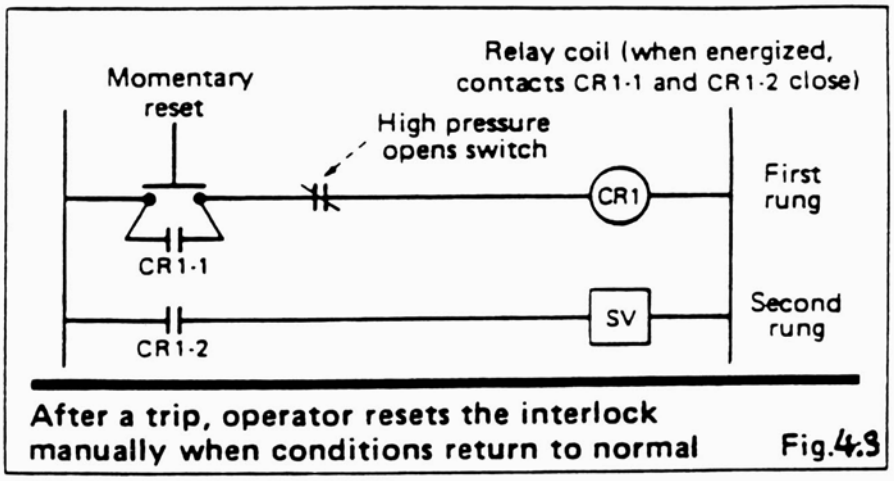
This is simple interlock and may not be suitable, for example, if the level is cycling around the level-switch setting. This might cause undue process oscillations and possible equipment damage.

Manual Reset Interlock - To prevent the problems associated with a self-cancelling interlock, a circuit is set up to require positive action by an operator in order to cancel the interlock, once conditions return to normal.
For example, in Fig. 4.3, suppose the energized solenoid normally keeps a feed valve open. A high-pressure signal will open the high-pressure switch, deenergizing the control relay CR1. Contact CRl-2, shown on the second rung, will open, deenergizing the solenoid, closing the feed valve and relieving the pressure. When the operator sees that all is clear, the momentary reset button may be pressed. The relay coil is energized, thus closing CR1-1 and CRl-2. The solenoid is reactivated and the feed valve is reopened. CR1-1 is a seal contact, to maintain the circuit when the reset pushbutton is released.

A strict protocol should be followed when an operator is manually resetting an interlock, Suppose that there is a reboiler with a steam-inlet control valve sensitive to the process controller as well as to a high-temperature interlock trip. While operating, there is a trip, due to high temperature. When the temperature drops, the operator might manually reset the interlock. But because the temperature is now low, the process controller calls for full steam. This may be dangerous. There should be a startup mode, whereby the steam builds up gradually.



Self-cancelling interlock clears itself as soon as the abnormal condition normalizes Fig.4.2



After a trip, operator resets the interlock manually when conditions return to normal Fig.4.3

In this case, a "lockout" feature can be incorporated to prevent reset of the interlock until the controller output satisfies a predetermined condition. The operator resets the controller to manual, reduces the output of the controller to some low value, manually resets the interlock, and the system is ready for startup.

Interlock with Bypass - Processes or equipment that are tripped on "low" conditions are often very difficult to startup, either initially or after a shutdown. To avoid this difficulty, a circuit is used that will bypass the low trip contact until the unit is running, and then clear itself so that the circuit will trip if an abnormal low condition arises (Fig. 4.4). This type of circuit is often used on compressor startup, when low speed will trip the unit.

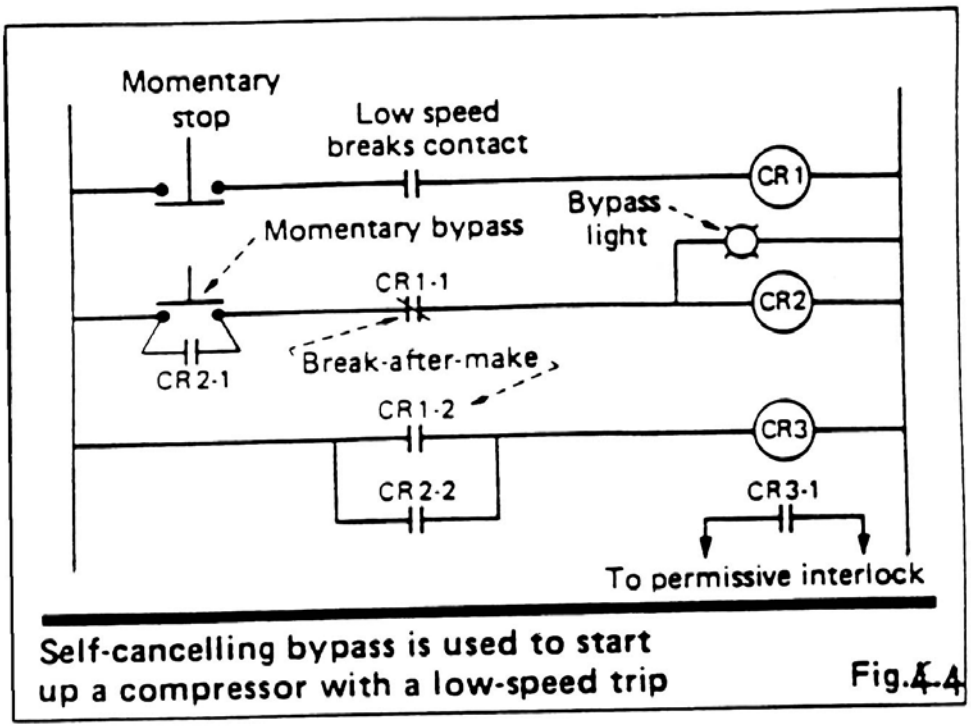
In Fig. 4.4, imagine that the compressor is shut down. The momentary-bypass pushbutton is pressed, energizing relay CR2. The light goes on to indicate that the bypass has been activated. The activated relay coil CR2 closes CR2-1, the seal contact across the pushbutton, and CR2-2, the bypass contact. This in turn energizes relay CR3, which closes the permissive contact CR3-1 located in the compressor start/stop circuit.

When the compressor speed increases above the low-speed setting, the low-speed contact closes, energizing relay CR1. This closes contact CR1-2 and opens CR1-1 in a make-before-break arrangement, maintaining the permissive contact CR3-1. Relay CR2 will then be deenergized and the bypass light will go off. However, relay CR3 will remain energized through contact CR1-2.

The bypass has now been cancelled, and the relay CR3 is maintained through the low-speed switch and relay contact CR-1. Should the compressor speed fall below the low-speed setting, the low-speed switch will open, stopping the compressor. The stop pushbutton is for emergency shutdown of the compressor.

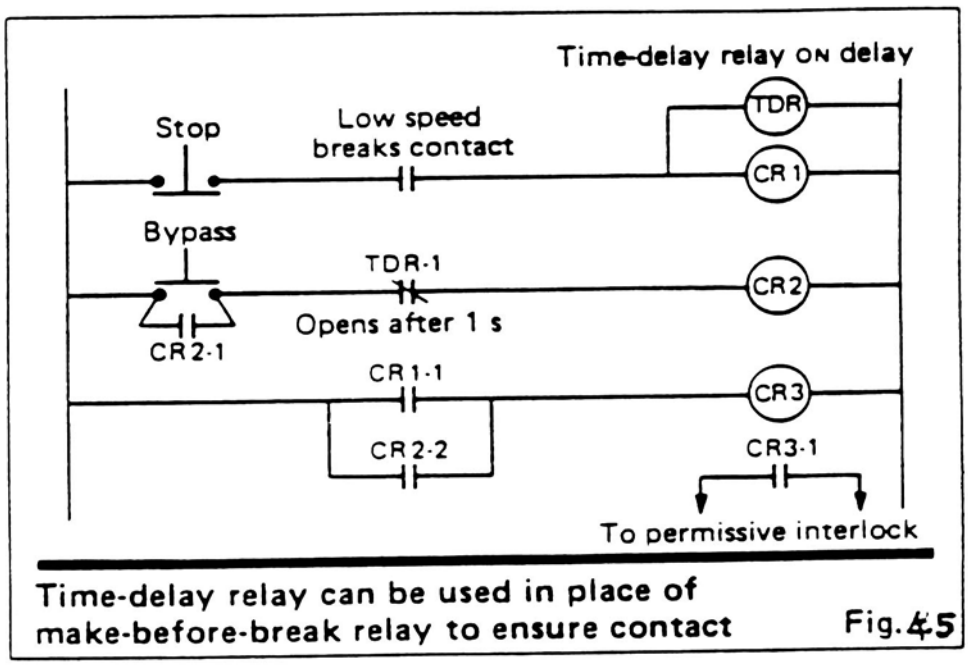
Time-delay action - Time-delay action is used where a predetermined time is required to allow the process to obtain its operating level, e.g., for lube oil pressure to rise above a low-pressure shutdown level (Fig. 4.5).

Time-delay action may be either: On delay, in which the time-delay contacts will change after the time-delay relay is energized for a given time; or OFF delay, in which the contacts will change state after the relay is deenergized for a given time. Referring to the previous circuit, the make-before-break contacts could be replaced by a time-delay (Fig. 4.5).



Self-cancelling bypass is used to start up a compressor with a low-speed trip

Fig. 4-4



Time-delay relay can be used in place of make-before-break relay to ensure contact

Fig. 4-5

After startup, and when the compressor speed is increasing, the low-speed contact closes, energizing CR1 and TDR. CR1-1 closes, and TDR-1 opens after one second, assuring the overlap of contact that was achieved in the previous circuit by the make-before-break arrangement.

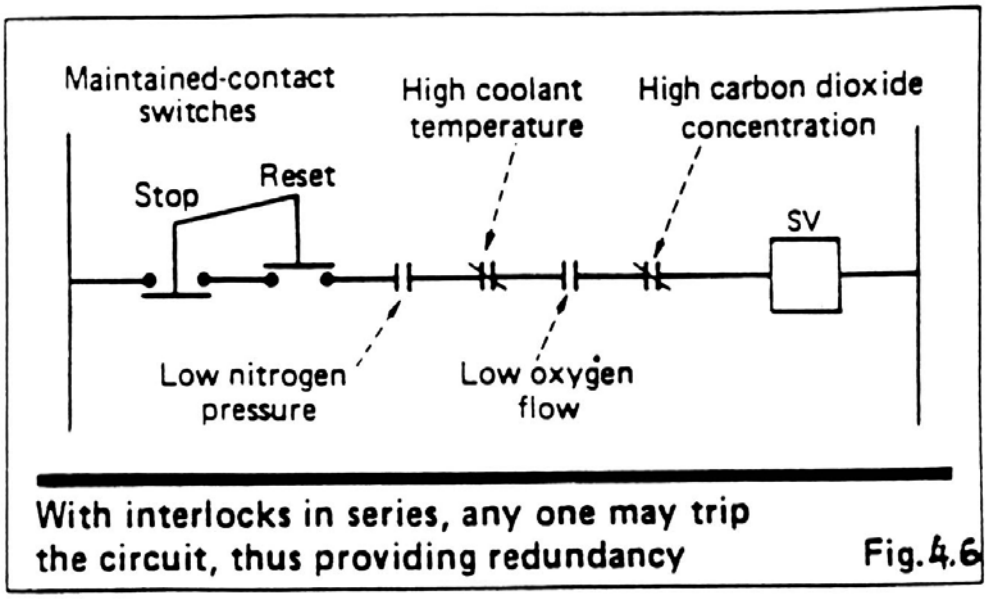
Interlock Chains - These may be of two kinds - series or parallel. With many safety interlocks, any one of several initiating contacts may trip the same circuit (see Fig. 4.6). The inlterlocks are placed in series (OR) configuration. Contacts are also arranged in series where redundancy is required, so that if one fails the other will actuate the interlock.

Where interlocks are designed so that more than one contact must trip to activate a shutdown, they are arranged in parallel (AND) configuration (see Fig. 4.7).
Voting circuit - This is a form of redundancy designed to increase the reliability of a trip system. It may be required where failure of the system will result in a potentially hazardous condition. However, this duplication introduces the possibility of an increased number of spurious shutdowns because of the failure of the trip system. To preclude this, the interlocks are interconnected so as to reduce the failure rate to an acceptable level without reducing overall reliability.

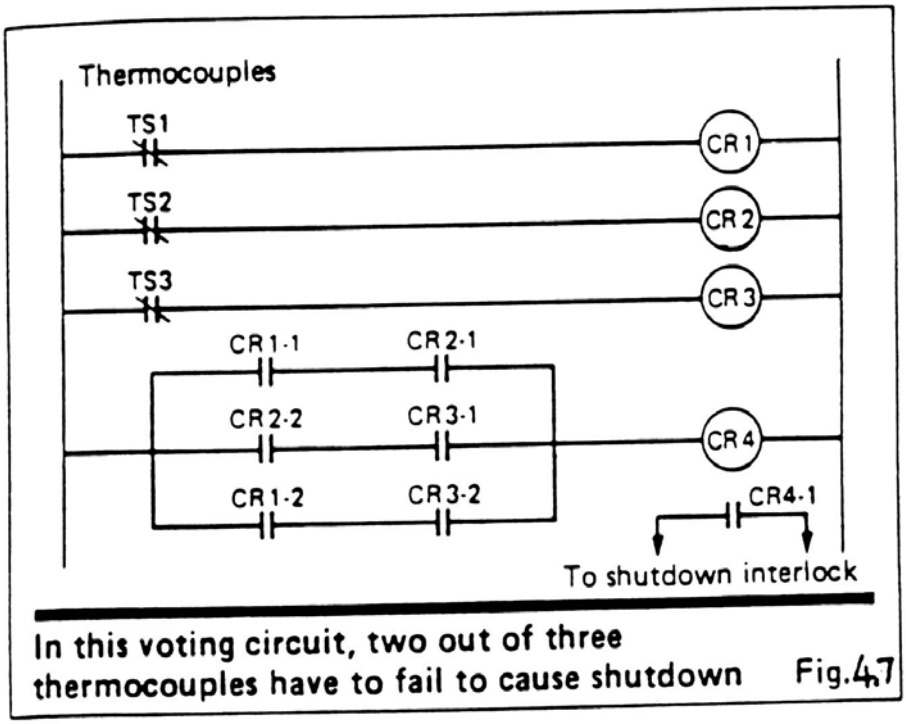
For example: Thermocouples are unreliable and burn out frequently; thus, unwarranted trips will occur if each thermocouple is linked to a unique interlock. To ensure that it has been a process upset (high temperature) that initiated the trip, and not just another burned-out thermocouple, a voting circuit is used, in which more than one sensor measuring the same variable (e.g., two out of three) is required to detect an alarm condition that will activate the trip.

In Fig. 4.7, three thermocouples measuring the same variable are normally linked through relay coils to the interlock system. If TS1 opens, CR1 is deactivated. This opens CR1-1 and CR1-2, but does not deactivate CR4. If TS2 also opens, CR2 is

deactivated. This opens CR2-2 and CR2-1. Now all three parallel circuits are broken and, although TS3 is still operating, CR4 is deactivated and the shutdown interlock is brought into play. TS1 and TS2 must be replaced if burned out.



With interlocks in series, any one may trip the circuit, thus providing redundancy Fig.4.6



In this voting circuit, two out of three thermocouples have to fail to cause shutdown Fig.4.7

Alarms

Alarm indication is one of the most important aspects of process instrumentation. The operator's reaction and subsequent action, when an alarm or series of alarms sounds, is directly related to three things: the information that is displayed, the order in which the alarms go up, and the operator's perception as to what has occurred.

Alarm management has become one of the most difficult areas of process system control and design, due to the increasing number of alarms that have become necessary, the confusion that arises when successive alarms are initiated other than those that indicate abnormal process conditions, and inconsistent alarm-design philosophy.

The complexity of processes, severity of their upsets, automating of the interlock systems, and their interaction with the operator -all impose a larger burden on the annunciator system than ever before. Typical problems are: What type of display should be used?, e.g., red or green?; what arrangement should be used?, e.g., which lights go on the top row and which go on the bottom, the left and the right, and in what sequence should the alarms appear.

Whether the alarm systems used are computerized or the more conventional kind, they should be designed so that they stand alone with a high degree or reliability.
Alarms may be in the form of either conventional backlighted legends or cathode-ray-tube (CRT) displays. There are many different alarms sequences to alert the operator when a process variable or operating condition is off-normal and to indicate when the process has returned to safe operation. The two most commonly used sequences are the standard and First-Out. In the Standard sequence, it is not possible to tell which alarm was initiated first.

The First-Out sequence is used where several alarms are initiated practically simultaneously and it is desirable to determine which was the first one to respond.

It should be noted that the first alarm to respond may not be the one that initiated the trip. All of the process variables, pieces of mechanical equipment, and interlock components have a speed of response and detection. For example, imagine that there is a blocked outlet valve on a pressure vessel that is being fed by a compressor. It is possible that an interlock on the compressor might trip before a pressure switch on the vessel would. The alarms will indicate "High pressure," "Compressor shutdown," but will not indicate that the trouble is a blocked valve on the other side of the vessel.

Conventional annunciators are usually of the relay type. However, solid-state switching is also available and is used with electronic instrumentation or in computer applications. Annunciators should be provided with their own circuits, power supplies, disconnects and overload protection. Auxiliary contacts on alarm relays are usually included, for example, to connect a horn. These contacts should never be used to initiate a trip or shutdown, because alarm and interlock circuits should be independent of one another.

Annunciators are powered from their individual power supplies that are wired to alarm contacts. If these components were coupled to the interlock system, and there should be a failure of the alarm power supply, relay or external contacts, there would result an unwarranted shutdown of the process.

Interlock and alarm system do, however, act in concert, for when an interlock is actuated, the alarms serve as indication to the operator that an upset is present.
CRT-displayed alarm sequences are determined by the "firmware" (preprogrammed software) supplied by the vendor. They are limited as to various sequences that are a vailable and should be used for early-warning alarms only.

Another technique, that is sometimes used, is to employ a multiplexer. In this way, a number of different alarm contacts may be scanned in a timed manner, so that their

status is determined at regular intervals. Variables that must be monitored continuously should not be assigned to multiplexers.

There are two types of alarms:

1. Those that indicate an abnormal condition (although one that does not cause a shutdown) and also indicate the status of equipment.
2. Those that are concurrent with an interlock trip.

Alarms listed under (1) above would be precursor or early-warning alarms and, as such, are not as critical as those initiating an interlock. An indirect method of actuating these alarms, using transmitter outputs or contacts built into instruments, may be considered, since any failure of these alarms would not necessarily jeopardize plant safety or operation. However, direct-operating switches are preferred.

However, due to the critical nature and high reliability required of the alarms and interlocks listed under (2) above, they should be independent of instruments that use an indirect means of actuating trip contacts. For example, alarms that use transmitter outputs, or contacts built into instruments operated by external power, should be avoided. (In these cases, if the transmitter fails, or if the external power fails, then the alarm fails as well).

Where it is impractical or impossible to avoid this (e.g., due to instrument circuitry), redundant instruments or uninterruptible power supplies should be provided.

Direct-operating process switches should be used wherever possible for alarms. These switches are usually mechanical and do not require external power. Double-pole, double-throw isolated contacts are provided: one contact for alarm, the other contact for interlock. They will be reliable, provided they are suitable for the process and the environment and are maintained regularly. Where ultimate reliability is required, alarm redundancy may be considered, using separate alarm switches.

If electronic alarms are used, the contacts should open on power failure, on loss of signal, or when in alarm condition.

Power Distribution

Power failures may be the result of the loss of power for the total plant, at distribution panels, or at individual users. Each type of failure must be reviewed to determine the effect on the process and on system interlocks.

Instrument power distribution should be so divided that power failure at the distribution panel or components will initiate a trip. This should be consistent with the concept stated above, whereby all components should deenergize when the interlock trip is activated. Where a supply circuit may be overloaded due to the number of components in an interlock system, another circuit must be used. In this case, a means for monitoring the power supply to each of the two supply circuits should be provided, so that power failure of either one will trip the other, ensuring that the entire system will fail safe.

MODULE 6:

SUPERVISORY CONTROL AND DATA ACQUISITION SYSTEM (SCADA)

MANAGEMENT INFORMATION SYSTEM (MIS) AND SUPERVISORY CONTROL

A management information system (MIS) is provided to support administrative control functions and essentially consists of individual personal computer work stations which are interconnected through a high speed local area network (LAN) to allow sharing of resources and automatic operation of associated functions. The entire system is supported and under the direct control of the LAN server also located in the control building for system security purposes.

SUPERVISORY CONTROL & DATA ACQUISITION (SCADA) SYSTEM

WHAT IS SCADA?

SCADA is the technology that enables a user to collect data from one or more distant facilities and/or send control instructions to those facilities. SCADA makes it unnecessary for an operator to be assigned to stay at, or to visit, remote locations in the normal operation of that remote facility.

DEFINITION OF SCADA

SCADA is an acronym that is formed from the first letters of supervisory control and data acquisition. Except that it does not refer to the factor of distance, which is common to most SCADA systems.

A SCADA system allows an operator, in a location central to a widely distributed process such as an oil or gas field, pipeline system, or hydroelectric generating complex, to make set point changes on distant process controllers, to open or close valves or switches, to monitor alarms, and to gather measurement information.

When the dimensions of the process become very large hundreds or even thousands of kilometers from one end to the other the benefits in terms of reduced cost of routine visits can be appreciated. The value of these benefits will grow if the facilities are very remote and require extreme effort (e.g. helicopter access) to visit.

APPLICABLE PROCESSES

SCADA technology is best applied to processes that are spread over large areas, are relatively simple to control and monitor, and require frequent, regular, or immediate intervention. The following examples of such processes should aid in visualizing the range of types.

A. Groups of small hydroelectric generating stations that are turned" on and off in response to customer demand are usually located in remote locations can be controlled by opening and closing valves to the turbine, must be monitored continuously, and need to respond relatively quickly to demands on the electric power grid.

B. Oil production facilities including wells, gathering systems, fluid measurement equipment, and pumps are usually spread over large areas, require relatively simple controls such as turning motors on and off, need to anther information regularly, and must respond quickly to conditions in the rest of the field.

C. Pipelines for gas, oil, chemicals, or water have elements located at varying distances from a central control point, can be controlled by opening and closing valves or starting and stopping pumps, and must be capable of fast response to market conditions and to leaks of dangerous or environmentally sensitive materials.

D. Electric transmission systems may cover thousands of square kilometers, can be controlled by opening and closing switches, and must respond almost immediately to load changes on the lines.

These examples are just that examples, SCADA has been successfully installed on each of these types of processes as well as many others.
The types of control noted on these examples may give the mistaken impression that more complex control is not possible.

As will be described later, the complexity of possible remote control has grown with the maturing of the technology.

Typical signals gathered from remote locations include alarms, status indication, analog values, and totalized meter values. However, with this apparently limited menu of available signal types, a vast range of information can be gathered.

Similarly, signals sent from the central location to the remote site are usually limited to discrete binary bit changes or analog values addressed to a device at the process. An example of a binary bit change would be an instruction ordering a motor to stop. An analog value example would be an instruction to - change a valve controller set point to 70 percent. Given simple signal types like these and some imagination, many control changes can be effected.

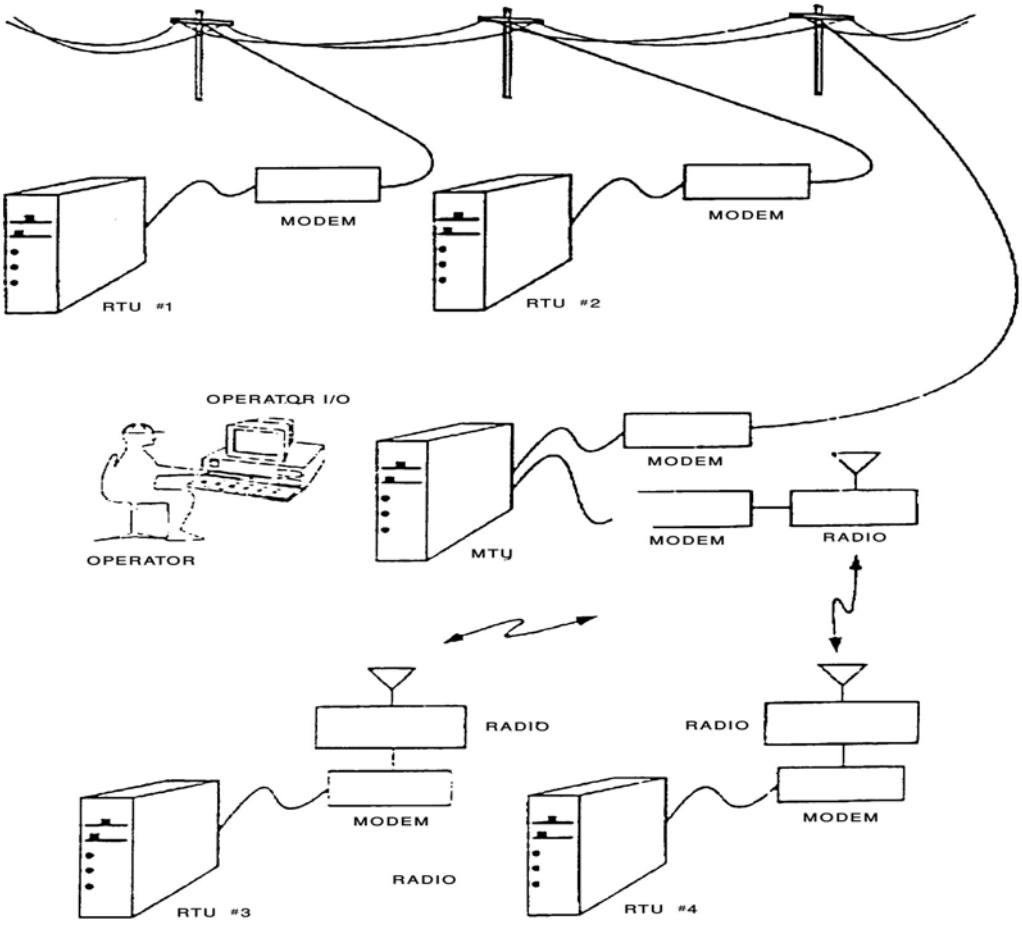
ELEMENTS OF A SCADA SYSTEM

Fig. 6.1 shows the major components of a SCADA system. At the center is the operator, who accesses the system by means of an operator interface device, which is sometimes called an operator I/O (for input/output).

The operator output, which really means system output to the operator, is usually a CRT (cathode ray tube), sometimes called a VDU (video display unit).
For very simple lystjms, it may be sufficient to have a set of annunciator windows that mimic the condition of the remote process. Often, an audible signal will be included.

The operator input is usually a computer keyboard, although pointing devices such as trackballs and mice are gaining in popularity. For very basic systems, a set of simple electrical switches may be sufficient.

The operator interfaces with the MTU (master terminal unit), which is the system controller. It is almost always a computer. It can monitor and control the field even when the operator is not present. It does this by means of a built-in scheduler that can be programmed to repeat instructions at set intervals, for example, it may be scheduled to request an update from each RTU (remote terminal unit) every six minutes.



SCADA SYSTEM LAYOUT

MTUs must communicate with RTUs that are located away from the central location. There are two common media of communication, land line, in the form of optical fiber cable or electrical cable and either owned by the company or leased from a telephone utility is one; the other is radio. In either case, a MODEM, which modulates and demodulates a signal on the carrier, is required. Some large systems may use a combination of radio and telephone lines for communication. Because the amount of information moved over a SCADA system tends to be rather small, the data rate at which the modem works is small. Often 300 bps (bits of information per second) is sufficient. Few SCADA systems, except for those on electric utilities, need to operate at data rates above 2400 bps. This allows voice-garde telephone lines to be used, and it does not overload most radio systems.

Normally, the MTU will have auxiliary devices (e.g. printers and backup memories) attached. These devices are considered to be part of the MTU.

In many applications, the MTU is required to send accounting information to other computers or management information to other systems. These connections may be either direct and dedicated or in the form of LAN (local area network) drops.
In a few cases, the MTU must also receive information from other computers. This is particularly true of the newer systems in which applications programs, operating on other computers and connected to the SCADA computer, provide a from of supervisory control over SCADA.

Fig. 6.2 shows an RTU and its various connections. The RTUs, as has been mentioned, communicate with the MTU by a modulated signal on cable or radio. A system can contain as few as one or up to several RTUs. Each must have the capability to understand that a message has been directed to it, to decode the message, to act on the message, to respond if necessary, and to shut down to await a new message. Acting on the message may be a very complex procedure. It may require checking the present position of field equipment, comparing the existing

position to the required position sending an electrical signal to a field device ordering it to change states, checking a set of switches to ensure that the order was obeyed, and a message back to the MTU to confirm that .the new condition has been reached. Because of this complexity, most RTUs are based on computer technology.
Connections between the RTU and the field devices are most often made with electrical conductors wires.

A TWO-WAY SYSTEM

One of the things that distinguishes SCADA from most telemetry systems is that SCADA is a two-way system. With SCADA, it is possible not only to monitor what is going on at a remote location, it is also possible to do something about it. The supervisory control part of SCADA takes care of that.

SCANNING

COMMUNICATION ACCESS

There are several means by which electronic machines can talk among themselves. Depending on the-purpose of their conversation, the speed requirements, and the machines status relative to each other, different access methods may be used. The communication requirements both determine and are controlled by the communications protocol selected.
This is not test on communications, so it will not describe many of the communications access methods that exist.

The communications method used by most SCADA systems is called "master-slave". In a master- sisal arrangement, only one of the machines (in this case the MTU) is capable of initiating communistic. The MTU calls one RTU, gives instruction, asks for information updates, and orders the RTU to respond. The MTU then listens for the answer. The RTU answers as soon as the MTU has finished talking, then stops talking

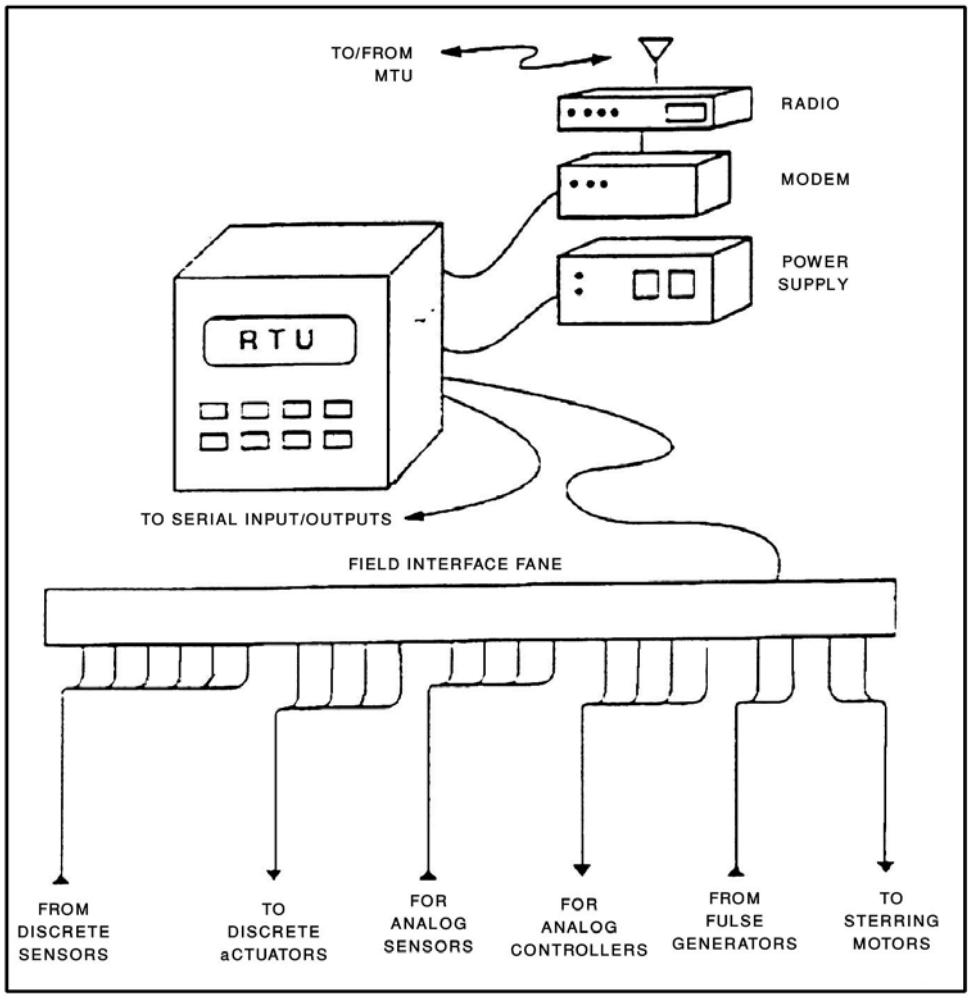
and listens for more orders. The MTU moves to the second RTU and goes through the same procedure. The MTU talks to each RTU, then returns to the first. The RTU cannot initiate a message; it can send a message only when specifically ordered by the MTU to do so.

The process of talking to each RTU in order and then going back to the first RTU to begin the cycle all over again is called "scanning".

DETERMINING SCAN INTERVAL

If control is not to be compromised by excessive time delay and yet constraints are imposed by the rate at which data can be transferred to and from the RTU, it follows that there is a "bout rate" at which to scan the RTUs for data.

Usually, the RTU supplies the electrical power for both actuators and sensors. Depending on the process, reliability requirements may necessitate an UPS (uninterruptible power supply) to ensure that utility power failures do not result in process or safety upsets.



AN RTU and its connection

In the same way that the- MTU scans each RTU, the RTU scans each of the sensors and actuators that are wired into it. This scanning is done at a much higher scan rate than the MTU scanning.

One of the factors that determines scan interval must be the number of RTUs that must be scanned. An estimate, made early in the design phase, of the likely number of RTUs will probably be sufficient for this determination.

A second factor to be considered is the amount of data that must be passed on each conversation. Depending on the size of the facility at each remote site and the amount of independence the remote site control is capable of exercising, the data to be gathered can be as little as one status or as much as several hundred status and alarm points, plus several dozen meter totalizer points and several dozen analog values. Each status or alarm point requires one bit of data to communicate. Each meter or analog point, since it will be transcribed to a binary word, requires about sixteen bits. (This will vary with equipment but is a close enough number for this calculation).

For simplicity, and to allow a safety factor, it is best to select the largest RTU when evaluating points. Multiply this point count by the total number of RTUs to get the count of all data back from all RTUs.

Remember that a conversation is usually a transfer of data in both directions. It is important to include the time taken for the MTU to talk to each RTU. This will include both the time for the MTU to ask the RTU for information and the time for the MTU to give other instructions to each RTU. Agar evaluating this point count for the largest outgoing message and multiplying by the number of RTUs is the best way to do it at the design stage. This should provide a conservative result because the messages from MTU to RTU are usually shorter than the messages from RTU to MTU. Evaluating each RTU may be beneficial if the exercise is being done on an existing system.

The third factor is the data rate. The number of bits per second that can be transmitted over the communication medium is important in determining scan interval, but at the early design stages this number may be flexible; these items may be traded back and forth to develop an optimum. For this part of the exercise, consider that there are two data rate groupings. The first which is used on voice-grade telephone lines and most UHF radio-modem communication systems is

between 300 and 2400 bits per second. Using 1200 bps in the calculation will result in a good first estimate. The second data rate grouping applies if a special communication medium is being considered.

TELEMETRY SYSTEM OVERVIEW

Telemetry Overview

Telemetry is the technology which enables a user to collect data from several measurement points at inaccessible or inconvenient locations, transmit that data to a convenient location, and present the several individual measurements in a usable form. This unit discusses some of those "inconvenient locations" where telemetry is used and shows all the elements of a generic telemetry system for collection, transmission, and presentation of data.

Learning Objectives — when you have completed this unit, you should:

A. Have a general understanding of how telemetry is used.
B. Know the elements of a generic telemetry system.
C. Understand the terms which relate to major system elements (subsystems).
D. Know which of the basic types of telemetry is best for a given application.

System Applications

In the preceding definition of telemetry, the term ''inaccessible or inconvenient location" is used to define the source of telemetry data. By implication, then, a potential telemetry system application exists when something of interest, which can be measured electronically, occurs at a location which is inconvenient or totally inaccessible to the one who needs the information.

An aircraft on a test flight has many parameters which change during the flight: fuel flow, amount of fuel remaining, engine performance, stress on the wings, vibration of critical parts, performance of the avionics systems, and temperature of various measurement parameters. The test pilot can monitor some of these, of course, by

reading instruments on the panel, but time permits reading only those instruments critical to the safety of the flight, and even those are not sufficiently accurate for detailed flight analysis. If the team of engineers which designed the plane could go along on test flights, analysis would be more thorough. If a computer with sophisticated data displays could be taken on the flight, analysis might be satisfactory. It is, of course, inconvenient or impossible to do this, so flight tests on newly designed or redesigned aircraft constitute a major use of telemetry. The team of designers and the computer remain in a laboratory on the ground, and the data is brought to them by telemetry.

The rocket or unmanned spacecraft presents a more obvious need for telemetry. The vehicle in this case is too small to carry even one person, much less the entire design team and its computer. Here, telemetry monitors all information which enables the team to evaluate performance of the test vehicle, temperatures, pressures, strains, vibration, electronic guidance system, and so on, while the flight is in progress.

A nuclear power generation facility presents other reasons for the use of telemetry in monitoring performance. Here, certain areas are unsafe for humans because of the radiation hazard. The same and other areas are too hot for comfort. Yet the operating conditions, temperatures, pressures, flow of coolant water, radiation levels, and so on, are critical in safety monitoring.

Similarly, a conventional fossil-fueled power generation facility requires constant monitoring to detect conditions which signal safety or reliability problems. There is no radiation hazard, but the plant is too hot and noisy for prolonged human occupancy.

In the electric utility company, power distribution must be monitored by telemetry because of the inconvenient location of load switching terminals. An operator at a central location observes increased or decreased demands and sends commands to switching sites as necessary to distribute the available power properly.
Water supply monitoring, waste water monitoring, weather monitoring at remote sites, and subway system "health monitoring," are other uses of telemetry in today's world.

In the vehicular test category, new and redesigned automobiles are tested for performance and for safety .Farm tractors are evaluated, also, and even new snowmobiles.

In cases involving evaluation of a newly designed product (aircraft, rocket, or automobile, for example), the telemetry system almost always includes provision for data storage. This gives the designers an opportunity to replay magnetic tapes (the typical storage medium) for detailed evaluation, even weeks or months after the test run is completed. The power plant monitor's weather telemetry system, on the other hand, transmits data which is of value as it happens but serves no continuing purpose for the user. These systems seldom include data storage equipment.

System Configuration

While every telemetry system is designed to meet the unique needs of a specific customer, the overall block diagram of any system has certain elements in common with that of any other system. This configuration commonality is shown in Fig. 2-1.

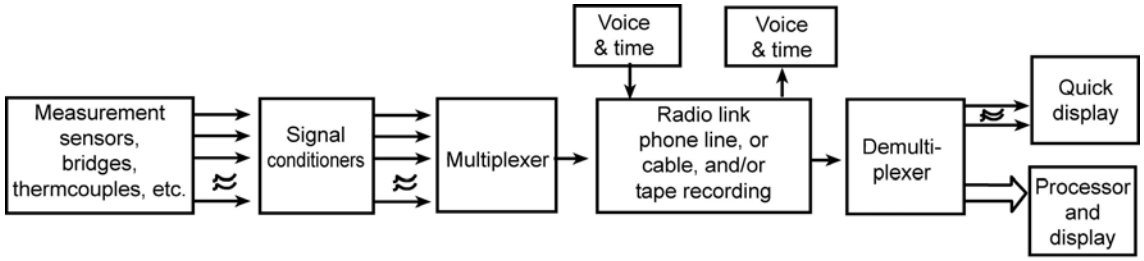


Fig. 2.1. Generic Telemetry System

Electrical data originates at the sensors or transducers, each of which converts some physical condition (temperature, pressure, acceleration, etc.) into a proportional electrical voltage. Typical sensor types are thermocouples, resistance-temperature devices, bridges, potentiometers, etc.

A typical system includes several types of signal conditioners, each used to convert the output of a specific type transducer to data with a range of 5 to 10 volts. One extreme of voltage corresponds to the lowest temperature, pressure, etc., expected at the measurement point; the other extreme corresponds to the highest measurement expected.

Obviously, if a transducer has a self-contained signal conditioner with an output range adjustable to 5 to 10 volts, or if the measurement is already in that range without signal conditioning, the signal conditioner can be bypassed by that measurement.

The multiplexer's task is to combine several measurements into a single output stream, so that they can be transmitted over a single radio channel, coaxial cable, or telephone line, and/or can be recorded on a single track of a tape recorder.
Obviously, when measurements are mixed in this way for transmission, they must be identified in that mixture ("multiplex") such that they can be separated properly at the receiving station. This is accomplished by placing each measurement at a known place in frequency or time, as we will see in later units.

The next link in a system is the transmission-reception medium (radio, coaxial cable, or telephone) and/or magnetic tape recorder. Voice annotations and time are generated at this stage or are recovered on tape playback.

Often, telemetry data goes to an analog display in raw form, as a reproduction of the data voltage or current which is generated by the measurement transducer. Such display can be a panel meter calibrated in current, voltage, or other physical units of measurement. More often, it is a data-versus-time trace on a chart recorder, or a data-versus-data trace on an X-Y recorder.

Most telemetry users eventually put their data into a digital computer for detailed analysis. In the old days, this was a laborious process, involving the preparation of tens of thousands of punched cards per hour of testing and often taking weeks to complete. This progressed to the automatic tape formatter system, in which computer-compatible magnetic tapes were generated in real time or on playback of the test data and these, instead of cards, provided the computer entry medium. Even this was too slow for many users, however. Now, modern equipment at reasonable prices enables a user to enter modest to high quantities of telemetry data into a computer in real time (as the test is taking place), process the data, observe computed test results, and make instant decisions concerning the continuation or termination of the test. In this manner, an aircraft (for example) can go into a test maneuver, the results of which are telemetered to flight test engineers at the ground station and analyzed within two or three minutes after maneuver completion. After real-time analysis, the maneuver can be repeated, the next maneuver performed, or the test pilot instructed to return to base for safety reasons.

The Dedicated Computer

Up until a few years ago, most telemetry users were unable to get processed data in real time. They could observe raw data on strip charts, bar charts, and panel meters, but for processed data they had to wait their turns at the large general-purpose computer at the computer center. The lucky ones had automatic tape formatters and a high priority at the computer and could be reading test results within a day or so. The unlucky ones might wait a month or more.

In 1968, the world's biggest aircraft (Lockheed's C5A for the U.S. Air Force) was flight-tested using a new concept. A minicomputer was dedicated to the job of processing telemetry data; finally flight engineers were able to make real time decisions as the flight progressed. There were three reasons for use of this dedicated computer on the C5 A: the aircraft was large and had a larger number of measurements than any other test vehicle; the price of mini-computers arid software was lower per computer measurement per second than had ever been the case before; and test engineers were able to get much more productive testing per hour of flight with on-line processing.

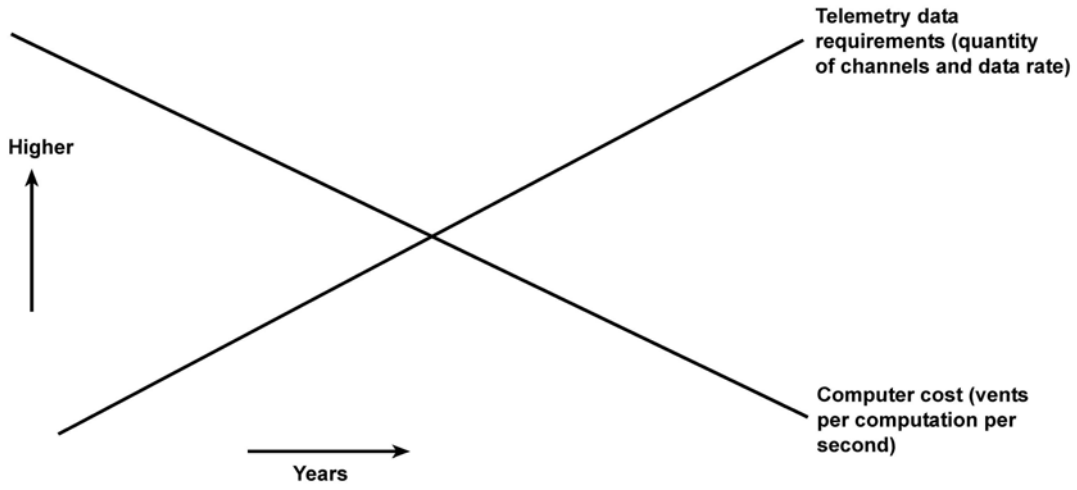


Fig. 2.2. Computer Use

Such a trend continues. Test demands are increasing, both in numbers of data points and in the data frequency of those measurements. And although the price of computers increases with the general trend of the economy, performance increases far more rapidly than does price. The trend is illustrated graphically in Fig. 2-2.

The telemetry computer system of today is practical for almost any user, with prices starting at less than $40,000 for a limited-capability microcomputer system and extending to 50 times that amount for a maximum-capability system.

Telemetry Standards

Fortunately for all telemetry users, a committee composed of aerospace telemetry engineers from all the major test ranges of the United States has been active since the early 1950s, formulating standards which establish the basis for design of telemetry equipment and systems. These Inter-Range Instrumentation Group (TRIG) standards are revised frequently (typically every two or three years) and therefore represent a reasonably current outline of the state of the art.

The IRIG Standard for telemetry is IRIG 106-xx, where "xx" denotes the year of latest issue (as IRIG 106-80). This document contains the following subjects, by section:

1. Introduction
2. Transmitter and Receiver Systems
3. Frequency Division Multiplexing Telemetry Standards
4. Pulse Code Modulation (PCM) Standards
5. Pulse Amplitude Modulation (PAM) Standards
6. Magnetic Tape Recorder/Reproducer Standards
7. Magnetic Tape Standards
8. Transducer Standards

In addition, certain supplementary material is included in the several appendices;

A. Frequency Management Plan for UHF Telemetry Bands
B. Use Criteria for Frequency Division Multiplexing
C. PCM Standards (Additional Information and Recommendations)
D. PAM Standards (Additional Information and Recommendations
E. Magnetic Tape Recorder/Reproducer Information and Use Criteria
F. Available Transducer Documentation

The complete document is available from the IRIG committee.

The IRIG Standard has no official status among industrial users, nor among users outside the U.S.A. However, it is held in high regard by all telemetry users for two reasons. First, it represents the best efforts of a large committee of experienced and unbiased telemetry users. Second, it affects the product designs of all telemetry equipment manufacturers and, therefore, defines equipment which is readily available.

Frequency Modulation (FM) Telemetry

One of the earliest techniques for mixing ("multiplexing") data channels in a telemetry system is frequency modulation (FM). This still accounts for about 20% of the new telemetry market, for reasons which we will see later.

An FM system is shown in Fig. 2-3. Each transducer/signal conditioner output modulates the frequency of a voltage-controlled subcarrier oscillator. Several oscillators, each operating in a dedicated part of the frequency spectrum, are mixed for radio transmission. This can be likened to several FM broadcast stations in a city. Each is assigned a unique place in the frequency spectrum, and each can be modulated within the assigned band without interfering with the others.

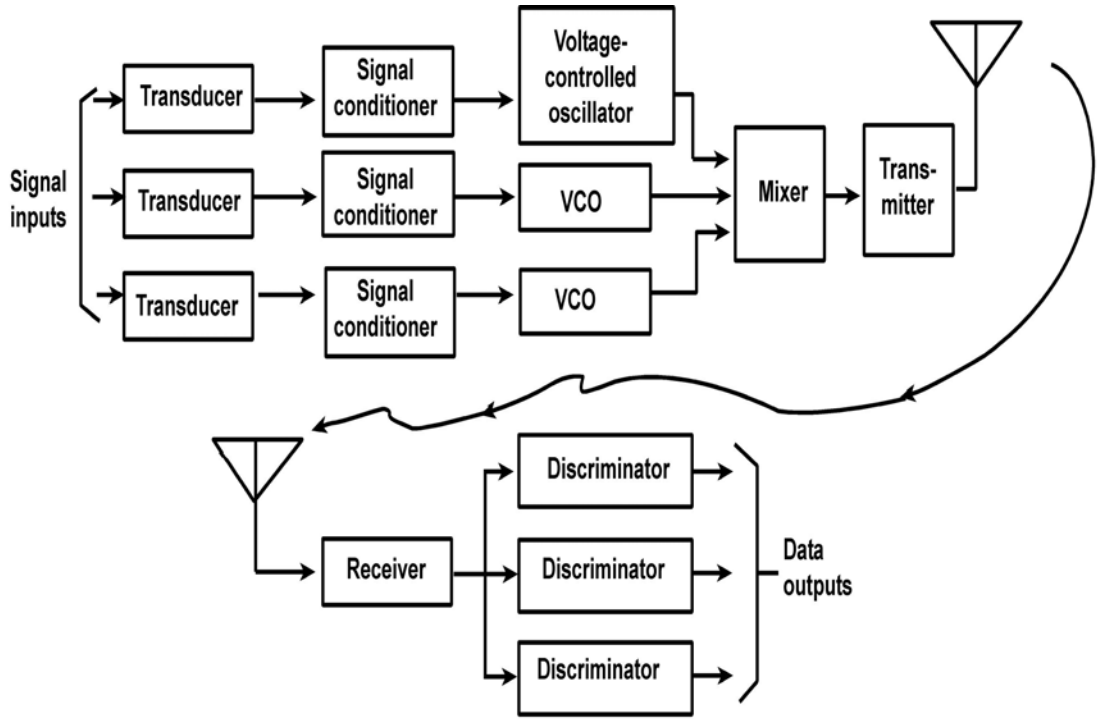


Fig. 2.3. Basic FM System

At the receiving station, an FM demodulator (the common term: "discriminator") is tuned to the frequency of each subcarrier and has a bandwidth equal to that of the modulated subcarrier. As the measurement value changes at the source, the discriminator output changes correspondingly.

Consider this example: A signal conditioner output is a signal with a maximum amplitude of +/—2.5 volts. The VCO is operating at a center frequency of 400 Hz and will deviate +/—7.5% from center frequency if a +/—2.5 volt signal is applied to its input. Therefore, the output of the VCO is a signal varying between 370 Hz and 430 Hz. If this channel is monitoring fuel level in a 600-gallon tank, the signal conditioner may put out —2.5 volts when the tank is empty and +2.5 volts when it is full. This will drive the VCO frequency to 3 70 Hz for the empty tank, 430 Hz for the full tank, and corresponding values in between (10 gallons causes a 1 Hz change in VCO frequency). The discriminator senses frequency variations and converts each frequency into the appropriate output voltage. The 370 Hz frequency may cause —

2.5 volts output, and 430 Hz may cause +2.5 volts output. Thus, the telemetry system becomes a 1:1 link between the signal conditioner output at the transmitting site and the discriminator output at the receiving site. By calibrating the —2.5 to +2.5 volt output of the discriminator into a 0-gallon to 600-gallon display, one can get a good indication of a liquid level, even from thousands of feet or thousands of miles away.

Pulse – Amplitude Modulation (PAM) Telemetry

Because bandwidth is at a premium and system requirements often call for a higher channel capacity than can be easily offered by FM, a method known as "time-division multiplexing" is employed.

Information theory tells us that signals need not be monitored continuously in order to have accurate data. Figure 2-4 reveals the basic theory behind time-division multiplexing. All channels use the same portion of the frequency spectrum, but not at the same time. The signal in each channel is sampled in sequence by a commutator, and the amplitude of each is an indication of the instantaneous data value at that point. When all channels have been sampled, the sequence starts over at the first channel. Thus, samples from a particular channel are interleaved in time with samples from all the other channels, and the amplitude of each is modulated by its data input.

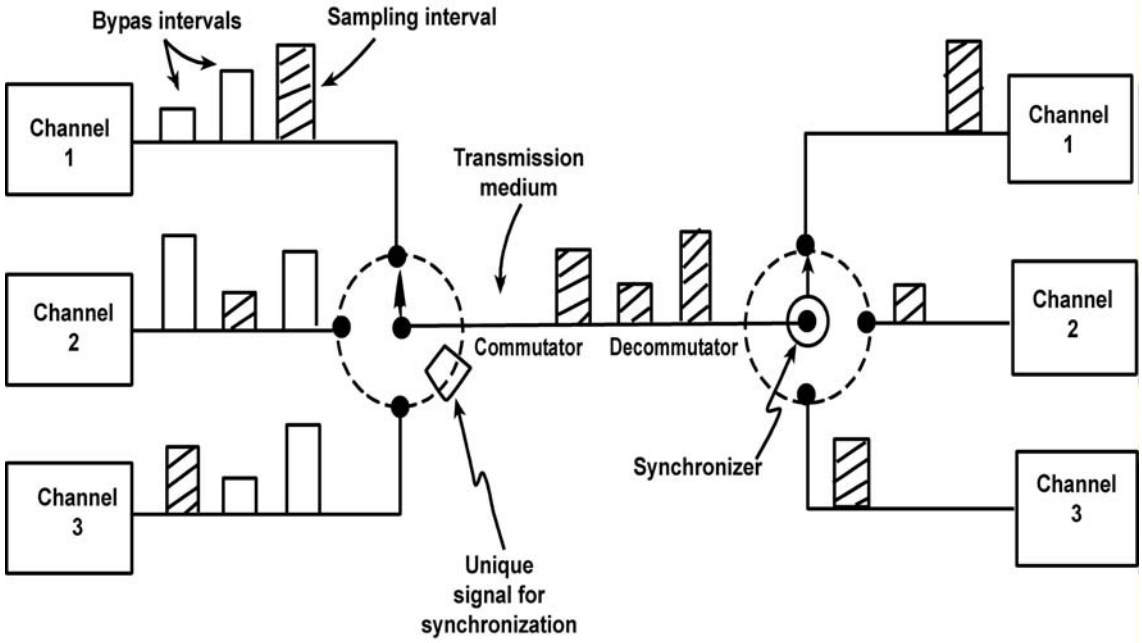


Fig. 2.4. Basic Time-Division Multiplex System

Since no channel is monitored continuously in a time-division system, the sampling must be rapid enough that the signal amplitude in any channel does not change too much between sampling intervals. Practical telemetry systems use high sampling rates to preserve all the information in the original signal .The sampling rate in a typical telemetry system is about five times the highest frequency component in the sampled signal. For example, if the highest frequency component in a particular channel is 40 Hz, the channel is sampled about 200 times per second. If there are eight such channels in a system, the commutator must take at least 1600 samples per second.

At the receiving end of the system, a decommutator operating at exactly the same frequency as the commutator distributes the parts of the multiplexed signal to the proper output channels. Since a time-division system is based on precise timing, it is vitally important that the decommutator be synchronized exactly with the commutator. Otherwise, information on fluid flow, for example, might be misinterpreted as temperature information.

PAM is the simplest form of time-division multiplexing because the samples are transmitted without being modified. As stated previously, it is vitally important in a time-division system that the decommutator be synchronized exactly with the commutator. Synchronization channels must be introduced as part of the commutation scheme as shown in Fig. 2-5. On the receiving end, no data is decommutated until these channels have been properly recognized and "time-tagged," at which time the decommutation process can begin.

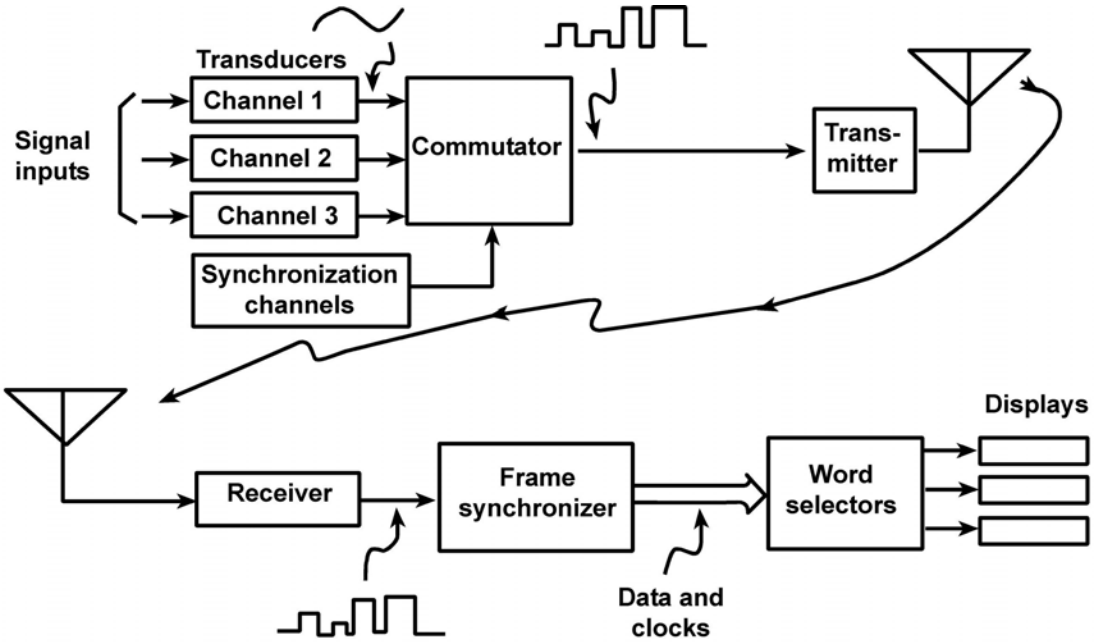


Fig. 2.5. Basic PAM System

Pulse Code Modulation (PCM) Telemetry

At first glance, a pulse-code modulation (PCM) telemetry system (Fig. 2-6) may appear to be like a PAM system. Here, as before, we use "time-division multiplexing" to sample all the measurement points for a test sequentially with a commutator. However, a closer look shows another element in the transmitting system, an "encoder," has been added. This device accepts each PAM sample in turn, converts the amplitude of that sample into a binary number, and shifts the bits of that number

out serially. The encoder may convert a zero-amplitude pulse into the binary number "0," the full-scale pulse into the binary number "1023," and pulses between those extremes into appropriate binary numbers so that each number is almost exactly proportional to the instantaneous amplitude of its measurement point.

The receiving station must synchronize on the serial data stream, identify the sequence of bits which make up each binary number, and convert those bits sequences or "words" into computer data, analog values, or other useful outputs.
Since the system makes binarily weighted "codes" of the measurement data, we call it a pulse-code modulation (PCM) system.

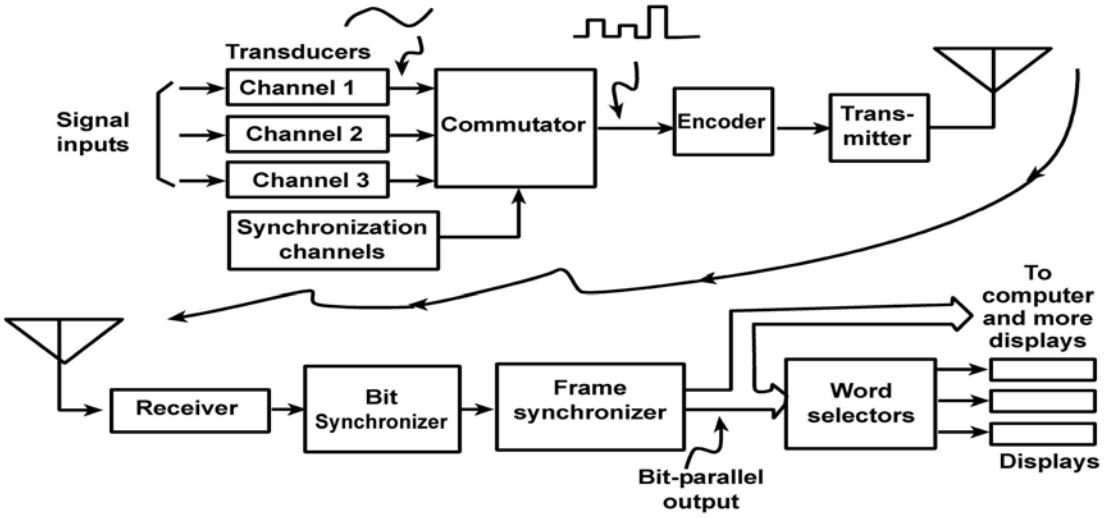


Fig. 2.6. Basic PCM System

Comparison: FM, PAM, and PCM

In summary, telemetry systems for general use employ one of three methods of data multiplexing: FM, PAM, or PCM.

There is not a clear-cut choice between FM, PAM, and PCM (otherwise everyone would drop two and concentrate on the third!).
However, for each application there is likely to be a good reasor for choosing one to the exclusion of the other two.

FM has one clear-cut advantage, it is about twice as efficient at PCM in translation of input data bandwidth into multiplex bandwidth; for example, it enables a user to put ten 1000 Hz data measurements into a multiplex with a top frequency of about 125 kHz, while the same measurements encoded into PCM would occupy a spectrum of about 250 kHz. This means twice as much data on a limited-bandwidth radio link or operation of a tape recorder twice as long as with PCM.

Also, in a system of just a few channels, FM is generally less expensive than PCM.
PAM is used in several Navy missile programs in which the low complexity and small size of the encoder lend themselves to small missile applications. Also, the bandwidth efficiency is even better than FM. However, it is a relatively inaccurate form of data transmission, and this makes it unpopular except in rare cases.

On the other hand, PCM has better accuracy, greater dynamic range, and less noise. And, if the system has more than about 14 data channels, PCM is generally less expensive per channel than FM. In those cases in which the eventual destination of data is a digital computer, there are many advantages of using PCM multiplexing, transmission, and storage.
The preceding comparisons are presented in tabular form in Table 2-1.

Content of this Course

Since PCM and FM account for about 85% of the telemetry use at this time, this ILM concentrates on those two methods from this point.

	FM	PAM	PCM
Efficiency in use of radio or tape recorder bandwidth	Medium	Best	Worst
Cost of a small transmitting system	Lowest	Low	Highest
Size	Smallest	Small	Largest

Cost of a large transmitting system	Highest	Lowest	Medium
Size	Largest	Smallest	Large
Cost of a small receiving system	Lowest	Higher	Highest
Cost of a large receiving system	Highest	High	High
Accuracy	Poor	Poor	Excellent
Percent of use (approximate)	20	15	65

Table 2 -1. Comparison of FM, PAM, and PCM

INTRODUCTION TO ADVANCED TRENDS IN COMPUTER CONTROL

Definition

A) Today' s Process Industries run on a new kind of fuel information because prompt answer to key business questions mean greater yields, lower costs, better designs and more accurate trouble shooting. The implementation of computing and control systems for the execution of the advanced control algorithms, data inter change of setpoints, process variables and control variables. The testing auditing and improving of implemented strategies.

B) The differentiation between regulatory and advanced control are:

1- Regulatory Controls:-

- An integral part of any DCS system
- Enable implementation of Process / Design constraints
- Operators control the process within the constraints the comfort level settings

2- Advanced Controls:-

- Operate the unit against the constraints
- The ultimately safe and maximum production settings

3- Advanced Control Benefits

Improved plant responsiveness, control and high return on investment. The identification of the optimum unit or plant wide setpoints based on :-

- Defined Economic Constraints
- Utility cost, availability and distribution
- Unit performance curves

- Plant equipment status
- Feedstook variations
- Production planning

Enhanced planning capability "WHAT IF?" case studies improved ability to troubleshoot equipment and instrument problems

Engineering / design studies

Operator training tool

Constraint monitoring

Trending and off-line application

Using the same model and user interface

3.3- Application

A) Process Monitoring

To analyze on-line plant data

Tracking equipment performance

Detecting instrument errors or malfunction

B) Process Control and Optimization

The type of plant optimization can be performed on-line or off-line in open loop or closed loop mode

C) Equipment Performance Analysis

The ability of identifying true equipment performance trends and resulting effects on the entire plant

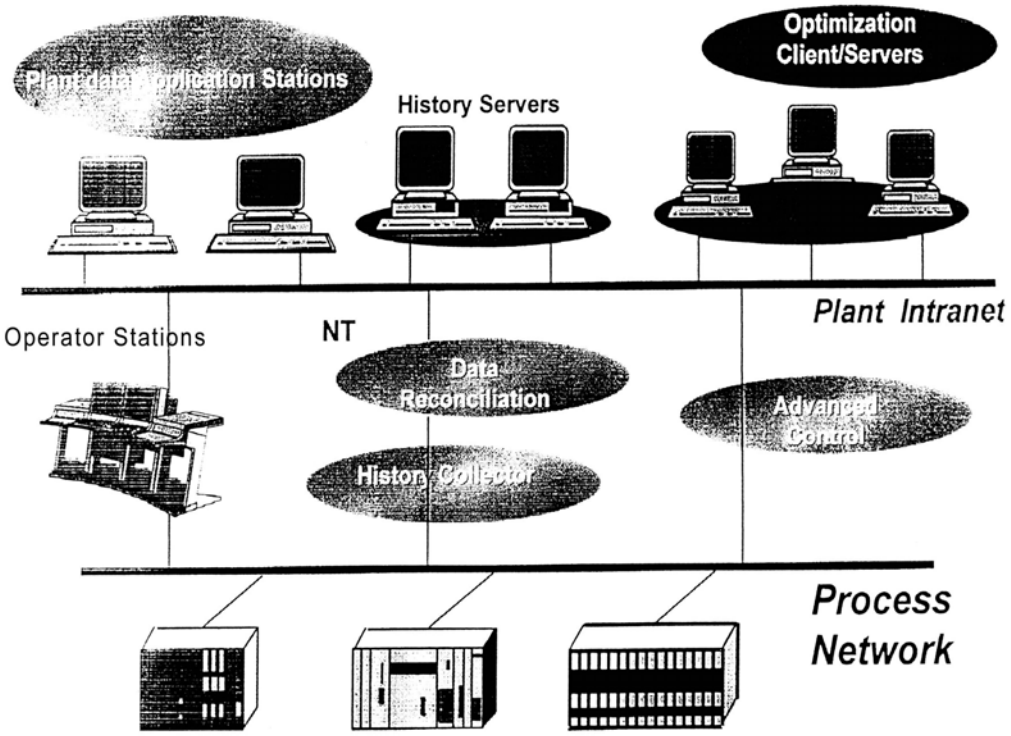
D) Instrument Maintenance

The ability of identifying the instruments need recalibration or repair:-

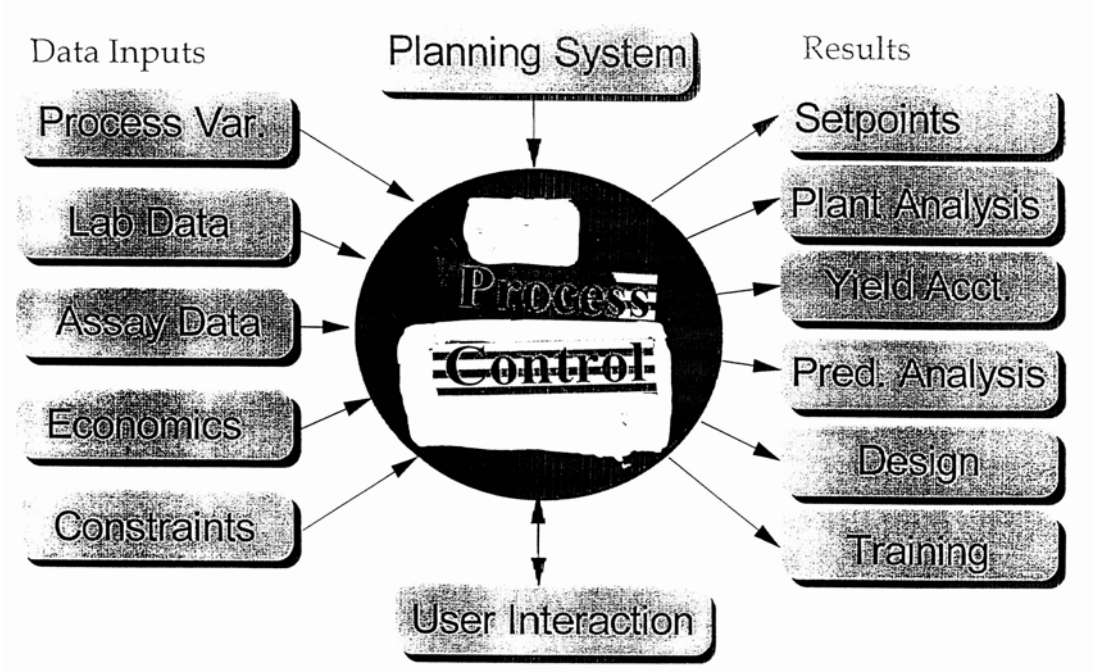
Troubleshooting and debottlenecking plant processes

Modernizing and improving plant yield and profitability

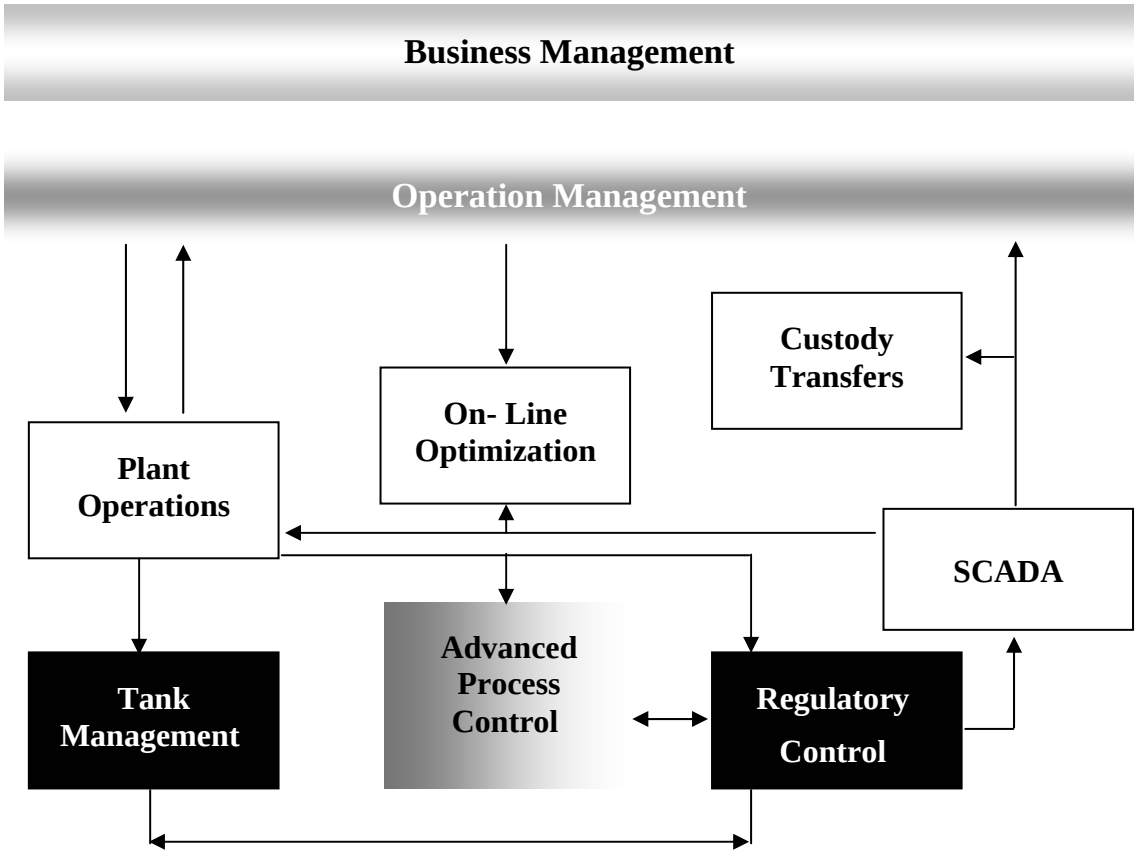
3.4- Overview of plant control architecture



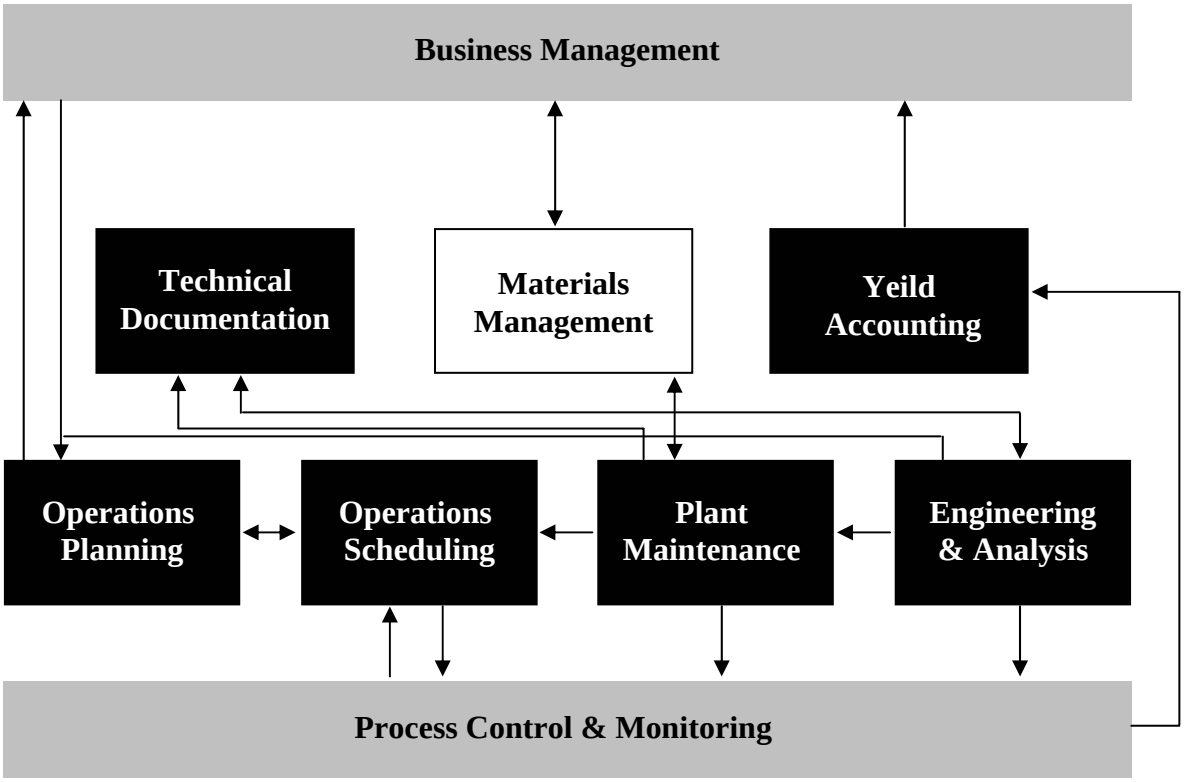
3.5- Plant Optimization Data Interchanges



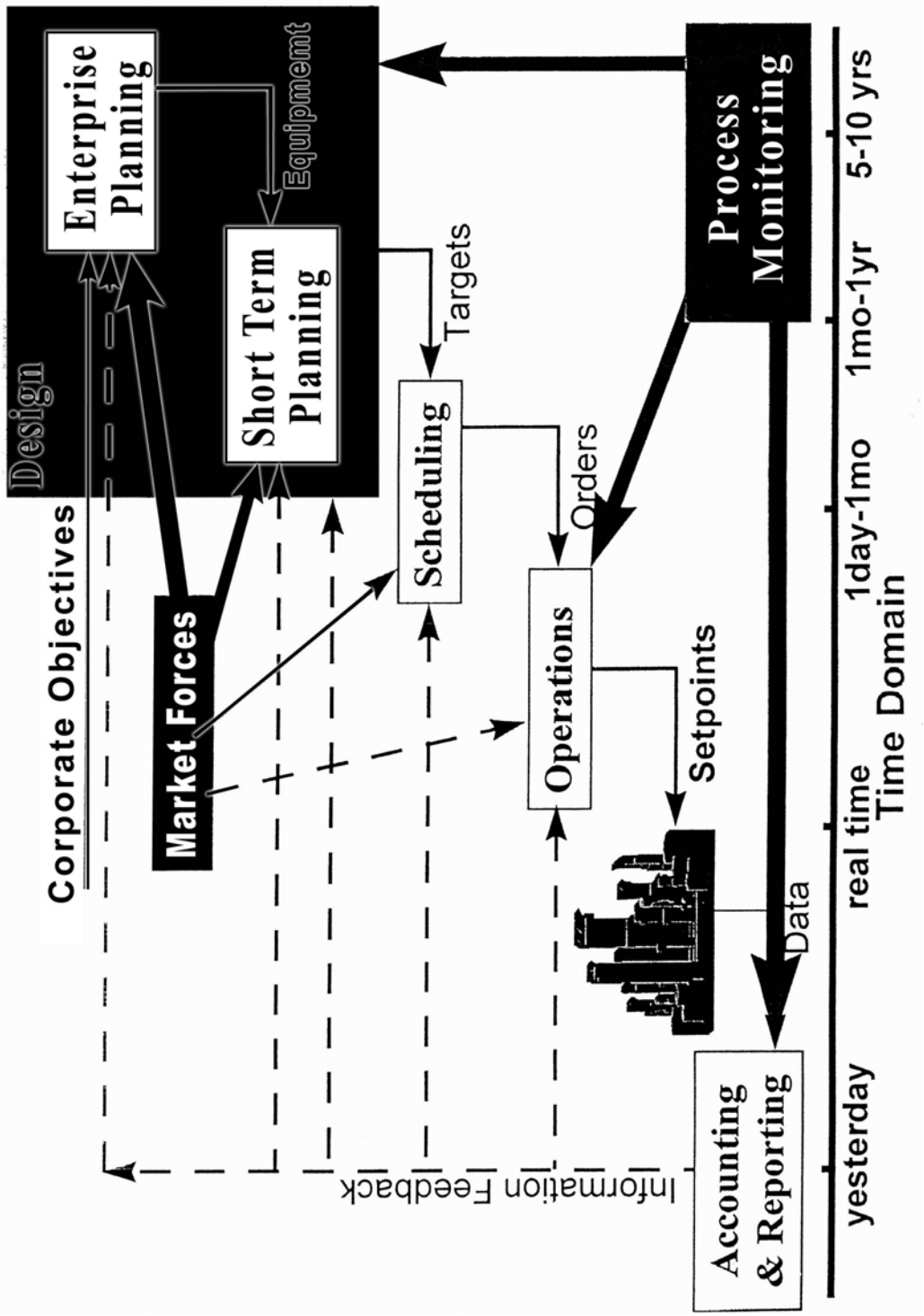
3.6- Operations Management Overview



3.7- Operations Management Overview



3.8 Implementation Of Plant Model Overview



ADVANCED OPERATOR INTERFACES

OPERATOR INTERFACES

Introduction

The control and communication equipment described in the previous chapters performs the bulk of the automated functions that are required for operating an industrial process. For this automated equipment to be used in a safe and effective manner, however, it is absolutely necessary to have a well-engineered human interface system to permit error-free interactions between the humans and the automated system. Two distinct groups of plant personnel interact with the control system on a regular basis:

1. ***Instrumentation and Control System Engineers***—these people are responsible for setting up the control system initially and adjusting and maintaining it from time to time afterwards;
2. ***Plant Operators***—these people are responsible for monitoring, supervising, and running the process through the control system during startup, operation, and shutdown conditions.

As the generalized distributed control system architecture in Figure 1.4 shows, a human interface capability can be provided at one or both of two levels:

1. through a low-level human interface (LLHI) connected directly to the local control unit or data input/output unit (DI/OU) via dedicated cabling;
2. through a high-level human interface (HLHI) connected to an LCU or DI/OU only through the shared communications facility.

The low-level human interface equipment used in distributed control systems usually resembles the panel board instrumentation (stations, indicators, and recorders) and tuning devices used in conventional electric analog control systems. The HLHI equipment makes maximum use of the latest display technology (e.g.,

CRTs or flat-panel displays) and peripheral devices (e.g., printers and magnetic storage) that are available on the market; it is configured in a console arrangement that allows operator and engineer to be seated during use.

When it is included in the system configuration, the LLHI generally is located geographically close to (within 100-200 feet of) the LCU or DI/OU to which it is connected. On the other hand, the HLHI can be located anywhere in the plant, including the central control room. The needs of the application will determine whether the particular installation has one or both levels of interface. Figures 6.1, 6.2, and 6.3 show some examples of typical installations and their corresponding equipment configurations.

Figure 6.1 illustrates a relatively small and simple installation. A single LCU located in the plant equipment room (sometimes called the relay room) performs all of the required control functions. Low-level human interface units located in the equipment room and the plant control rooms provide the complete operator and instrument engineer interface for the control system. This type of equipment configuration is typical of a standalone control system for a small process or of a small digital control system installed in a plant controlled primarily with conventional electrical analog or pneumatic equipment.

Figure 6.2 shows a typical structure of a complete plantwide control system. Several LCUs are used to implement the functions required in controlling the process; therefore, the control '^functionally distributed. However, the LCUs are all located in a central equipment room area, and so it is not a geographically distributed control system. Both high-level and low-level human interface devices are located in the control room area for operational purposes. Most of the operator control functions are performed using the high-level interface; the low-level interface is included in the configuration primarily to serve as a backup in case the high-level interface fails. A high-level human interface is located in the instrument engineer's

area so that control system monitoring and analysis can be done without disturbing plant operations. This type of installation is typical of early distributed control system configurations in which equipment location and operator interface design followed conventional practices.

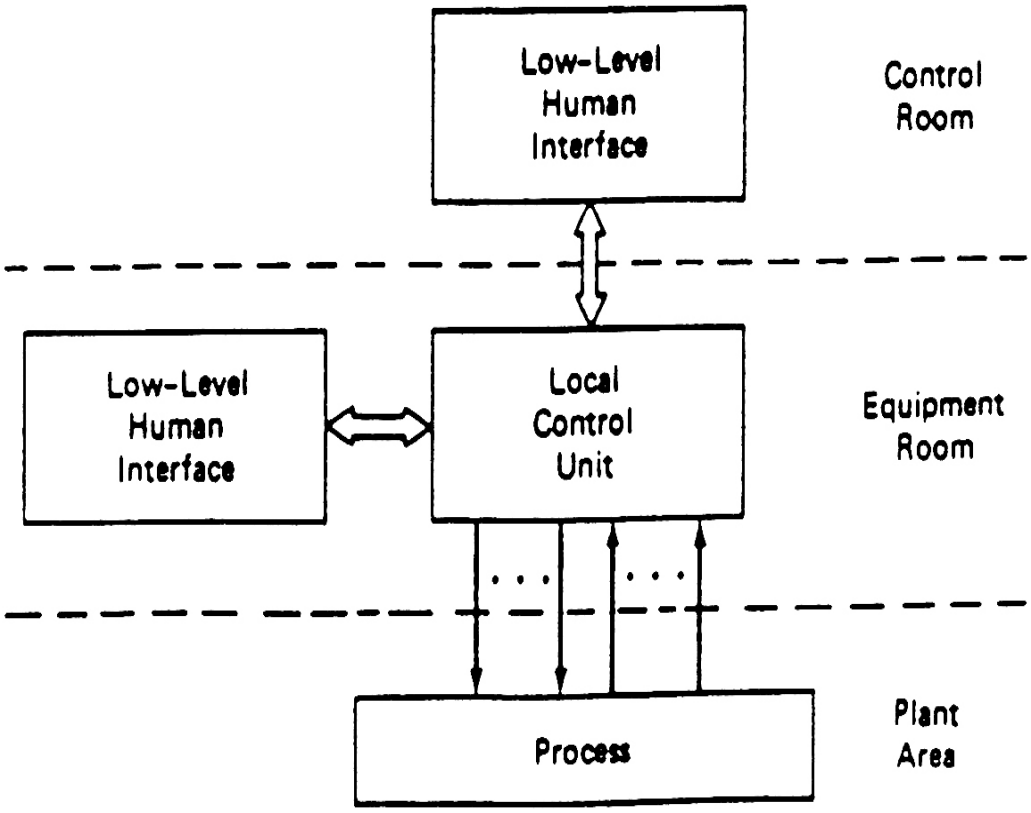


Figure 6.1 Stand- Alone Control Configuration

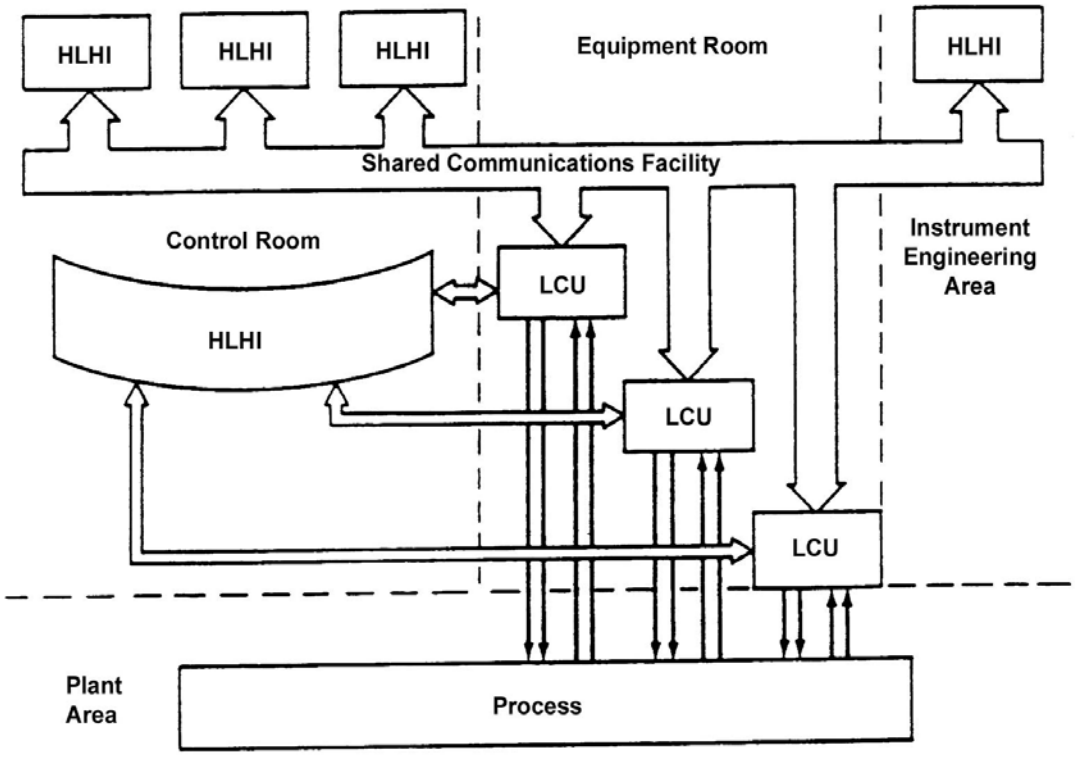


Figure 6.2. Geographically Centralized Control Configuration

Figure 6.3 shows a fully distributed control system configuration. In this case, each LCU is located in the plant area closest to the portion of the process that it controls. Associated low-level human interface equipment (if provided) is also located in this area. The control room and instrument engineering areas contain high-level human interface units, which are used to perform all of the primary operational and engineering functions. The low-level units are used only as manual backup controls in case the high-level equipment fails or needs maintenance. This configuration takes advantage of two areas of equipment savings that result from a totally distributed system architecture: (1) reduction in control room size (by eliminating panel board equipment), and (2) reduction in field wiring costs (by placing LCUs near the process).

These examples of system configurations illustrate the point that human interface equipment in a distributed control system must be designed to meet a wide range of applications:

1. Large as well as small systems;
2. Centralized equipment configurations (often used in retrofit installations made long after original plant construction) as well as distributed ones (likely in "grass roots" installations made during plant construction);
3. Variety of human interface philosophies, ranging from accepting CRT-only operation to requiring panel board instrumentation in at least a backup role.

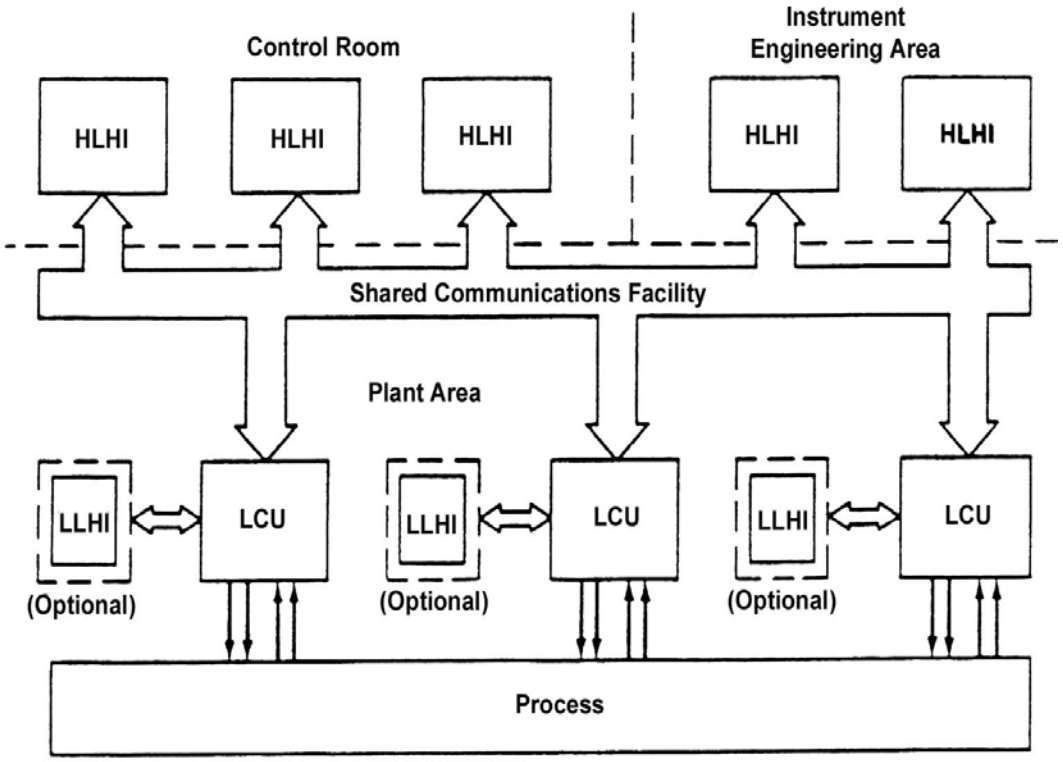


Figure 6.3. Geographically Distributed Control Configuration

This chapter and the next provide an overview of the major issues to consider when evaluating or designing human interface equipment in a distributed control system. As in the previous chapters, the discussion will not be a detailed analysis but instead will address only the significant points; the references will permit the reader to go into any selected area in greater depth. This chapter discusses operator interface

requirements and design issues; Chapter 7 will deal with instrument engineer interfaces. Section 6.2 summarizes the key requirements of an operator interface system; Sections 6.3 and 6.4 describe and evaluate alternative design approaches to implementing these requirements for the low-level and high-level operator interfaces, respectively.

Operator Interface Requirements

Despite the continuing trend toward increased automation in process control and less reliance on the operator, the basic responsibilities of the operator have remained largely the same in the last fifty years. Most of the changes have come in the relative emphasis on the various operator functions and the means provided to accomplish them. As a result, the operator interface in a distributed control system must allow the operator to perform tasks in the following traditional areas of responsibility: process monitoring, process control, process diagnostics, and process record keeping. In addition, it is important to design the operator interface system using human factors design principles (also called ergonomics) to ensure that the operator can perform these tasks in an effective manner with minimum risk of confusion or error. The following paragraphs provide a discussion of the key functional requirements in each of these areas. References 6.1-6.8 give additional information on advances in operator interfaces.

Process Monitoring. A basic function of the operator interface system is to allow the operator (whether one or more) to observe and monitor the current state of the process. This function includes the following specific requirements:

1. The current values of all process variables of interest in the system must be available for the operator to view at any time. This includes both continuous process variables (e.g., flows, temperatures, and pressures) and logical process variables (e.g., pump on/off status and switch positions). The operator must have rapid access to any variable, and the values displayed must be accurate and current. If the information provided is not valid for some reason (e.g., a sensor has failed or has been taken out of service for maintenance), this condition should be readily visible to the operator.
2. Each process variable, rather than being identified by a hardware address only, must be identifiable by a "tag" or name assigned by the instrument engineer; a descriptor that expands on and describes the tagged variable must be associated with the tag. The tag and descriptor give the variable a meaning relative to the process; an example might be to label a certain temperature with a tag of TT075/ B and a corresponding descriptor **"COLUMN TEMPERATURE 75 IN AREA B."**
3. The value of the process variable must be in engineering units that are meaningful to the operator, and those units must be displayed along with the variable values. In the temperature example just given, the engineering units might be in degrees Fahrenheit or Celsius.
4. In many cases, the operator is interested in variables that are functions of or combinations of the basic process variables being measured (e.g., an average of several temperatures, a maximum of several flows, or a computed enthalpy). The operator must have these computed variables available at all times in the same formats as the basic variables (i.e.. tags, descriptors, and engineering units).

Another monitoring function of the operator interface is to detect abnormalities in the state of the process and to report them to the operator. In its simplest form, this is the familiar function of alarming. Some of the specific requirements of this function are the following:

1. The control and computing hardware in the distributed system identifies the alarm statuses of individual variables in the process. The operator interface system must report these statuses to the operator in a clear manner. Types of alarms for each variable-such as high, low, and deviation (from a nominal value)-must be differentiated clearly. True alarms must be differentiated from indications of process equipment status that do not denote an abnormal condition requiring operator action.
2. The operator interface must also report similar alarm statuses for computed variables.
3. The operator interface must either display the alarm limits along with the process variable or make them easily accessible to the operator.
4. When the system has detected an alarm condition, the interface must alert the operator to this condition in unambiguous terms and require the operator to acknowledge the existence of the alarm.
5. If the system detects multiple alarm conditions within a short time period, the operator interface must inform the operator that multiple alarms have occurred, preferably with some indication of the priority of the various alarm conditions.
6. In some processes, "abnormal operation" can be detected only by looking at a combination of several process variables and noting if this combination is within an allowable region of operation. In this case, the operator interface system must provide an appropriate mechanism to allow the operator to view this multivariable alarm status condition and interpret it properly.

When monitoring the process, an operator is interested in not only the current value of a process variable but also its trend in time. This gives the operator an idea of the direction in which the process is moving and whether or not there is trouble ahead. For this reason, the operator interface system must provide the operator with fast access to the recent history of selected (not necessarily all) process variables in the plant; these variables are called trended variables. Some specific requirements in trending are that:

1. It must be possible to group the trended variables by related process function as well as by similarities in time scale of interest. For example, it might make sense to group all temperatures that are associated with a particular portion of the process.
2. The trend graph must clearly label the engineering units, time increments, and absolute time of day of the trended variables.
3. The operator must be able to obtain a precise reading (in engineering units) of both the current value as well as past values of the trended variable.
4. If at all possible, the same graph displaying the trend should also show auxiliary information that would help the operator evaluate the status of the trended variable. This information might include the nominal value of the variable, the set point of the associated control loop, the allowed range of the variable, or the allowed rate of change.

Process Control. The process monitoring capabilities just described provide the necessary information for the operator's primary function— process control. The following specific operator interface requirements come under the category of process control:

1. The operator interface must allow the operator to have rapid access to all of the continuous control loops and logic sequences in the process control system.
2. For each continuous control loop, the interface must allow the operator to perform all of the normal control functions: changing control modes (e.g.,

automatic, manual, or cascade), changing control outputs in manual mode, changing set points in automatic mode, and monitoring the results of these actions.

3. The interface must allow the operator to perform such logic control operations as starting and stopping pumps or opening and closing valves. If interlocking logic is included in these operations, the interface must allow the operator to observe the status of the most recently requested command, the current logic state of the process, and the status of any permissives (interlocking signals) that may be preventing execution of the requested command.
4. In the case of a batch control sequence, the operator interface must allow the operator to observe the current status of the sequence and to interact with it to initiate new steps or halt the sequence, as required.
5. In both the continuous and sequential control cases, the interface system must allow the operator to have access to and be able to manipulate the control outputs despite any single-point failure in the equipment between the operator interface and the control outputs.

Process Diagnostics. Monitoring and controlling the process under normal operating conditions are relatively simple functions compared to operation under abnormal or hazardous conditions caused by failures in plant equipment or in the instrumentation and control system. The operator interface system must provide enough information during these unusual conditions to allow the operator to identify the equipment causing the problem, take measures to correct it, and move the process back to its normal operating state.

The first step in this sequence is to determine whether it is the instrumentation and control equipment that is causing the problem. To this end, the distributed control system should provide the following diagnostic features and make the results of the diagnostic tests available to the operator:

1. Ongoing tests and reasonableness checks on the sensors and analyzers that measure the process variables of interest;
2. Ongoing self-tests on the components and modules within the distributed control system itself: controllers, communication elements, computing devices, and the human interface equipment itself.

These diagnostic features also are essential to aid the work of the instrumentation engineer. Chapter 7 will discuss diagnostics from that point of view.

Historically, diagnosing problems within the process itself has been a manual function left to the operator. Operator interface systems have been designed to display all of the available process information (both relevant and irrelevant), and the operator has had to sort it all out and come up with the right diagnosis. This was not a bad approach when the operator had to contend with small processes, those characterized by only a few hundred process variables. More recently, however, processes have grown to such a size that describing them takes 5,000-10,000 process variables (many of which may be strongly interacting). It has become extremely difficult for an operator to identify the source and nature of a fault in an item of process equipment in this environment. A conventional alarming system, for example, may indicate the most immediate failure symptom but provide few clues as to the original source of the alarm condition. As a result, diagnostic functions that automatically detect process faults are now often required in distributed control systems. These functions may include:

1. First-out alarming functions, which tell the operator which alarm in a sequence occurred first;
2. Priority alarming functions, which rank the current alarms by their importance to process operation, allowing the operator to safely ignore the less important ones, at least temporarily;

3. More advanced diagnostic functions that use a combination of alarming information and data on process variables to identify the item of failed process equipment and (in some cases) the mode of failure.

Many of these advanced alarming and diagnostic functions are application-oriented ones that the designer must configure for the specific process of interest; however, the distributed control system must support the implementation of these functions.

Process Record Keeping. One of the more tedious duties that operating people in a process plant must perform has been to walk the board; that is, to take a pencil and clipboard and periodically note and record the current values of all process variables in the plant. Depending on the process, the frequency for doing this has ranged from once an hour to once every several hours. This logged information, along with the trend recordings obtained automatically, serves as a useful record of plant operating status during each shift. The record-keeping burden has increased significantly in recent years due (in part, at least) to governmental reporting requirements related to pollution monitoring, product liability, and worker safety regulations.

Record-keeping was one of the first functions to be automated using conventional computer systems. In state-of-the-art distributed control systems, this function often can be implemented in the operator interface system without the use of a separate computer. Specific record-keeping requirements include the following:

1. ***Recording of Short-Term Trending Information*** — the earlier section on process monitoring described this requirement.
2. ***Manual Input of Process Data*** — The operator must be able to enter manually collected process information into the system for record-keeping purposes. This information includes both numeric data and operator "notes" or journal entries.
3. ***Recording of Alarms*** — These are logged on a printer, a data storage device, or both, as they occur. Often, the return-to-normal status and operator

acknowledgments must also be logged. The information recorded includes the tag name of the process variable, the time of alarm, and the type of alarm (high, low, or deviation). There must also be a mechanism that allows convenient review of the alarm information.

4. ***Periodic Records of Process Variable Information***—The values of selected variables are logged on a printer, data storage device, or both, on a periodic basis: every few minutes or every hour, depending on the dynamics of the variable. The operator or instrument engineer may decide to store an averaged value over the sampling period instead of the instantaneous value.
5. ***Long-Term Storage and Retrieval of Information***—The alarms and periodic logs as described above are accessible for short periods of time, commonly, for a single eight-hour shift or a single day. In addition, the same information, or a smoothed or filtered version of it, must be stored on a long-term basis (months or years). The system must include a mechanism for easy retrieval or "instant replay" of such information.
6. ***Recording of Operator Control Actions***—Some process plants require the actions of the operator affecting control of the process to be recorded automatically. These include changes in control mode, set point, manual output, or logic command. Clearly this recording function must be implemented in such a way that the operator cannot deactivate it.

Guidelines for Human Factors Design. In the past, equipment used for operator interfacing has often been designed more for the convenience of the equipment vendor or architect-engineer than for ease of use by the operator. In recent years, it has become clear that a small investment made in the proper design of human interfacing equipment pays handsome dividends: fewer operator errors (which can cause plant downtime or damage to equipment), less operator fatigue (which can cause a loss in productivity), and more efficient use of operating personnel. Compiling an exhaustive list of requirements in the area of human factors design is beyond the scope of this chapter. (For such a list, see References 6.32-6.45, for example.) However, some general

design guidelines for designing operator interface systems for industrial control include the following:

1. Consider the full range of expected operator population (e.g., male and female, large and small, right-handers and left-handers).
2. Take into account common minor disabilities in operators (e.g., color blinded ness and nearsightedness).
3. Design the system for operators, not for computer programmers or engineers.
4. Allow rapid access to all necessary controls and displays.
5. Arrange equipment and displays to make sense from an operational point of view; cluster with respect to process unit, functional operation, or both.
6. Make consistent use of colors, symbols, labels, and positions to minimize operator confusion.
7. Do not flood the operator with a lot of parallel information that is not structured in any way; the information should be prioritized, organized in a meaningful manner, and reported only when it changes significantly.
8. Ensure that the operator's short-term memory is not overtaxed when performing a complex sequence of operations: provide aids such as operator guides, menus, prompts, or interactive sequences for assistance in these operations. These aids are particularly important in stressful situations, during which short-term memory is not a reliable source of operating information.
9. As much as possible, design the system to detect and filter out erroneous operator inputs; when an error occurs, the system must tell the operator what the input error was and what to do next.
10. Make sure the control room environment (e.g., light, sound levels, and layout) is consistent with the selection and design of the control room equipment.

In some respects, these guidelines may only seem to state obvious, common-sense design principles; however, a glance at existing operator interface designs shows that these principles are very often violated, either for design expediency or through ignorance of these ergonomic issues. In later sections of this chapter, discussions of

operator interface design features will include commentary on the degree to which these features meet the intent of the guidelines.

Low-Level Operator Interface

As the introduction to this chapter indicated, the low-level operator interface (LLOI) in a distributed control system is connected directly to the LCU and is dedicated to controlling and monitoring that LCU. This contrasts with the high-level operator interface (HLOI), which can be associated with multiple LCUs and is connected to them through the shared communication facility. LLOIs are used in a variety of applications, in some cases in conjunction with high-level operator interfaces (HLOIs) and in others in place of them. In some applications, all operator functions are performed through the HLOI and no low-level interface is required except during emergency or failure conditions. There are a number of motivations for using an LLOI:

1. It provides an interface that is familiar to operators trained to use panelboard instrumentation, since it is usually designed to resemble that type of instrumentation.
2. It is usually less expensive than an HLOI in small applications (say. less than 50 control loops).
3. It can provide manual backup in case the automatic control equipment or the HLOI fails.

LLOI instrumentation usually includes the following devices: control stations, indicator stations, alarm annunciates, and trend recorders. In some distributed control systems, the vendor offers exactly the same type of instrumentation as used in his conventional analog and logic control systems. More often, however, the vendor supplies smart (microprocessor-based) instrumentation, which offers the user functionality beyond that available in conventional panelboard instrumentation. The following paragraphs describe several of these smart devices.

References 6.38 and 6.43 provide many helpful suggestions regarding the proper ergonomic design of this type of operator interface.

Continuous Control Station. One type of panelboard instrumentation used in process control systems is the manual/automatic station associated with a continuous control loop. The stations discussed here are split stations; that is, they are separate from the LCU. Figure 6.4 illustrates a typical version of a smart continuous control station in a distributed control system. As in the case of most conventional control stations, this one has bar graph indicators that display the process variable ("TV"), associated set point ("SP"), and the control output as a percent of scale ("OUT"). In addition, however, the smart station includes a shared digital display to provide a precise reading of each of these variables in engineering units. The units used are indicated in an accompanying digital display or are printed on a removable label (in this example, "DEGF" or "%"). The shared digital display also can be used to indicate the high and low alarm limits ("HI ALM" and "LO ALM") on the process variable when selected by the operator. Pushbuttons allow the operator to change the mode of control (e.g., manual, automatic, or cascade) and to ramp the set point ("SET") or control output ("OUT"), depending on the mode. Usually, both fast and slow ramping speeds are provided for the convenience of the operator. Other indications the control station often provides include any alarms associated with the process variable being controlled and an indication of the operational status (whether "healthy" or not) of the associated LCU.

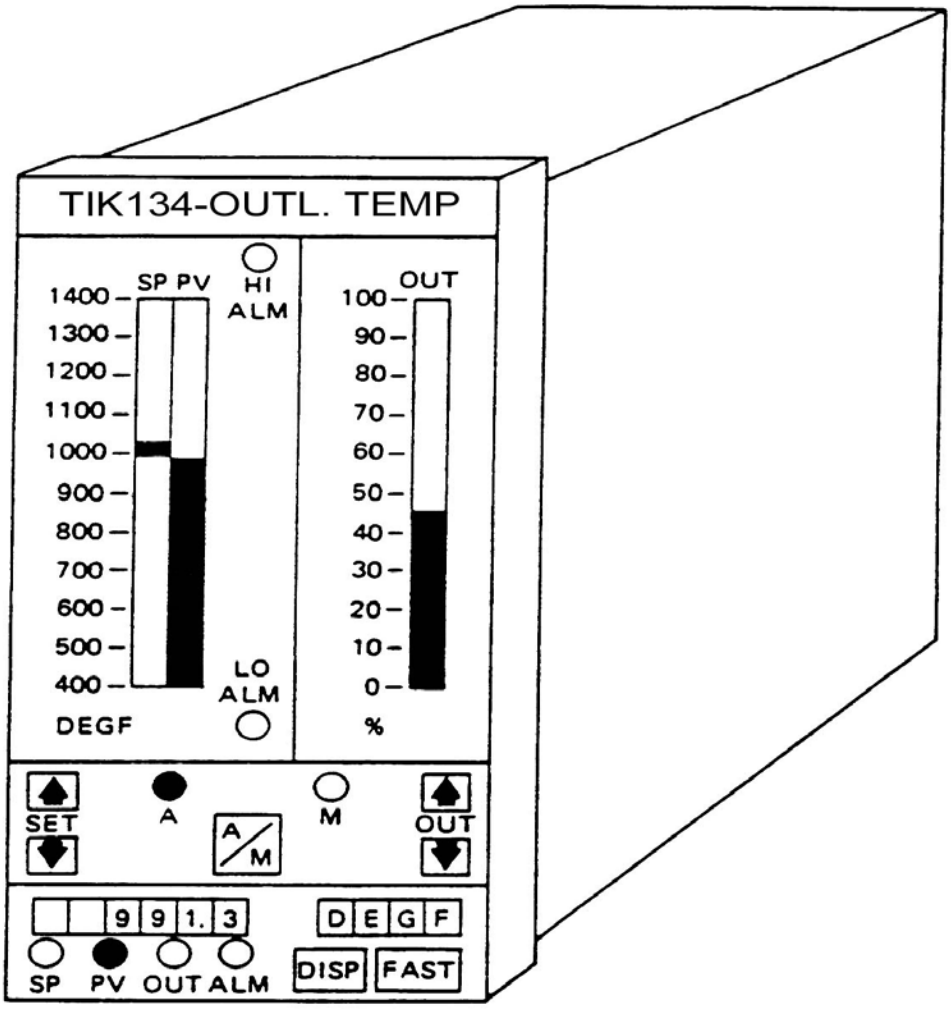


Figure 6.4. Continuous Control Station

To minimize requirements for spare parts, one basic control station should be used for all types of associated control loops: standard PID, cascade, ratio, or bias. The station can be customized through the configuration of options in the electronics (using jumpers or switches) and on the front plate indicators and switches (using different overlays or faceplates as appropriate).

To be effective as a manual backup station in addition to its role as a single-loop operator interface, the control station must be connected directly to the control output section of the LCU or to the associated field termination panel. In this arrangement, a "hard" control output (e.g., a 4-20 ma signal) generated by the station can pass directly to the process as a backup control signal in case the LCU fails or is

undergoing maintenance. The direct connection also allows the process variable input to come into the station for indication to the operator during manual control. To keep the control output from going through a step change in value when the backup mode is initiated or concluded (automatic bump less transfer), the station and the LCU must be aware of each other's nominal control output signal. These signal values and other useful information (such as alarm and diagnostic status signals) are often sent over a direct serial communication link between the LCU and the station. (See Figure 4.8 and the discussion in Section 4.2.5 for more details.)

Manual Loader Station. Some applications use the HLOI as the primary control station and don't require a full-blown continuous control station for each loop. However, a device is still needed to hold the 4-20 ma control output signal if the LCU fails or is taken off-line for maintenance or other reasons. In this situation, a simple manual loader station is a low-cost alternative to the continuous control station for the purposes of backup. The manual loader station is plugged in at the same point as the continuous control station but only allows the operator to run the loop in manual mode. Sometimes the process variable is displayed: more often it is not. Any balancing of the control output to accomplish bumpless transfer to or from backup is accomplished manually.

Both the continuous control station (if used for backup) and the manual loader station should be powered from a different supply than that used for the LCU. to ensure continuous backup in case of an LCU power failure.

Indicator Station. If the operator must be able to monitor process variables not associated with control loops, a panel board indicator station can be provided as a part of the LLOI family of products. The indicator station is similar to the control station in that it provides both bar graph and digital numeric readouts of the process variables in engineering units. Of course, an indicator station requires no control push buttons. However, it often provides alarming and LCU diagnostic indications, as does the continuous control station.

Logic Station. Figure 6.5 illustrates a control station for a logic control or sequential control system. It consists simply of a set of pushbuttons and indicating lights that are assigned different meanings (through labels) depending on the logic functions being implemented. This type of station is used to turn pumps on and off, start automatic sequences, or provide permissive or other operator inputs to the logic system. In some systems, the logic control station performs a manual backup function similar to that performed by the continuous control station in case of a failure of an LCU. More often, however, the logic station acts simply as a low-cost operator interface; if the LCU fails, the logic outputs revert to their default or safe states, as described in Section 4.2.4.

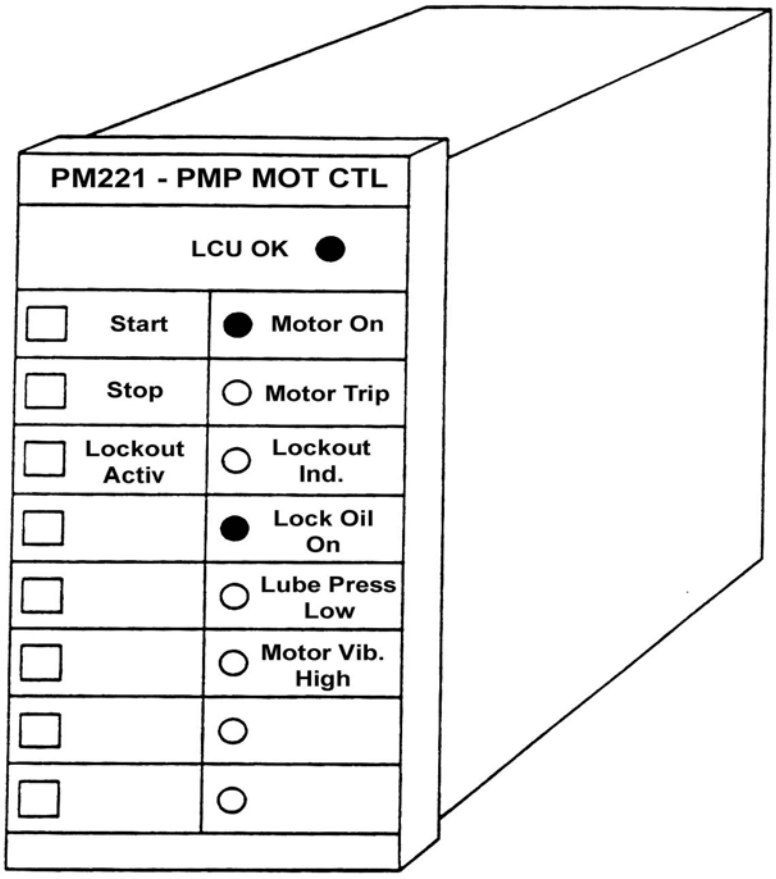


Figure 6.5. Logic Control Station

Smart Annunciators. Alarm annunciators in distributed control systems are often microprocessor-based, providing a level of functionality beyond the capability of conventional hard-wired annunciator systems. These smart annunciators can provide such functions as:

1. ***Alarm Prioritization*** — the annunciator differentiates between status annunciation and true alarms (and how critical the alarms are);
2. ***Annunciation and Acknowledgment Mode Options*** — The operator receives a variety of audible and visible alarm annunciation signals (e.g., horns, buzzers, flashing lights, and voice messages). A range of alarm acknowledgment and silencing modes also can be provided.
3. ***First-out Annunciation*** — the annunciator displays the first alarm that appears within a selected group.
4. ***Alarm "Cutout"*** — The annunciator suppresses an alarm condition if other specified status conditions are fulfilled.

The last function is valuable in minimizing meaningless "nuisance" alarms, a problem that References 6.14 and 6.36 describe in some detail. For example, if a pump fails and triggers an alarm, the pump-failed status signal can be used to lock out other related alarms such as "low flow" or "pump speed low," since they are not meaningful given the failed operating status of the pump.

Of course, the four alarm logic functions previously listed also can be accomplished within the LCUs in the distributed system itself; however, in some applications it may be convenient to incorporate the functions externally in the annunciators.

Chart Recorders. Although conventional round chart or strip chart recorders are often used to record process variables in a distributed control system, digital recorders which use microprocessors are becoming more cost-effective and popular. The digital recorder gathers trend data in its memory and displays the data to the operator using a liquid crystal panel or other flat display device. For hard-copy output, the recorder uses an impact- or heat-type of printing mechanism instead of pen-and-ink to record the information. In some models, the recorder draws the chart scales as it is recording, so that plain paper instead of chart paper can be used. The recorder often provides such functions as automatically labeling time and range of

variable directly on the chart, using alphabetic or numeric characters. Also, each process variable can be recorded using a different symbol or color to allow the operator to distinguish between the variables easily. Because of the flexibility of the printing mechanism and the memory capabilities of the recorder, intermittent printing of the process variables can supplement the display output without losing any of the stored information.

Selection of Station Components. To a considerable degree, the type of display and pushbutton components selected determines the reliability and ease of use of the LLOI equipment. This equipment must be designed to meet the exacting needs of the process control application: components that are suitable for home or office environments may not be at all appropriate in a process plant or factory. Some of the requirements the designer should meet include the following:

1. Displays and pushbuttons should be sealed against the atmosphere to avoid contamination (from dirt or corrosive gases, for example);
2. Displays should be selected for high visibility in the expected ambient light environment;
3. Each pushbutton when depressed should provide tactile (touch) feedback to the operator to minimize potential errors.

A variety of display types has been developed for industrial use. The workhorse of the discrete display world is the light-emitting diode (LED). This component type is used in on/off status, bar-graph, and alphanumeric displays. It is quite suitable in most applications. However, bar-graph displays tend to be somewhat low in resolution when composed of LEDs rather than other display components. Gas-discharge and gas-plasma displays provide high-resolution and high-visibility bar graphs. The declining cost trend for these devices has made them an attractive display alternative to LEDs. Liquid crystal displays (LCDs) are very flexible and have been used in a wide variety of display configurations (mixtures of bar graphs, alphanumerics, and status displays). However, LCDs are not as visible as LEDs or

gas-discharge displays, especially in low ambient lighting situations.

In the operator input area, the two most common types of push-button and switch inputs are:

1. ***Spring-Loaded Plungers***—This component, often used in type writer like keyboards, can make use of a number of different switch-sensing mechanisms—magnetic, optical, or capacitive, for example. It is not used very often in panelboard equipment since its mechanical configuration makes it difficult to isolate the sensing mechanism from the environment.
2. ***Membrane or Dimple Switches***—These are inexpensive mechanical switching components that come in a flat configuration suitable for direct mounting on a printed circuit board. An overlay sheet of mylar or plastic protects the switch assemblies from the environment. If packaged properly, they provide tactile feedback to the operator when depressed.

There are many other types of displays and switches suitable for use in panelboard equipment. For a survey of these types and their relative advantages and disadvantages, consult references such as 6.21 and 6.25.

Application in Distributed Systems. In the early installations of distributed control systems, both suppliers and users were concerned whether the operating personnel in the plants would accept the new technology. This involved two aspects of the new technology: (I) using microprocessors to implement closed-loop control functions, and (2) using new human interface hardware such as CRTs as the primary operating tool. As a result, those responsible for many of these installations took a "belt and suspenders" approach to the design of operator interface equipment by providing both panelboard instrumentation and CRT consoles. Operators were first trained to use the panelboard-type equipment, then were gradually weaned from it in favor of the CRT consoles. As the next section of this chapter will discuss, few industrial users now have any hesitation in providing CRT consoles as the primary operator

interface tool. The LLOI equipment described in this section does, however, find continuing application in small control systems and as a backup mechanism for critical process control loops.

High-Level Operator Interface

In contrast with the low-level operator interface described in the previous section, the high-level operator interface in a distributed control system is a shared interface that is not dedicated to any particular LCU. Rather, the HLOI is used to monitor and control the operation of the process through any or all of the LCUs in the distributed system. Information passes between the HLOI and the LCUs by means of the shared communications facility, as described in Chapter 5.

While the LLOI hardware resembles conventional panelboard instrumentation, HLOI hardware uses CRT or similar advanced display technology in console configurations often called video display units (VDUs). The HLOI accepts operator inputs through keyboards instead of the switches, push-buttons, and potentiometers characteristic of conventional operator interface panels. Other digital hardware prints, stores, and manipulates information required in the operator interface system. This section will discuss many of the issues involved in the design of the HLOI; References 6.9-6.14 provide additional information on these issues.

In general, the use of microprocessor-based digital technology in the design of the HLOI system allows the development of a hardware configuration that departs radically from the design of panelboard-based operator interfaces. This configuration provides the user with several significant advantages over previous approaches:

1. Control room space is reduced significantly; one or a few VDUs can eplace panelboards several feet to 200 feet in length, saving floor space and equipment expense.
2. One can design the operator interface for a specific process plant in a much

more flexible manner. The hardware operations peculiar to panelboard design and implementation (e.g., selecting control stations and cutting holes in panels) are eliminated. Instead, CRT display formats define the key operator interface mechanisms. One can change these formats during startup and duplicate selected information displays wherever necessary. Being able to do this leads to a much more usable operator interface configuration than conventional approaches.

3. Using microprocessors permits cost-effective implementation of functions that previously could be accomplished only with expensive computers. These include color graphic displays that mimic the organization of the process; information presented in engineering units: and advanced computing and data storage and retrieval functions.

However, one must take great care in designing the HLOI to achieve these benefits while minimizing any negative effects or concerns on the part of the operating personnel. It became evident during the early introduction of this technology to the marketplace that operators accept properly designed HLOIs very quickly. This is especially true of younger operators who have been preconditioned and pretrained by video games and personal computers.

Architectural Alternatives

All high-level operator interface units in distributed control systems are composed of similar elements: operator display; keyboard or other input device; main processor and memory; disk memory storage: interface to the shared communication facility; and hard-copy devices and other peripherals. However, the architectures of the various HLOIs on the market vary significantly depending on the way in which these common elements are structured.

Figure 6.6 shows one architecture commonly used in computer-based control systems and early distributed systems. In this architecture, there is a single central processing unit (and associated random-access memory) that performs all of the calculations, database management and transfer operations, and CRT-and-keyboard interfacing functions for the entire HLOI system. A separate communications controller interfaces the central processor with the shared communications facility.

There are several advantages to this configuration. First, there is a single database of plant information that is updated from the communication system. As a result, each of the CRTs has access to any of the control loops or data points in the system. This is desirable, since it means that the CRTs are all redundant and can be used to back each other up in case of a failure. Another advantage is that the peripherals can be shared and need not be dedicated to any particular CRT/keyboard combination. This can reduce the number of peripherals required in some situations.

The disadvantages of this configuration are similar to those of all centralized computer systems:

1. It is an "all eggs in one basket" configuration, and so is vulnerable to single-point failures. In some cases, redundant elements can be provided; but this approach can lead to complex peripheral-switching and memory-sharing implementations.
2. Any single-processor, single-memory configuration has limitations on the number of loops and data points it can handle before its throughput or memory capacity runs out. In many operator interface systems of this type, the display response times are long and the size of system that can be handled is severely limited.
3. The centralized architecture is not easily scalable for cost-effectiveness: if it is designed properly to handle large systems, it may be too expensive for small ones.

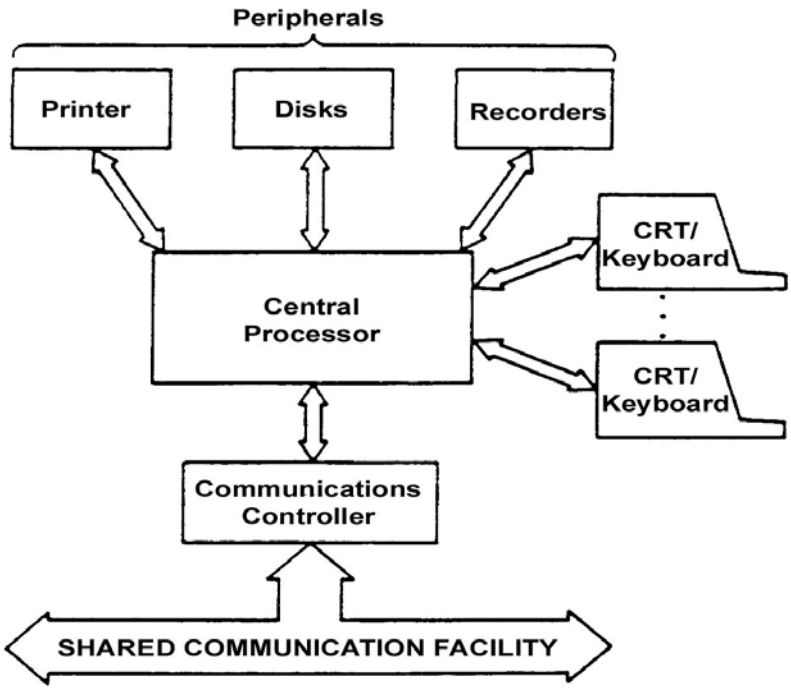


Figure 6.6 Centralized HLOI Configuration

Because of these limitations, most distributed control systems use a decentralized HLOI design. That is, several HLOI units provide the operator interface for the entire system. In this context, an HLOI unit refers to a single element or node that makes use of the shared communication facility. Each unit may include one or more CRTs, keyboards, or peripherals such as printers or disk memories.

When the operator interface is distributed in this manner, the issue arises of how to partition the responsibilities of each unit to cover the entire process. Usually, each unit is designed to be cost-effective when monitoring and controlling a relatively small process (say, 400 control loops and 1,000 data acquisition points). However, this means that several of these units must be used to monitor and control a larger process (say, 2,000 control loops and 5,000 data acquisition points). For example, five units of the capacity indicated (400 loops and 1,000 points) could just cover the 2,000-loop system. In this case, however, if one of the control units failed, the operator interface for one-fifth of the process would be lost. To avoid this situation, the HLOI units usually are configured for significant overlap in the portions of the process each unit covers. Figure 6.7 illustrates a two-to-one overlap configuration, in which

three HLOI units control and monitor a 600-loop process. With this approach, the loss of any single HLOI unit does not affect the capability of the operator interface system to control the process.

Overlap obviously is not an issue if each HLOI is designed to be large enough to accommodate all of the points in an installation (say, 5,000 to 10,000 points). This design approach results in a more expensive version of an HLOI than one designed to handle a smaller number of points. However, in this case each HLOI unit is capable of backing up any other unit in the system.

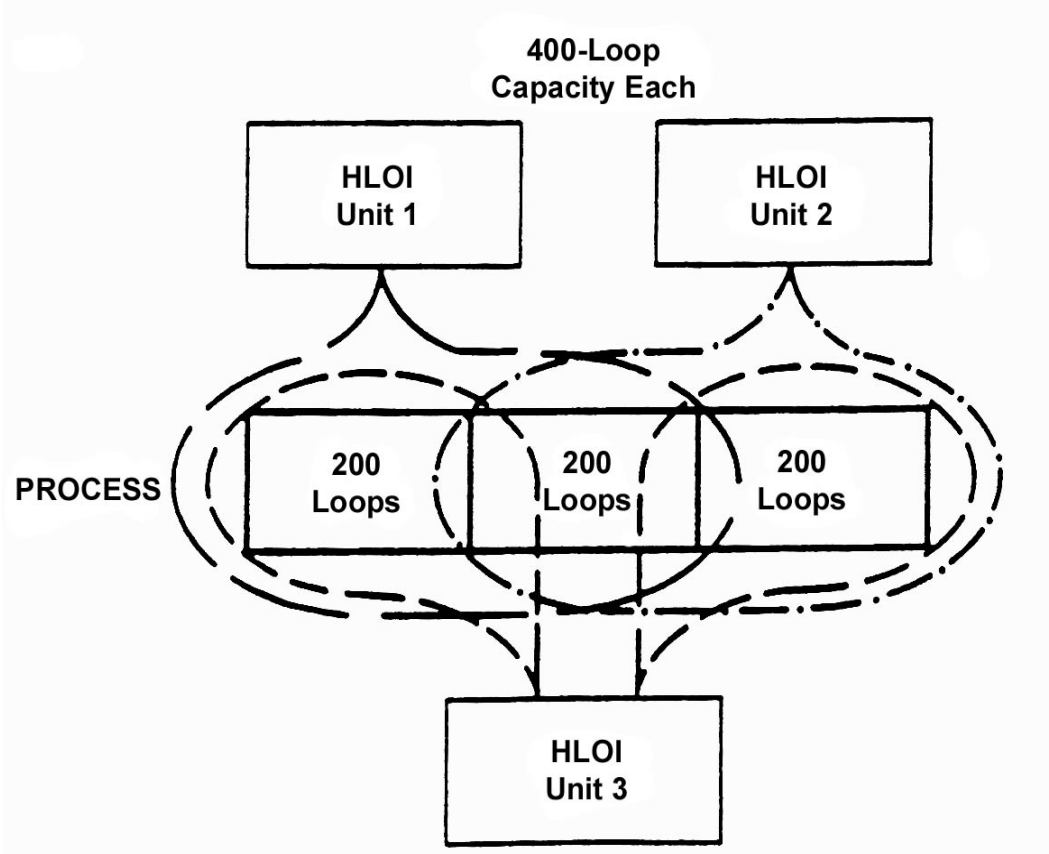


Figure 6.7. Overlap in HLOI Scope of Control

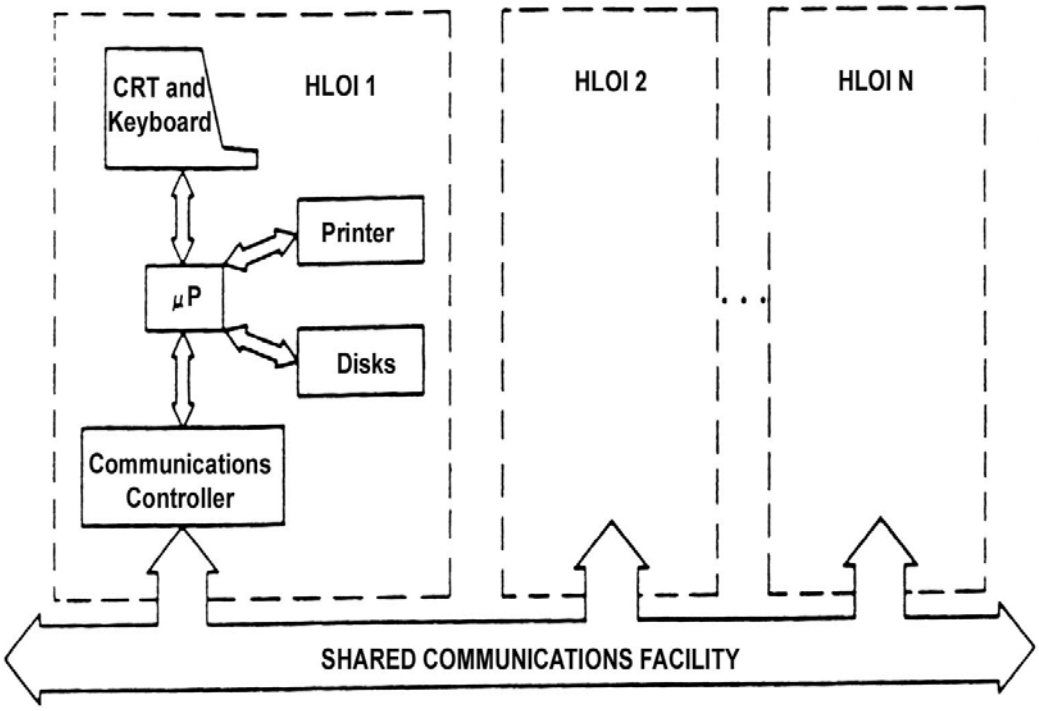


Figure 6.8. Fixed HLOI Configuration

The first versions of distributed HLOI systems introduced to the marketplace had a relatively fixed configuration of elements, such as that shown in Figure 6.8. That is, a single HLOI unit consisted of a communications controller, main processor, CRT and keyboard, and associated mass storage. The only option for the user was whether to include a printer or other hard-copy device. Because of this fixed configuration of elements, the scope of control and data acquisition of the HLOI unit also was fixed.

Later versions of HLOI units have been designed to be modular: the user can buy the base configuration at minimum cost or expand it to handle a larger number of control loops and data points. Figure 6.9 shows one example of a modular HLOI configuration. The base set of hardware in this case is a communications controller, main processor, single CRT and keyboard, and mass storage unit. However,-the system is designed to accommodate optional hardware such as:

1. One or more additional CRTs to allow monitoring of a portion of the process (for example) while the primary CRT is being used for control purposes.
2. Additional keyboards for configuration or backup purposes.
3. Hard-copy devices such as printers or CRT screen copiers.
4. Additional mass storage devices for long-term data storage and retrieval.
5. Interfaces to trend recorders, voice alarm systems, or other external hardware.
6. Interface ports to any special communication systems such as backdoor networks to other HLOI units or diagnostic equipment.
7. Backups to critical HLOI elements such as the main processor, communications controller, or shared memory.

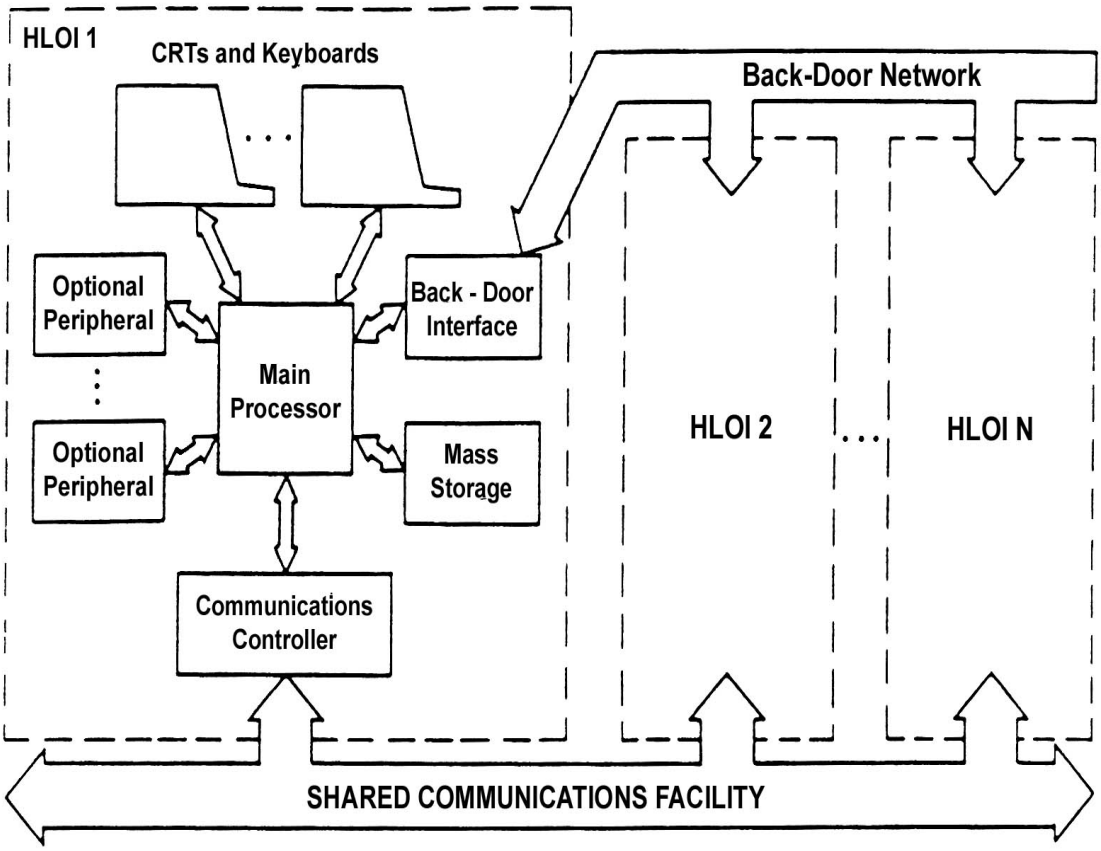


Figure 6.9. Modular HLOI Configuration

The modular approach to HLOI unit design significantly improves configuration flexibility. The user can select the base configuration for small applications and add the optional hardware as the user sees fit. Of course, the performance of the main

processor and other hardware must be adequate to provide the display update and response capability required, even with the maximum size hardware configuration.

Figure 6.10 shows a more detailed block diagram of the modular HLOI configuration. Usually, an internal bus is used to allow communication of information among the modular elements in the HLOI. Note that this configuration uses a direct memory access (DMA) port to allow the communications controller to transfer data directly into the shared memory of the HLOI. The other elements of the HLOI then can obtain access to this information over the internal bus.

Hardware Elements in the Operator Interface

Since the HLOI system is based on digital technology, many of the hardware elements that go into the system are similar to those used in other portions of the distributed control system (e.g., the microprocessors and the memory and communications components). However, the performance requirements of the HLOI place special demands on its elements; also, the on-line human interface functions performed by the HLOI require display and input hardware that is unique to this subsystem. An exhaustive survey of HLOI hardware requirements and alternatives is beyond the scope of this chapter; rather, some of the key factors to consider in selecting or evaluating operator interface hardware are summarized. References 6.21-6.31 provide additional information.

Microprocessor and Memory Components. The high-level operator interface is the most complex subsystem in the distributed control hierarchy. As a result, the microprocessors used in its implementation must be faster and more powerful than microprocessors used elsewhere in the distributed system. The microprocessor hardware selected for use in the HLOI in commercially available systems tends to have the following characteristics:

1. It uses one or more standard 16- or 32-bit microprocessors available from multiple vendors; these are not single-sourced special components.
2. Its processors are members of a family of standard processors and related support chips.
3. It is designed to operate in a multiprocessor configuration, using a standard bus for communication between processors and an efficient real-time operating system as an environment for application software.

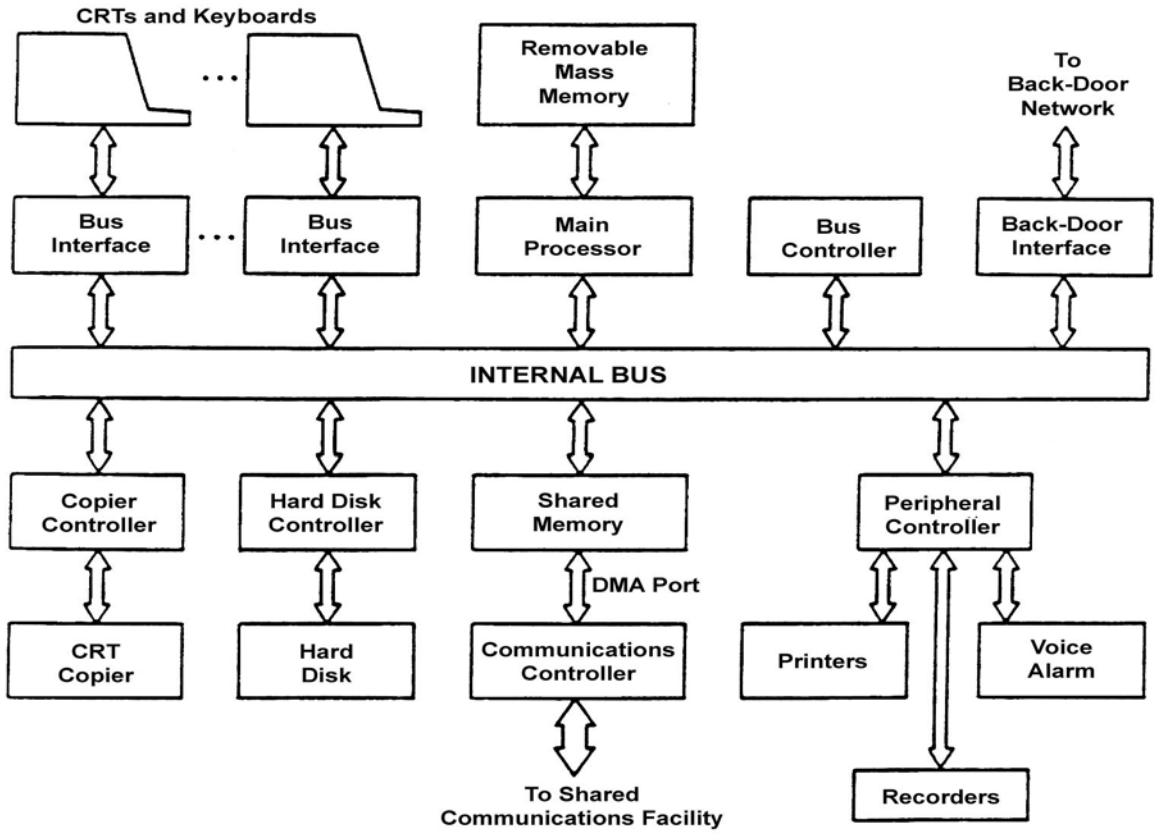


Figure 6.10. Modular HLOI Configureation in Detail

The last characteristic is important, since to accomplish the desired computing and data control functions at the speeds required by the real-time operating environment, most HLOI configurations must use multiple processors.

The HLOI requires the same types of semiconductor memory devices as used in the LCUs and the shared communications facility: RAM for temporary storage of changing data (e.g., current values of process variables); ROM for storage of predetermined, unchanging information (e.g., standard display formats and computational algorithms); and nonvolatile, alterable memory for storage of data that changes only infrequently (e.g., custom graphic display formats). The nonvolatile memory options for the HLOI are the same as those discussed in Chapter 2 for the LCU: battery-backed RAM, electrically erasable programmable read-only memory (EEPROM), and bubble memory. The main differences between the memory requirements for the HLOI and those for the LCU are that the HLOI requires shorter access times and greater amounts of memory to meet its high performance and large data storage requirements.

Operator Input and Output Devices. The primary function of the HLOI subsystem is to allow communications between the operator and the automatic portions of the distributed control system. Therefore, the particular data input and output devices selected are vital to the usability of the system and its acceptance by operating personnel.

A great variety of operator input devices have been considered for use in industrial control systems:

1. ***Keyboard***—there are two kinds of keyboard in use: the conventional electric typewriter type and the flat-panel type;
2. ***Light Pen***—this device allows a person to "draw" electronically on a CRT screen. In industrial control systems, the light pen is more often used by the operator to select among various options displayed on a CRT screen.
3. ***Cursor-Movement Devices***—These allow the user to move a cursor (position indicator) around on a CRT screen. Types of devices used include joy sticks (such as those used in video games), track balls (imbedded in a keyboard), and

the "mouse" (a hand-held device that the operator moves around on a flat surface).

4. ***Touch Screens***—there are CRTs with associated hardware that allows the user to select from menus or to "draw" on the screen by directly pointing with a finger.
5. ***Voice-input Devices***—these devices allow a user to speak directly to the HLOI and be understood. They are most useful for specific commands such as selecting displays or acknowledging alarms.

Many of these devices were originally developed for use in other applications, such as military or aerospace systems or in terminals for computers or computer-aided design systems. In evaluating these devices, the designer of an industrial control system must remember that the needs, background, and training of an operator in a process control environment differ substantially from those of an astronaut, military aviator, or engineer. For example, the use of hand-held devices such as light pens or mice in industrial applications has been criticized on the grounds that operators prefer not to use any devices that require a separate operation—removal from or return to storage—for their use. As a result, the most popular input device technologies in industrial control systems are keyboards and touch screens. The next section will discuss these in more detail.

As with input devices, several types of display and output devices have been considered for use in HLOI systems:

1. ***CRTs***—This still is the dominant technology in the area of operator displays; CRTs come in either color or monochrome versions.
2. ***Flat-panel Displays***—These include gas plasma, vacuum fluorescent, liquid-crystal, and electroluminescent displays; most are monochrome only.
3. ***Voice-output Devices***—Usually these are used to generate alarm messages that the operator will hear under selected plant conditions. The messages can be prerecorded on tape or computer-generated by voice synthesizer.

In industrial distributed control systems, the CRT has been and continues to be the predominant visual display device used in HLOI systems. Some flat-panel display devices have been used in aerospace and military applications (see Reference 6.30); however, in general they have not yet reached the point of development at which their performance-cost ratio seriously threatens that of the CRT. The main features that differentiate the various CRTs on the market include:

1. Size of screen;
2. Number of colors available;
3. Resolution of pixels on the screen;
4. Character-oriented versus bit-mapped displays.

A screen size with a 19-20" diagonal is adequate for most purposes; occasionally special-purpose consoles that include a large amount of panel-board equipment use a 25" screen. Ideally, the design of the HLOI should accommodate a range of CRT sizes, ranging from 19" to 40". Most video display units employ a range of eight to 16 colors. An operator generally finds it difficult to discriminate among a larger number of colors.

The third feature listed above can be thought of as the graininess of the displays generated on a particular CRT screen. Each display is, of course, composed of a large number of individual pixels (dots) that are controlled by a processor in the VDU. The display will be quite grainy if the number of pixels per square inch is small. A larger number of pixels per square inch results in a better-quality display. A medium-quality, 19-inch CRT might have an array of 640 by 512 pixels; higher-quality CRTs might have twice the number of pixels in each direction—say, 1280 by 1024. Of course, the computing requirements (and therefore the cost) on the display processor go up significantly as the density of pixels increases.

The fourth feature refers to the method by which the display processor controls the pixels to form images on the CRT screen. In bit-mapped designs, the on/off status and the color of each pixel are controlled by the processor on an individual pixel basis. In character-oriented designs, the pixels are grouped into rectangular clusters (usually eight high by six wide) called characters. Each character depicts a letter, a number, or a special graphical symbol. In this approach, the on/off status and the color of the pixels in each type of character are controlled by the processor on a character-by-character basis instead of on an individual pixel basis. Figure 6.11 illustrates these two approaches to display generation. The upper portion of the figure shows typical characters that are defined and used in combination to create a complete display. The lower portion of the figure shows a segment of a display created using the bit-mapped approach.

It is clear from the figure that the two approaches will produce similar quality results if only alphabetic or numeric information has to be displayed. However, there can be a significant difference in appearance when generating displays such as trend graphs or piping and instrumentation drawings. The character-oriented approach requires that individual characters be linked together to form the graphic display. If the types of characters available match those needed to construct the desired display, the result will be acceptable; otherwise, the display will look ragged. On the other hand, the increased flexibility and generality of the bit-mapped approach allows a smooth picture to be drawn in almost all cases.

The advantage of the character-oriented approach is that it simplifies the hardware and software requirements on the display processor, resulting in a relatively low-cost product. The bit-mapped approach is relatively expensive since it requires significantly more computing power. As a result, HLOI systems have more frequently used the character-oriented technique. This situation is changing rapidly, however; with the emergence of special graphical processing chips as auxiliaries to the main display processors, the cost of computing power continues to go down.

Each user or designer must select the level of performance that is appropriate to the application, keeping in mind the ever-present trade-off with regard to cost of the HLOI.

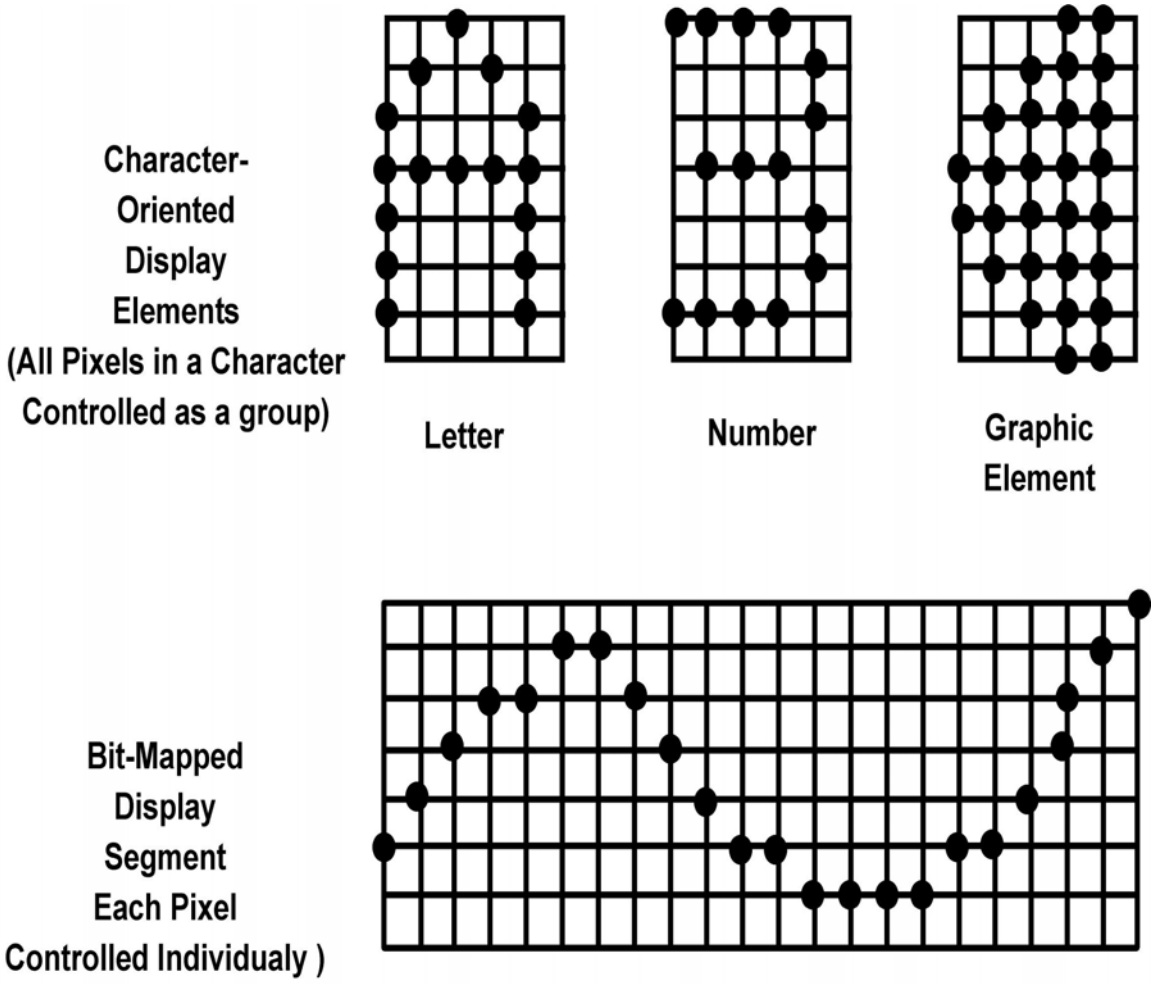


Figure 6.11. Two Types of Display Construcation

Peripherals. In addition to the processing, memory, input, and display devices required to perform the basic operator interface functions, the HLOI configuration must include the following additional peripheral devices to implement the full range of functions:

I. ***Fixed Disk Drives*** — This type of mass memory uses non-removable memory media to store large amounts of information that must be accessed rapidly

(such as standard and graphical display formats) or to provide a location for temporary storage of historical data. One example of such a memory device is the Winchester disk, a magnetic memory disk with a capacity in the 5-100 Mbyte range. Another is the read/write optical disk, which is used for even higher density storage (500-1,000 Mbytes).

2. ***Removable mass Memory***—This type of mass memory, typified by the floppy disk, uses removable memory disks for storage of information that is to be downloaded to other elements in the distributed system (such as control configurations, tuning parameters, and custom programs). It also can be used for long-term storage of small quantities of historical data; magnetic tape can be used for large quantities. Read-only optical disks, such as those in commercial video applications, are useful for storage of large amounts of unchanging information, such as text used in operator guides.

3. ***Printers and Plotters***—At least one printer is required to implement the logging and alarm recording functions. The printer can also provide a hard copy of the CRT displays; or a separate printer can be dedicated to that function. A black-and-white dot matrix printer is the most prevalent type used, since it can implement either function. Pen plotters, ink-jet printers, or thermal transfer printers can provide full-color hard copies of CRT displays. (See Reference 6.27 for a survey of these devices.) Since these devices tend to be slow, it is important that the HLOI be designed so that the keyboard and CRT are active and available to the operator while the printing or plotting process is going on.

Modular Packaging Approach. A few suppliers of distributed control systems provide their HLOI units in a split configuration, in which a table-mounted package houses the CRT and keyboard and a separate cabinet is used to mount the driving electronics and peripherals. However, most vendors use a modular packaging approach, which provides the user with maximum configuration flexibility without requiring the use of any separate furniture. Figure 6.12 shows an example of such a family of module that can be used to form an integrated HLOI unit. In this family, the base module houses a CRT, a keyboard, driving electronics, and mass memory peripherals. Together they form a stand-alone, single-CRT HLOI unit. The following additional modules can be added to expand the capabilities of this base unit:

1. ***CRT Module***—For additional display:
2. ***Alarm Panel Module***—For dedicated alarm indicators in addition to those provided on the CRT;
3. ***Panelboard Instrumentation Module***—To provide a space for mounting trend recorders, indicators, manual backup stations, telephones, and other auxiliary equipment.
4. ***Work Space Module***—To provide a place for operator documentation and large peripherals such as printers or plotters.

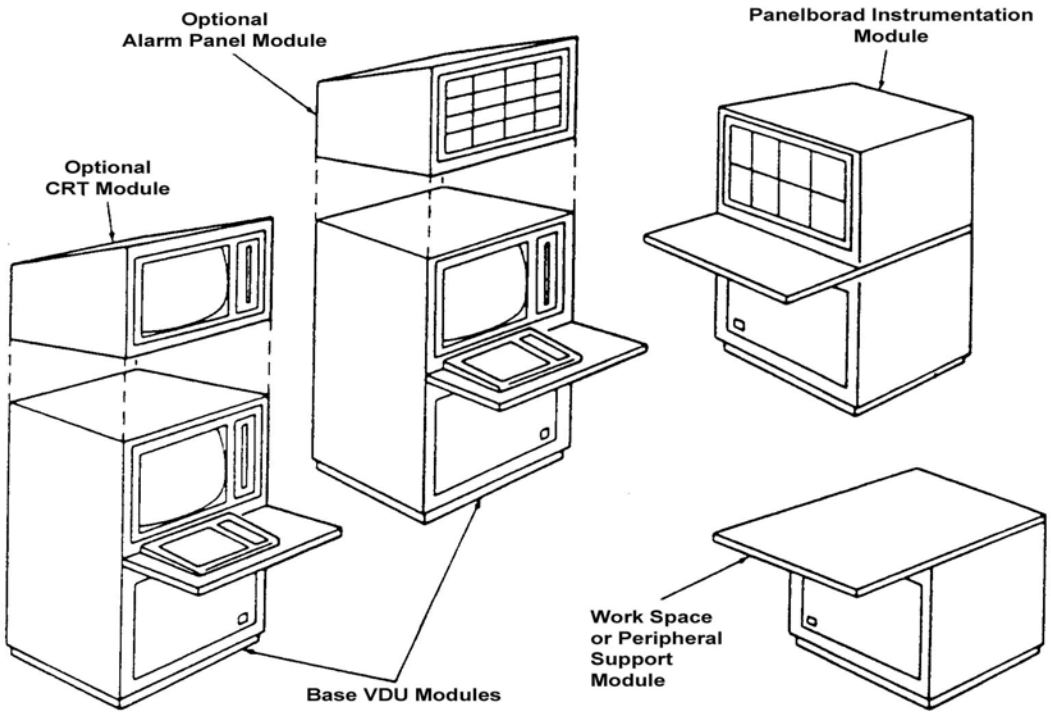


Figure 6.12. Family of HLOI Modules

Many versions of this modular approach to packaging have been brought to the marketplace, and no two are exactly alike. However, the well-designed versions have a number of characteristics in common:

I. ***Good Anthropometric Design*** — Attention is paid to the height, positioning, and orientation of the CRT and keyboard arrangement so that it is suitable for long-term use by the full range of operational personnel and in the operating positions expected (i.e., seated, standing, or both). Ideally, the design allows for adjustment of the arrangement to suit an individual's preferences (as is done in the case of a tilt steering wheel in an automobile, for example).

2. ***Elimination of Glare--*** — The console is designed in such a way that lighted control room objects reflecting in the CRT do not interfere with the operator's ability to view the displays. This usually is accomplished through a combination of approaches: by integrating the design of the ambient lighting in the control room with the design of the console; by adjusting the orientation

and location of the display screen in the console; and by using glare-reducing screens to cover the CRT face.

3. ***Easy accessibility of Peripherals***—If the operator must use floppy disks, for example, the disk drives are made easily accessible to the operator while in a seated position. Printers and copiers are designed for ease of use and maintenance (including routine operations such as paper replacement).

4. ***Simple Interconnection of Modules***—The various modules in the HLOI system are interconnected with wiring to carry both data signals and electrical power. An effective design requires no special engineering to accomplish this function and uses standard cables and connectors.

Operator Displays

The panelboard in a conventional control room uses many square feet of dedicated instruments to provide the operator with the information and mechanisms needed to control the plant. In theory, the operator has simultaneous access to all of these instruments at one time, since they all are physically located in the same room. In practice, of course, the operator must move about the room to be able to see the indicators and manipulate the various stations needed to control the plant. The video display unit in an HLOI system, in contrast, provides a "window" to the process that allows the operator to see only a relatively small amount of information at any one time on one or more CRT displays. The operator is able to monitor and control the whole process only by calling up a number of these displays, which usually are arranged in a fixed logical structure or hierarchy. If this display structure and the associated display access mechanisms are designed properly, the HLOI will provide the operator with much faster access to the needed information than is possible by moving around a panelboard (see Reference 6.8). If they are designed poorly, however, the operator will view the HLOI as an impediment to access of this information and a step backwards from the "good old days" of panelboard

operation. This section describes some of the elements that go into a well-designed structure of CRT displays in an HLOI system. The next section will discuss display access mechanisms in the context of operator input hardware.

Typical Display Hierarchy. The flexibility of CRT display technology makes it possible to conceive a great number of different display structures that would be appropriate for industrial control systems. Since the introduction of distributed control systems in the mid-1970s, however, a de facto standard display hierarchy has evolved over the years through the pioneering efforts of Dallimonti (6.2) and others after him (see References 6.4-6.8). A typical version of this hierarchy, illustrated in Figure 6.13, is composed of displays at four levels:

1. ***Plant level***—Displays at this level provide information concerning the entire plant, which (if large enough) can be broken up into several areas of interest.
2. ***Area level***—Displays at this level provide information concerning a portion of the plant equipment that is related in some way. e.g., a train of separation processes in a refinery or a boiler-turbine-generator set in a power plant.
3. ***Group level***—Displays at this level deal with the control loops and data points relating to a single process unit within a plant area, such as a distillation column or a cooling tower.
4. ***Loop level***—Displays at this level deal with individual control loops, control sequences, and data points.

With some variations, the VDUs that most distributed control system vendors offer follow this general structure. (Some vendors provide an option that allows the user to define a customized display hierarchy and the allowed movements within the hierarchy.) There are several types of CRT displays that generally are associated with each level, and these will be described in the following paragraphs.

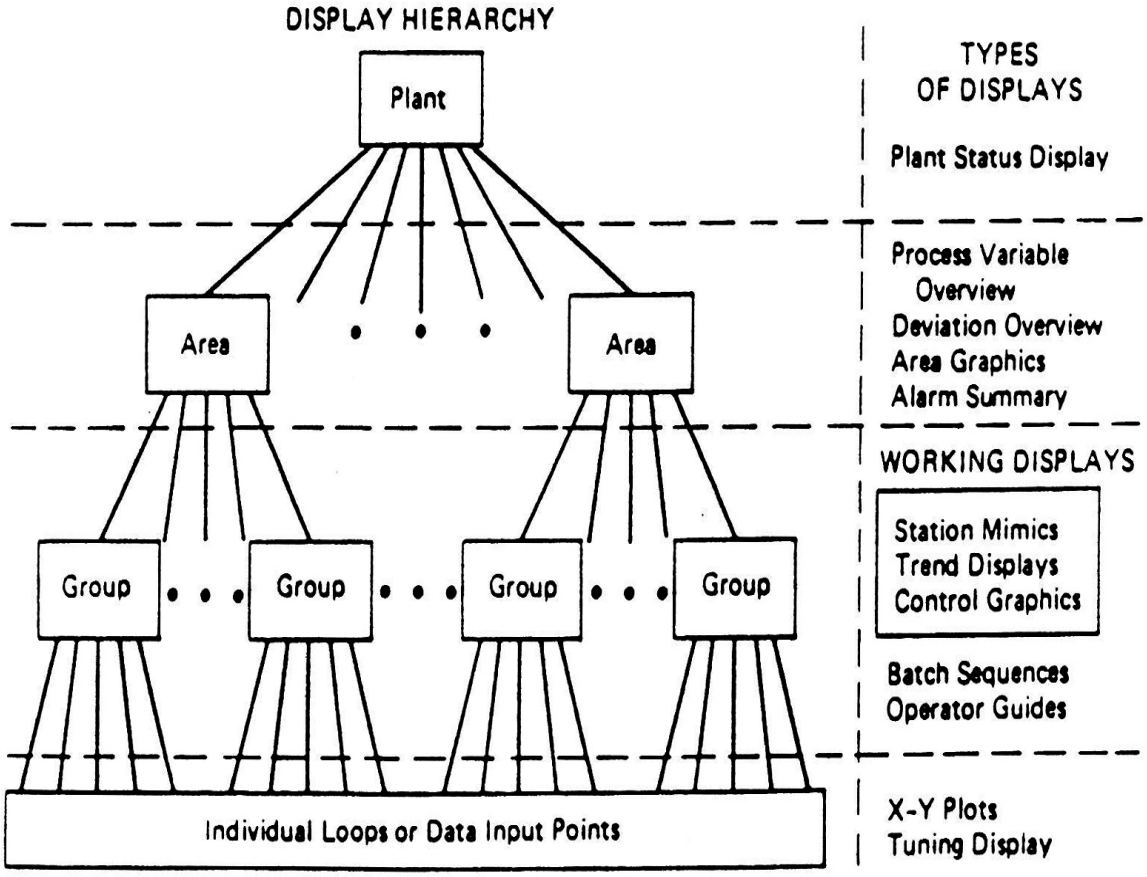


Figure 6.13. Typical Display Hierarchy

This general display structure is attractive from several points of view. First, it covers the full range of detail of information that is likely to be of interest to the operator, from overall plant conditions to the status of each loop in the plant. Also, it allows for the grouping of available information in a way that matches the structure of the process itself. Finally, it provides a mechanism that allows the operator to form a mental model of the relationships between the various pieces of information in the plant. This is similar to the mental model of a panelboard that develops in the mind of an operator after gaining experience with its layout. After a period of weeks or months, the operator no longer has to refer to the labels on the panelboard to find a particular instrument, but moves to it instinctively. Similarly, a meaningful display structure such as the one shown in Figure 6.13 allows the operator to learn to move from one display to the next in a smooth and efficient manner.

Plant-Level Displays. Typically, at the top level of this structure is a single type of plant status display (perhaps consisting of several pages), as Figure 6.14 illustrates. This display summarizes the key information needed to provide the operator with the "big picture" of current plant conditions. This example shows the overall production level at which the plant is operating compared to full capacity. It also indicates how well the plant is running (e.g., by plotting efficiency of energy usage). In addition, some of the key problem areas (e.g., equipment outages or resource shortages) are displayed. A summary of the names of the various areas in the plant serves as a main menu (index) to the next level of displays. At the top of the plant-status display is a status line of information provided in all operating displays. This line shows the current day of the week, the date, and the time of day for display labeling purposes. In addition, it provides a summary of process alarms and equipment diagnostic alarms by listing the numbers of the plant areas in which outstanding alarms exist. (The subject of alarming is discussed further in Section 6.4.5.)

Area-Level Displays. After obtaining a summary of the plant status from the top level of the display hierarchy, the operator can move down to the next level to look at the situation in a selected plant area. This can be done by means of several types of displays; Figure 6.15 shows a composite of four of these types. The top line of the display is the system date and status line described previously. The upper left quadrant illustrates an area display type known as deviation overview, which displays in bar graph form the deviation of key process variables from their corresponding set points. The deviations are usually normalized to reflect a percentage of total span, and are clustered into a number of groups within each area. If the absolute value of deviation exceeds a predetermined level (e.g.. 5 percent of span), the process variable enters a deviation-alarm status condition and the bar graph for that variable changes color. This approach to overview display derives from the green-band concept in conventional analog instrumentation, in which the

manual-auto stations for continuous control loops are arranged side by side in a row on the panelboard. For each loop, if the process variable is within a small percentage of the set point, the analog pointer for that variable remains hidden behind a green band on the station face. The operator then can determine which loops are upset by simply scanning the row of stations and seeing which pointers are outside the green band. The deviation overview display provides the operator with the same information in a CRT display format.

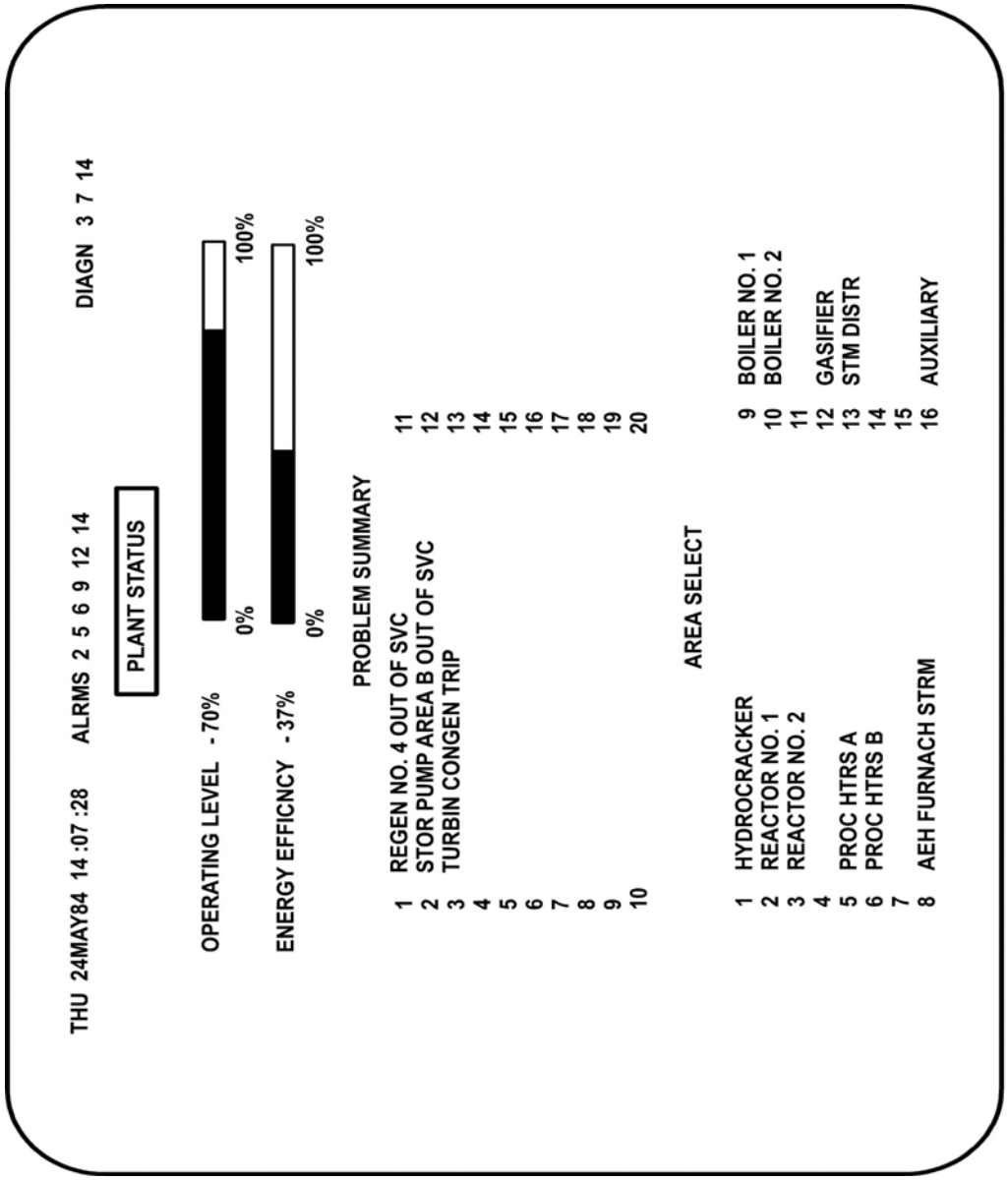


Figure 6.14. Example of a Plant status Display

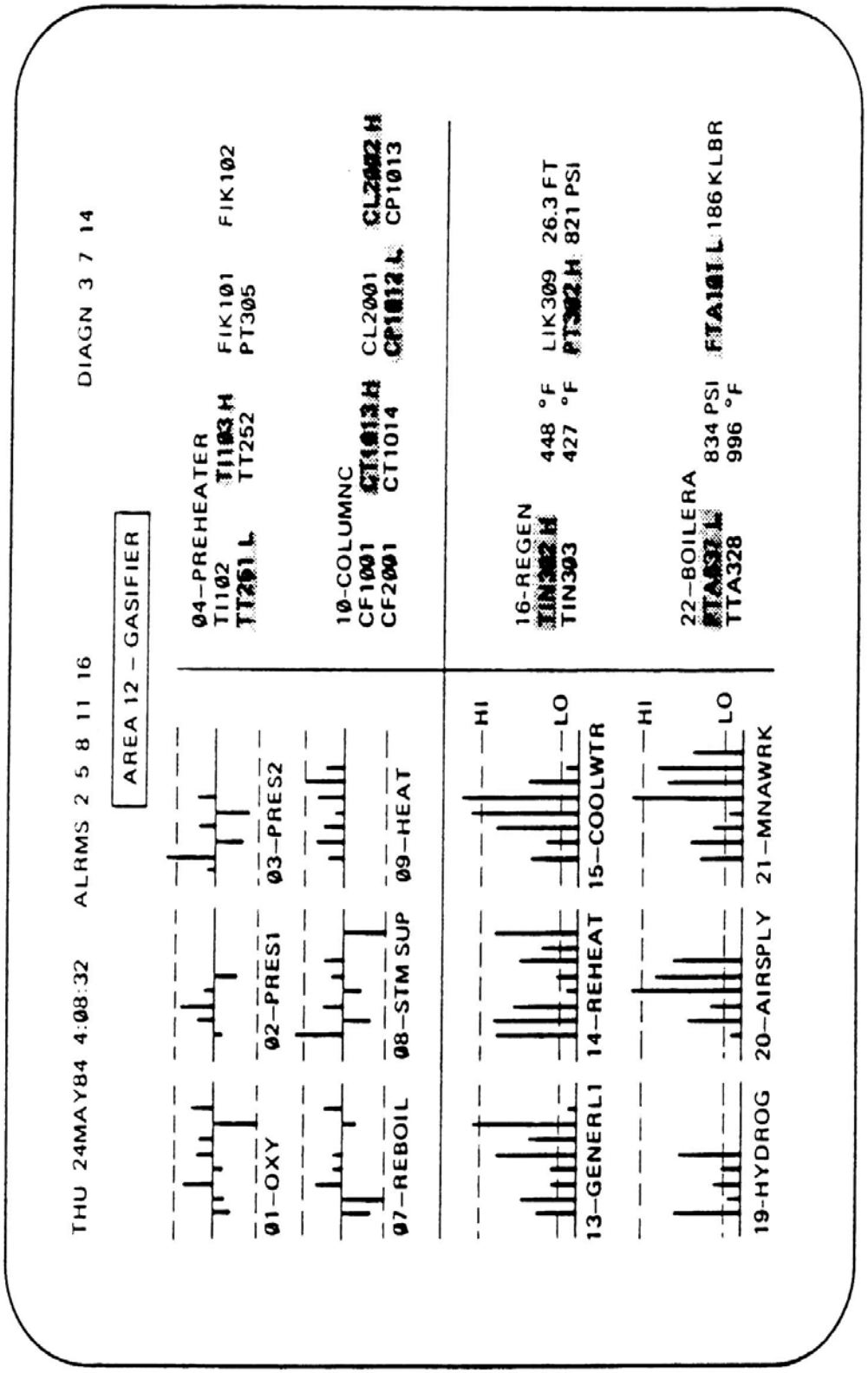


Figure 6.15. Examples of Area Overview Display

The lower left quadrant of Figure 6.15 shows a variation of this approach in which a bar graph indicates the absolute value of the process variable instead of its deviation from set point. Some versions of this display also show the set point and the high and low alarm limits on the process variable. When one of these limits is exceeded, the bar graph changes color as in the case of the deviation display. This method is more general than the previous one in that it accommodates the alarming of process variables that are not used in control loops as well as those that are.

The two display types just described essentially mimic the analog portions of a conventional panelboard. The upper right-hand quadrant of Figure 6.15 shows another approach to the area overview display. Here the tag numbers of the various loops and process variables are arranged in clusters by group. If the point associated with a particular tag is not in alarm, its tag number is displayed in a low-key color. If it does go into alarm, it changes color and starts flashing to get the attention of the operator. Underlining also can be used under the tag number, so that a colorblind operator still can see the alarm state clearly. This format of an overview display is similar to that of an alarm annunciator panel in a conventional panelboard.

The lower right-hand quadrant shows a variation of this display. In addition to the tag number itself, the current value of the process variable is displayed in engineering units to the right of the tag. This provides the operator with information on the values of the key variables in a group in addition to their alarm status.
In some implementations of area overview displays, several of these approaches may be intermixed in a single display. Also, two other types of area-level display often are provided:

1. ***Area Graphics Display***—This display is similar to a piping-and-instrumentation diagram (P&ID) or mimic panel used on conventional paneiboards to illustrate the process equipment and its associated instrumentation. It usually is designed to provide the same type of information

that other area displays give: alarm status and perhaps current values of key process variables. The capabilities and use of graphics displays are discussed in more detail later in this section.

2. ***Alarm Summary Display***—This display is simply a listing of the most current alarms that are still outstanding in the area. Its format is similar to that of an alarm log produced by a computer, and would include the following information on the points in alarm: tag number and description of point in alarm, time of alarm, type of alarm (e.g.. deviation, high, and low), current value of point, and current alarm status (e.g., in alarm or not in alarm, returned to normal, acknowledged or not acknowledged).

Group-Level Displays. The displays at the plant and area levels of the hierarchy in Figure 6.13 are designed to provide the operator with information on the alarm and operational status of the key process variables in the plant. To perform control operations, however, it is necessary to use the displays at the next lower level in the hierarchy—the group level. As in the case of the higher-level displays, many of the display formats at the group level are patterned after the layout of panelboard instrumentation designed to accomplish similar functions. Figure 6.16 shows one example of a typical group display. Mimics of manual and automatic stations for continuous control loops occupy the upper left-hand corner of the display. These mimics include all of the elements contained in a similar panelboard station: bar graphs showing values of set point, control output, and process variables; manual, automatic, and cascade mode indicators; high and low alarm levels; and other information as needed for the type of station implemented. (The next section will discuss the mechanisms that allow the operator to interact with these stations.) The upper right section of Figure 6.16 shows indicator stations that let the operator view (but not control) selected process variables. Also in this section is a logic station that allows the operator to perform logic operations such as opening and closing valves, as well as starting or stopping sequential control sequences (e.g., for batch

processes). The bottom half of the display is devoted to plotting the trends of one or more process variables as a function of time, mimicking the operation of a trend recorder on a panelboard. In some operator interface systems, each screen "page" of a group display can use only one type of station or trend recorder: others provide much more flexibility by allowing the user to mix and match the types on each display.

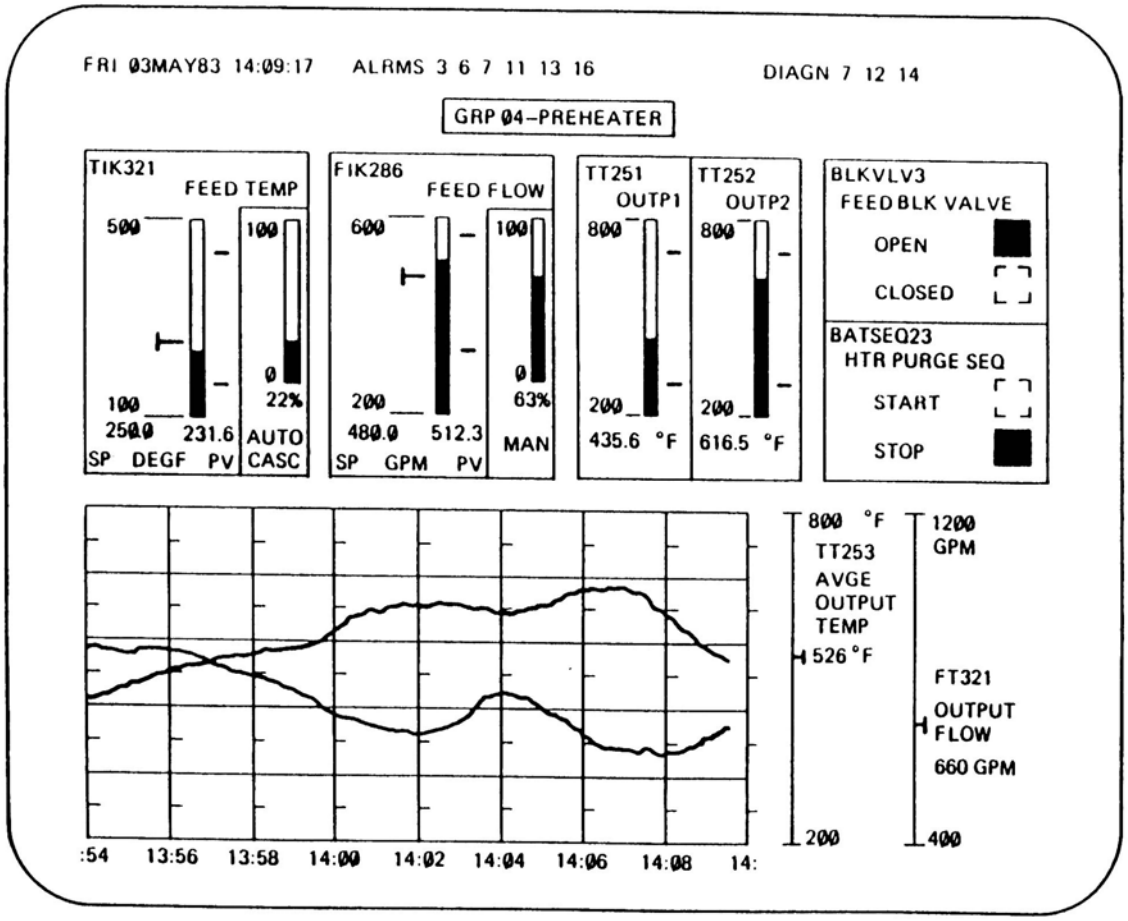


Figure 6.16. Example of a Group Display

The type of group display shown in Figure 6.I6 can be viewed as the equivalent of a section of panelboard in a conventional type of operator interface. Switching from one group display to another is the equivalent of having the operator move around a panelboard to accomplish the monitoring and control functions. The CRT-based "panelboard" offers the user some significant benefits over the conventional

panelboard, however. First, stations and recorders can be added to or removed from the CRT "panelboard" by reconfiguring displays rather than cutting or patching real holes in a panel and procuring additional instrumentation hardware. This provides a significant flexibility advantage during initial plant startup, at which time the user often discovers that additional stations or recorders would be very helpful in plant operations. Another benefit is that one can duplicate stations and recorders in several displays without any additional hardware. This duplication capability can be a significant aid to improving plant operations—one whose cost could not be justified in a conventional panelboard.

Of course, the capabilities of a VDU permit the configuration of operator displays that go well beyond simple replacement of panelboard functions. One example of this is the graphic display for a piping-and-instrumentation diagram (P&ID). shown in Figure 6.17. This differs from the area level P&ID display described previously in two respects:

1. The scope of process covered is smaller—a group rather than an entire area.
2. Control capability is included in addition to the monitoring capability provided in the area P&ID.

The controller station shown on the right side of the display allows the operator to perform control functions. The operator is able to select one of the control loops shown on the graphic through one of several possible mechanisms described in the next section: the controller station then becomes active for that loop and can be manipulated from the operator's console. Similarly, a logic station can be used to perform a sequence control or batch control operation using the graphic display.

It should be noted that in some systems, selected controller stations or logic stations on a particular console can be designated as monitor only: the operator cannot perform any control actions but can only monitor the station variables on the

display. This capability is useful in ensuring that operator control actions are coordinated when the consoles are physically distributed in several locations in the plant. Two other types of displays also have proved useful at the group level:

1. ***Batch Control Displays***—These are menu-oriented displays that allow an operator to observe the progression of a batch recipe (such as that shown in Figure 3.7 in Chapter 3) and interact with the sequence: stan it. stop it, provide permissive to allow it to continue, and so forth. This class of displays also allows the operator to diagnose problems in executing a sequence, such as identifying the part of the process that is preventing the sequence from continuing.

2. ***Operator Guides***—These are advisory displays that provide the following kinds of information to the operator: problems diagnosed by the automatic system, suggested alternative courses of action in an emergency, or step-by-step startup and shutdown procedures for a piece of plant equipment. These displays may combine alphanumeric and graphic information. One can think of them as CRT-based substitutes for a set of plant operating manuals. They differ from manuals in that they can take current plant conditions into account as well as simply provide standard operating procedures to the operator.

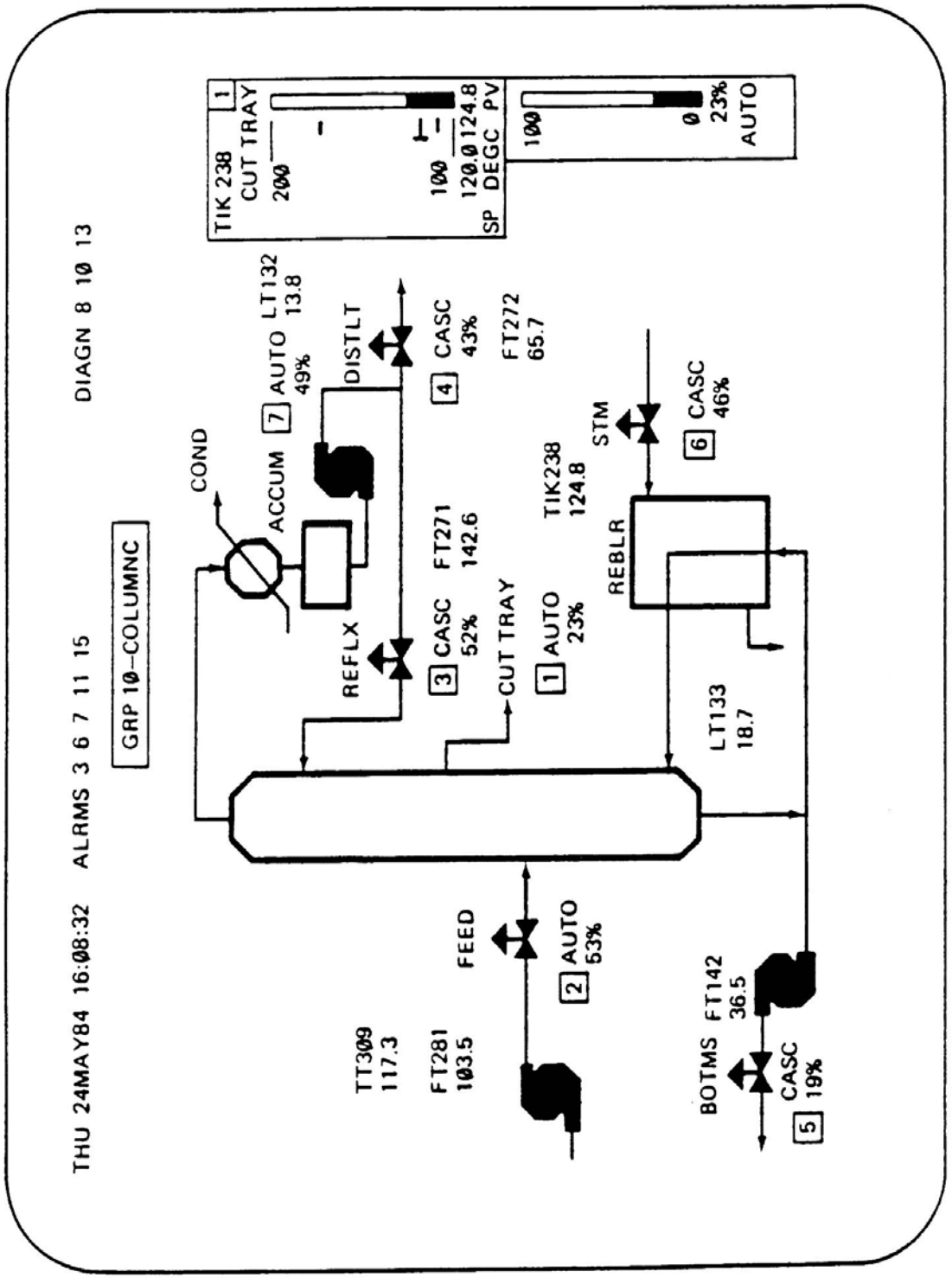


Figure 6.17. Example of a Control Graphics Display

Loop-Level Displays. The displays at the group level are the operator's primary working displays. The operator uses a few types of displays dealing with single loops or data points for control and analysis purposes. Figure 6.18 shows one example, the X-Y operating display. Here one process variable is plotted as a function of another to show the current operating point of this pair of variables. The operator then can compare this operating point against an alarm limit curve or an operating limit curve. In the example shown in the figure, the combination of temperature and pressure for a particular portion of the process may be critical to safety. Therefore, this pair of variables is made available to the operator in the X-Y format shown. If this pair can be controlled directly, manual/automatic stations also can be included in the display for direct operator manipulation. This approach to control and display is not possible using standard panelboard instrumentation. The CRT format makes it feasible and cost-effective.

Figure 6.19 shows an example of a tuning display, another single-loop display that is of use to both operating personnel and instrumentation engineers. This display includes several elements that make the tuning function possible:

1. A "fast" trend-plotting capability;
2. A manual/automatic station to allow the operator to control the loop;
3. A list of the tuning parameters (e.g., proportional band, reset rate, and derivative rate) associated with the loop.

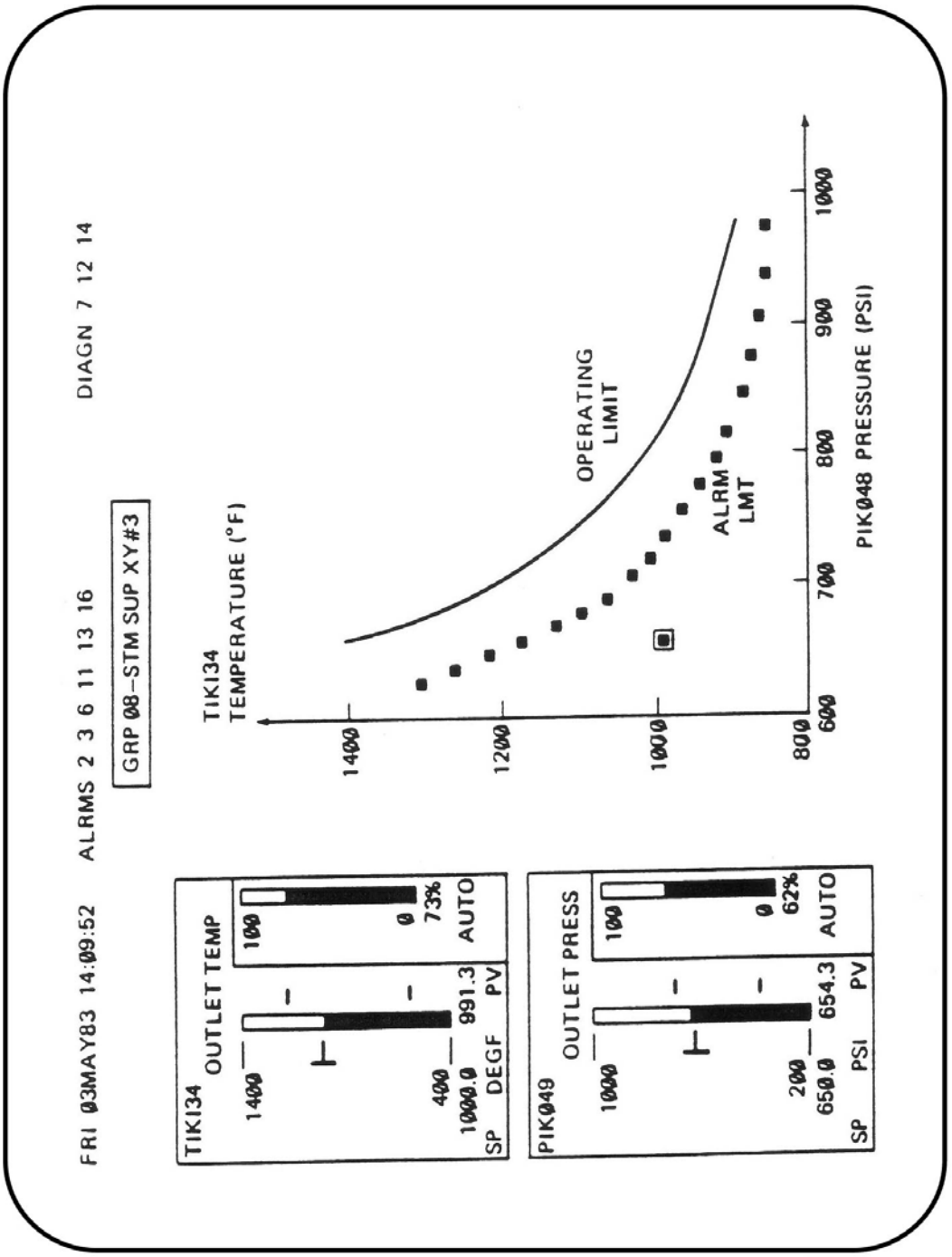


Figure 6.18. Example of an X–Y Operating Display.

The trend graph is used to plot set-point changes (in automatic mode), manual control output changes (in manual mode), and the resulting responses of the process variable being controlled. Based on these responses, the operator or instrumentation engineer can make on-line adjustments to the tuning parameters to improve the performance of the control loop. This example of integrating control, trending, and

tuning functions is one illustration of the ability of the CRT-based operator interface to provide the operator with a very usable and convenient tool for plant operation.

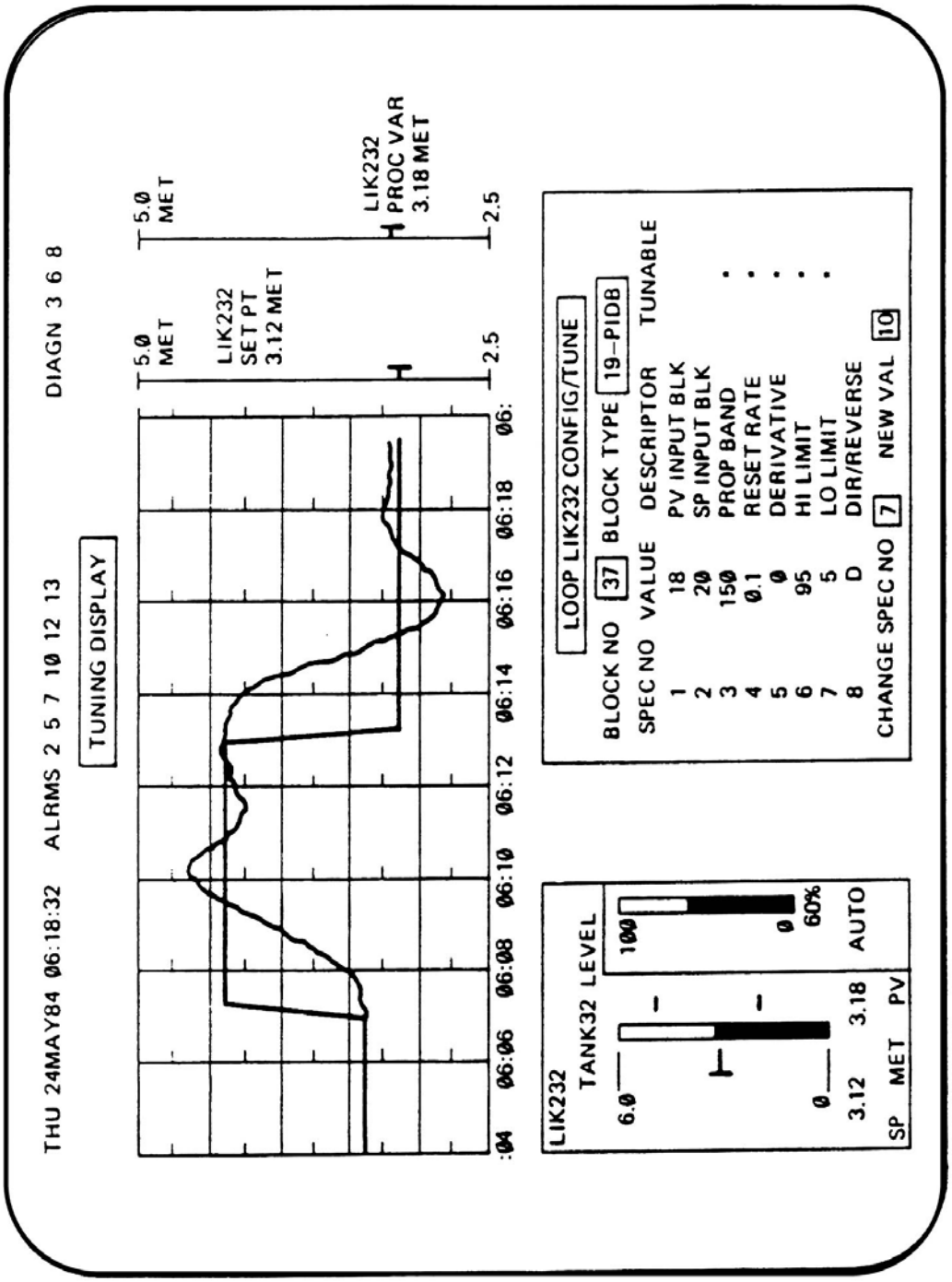


Figure 6.19. Example of a Tuning Display.

Graphics Displays. The concept of user-generated custom graphics displays was introduced above in the context of their application to P&IDs. Operators have accepted the graphics P&ID display both for monitoring the process at the area level of display and for controlling the process at the group level. This type of display

helps an operator (especially an inexperienced one) maintain an accurate mental image of the effect that any control actions will have on the process, thereby minimizing errors that could have an adverse effect on its operation. This is especially important given the typical situation in a process plant with regard to operating personnel: high turnover, minimal opportunity for training, frequent reassignments, and steady increase in workloads. It is difficult for an operator to maintain a good mental model of the process under these conditions; as a result, graphic aids to visualization of the process are very helpful, both at the area and the group level of displays. Process mimic diagrams have been used in the layout of panelboard instrumentation in the past; however, they have proved to be too expensive, space-consuming, and difficult to update as the process changes. The graphics approach to generating control-oriented P&IDs has been very effective in overcoming these difficulties while retaining the benefits of the mimic concept.

The graphics display capabilities provided in most HLOI systems can be used to generate a variety of displays other than P&IDs. The types of displays generated are limited only by the imagination of the user. Typical graphics display features include:

1. ***Static Fields***—These provide a background for the dynamic portion of the display. They include labels, symbols, and other elements that do not change.
2. ***Data Fields (in Engineering Units)***—These display process information that is updated automatically.
3. ***Dynamic Display Elements***—These change size, color, or shape as a function of changing process conditions (e.g., line drawings, process equipment symbols, bar graphs, or pie charts). They can include both user-defined elements and other elements (e.g.. station faces, trends, or process equipment symbols) that are provided as a part of the standard display product.

4. ***Ability to build a graphics display that is several times larger than a single CRT screen*—**The operator uses a mechanism to pan across this display or "zoom" in on portions of interest.

References 6.15-6.20 provide additional suggestions regarding the proper design of graphics displays.

Design Considerations for Displays. Whether evaluating standard display formats or configuring custom graphics displays, the user of an HLOI system needs to be aware of the qualities that differentiate good from bad displays. Of course, this evaluation can be very subjective, since there are no hard and fast rules in this new area of human interface design. However, as References 6.32-6.45 describe in detail, guidelines to keep in mind during such an evaluation include the following:

1. Displays should not be cluttered, but kept as simple as possible. Often, displays are designed to cram as much information as possible on one screen; this is counterproductive if it confuses rather than helps the operator. Some systems solve this problem by allowing the operator to select between a simple version and a detailed version of the same display. The simple version is used for most operations: the operator presses a detail key to get more information when needed.
2. Displays should not be overly "flashy" or have light-colored backgrounds. Such displays may look impressive at demonstrations and trade hows but can be very tiring and annoying to an operator who must ait and look at them all day while trying to run a plant.
3. As described previously, the top line or two of each display should contain common information of interest to the operator, such as the date and time of day as well as an overview of the alarm status situation. The bottom line or two of each display should be reserved for communications (e.g., prompts or error messages) between the HLOI system and the operator.

4. Color should be used in a consistent way throughout all displays to minimize operator confusion; for example, certain colors should be reserved for the static portions of the display, dynamic fields in the display, or alarming information. If at all possible, the color conventions of the industry in which the system is being applied should be followed. The user should be able to select or change colors in both standard and custom displays to meet the needs of the application.
5. Color should not be the sole means for communicating with the operator in the case of critical functions such as alarming. The incidence of color-btindedness among the operator population is too great to permit this approach. Instead, other mechanisms such as blinking or underlining should supplement color to ensure that proper communications takes place.

As in the case of other guidelines in the human factors area, these common-sense rules appear to be obvious but often are neglected in actual practice.

Design Considerations for Operator Input

The CRT in a high-level operator interface unit is the primary way the automatic control system transmits information to the operator. The HLOl also must provide a way for the operator to input the following types of information into the automatic system:

1. ***Display-select Commands***—To allow the operator to move about the display hierarchy described in the previous section;
2. ***Cursor-movement Commands***—To allow the operator to move a cursor (position indicator) from place to place on any one display;
3. ***Control-input Commands***—To allow the operator to interact with the station mimics and other control-oriented displays in the hierarchy;
4. ***Data Inputs***—To allow the operator to enter numeric information (e.g., set points and measurement values obtained manually) into the automatic control

system.

As mentioned in Section 6.4.2, there are a variety of hardware devices that can implement this input function. The one device that is common to all commercially available HLOI systems is one version or another of an operator's keyboard. In more recent years, the CRT touch screen has developed into a cost-effective and reliable device for operator use. Devices such as light pens and "mice" have been found to be less suitable for two reasons: (1) they are more prone to failure in an industrial environment, and (2) they must be stored and retrieved from some location within the console.

Figure 6.20 illustrates the use of a touch screen as an operator input device. (See References 6.25 and 6.28 for more information on touch screens.) A fine grid of sensing areas on the CRT screen allows the HLOI to sense the touch of the operator's finger on any portion of the screen. The grid of sensing areas usually is implemented using one of two approaches: (1) an array of infrared emitters and receivers located on the periphery of the CRT tube, or (2) a pattern of transparent touch wires, thin-film conductors, or capacitance-sensing panels overlaid on the screen itself. The HLOI relates the screen location of the operator's touch to the corresponding segment of the display on the screen. In this example, one can use the touch screen both to select and to manipulate a control loop. To do this, the operator need only touch the loop-select segment of the display ("LOOP SEL" on right-hand side) then touch the desired control loop on the P&ID graphic. At this point the control station on the right side of the display becomes active, and the operator can change modes, raise and lower set points, and perform other functions by touching the relevant portion of the station mimic display. This is done in the same way as in operating a physical station on a panelboard.

There are two design considerations to keep in mind when evaluating or designing the layout of a touch screen and the corresponding CRT displays for use as an operator input device:

1. The spaces between the keys or control segments on the CRT must be large enough to meet the needs of a "gloved hand operator" described by Herb (6.12).
2. Audible feedback (such as a beep or tone) must be provided to confirm that the operator has depressed a control segment on the CRT. This feature is essential to minimizing operator errors.

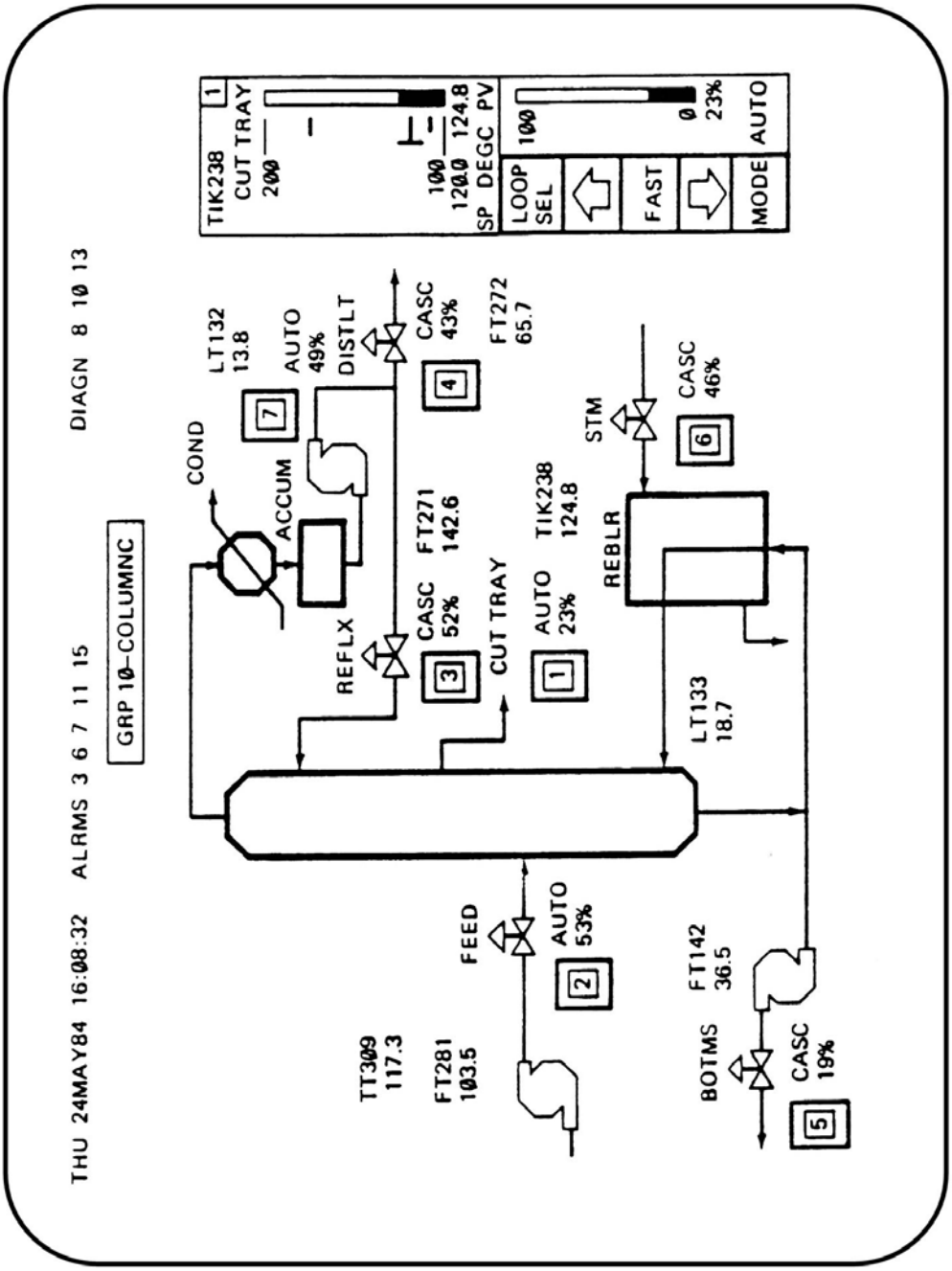


Figure 6.20. Example of Touch Screen Operation.

While useful, the touch screen is not adequate by itself to provide the operator with the full range of input functions needed in an HLOI. It is most convenient for implementing cursor-movement and control-input functions. A keyboard usually is necessary for display-select commands and data inputs. Two versions of keyboard hardware are in common use: (1) the conventional push-button type of keyboard found in electric typewriters; or (2) the flat-panel type using membrane switches or similar hardware to implement the push buttons under a flat layer of Mylar. Generally, the flat panel type is preferred for use in industrial control systems because of its ruggedness. The layer of mylar protects the key contacts from a contaminated atmosphere and eliminates the infamous "coffee-spill problem." Because of the limited key action that the flat-panel type allows, it is not suited to touch typing or other fast operations. However, since most operator actions do not require extremely fast action, the flat-panel approach is quite acceptable in this application. Just as in the case of the touch screen, audible feedback to push-button operations helps to minimize errors. In the case of the keyboard, proper selection of keyboard components also can provide tactile feedback to the operator.

The layout of the flat-panel keyboard is crucial to ensuring its ease of use in an industrial operating environment. Since the operator is not a typist or computer technician, providing a general-purpose type writer like keyboard is not appropriate. Instead, the keyboard should be partitioned into dedicated functional areas. Figure 6.21 is an example of such partitioning. The following paragraphs summarize the purpose and layout design of each of these areas.

Display Select Area. As its name implies, this keyboard area allows the operator to select a particular display of interest within the hierarchy. In some cases, this selection can be done using the touch screen on the CRT; in others, using the keyboard is more convenient. The keys in this area should allow the operator to move to any display in the hierarchy with a minimum number of keystrokes (two or three at most). The operator should have the option of moving either by making

selections from menus or by calling up the area, group, or loop by name or number. One means for providing the operator with a flexible input capability (for display selection or other purposes) is through the use of soft keys. These are selected keys in the keyboard whose assigned functions can change depending on the current display on the screen. One approach to implementing this concept is to mount a set of blank keys on the keyboard as close as practicable to the CRT display area. The current display then defines the operational meaning of these keys through messages or symbols displayed near the corresponding keys. Another approach is to mount a small CRT or flat-panel display in the keyboard area and overlay it with touch-sensitive sensors. Then the "keys" shown on the display can be labeled dynamically, depending on the particular human interface situation.

The advantage of the soft-key approach is that it can reduce the number of dedicated function keys required on the keyboard. However, the use of soft keys also has been criticized on the grounds that in crisis situations the operator may become confused over their function when moving quickly from one display to another.

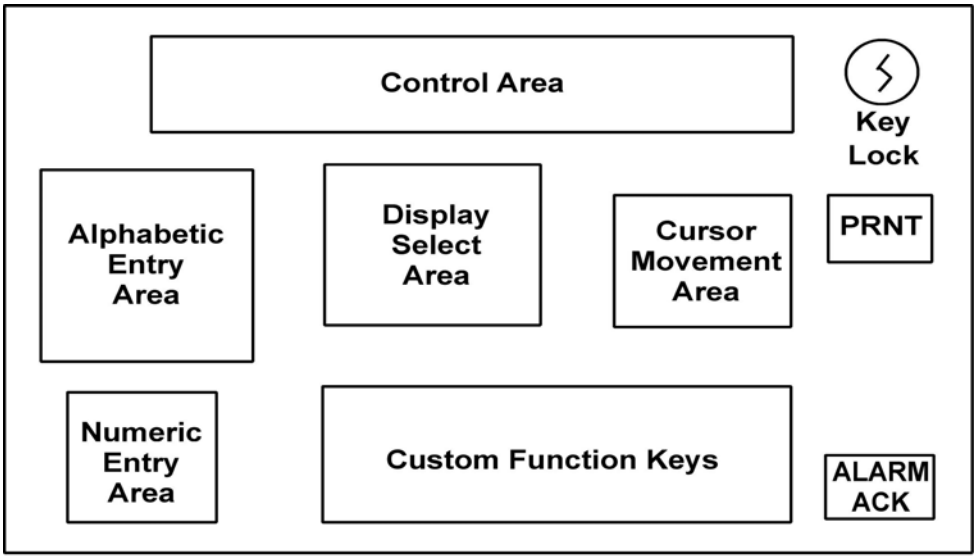


Figure 6.21. Typical Partitioning of a keyboard.

Alphabetic and Numeric Entry Areas. In the process of calling up various displays by name or identification number, it may be necessary for the operator to enter alphabetic and numeric information into the system through the keyboard. As mentioned previously, a process operator is not a typist and therefore is likely to enter this information through the hunt-and-peck method. To make this as painless as possible, the standard QWERTY typewriter keyboard layout is not used, since its configuration has no recognizable pattern as far as the average operator is concerned. Instead, the keys in the alphabetic entry area usually are laid out in alphabetic order. Very often, the numeric entry area is configured like the keypad on a touch-tone telephone, since the operator is likely to have some familiarity with that arrangement.

Cursor Movement Area. After selecting a particular display, very often it is necessary for the operator to move a cursor around the display to perform certain operations (such as activating a control station or selecting an item from an on-screen menu). In the cursor movement area of the keyboard are arrow keys for moving the cursor in the direction indicated. Some keyboard implementations replace the arrow keys with such devices as a joy stick, a track ball, or a mouse. However, the arrow keys have been preferred in most vendor offerings for reasons of reliability and cost.

Control Area. To control the process through the HLOI, the operator first selects a particular loop for manipulation or a logic operation for initiation through one of the mechanisms described in preceding sections. Depending on how the operator interface system is designed, the operator then works through the CRT touch screen or through the control area of the keyboard to execute the control operation. In either case, the operator input system accommodates the following continuous control actions: selecting manual, automatic, or other modes (e.g., cascade, ratio, or supervisory control); increasing or decreasing the set-point value or control-output value in either a slow or fast mode; and entering a desired set-point or control-

output value in engineering units. Except for the last action, these functions are the same as those available using panelboard instrumentation. In the case of a logic or sequencing operation, the operator input system allows execution of start, stop, and permissive commands. These are usually simple push-button operations and can be implemented using either the touch screen or the keyboard.

The layout of the control area of the keyboard varies widely from one distributed control system to another. In some systems, the layout is similar to that of the vendor's corresponding panelboard instrumentation. This is an advantage in applications that include both panelboard instrumentation and an HLOI, since it minimizes operator confusion when moving from one human interface device to another. In some layouts, only one set of control operation keys is provided for multiple loops. The operator first selects the loop to be controlled and then works through that set of keys. In other systems, multiple sets of control operation keys are provided to match the layout of a particular group display (e.g., eight sets of control keys for an eight-loop group display). After selecting the group, the operator can manipulate any of the loops in the group at the same time without having to select each loop first. No conclusive ergonomic results have been obtained to determine which approach is better; the multiple-loop method provides slightly faster access to loops within the same group, at the expense of cluttering up the keyboard with a larger number of keys.

Some vendors mount panelboard instrumentation in the keyboard itself to serve as the control input hardware. The intent is to provide the operator with familiar hardware, but in practice this approach leads to keyboards that are large and cumbersome. In addition, the operator then must switch back and forth between this panelboard instrumentation and the rest of the keyboard, which can cause confusion. The more prevalent practice is to keep the keyboard as simple and consistent as possible to minimize such confusion.

Miscellaneous Keyboard Functions. In addition to the dedicated functional areas just described, there are several other operator functions that the keyboard hardware must accommodate. Many distributed control system vendors are now offering a set of keys that the user can configure to perform custom functions. These keys can provide direct call-up of a selected set of displays critical to the particular process. Direct call-up bypasses the normal display hierarchy, giving the operator faster access to these displays. Keyboard hardware used to implement other miscellaneous operator interface functions include the following:

1. ***A Print Key*** — This allows the operator to obtain a hard copy of any display that is currently on the screen. The print function must be designed in such a way that the CRT and keyboard are still active while the printing process is going on;
2. ***An Alarm Acknowledge Key*** — This allows the operator to inform the HLOI that he or she has recognized a particular alarm situation and acknowledges it;
3. ***A Key Lock*** — this sets the status of the console to one of three modes: (a) off-line, in which console reconfiguration and maintenance functions are performed; (b) operational, in which the normal operator functions are activated; or (c) engineering, in which control system tuning and modification can be accomplished.

System Design Issues

The hardware elements and the display capabilities described in the previous subsections combine to form a total high-level operator interface system, which the operator uses to perform the functions summarized in Section 6.2; monitoring, control, diagnostics, and record keeping. The methods used in implementing many of these functions have already been described in the preceding paragraphs in the context of particular hardware elements and display types. The following subsections discuss other implementation issues involving the total operator interface system.

Distributed Database Considerations. The database associated with each point (control loop, analog input, or digital input) in a distributed control system consists of the current value of that point along with a great deal of auxiliary information. Some of this information relates to the identification of the point in the system, such as the following:

1. Physical hardware address of the point (cabinet, module, or input channel);
2. Classification of point (analog or digital);
3. Tag name and descriptor (if any) associated with the point;
4. Engineering units associated with the point.

Other information relates to the monitoring, alarming, and trending functions performed on the point:

1. High and low alarm limits;
2. Alarm status (not in alarm, in high alarm, in low alarm, or returned to normal);
3. Alarm acknowledgment status (alarm acknowledged or not acknowledged by the operator);
4. Past values of the point (used in trending):
5. Computed functions of past values of the point (e.g., average value, maximum value, minimum value, and smoothed or filtered value over some time period).

Each vendor's distributed control system stores this database for each point in different locations. In many systems, the local control unit stores only the hardware address and the current value of the point in engineering units. The other information is distributed throughout the system in higher-level devices such as the HLOI units or computers. This distributed approach to the structuring of the database has a number of negative consequences, including the following:

1. If there are multiple HLOIs that display the same set of points (which is usually the case), each HLOI must store the tag name, descriptor, alarm limits, and other information associated with each point. This is expensive in terms of memory requirements on the HLOI.
2. Each HLOI must compute the alarm status for each point, since each interface must display this status. This procedure is expensive in terms of computing requirements on the HLOI.
3. If the operator changes an alarm limit for a point, he or she must either change it manually in all HLOIs or there must be a mechanism to change the corresponding alarm limits automatically in the other HLOIs. The latter leads to complications if, for example, one of the HLOIs is temporarily off-line.
4. The same problem as in item 3 holds for alarm acknowledgments.

A better approach is to store all of the information associated with a point in the same LCU as the point value itself. Then there is a single location that all HLOIs and other devices in the system can query for this information. This approach also implies that such computational functions as alarm checking and averaging must be done in the LCU. This used to be impractical to implement due to the limited memory and computational resources of the LCUs. With the continuing reductions in cost and increases in performance of the hardware in the LCUs, this approach has become feasible.

One result of performing the trending and alarming functions in the individual LCU is that each LCU must then keep track of the time of day. Since the HLOI units must sort out and display trend and alarm data from a large number of LCUs in a consistent manner, each data point must be tagged with its associated time of occurrence. The mechanism for providing synchronized time clocks in each LCU varies from system to system. The simplest approach is to have one high-level element in the system maintain the master time base. That element can periodically update the other clocks in the system by sending messages over the shared communications facility (for example).

Alarming. In conventional control rooms, the operator relied on the annunciator panel to provide timely information on alarms in the system. The design of the HLOI has replaced this panel with an alarming function distributed among the hardware in the distributed control system and among the displays in the HLOI. As described in the previous paragraph, the alarm-checking function is often performed by the LCU, which then makes the alarm status of each point available to other elements in the distributed system. Usually there is one display or set of displays in the HLOI dedicated to the alarming function. It generally takes the form of a chronological list of the previous N alarms. In large systems, a complete HLOI unit can be dedicated to displaying this set of alarms. If the HLOI includes several CRTs, one of them can display this alarm list.

However, it is important that the operator be made aware of the alarm status of the plant on each display in the HLOI system, since at any particular time he or she may be working with and concentrating on a display other than the alarm display. For this reason, in each operating display in the HLOI the top line is usually a time/date/alarm summary, as described in Section 6.4.3. The design of each display also incorporates alarming functions by giving the alarm status, alarm type, and (sometimes) the alarm limits along with the value of each point.

When a point first goes into alarm, its new alarm status usually is signaled when the tag name of the alarmed point changes color and starts to blink. Often an audible signal such as a bell or a tone accompanies the visible change in alarm status. When the operator acknowledges the alarm, the blinking stops and the audible signal is silenced but the color change on the display remains as long as the point is in alarm. In some systems, the alarm status of each point can be placed into one of several levels of priority when the system is first configured. This priority level is displayed along with the alarm status itself, usually through the color of the alarm indication.
The mechanism used for acknowledging alarms differs from one distributed system

to another. An alarm acknowledgment key is always included in the keyboard, but the scope of the alarm points acknowledged by the keystroke ranges from a single point to all the points in the area being displayed. Usually the design forces the operator to move to a group display that includes the point in alarm before the system will accept the operator's acknowledgement. This mechanism is intended to force the operator to review the nature of the alarm and act accordingly instead of simply hitting the acknowledge button as a reflex action to silence the audible signal. References 6.14 and 6.36 provide additional information on alarm management in distributed control systems.

Trending. The data storage and retrieval capabilities of the HLOI system provide the operator with much more flexibility and accuracy in trending than ever possible with conventional strip chart or round chart recorders. Of course, many implementations of the HLOI provide an analog output capability that can be used to drive conventional recorders. However, the real benefits of advanced trending accrue when trend data are stored digitally within the distributed control system (either in the LCUs, in the HLOI unit, or in a separate trending box). The following trending features are commonly available in state-of-the-art distributed control systems:

1. Trend data for each point are stored at a resolution the user selects to match the dynamics of the point being trended (e.g., once a second for flows, once every 30 seconds for temperatures).
2. The operator receives the data in graphical form by means of CRT displays designed to look like conventional trend graphs (as described in Section 6.4.3.). Usually, each display can share trend graphs of several points.
3. The data shown on the CRT screen at any one time are usually a small fraction of the data available in storage. This allows the operator to pan through the data or zoom in on the portion of time that is of particular interest. Of course, the maximum time resolution achievable in the use of these functions is limited by the frequency of data collection originally specified for the point.

4. The amplitude of the trend graph can usually be changed on-line by the operator to magnify the graph, making it easier to read. Of course, the resolution of the particular CRT used will limit the amplitude resolution of the trends. Also, a digital readout usually is provided to display the value of the point at any time along the trend graph, along with a cursor function that allows the operator to select the time to which the digital readout applies.
5. Pressing the print-display key produces a hard copy of the trend graph.
6. In addition to providing the operator with trend displays preconfi-gured to include certain points, some systems have a "wild card" display that allows the operator to select a unique set of trends that are of particular interest to him.

Trend data are usually stored in the distributed system elements for only a limited amount of time (typically 10 to 12 hours) to allow some overlap between operator shifts. The long-term data storage and retrieval function (described in the following subsection) provides storage over a longer period of time.

Long-Term Data Storage and Retrieval. This feature is the process control equivalent of the flight recorder, which commercial and military aviation uses to store selected data points as a function of time and which allows later review for investigative or historical purposes. For example, this information can be quite valuable for determining the root cause of an equipment failure, for analyzing the dynamic characteristics of a process, or for providing proof of a product's proper manufacture. The information stored commonly includes a record of key process variables, alarms, and operator actions (for example) as a function of time. Tradi-tionally, this feature has been implemented using a computer. The significant expansion of processing and memory capabilities in an HLOI unit has made it feasible and cost-effective to implement this feature in the HLOI itself.

In its basic form, long-term data storage and retrieval involves the following sequence of operations:

1. A sequence of process data points or events over a selected time period is recorded.
2. Each data point is labeled with the appropriate tag name, engineering units, and time at which it was recorded.
3. The sequence is written onto a memory medium, such as a floppy disk or magnetic tape, that can be removed from the HLOI and placed in storage.

As a minimum, the long-term data storage and retrieval implementation in a high-level operator interface must allow the operator to run an "instant replay" of selected data sequences on the HLOI in the form of trend graphs. Other features to consider in evaluating or designing a long-term data storage and retrieval system are:

1. ***Compatibility with on-line Displays***—To minimize operator confusion, the HLOI should allow the replay of long-term storage data by means of the same trend displays used for on-line trending. It should not be necessary to take the HLOI off-line to perform the replay function.
2. ***Tabular Format Capability***—In addition to allowing replay in a trend graph format, the HLOI should be designed to permit display of the data in a tabular format. As a minimum, this format should include the tag name of the point, the time associated with the point value, and the point value itself in engineering units. It also should be possible to produce a hard copy listing of selected time sequences of points in long-term storage.
3. ***Data Storage Format***—The format of data storage on the floppy disk or magnetic tape should allow computers or digital devices other than the vendor's HLOI to read and manipulate the data. In many cases, the off-line data analysis operations that must be performed require the use of a general-purpose digital computer. Convenience in accessing the data is essential.
4. ***Density of Data Storage***—Data compression techniques should be used to maximize the amount of information stored on a single floppy disk or magnetic tape cartridge or cassette. This will minimize the amount of human interaction

(i.e., disk or tape loading and unloading) required to support the long-term storage function. The operator or instrument engineer should not have to be involved in this function any more often than once a day. Appropriate prompts and alarms must be provided to ensure that the magnetic medium is replaced on schedule and that no data are lost.

In some distributed control systems, the long-term data storage and retrieval feature is not included in the HLOI. Rather, it is implemented in a separate device, sometimes called a. process historian, which gathers data using the plant communications facility.

Logging. The third record-keeping function (in addition to trending and long-term data storage and retrieval) that distributed control systems provide is logging. The primary objective of the logging function is to produce a hard-copy record of process data and events on a printer. Like the long-term storage and retrieval function, logging traditionally has been implemented in a computer system. In current distributed control systems, logging is implemented either in the high-level operator interface or in a separate process historian.

Logging Functions fall into two Categories: periodic and event-driven. A periodic log is simply a printed record of the values of a particular process point or points at regular time intervals. Many early computer systems performed this data-logging function. It was mainly successful in generating reams of printout paper that was rarely looked at again unless a process problem developed. The periodic logging function has largely been supplanted by long-term data storage and retrieval. Recording the information on magnetic media has proved to be much more efficient than producing hard copy.

Most hard-copy logging functions are now event-driven, and include the following examples:

1. ***Process Alarms***—High. low. and deviation alarms are logged when they occur: often, operator acknowledgments and alarm return-to-normal conditions also are logged.
2. ***Equipment Alarms***—The failures of devices (e.g., sensors, transmitters, and controllers) within the instrumentation and control system often are detected through on-line diagnostics and are recorded on a hard-copy log.
3. ***Operator Control Actions***—Controller mode changes, set-point and control-output changes, sequence initiations, and other control actions performed by the operator often are logged. Some systems also allow the operator to store notes or journal entries that explain or expand on the record of his or her actions.
4. ***Sequence-of-Events (trip) logs***—In many distributed control systems, dedicated devices called sequence-of-events recorders monitor the occurrence of discrete events (e.g., switches opening or closing or a variable exceeding a limit) dealing with a major piece of plant equipment. If this equipment fails or is tripped off-line, the recorder remembers the sequence of events that led to the failure or trip event. If the recorder is integrated into the distributed control system (instead of being a stand-alone device), the sequence can be transmitted to the logging device for recording on the logging printer.

One of the issues involved in implementing the logging function is what format the log printout should use. Usually, the vendor provides a standard format that includes the time the logged event occurred, the type of event that occurred, and the appropriate tag number or device number associated with the event. Some logging systems allow the user to customize this format to his or her own needs; others are not as flexible.

In most distributed control systems, at least one printer is dedicated to the logging function. This approach minimizes the mixing of log printouts with other system functions, such as hard copy of CRT displays and printing of data from long-term storage.

This concludes the review of operator interface design issues in distributed control systems. The discussion in Chapter 7 moves to the design of human interfaces for instrument engineers and other support personnel.

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MODULE 7:

PROGRAMMABLE LOGIC CONTROLLERS (PLCS)

PROGRAMMABLE LOGIC CONTROLLERS (PLCS)

Objectives:

At completion of this module, the developee will have an understanding of:

1. Programmable logic controllers basic architecture.
2. Programmable logic controller main parts.
3. Power supply module function and circuits.
4. Input / Output modules types and function.
5. Processor module typical function.
6. Memories and memory allocation.
7. Programming devices.
8. PLC program languages.
9. Writing programming sheets.
10. Storing the program.
11. Entering the program.
12. Programmable Controller hardware; racks & modules.
13. Wiring connections of Input / output modules.
14. Multiplexing system and its advantages.
15. Attached: ESD PLC of USP.

Related HSE Regulations for Module 7:

Juniors have to be familiarised with the following SGC HSE regulations, while studying this module:

Regulation No. 6: Work to permit system.

Regulation No. 7: Isolation

7.18 (1-10) control systems procedures and isolations

Regulation No. 19 & 20 "Working with Electricity"

19.7.4 Precautions on Low Voltage Systems

19.7.5 Precautions on Extra Low Voltage Systems

19.11 Static Electricity Precautions

INTRODUCTION

For a number of years industrial control had been achieved by electromechanical devices such as relays, solenoid valves, motors, linear actuators and timers. These devices were used to control manufacturing operations, industrial processes, and heavy equipment where only switching operations were necessary. Most control was of the two-state type, which simply called for a machine to be turned on or off. The control circuitry was hardwired to the machine and considered to be a permanent installation. Modification of the system was rather difficult to accomplish and somewhat expensive. In industries where production changes were frequent, this type of control was rather costly. It was, however, the best way and in many cases, the only way that control could be effectively achieved.

In the late 1960s, solid-state devices and digital electronics began to appear in industrial controllers. Circuitry that utilised this type of control was designed to replace the older electromechanical devices. The transition to solid-state control has, however, been more significant than expected. Solid-state devices, digital electronics, integrated circuit technology and computer-based systems have lead to the development of programmable controllers or PCs. These devices have capabilities that far exceed the older electromechanical controllers. Programmable controllers permit flexible circuit construction techniques, have reduced downtime when making changeovers, operate with improved efficiency, and can be housed in a very small space.

The first programmable controllers could only perform a limited number of functions. Two-state control, AND, OR, and some limited timing functions were the extent of the control capabilities. Today, this type of controller can perform all logic functions, do arithmetic operations, and can sense analogue changes in a manufacturing operation. It can accept a millivolt signal from a thermocouple, multiply it by a constant, and display the results in degrees Celsius. The resulting control operation can be stored in memory for future use, displayed on a cathode-

ray tube, or used to energise an alarm. The unique features of a modern PC are flexibility, operational efficiency, and versatility.

PLC Definition

A programmable controller is defined as a "**digital electronic device that uses a programmable memory to store instructions and to implement specific functions such as logic, sequence, timing, counting, and arithmetic to control machines and processes**"

Characteristic Function of a PLC

Seven of the most important characteristics of a PLC include the following:

1. **It is field programmable by the user**. This characteristic allows the user to write and change programs in the field without rewiring or sending the unit back to the manufacturer for this purpose.

2. **It contains pre-programmed functions**. PLCs contain at least logic, timing, counting, and memory functions that the user can access through some type of control-oriented programming language.

3. **It scans memory and inputs and outputs (I/O) in a deterministic manner**. This critical feature allows the control engineer to determine precisely how the machine or process will respond to the program.

4. **It provides error checking and diagnostics**. A PLC will periodically run internal tests of its memory, processor, and I/O systems to ensure that what it is doing to the machine or process is what it was programmed to do.

5. **It can be monitored**. A PLC will provide some form of monitoring capability, either through indicating lights that show the status of inputs and outputs, or by an external device that can display program execution status.

6. **It is packaged appropriately**. PLCs are designed to withstand the temperature, humidity, vibration, and noise found in most factory environments.

7. **It has general-purpose suitability**. Generally a PLC is not designed for a specific application, but it can handle a wide variety of control tasks effectively.

Parts of a Programmable Controller

All programmable controllers have the same basic parts and characteristics.
The four basic parts of the programmable controller are:

1. **The power supply,**
2. **Input/output interface sections,**
3. **Processor section, and**
4. **Programming section.**

These basic parts are illustrated in Figure 10-1.

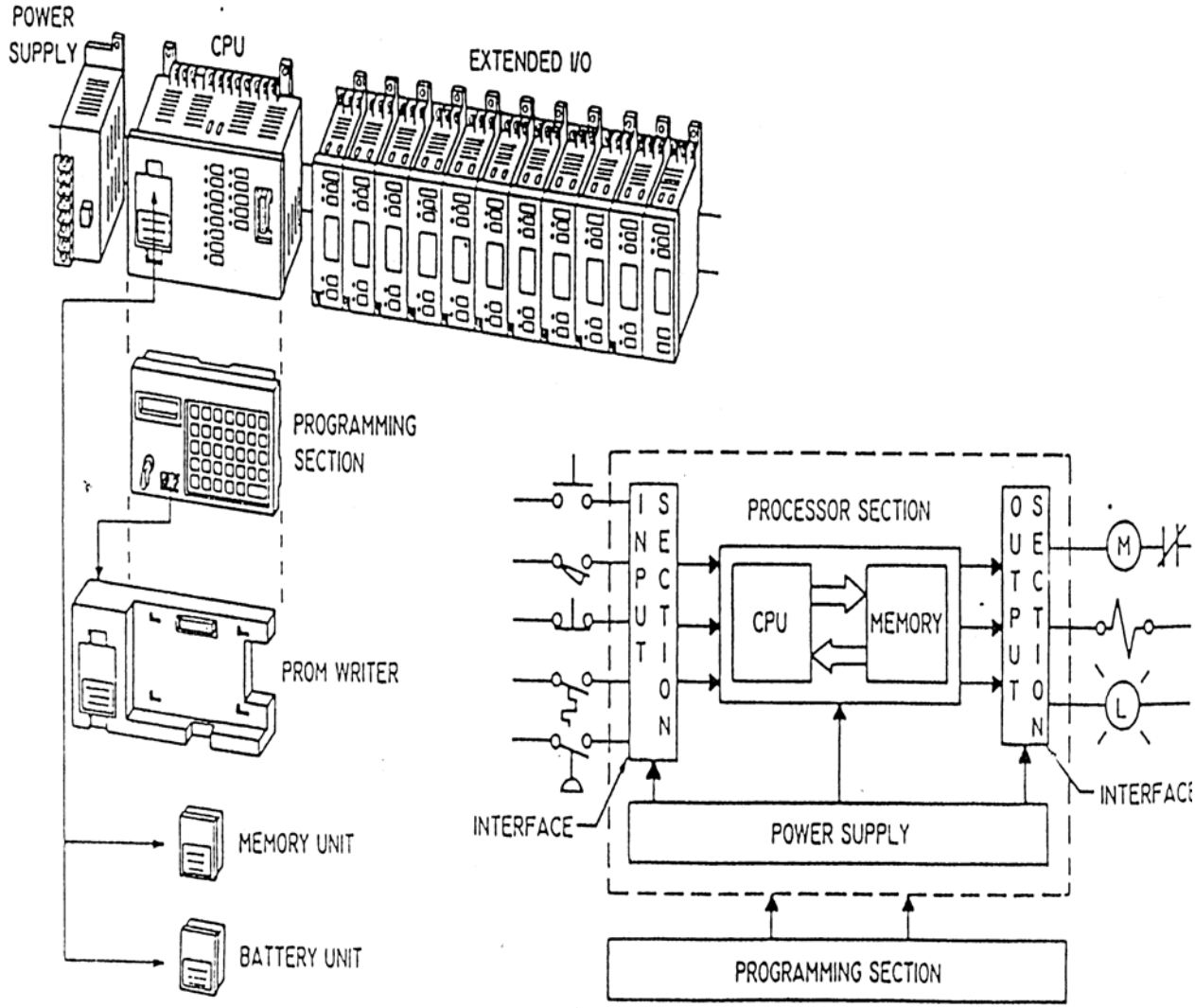

Figure 10-1 the four basic parts of a programmable controller include the power supply, input/output interface section, processor section and programming section.

Programs are stored and retrieved from memory as required. Sections of the programmable controller are interconnected and work together in order to:

1. Allow the programmable controller to accept inputs from a variety of sensors,
2. Make a logical decision as programmed, and
3. Control outputs such as motor starters, solenoids, valves, and drives.

1) The Power Supply Section

The power supply provides all the necessary voltage levels required for the programmable controllers' internal operations. In addition, the power supply may provide power for the input/output modules. The power supply can be a separate unit or built into the processing section. Its function is to take the incoming voltage (usually 120 or 240 VAC) and change this voltage as required (usually 5 to 32 VDC).

The Power Supply must provide:

a. Constant output voltage free of transient voltage spikes and other electrical noise.
b. Charges an internal battery in programmable controllers to prevent loss of memory when external power is removed. Memory retention time may vary from hours up to 10 years on many programmable controllers.

2) The Input / Output Interface Section

The input and output interface section functions as the eyes, ears, and hands of the programmable controller.

- The input section is designed to receive information from pushbuttons, temperature switches, pressure switches, photoelectric and proximity switches, and other sensors. The input section receives incoming signals (usually at a high voltage level) and interfaces the signal to the low power digital processor section. The processor can then register and compare the incoming signals to the program.

- The output section is designed to deliver the output voltage required to control alarms, lights, solenoids, starters and other supports. The output section receives low power digital signals from the processor and converts them into high power

signals. These high power signals can drive industrial loads than can light, move, grip, rotate, extend, release, heat, and perform other functions.

Discrete Input. The most common type of inputs and outputs are the discrete type. This type of I/O uses bits, with each bit representing a signal that is separate and distinct, such as ON/OFF, open/ closed or energised / de-energised. The processor reads this as the presence or absence of power.

Examples of discrete inputs are pushbuttons, selector switches, joy sticks, relay contacts, starter contacts, temperature switches, pressure switches, level switches, flow switches, limit switches, photoelectric switches, and proximity switches. Discrete outputs include lights, relays, solenoids, starters, alarms, valves, heating elements, and motors.

Input Module Circuitry

The input module of a PC is extremely important in the operation of a circuit. It is responsible for connecting an external input source to the PC so that it modifies the operation of the processor. The input source usually necessitates some degree of electrical isolation to protect the delicate input of the processor. As a rule, input module circuitry has a prescribed operating voltage and current rating. This varies a great deal among different manufacturers. Typically, the input voltage has an operational range of several volts.

Some representative values are 20 to 28 volts ac/dc, 10 to 55 volts dc, 105 to 130 volts ac/dc. The current needed to actuate the input is generally of a nominal value. Operational ranges of 10 to 50 milliamperes are very common. Different modules are also made to accommodate a number of inputs, ranging from one to eight in current PCs.

Figure 10-2 shows a wiring diagram for a representative 24-volt ac/dc input module. This particular module will accommodate eight input circuits. Typical input devices include push buttons, limit switches, selector switches, and relay contacts. The input will interface either 24 volts ac or dc from an external source to the processor. The actuating voltage source is derived from circuitry outside of the input module. When the module accommodates a number of inputs, each one has a reference location or number designation. This number is the same for each module. It will vary however, according the working position of the module in the system. In practice, the" number has three or more digits. The least significant place value refers to the module circuit number. The next most significant place value usually denotes the input module location in the system. The most significant place value generally denotes the function of the module, such as input or output.

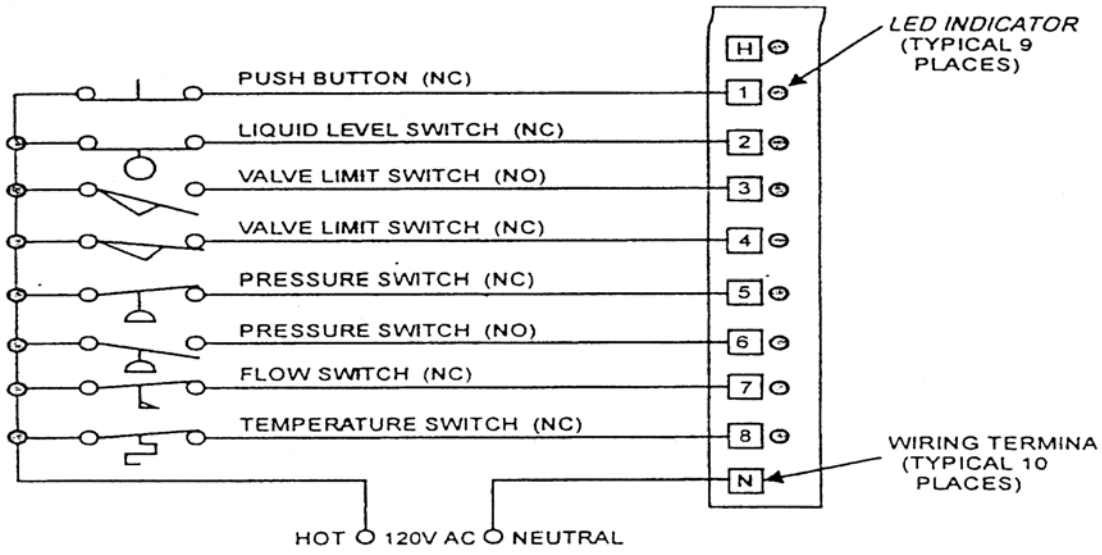

Figure 10-2 Wiring Diagram of Discrete Input Module

The circuitry of one-input of the previous module is shown in Figure 10-3. This circuit responds to either an ac or dc energy source. The circuit actuates an LED in its output.

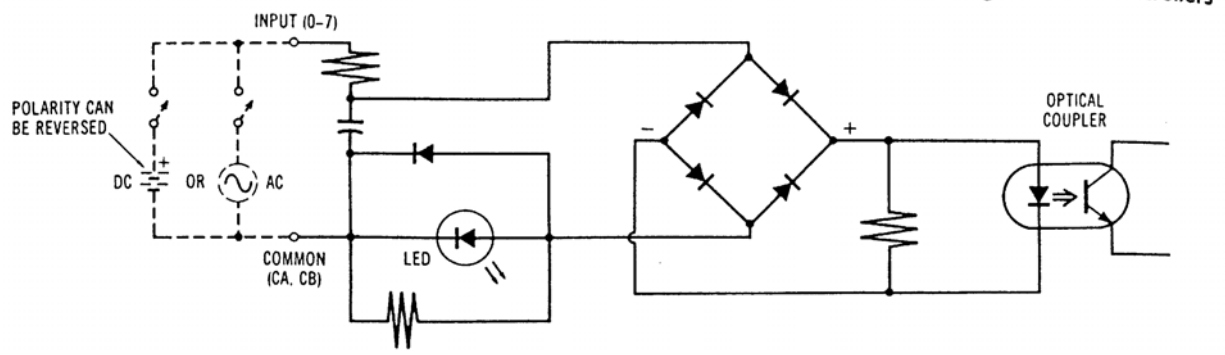

Figure10-3 Circuitry of an Input Module

With ac applied, the bridge will be energized and produce dc to energize the LED of the optical coupler. When the optical coupler is energized, output energy is transferred to the processor through a phototransistor. The coupler isolates the processor from the input source voltage. A similar response will be produced by either polarity of dc applied to the input. The LED on the left or line side of the circuit serves as an indicator to show when the input is being energized.

One circuit of the input module of Figure 10-3 uses 90 milliwatts of power from its source when actuated. A representative circuit might respond to 9 volts dc at 10 milliamperes. Each of the eight input circuits will use the same amount of power when actuated. All eight circuits in operation at the same time will use 8 X 90 or 720 milliwatts of power from the external energy source. Essentially, the operating energy of the input module does not detract from energy being supplied to the processor.

Data/Analogue Input. In many applications, more complex information is required than the simple discrete I/O is capable of. For example, measuring temperature (72 °F) may be required as input into the programmable controller and numerical data (001) may be required as an output. Figure 10-4 shows a wiring diagram for a representative analogue input module.

These types of inputs are called ***data inputs***. They may be an analogue type, which allows for monitoring and control of analogue voltages and currents, or they may be digital, such as BCD inputs.

When an analogue signal (such as voltage or current) is input into an analogue input card, the signal is converted from analogue to digital by an analogue to digital (A to D) converter. The converted value, which is proportional to the analogue signal, is sent to the processor section. Examples of data inputs are potentiometers, rheostats, temperature transducers, level transducers, pressure transducers, humidity transducers, encoders, bar code readers, and wind speed transducers.

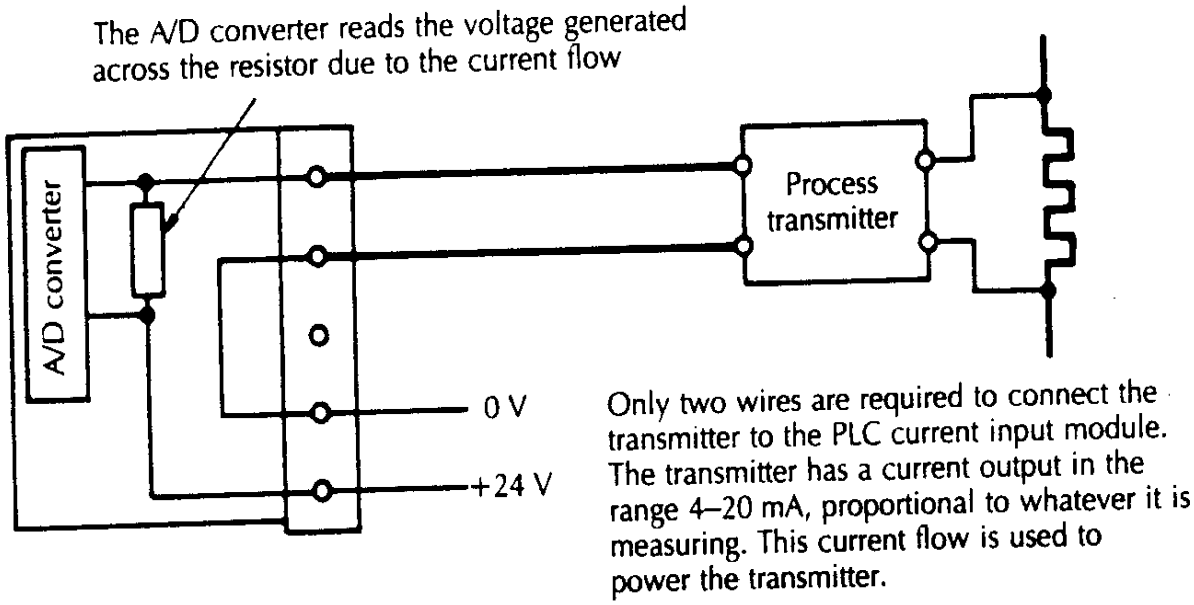

Figure 10-4 Current Loop Input

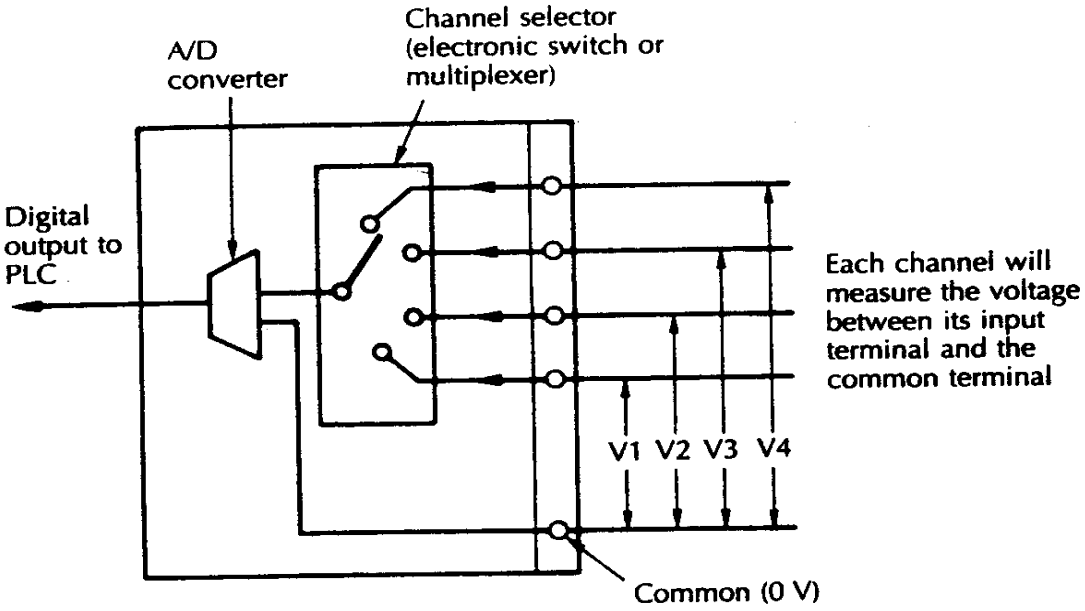

Figure 10-5 Single-ended A/D Converter

Discrete Output Module Circuitry

The output module of a PC is responsible for connecting the processor to a load device being energized by an outside source. These modules vary a great deal in their design and operation. Some units may house only one output circuit per module, while others may incorporate several individual output circuits in a single module. The circuitry of a module will vary a great deal among different manufacturers. The components of a module must be capable of sinking the energy source supplied to the load device. Sinking refers to the ability of a device to dissipate or give off heat. An output module usually contains a power control device such as a transistor. The power dissipation rating of this device determines its sinking capability. This function of the output module is dependent on ambient temperature. Most output modules have rated operating temperatures that must be followed in order to assure that the output device will not be damaged. Some modules fuse each circuit to protect the output device from damage. Figure 10-6

shows a wiring diagram of a typical output module. This particular module will accommodate four outputs. Note that these are numbered as 070, 071, 072, and 073. The loads are energized by a dc source of 5 to 24 volts dc.

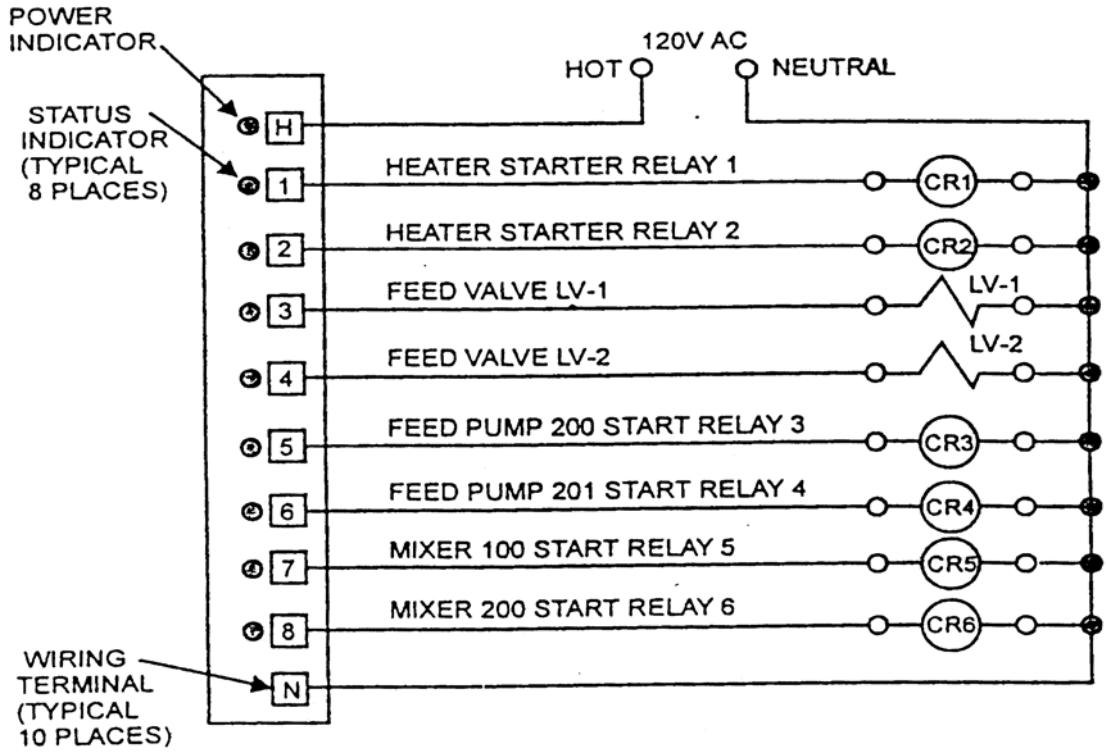

Figure 10-6 Wiring Diagram of an Output Module

The circuitry of one section of the output module is shown in Figure 10-7. This particular circuit has an optical coupler connecting the output of the processor to the module. This is used to isolate the processor from the power source of the load de-vice. The output device of the circuit is an N-chan-nel MOSFET of the depletion type. A string of zener diodes connected across the source/drain of the FET is used to regulate the voltage to a value that will not destroy the device. These diodes will conduct if the voltage rises above a prescribed value. The output of this circuit is also protected by a fuse. If the source/drain current exceeds a prescribed value, the output device will be protected from damage. An external source is used to energize

the output transistor. A signal from the processor actuates the gate of the FET causing it to conduct. The signal, in a sense, is the connecting link between the processor and load device operation. When an appropriate signal is applied to the input of the module, the output is energized. This action ultimately controls the load device. The circuitry of an output module varies a great deal among different manufacturers. The circuit shown here is only representative of that used by one manufacturer. The external source voltage and sinking capability of the module generally dictate the circuitry and the type of output device used. Mini PCs use low level output sinking circuits, while larger PCs may use heavy duty modules with high level sinking capabilities. As a rule, the output circuit develops only two states or conditions of operation on and off. The load device changes according to these conditions.

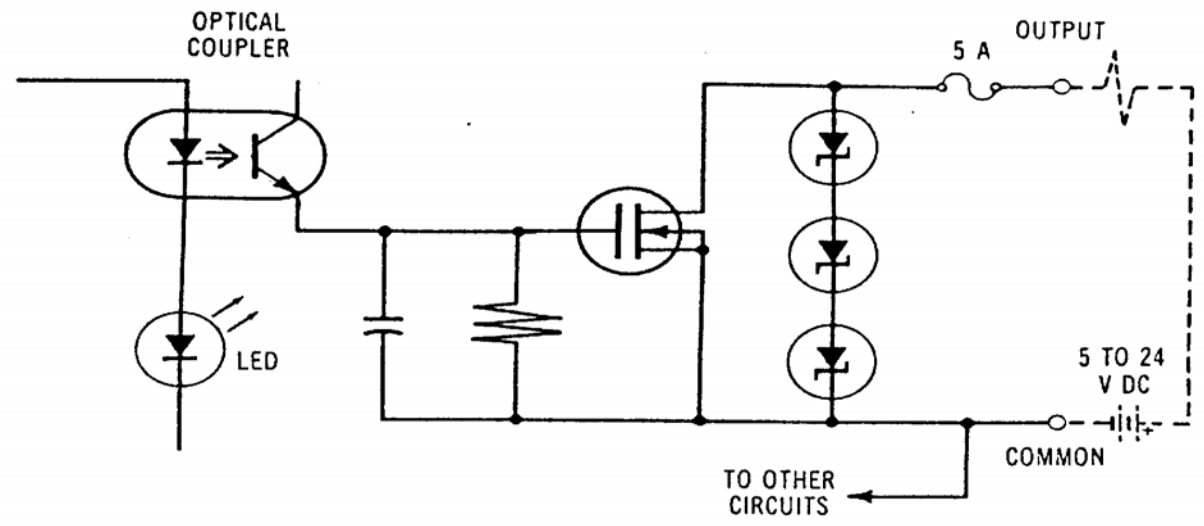

Figure 10-7 Circuitry of an Output Module

Data/Analogue Output. Figure 10-8 shows a wiring diagram for a representative analogue input module. These types of outputs are called ***data outputs***. After the processor has processed the information according to the program, the processor outputs the information to a digital to analogue (D to A) converter. The converted

signal can provide an analogue voltage or current output that can be used or displayed on an instrument in a variety of processes and applications.

Examples of data outputs are analogue meters, digital meters, stepping motor (signals), variable voltage outputs, and variable current outputs.

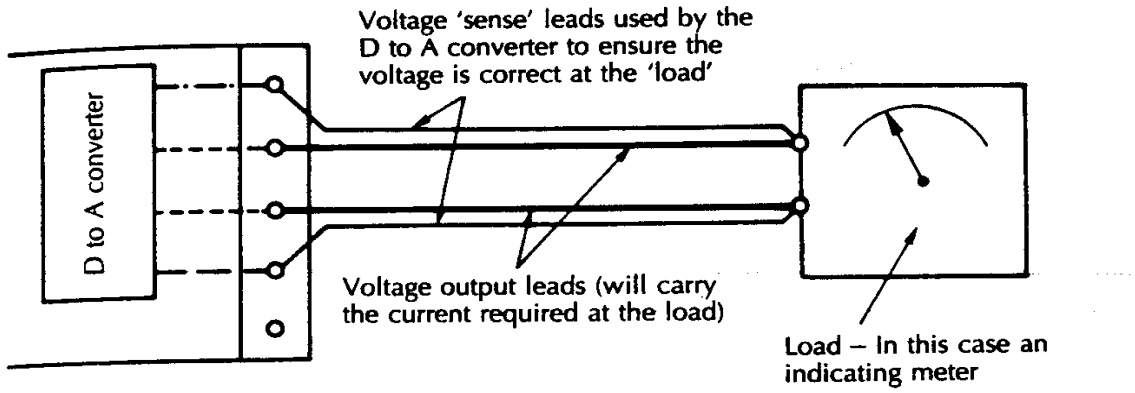

Figure 10-8 Analogue Output Connections

I/O Capacity

The programmable controller of Figure 10-9 is classified as a mini-PC. This type of unit is designed to control a small number of machine operations and a variety of manufacturing processes.

Mini-PCs is classified as systems that can economically replace as few as four relays in a control application. They are capable of providing timer and counter functions, as well as relay logic, and are small enough to fit into a standard 19-inch rack assembly. Most systems of this type can accommodate up to 32 I/O ports or modules.

Larger units can accommodate up to 400 I/O ports or devices. Mini-PCs, in general, can achieve control similar to that of larger units, but on a smaller basis. These units are less expensive, easier to use, smaller, and more efficient than the larger PC units. In the future, most relay applications of industry will be accomplished by mini-PCs.

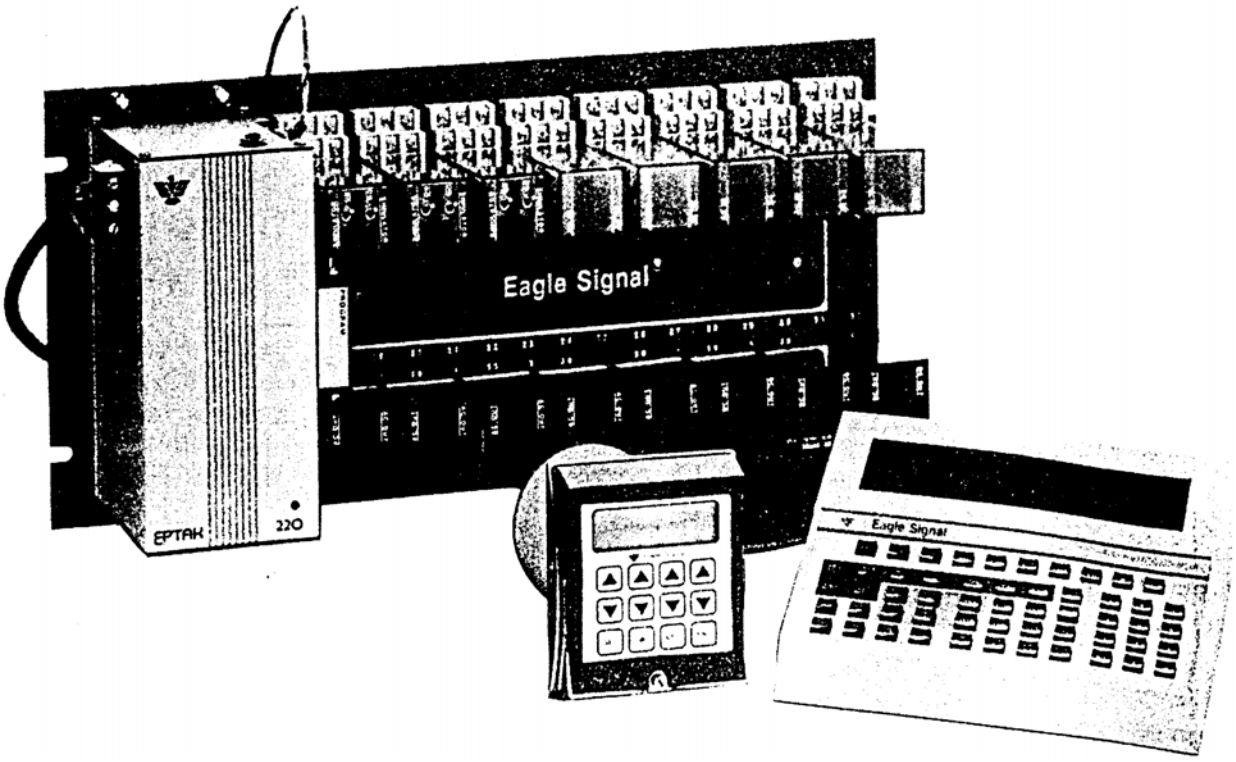

Figure 10-9 A Mini Programmable Controller

A factor that determines the size of a programmable controller is the controller's input/output and capacity. Typical input/output capacities of different size programmable controllers are listed.

1. **Mini/Micro**. Usually 32 or less I/O but may have up to 64
2. **Small**. Usually have 64 to 128 I/O, but may have up to 256
3. **Medium**. Usually have 256 to 512 I/O, but may have up to 1023
4. **Large**. Usually have 1024 to 2048 I/O, but may have many thousands more on very large units.

The inputs and outputs may be directly connected to the programmable controller or may be in a remote location. I/Os in a remote location from the processor section can be hard-wired back to the controller, multiplexed over a pair of wires, or sent by a fibre optic cable. In any case, the remote I/O is still under the control of the central processing section. Typical remote I/Os are of the 16, 32, 64, 128, and 256 size.

Fibre optic communications modules (Figure 10-10) route signals to and from I/Os to the processor section. Fibre optics communications modules are unaffected by noise interference and are commonly used for process applications in the food industry, petrochemicals, and hazardous locations.

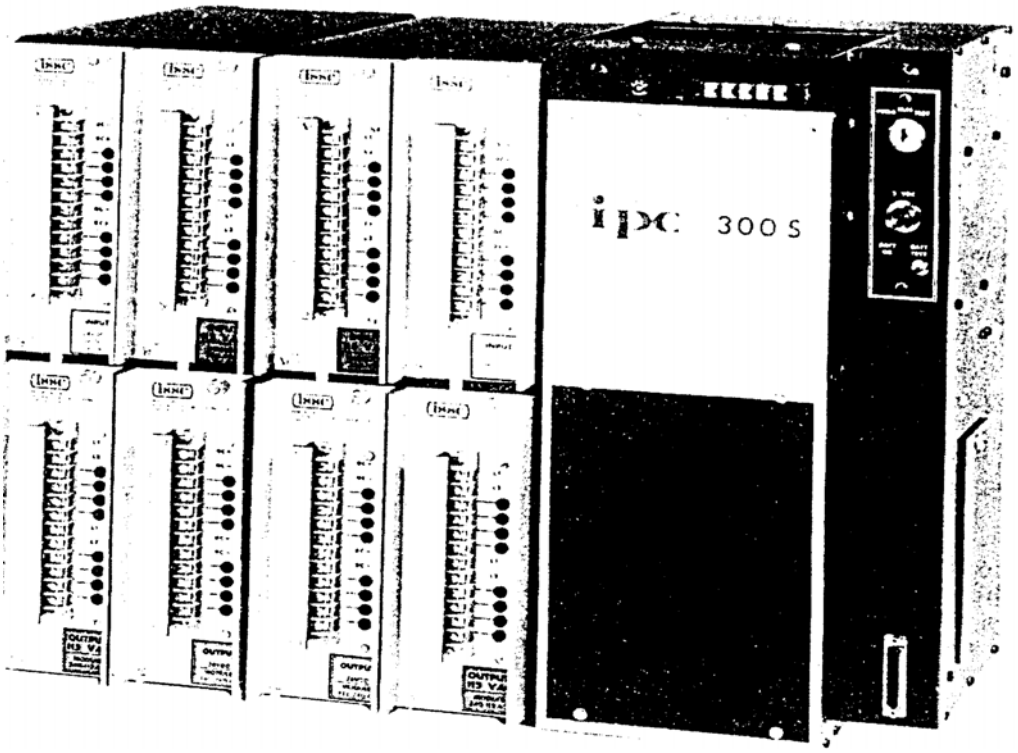

Figure 10-10 I/O modules connected with fibre optic cable provides transmission of data unaffected by noise interference.

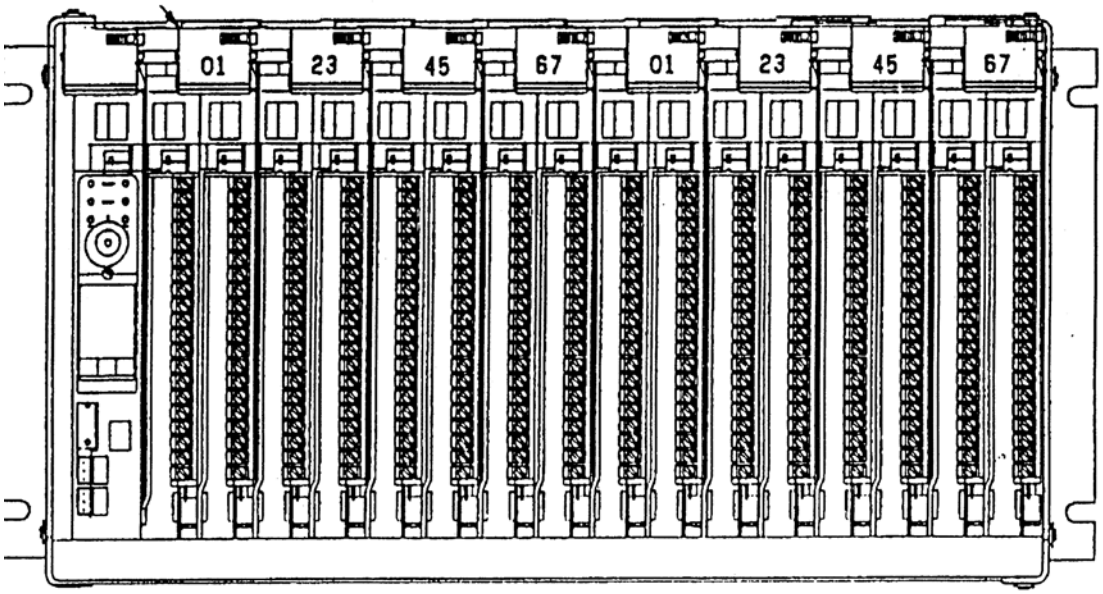

Figure 10-11 PLC Rack Layout

3) Processor Section

The processor section is the brain of the programmable controller. This section organises all control activity by receiving inputs, performing logical decisions according to the program, and controlling the outputs.

The processor section does the following:

a. Evaluates all input signals and levels.
b. This data is then compared to the memory in the programmable controller, which contains the logic of how the inputs are interconnected in the circuit. The interconnections are programmed into the processor by the programming section.
c. Based upon the input conditions and program the processor section then controls the outputs.
d. The processor continuously examines the status of the inputs and outputs and updates them according to the program.

Figure 10-12 illustrates some of the many functions the processor performs.

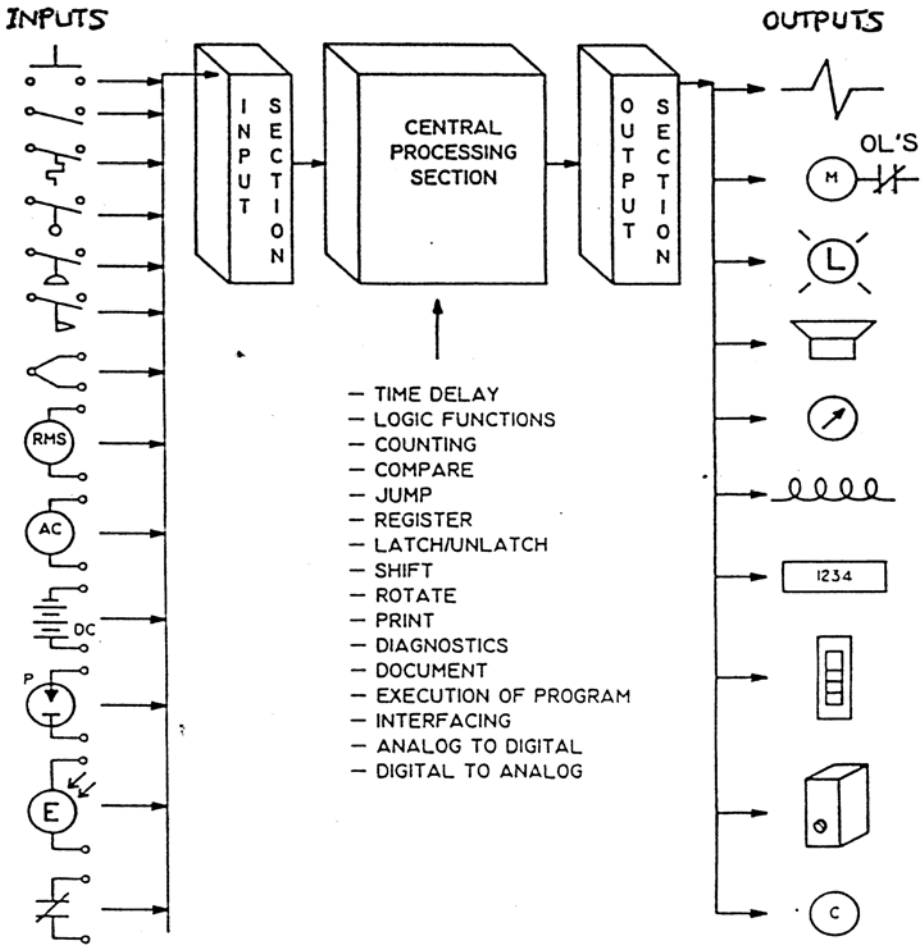

Figure 10-12 the processor section organizes all control activity by receiving inputs, performing logical decisions as programmed and controlling the outputs.

A view of a disassembled processor board is shown in Figure 10-14. This unit has a power supply, memory chips, signal conditioning units and the processor on a single board assembly. This type of board construction permits the chips to be inter-connected without using external wires. Digital signals applied to the assembly move along the foil lines of the printed-circuit board. Off-board components can be connected to the board by an interconnecting cable. Sixteen I/O modules of this system are attached to an off-board assembly by a ribbon connector. This type of construction permits a number of external components to be connected to the processor board using a minimum of conductors.

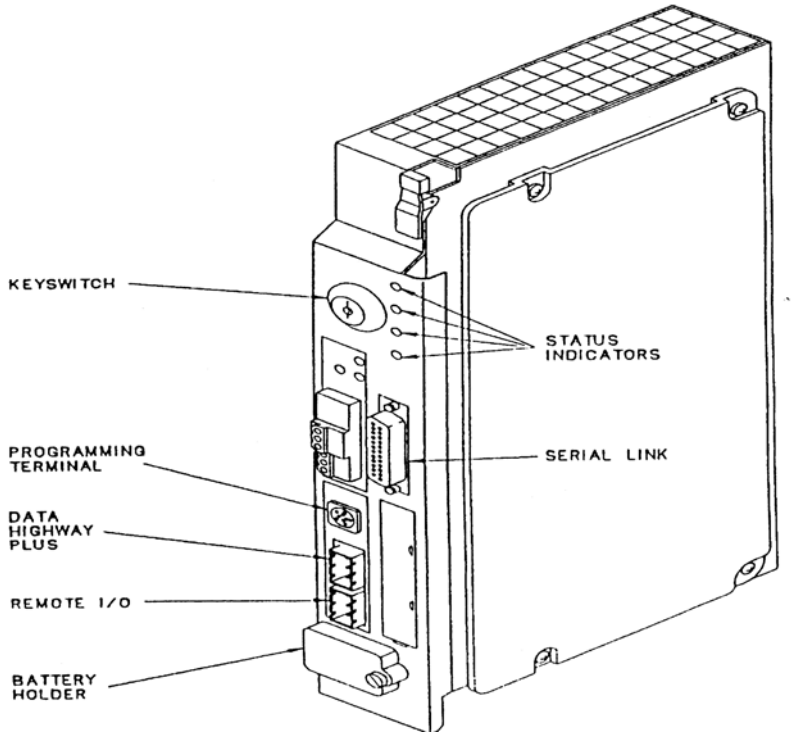

Figure 10-13 Processor Module Front Panel

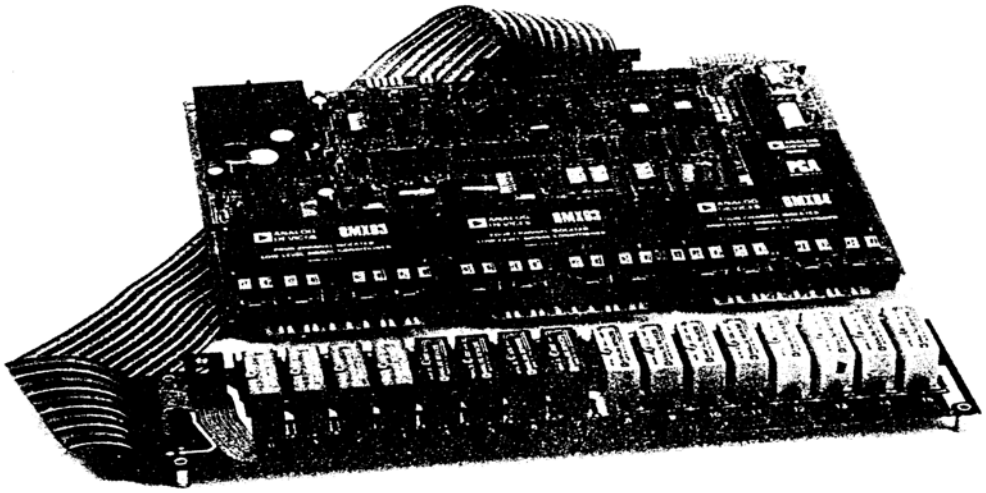

Figure 10-14 A View of a Disassembled Processor Board

Processor Operation

Operation of the processor of a PC is somewhat complex when compared to other instruments. A person does not need to know a great deal about the overall operation of the internal workings of the processor in order to use it effectively. In general, only a few of the basic operational facts of the PC are needed to make it functional.

The processor of a PC has a master program permanently stored in its memory. This program is needed to make the processor operational when it is turned on initially. In a sense, this program material is similar to the firmware of a computer-based system. Without this program material in the processor, the system would not function when it is energized. The master program or firmware does several things in the normal operational sequence. It must energize the system, tell the processor to send a startup menu to the CRT terminal or display, and open the operational channels to energize the keyboard. This permits the user to see an operational menu on the display unit and make the necessary selections to start programming of the keyboard. The firmware of a PC is generally called the executive, or ROM, program. This program also monitors the master switch of the PC panel. The switch is used by the programmer or PC user to determine whether the processor uses a program

keyed into memory, or if it operates from a program stored in memory. Some PCs have the option of selecting a test program from memory without energizing the outputs.

Alien Bradley calls the positions of their master select switch "run program," "run," "test program," and "program." Modicom, and other manufacturers use specific switch selections instead of a master key switch to set up the operation of their system. Once the selection is made, the processor begins running through the executive program automatically.

If the user selects the program switch position, the processor recognizes signals from the keyboard and begins the program development procedure. If the run or test switch position is selected, the processor begins to execute the user's program while looking for input signals, and develops appropriate output signals. To stop the program sequence but leave the PC powered up, the select switch is placed in the stop processor position. The operator of the PC has control over the switch selection procedure, and the processor responds according to its directions.

Memory

The memory of each processor is generally somewhat different because of the chip used and the operations it must perform in a PC. The differences are usually in the size of the storage space for programs, counters, timers, messages, and registers. Figure10-15 shows a layout of the memory allocation of a typical PC. Notice that this shows the memory divided into distinct parts. These parts are made of binary bits. Sixteen bits are grouped together to form words. Memory allotment is made in total words. The distribution of memory is divided into three groups called the data table, user pro-gram, and message storage. The data table uses 128 words for factory configured data and a variable amount of data space for preset data and file/ bit storage. The user program section houses the main program data and some of the subroutines of operator-controlled programs. The third group deals with message

storage. This space allotment is variable and stores messages pertaining to operating conditions. If more memory is needed for a particular operation or control function the system can be expanded to accommodate this.

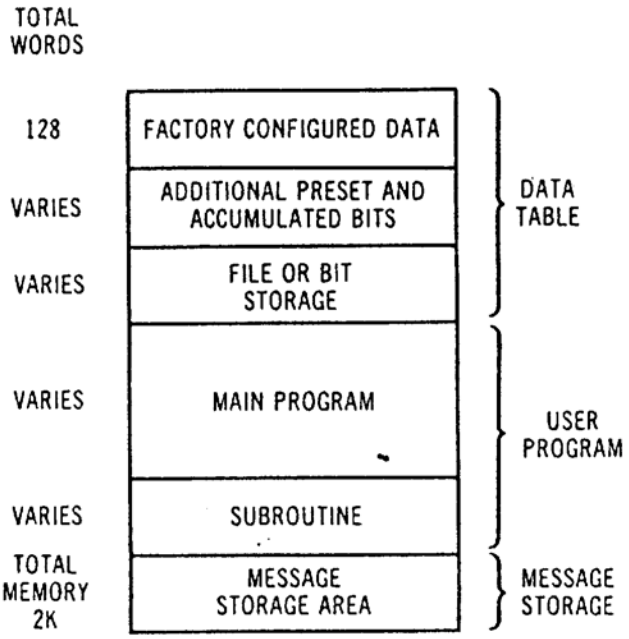

Figure 10-15 Memory Layout of a PC

The memory size of a PC can be expanded to meet the needs of the system where it is being used. A typical system is generally purchased with a rather small memory to control one machine with a limited number of I/O ports. If the need arises, the memory can be expanded to meet the demands of a larger operation. Most PC manufacturers feel that this is the best solution for memory selection. The system can be designed to fit any application by expanding memory according to the operations being performed.

The memory of a PC varies a great deal among different manufacturers. Presently, memory is specified in bytes. A group of eight bits is called a byte. Computer-based systems are usually described according to the word size of an instruction. A word is also eight bits in length. The terms word and byte can be used interchangeably. A 16-bit computer-based system is said to have a word size of two bytes. Microprocessors used in PCs can have an eight-bit word size or one-byte instructions. The letter K as

used with computers is a symbol equivalent to the numeral 1024. Memory is generally specified in one thousand bytes or 1K increments. Typical memory sizes of a PC are 1K, 2K, 4K, 8K, 16K, 32K, 64K, 128K, and 256K. As a rule, the more memory that a PC has, the more operations that it can perform.

Program Scanning

When a processor checks the memory of a PC and executes the program stored there, it must follow a procedure that tells what is stored at each memory location. This procedure is called program scanning. Some PCs scan the memory by vertical columns. A vertical scan usually starts at the top left corner and moves to the bottom of the first column, then up through the next column, down to the bottom of the next column, and continues through the remaining columns in the same manner. Horizontal scanning is similar to eye movement when reading a printed page. The memory elements are scanned across vertical columns in a horizontal line or one rung of the ladder diagram at a time. The scanning procedure used by a particular system is determined by the manufacturer.

Most PCs may have ten or eleven vertical columns in their format. The number of columns refers to the number of element functions of a rung in a ladder diagram. The Modicon 484, for example, can support 10 elements plus one output in each of its rungs. This element number is largely determined by the processor being used and the programming plan of the system. Obviously, the number of elements in a specific rung cannot exceed the number of columns established by the system.

The scanning procedure of a PC deals with the program material placed in memory. Essentially the entire program is scanned many times per second. Scanning rates vary from four milliseconds to several hundred milliseconds, depending on the memory size. A millisecond is one thousandth of a second. This means that a great deal of program scanning can occur in a very short period of time. Scanning is done to update the inputs, outputs, timers, counters, and math registers.

In mini PCs the scanning time is not of any major concern. In large PC systems scanning time may be an important issue. Data registers, for example, may not be updated rapidly enough to keep the data accurate in a long program.

In some systems, parts of the program can be skipped or subroutines may be developed that will make critical functions more accurate.

The process of evaluating the input/output status, executing the program, and updating the system is called ***scan***. Figure 10-16 illustrates PLC scan cycle.

The time it takes a programmable controller to make a sweep of the program is called the **scan *time***.

Scan time is usually given as the time per 1k byte of memory and typically runs in the 1 to 25 millisecond range. Scanning is a continuous and sequential process of checking the status of inputs, evaluating the logic, and updating the outputs.

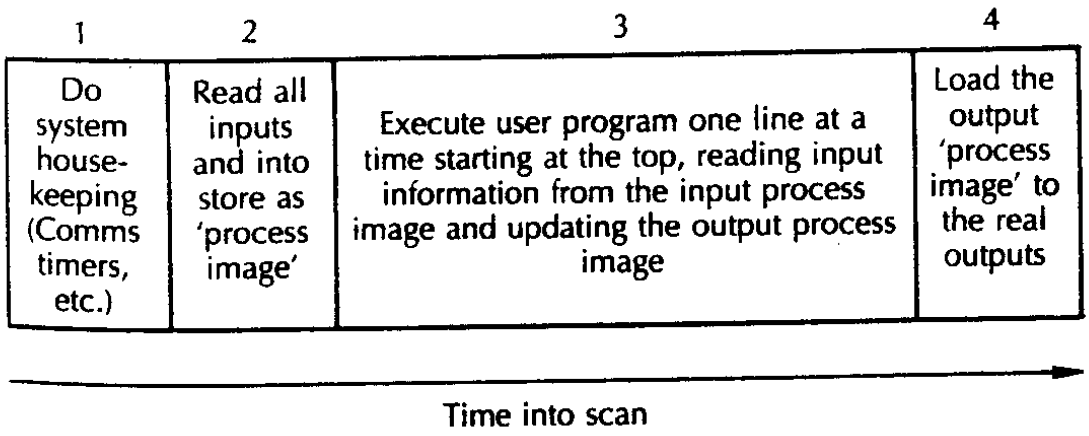

Figure 10-16. PLC Scan Cycle

4) The Programming Section

The programming section of the programmable controller allows input into the programmable controller through a keyboard. Even though the programmable controller has a brain (the processor section) it still must be told what to do. The processor must be given exact, step-by-step directions. This includes communicating to the processor such things as load set, reset, clear, enter in, move, and start timing.

Programming a programmable controller involves two components:

a. The first component is **the programming device that allows access to the processor**.
b. The second component is **the programming language that allows the operator to communicate with the processor section.**

a) Programming Devices.

Programming devices vary in size, capability and function. Programming devices are available as simpler, small, handheld units or complex colour CRTs with monitoring and graphics capabilities.

A programming device may be connected permanently to the programmable controller or connected only while the program is being entered. Once a program is entered, the programming device is no longer needed, except to make changes in the program or for monitoring functions.

Some programmable controllers are designed to use an existing personal computer, such as an IBM® for programming. Using a personal computer to program is called ***off line programming***. This permits the computer to be used for other purposes when not being used with the programmable controller.

b) Language of Programmable Controllers

The first programmable controllers used a language that was compatible with industry was the line (ladder) diagram. Line diagrams are still commonly used as a language for programmable controllers throughout the world. Other languages used are Boolean, Functional blocks, and English statement. Figure 10-17 illustrates the common program languages.

Line diagrams and Boolean are basic programmable controller languages. Functional blocks and English statement are higher level languages required to execute more powerful operations, such as data manipulations, diagnostics, and report generation. The line diagram is drawn in a series of rungs. Each rung contains one or more inputs and the output (or outputs) controlled by the inputs. The rung relates to the machine or process controls, and the programming instructions relate the desired logic to the processor.

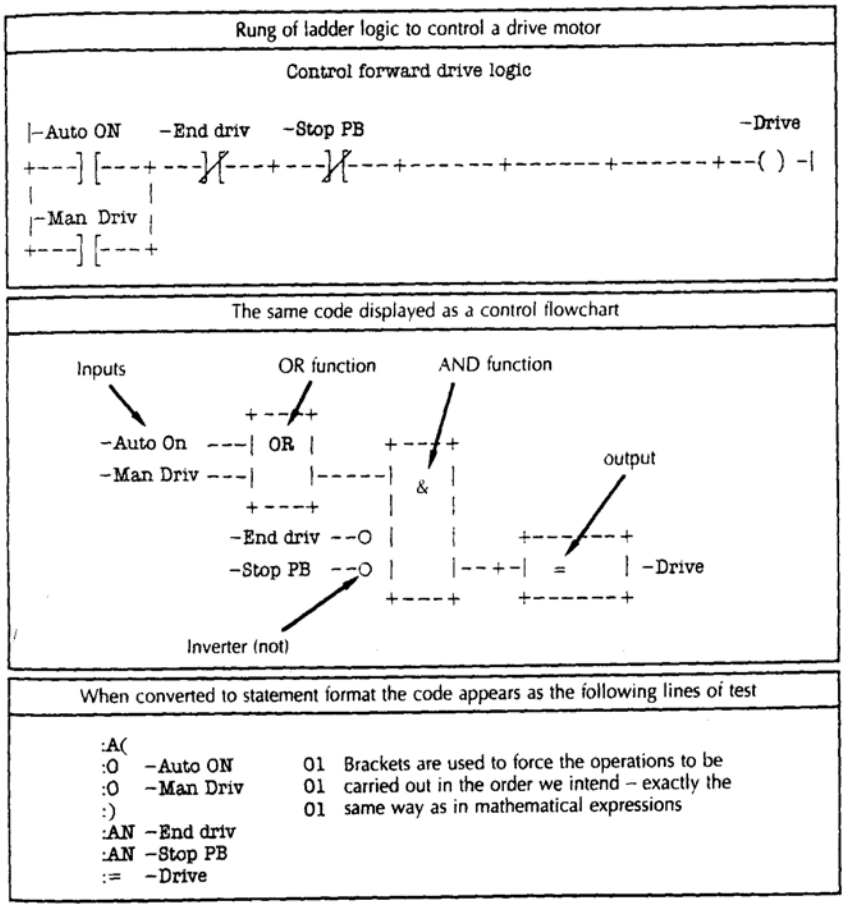

Figure 10- 17 three common program representations

PC Programming

Before a program can be entered into a programmable controller, several steps must be taken.

1. The first step is to develop the logic required of the circuit into a line diagram.
2. The second step is to take the line diagram and convert it into a programming diagram.
3. The third step is to enter the desired logic of the circuit into the controller. Every manufacturer will have a slightly different set of steps and functions to enter the program into the programmable controller.
4. The fourth step is to take the written program and enter it into the programmable controller. Once the program is entered, it can be tested.

Instructions for the operation of a programmable controller are given through push buttons, a keyboard programmer, or a magnetic disk drive assembly. Each PC has a special set of instructions and procedures to make it functional. How the unit per-forms is based on its programming procedure. In general, PCs can be programmed by relay ladder diagrams, logic diagrams or Boolean equations. These procedures can be expressed as language words or as symbolic expressions on a crt. One manufacturer describes these methods of programming as assembly language and relay language. Assembly language is used by the microprocessor of the system. Relay language is a symbolic logic system that employs the relay ladder diagram as a method of programming. This method relys on relay symbols instead of words and letter designations. We will use the relay ladder method of programming in this presentation.

Relay Logic

The processor of a PC dictates the language and programming procedure to be followed by the sys-tem. Essentially, it is capable of doing arithmetic and logic functions. It can also store and handle data and continuously monitor the status of its input and output signals. The resulting output being controlled is based on the response of the signal information being handled by the system. The processor is generally programmed by a key-board, program panel or CRT terminal. Figure 10-17 shows the layout of a relay ladder programmer. This particular panel uses a liquid crystal display. The display area is divided into three fields or areas. The top field shows the program statement number, error indicator, and power flow information. The middle field shows the alphanumeric de-tails of the contents of a statement number or the dynamic status of data registers, timers, or counters.

The bottom field displays the dynamic status of the input/output (I/O), control relay (CR), retentive control relay (RCR), shift registers (SR), and cam timers (CAM TMR).
In a relay language system, the basic element of programming is the relay contact. This contact may be normally open (NO) or normally closed (NC). Figure 10-18

shows the symbolic expression of these contacts. Note that the normally open contacts are on the left and the normally closed contacts are on the right. The line on the right side of the two lower contacts is for connection to branch circuits. This is generally an optional circuit possibility. Below each contact is a four-digit reference number. This number is used to identify specific contacts being used in the system. The contact is then connected in either series or parallel to form a horizontal rung of the relay ladder diagram.

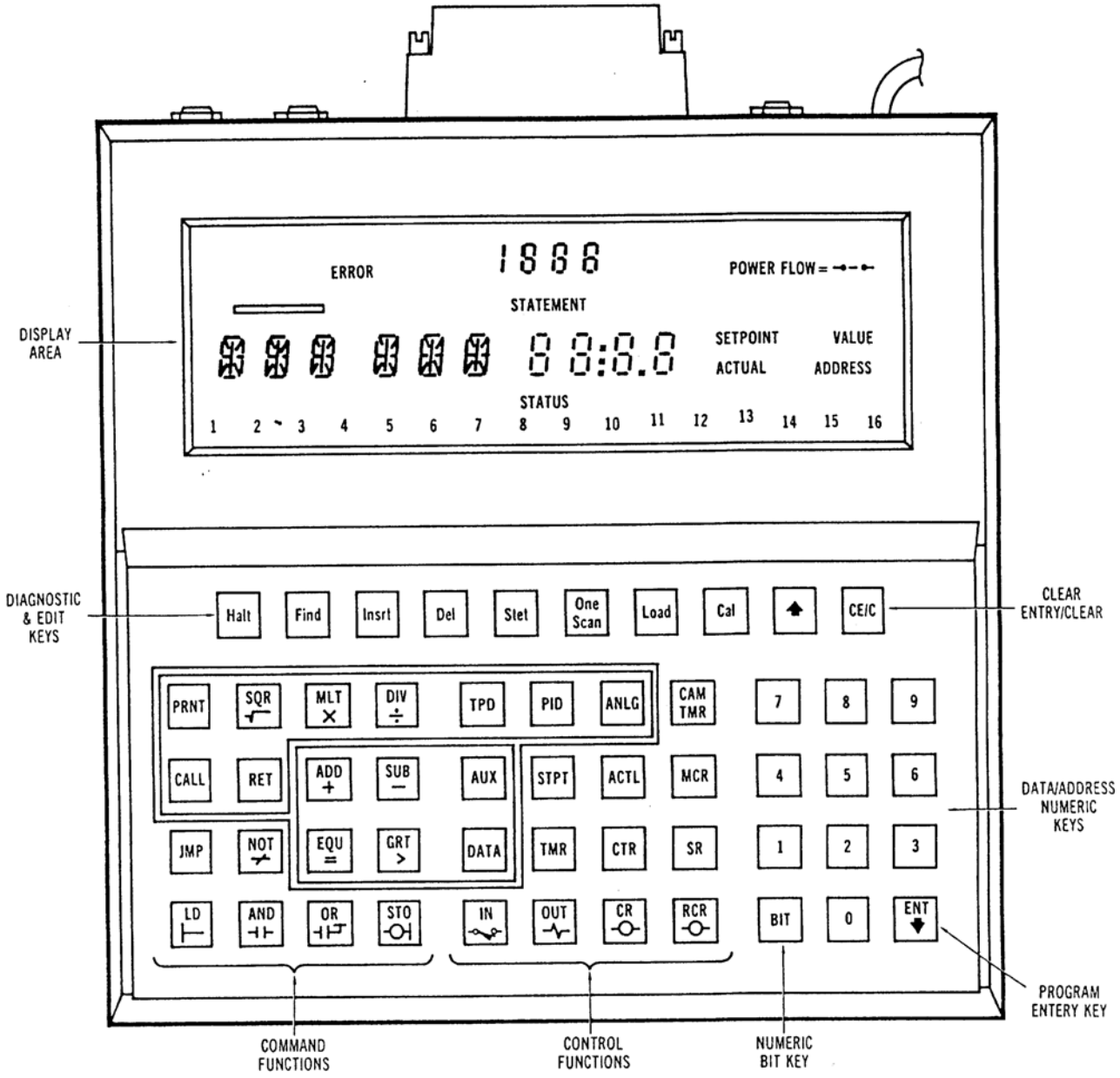

Figure 10-17. Relay Ladder Programmer Layout

Once the relay program has been entered into the PC, it can be monitored and modified if the need arises. For systems with a CRT, modification is accomplished on the display. This is achieved by simply moving the cursor of the CRT to the component being altered and making the change with a key-stroke. For systems without a CRT, modification is made by reviewing the program one step at a time. Changes are made by altering the program statement so that it conforms with the desired procedure. Most systems of this type may have simulator modules that can be placed at strategic locations to monitor program operation. The pro-gram can be stepped through to see if the sequence is correct.

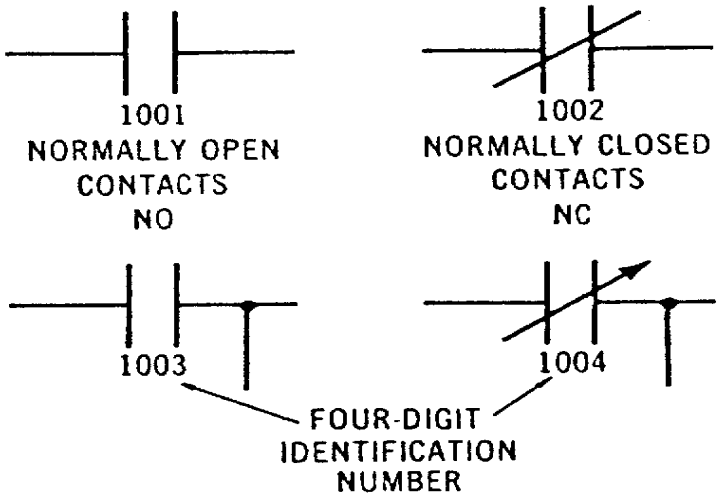

Figure 10-18 Programming Format of Relay Contacts

All the control components of a PC are identified by a numbering system. As a rule, each manufacturer has a unique set of component numbers for its system. One manufacturer has a four-digit numbering system for referencing components. The numbers are divided into discrete component references and register references. A discrete component could be used to achieve on and off control operations. Limit switches, push buttons, relay contacts, motor starters, relay coils, solenoid valves and solid-state devices are examples of discrete component references. Registers are used to store some form of numerical data or information. Timing counts, number

counts, and arithmetic data may be stored in register devices. All component references and register references are identified by the numbering system. Each manufacturer has some distinct way of identifying system components.

Each I/O module of a programmable controller has a distinct reference number or address to identify its location in the system.
The number in general identifies the location of the module in the system. If a track system is used, the reference number refers to a specific track location.
Some systems may identify an input module with a four digit number beginning with a one (1). The output module is then identified by a four digit number beginning with a zero (0). The prefix number can not be altered in the programming procedure.

Assume now that a relay diagram has been placed into the PC by selection of proper number data entries and symbol selections. The PC must then examine this network and solve the interconnected logic elements in the proper sequence. In doing this, the first rung or network of the ladder must be solved. Then networks 2, 3, and 4 must be solved in order. The solving of each sequence is achieved by a series of scanning pulses.

These pulse scans occur at a rather high rate. Each pulse passes through the network in a specific sequence. Scanning occurs in a PC when power is first applied and continues as long as the system is energized. This permits each network rung of the ladder to be solved from the left rail to the right and from the top to the bottom in an appropriate sequence. This assures that each network is solved according to its numerical step and not by the value assigned to a specific coil or contact.

Programming Basics

Programmable controllers are provided with the capability to program or simulate the function of relays, timers, and counters. Programming is achieved on a format of

up to 10 elements in each horizontal row or rung of a relay diagram, and up to 7 of these rungs connected to form a complete net-work (Figure 10-18). A network can be as simple as a single rung or a combination of several rungs as long as there is some interconnection between the elements of each rung. The left rail of the ladder can be the common connecting element. Each network can have up to seven coils connected in any order to the right rail of the ladder. These coil numbers can only be used once in the operational sequence. The quantity of discrete devices and registers available for use depends on the power or capacity of the system.

When programming a relay ladder diagram into a PC, the discrete devices and registers are placed in the component format of Figure 10-21. Each component in this case is assigned a four-digit identification number. The specific reference number depends on the memory size of the system. In a low-capacity system, number assignments could be 0001 to 0064 for output coils and 0258 to 0320 for internal coils. A system with a larger capacity might use number assignments of 0001 to 0256 for output coils and 0258 to 0512 for internal coils. Any coil output or internal coil can only be used once in the system.

References to contacts controlled by a specific coil can be used as many times as needed to complete the control operation. Output coils that are not used to drive a specific load can be used internally in the programming procedure.

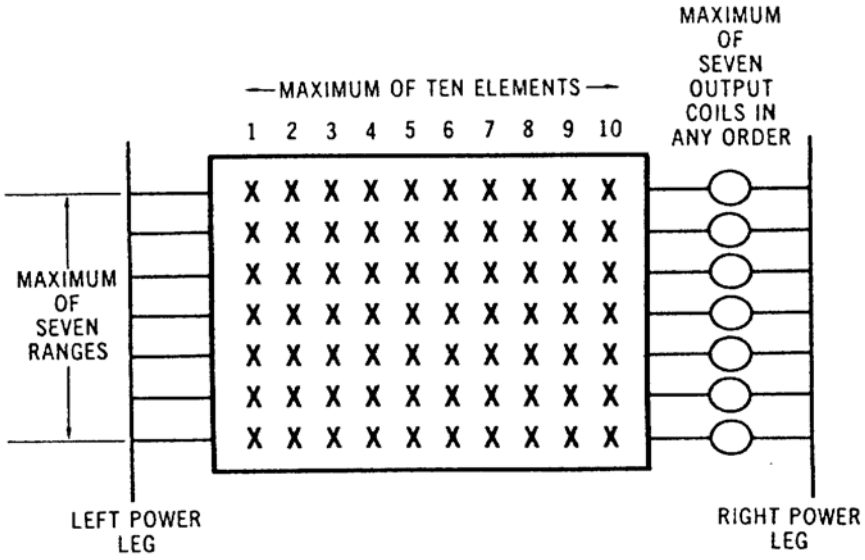

Figure 10-19 Relay Ladder Programming Format

When programming the response of a particular input module, it may be identified as a relay con-tact. In this regard, the symbol may be a normally closed contact or a normally open contact. The coil or actuating member of the contact takes on the same numbering assignment. The coil, however, is identified as a circle on the diagram and the contacts are identified by the standard contact symbol. Figure 10-18 shows some examples of the symbol identification procedure. The number designation is used to identify specific devices and contacts. The contacts can be programmed to achieve either the NO or NC condition according to its intended function.

Any external input that is considered to be normally closed, such as a safety switch, overload switch, or stop push button, must be treated differently. An external NC push button, for example, would not be entered on a CRT as closed contacts. It would produce the opposite effect internally from that of a NO contact. Inverting the external contact function, as well as its signal, constitutes a double inversion operation. It is for this reason that all normally closed external contacts or switches are programmed as normally open on the CRT.

Assume now that the simple start-stop motor controller of Figure 10-20 is to be connected by a programmable controller. The start and stop buttons are located externally. Pushing the start button energizes the relay coil. This action latches the relay coil by closing contacts CR1 across the start button. Contacts CR2 close at the same time, completing the energy path to the motor, thus causing it to run.

The motor continues to run as long as energy is supplied from the source. Pushing the stop button turns off the motor and removes the latch from the start button.

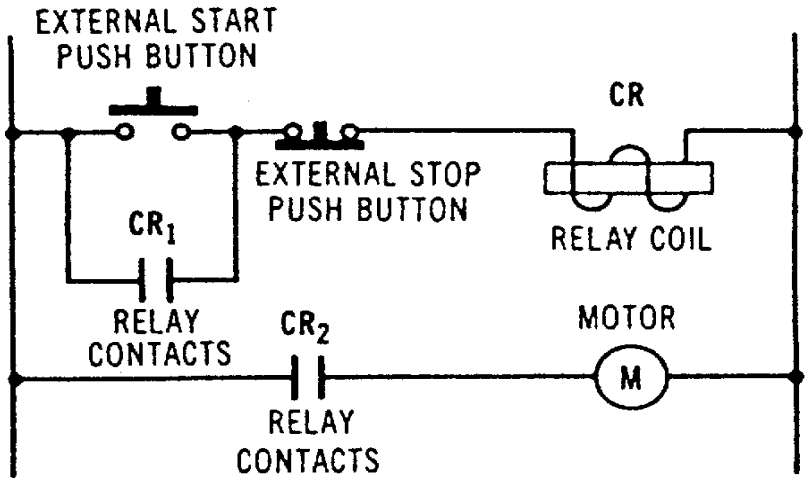

Figure 10-20 Simple Start-Stop Motor Controller Circuit

A programmable controller equivalent of the motor starter of figure10-20 is shown in figure10-21. The number assignments refer to the specific components of the PC. Input devices are numbered 1001 and 1002. This includes the input module and its resulting switching operation. The output device is numbered 0049. The start and stop buttons are externally connected and do not have a module number assignment. Operation of the PC equivalent circuit will achieve the same control procedure as the original relay ladder diagram.

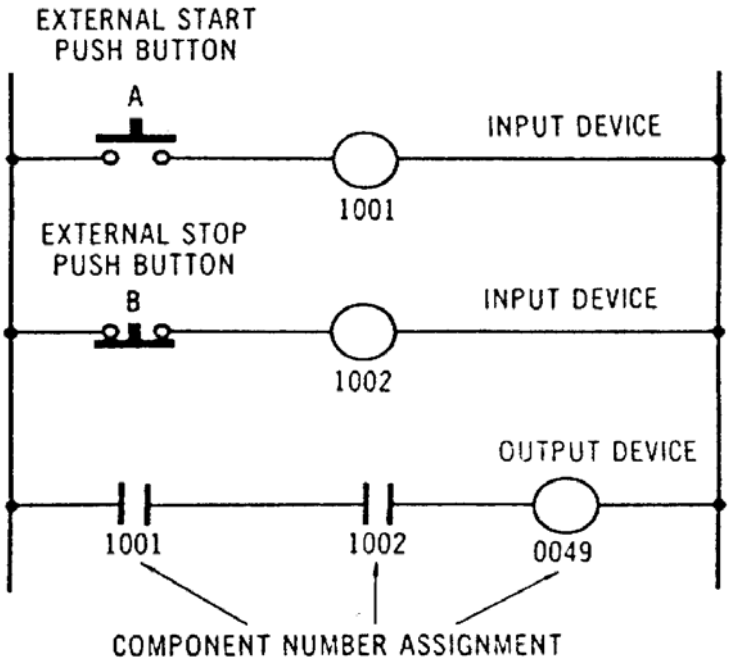

Figure 10-21 PC Equivalent of Motor Controller Circuit

The actual PC circuit for a motor start-stop control operation is shown in Figure10-22. This diagram shows how the I/O modules are interfaced with the processor. Note that the input modules and output modules are treated as independent parts of the system that are controlled by the processor. This circuit would be displayed on a CRT type of indicating system and could be modified with a few simple keystrokes. The PC equivalent is somewhat more complex than its ladder diagram equivalent. It is, however, more versatile and can be modified very quickly by a program change. Programming is simply a process of entering the appropriate component number assignment and then designating the function to be achieved by each component. The procedure can then be placed in memory and retained for future use, or used immediately according to the needs of the system.

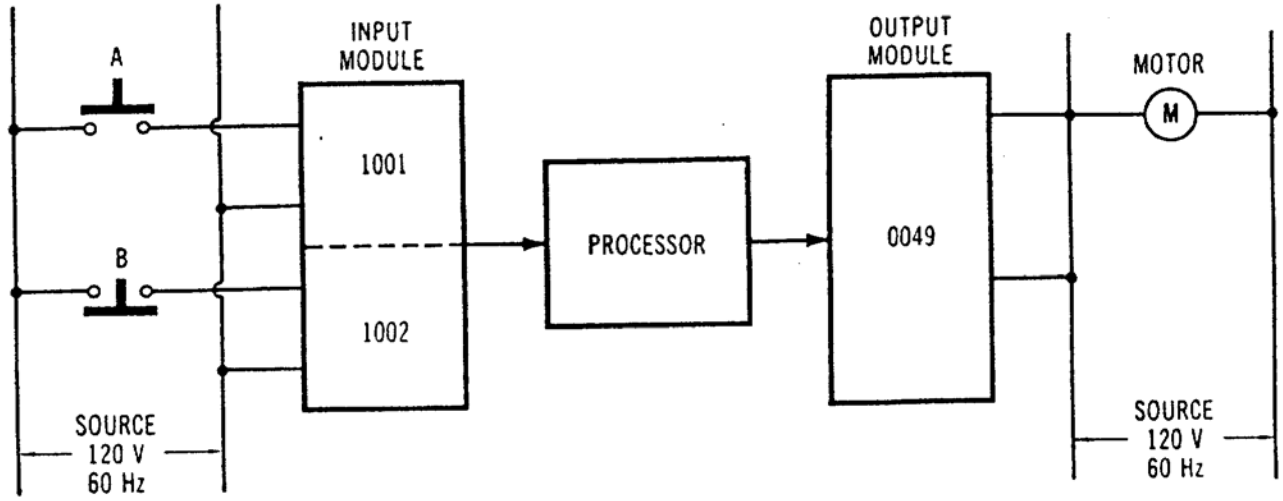

Figure 10-22 Actual PC Circuit of Motor Controller

Storing and Documentation

Once a program has been developed it may be necessary to store the program outside of the controller or document the program by printing it out (Figure 10-23). This allows for a means of storing and retrieving control programs, which makes for fast changes in a process or operation. Storage of a program is commonly achieved using a cassette tape recorder. When a change from one control program to another is required, one need only load the programmable controller with the correct tape to start the line for all the proper control settings.

Even if the programmable controller is not likely to ever have its program changed, the program should be stored on a tape. This ensures the safety of the program in the event of a problem.

Another method of storing a program is to use a read-only memory **(ROM),** random access memory **(RAM),** or erasable programmable read only memory **(EPROM)** type memory chip. This allows a permanently stored program (EPROM) to provide the memory storage. These chips allow for a program to be stored on them. Many original equipment manufacturers (OEM) use this type of storage to store the machine's program after it has been developed. They can be mass-produced and placed in the machine as needed.

Once a program has been entered into the programmable controller, connecting the controller to a printer can make a copy of the program. The printout can be used as a hard copy of the program for documentation and future reference.

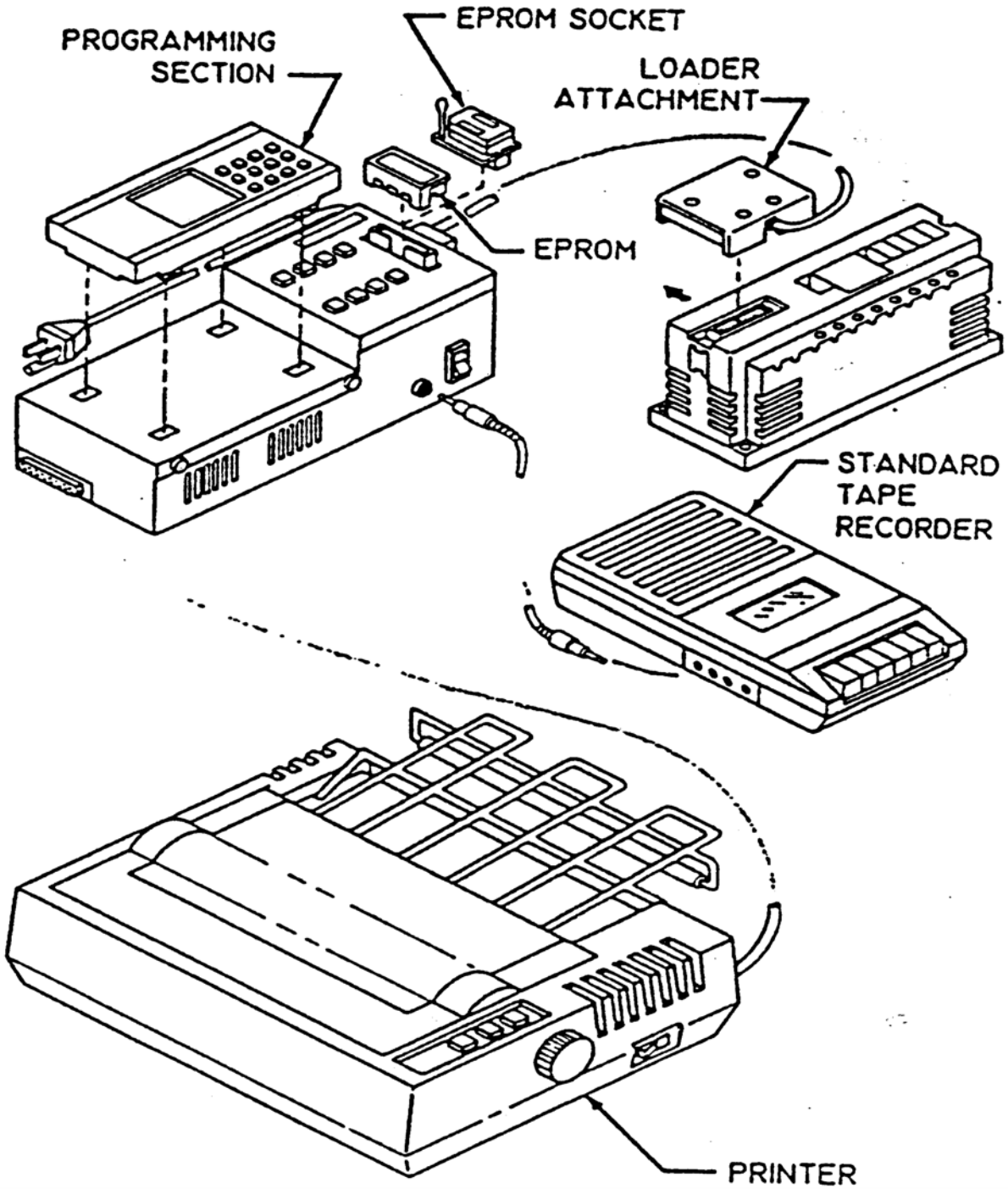

Figure 10-23 The program can be stored on a cassette for later use. Connecting the controller to a printer can make a copy of the program.

Programmable controllers can have many types of inputs, including pushbuttons, level switches, temperature controls, and photoelectric controls. Inputs such as pushbuttons and temperature controls are usually easy to input. However, more

complex solid state control inputs such as proximity and photoelectric inputs require special consideration because of their function.

Multiplexing

Multiplexing is a method of transmitting more than one signal over a single transmission system. As the distance increases between any transmitting and receiving point, the cost of multi-conductor cable with separate wires for each signal becomes very expensive through installation, maintenance and replacements. With multiplexing, a single-wire pair can serve multiple transmitters and receivers. A multiplexing system is ideal when used with programmable controllers, as all inputs and outputs can be connected with just one pair of wires. A multiplexing system is also called a ***two-wire system***.

Many advantages exist in using a multiplexing system for control. One of the main advantages is the elimination of costly hard wiring. Figure 10-24 illustrates how eight control switches are hard wired to control eight loads. A pair of wires connected through conduit is required for each control switch. This means that time and money would be wasted for even the shortest distance. As the distance between the control switches and loads increases, the cost of time and materials for the hard-wired circuit increases.

This same circuit can be connected using a multiplexing system. Only one pair of wires is required between the eight control switches and eight loads. Additional control switches to be added require no additional transmission wires. Additional transmitters, receivers, displays, or programmable controllers can all be connected to the same pair of wires.

As a control circuit increases in size and function, wiring it becomes more difficult. Figure 10-25 illustrates how a multiplexing system can send back a signal to indicate that the load is energised. The multiplexing system is much simpler than hard

wiring and can be expanded to almost any number of inputs and outputs, all controlled by a programmable controller. The programmable controller controls all inputs and outputs and makes timing, counting, and sequencing decisions and any other required logic decisions. Figure 10-18 illustrates how a programmable controller could be connected to the system.

The multiplexing system can be used to transmit both analogue and digital signals on the same two-wire system. This makes the system ideal for any instrumentation application, including the transmission and control of temperatures, BCD signals, rpm, voltage and current levels, and counts.

In addition, a 24-hour clock and printer can be added to the system for documentation. This addition would make it possible to print out the time of day when a certain event has taken place on the multiplexing system.

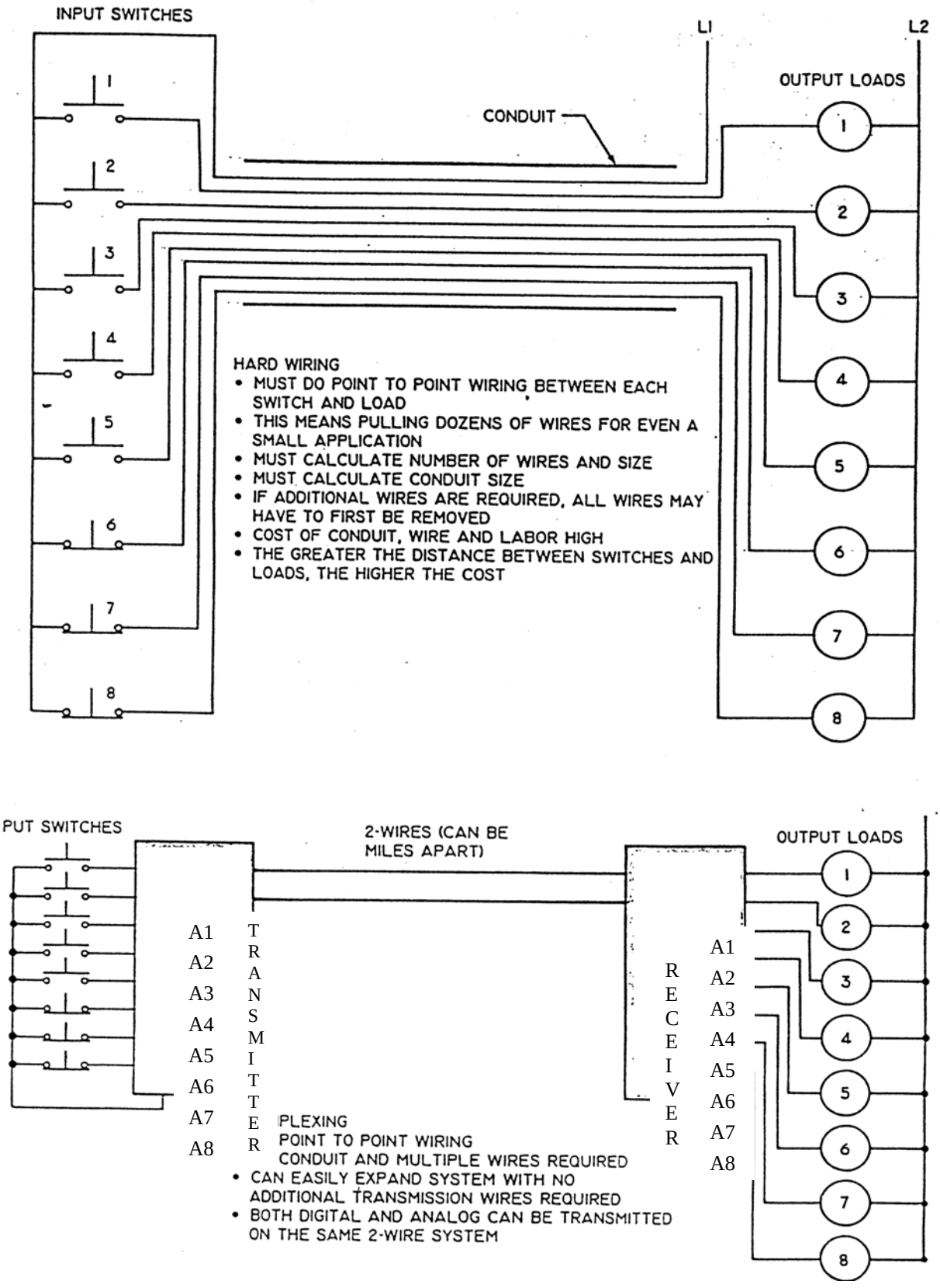

Figure 10-24 Multiplexing eliminates the need for costly hard wiring in a system

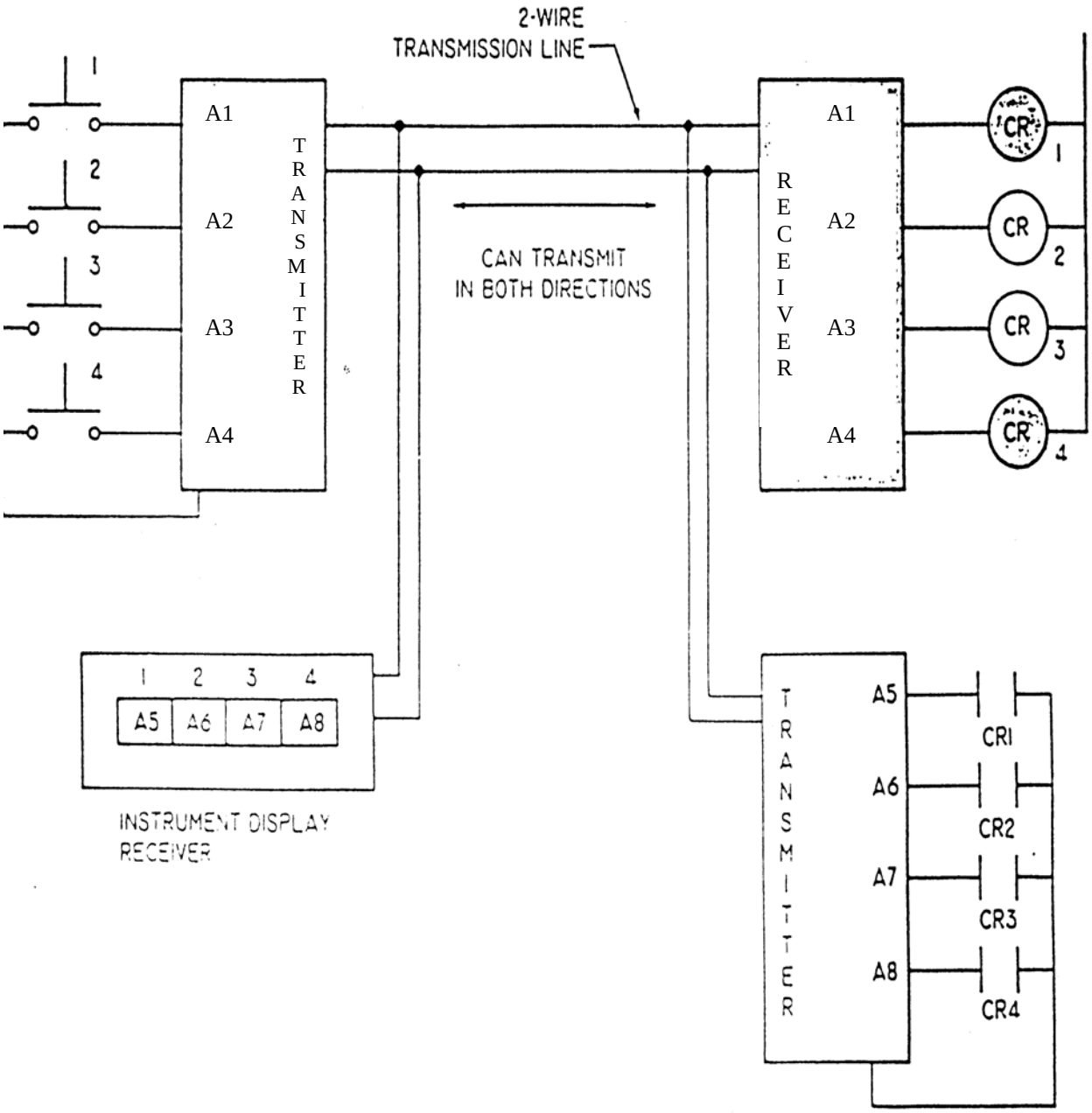

Figure 10-25. In this multiplexed system, four control switches to control four loads use signals sent back to indicate the loads are energised.

Applications of Programmable Controllers

Programmable controllers are useful in increasing production, and improving overall plant efficiency. Programmable controllers can control individual machines and link the machines together into a system. The flexibility provided by a programmable controller has resulted in many applications in manufacturing and process control. Process control has gone through many changes in the past few years. In the past, process control was mostly accomplished by manual control. Flow, temperature, level, pressure and other control functions were monitored and controlled at each stage, by production workers. Today, using programmable controllers, an entire process can automatically be monitored and controlled with little or no workers involved at all. Following is a list of a few process applications in which programmable controllers have been used.

1. Grain operations involving storage, handling, and bagging.

2. Syrup refinery involving product storage tanks, pumping, filtration, clarification, evaporators, and all fluid distribution systems.

3. Fats and oils processing involving filtration units, cookers, separators, and all charging and discharging functions.

4. Dairy plant operations involving all process control from raw milk delivered to finished dairy products.

5. Oil and gas production and refinement from the well pumps in the fields to finished product delivered to the customer.

6. Bakery applications from raw material to finished product.

SGC HSE Safety Regulation Related to PLC

Refer to HSE Regulation No. 19 & 20 "Working with Electricity"

19.7.4 Precautions on Low Voltage Systems

1. The consequences of shock, or serious burns, from short circuits associated with low voltage systems (50 - 1000V ac/120 - 1500V dc between conductors, or 50 - 600V ac/120 - 900V dc between conductor and earth) can be serious and often fatal. Whenever possible therefore, work on low voltage equipment and cables shall be carried out after they are proved **DEAD** by use of an approve instrument and where appropriate **EARTHED** using an Electrical Isolation certificate (refer to Paragraph 19.8).

2. If it is not possible to make **DEAD,** to prove **DEAD** and where appropriate **EARTH** low voltage systems, work on them shall be carried out as if they were **LIVE** using a Sanction For Test Certificate (refer to Paragraph 19.9).

19.7.5 Precautions on Extra Low Voltage Systems

1. Control and telecommunications plant operating at extra low voltage (< 50V ac/120V dc between electrical conductors or to earth) shall not be worked on without an Electrical Isolation Permit being issued. This is necessary to prevent the possibility of sparks in a hazardous area (refer to Paragraph 19.8).

2. Battery systems with high stored energy can be dangerous to personnel and therefore precautions should be taken when working with such systems. In particular flooded cells requiring electrolyte replacement are hazardous. Where these types of cells exist, a local procedure should be produced for work on battery systems.

19.11 STATIC ELECTRICITY PRECAUTIONS

Some of the risks from static electricity and suggested precautions are listed as follows:

a. Static electricity is readily generated by personnel clothing, especially man-made fibres. The human body can accumulate a static charge in excess of 10,000 volts, although when discharged, it is short-lived and of low temperature.

b. Hydrocarbons may become charged with static electricity from pumping, filtering, splash filling, or by settling out of water through them. High velocity flow rates increase static generation and reduce the opportunity for charge relaxation which may result in sparking. A low flow rate assists by reducing charge separation in the fluid (and hence charge accumulation) and may allow charge to migrate to earth, hence reducing the risk of sparking.

c. When pouring flammable low-conducting fluids from a container to a receptacle, e.g. taking crude oil samples, the container, receptacle and funnel, if used, must be bonded together and to earth. All equipment should be of metal. Recipient vessels and loading nozzles or hoses should be bonded to earth during transfer operations.

d. Where practicable, inert gas blankets should be maintained over the liquid in storage when filtering operations take place.

e. Other items in common usage which, may cause static electricity build up if not properly earthed are grit blasting and even fine water sprays used for fire fighting. Safeguards should include bonding of nozzles and the use of anti-static hoses.

f. Electronic equipment can be very sensitive to electrostatic discharge.

Suitable precautions such as the use of earthed wrist straps should be Used when handling electrostatic sensitive electronic equipment (including packing and unpacking). Wristband cords shall be checked prior to use.